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Shelton, IV et al.

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(54) **ELECTROSURGICAL ADAPTATION
TECHNIQUES OF ENERGY MODALITY FOR
COMBINATION ELECTROSURGICAL
INSTRUMENTS BASED ON SHORTING OR
TISSUE IMPEDANCE IRREGULARITY**

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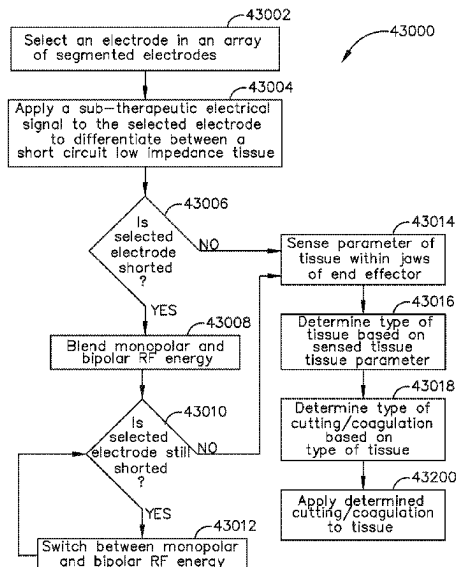
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(57) **ABSTRACT**

Disclosed is a method of adapting energy modality due to a
short circuit or tissue type grasped in the jaws of an end
effector of a surgical instrument and a system with a control
circuit configured to execute the method. The method
includes determining a first and second electrode of an array
of segmented electrodes are shorted to each other, blending
monopolar and bipolar RF energy to the array of segmented
electrodes, and determining that the first and second elec-
trodes remain shorted after the blending step. Additionally,
or alternatively, the method may include determining that
the first and second electrodes are no longer shorted after the
blending step, sensing an electrical parameter of tissue
grasped within jaws of the end effector, and determining an
energy sealing and cutting procedure based on the sensed
electrical parameter of the tissue.

20 Claims, 189 Drawing Sheets



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10,856,870 B2	12/2020	Harris	11,672,605 B2	6/2023	Messerly
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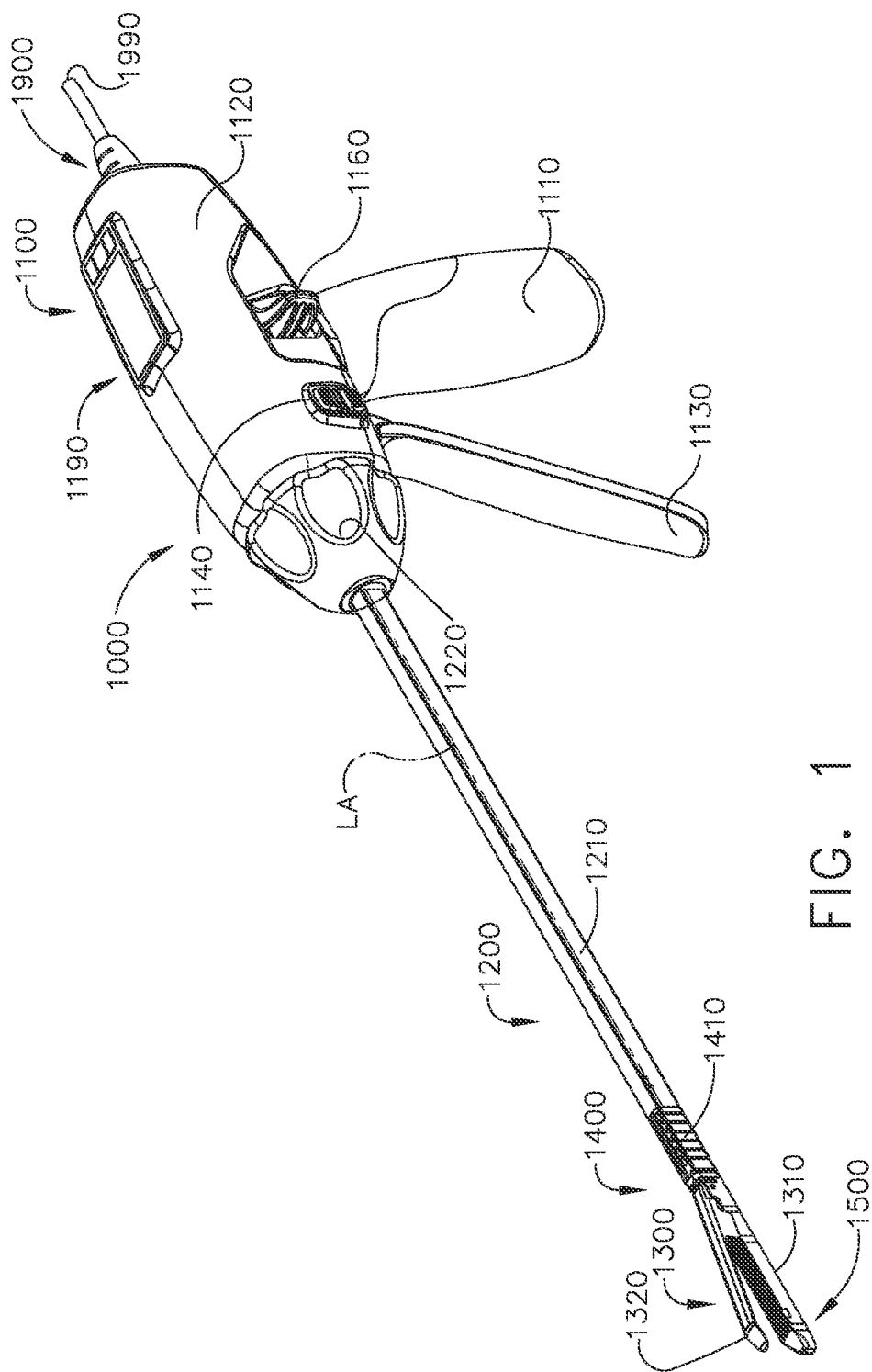


FIG. 1

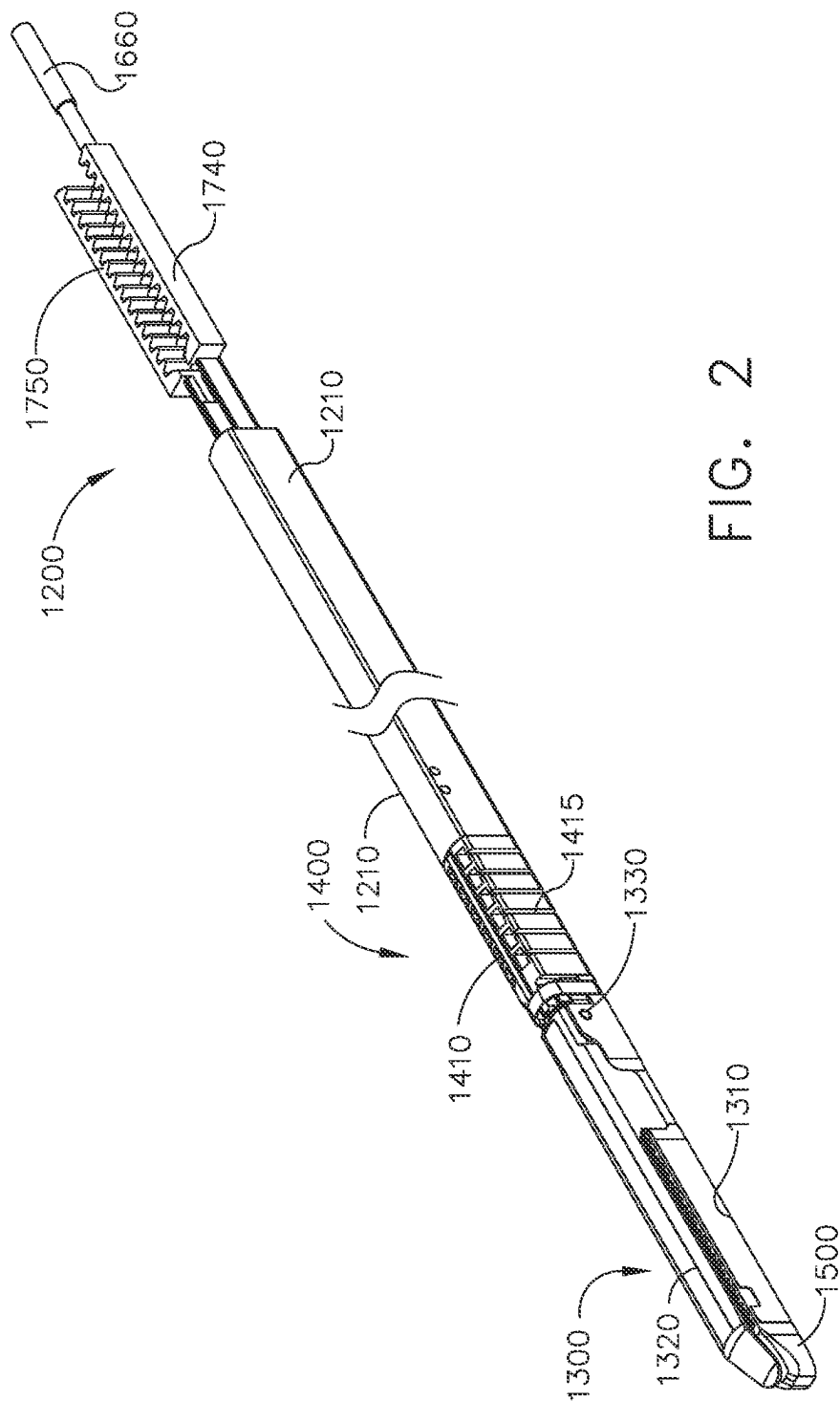
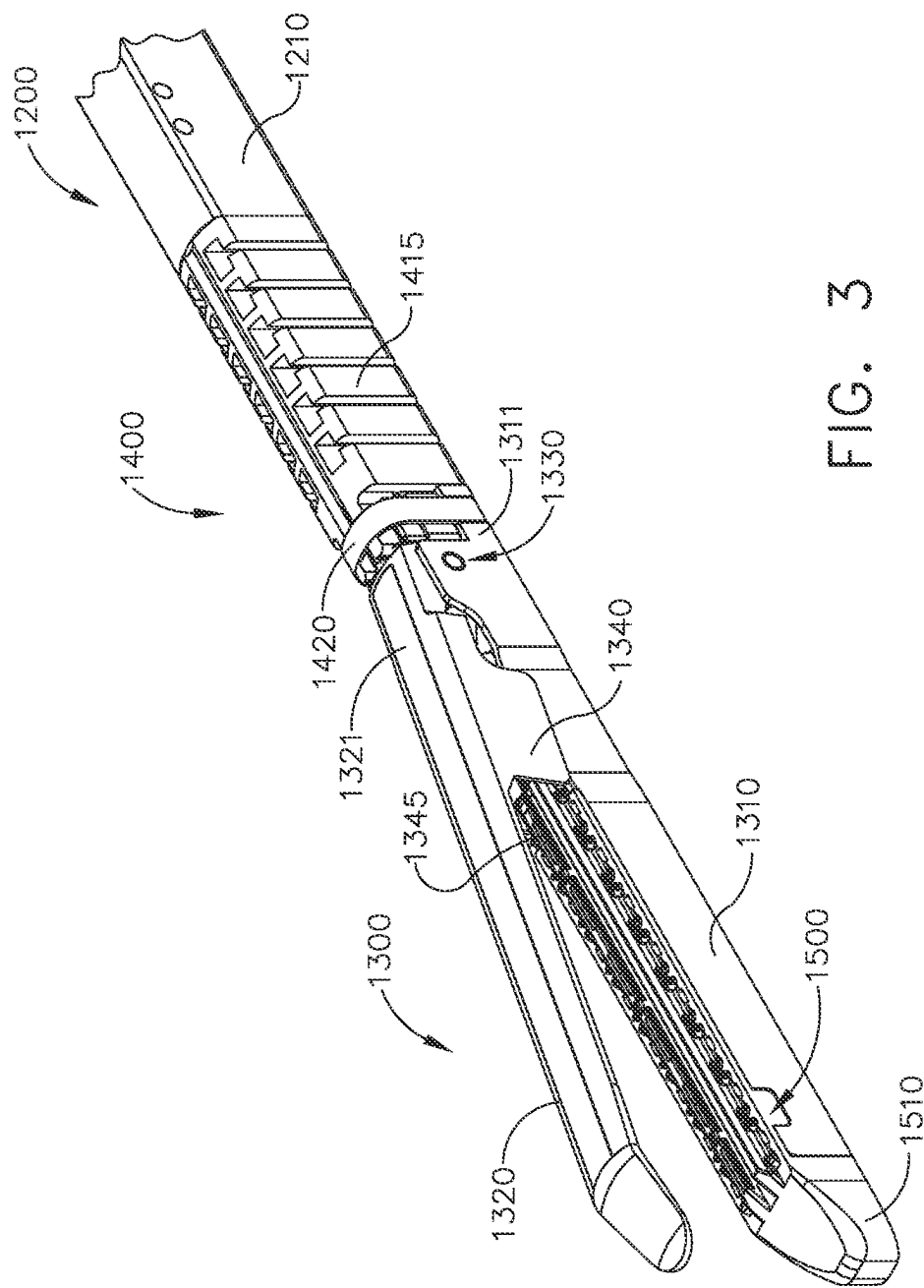


FIG. 2



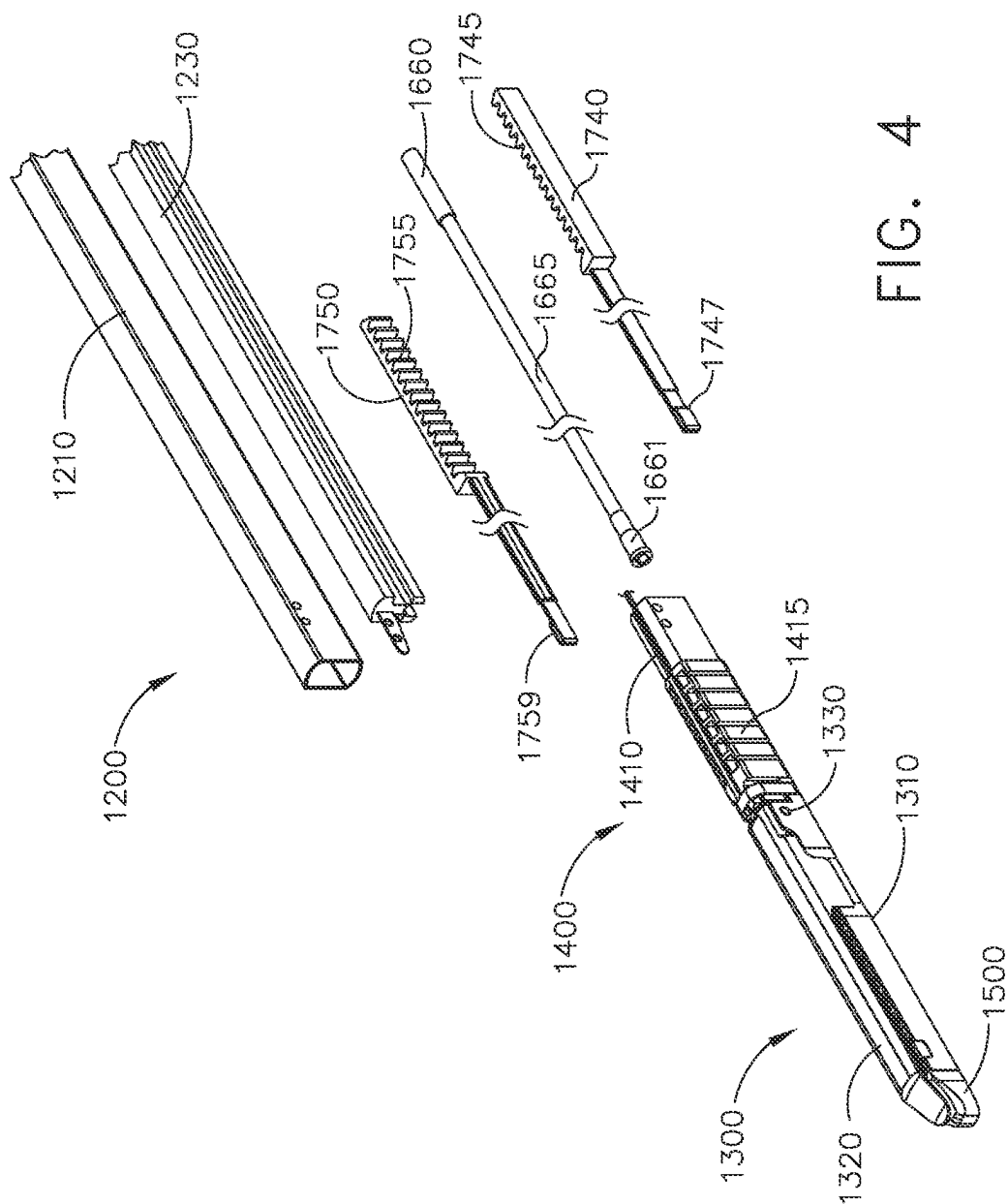
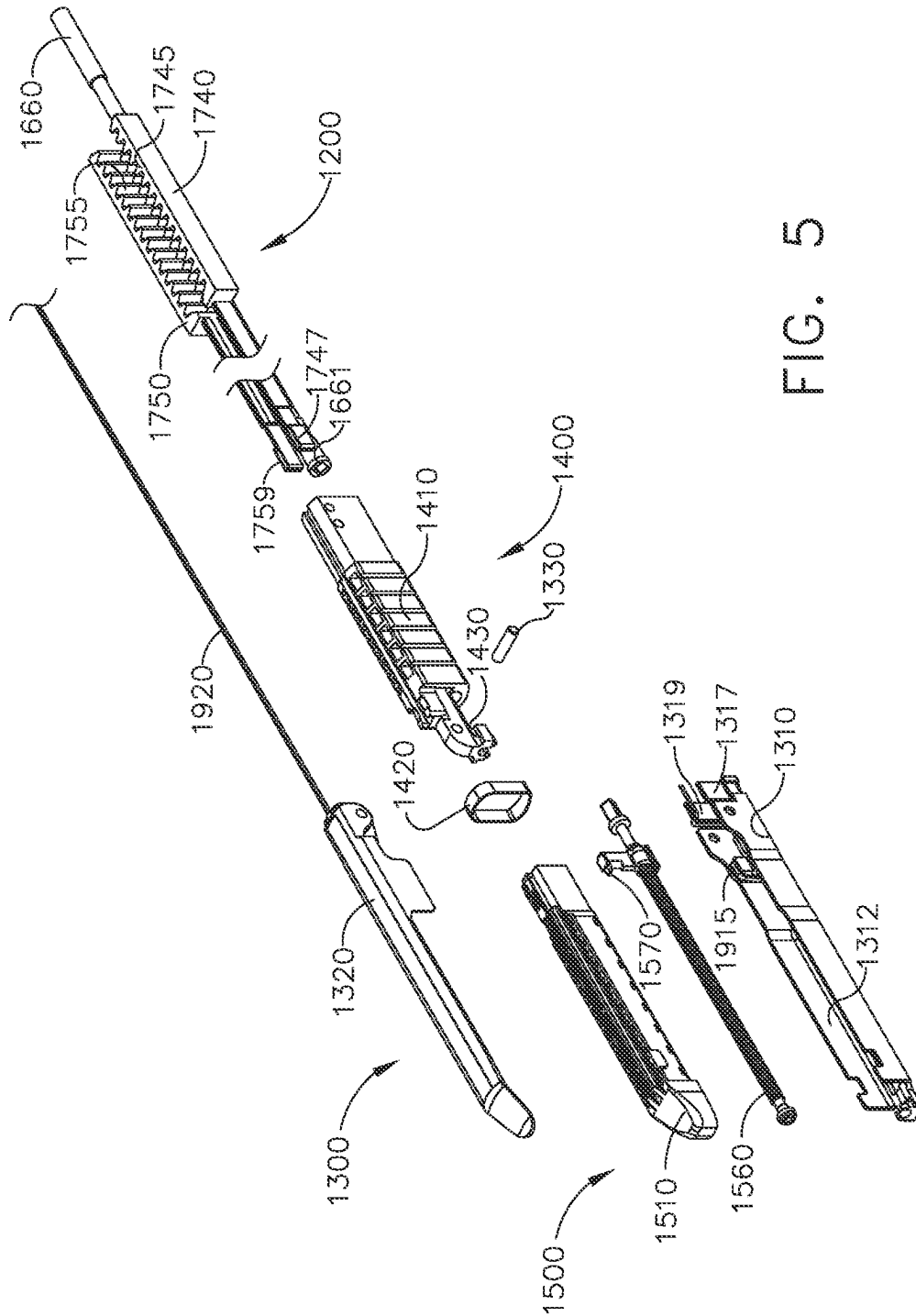


FIG. 4



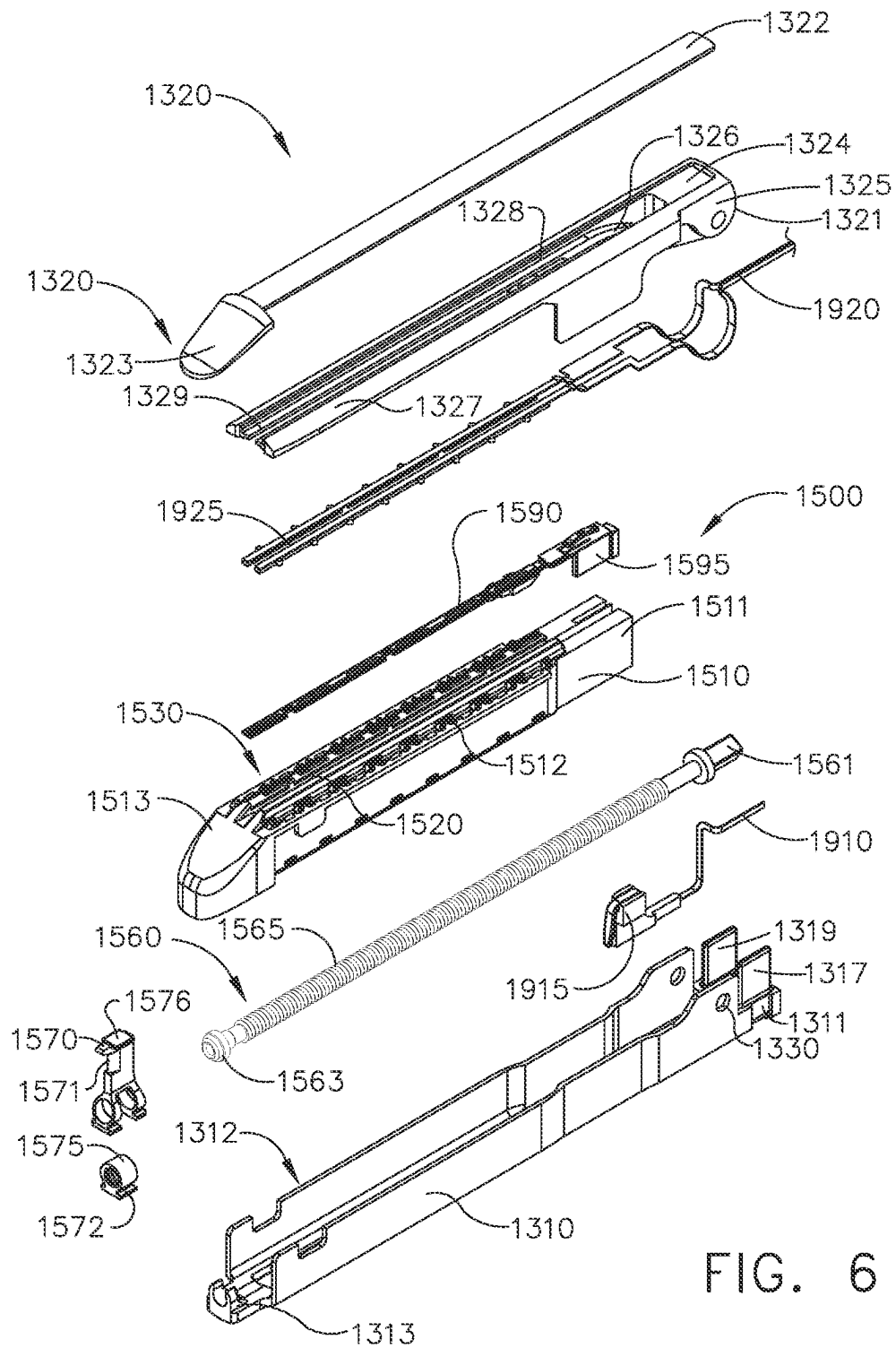


FIG. 6

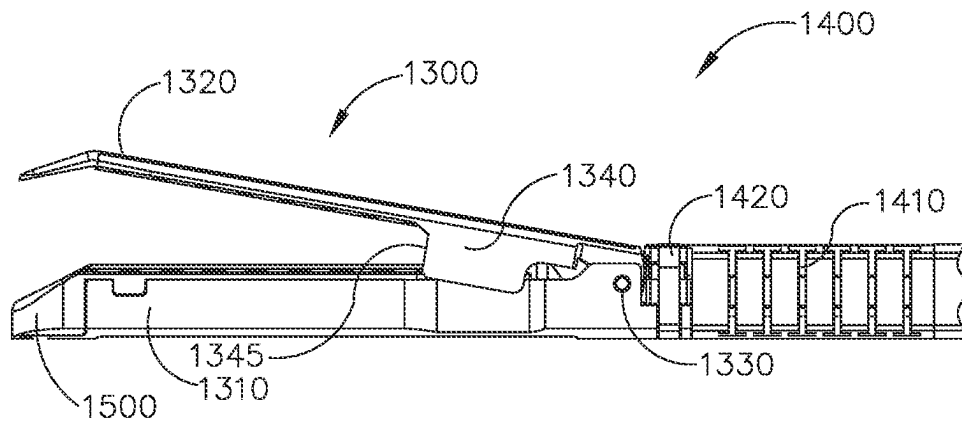


FIG. 7

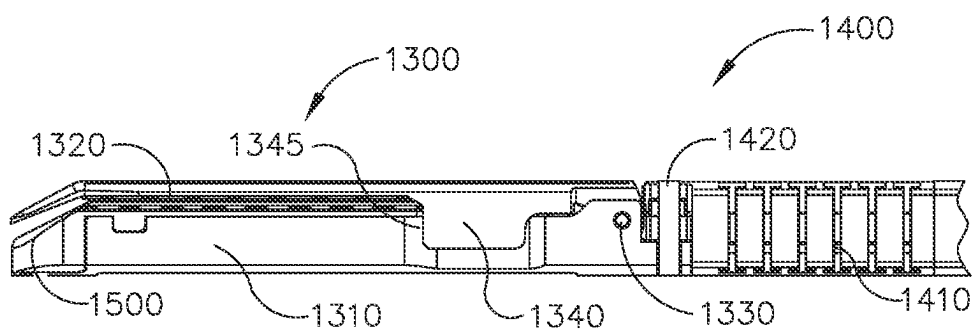


FIG. 8

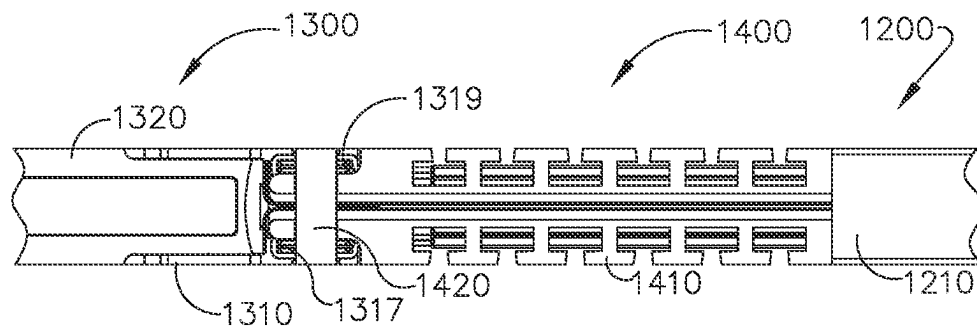


FIG. 9

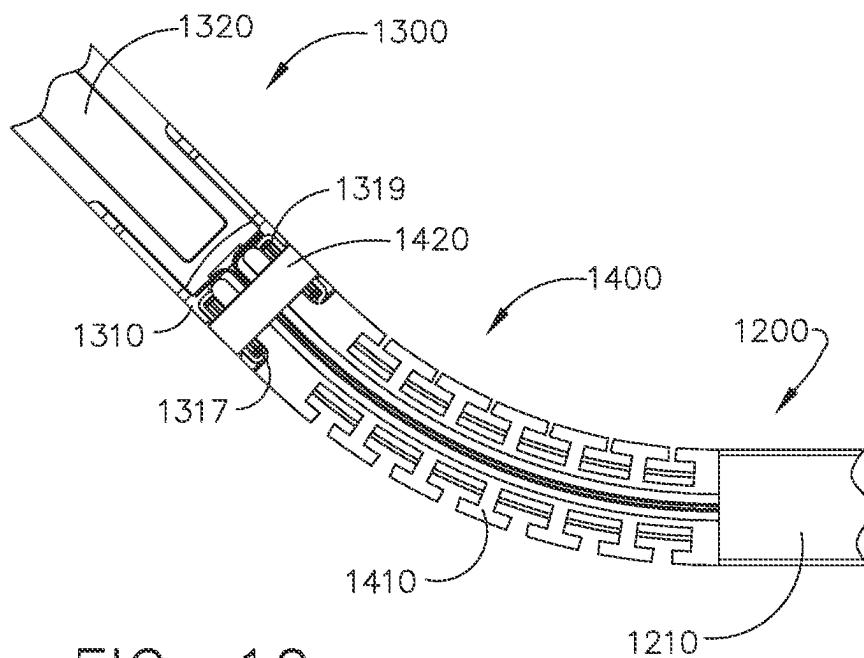


FIG. 10

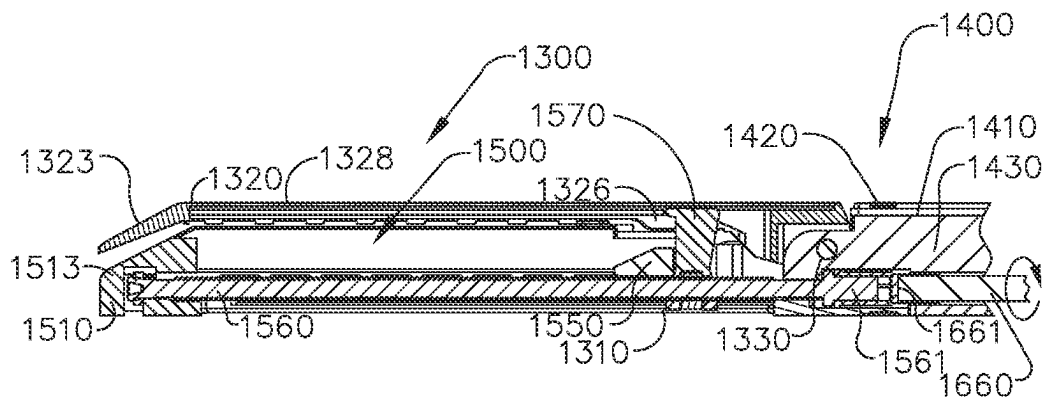


FIG. 11

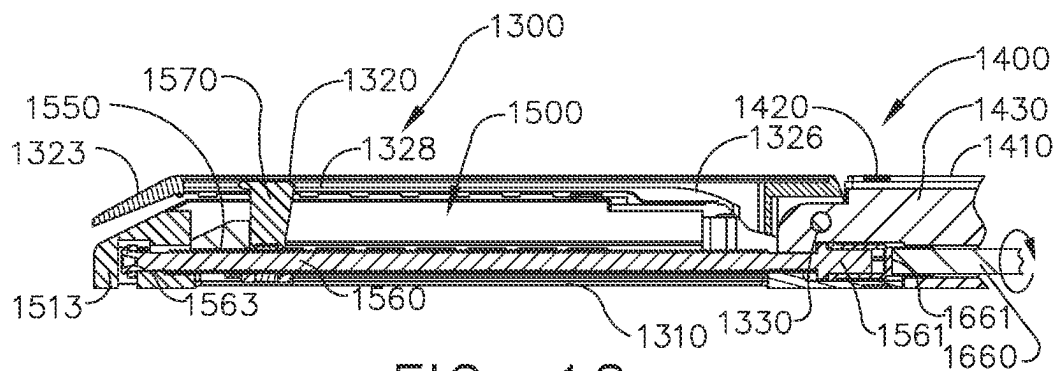


FIG. 12

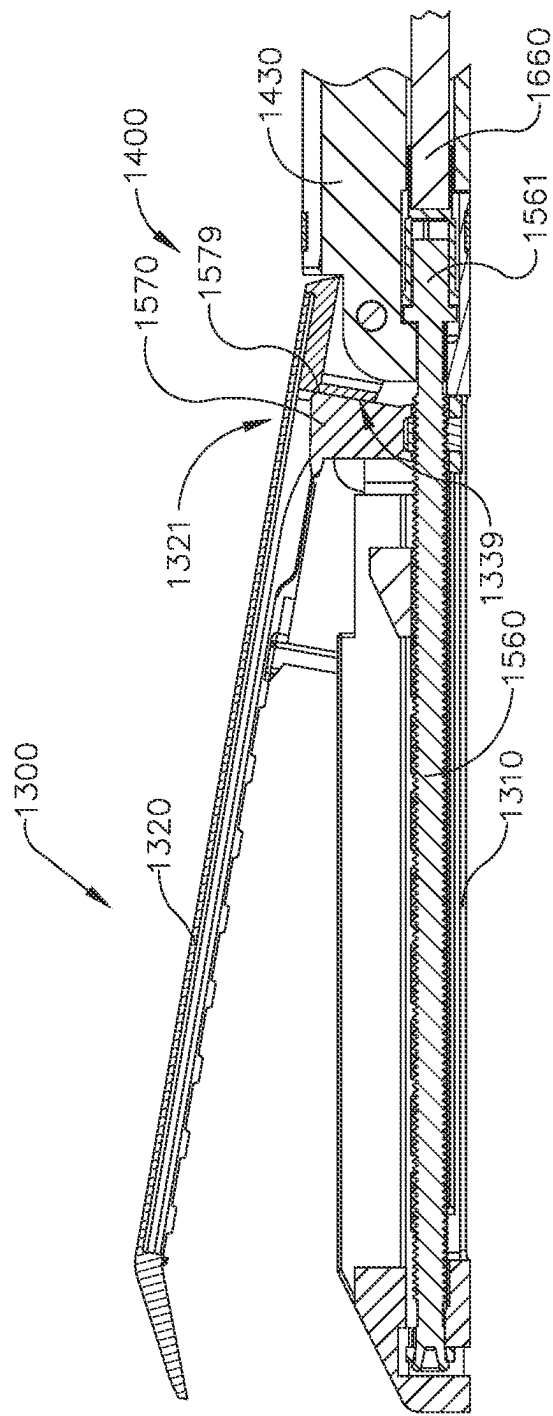


FIG. 12A

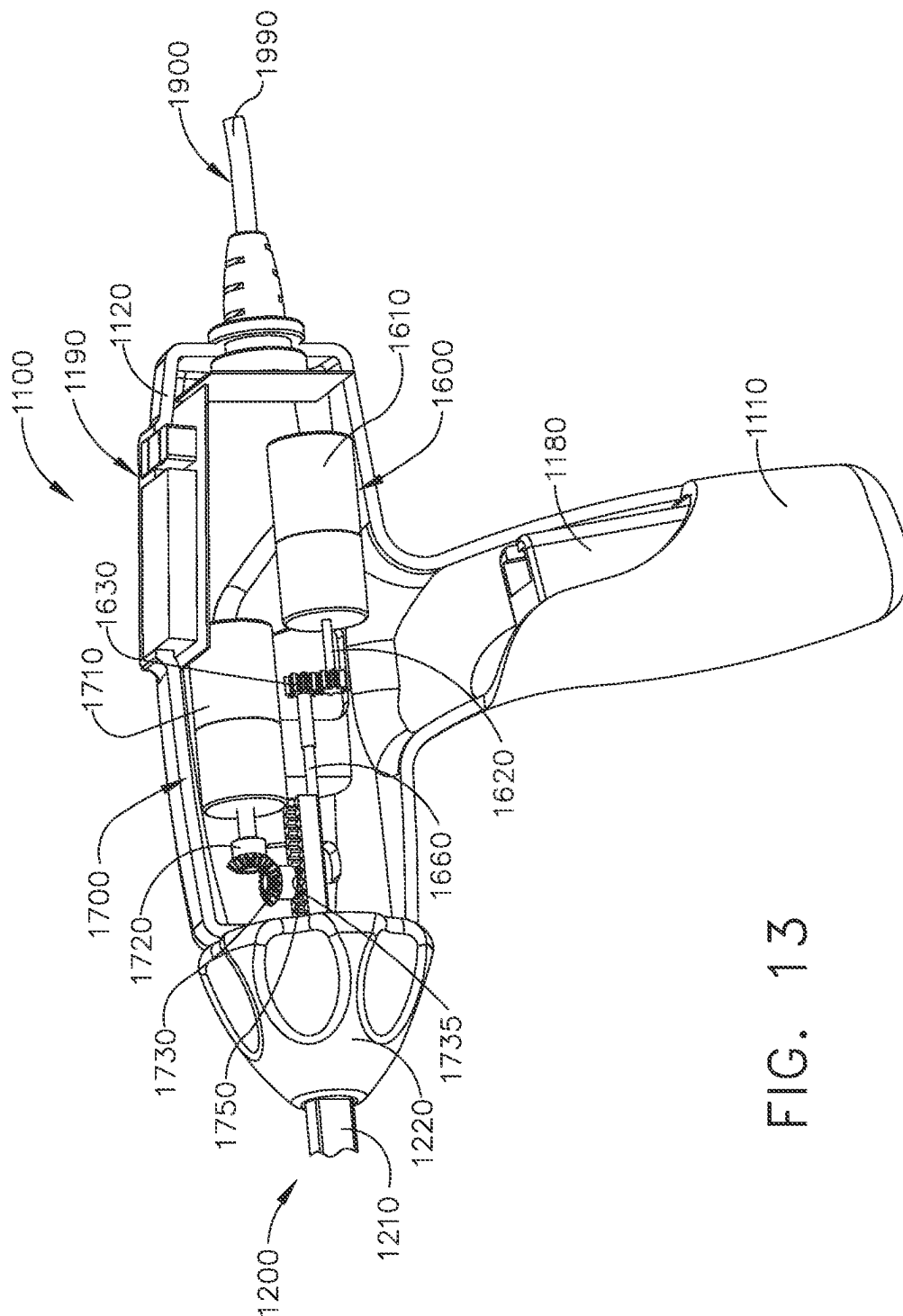


FIG. 13

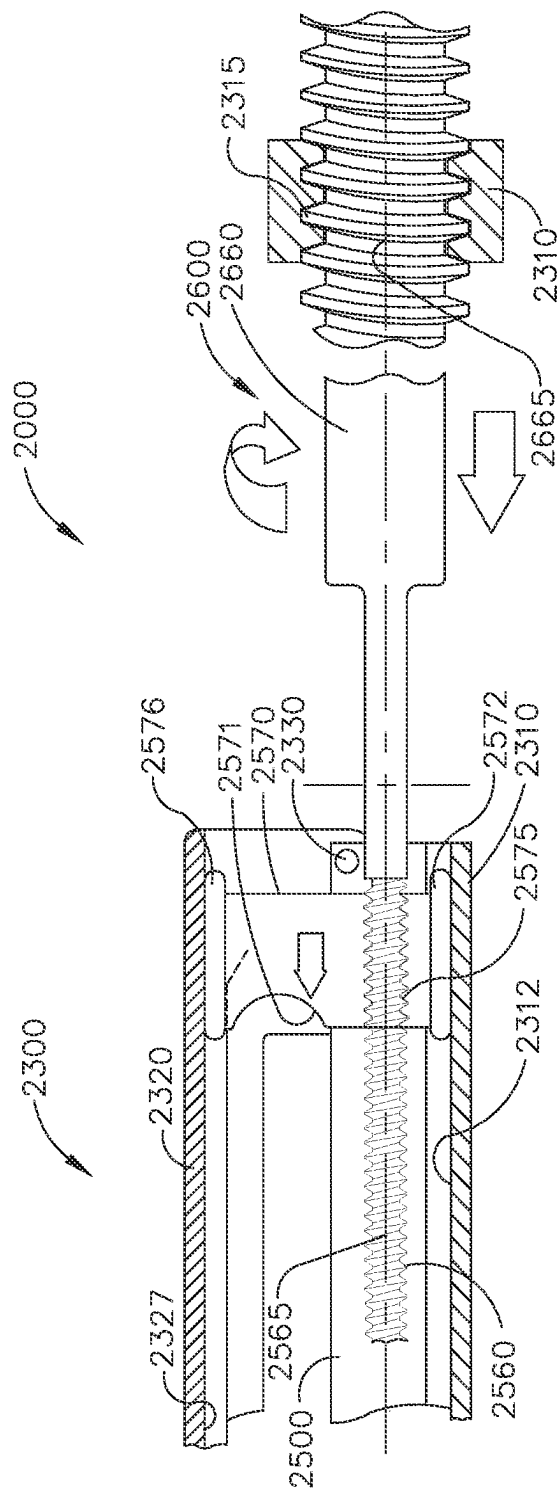


FIG. 14

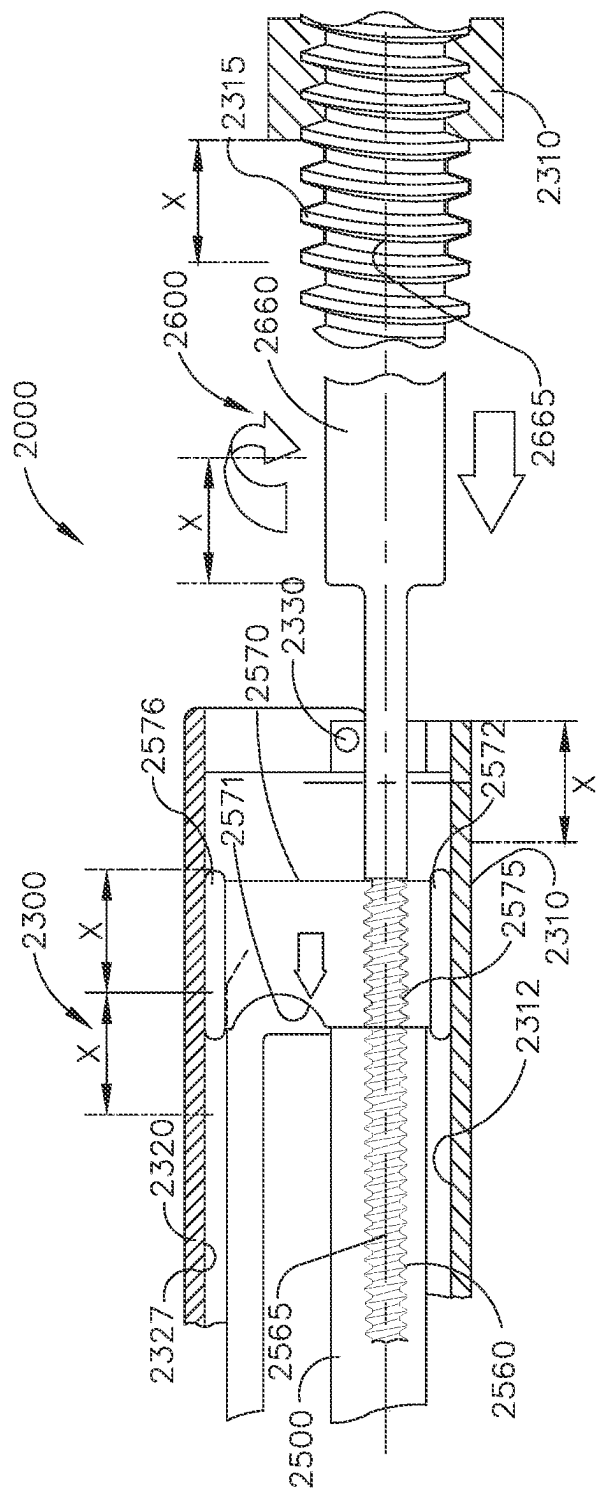


FIG. 15

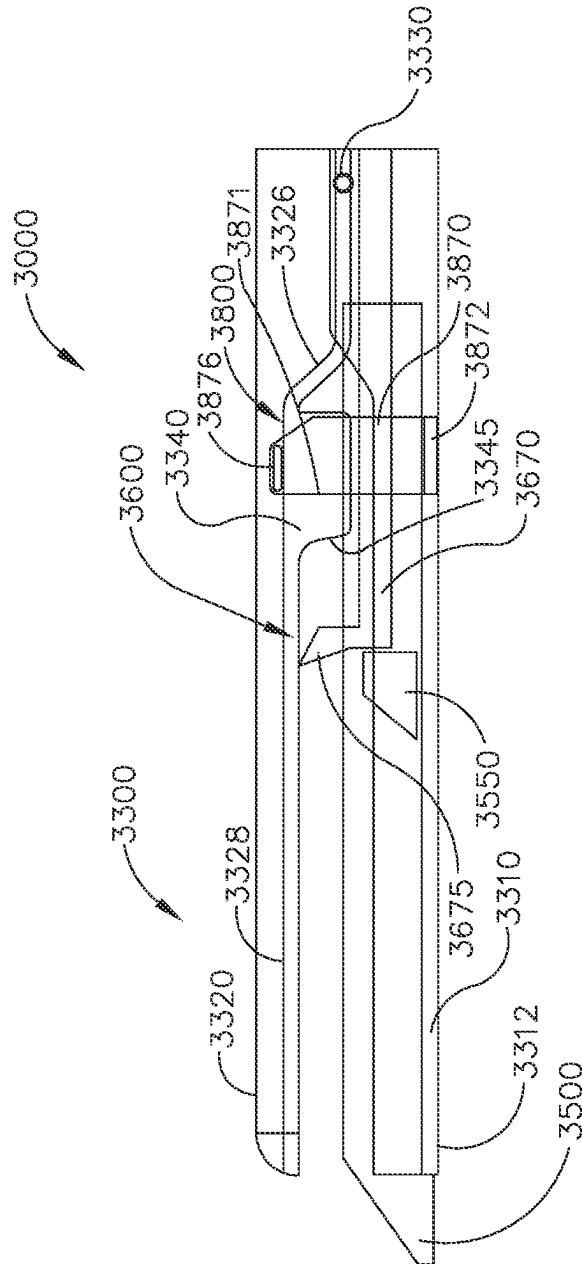
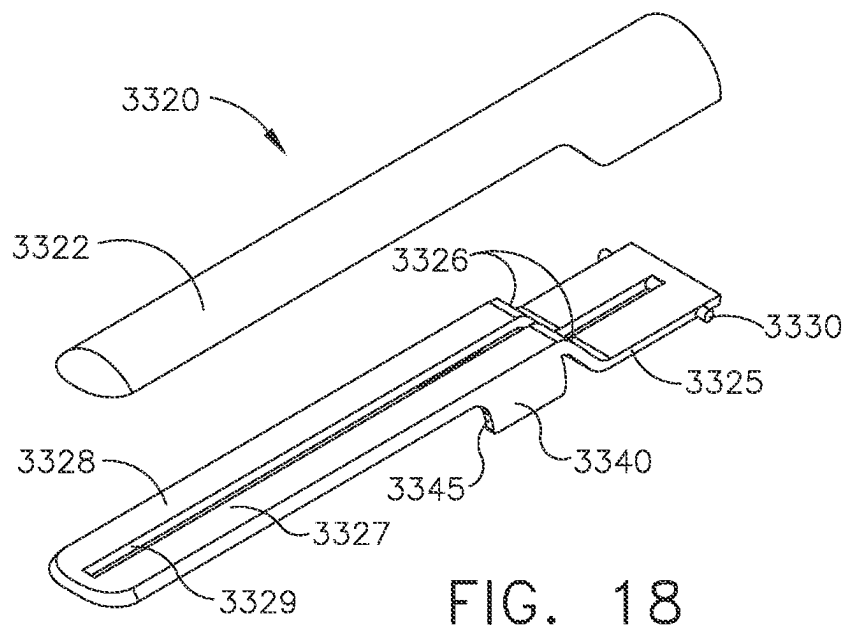
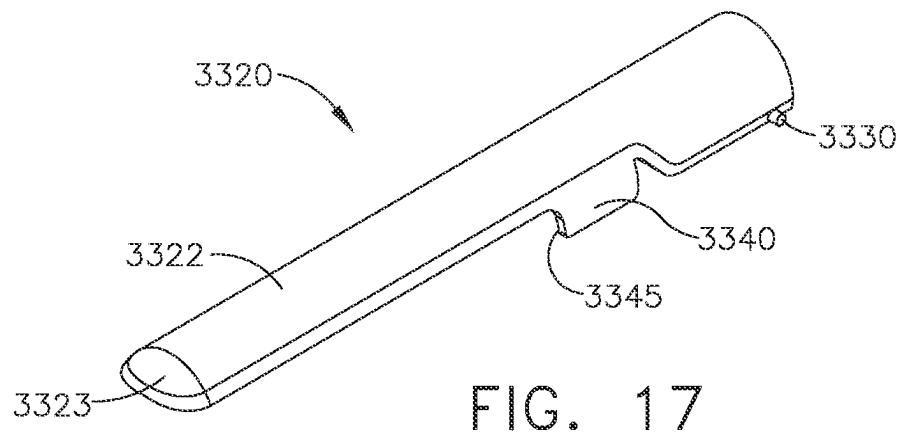


FIG. 16



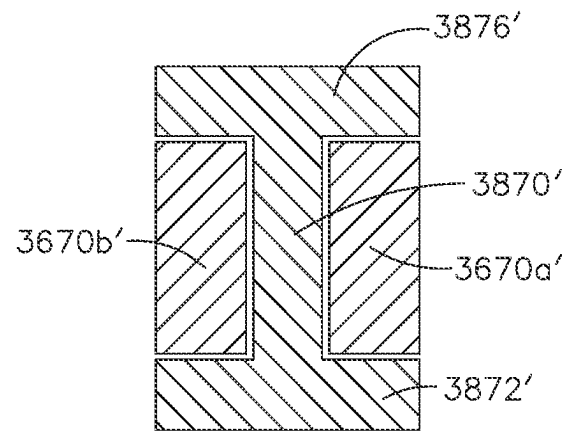


FIG. 20

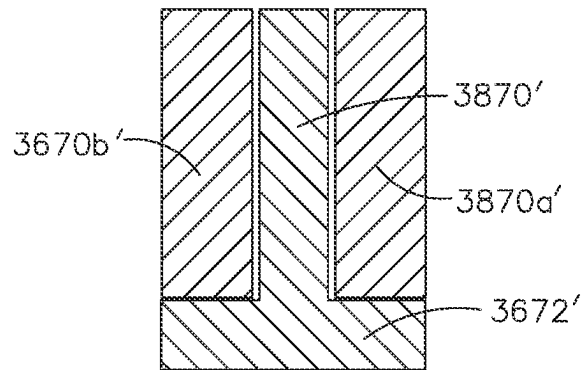


FIG. 19

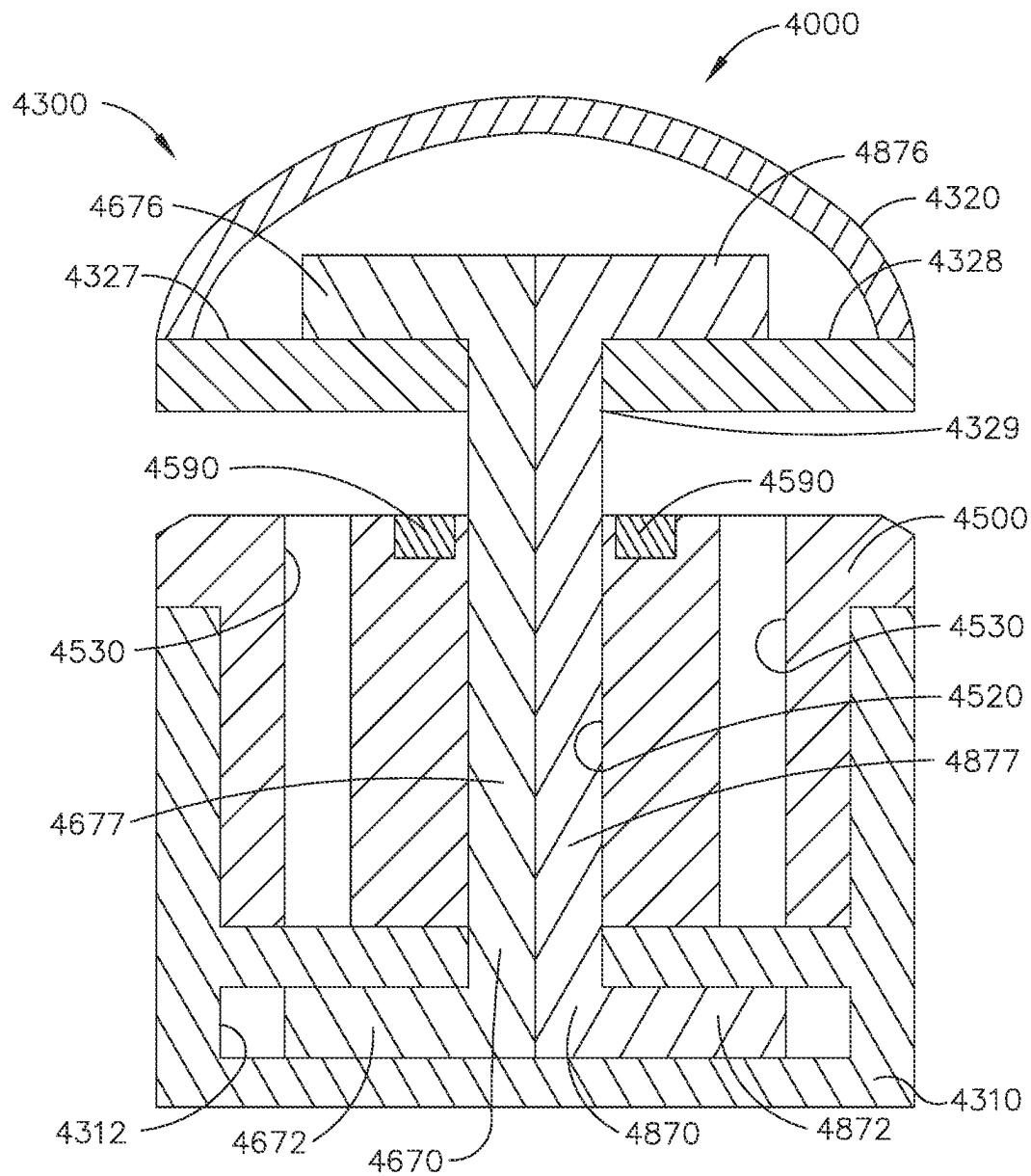


FIG. 21

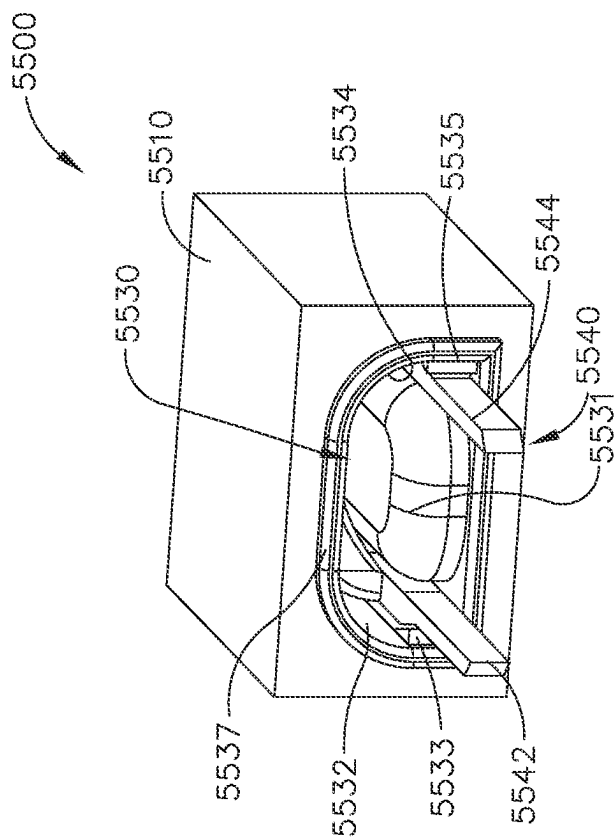
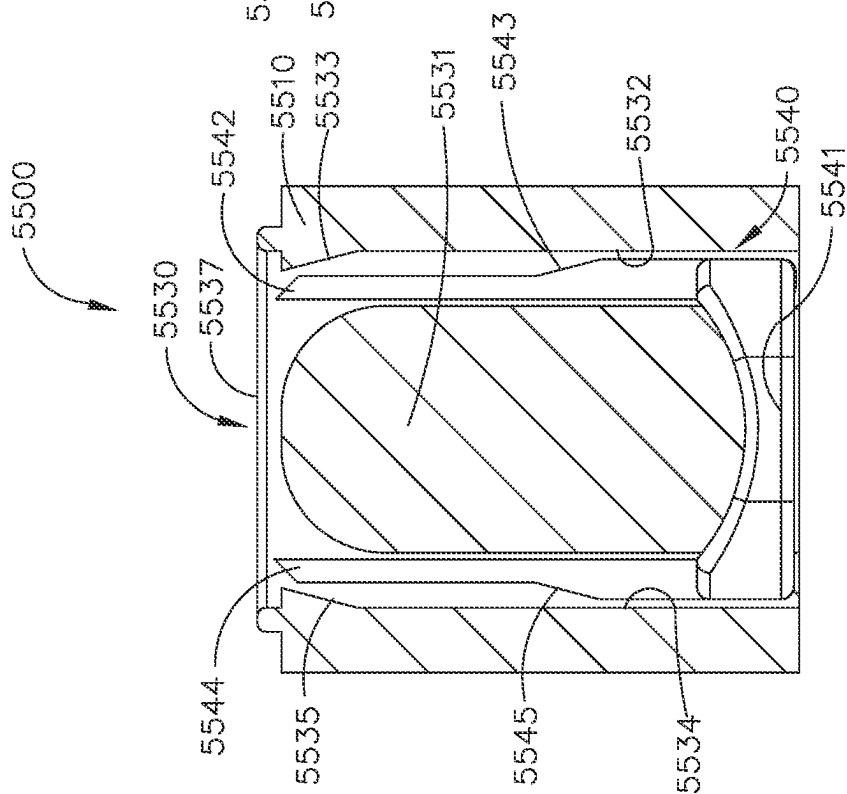
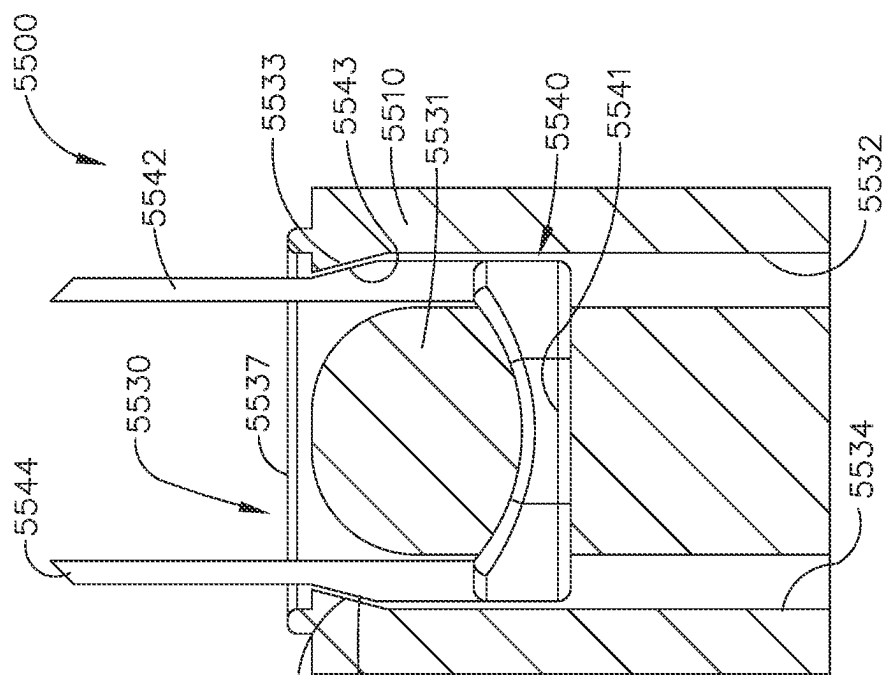


FIG. 22



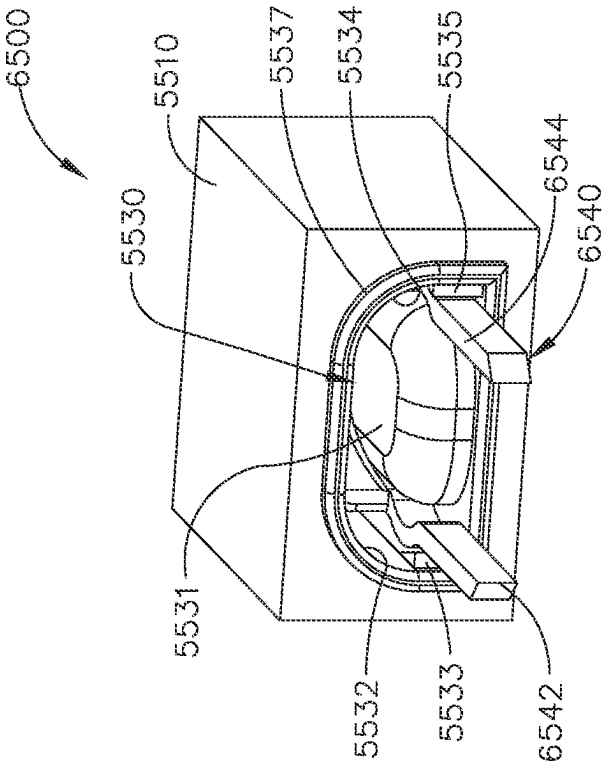


FIG. 25

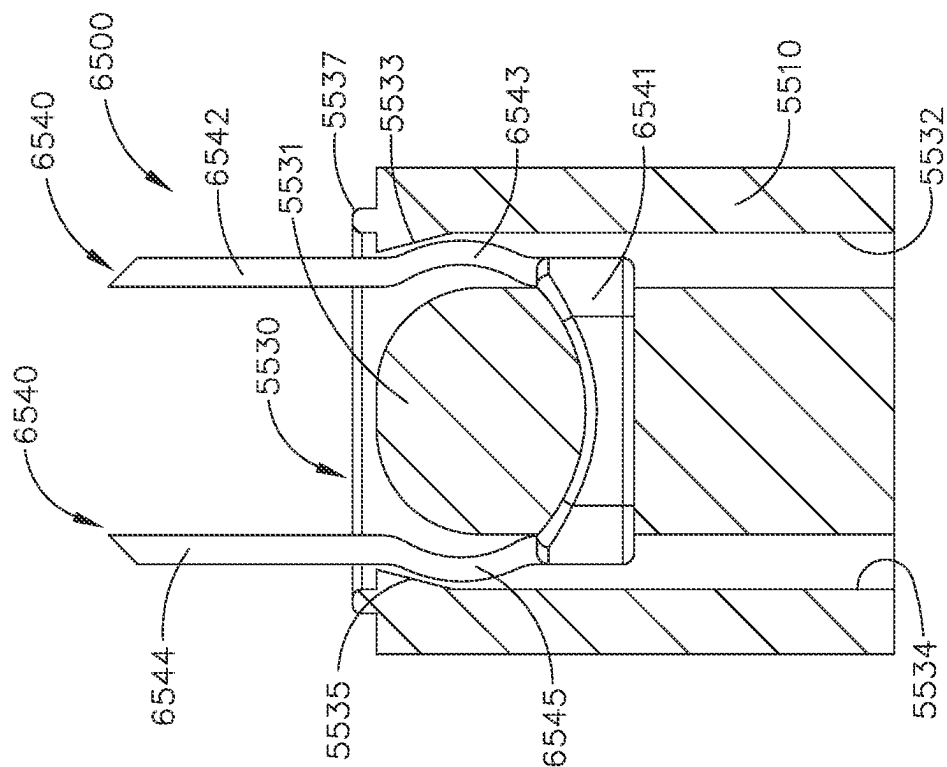


FIG. 27

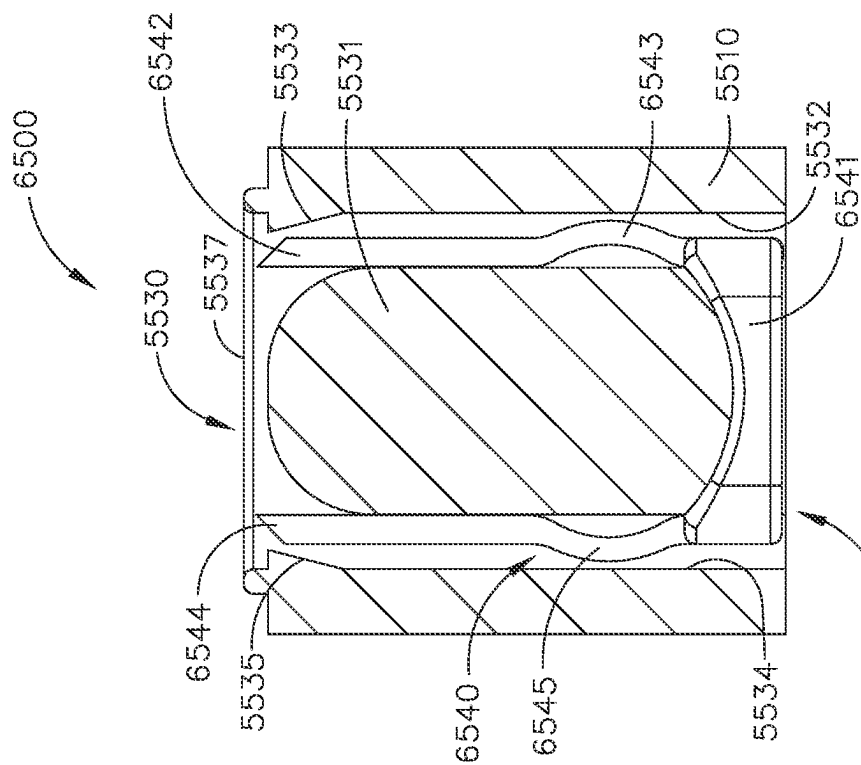
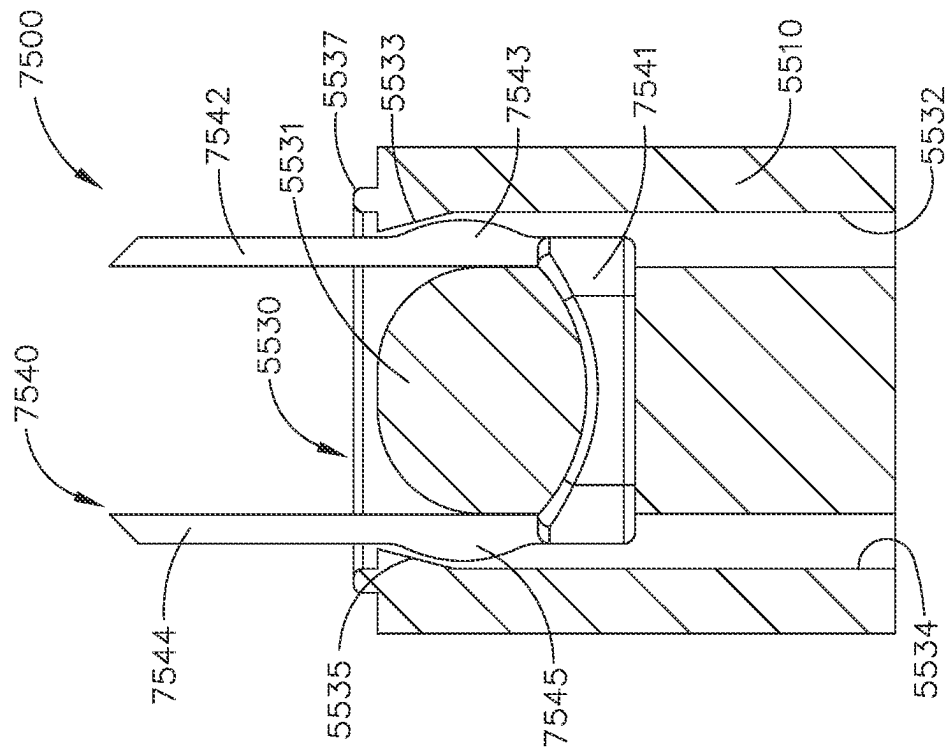
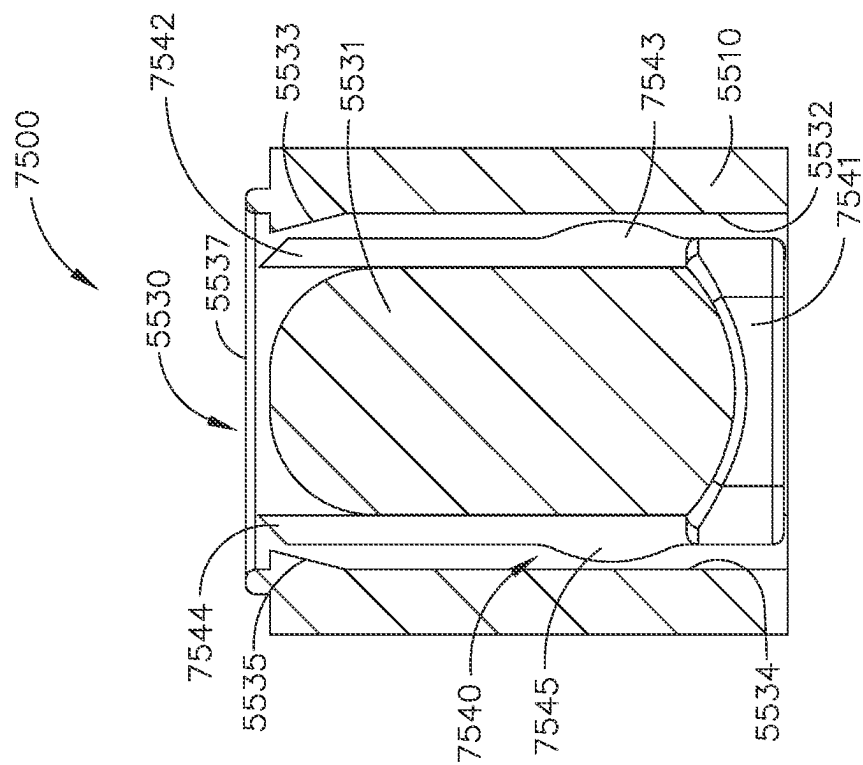


FIG. 26



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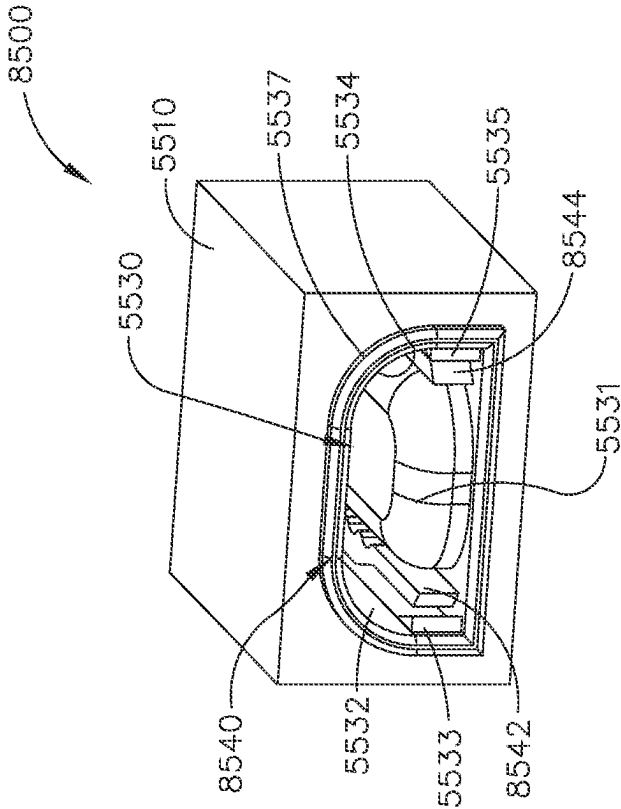


FIG. 30

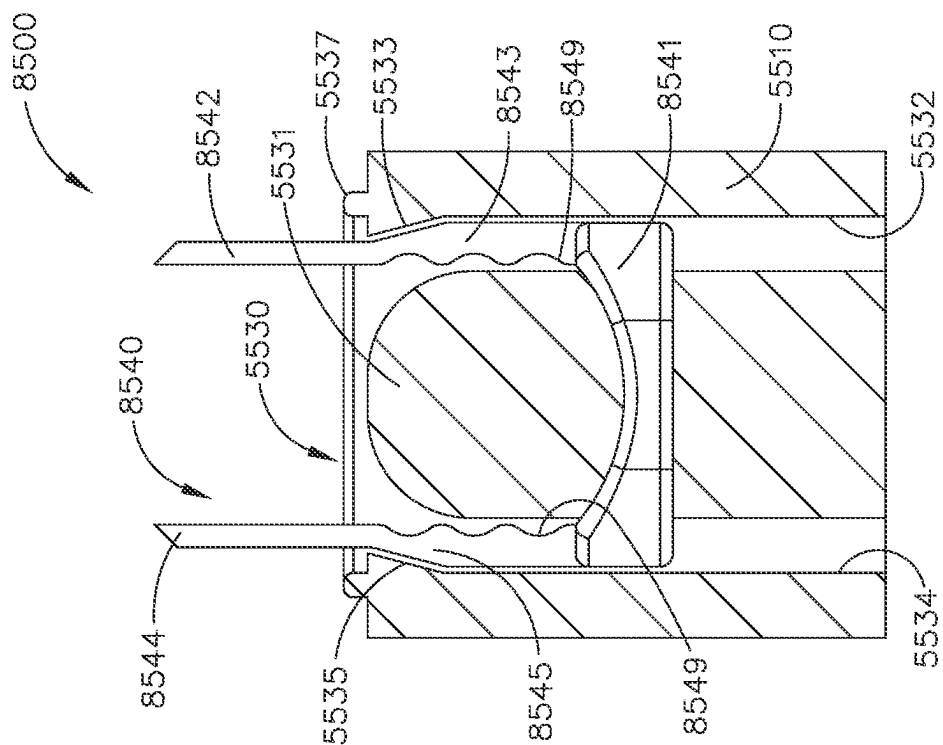


FIG. 32

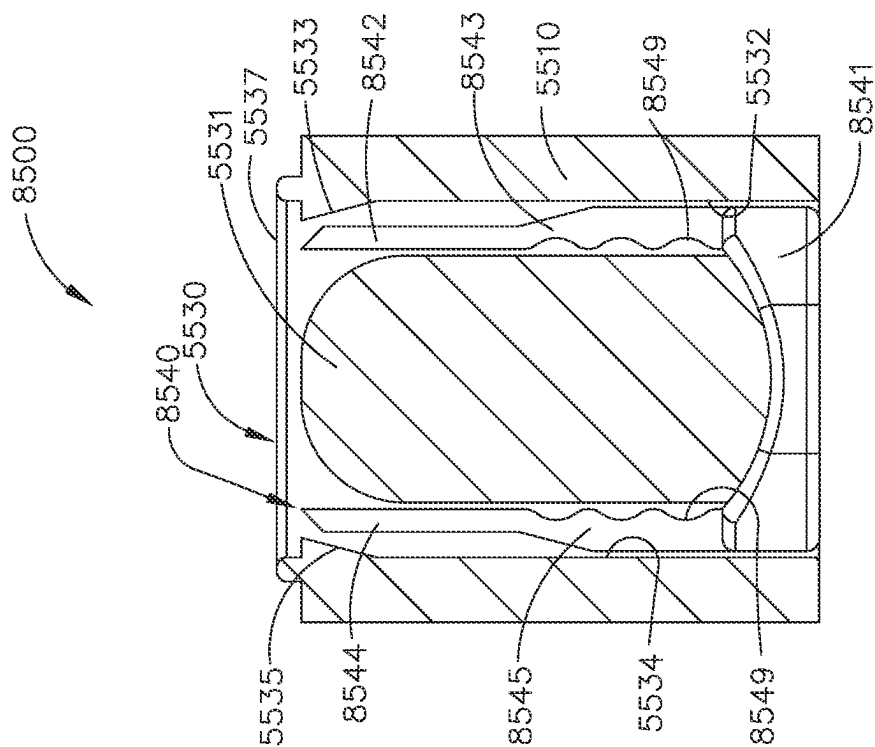
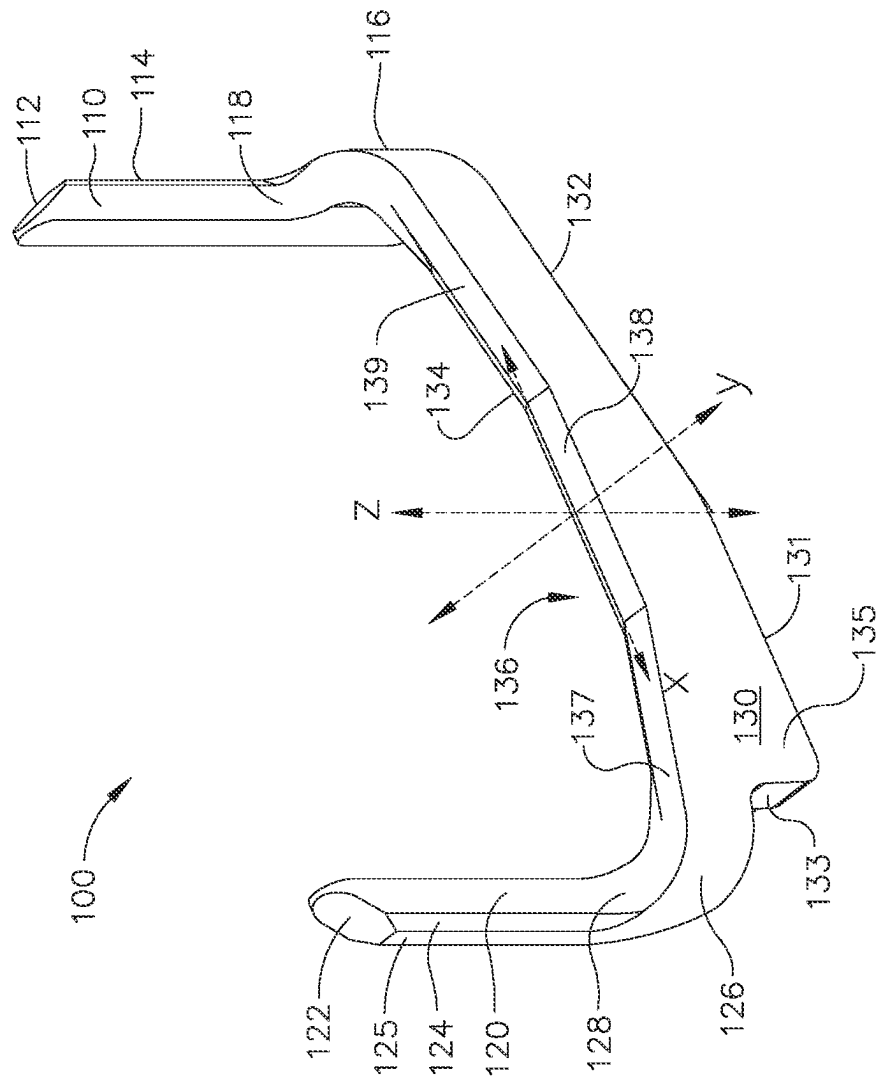


FIG. 31



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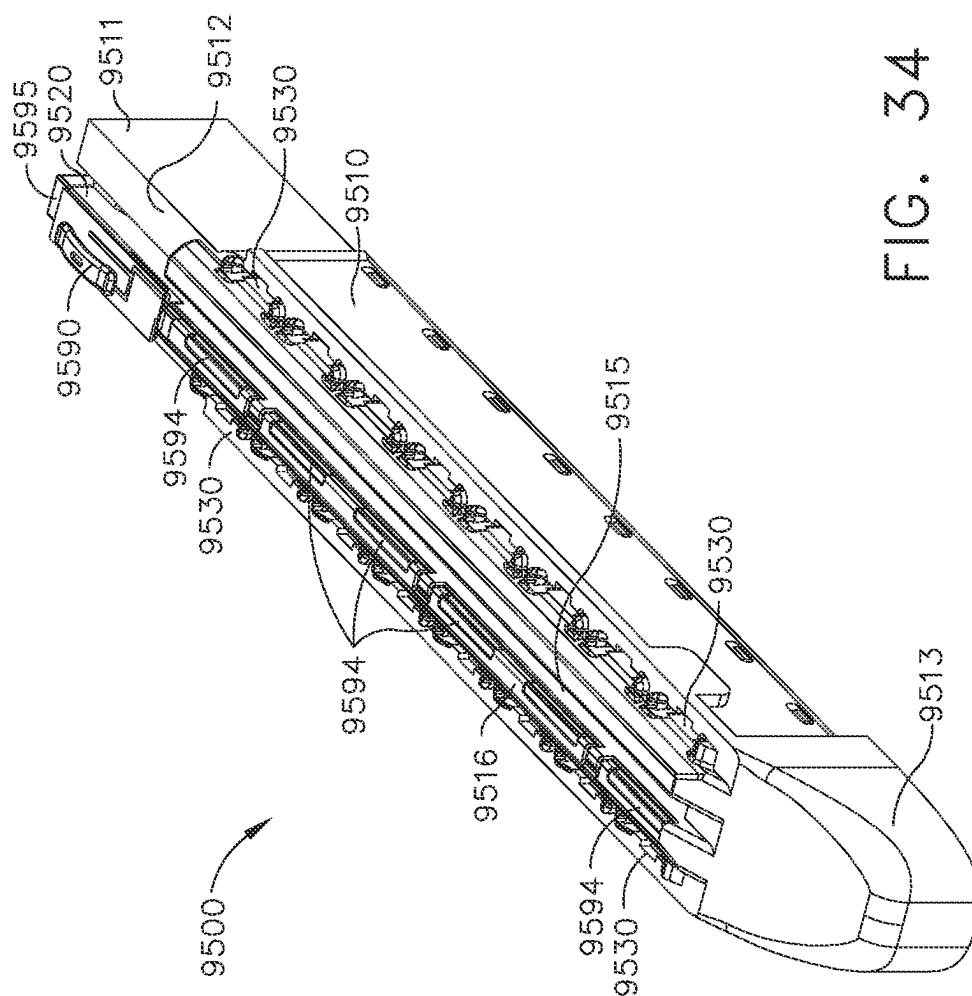


FIG. 34

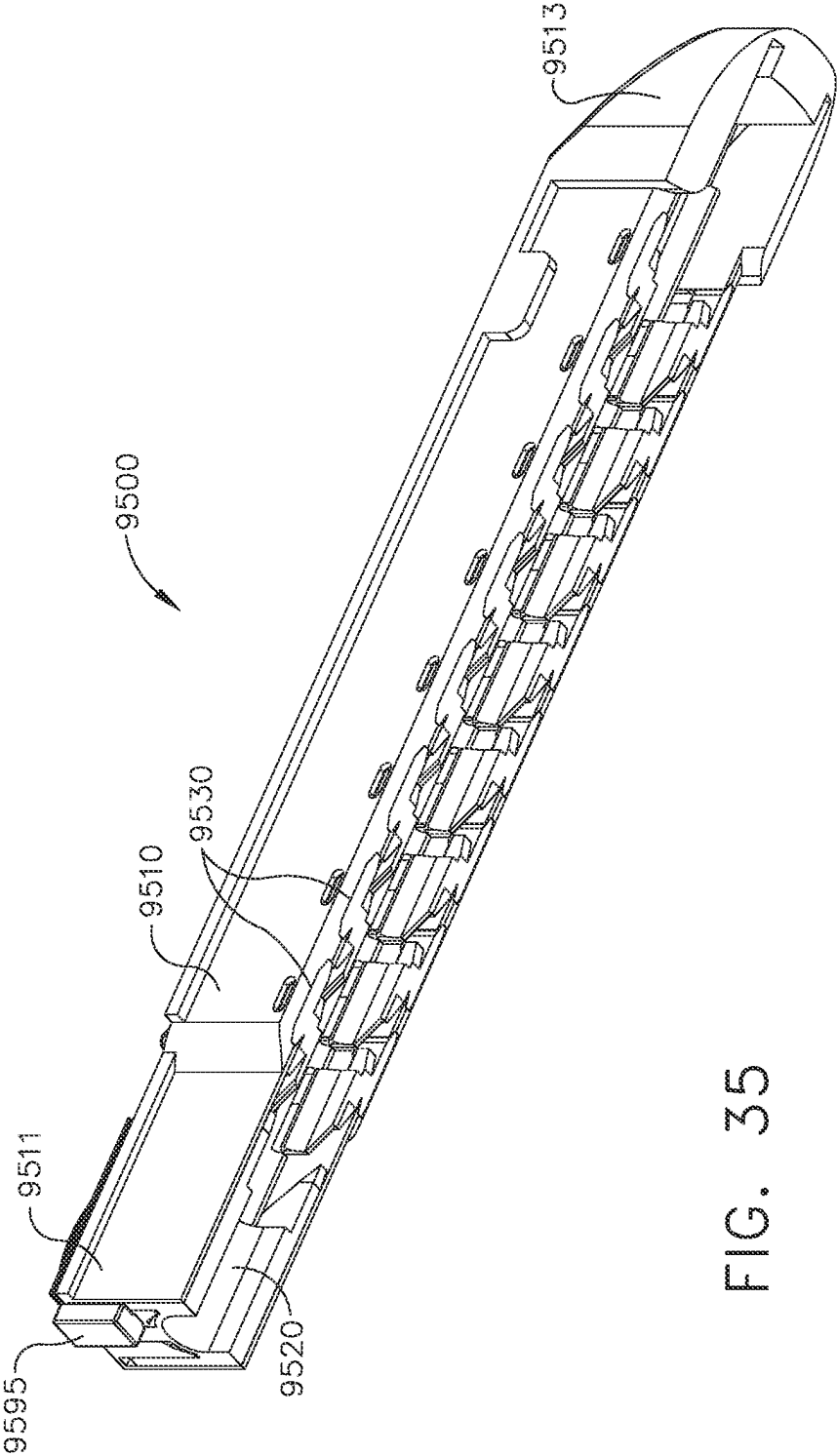


FIG. 35

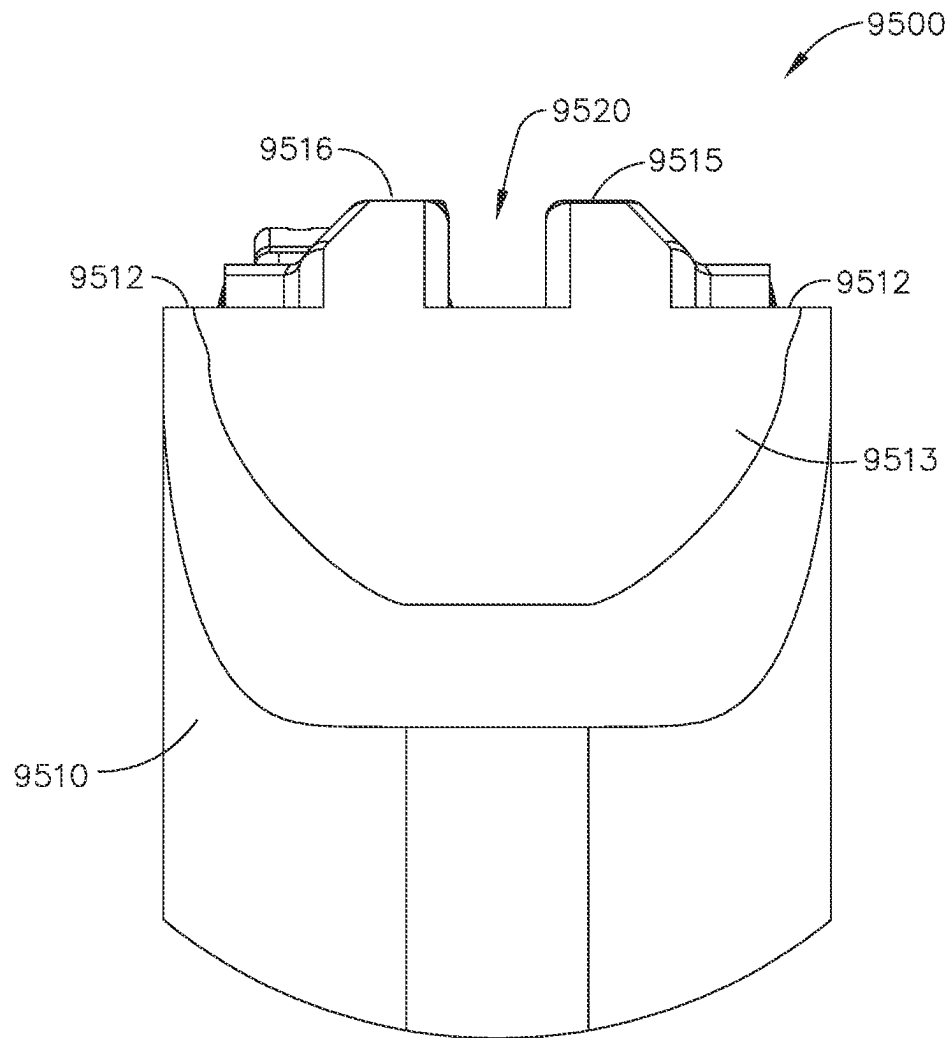


FIG. 36

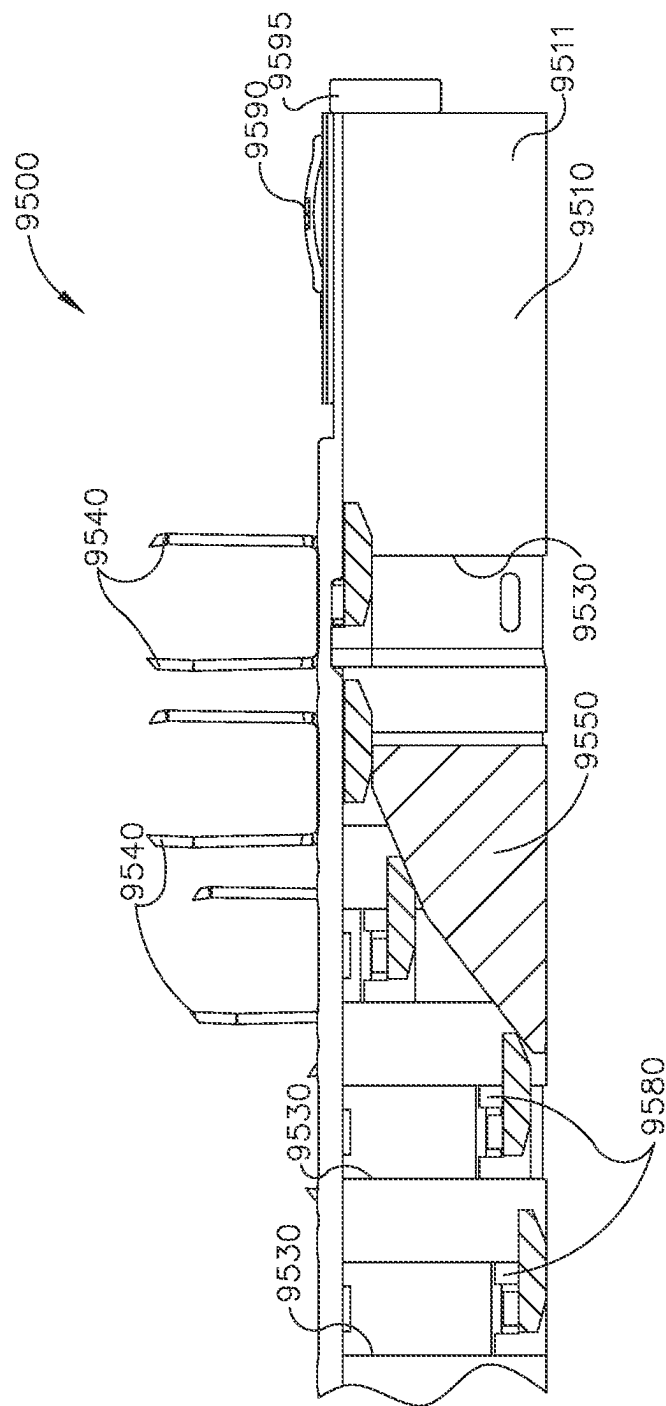


FIG. 37

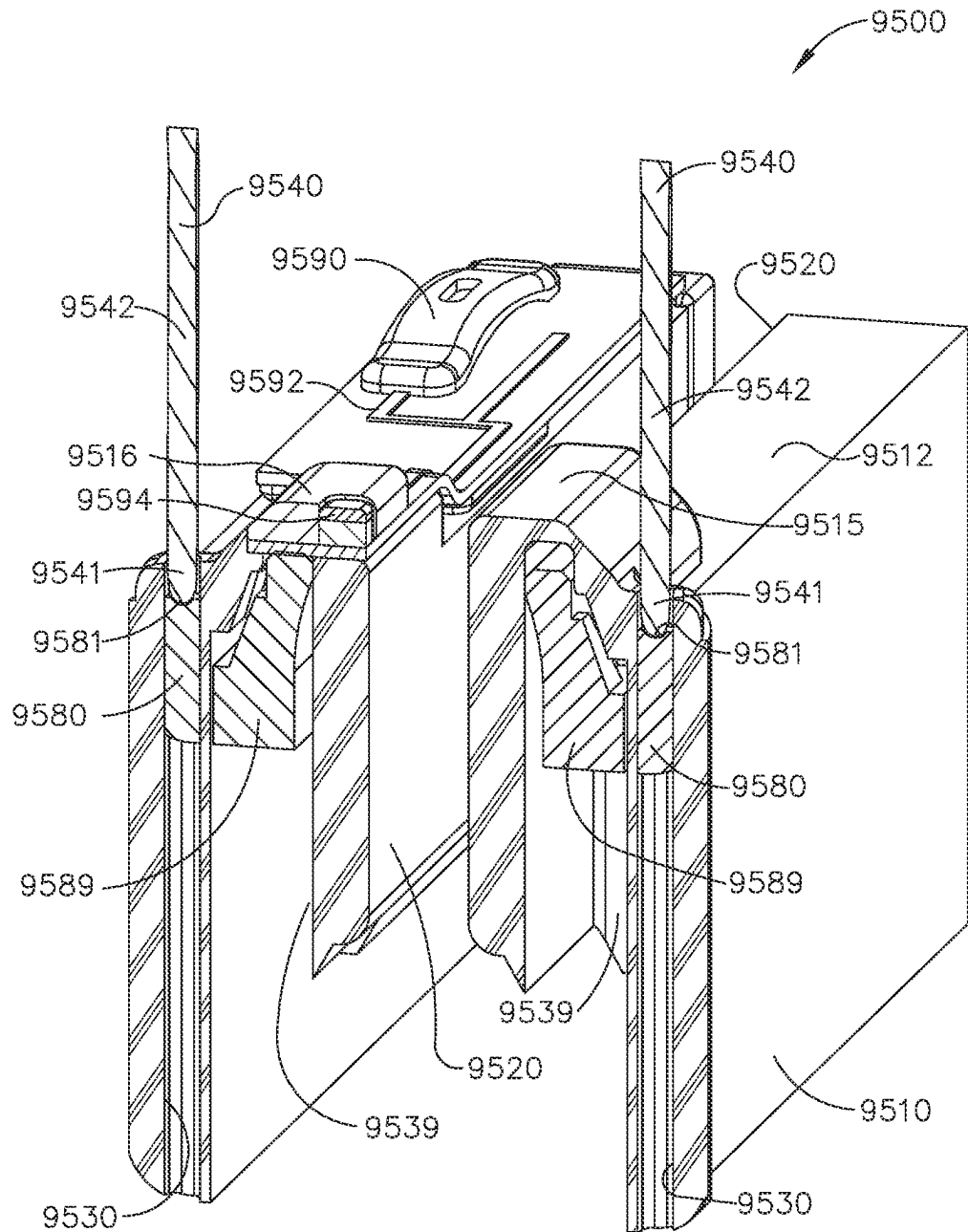


FIG. 38

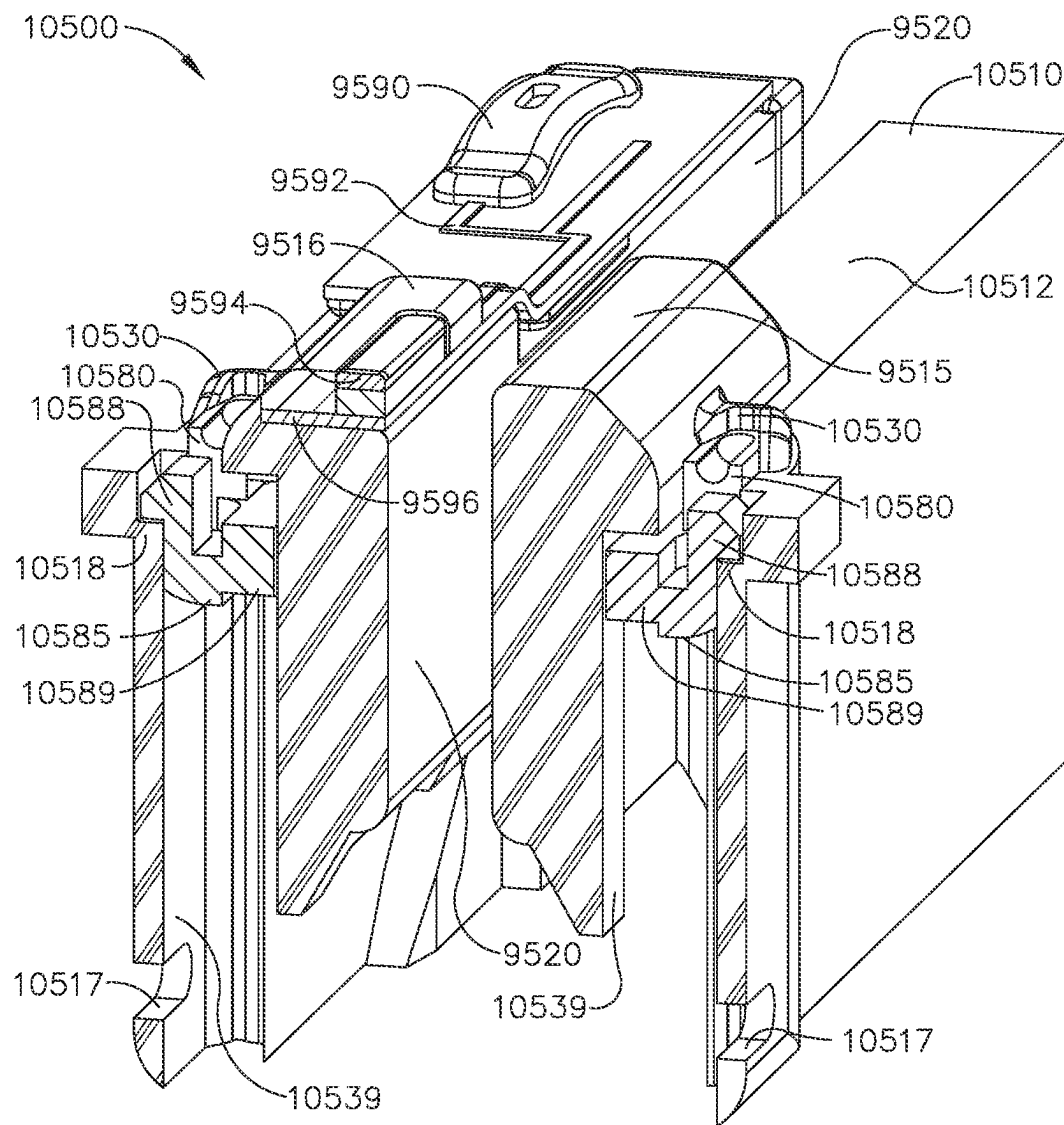


FIG. 39

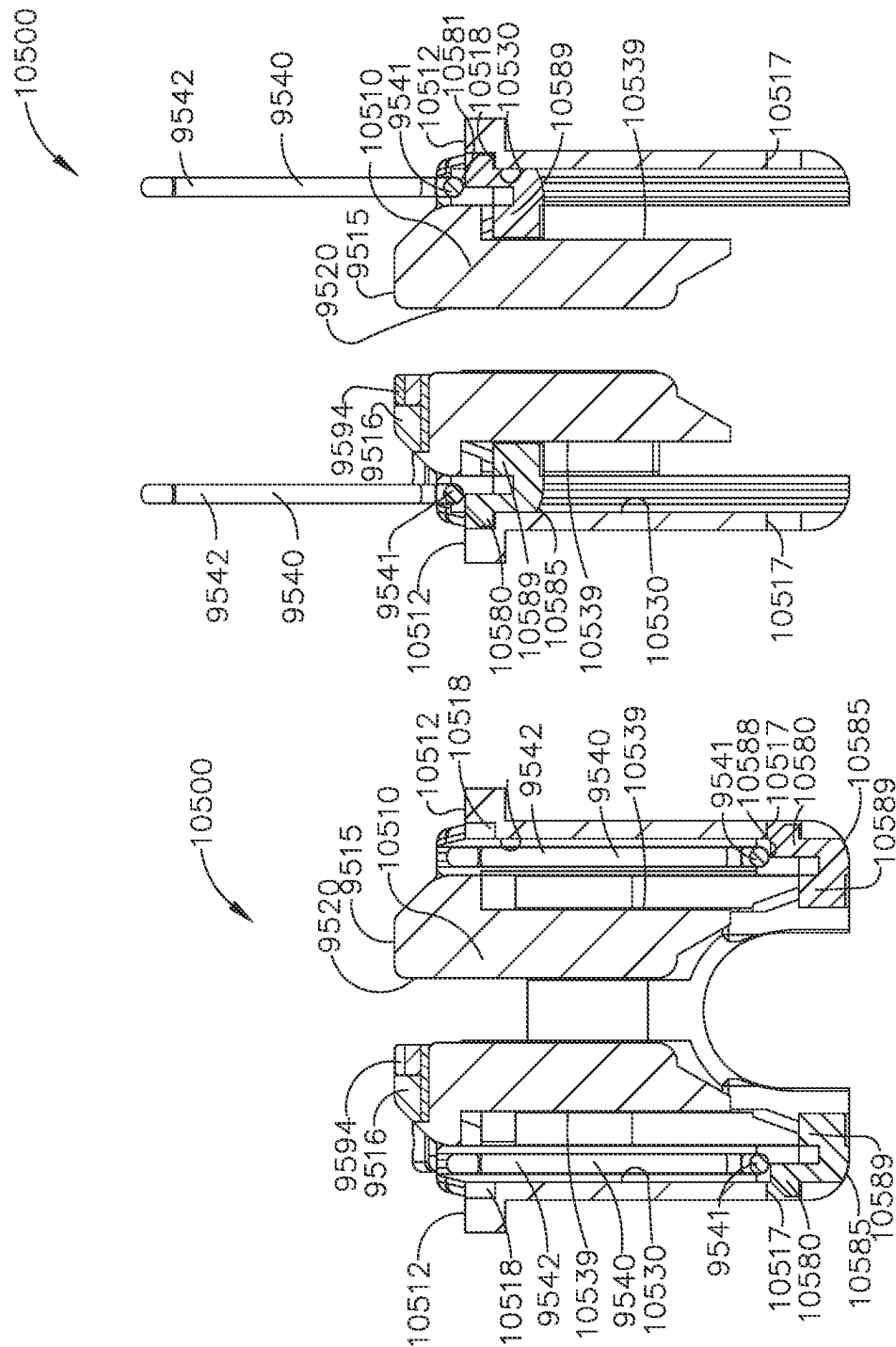
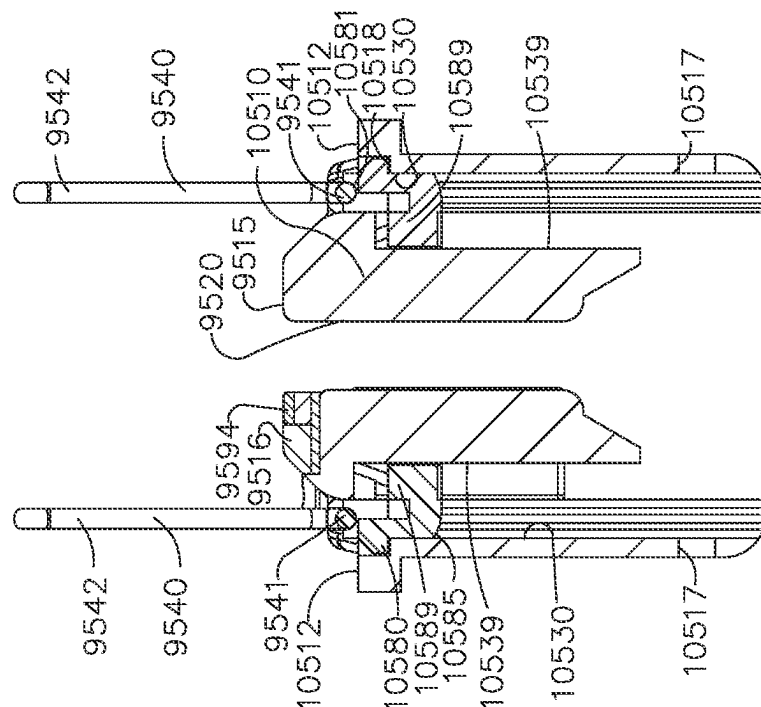
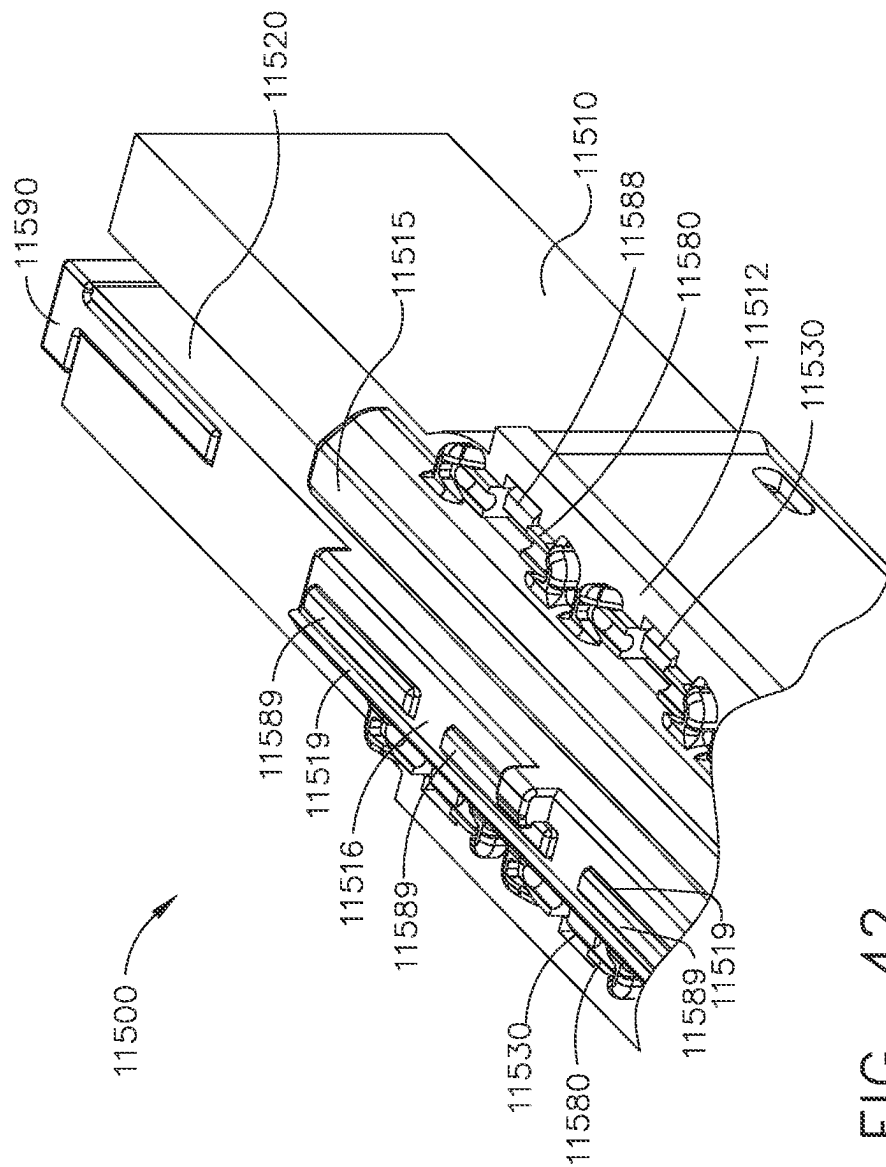


FIG. 40





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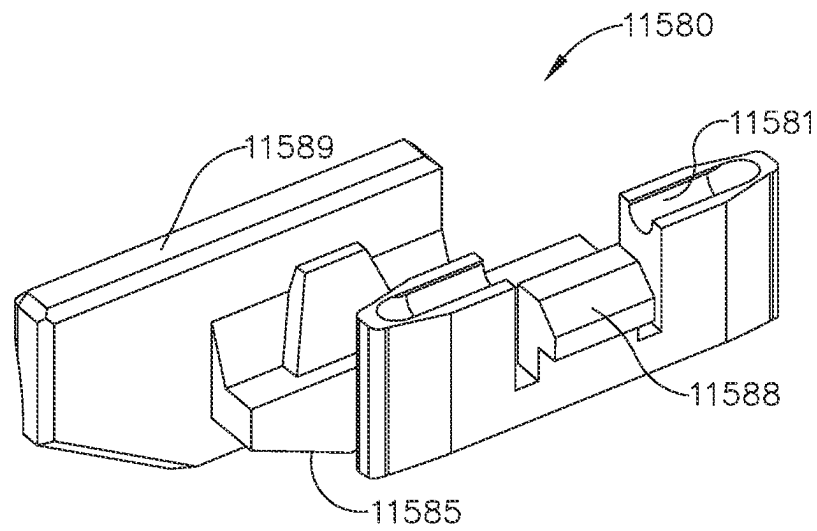


FIG. 43

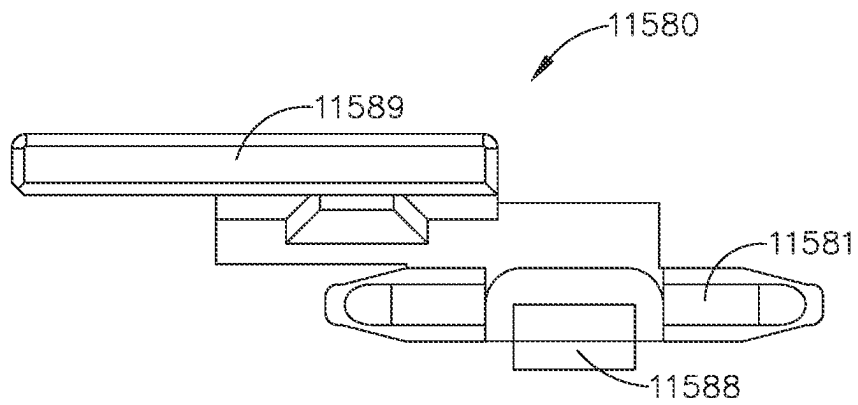


FIG. 44

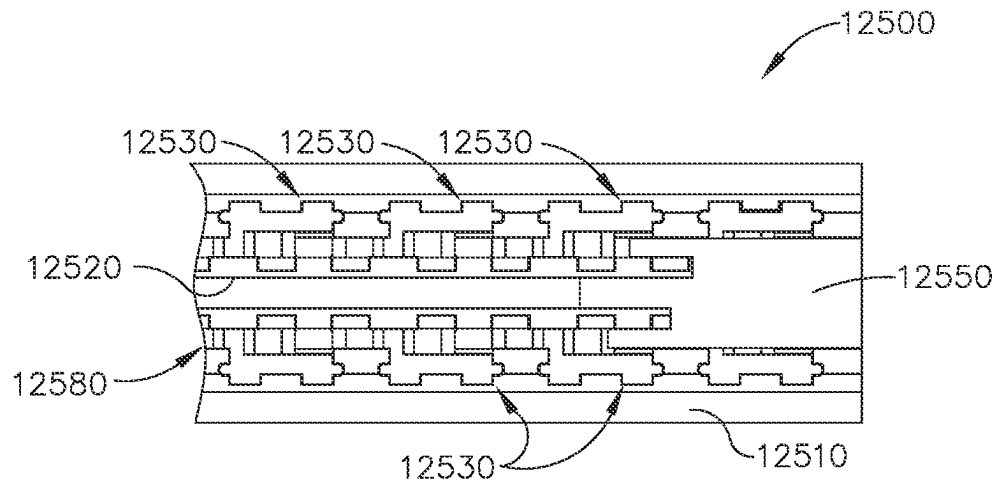


FIG. 45

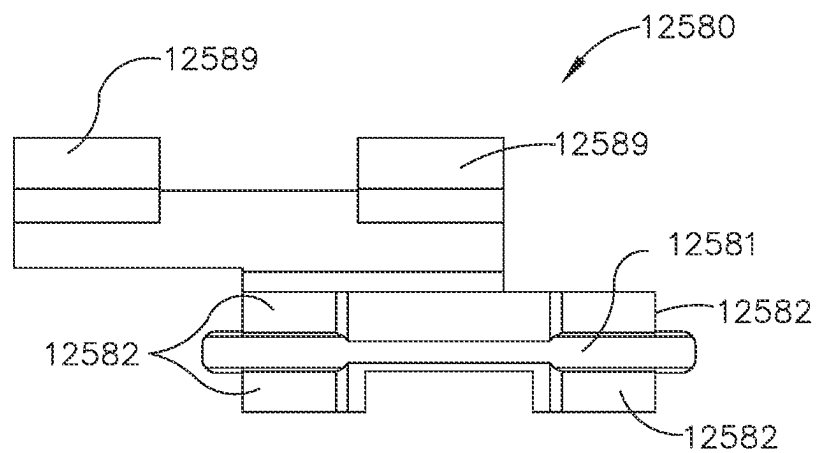


FIG. 46

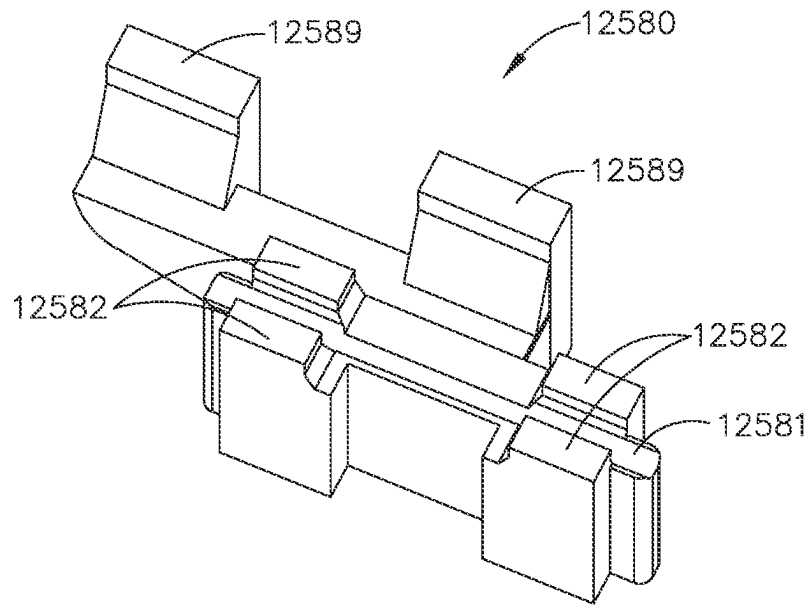


FIG. 47

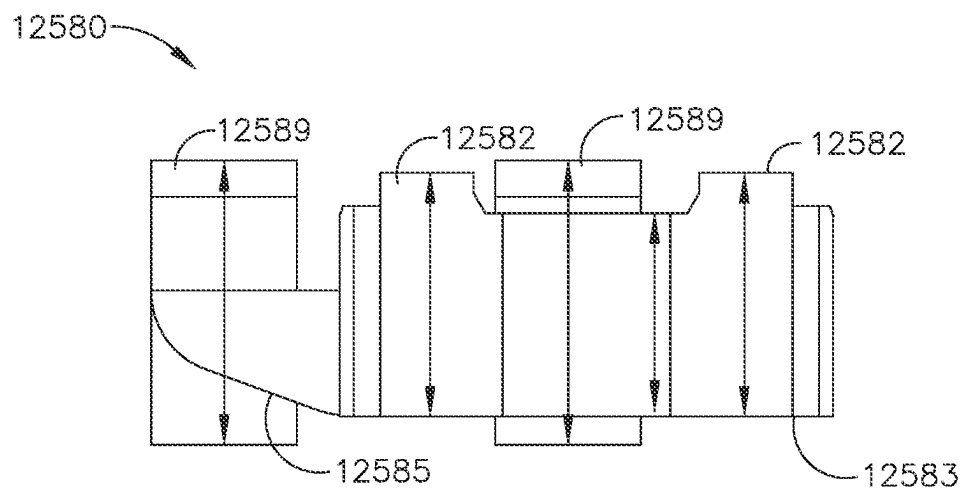
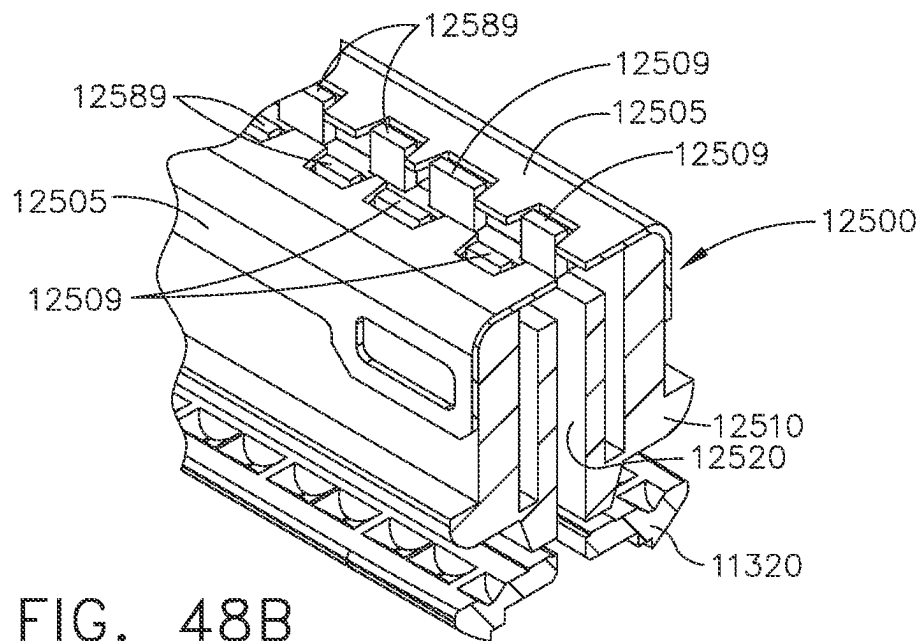
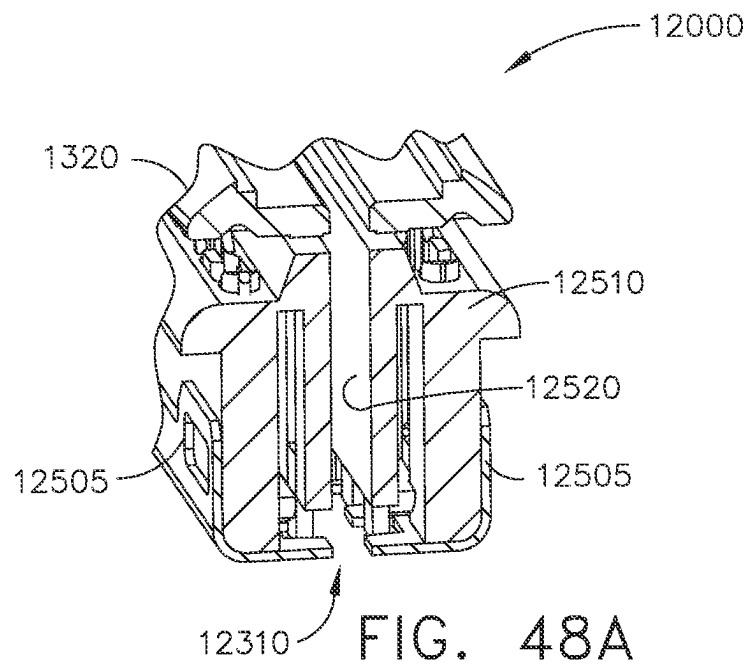
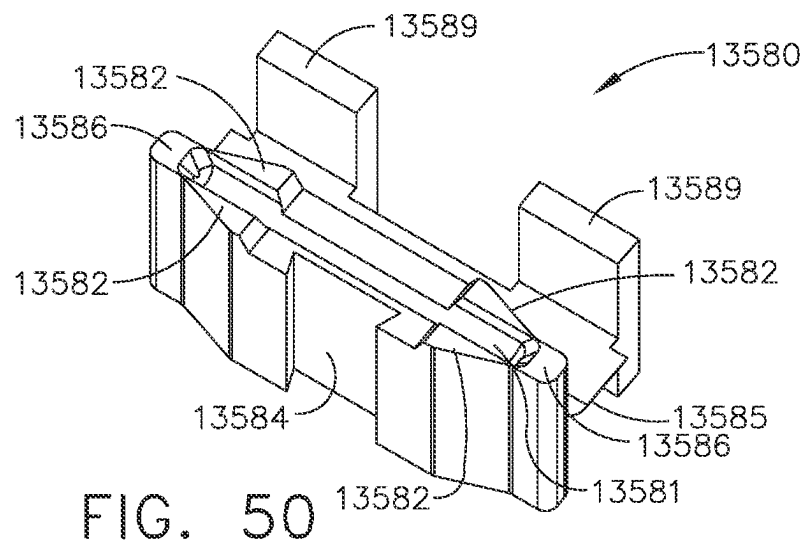
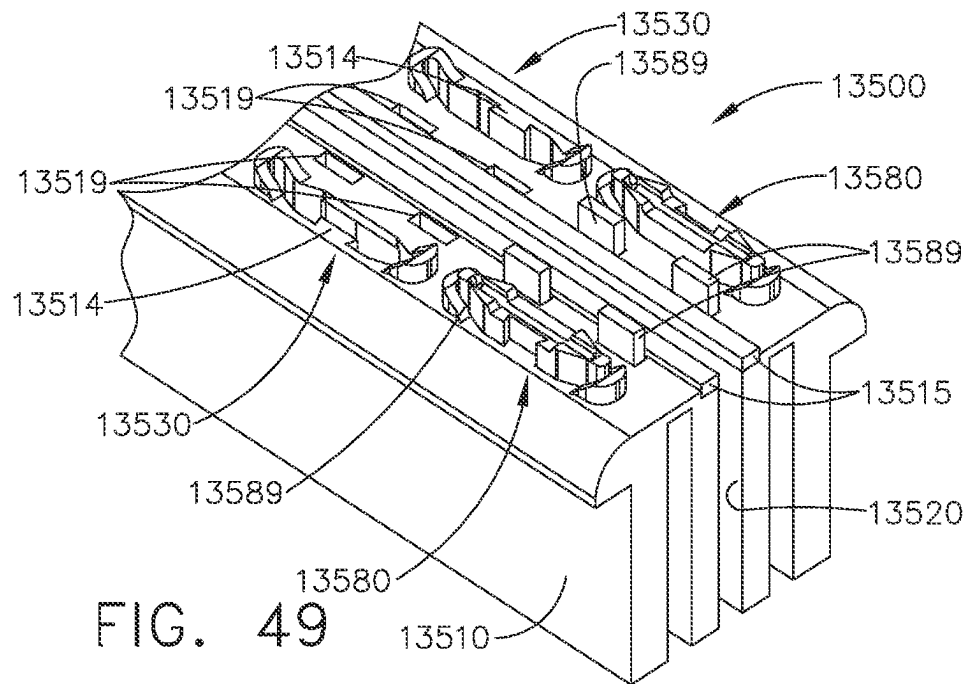
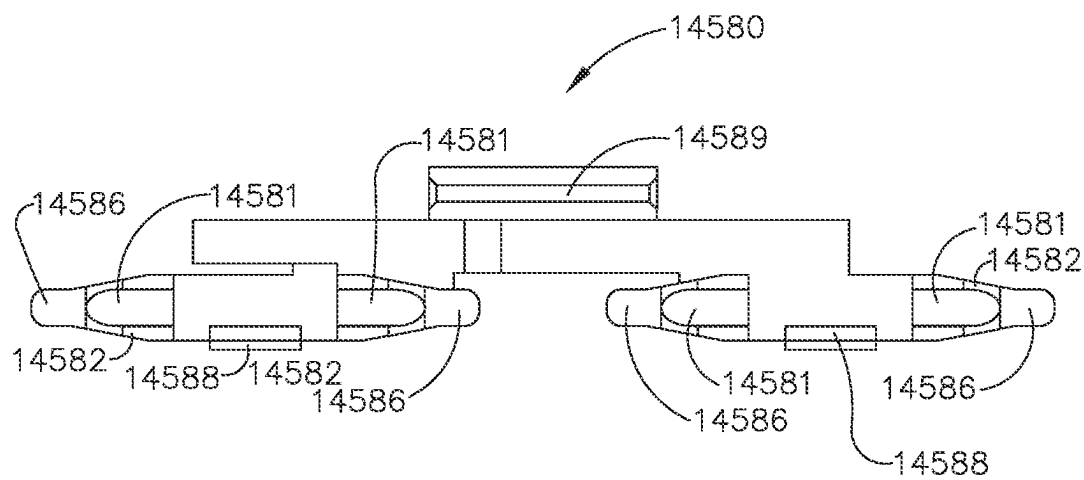
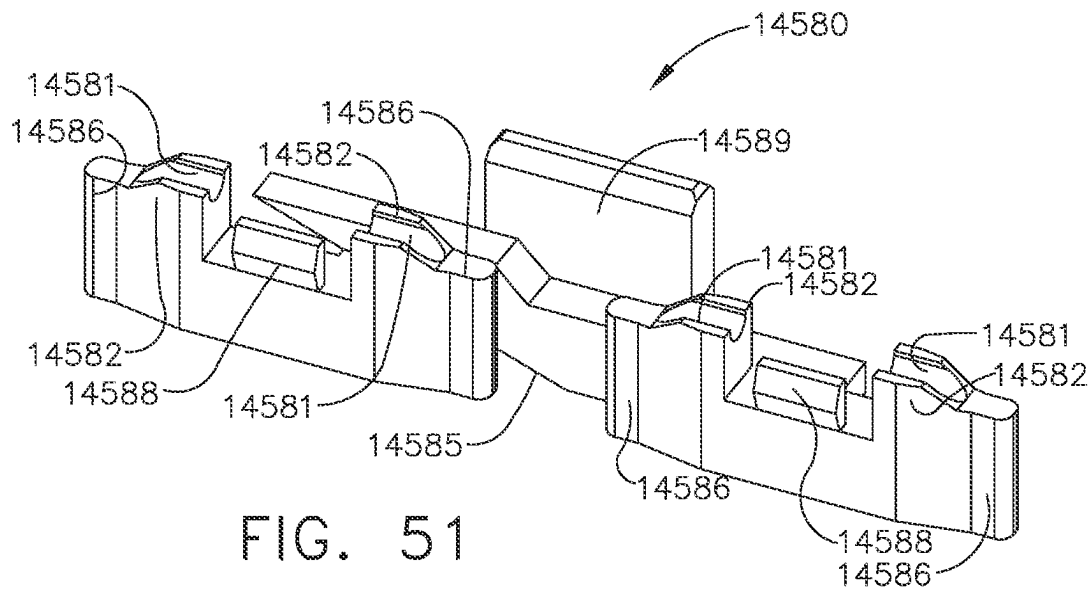


FIG. 48







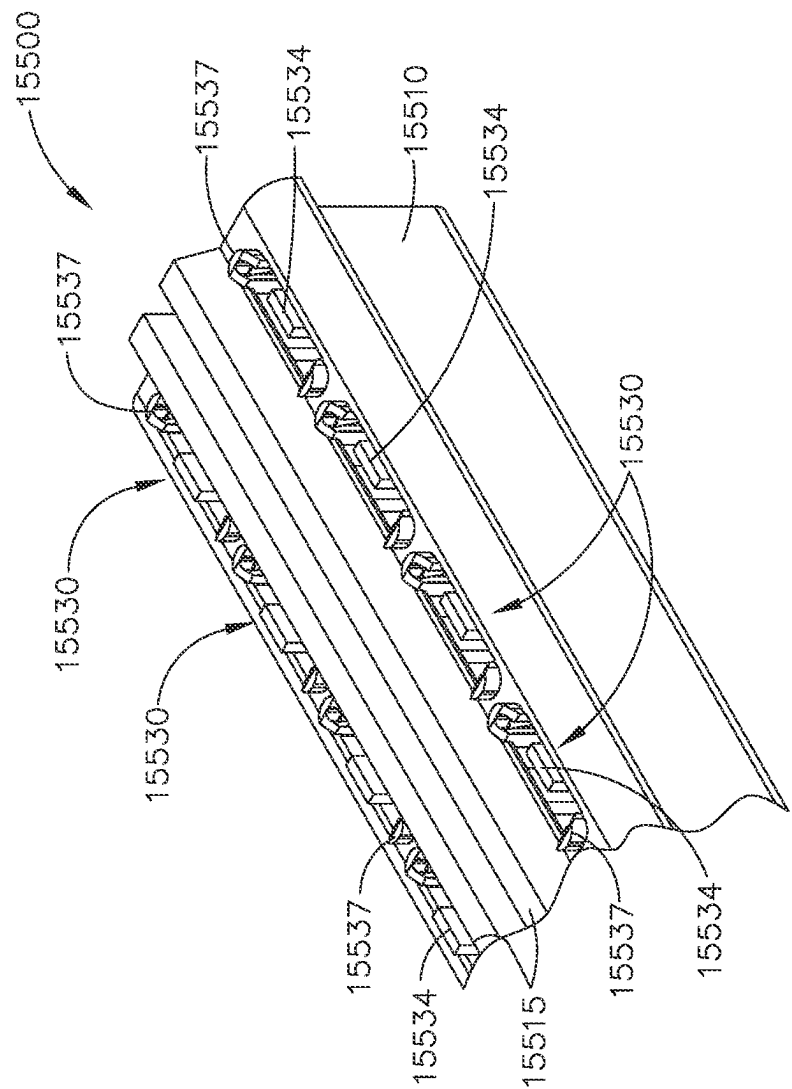


FIG. 53

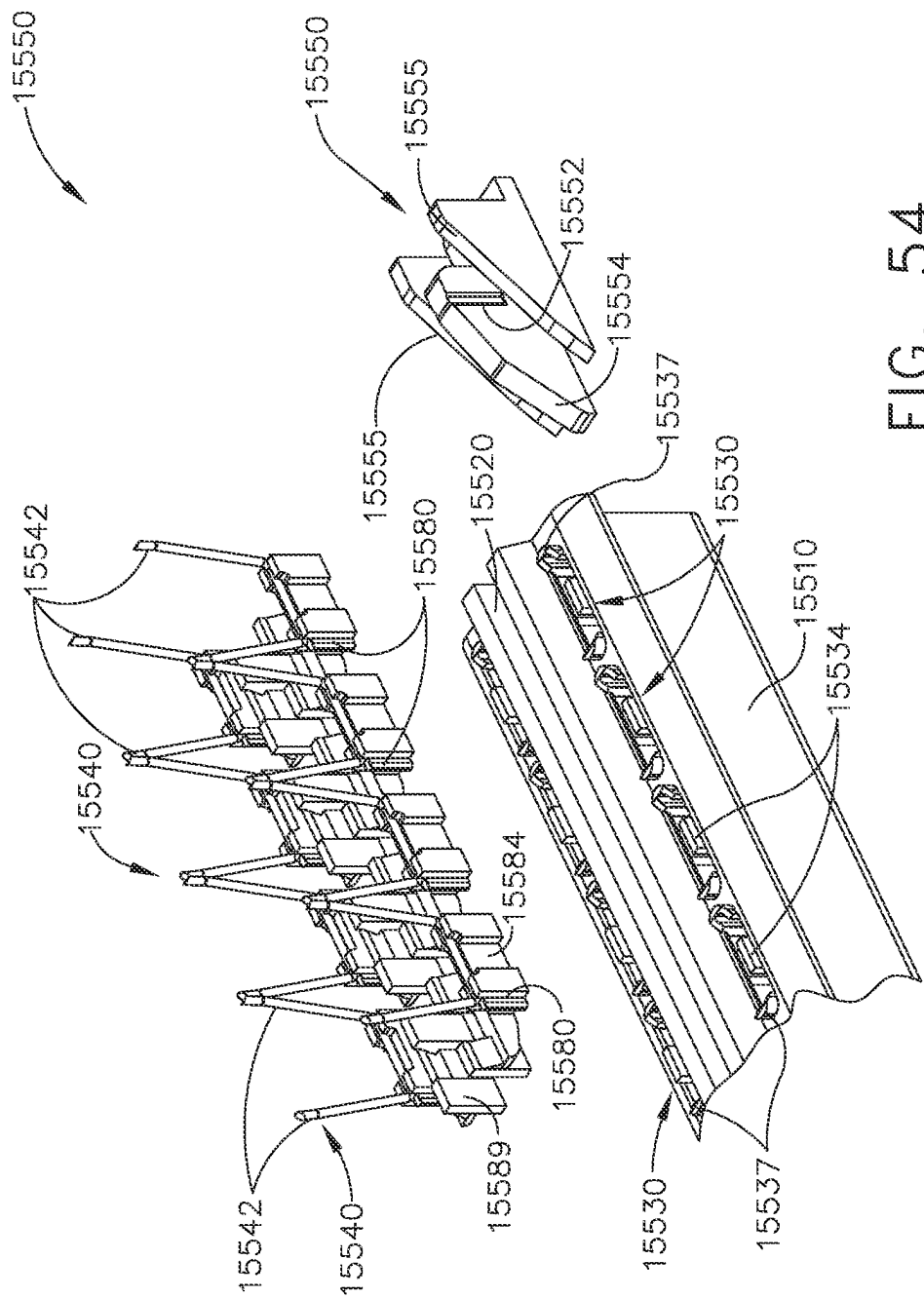
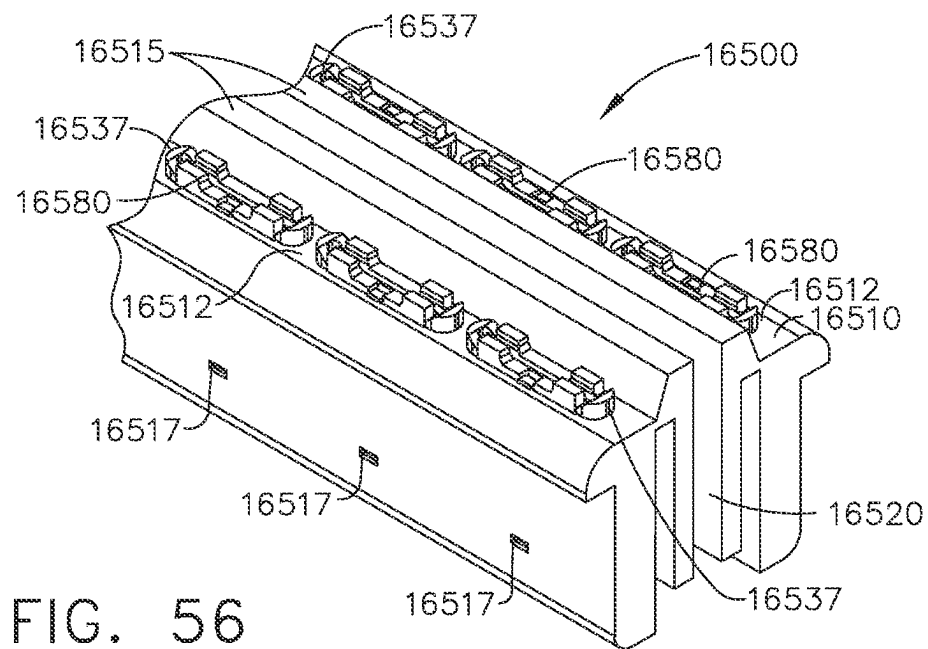
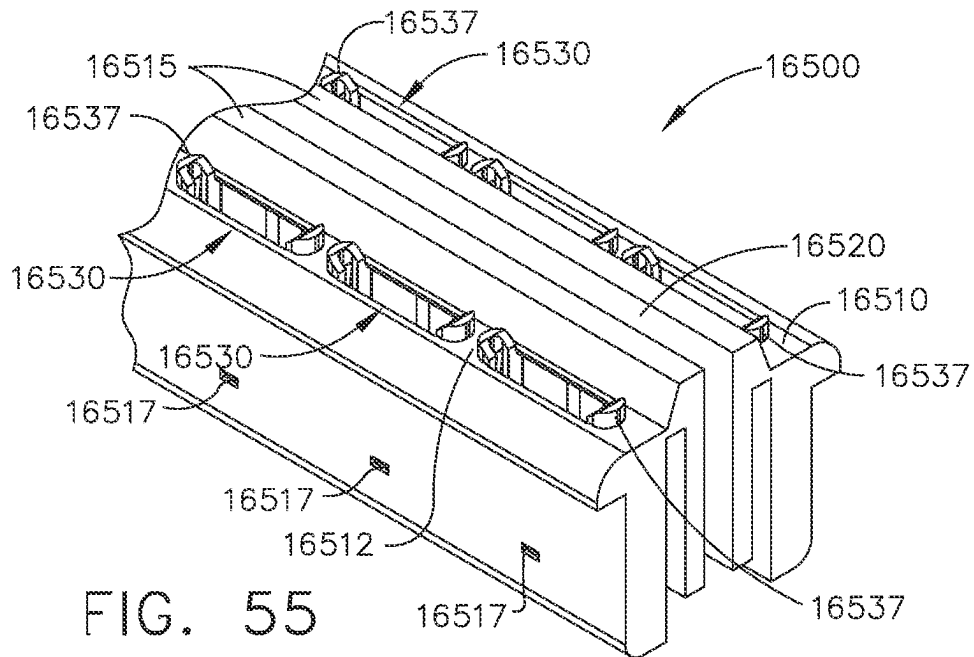


FIG. 54



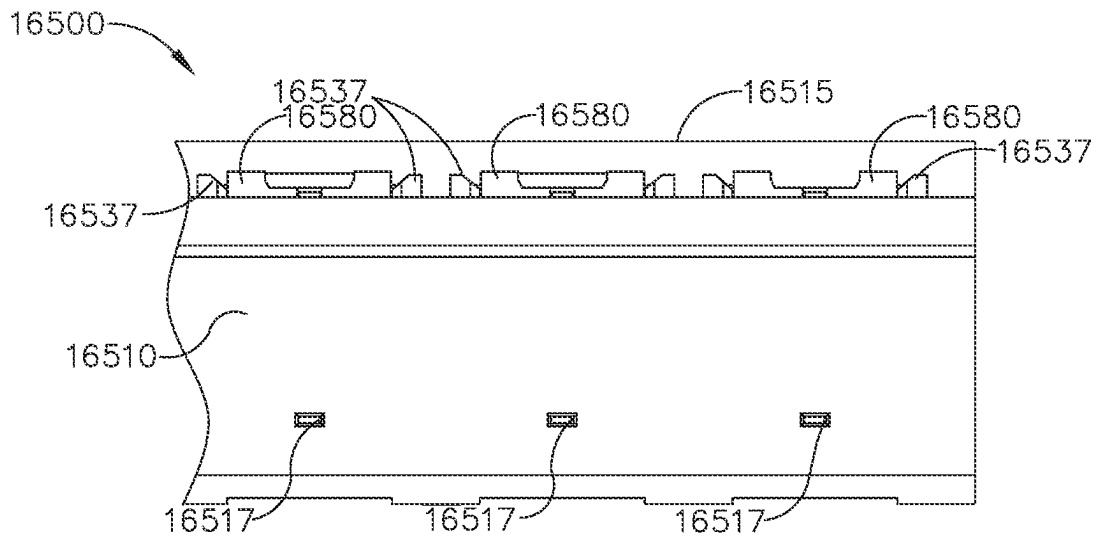


FIG. 57

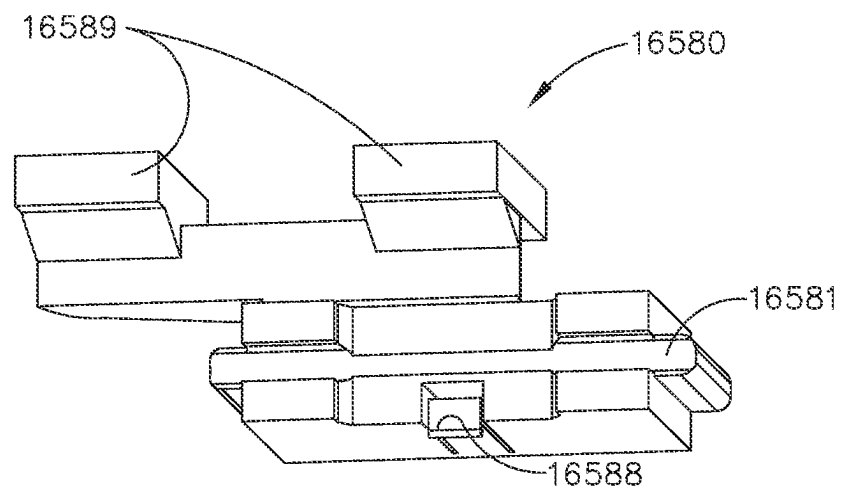


FIG. 58

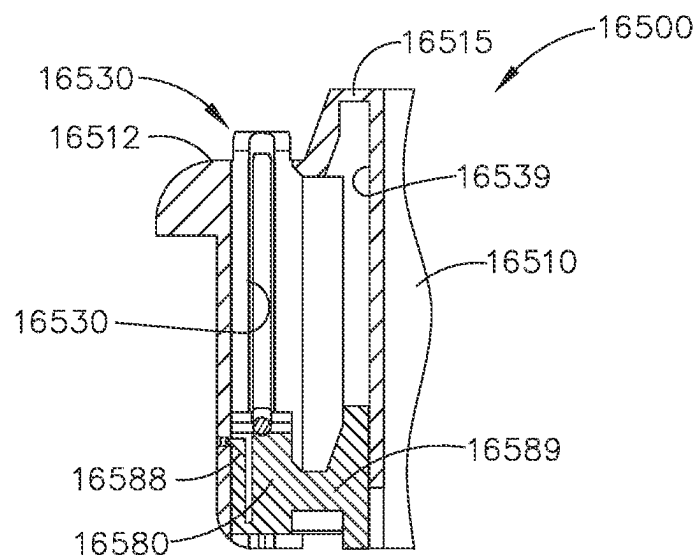


FIG. 59

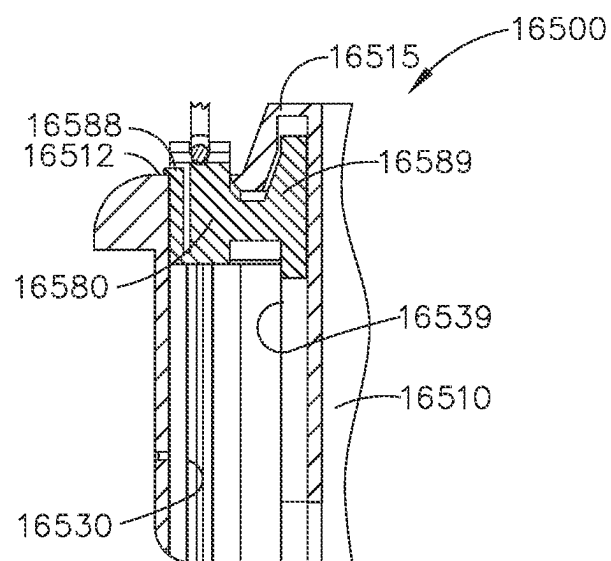
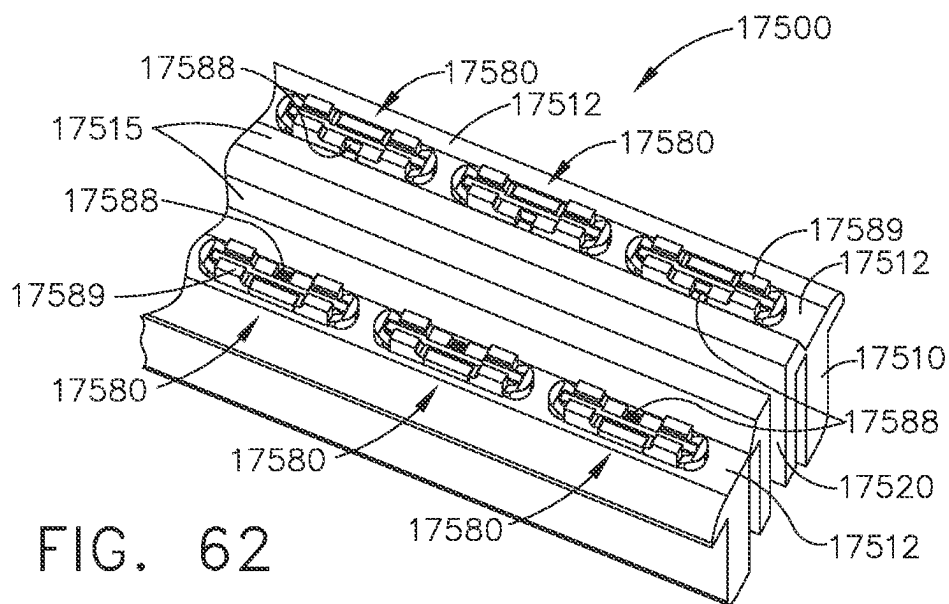
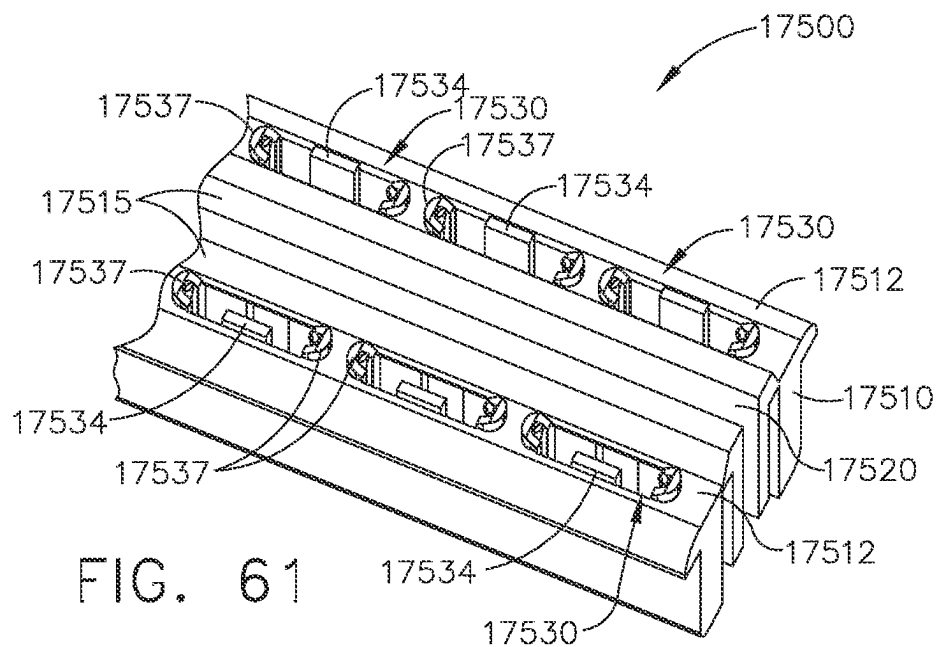


FIG. 60



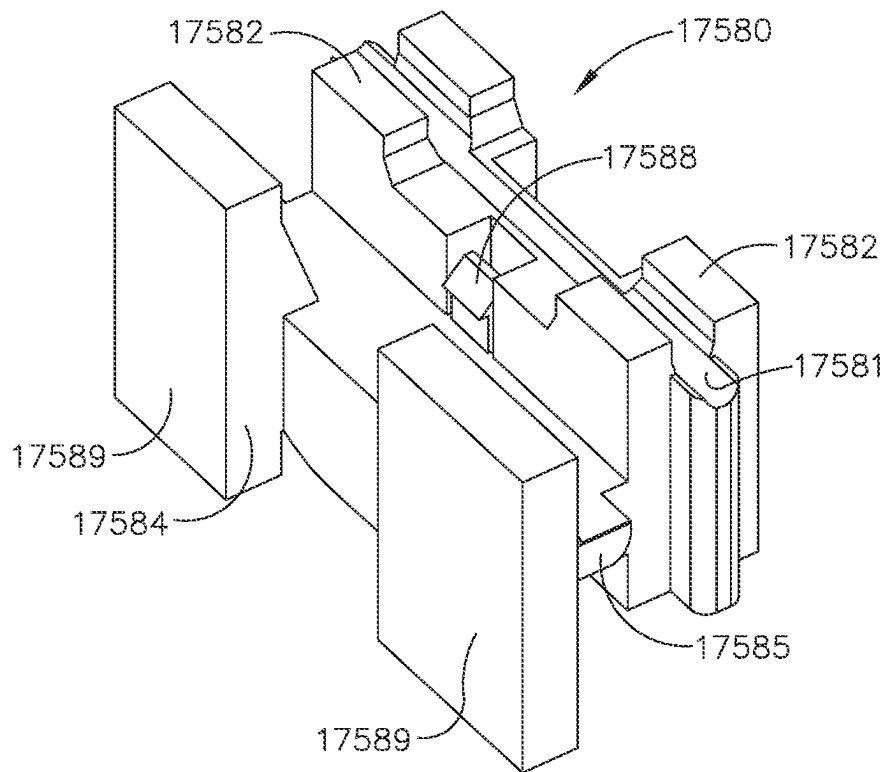


FIG. 63

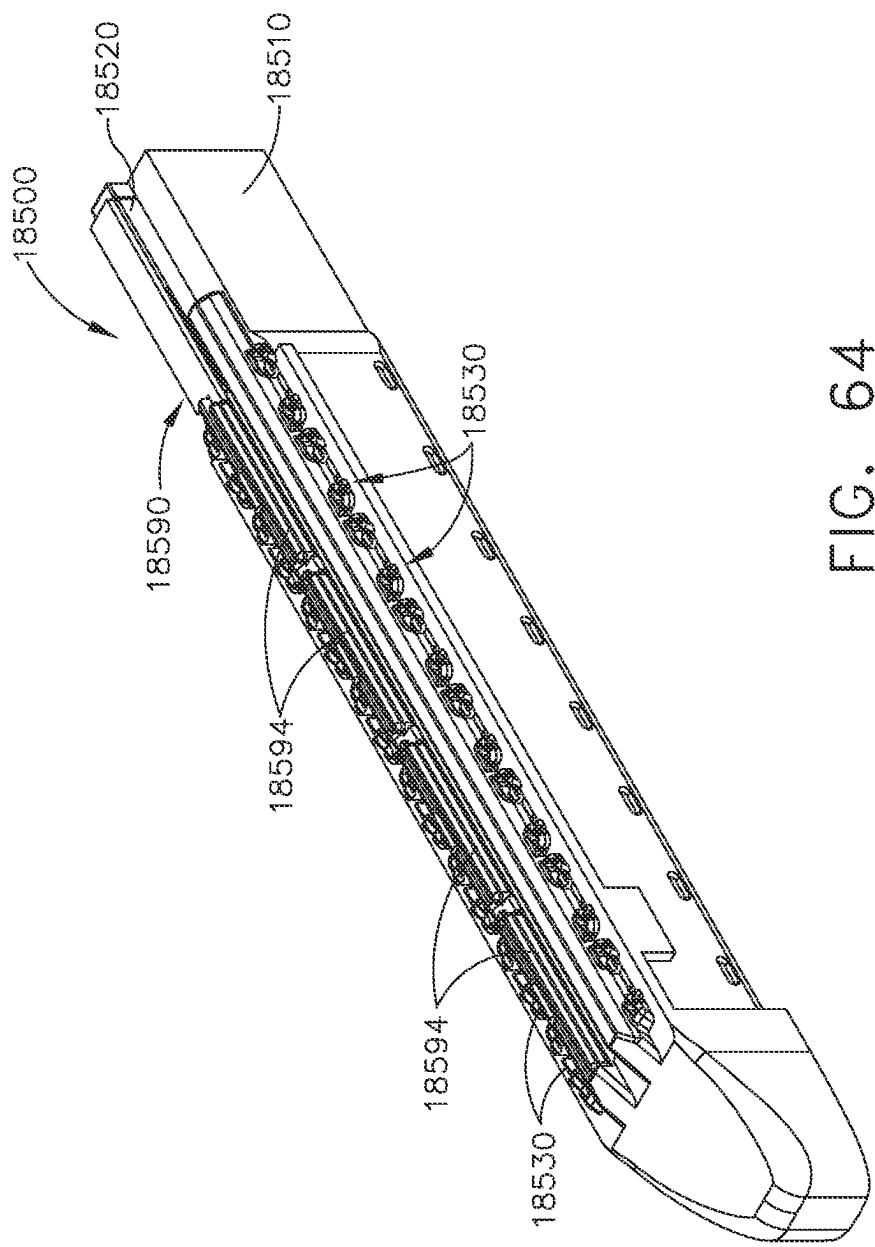


FIG. 64

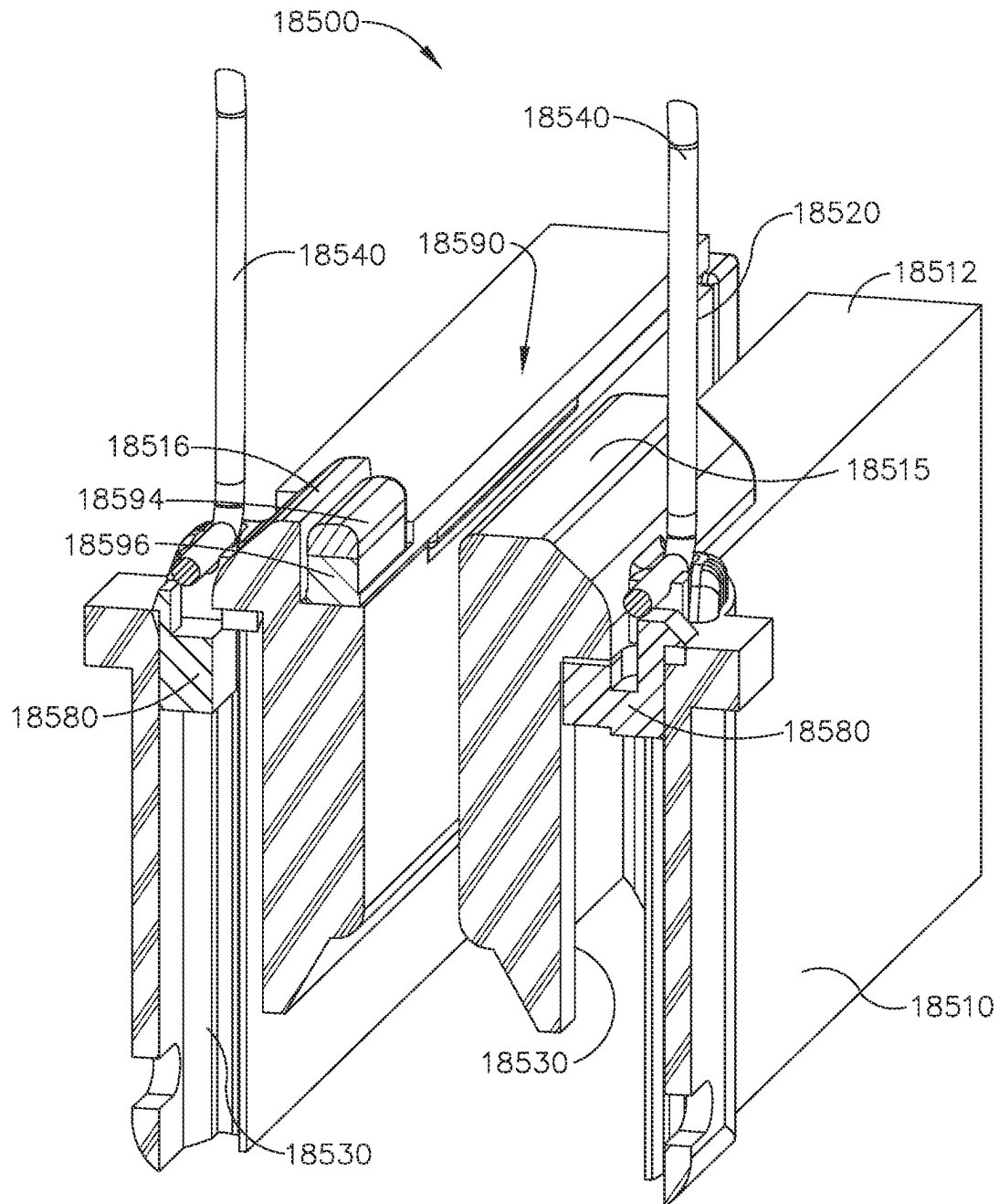


FIG. 65

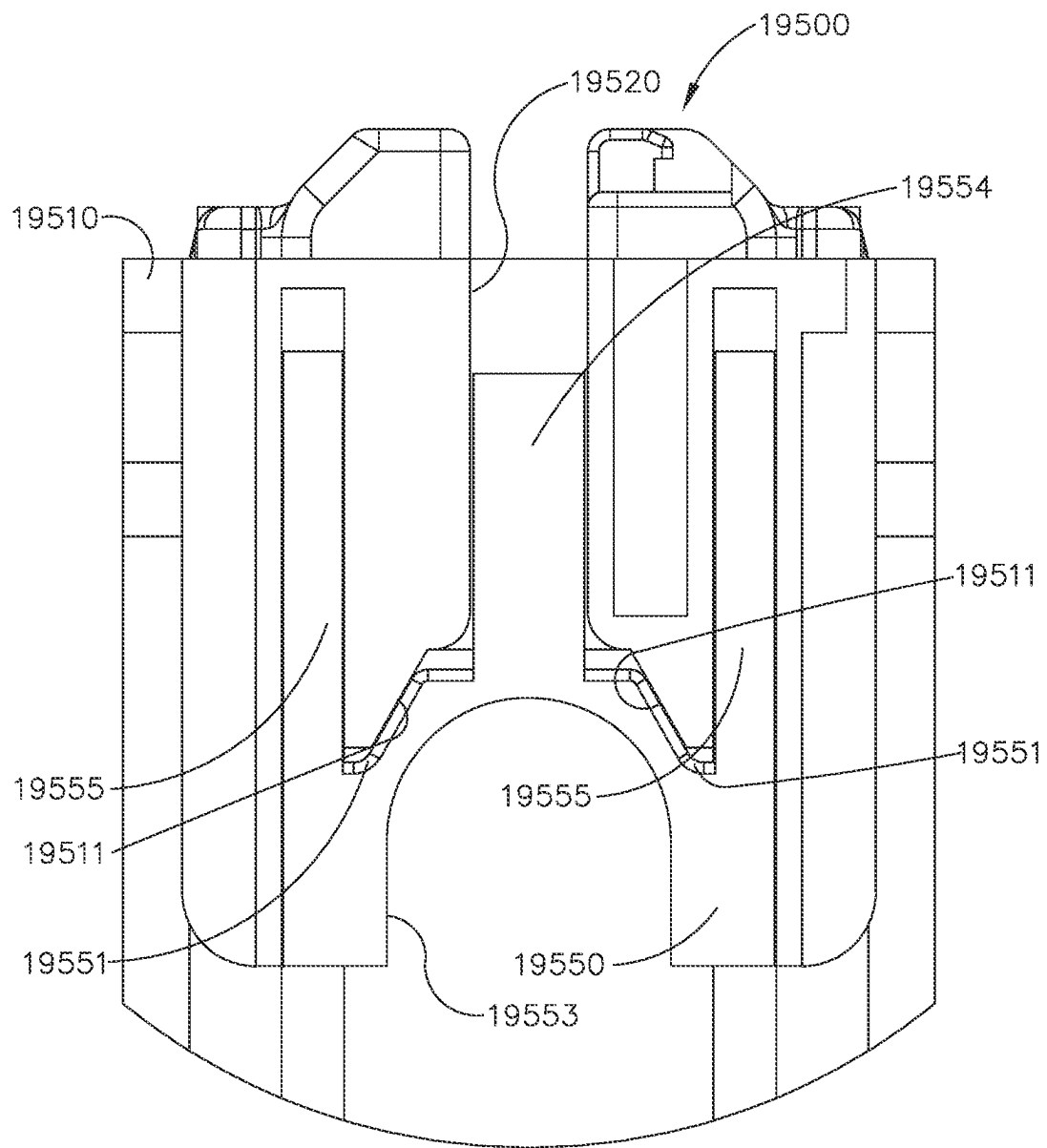
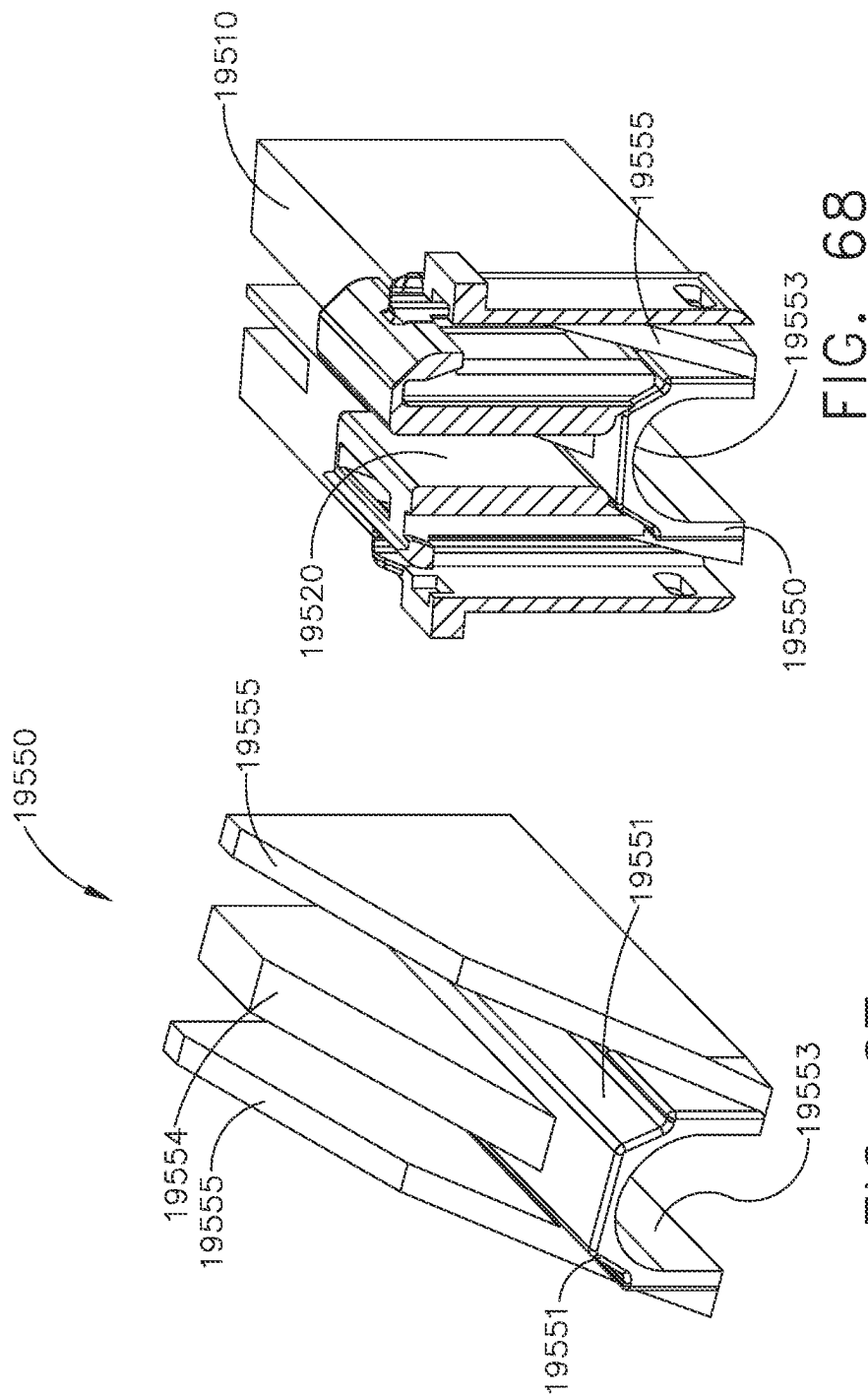


FIG. 66



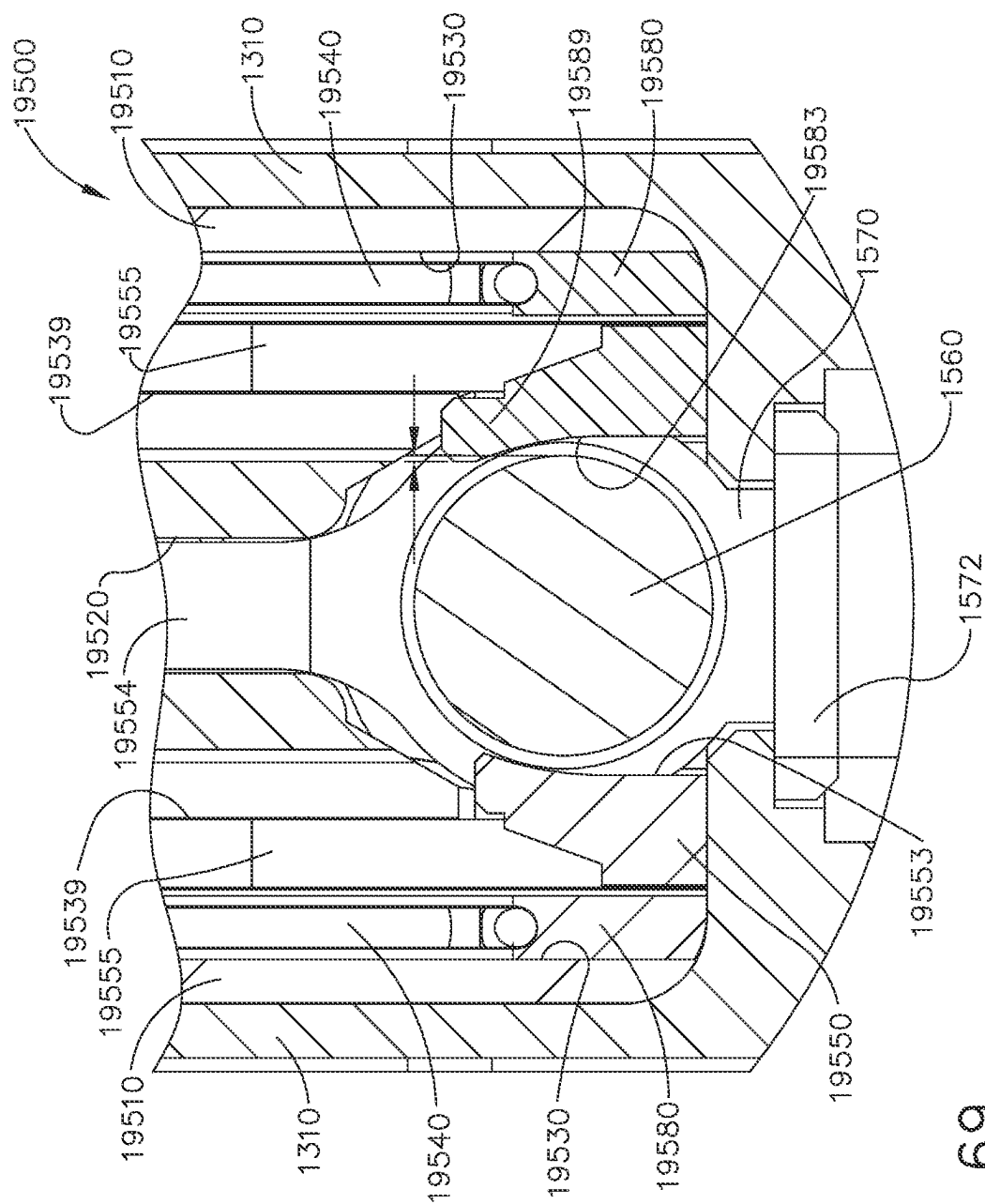


FIG. 69

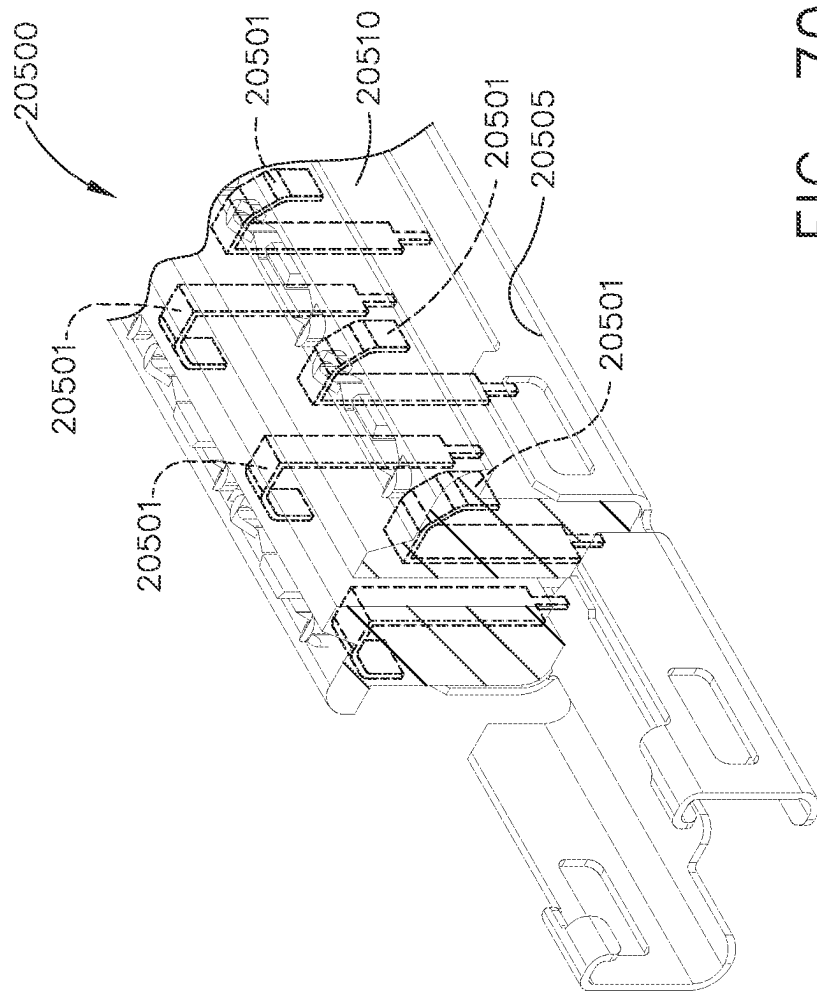
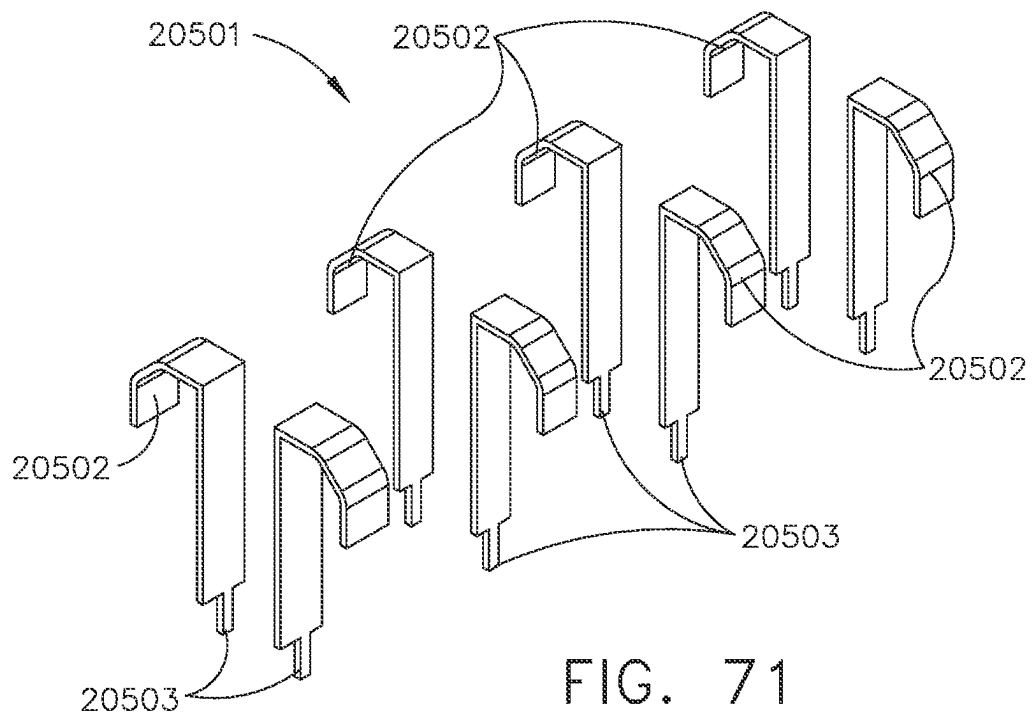


FIG. 70



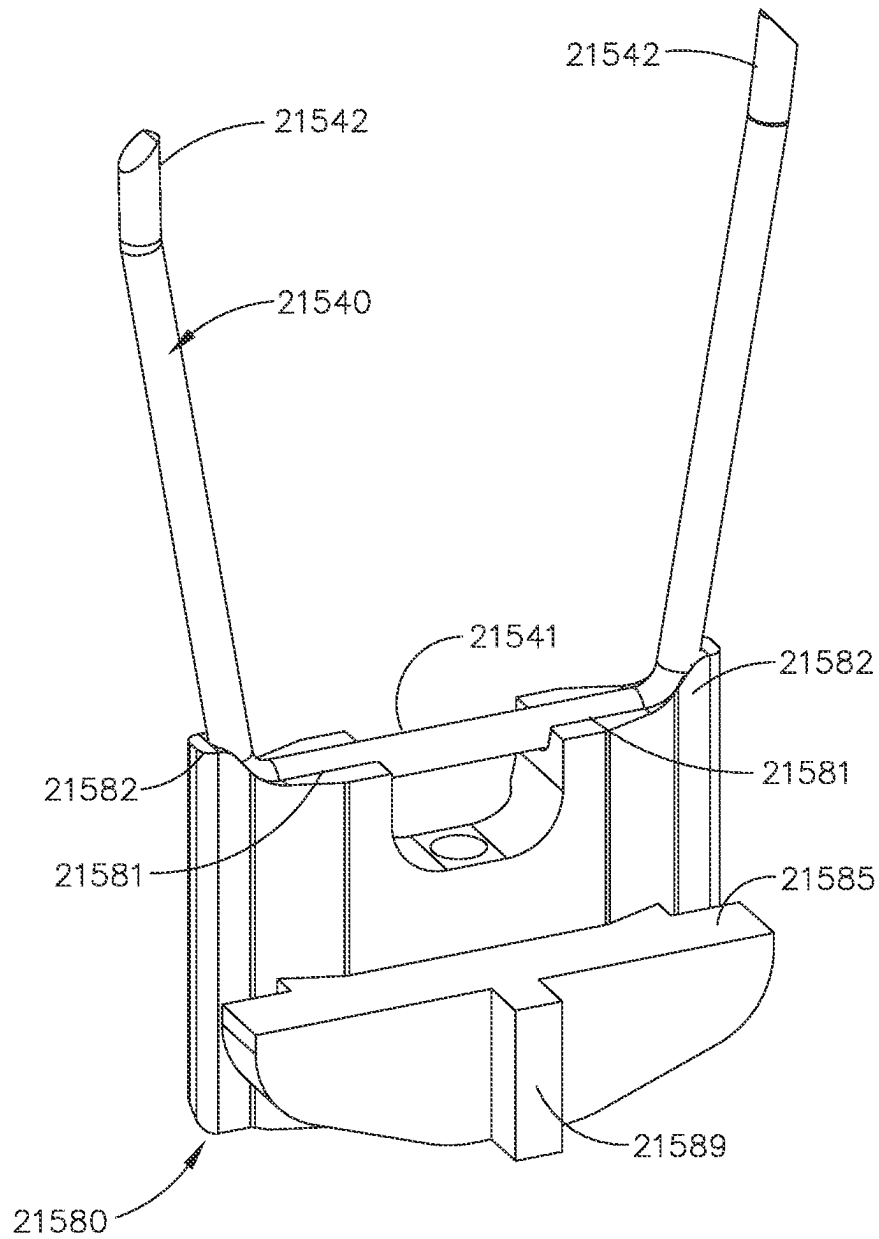


FIG. 72

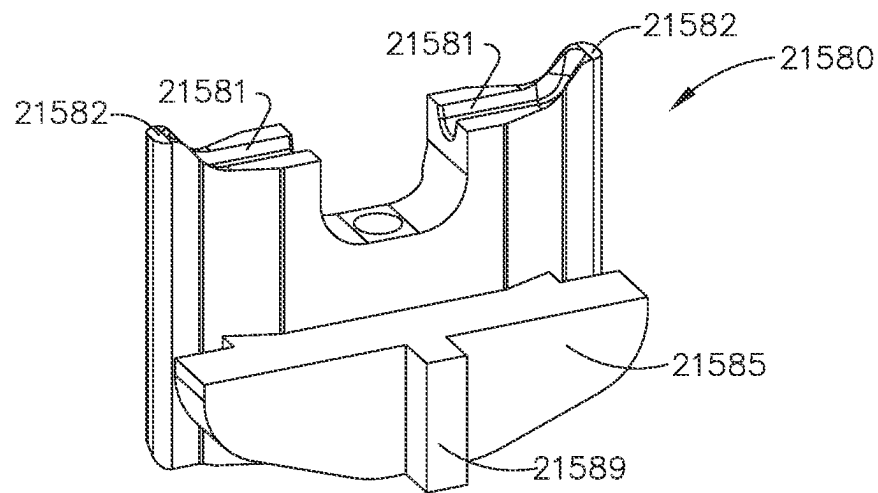


FIG. 73

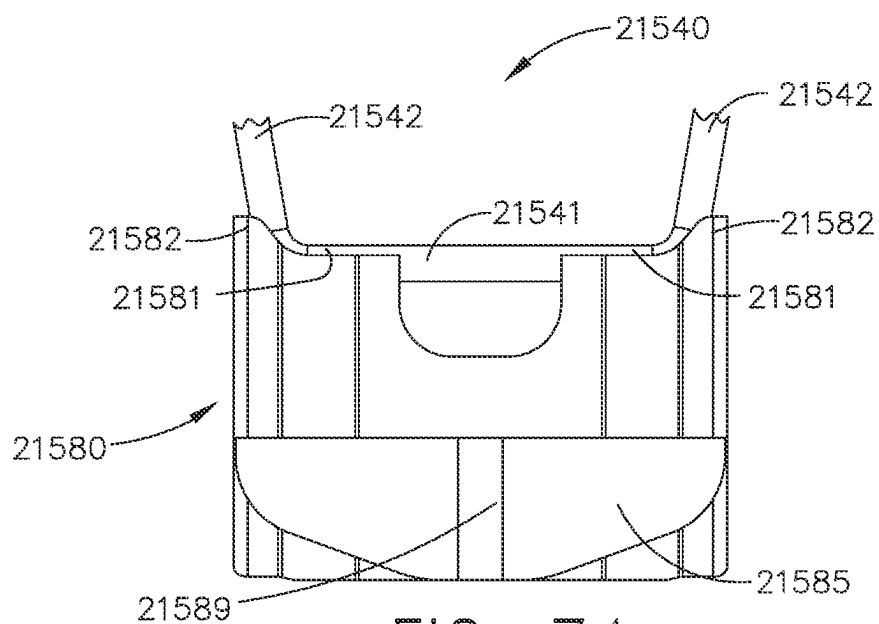


FIG. 74

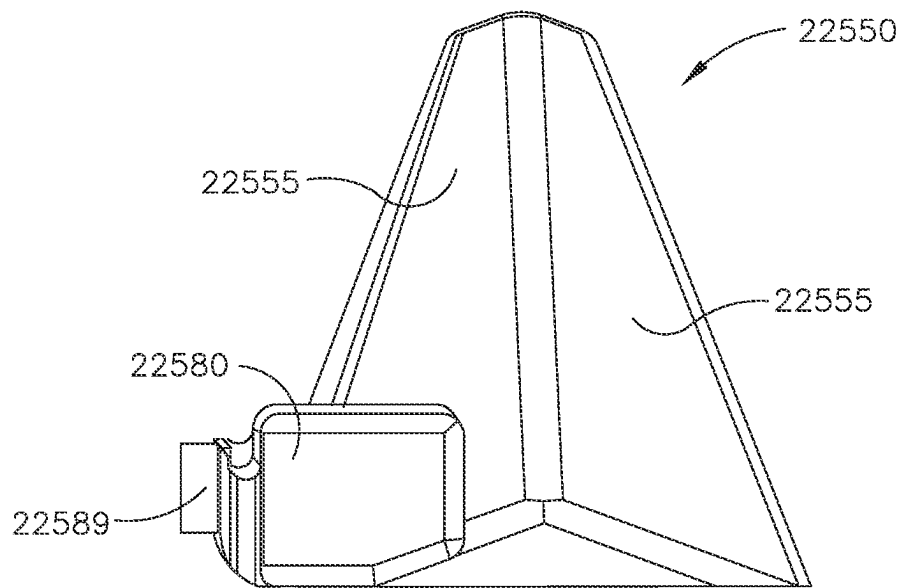


FIG. 75

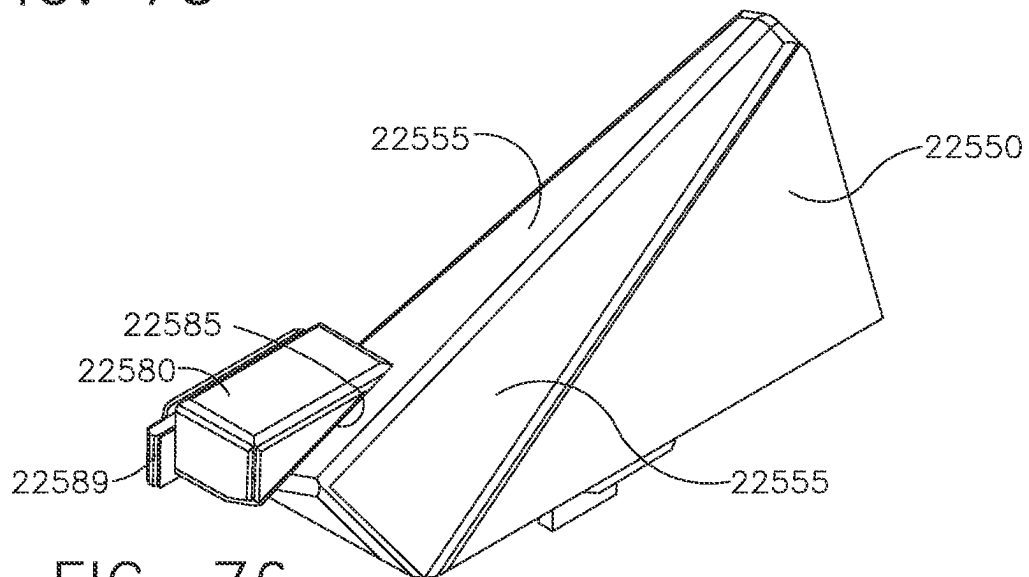


FIG. 76

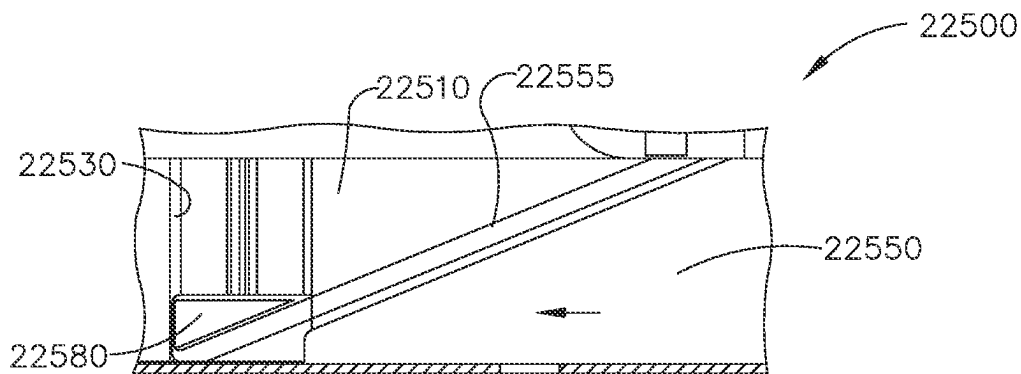


FIG. 77

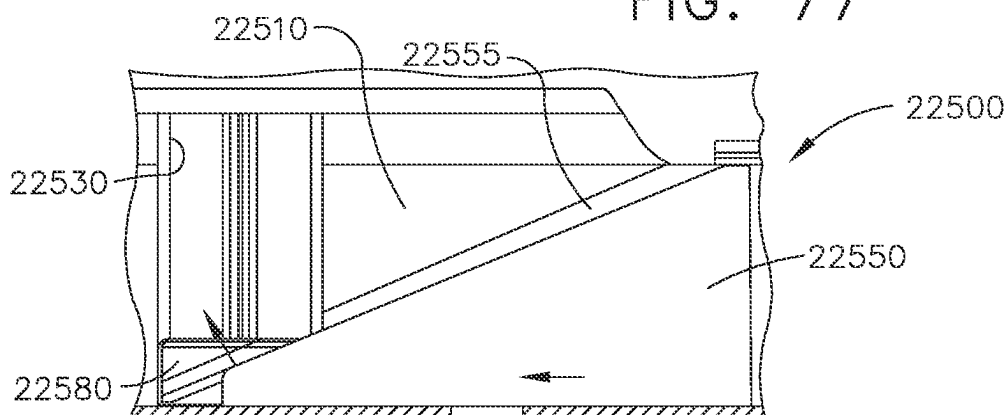


FIG. 78

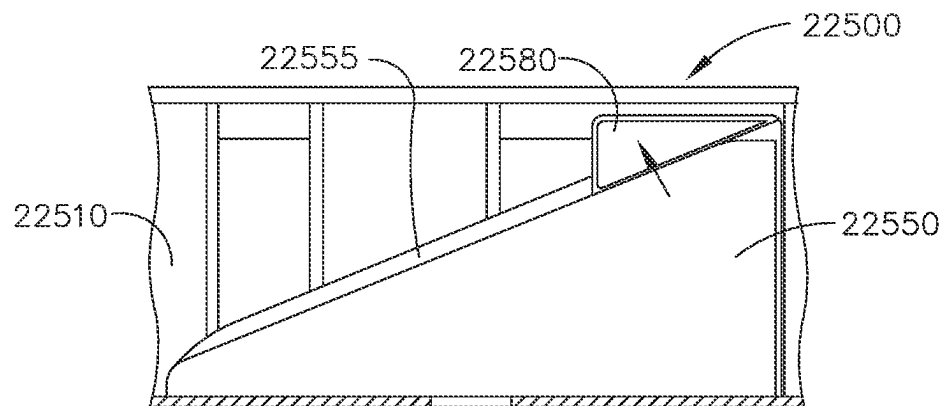
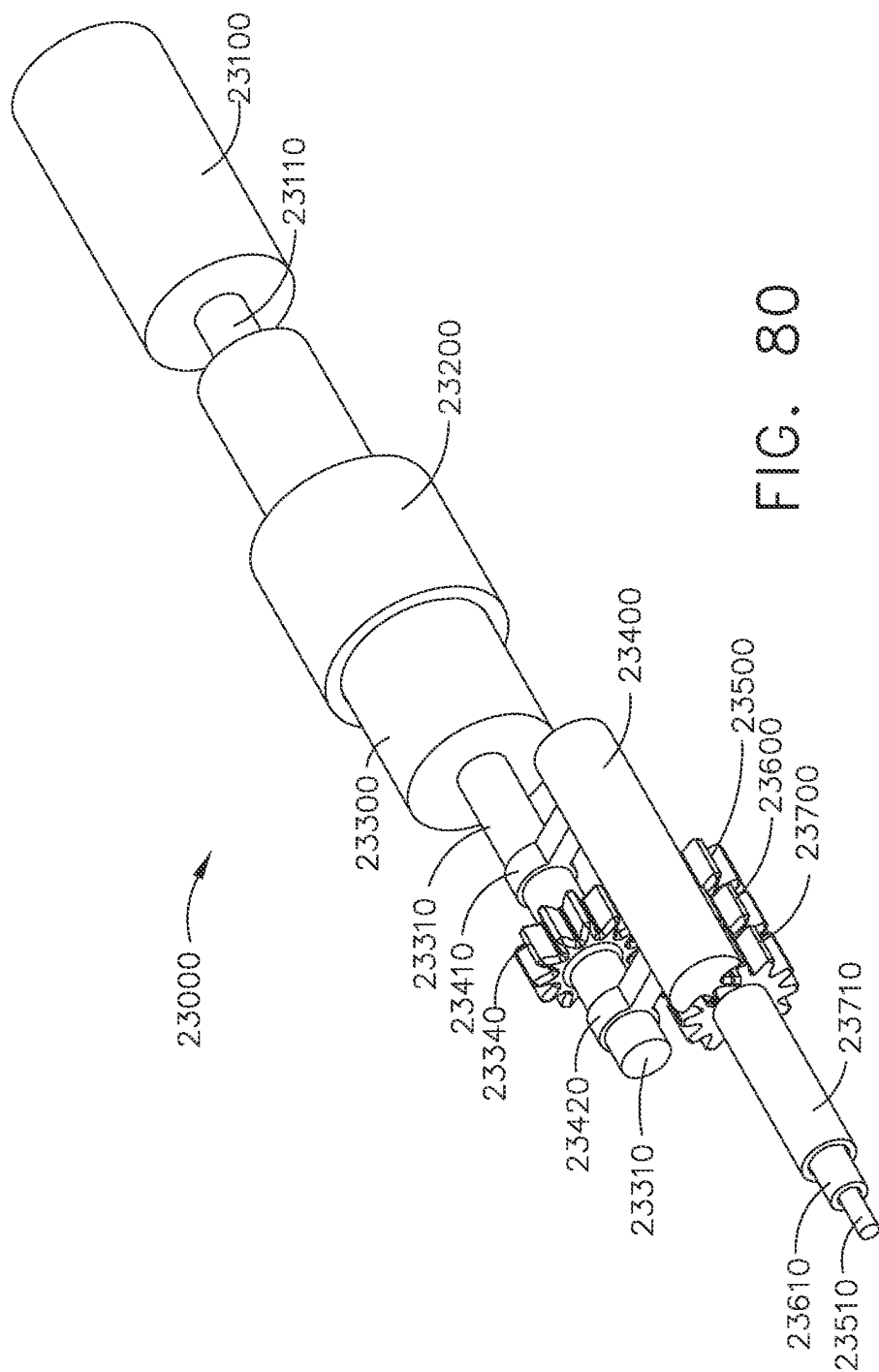
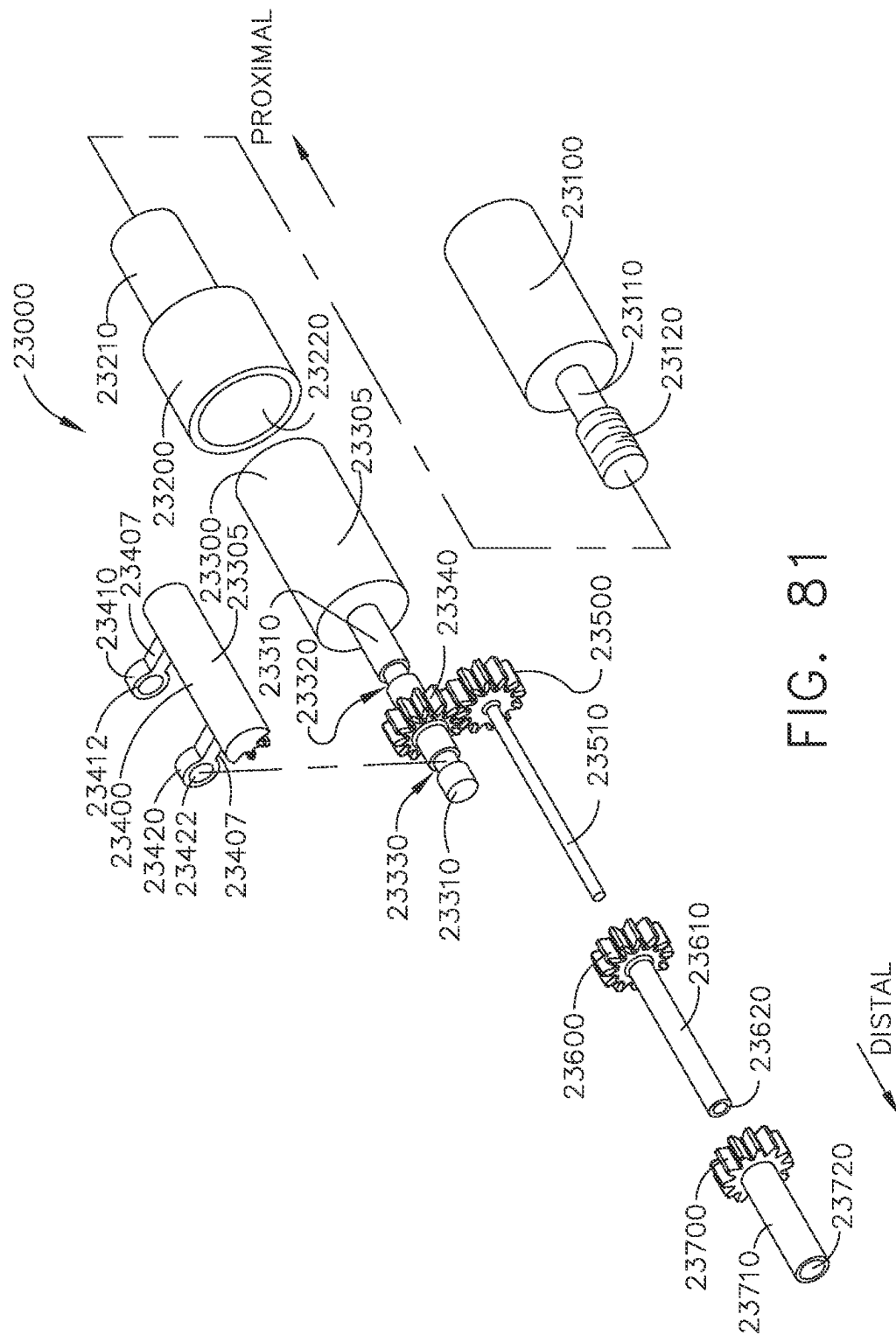


FIG. 79





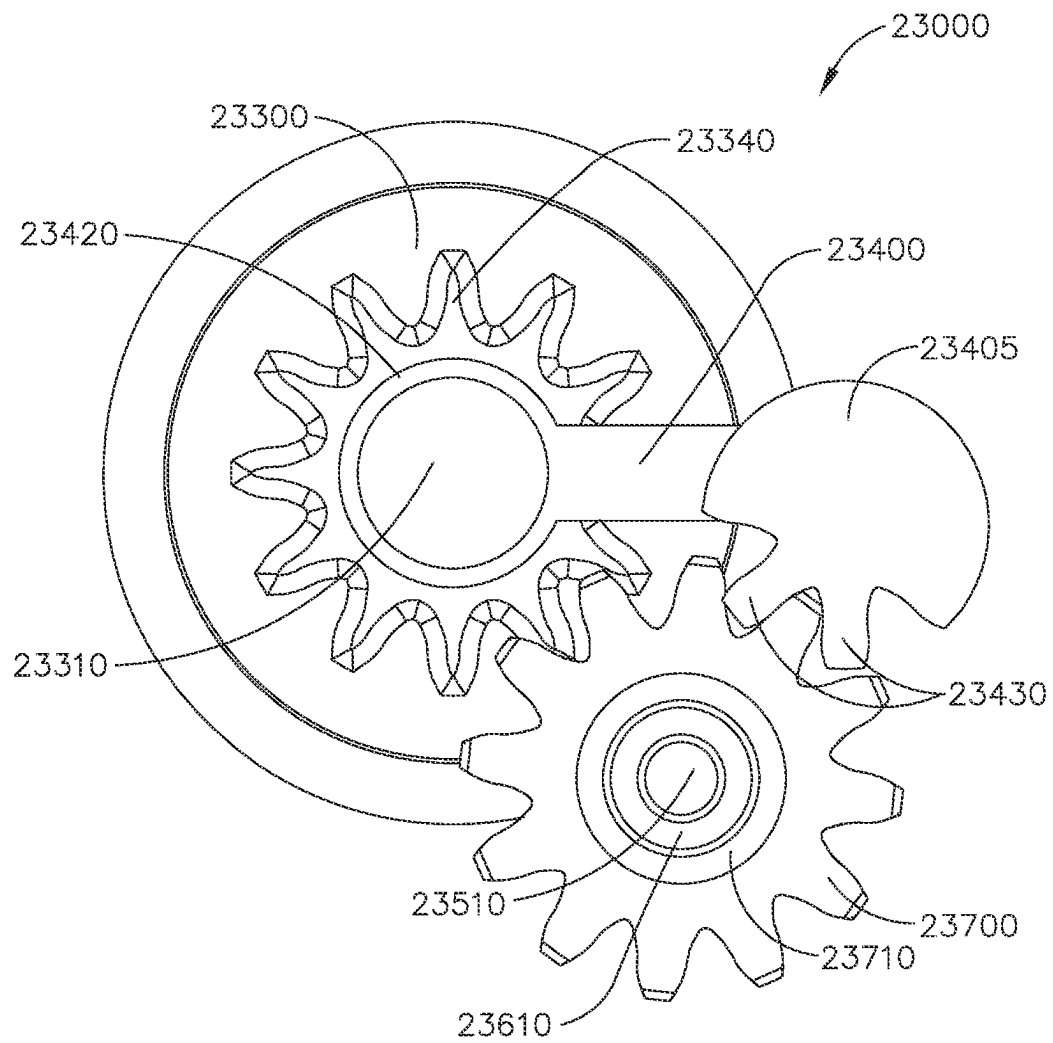


FIG. 82

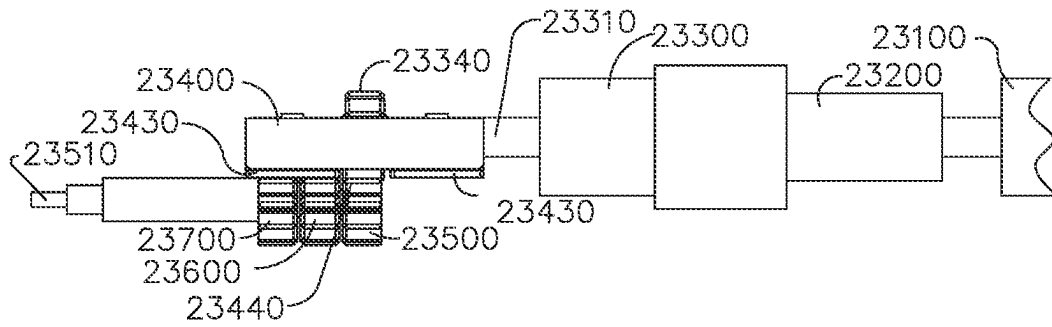


FIG. 83

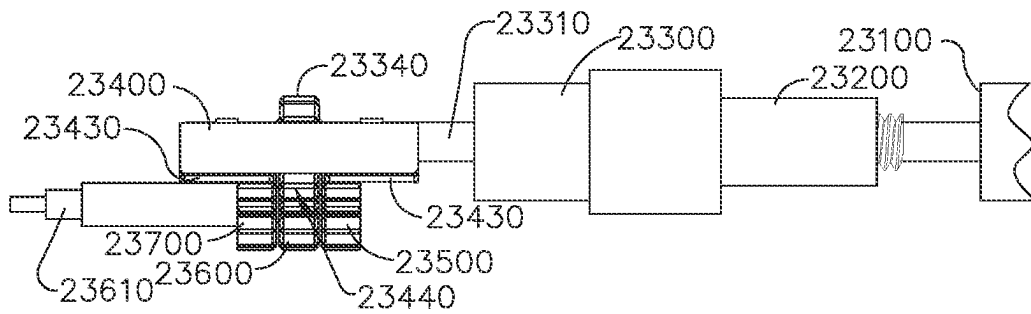


FIG. 84

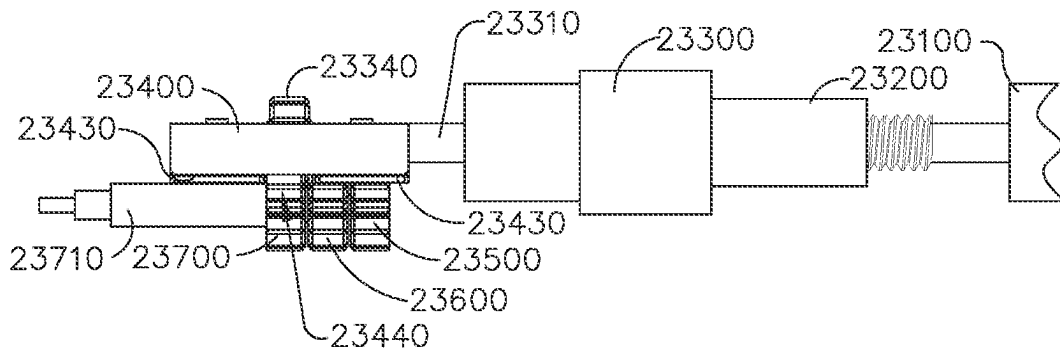
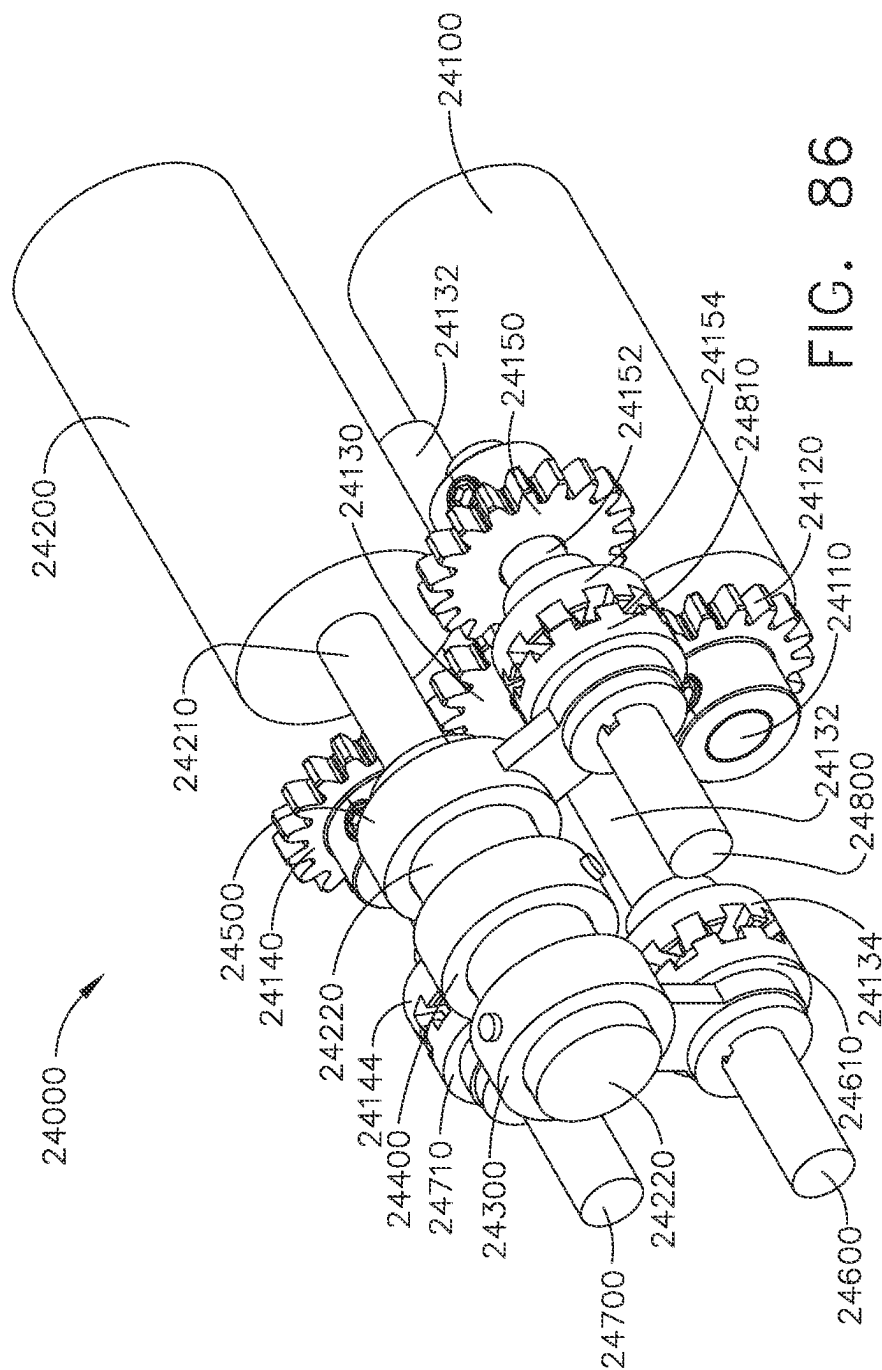


FIG. 85



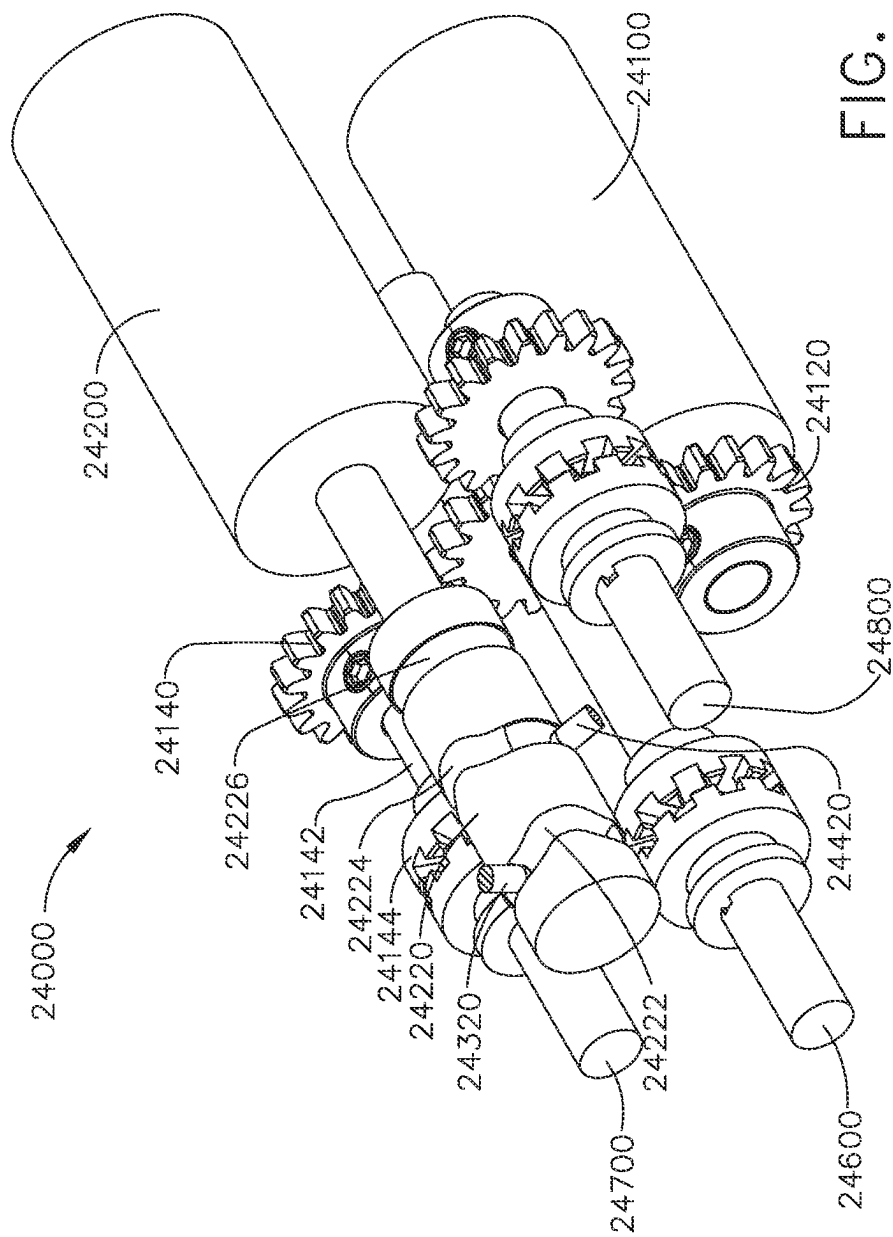


FIG. 87

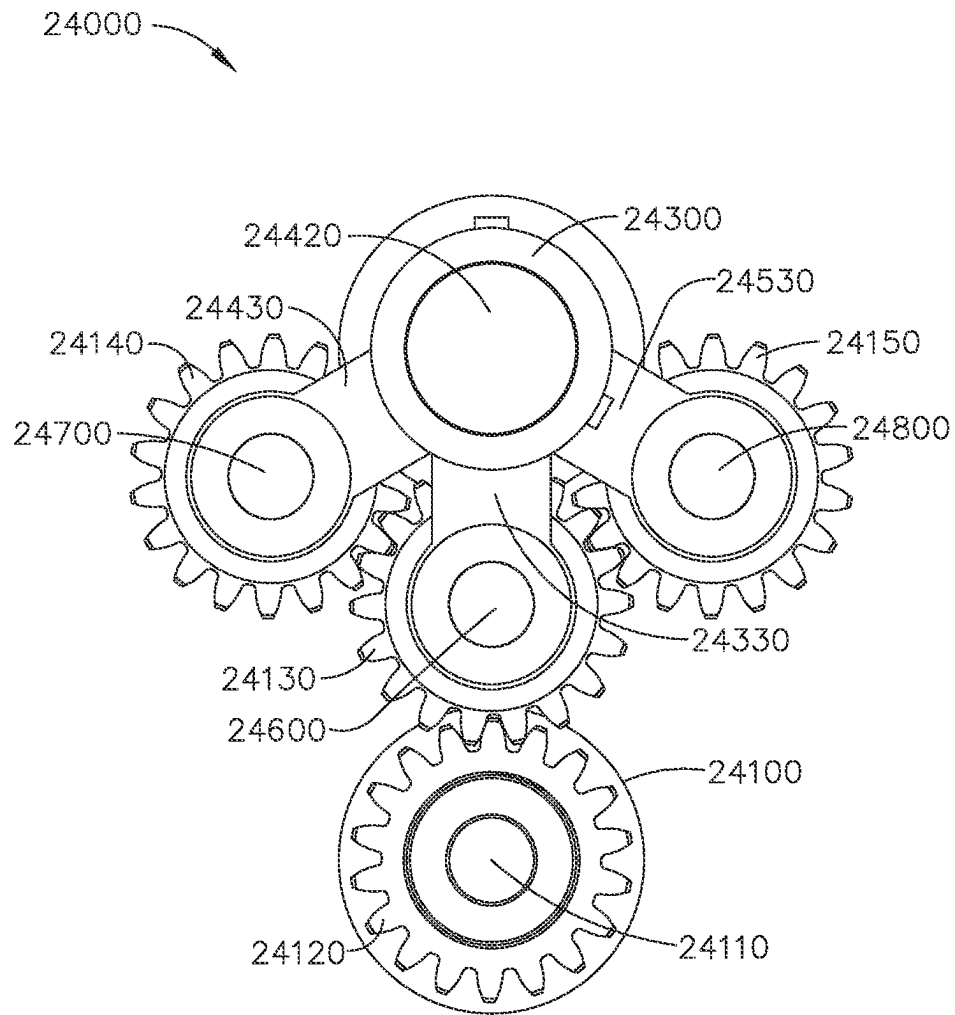
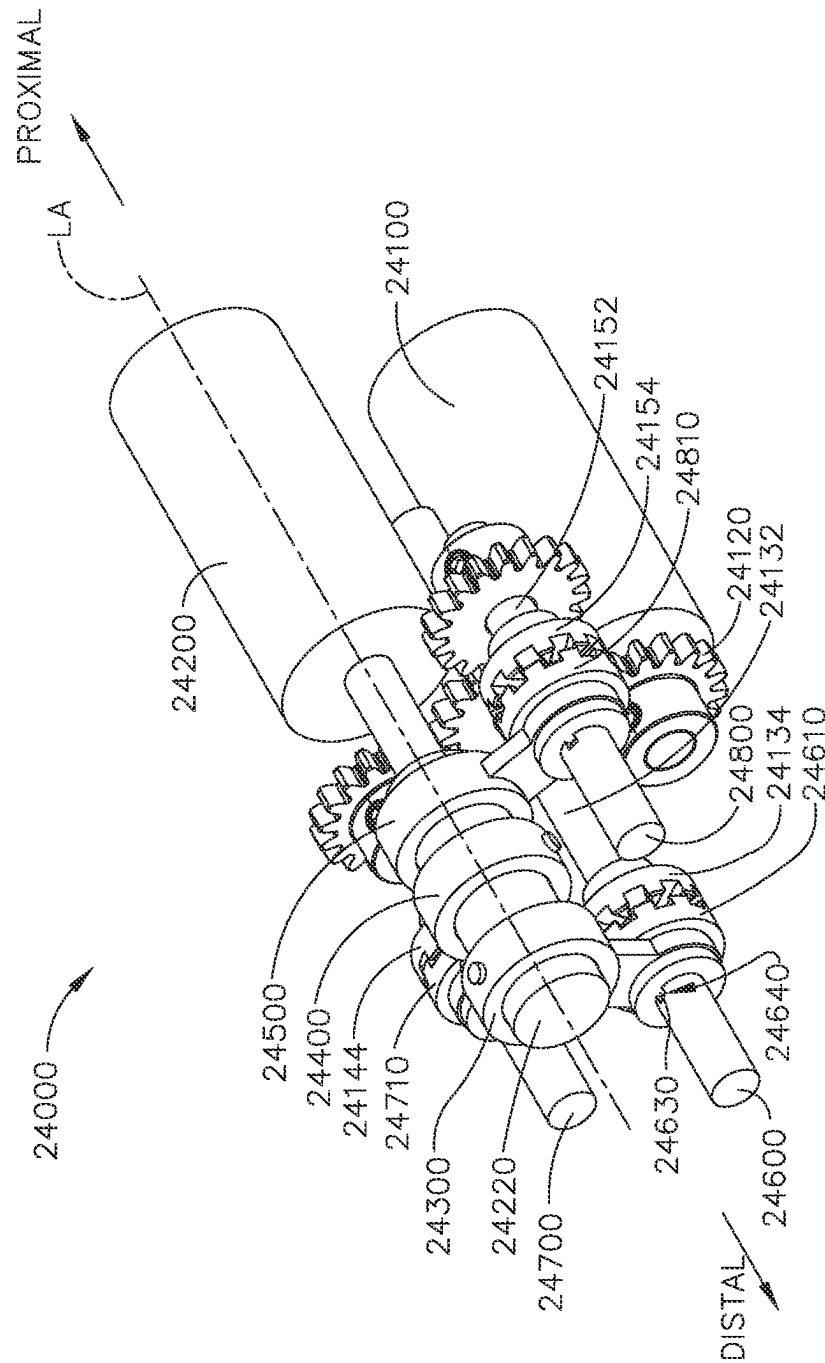


FIG. 89



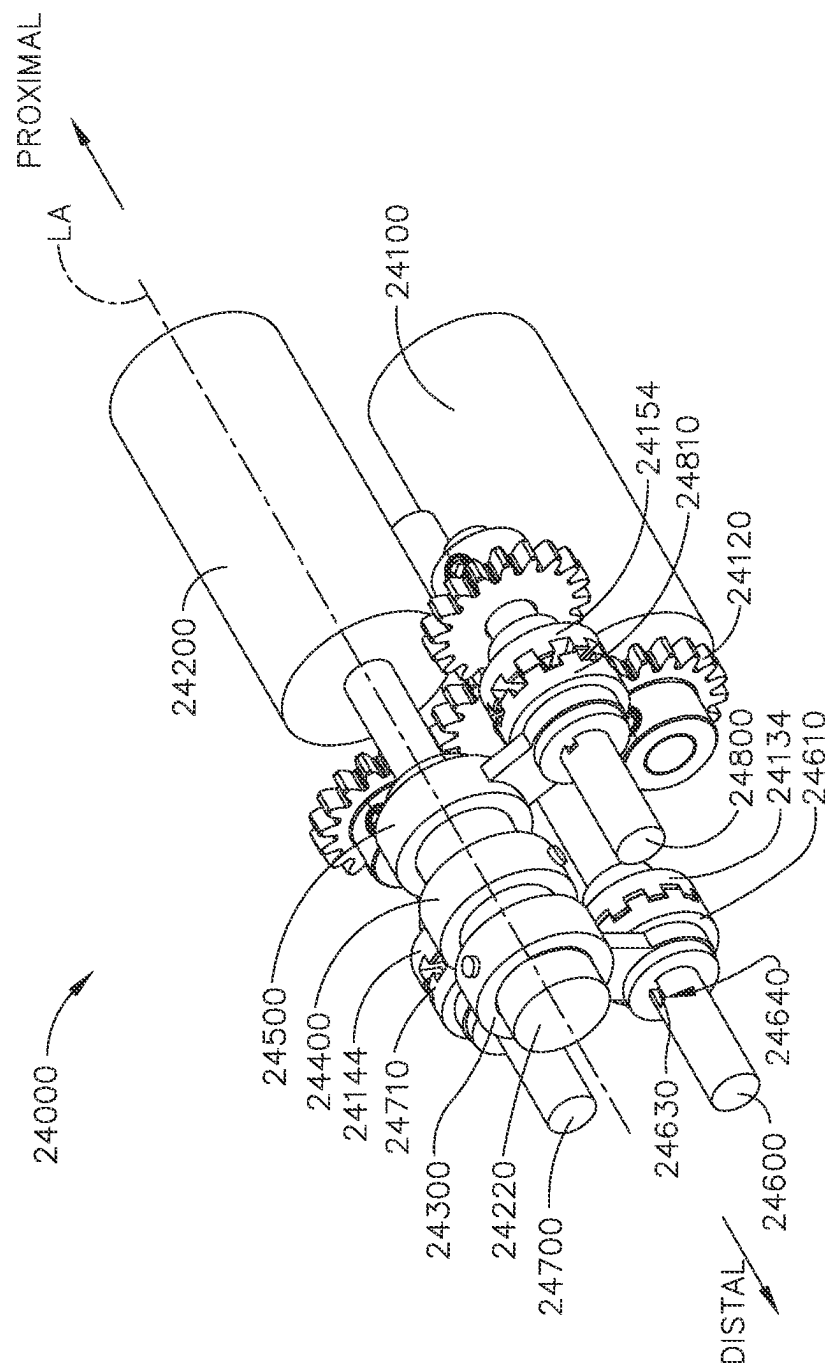


FIG. 91

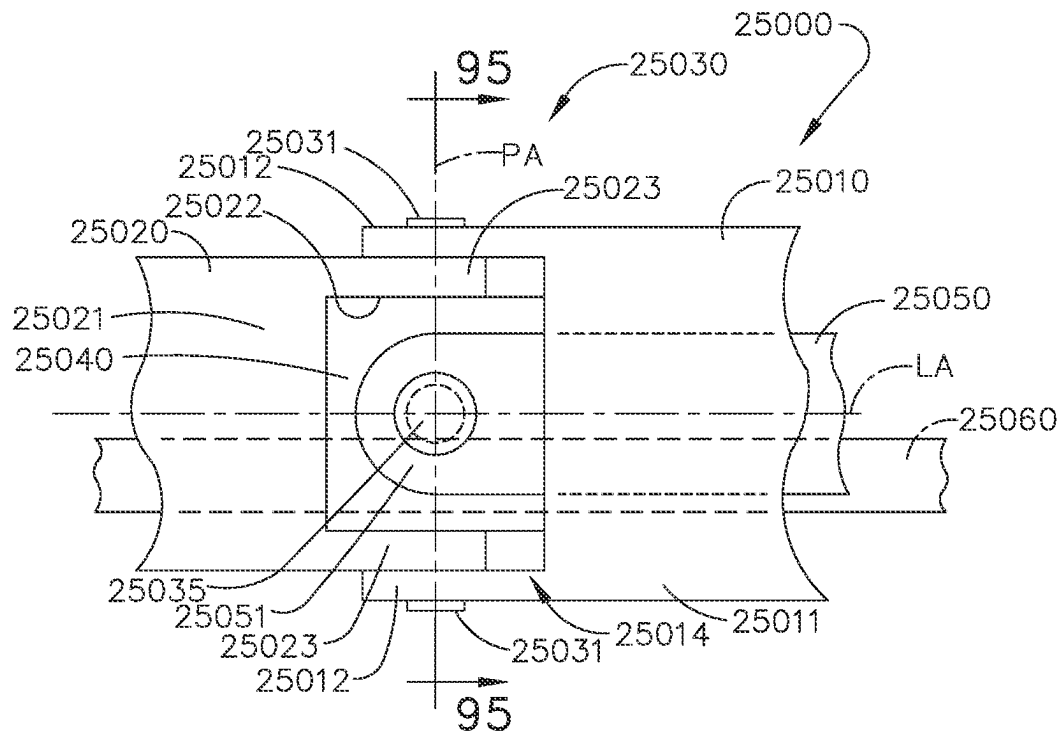


FIG. 93

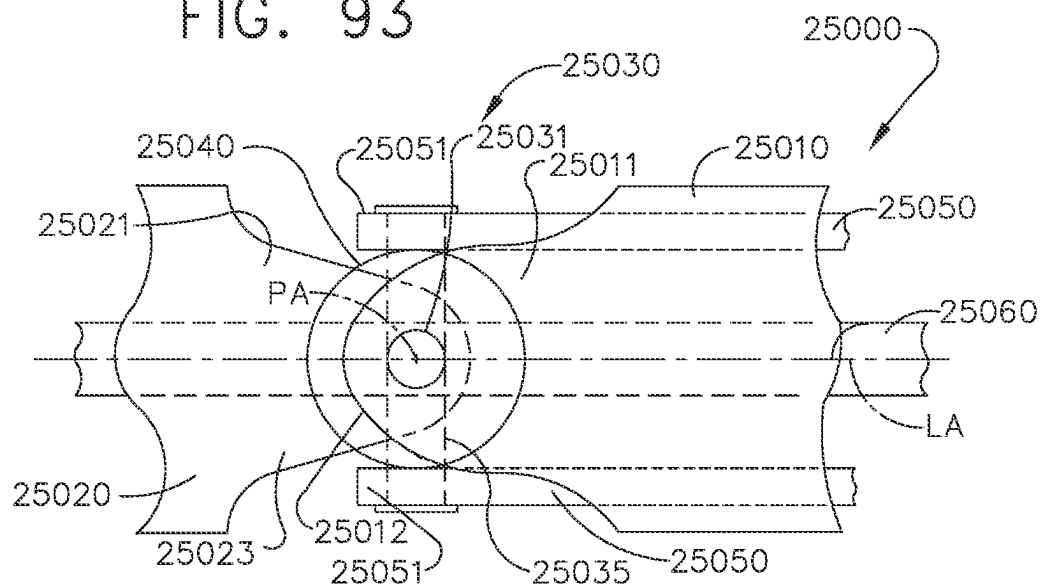


FIG. 94

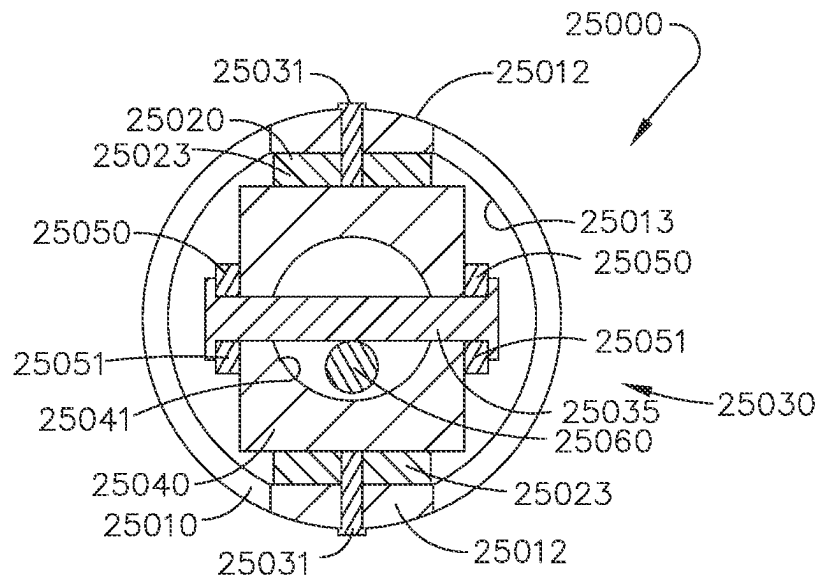


FIG. 95

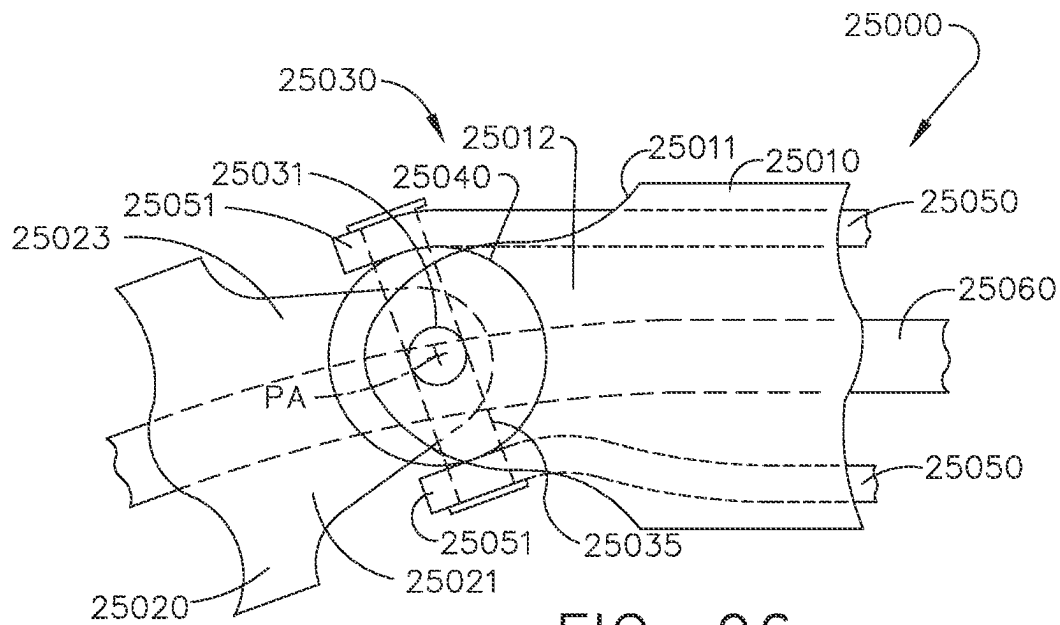


FIG. 96

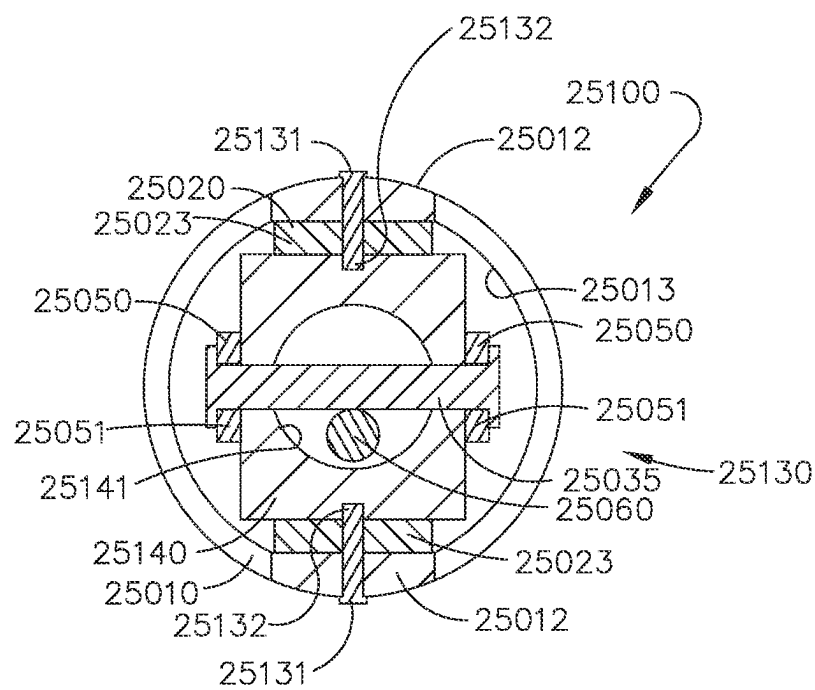


FIG. 97

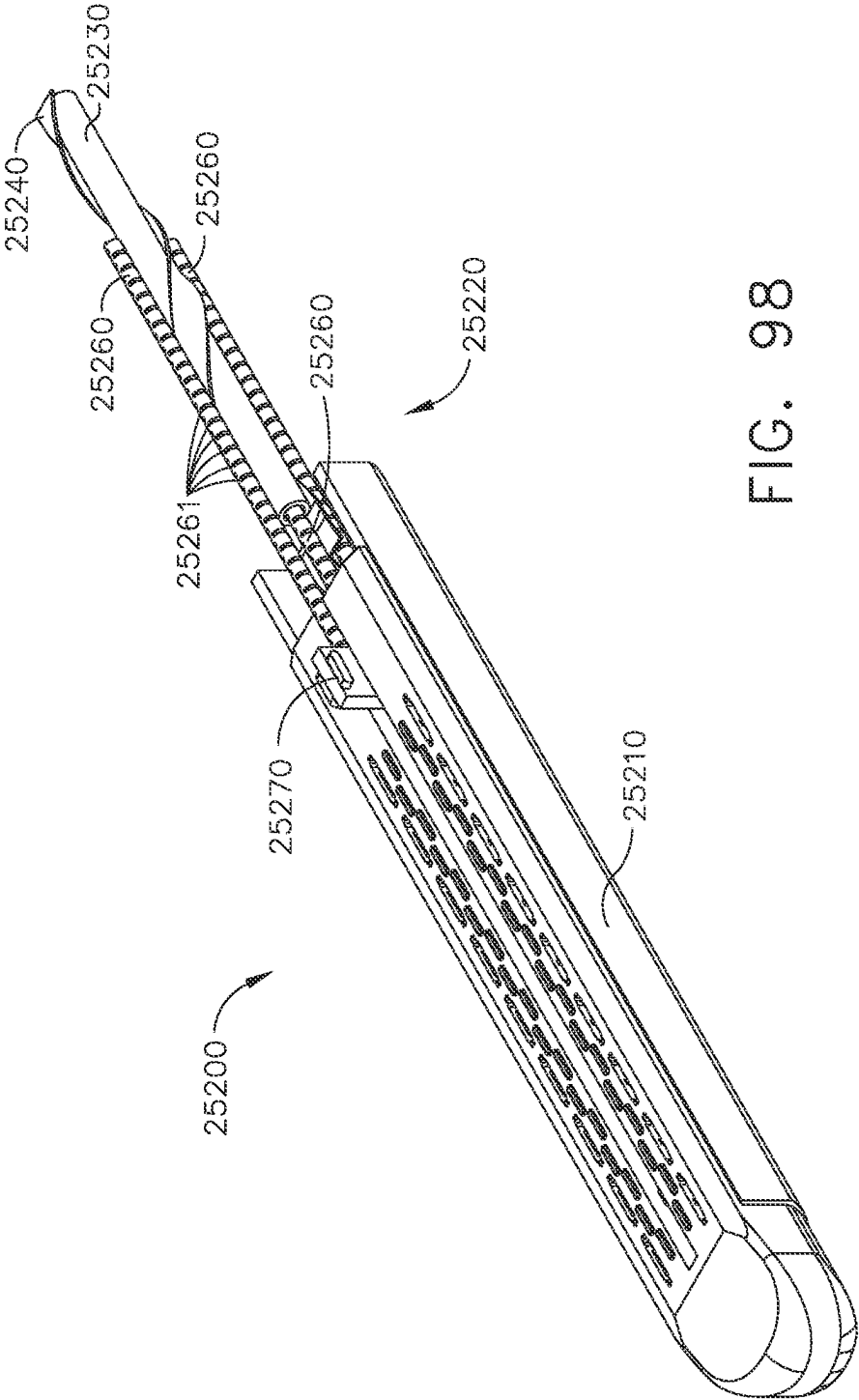


FIG. 98

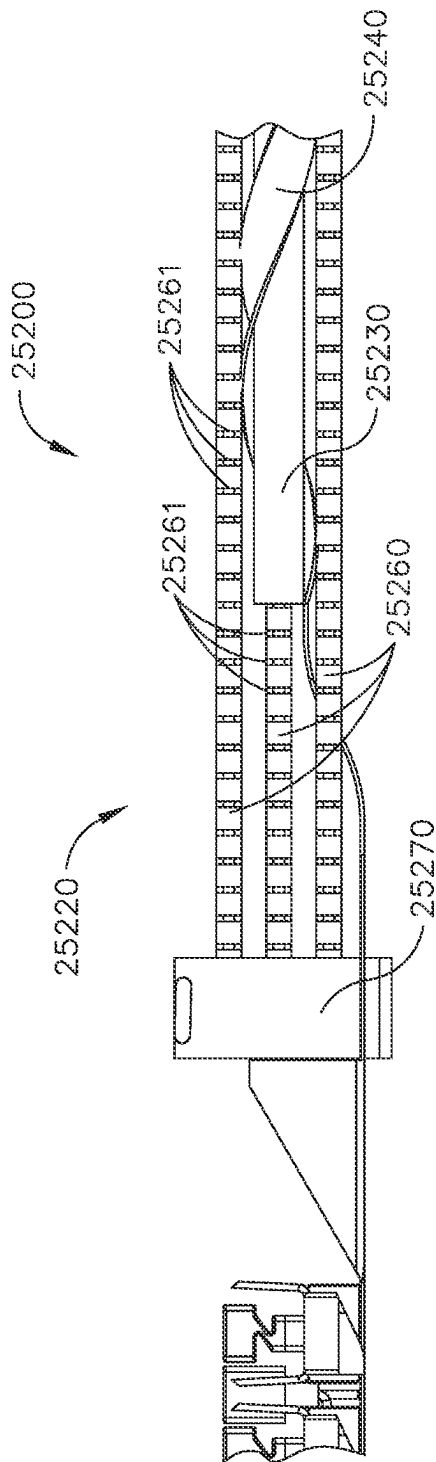


FIG. 99

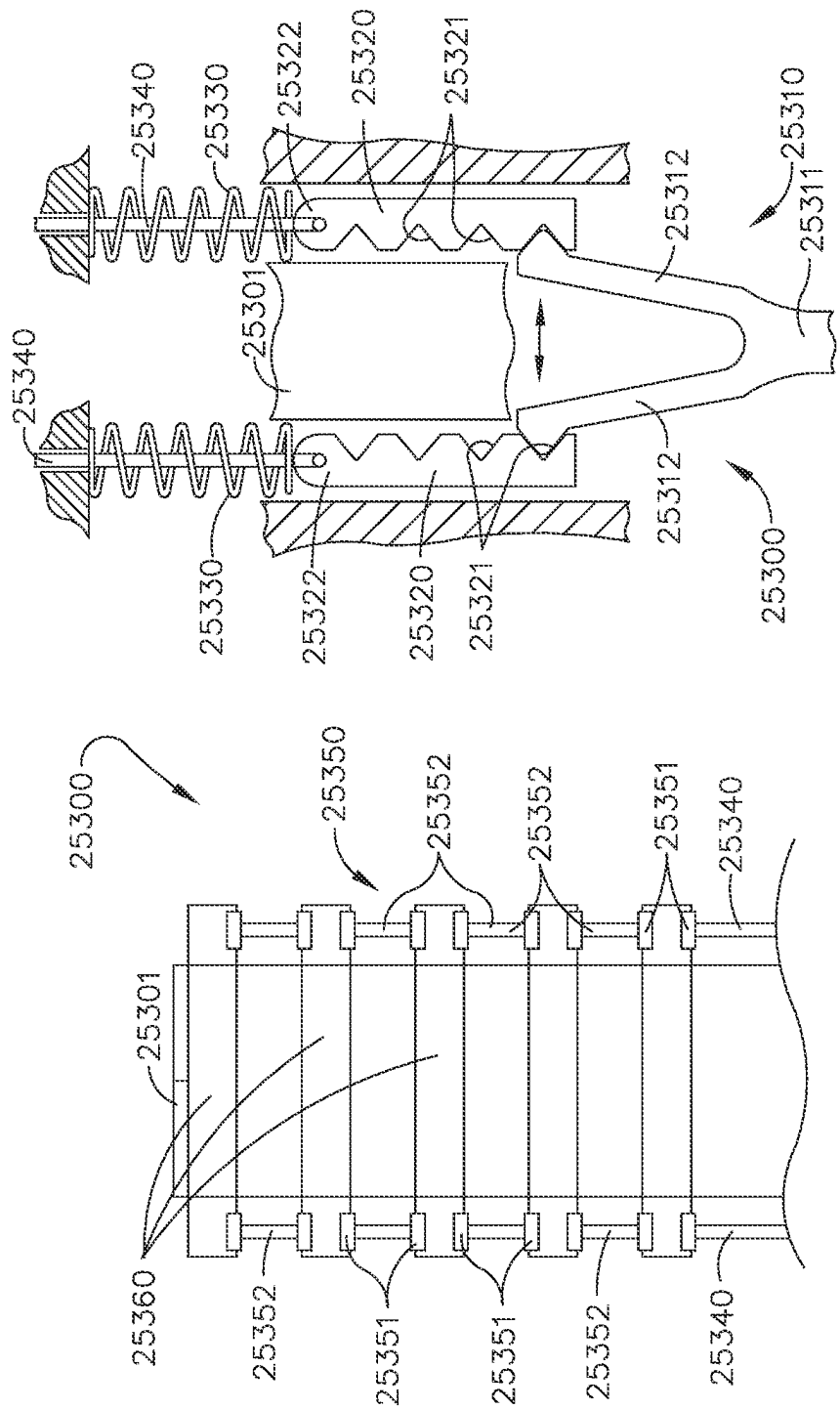


FIG. 100

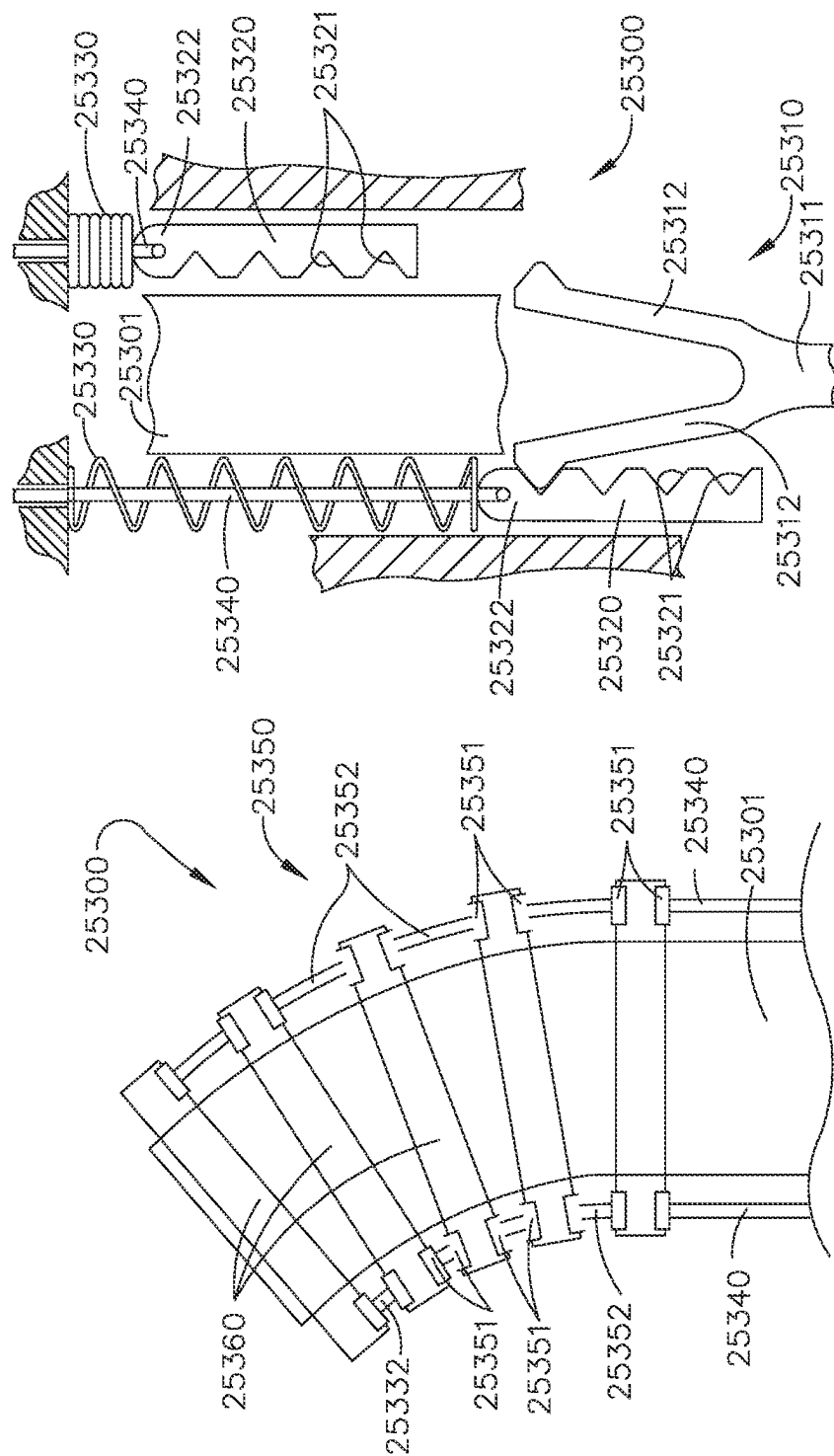
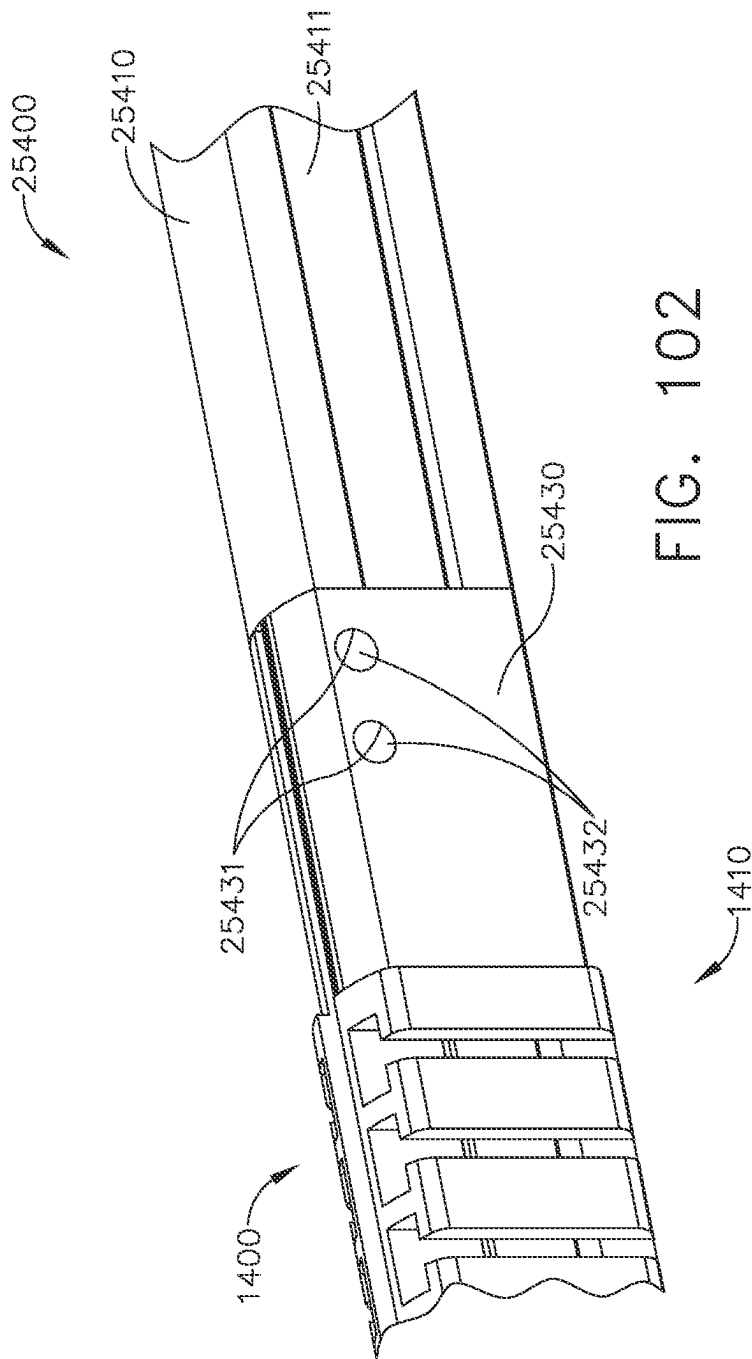


FIG. 101



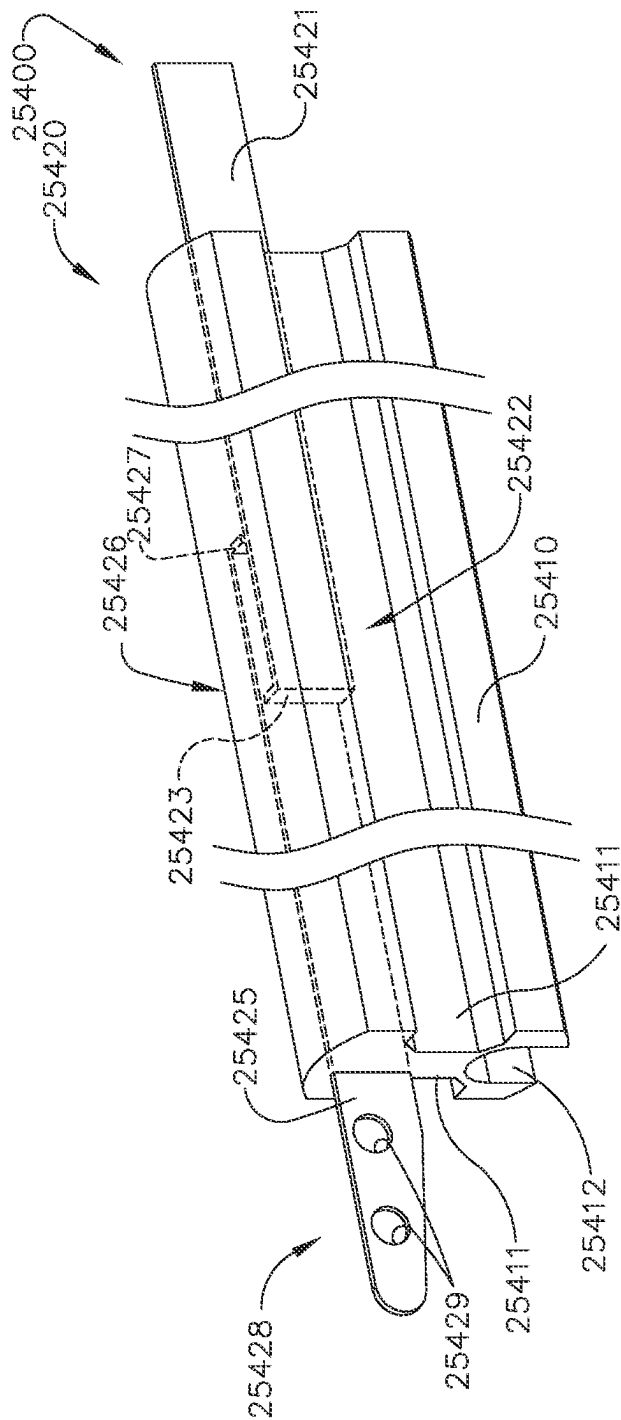


FIG. 103

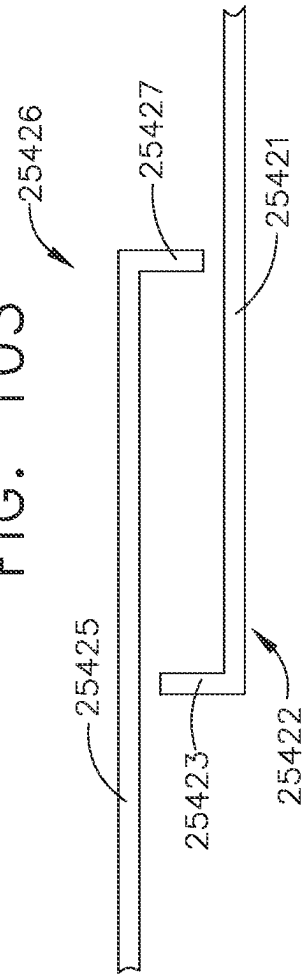


FIG. 104

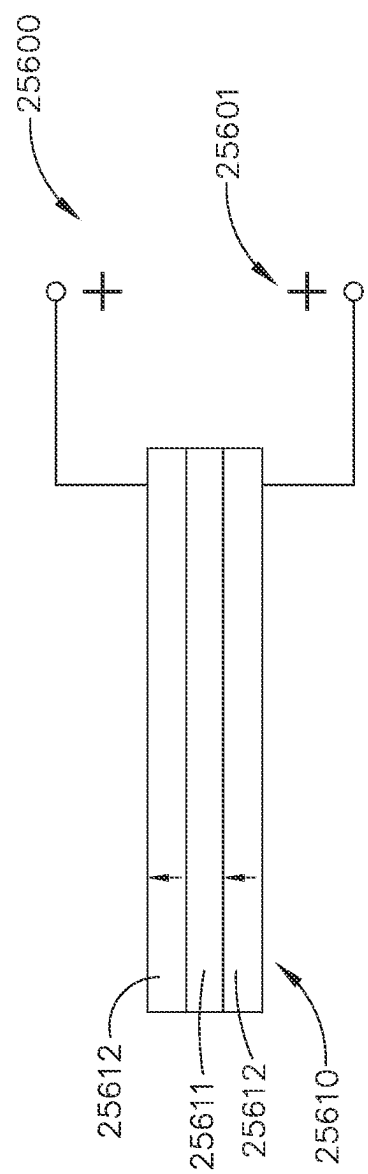


FIG. 105

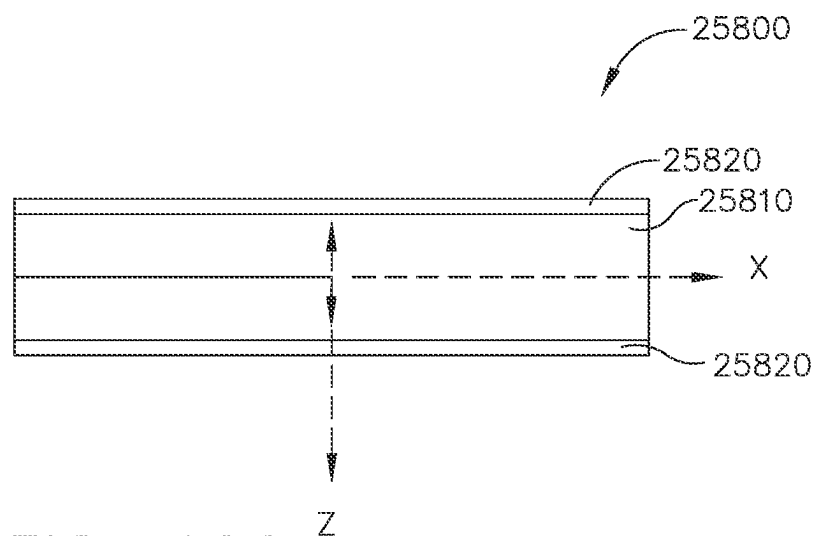


FIG. 106

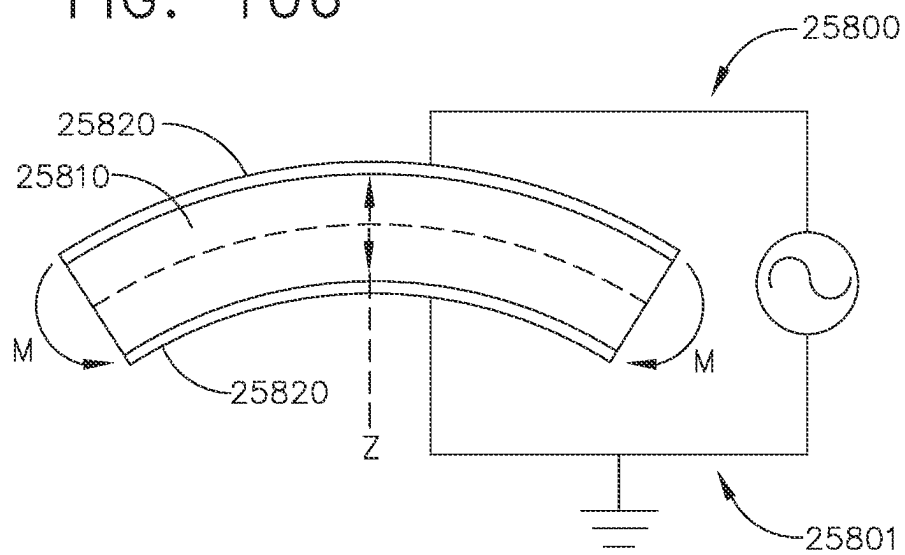


FIG. 107

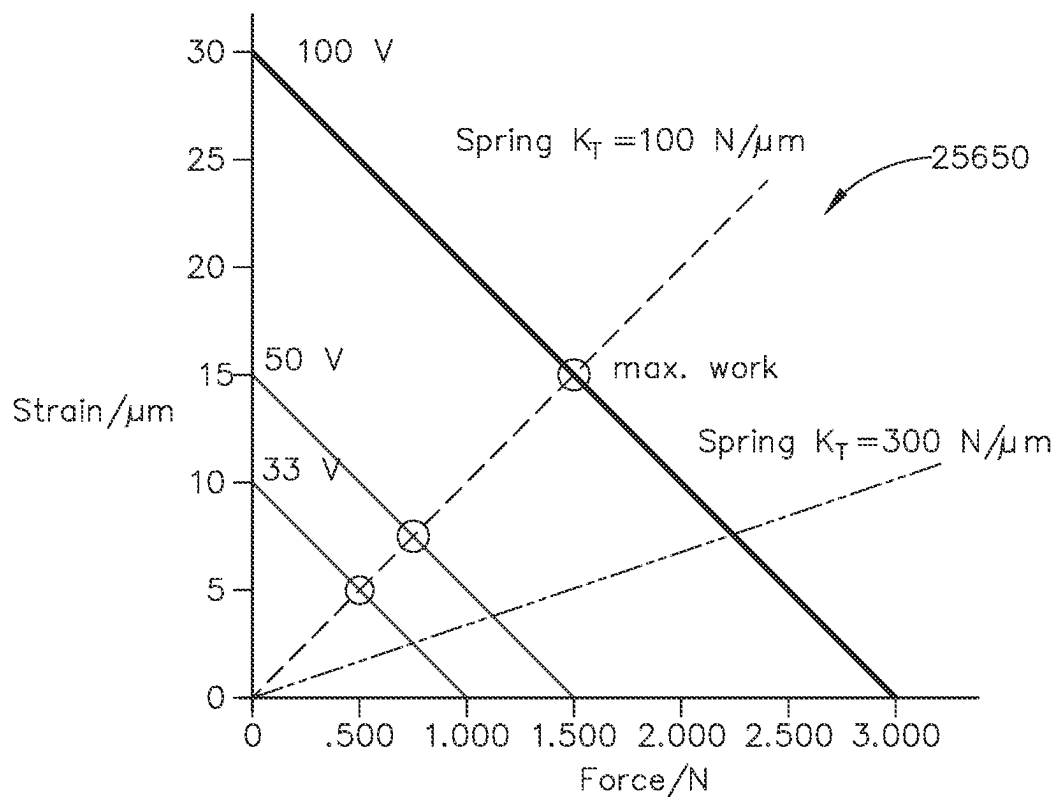


FIG. 110

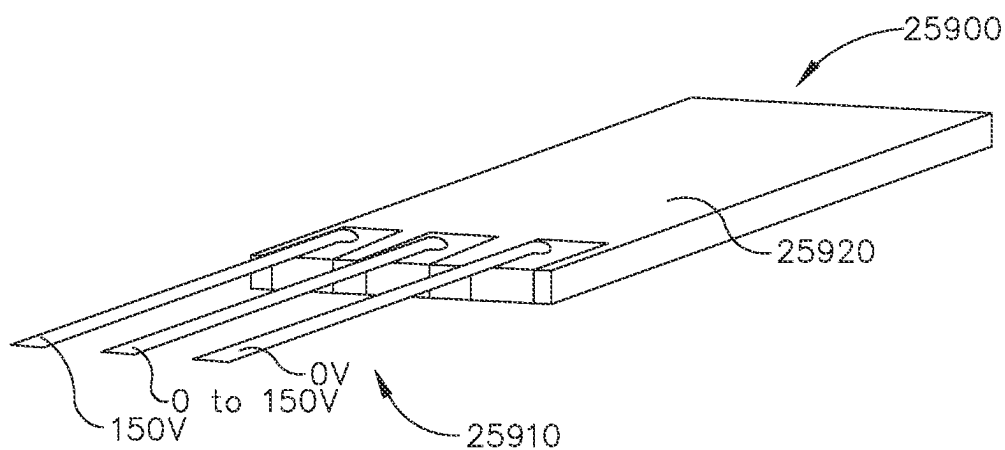


FIG. 108

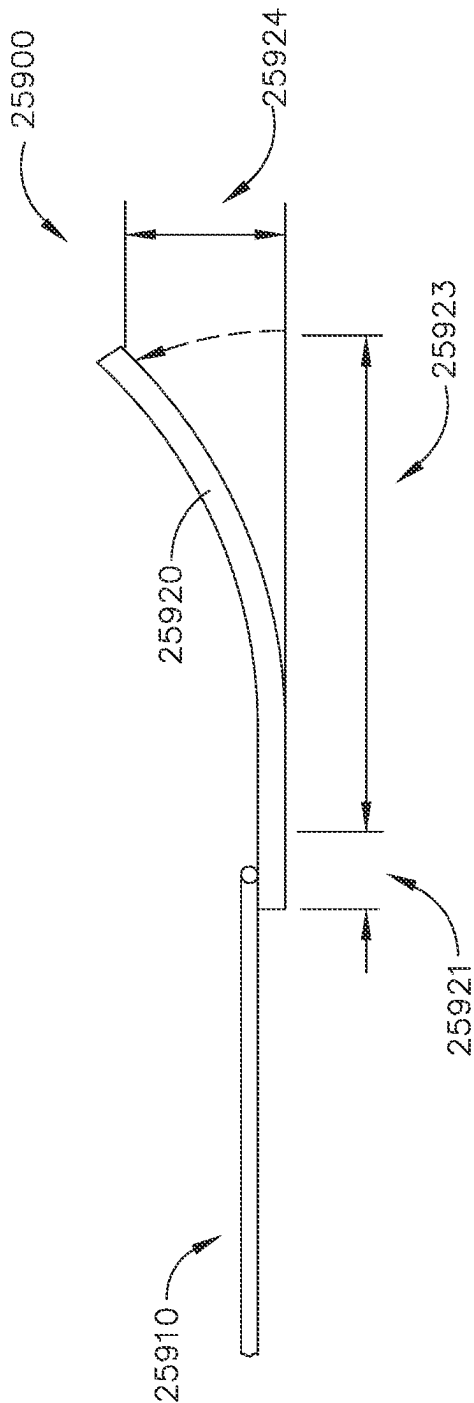


FIG. 109

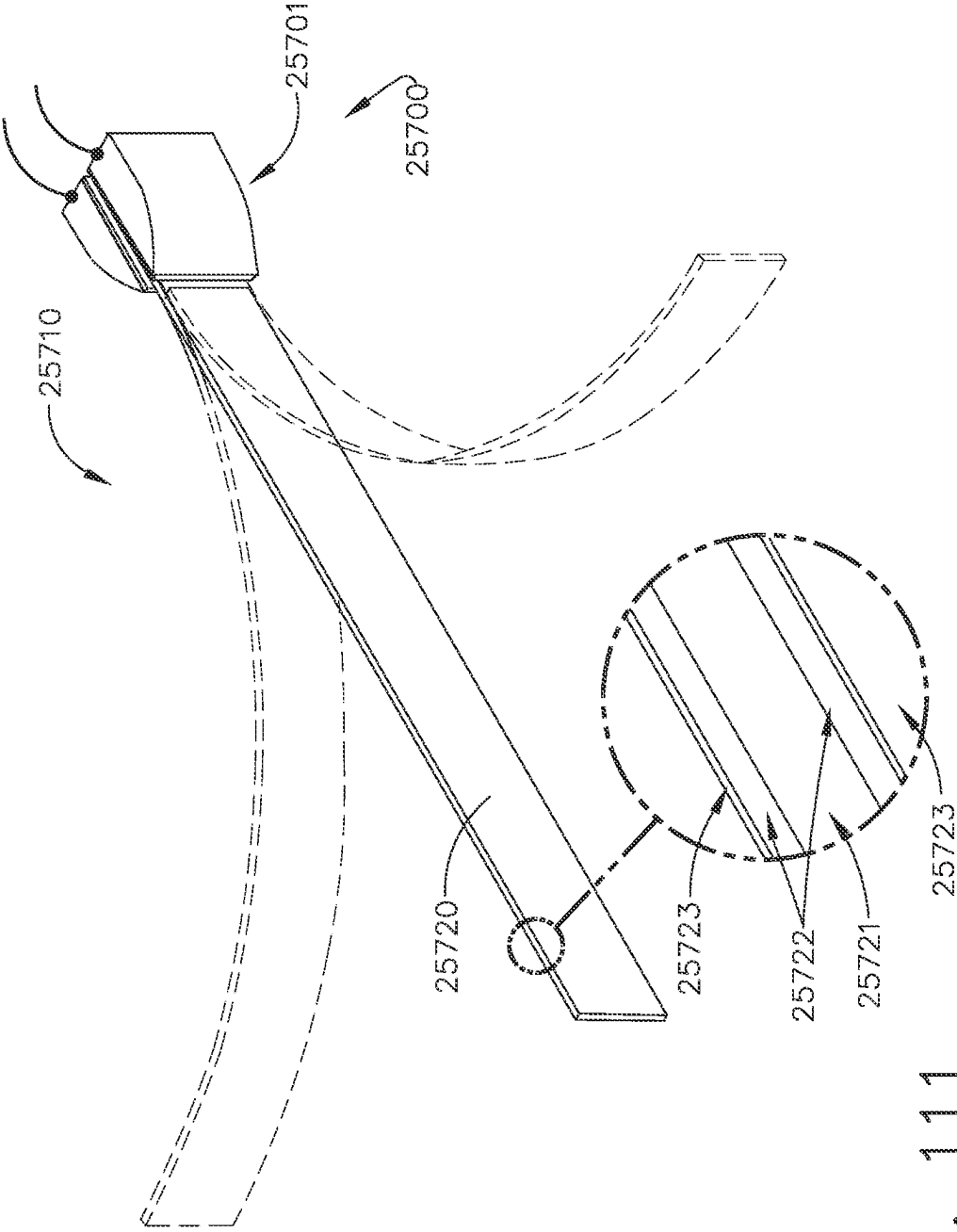


FIG. 111

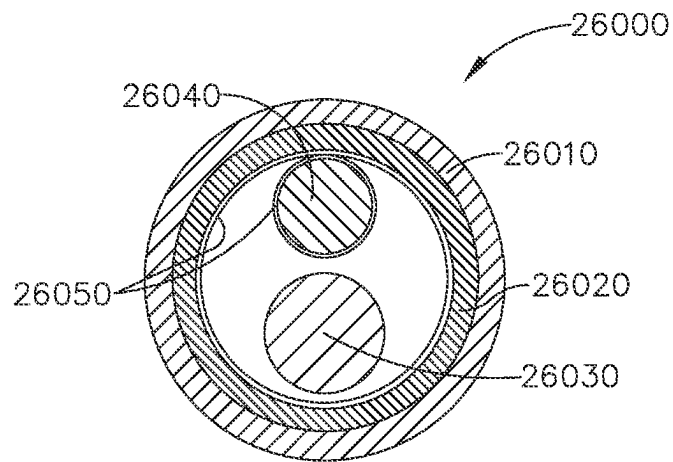


FIG. 112

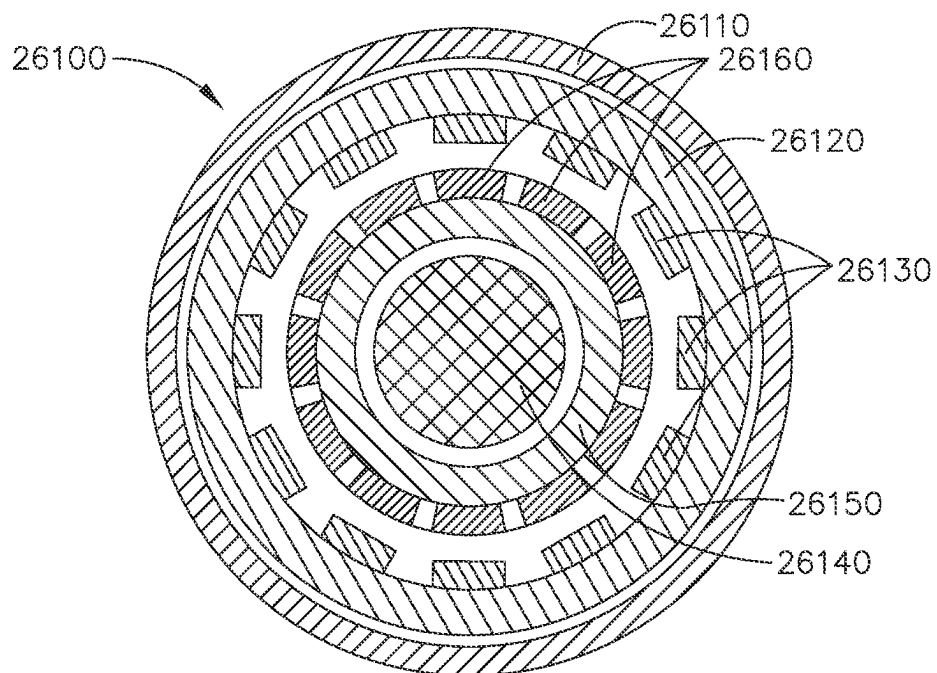
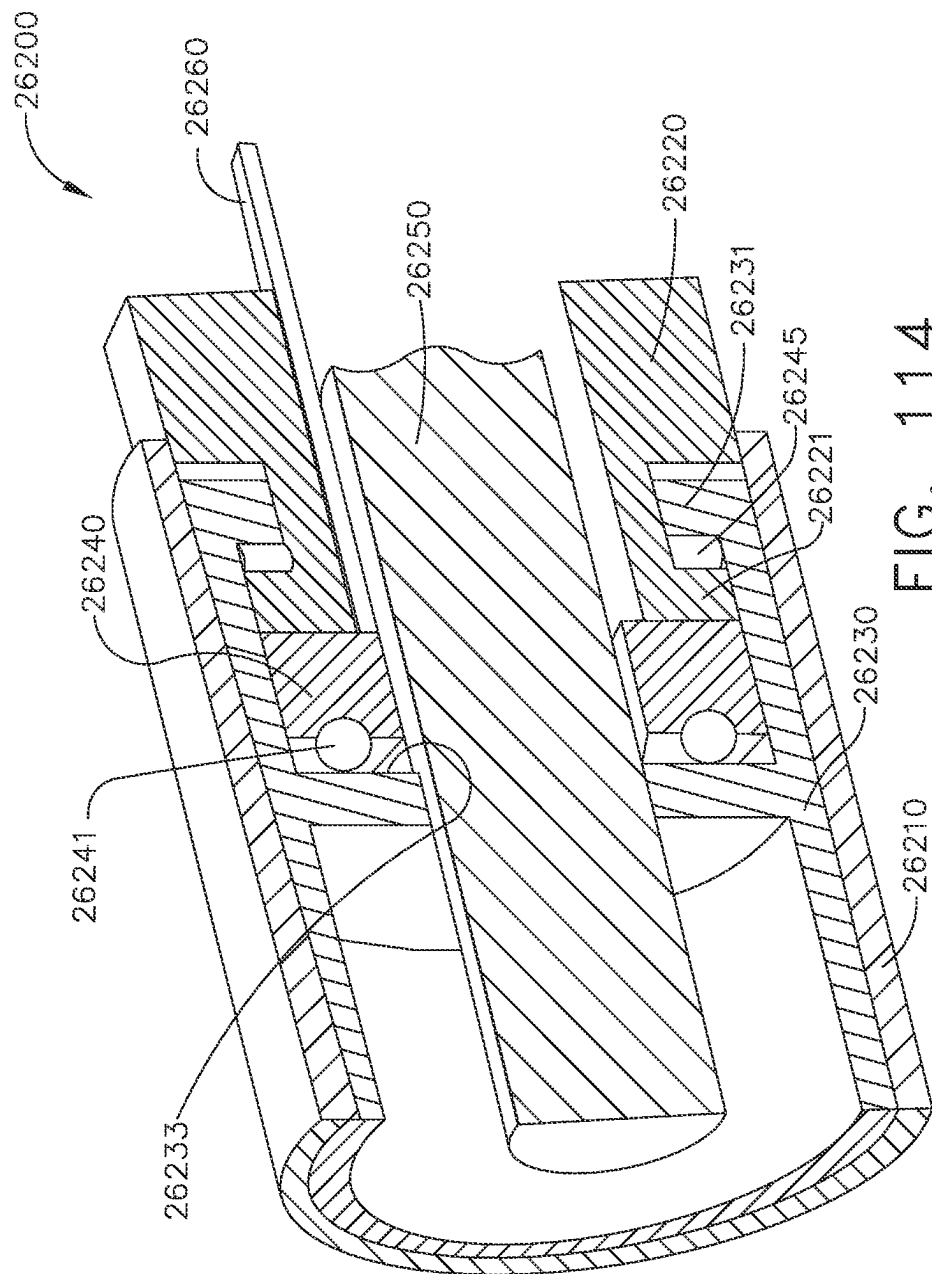


FIG. 113



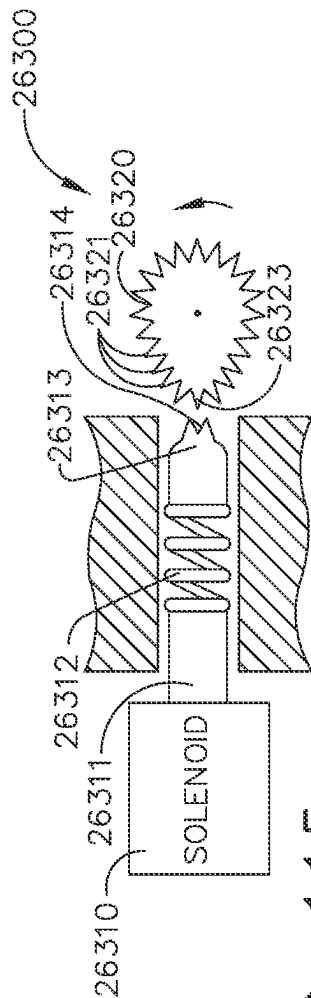


FIG. 115

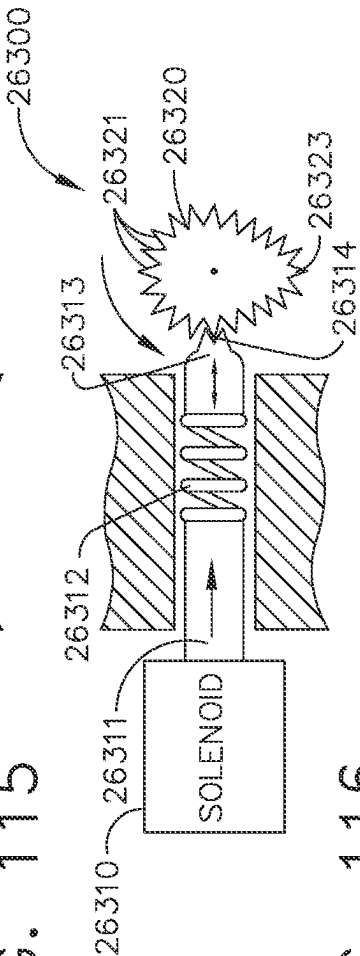


FIG. 116

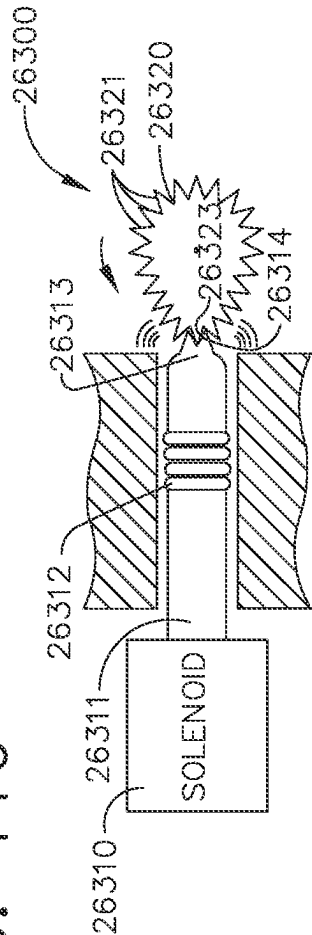
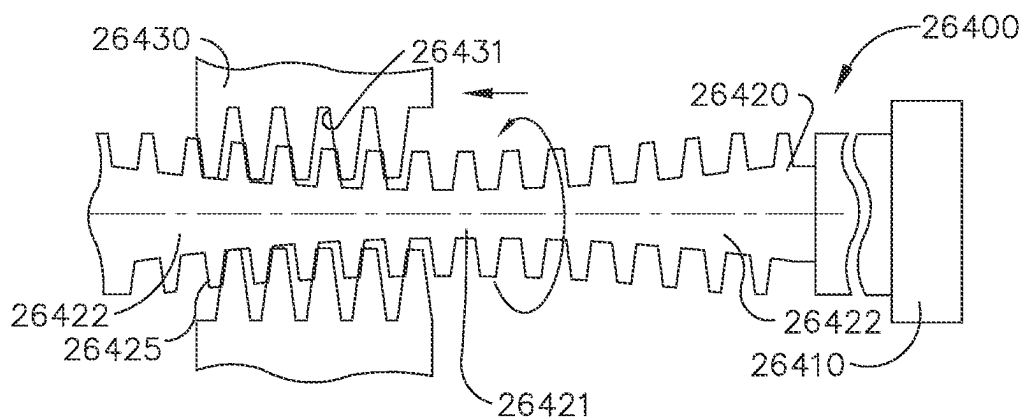
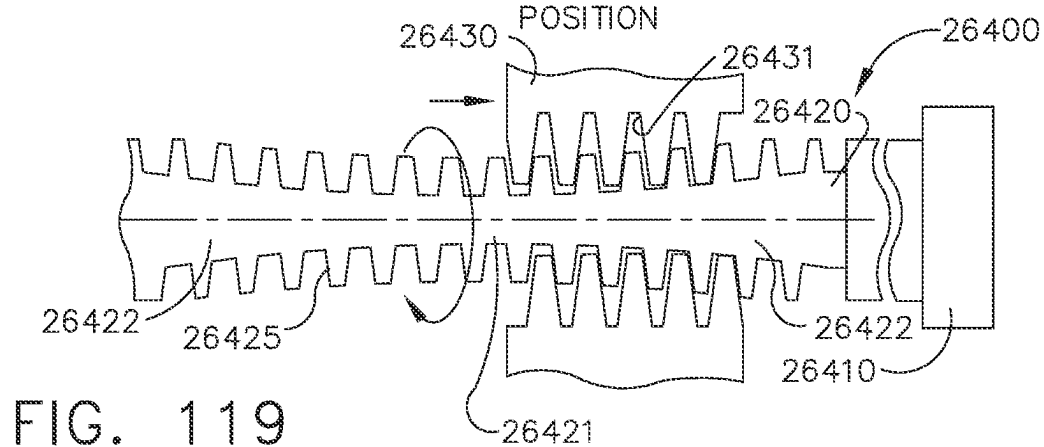
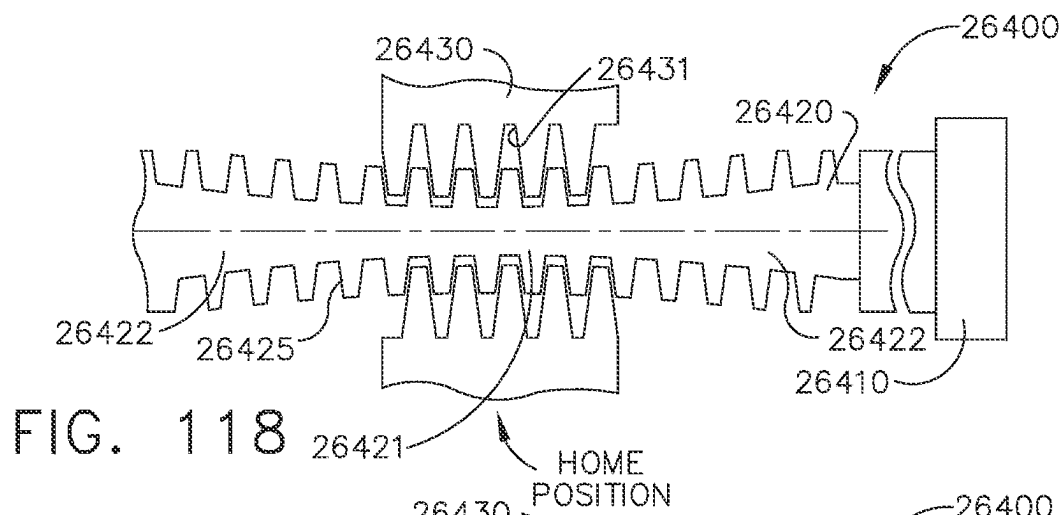


FIG. 117



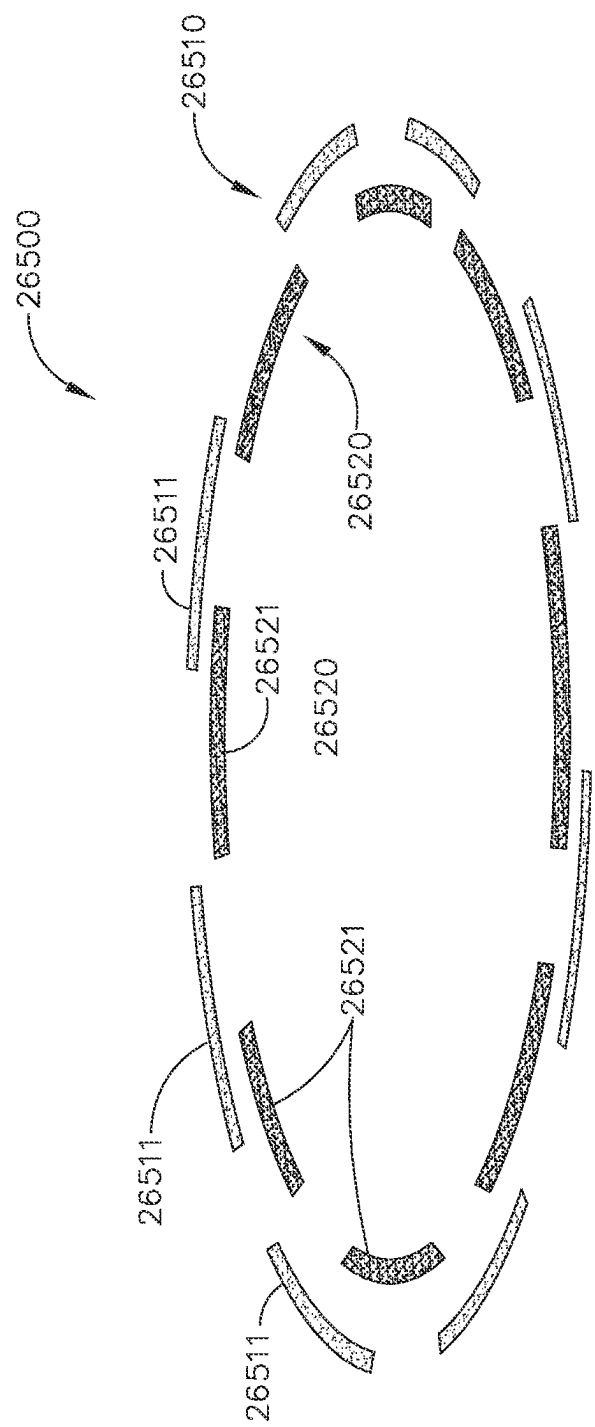
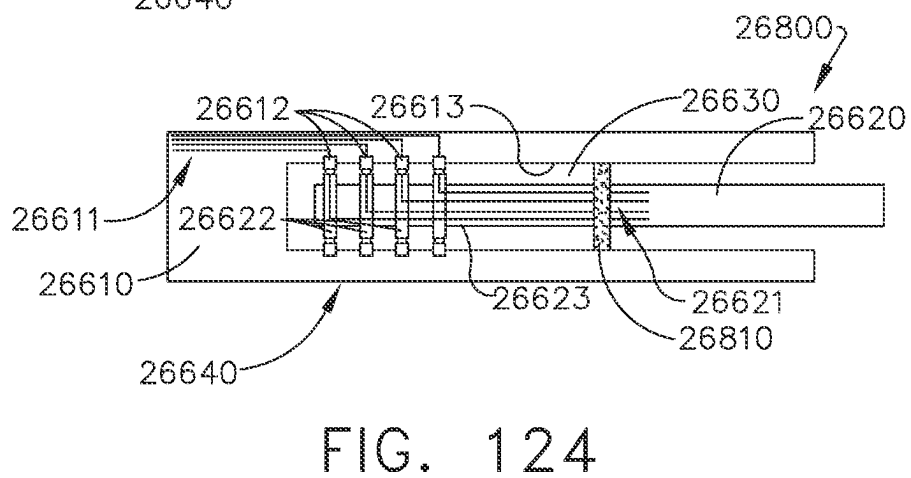
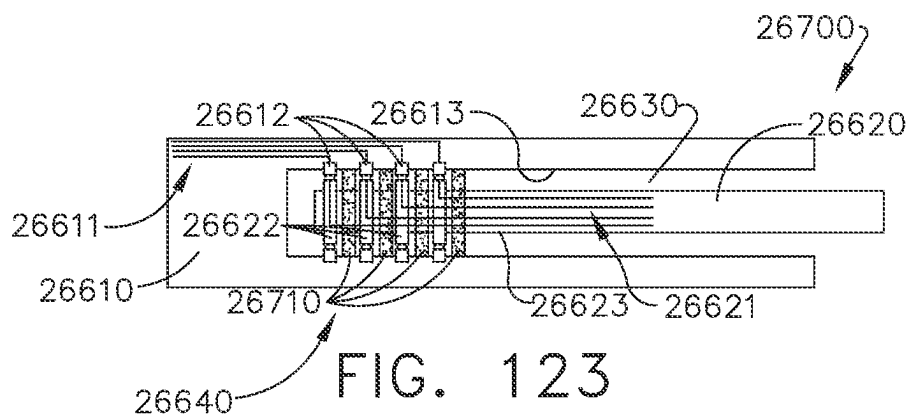
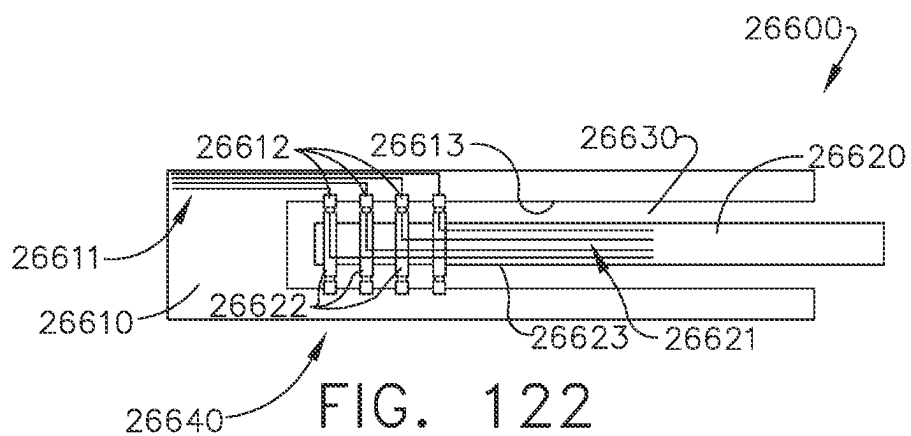


FIG. 121



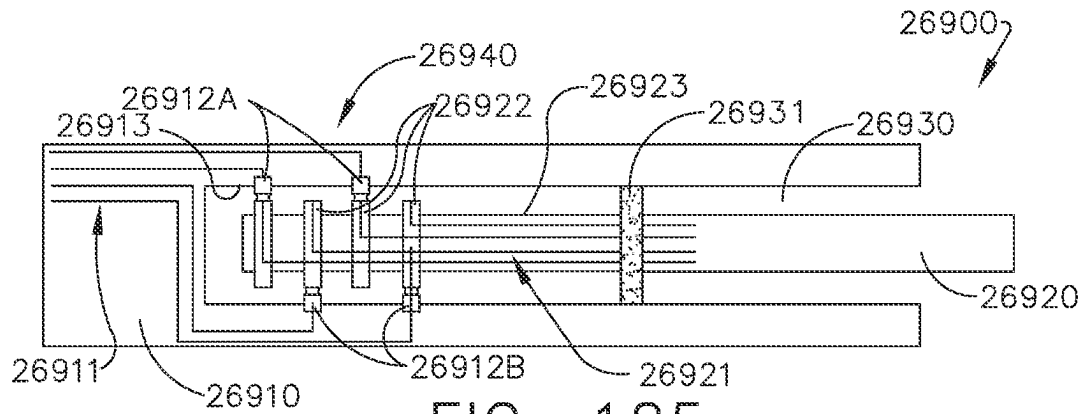


FIG. 125

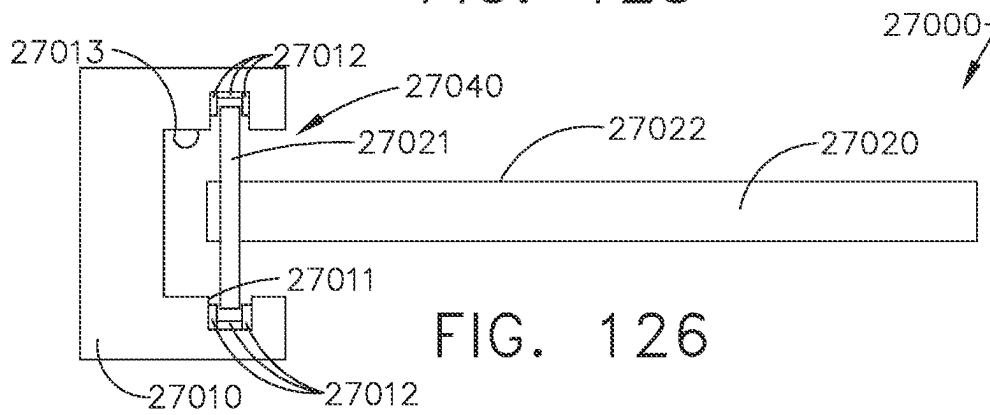


FIG. 126

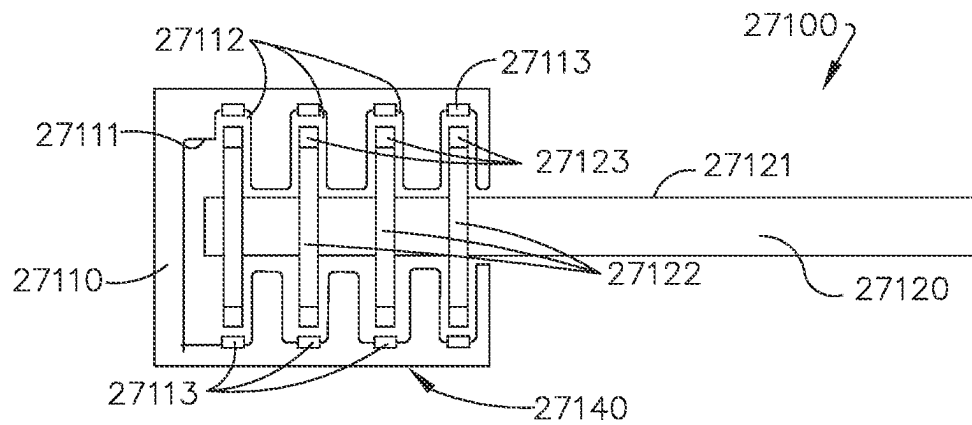


FIG. 127

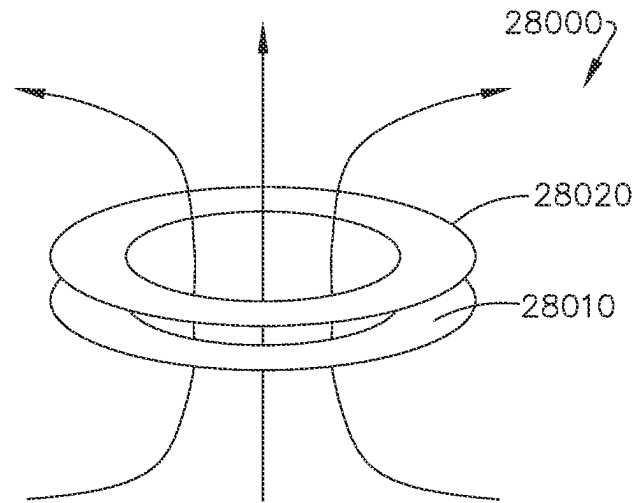


FIG. 128

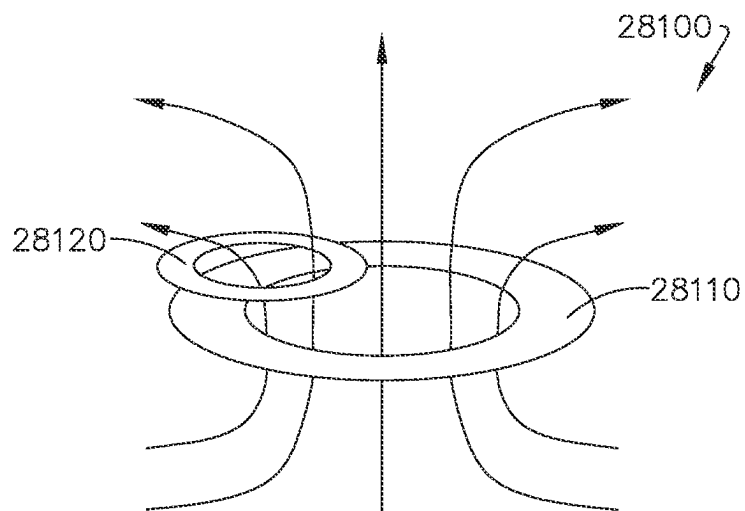


FIG. 129

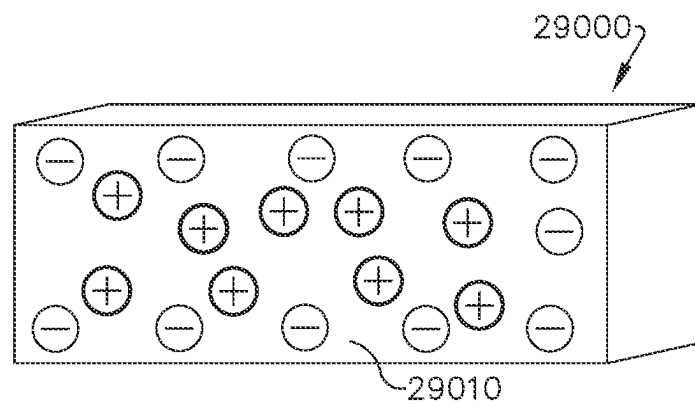


FIG. 130

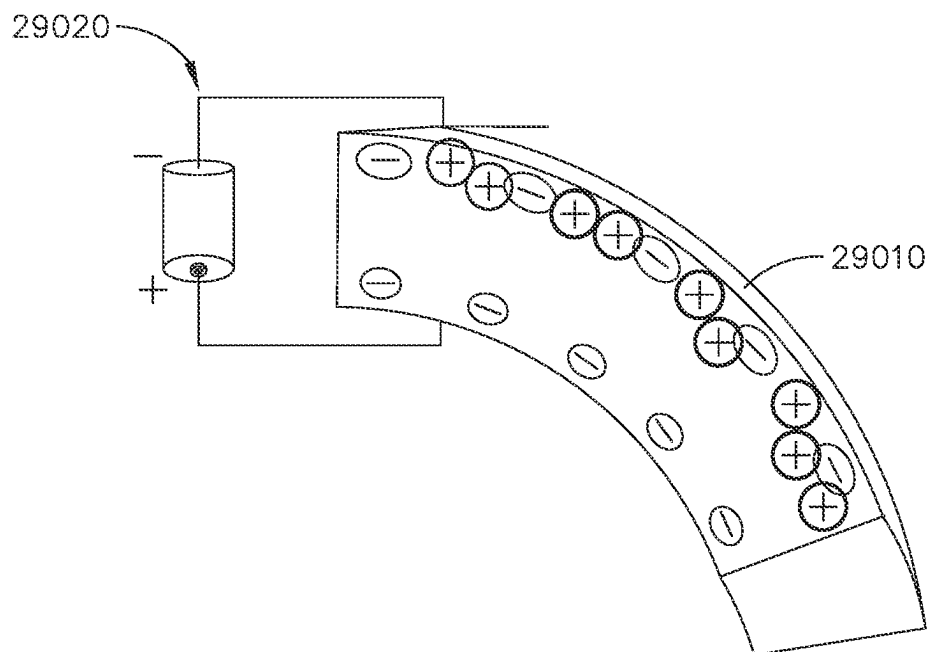


FIG. 131

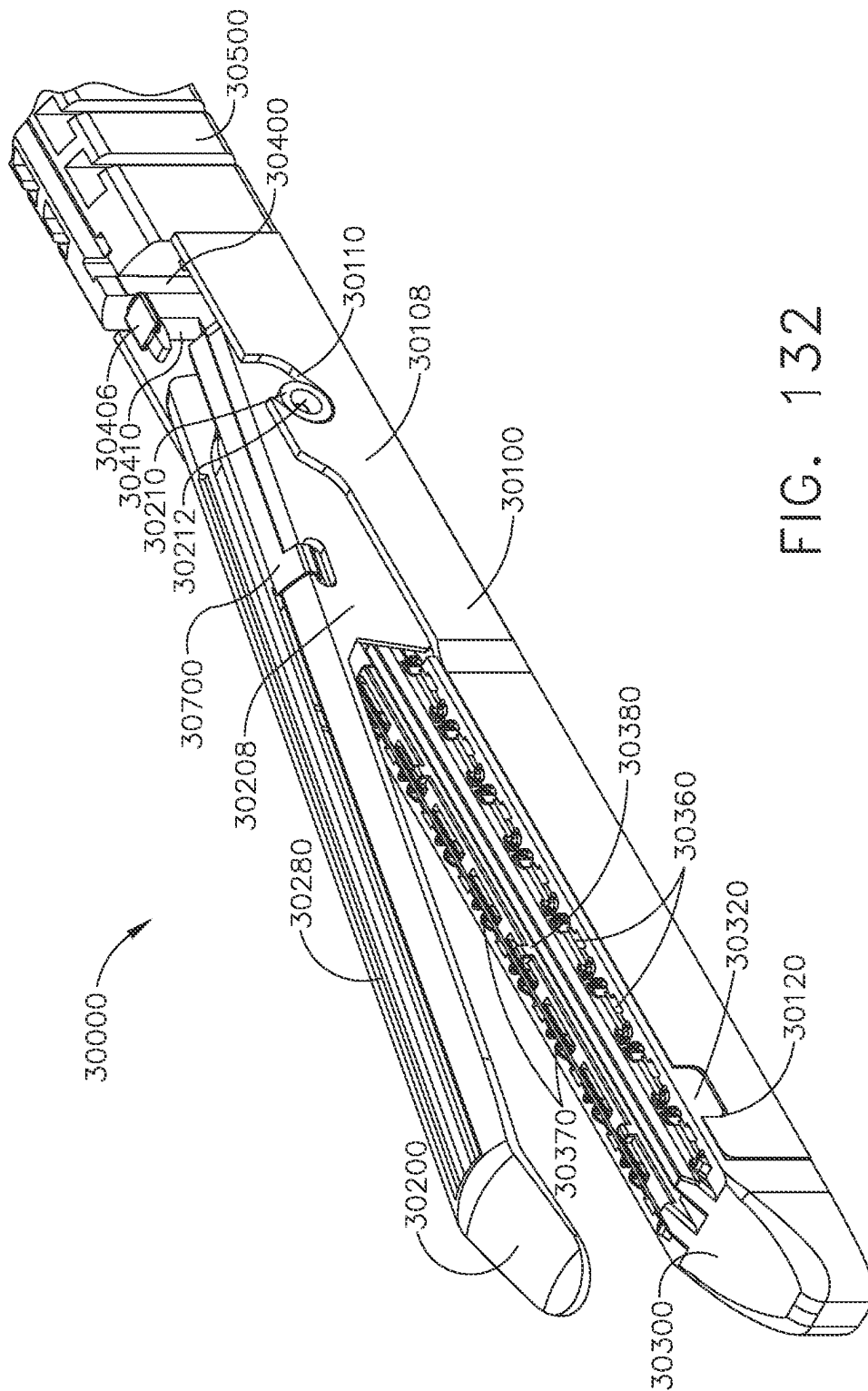
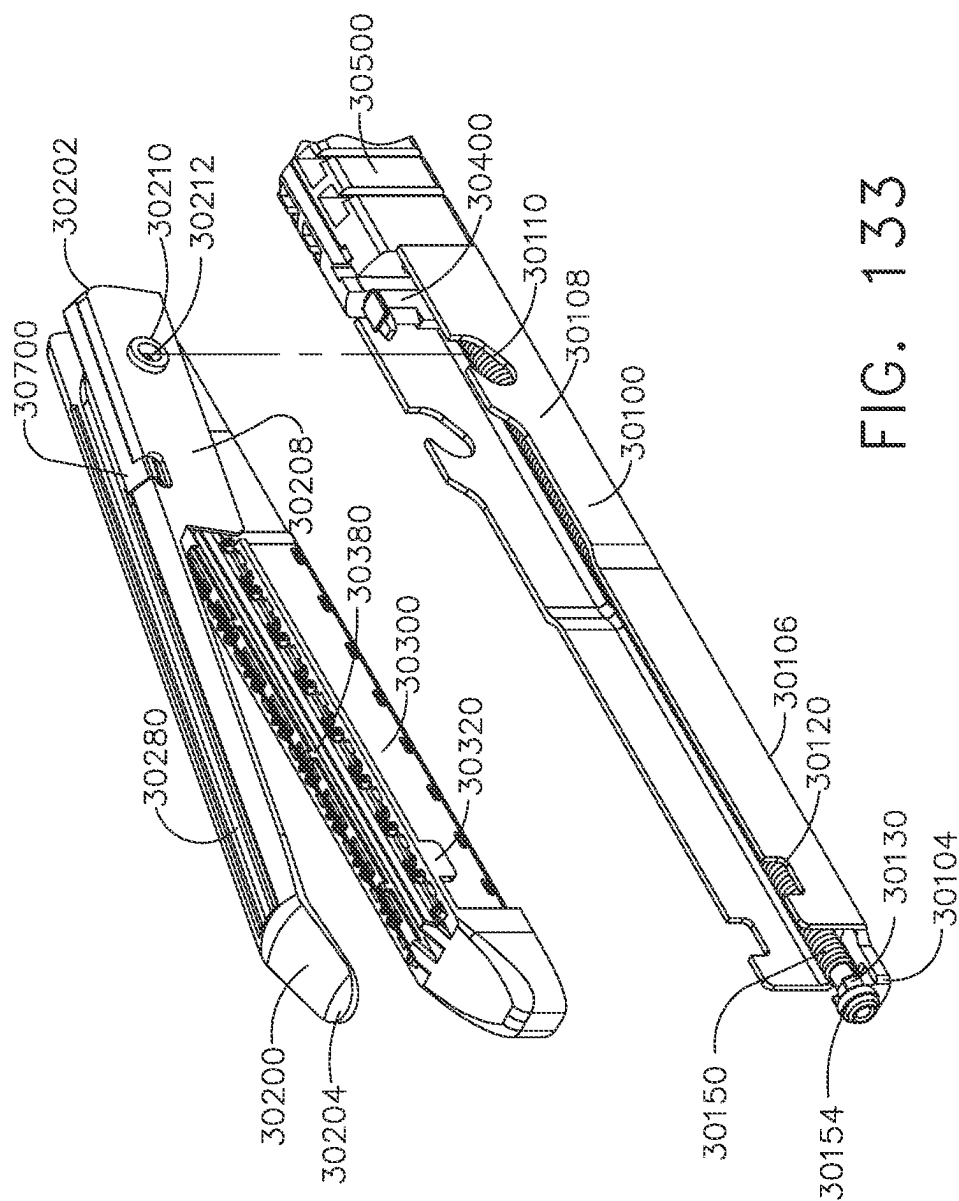


FIG. 132



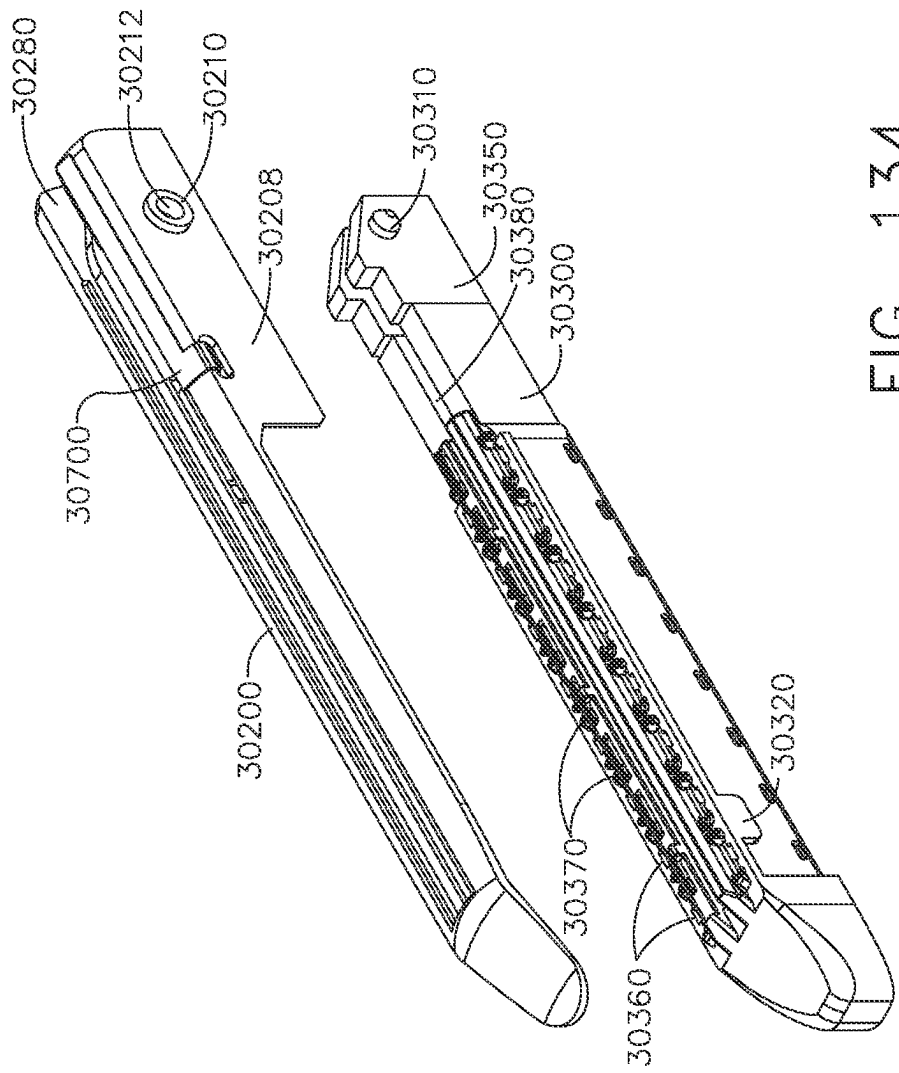


FIG. 134

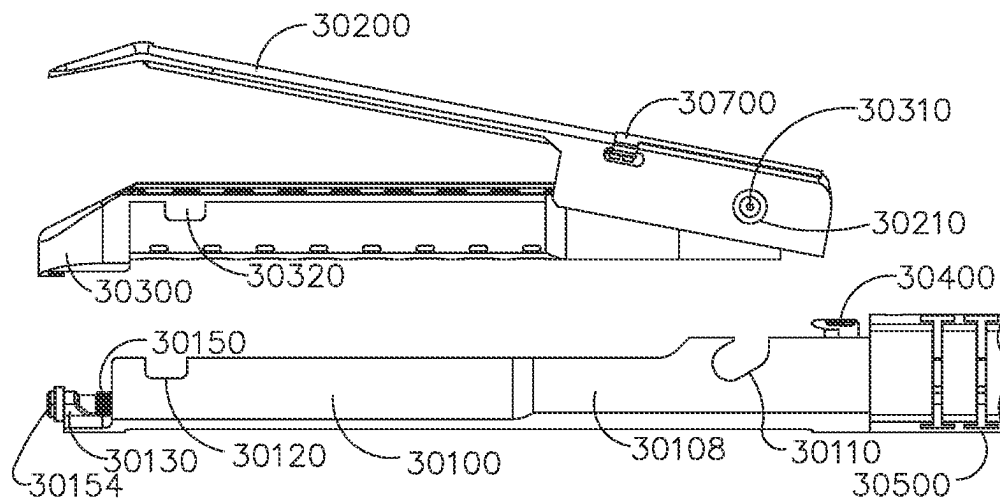


FIG. 135

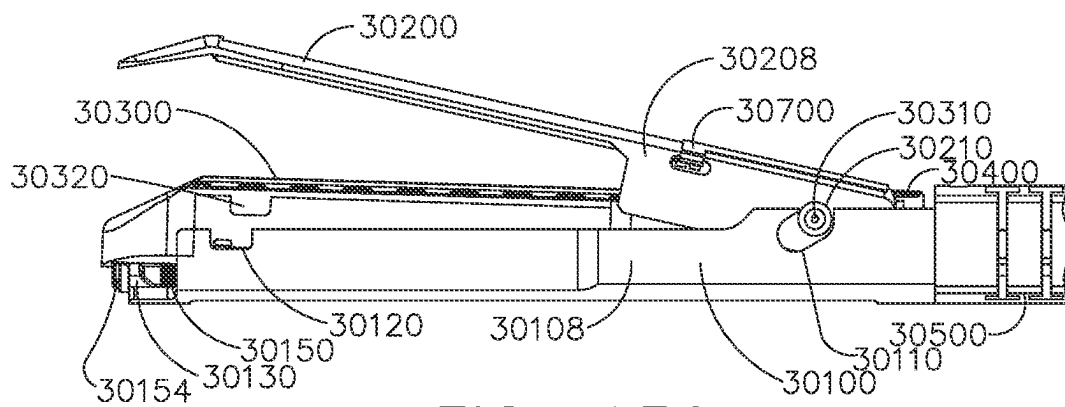


FIG. 136

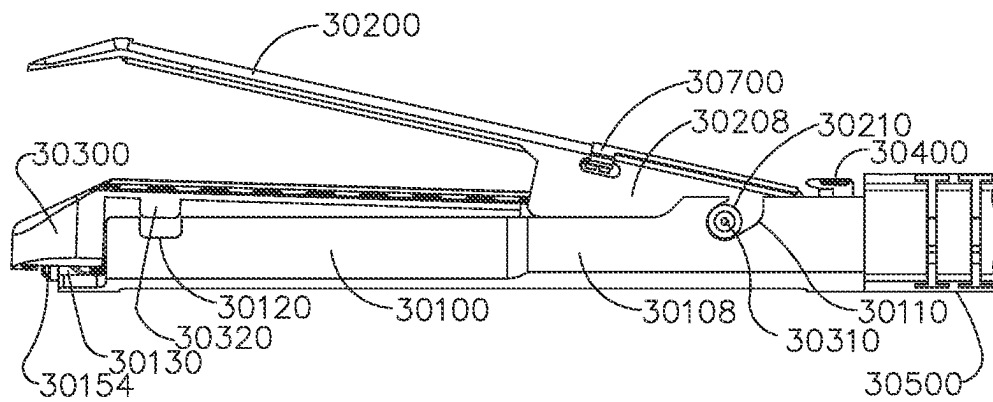


FIG. 137

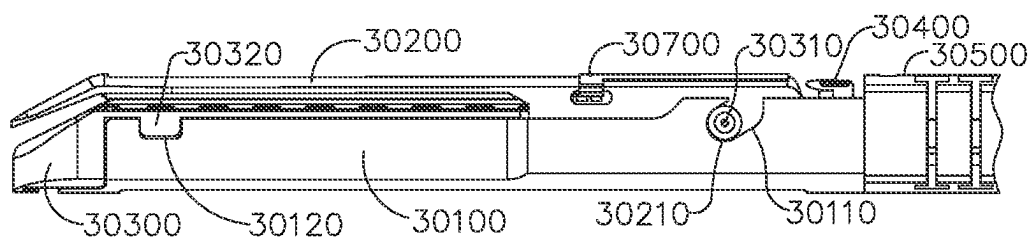


FIG. 138

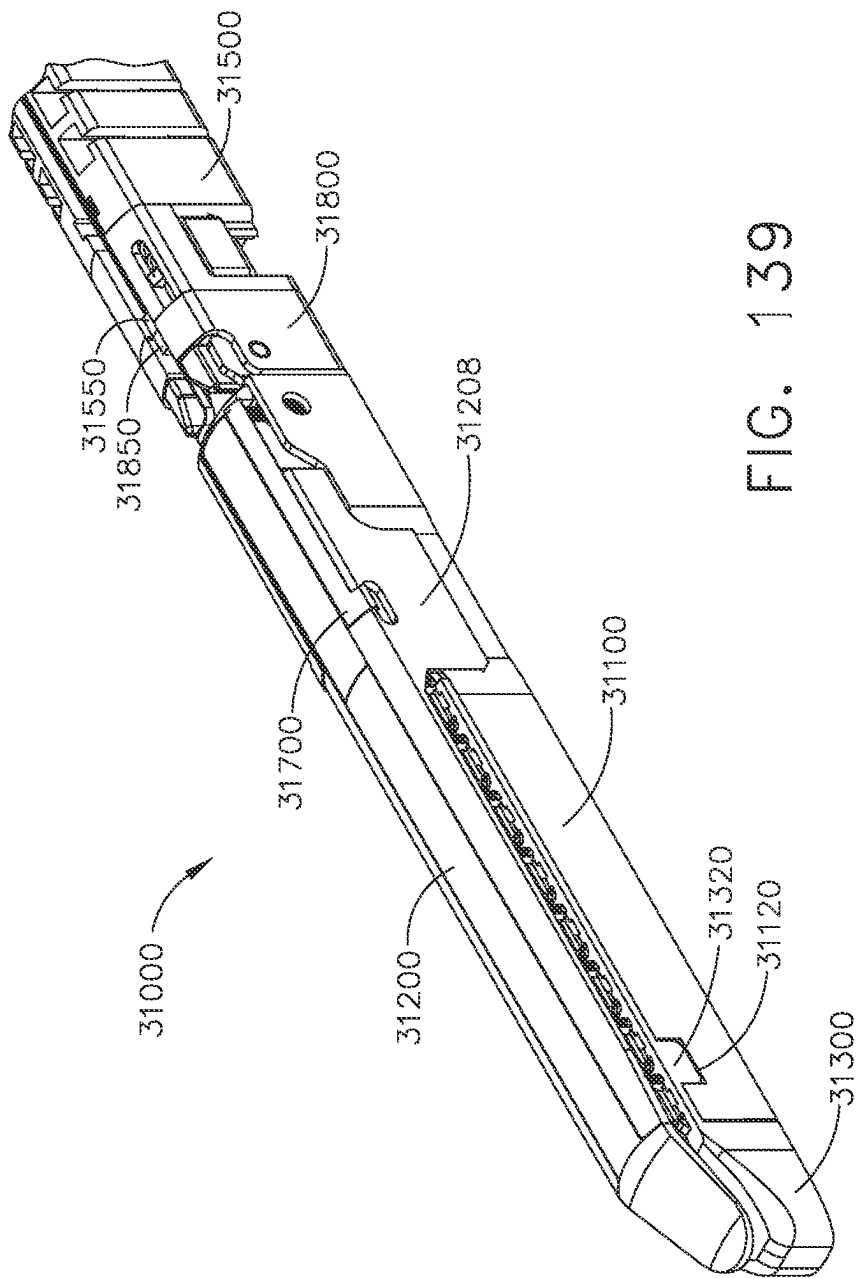
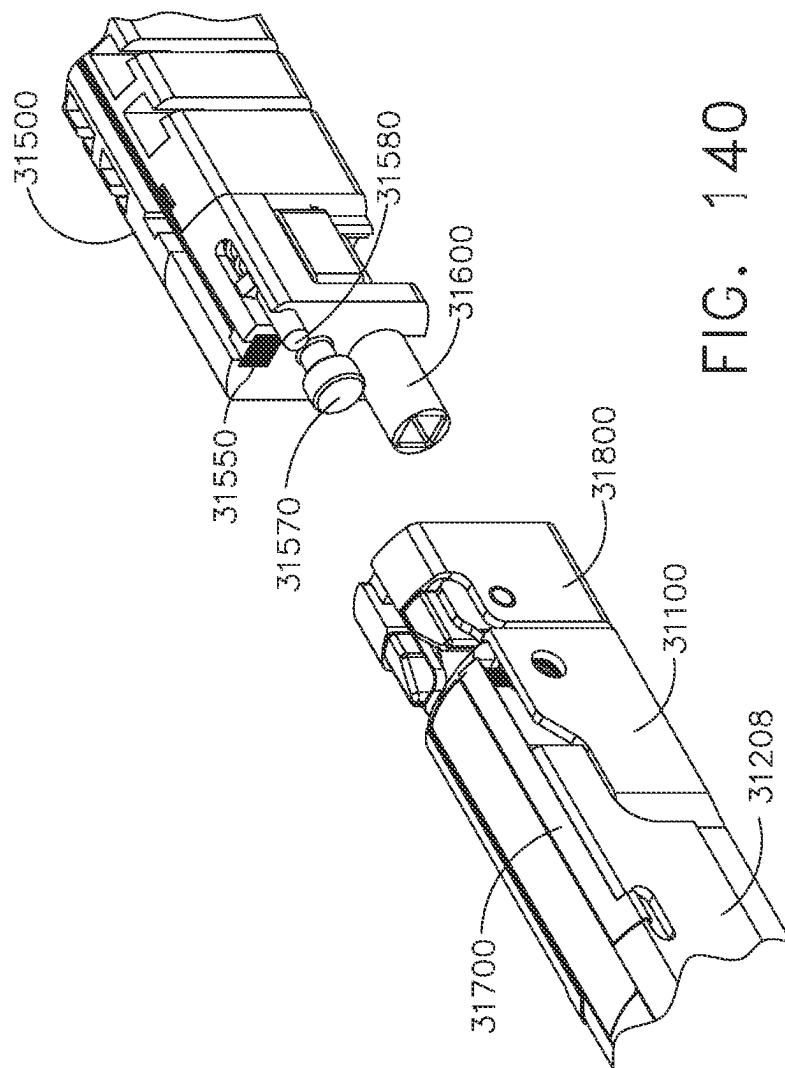


FIG. 139



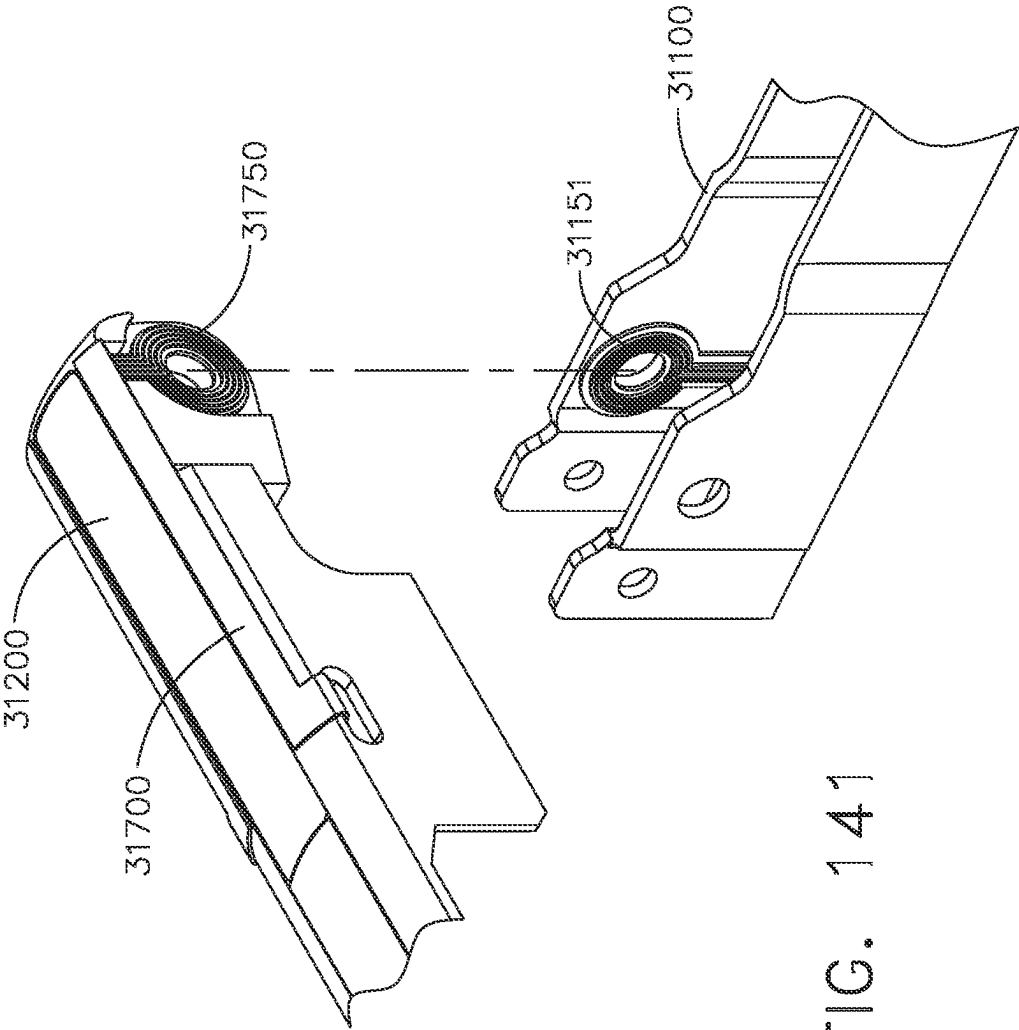
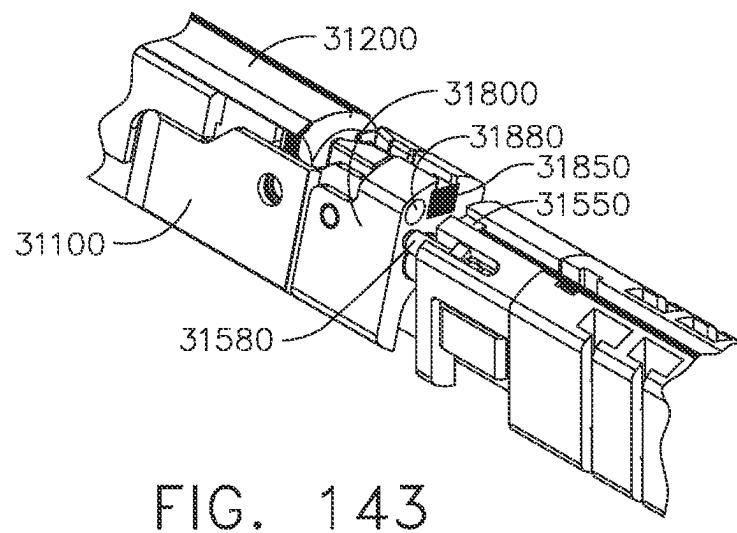
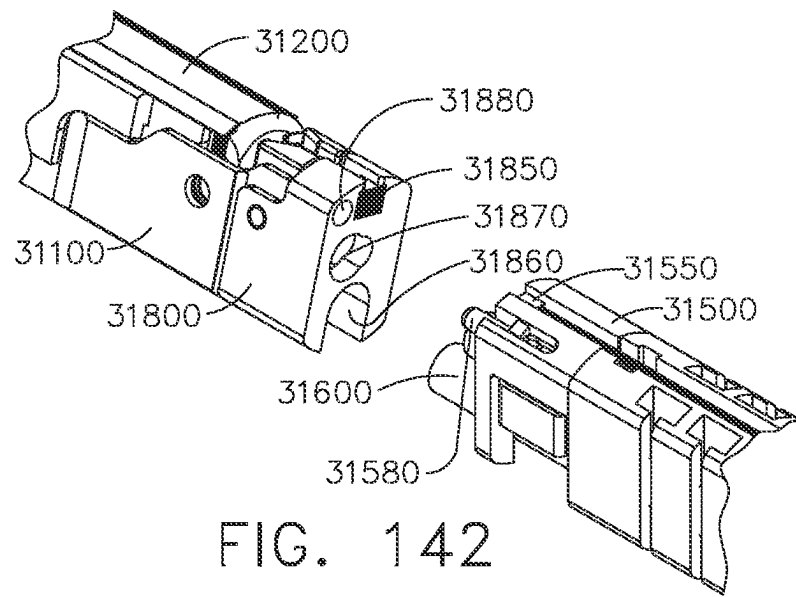
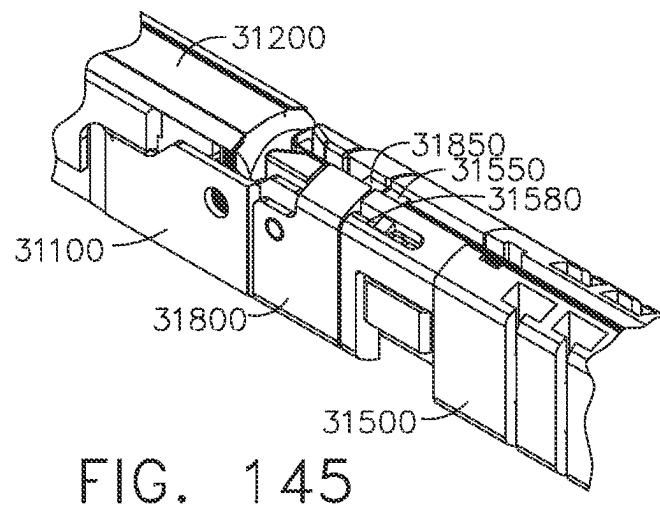
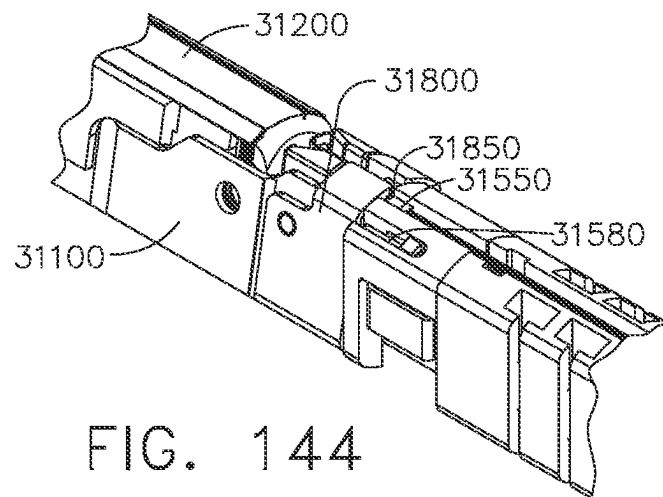


FIG. 141





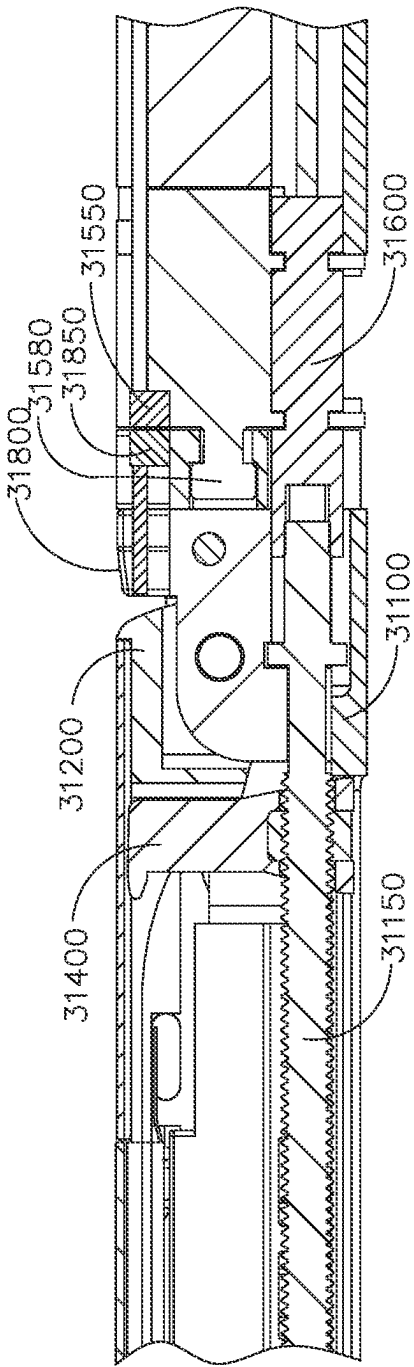


FIG. 146

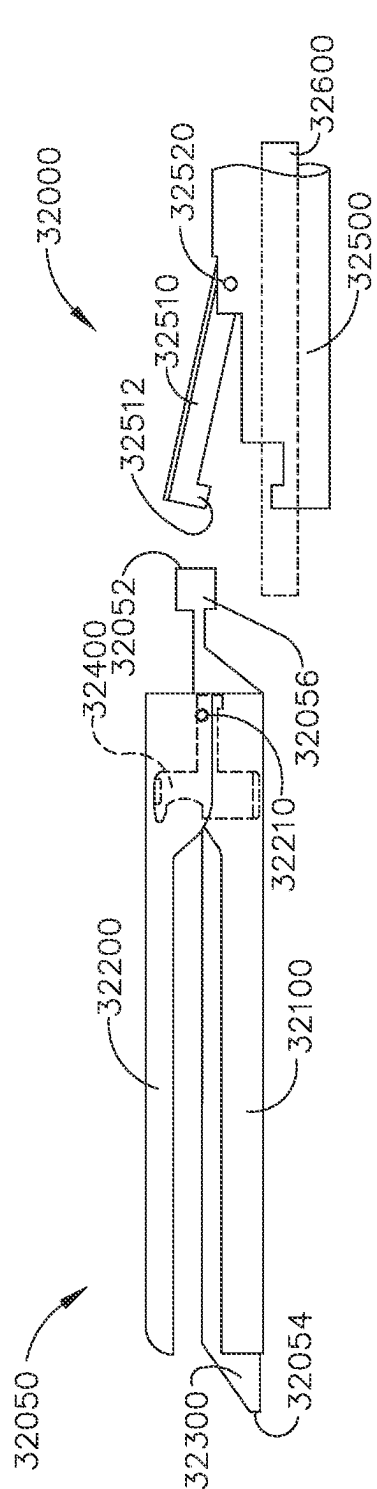


FIG. 147

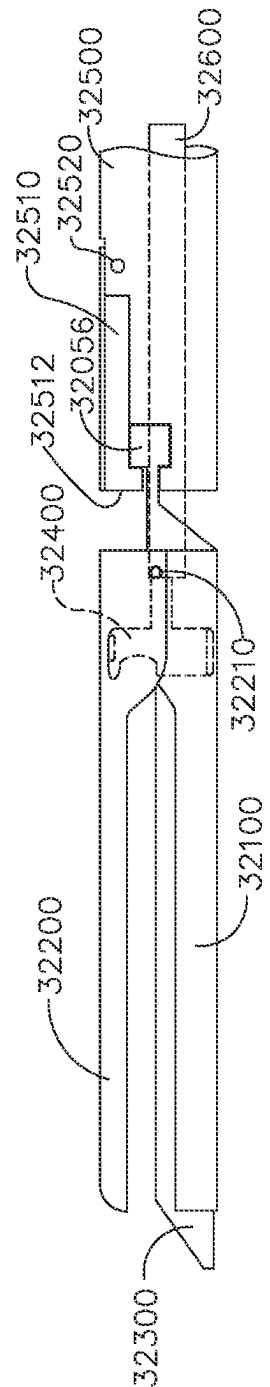


FIG. 148

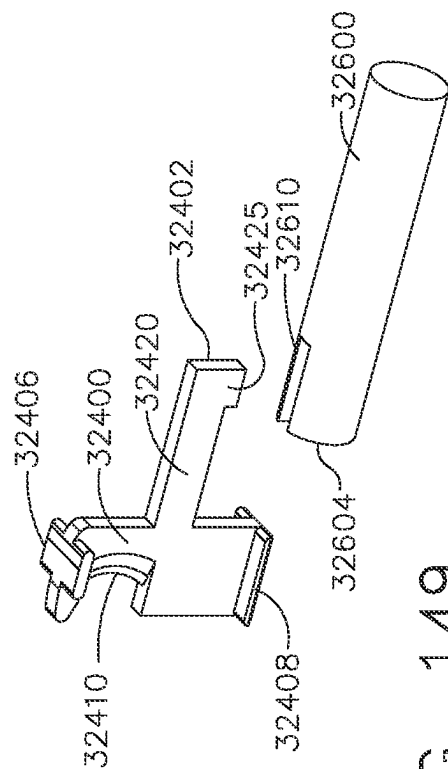


FIG. 149

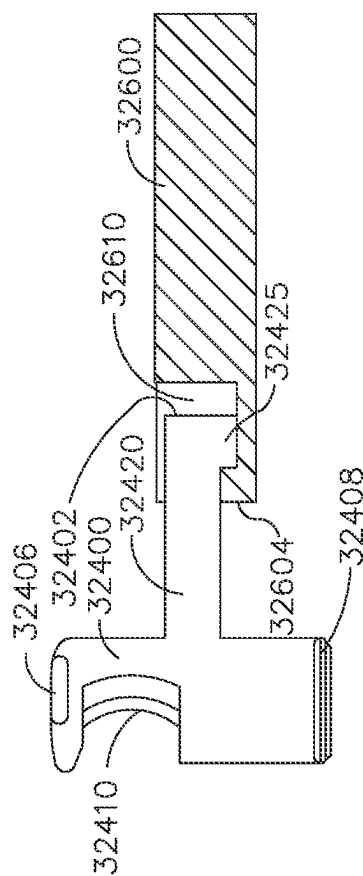
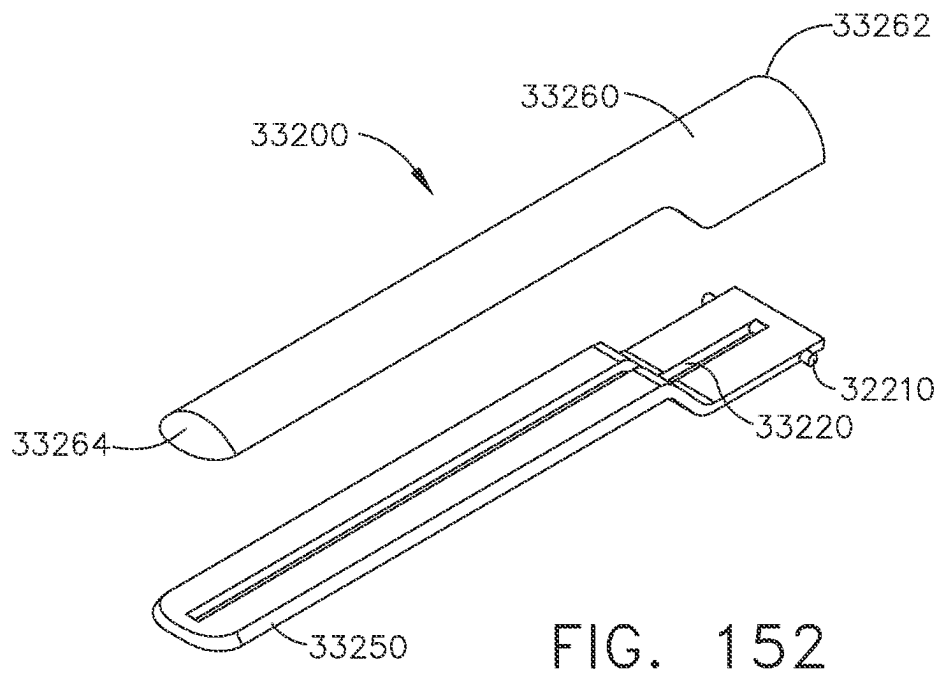
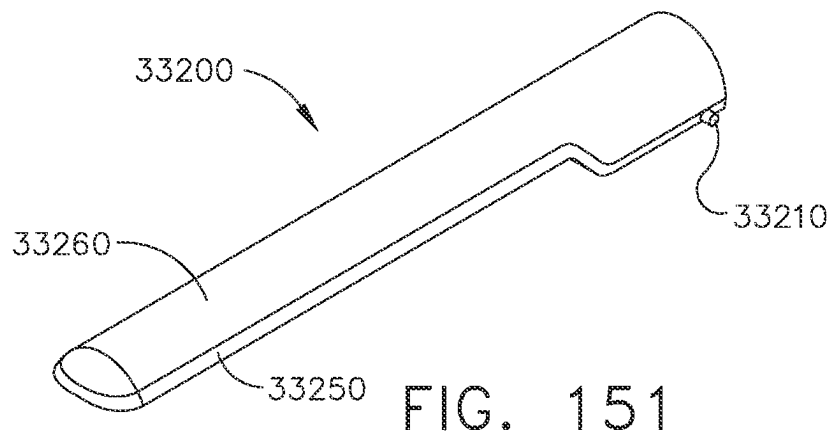


FIG. 150



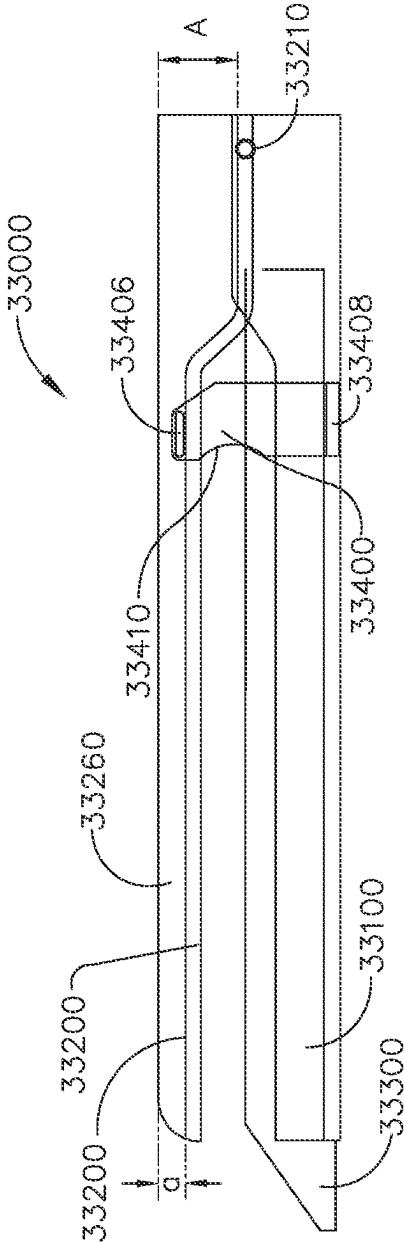


FIG. 153

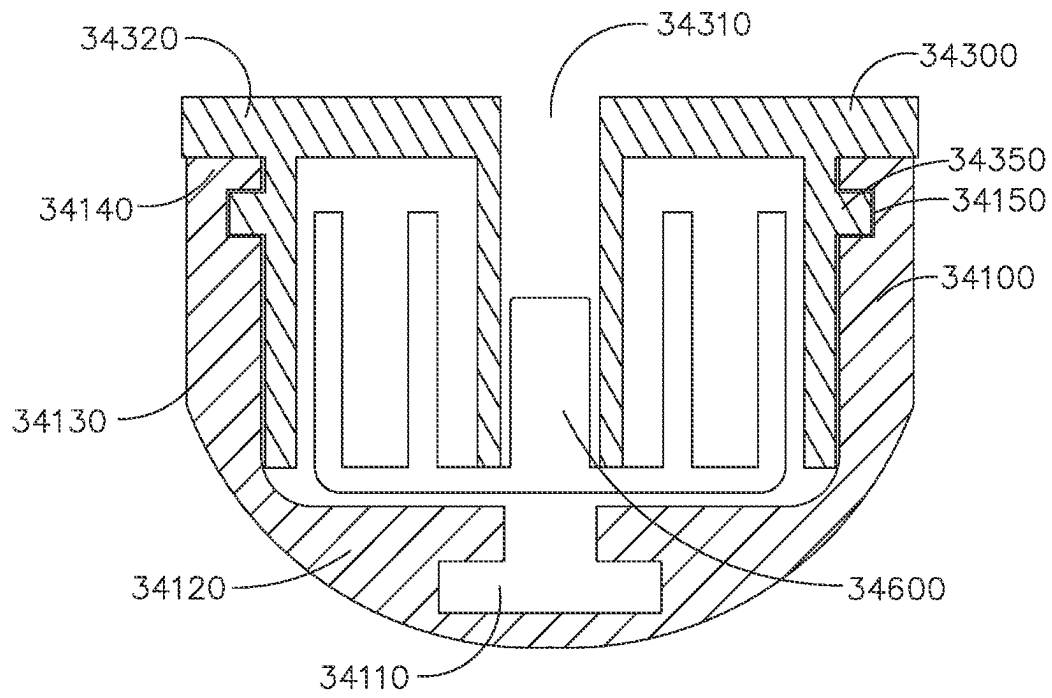


FIG. 154

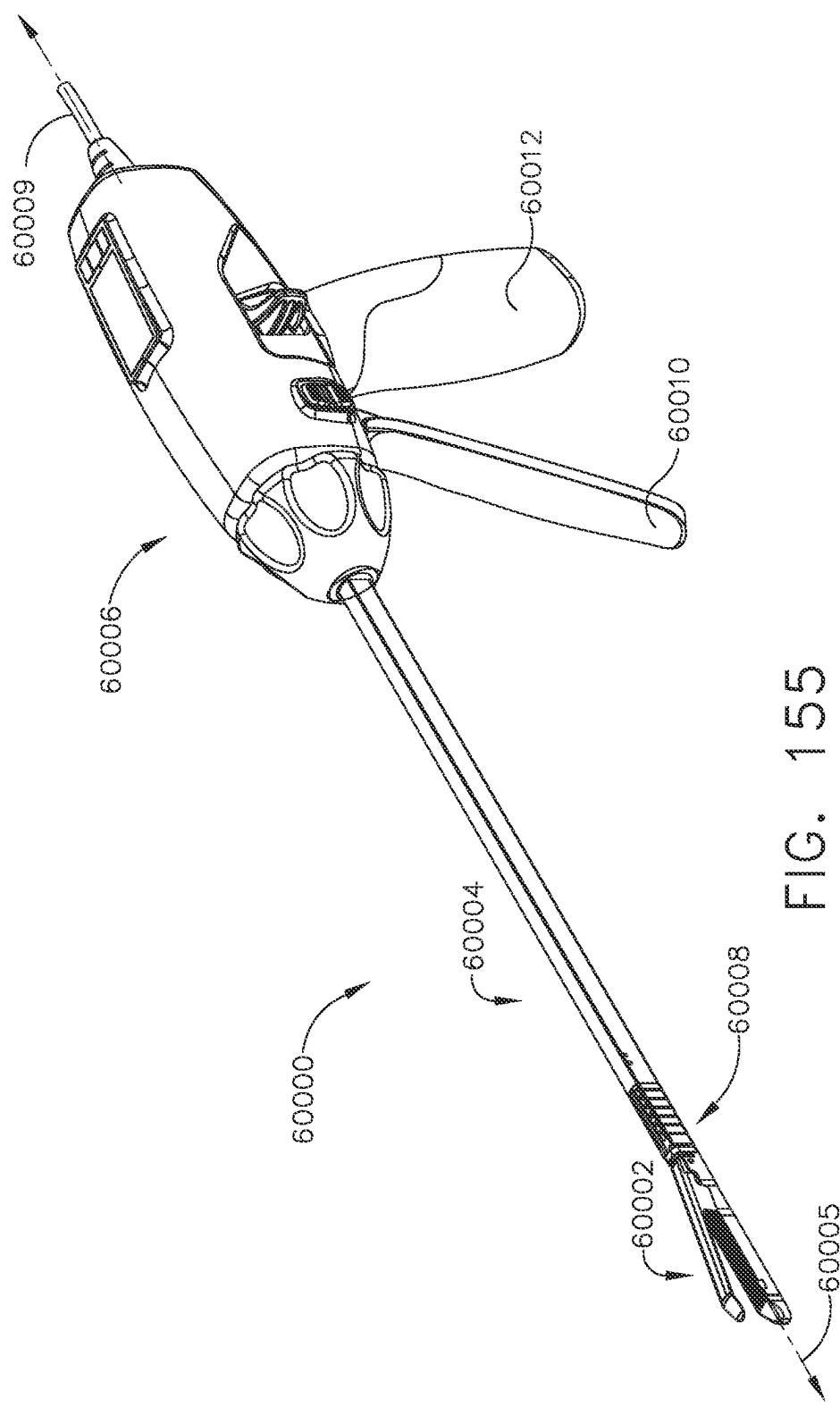
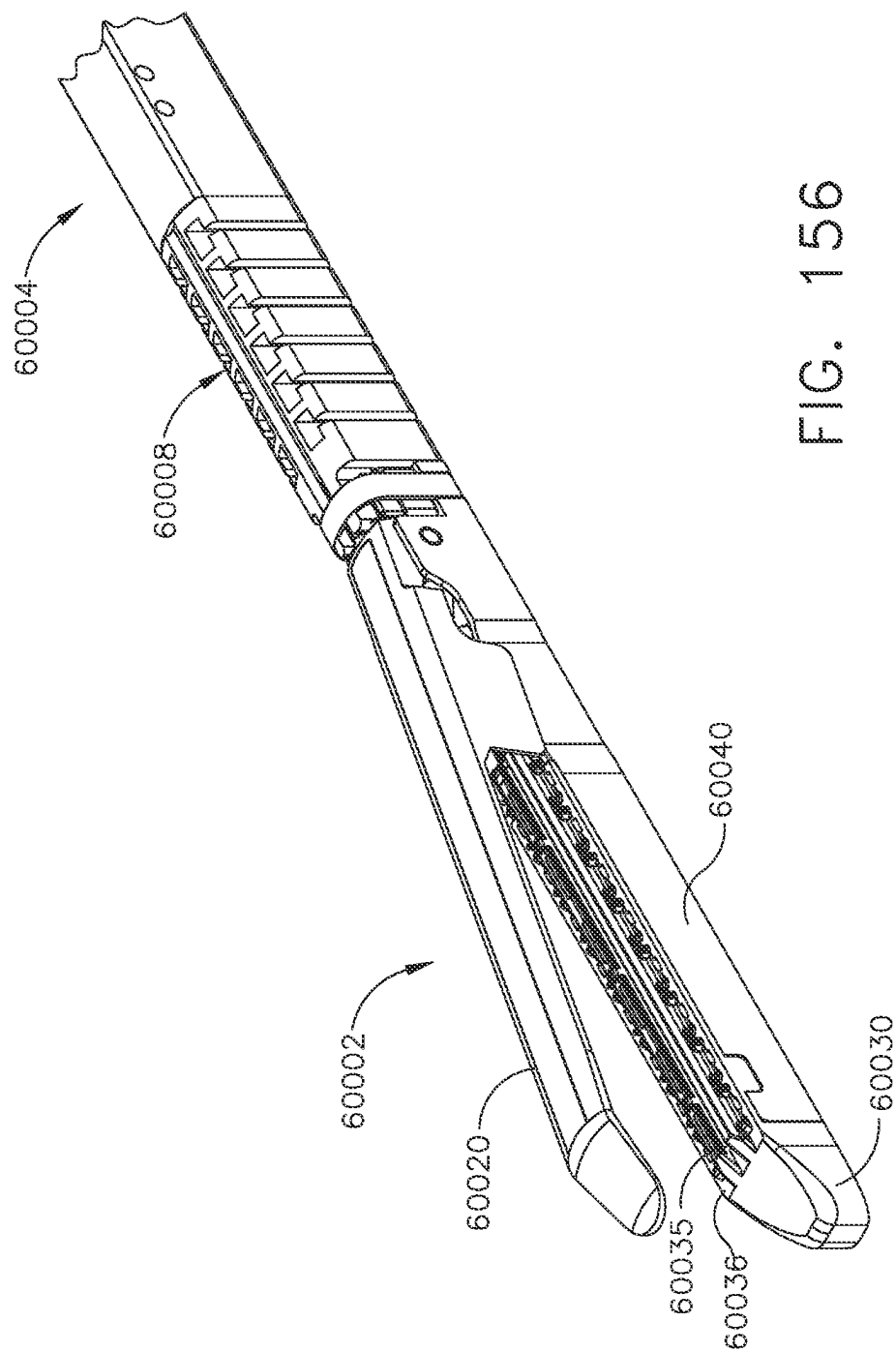


FIG. 155



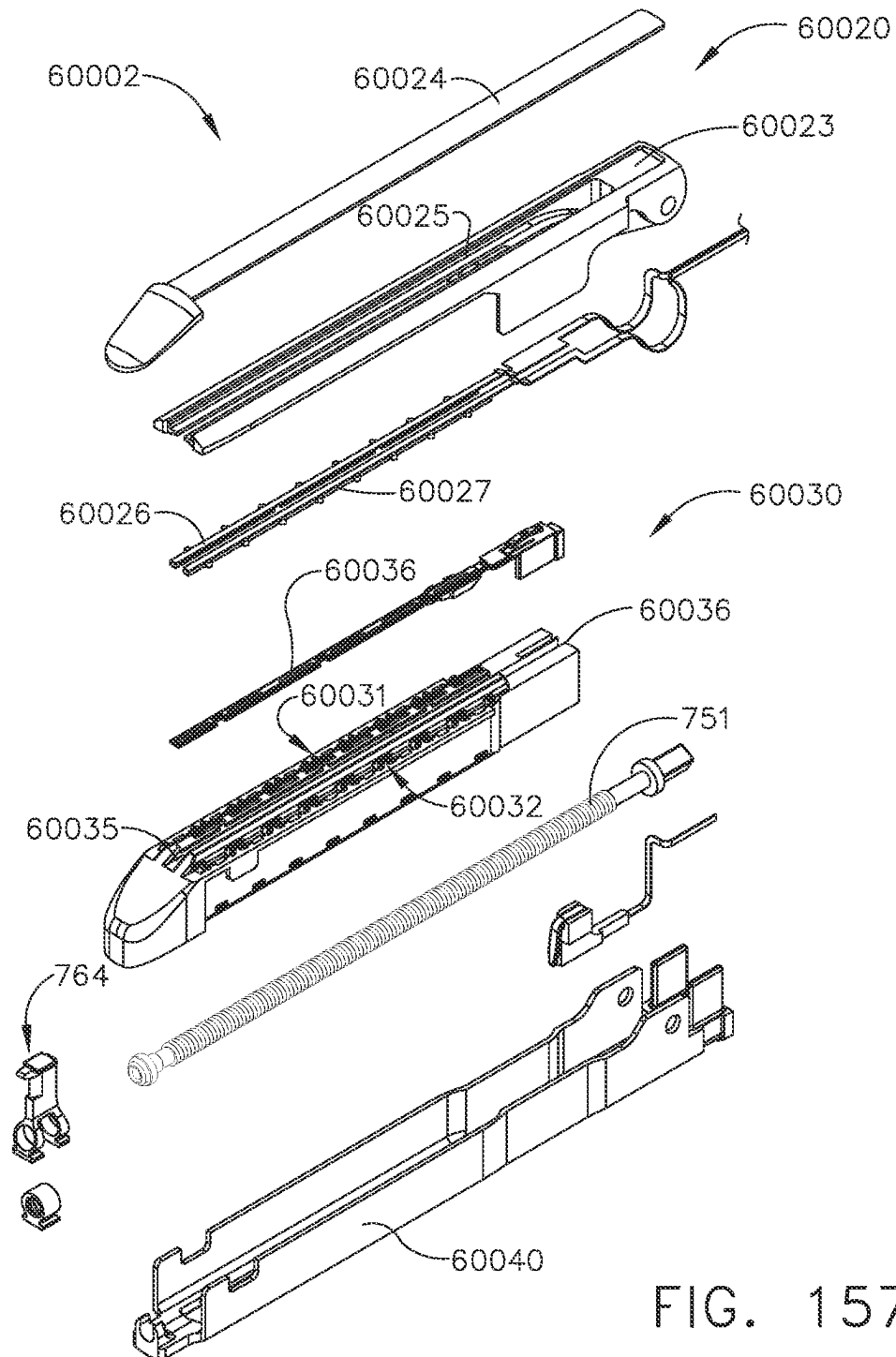


FIG. 157

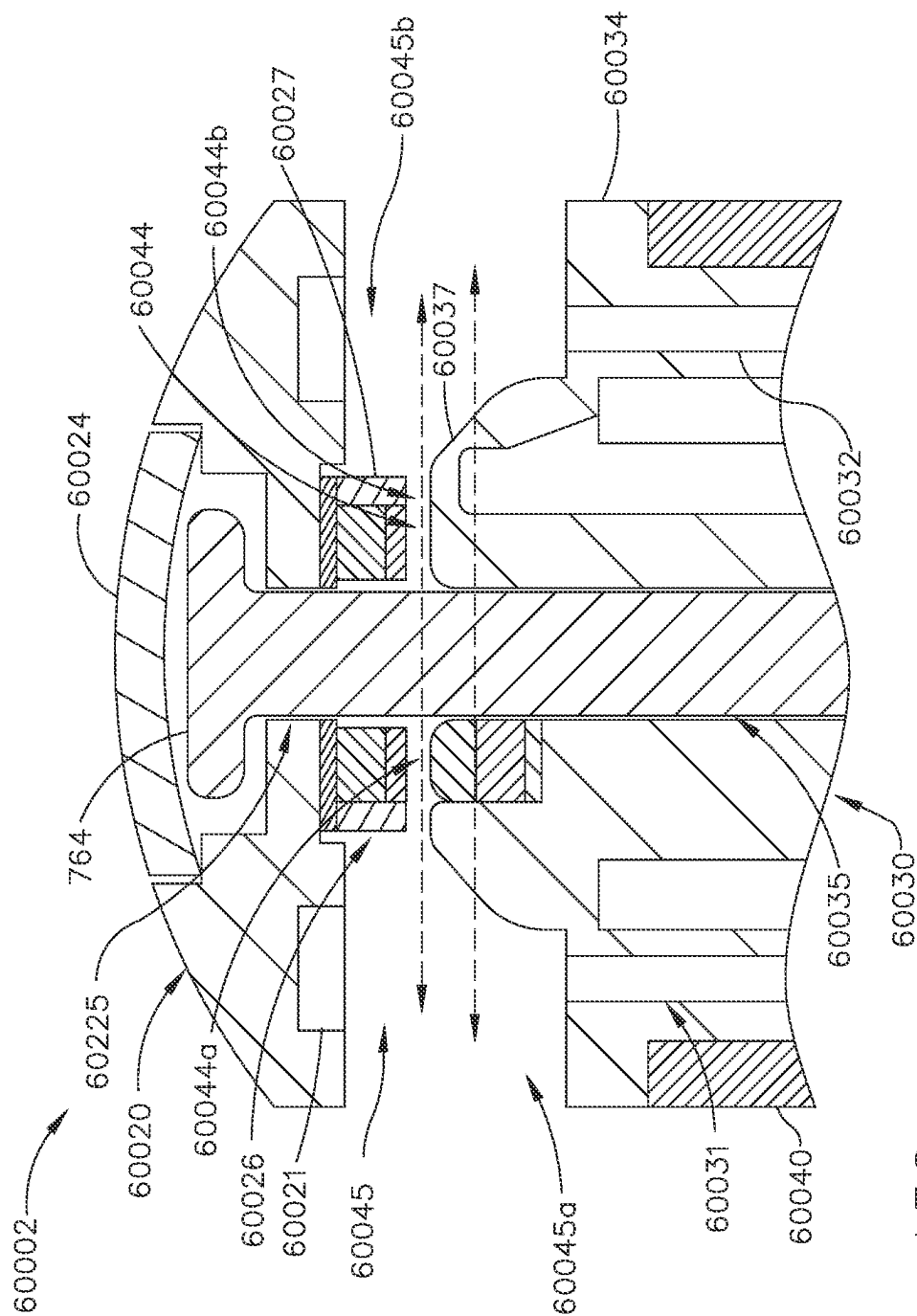
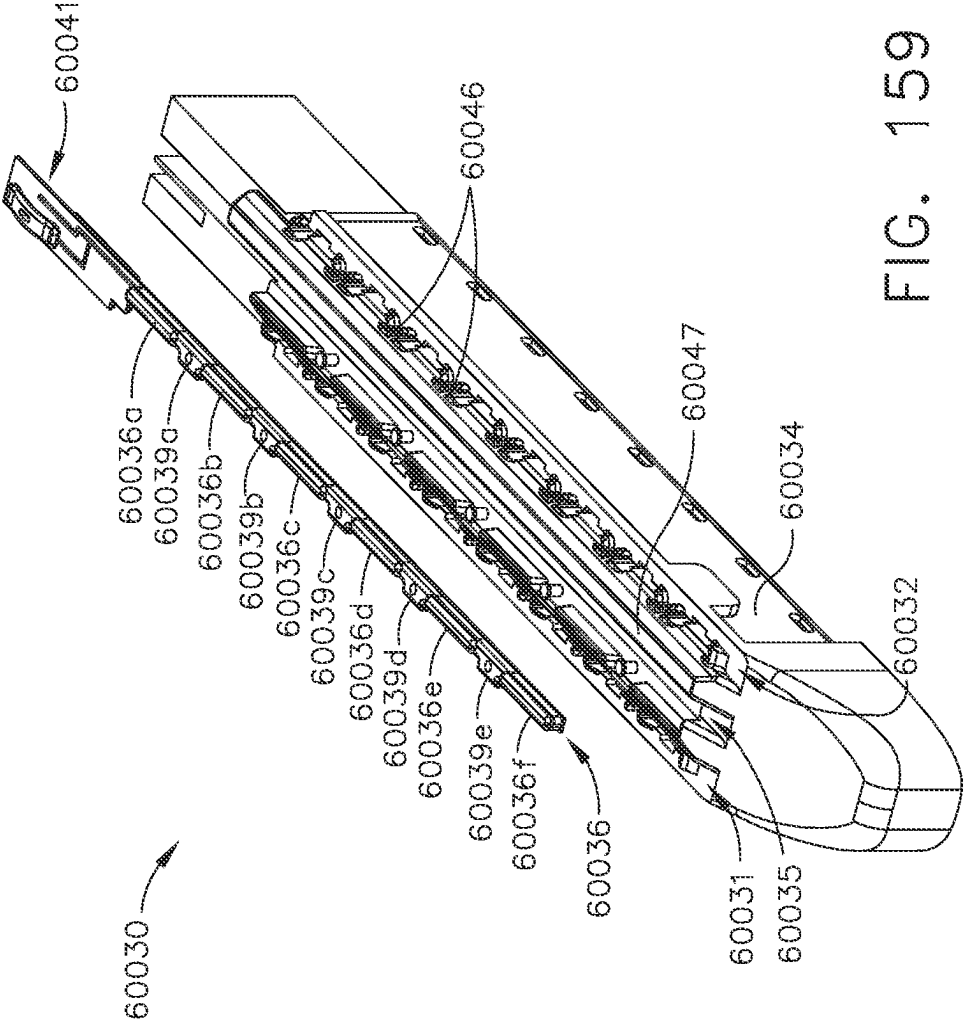


FIG. 158



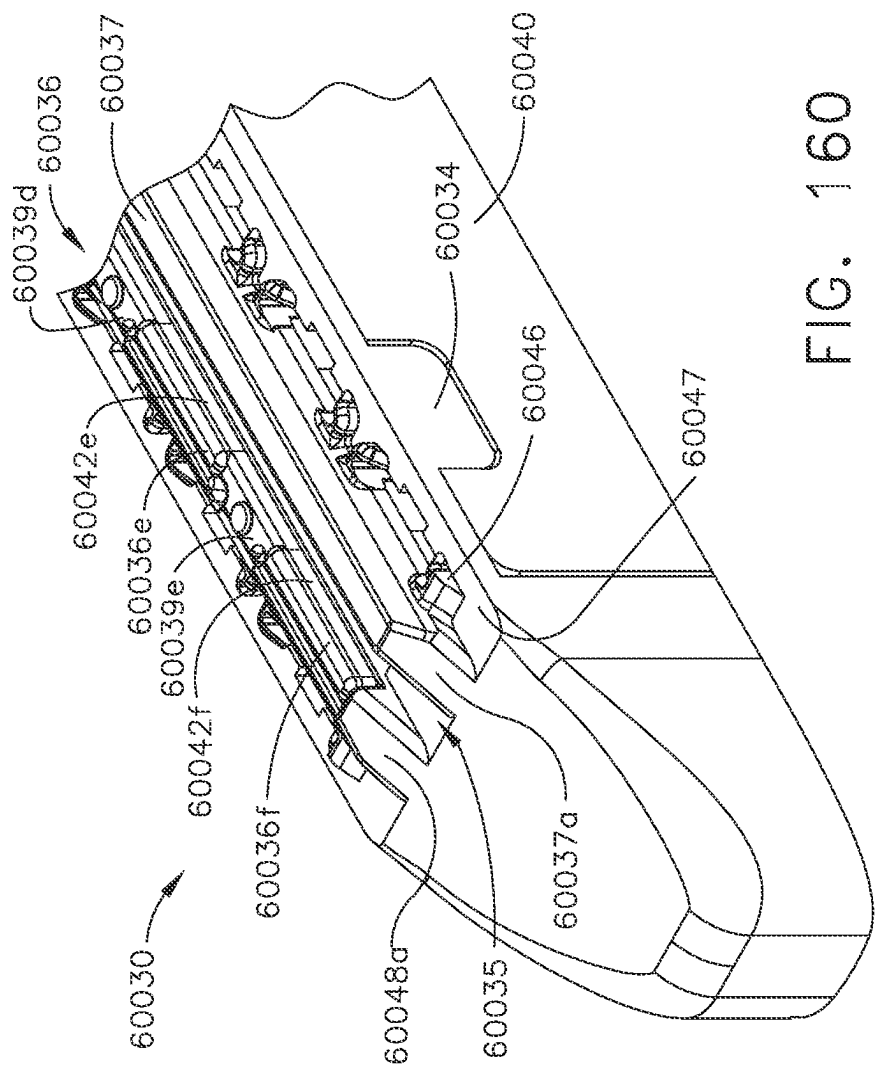
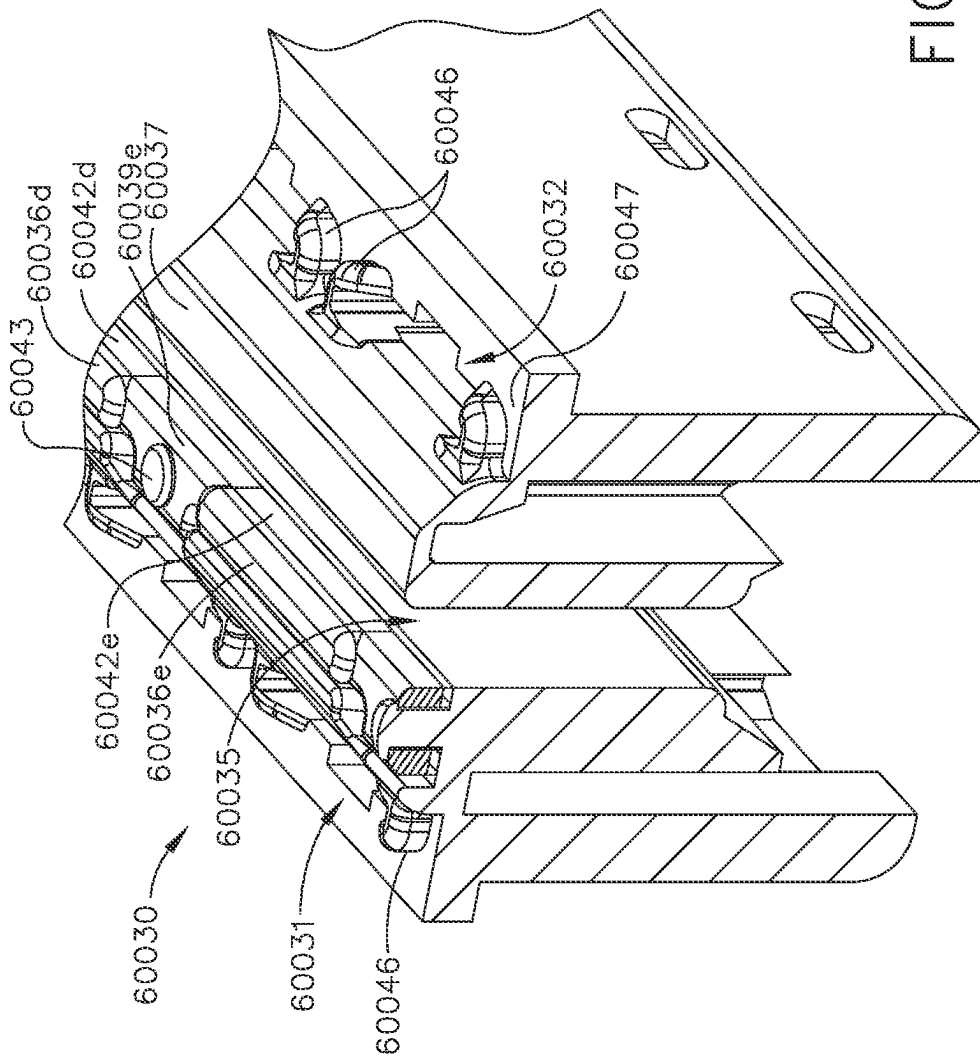


FIG. 160



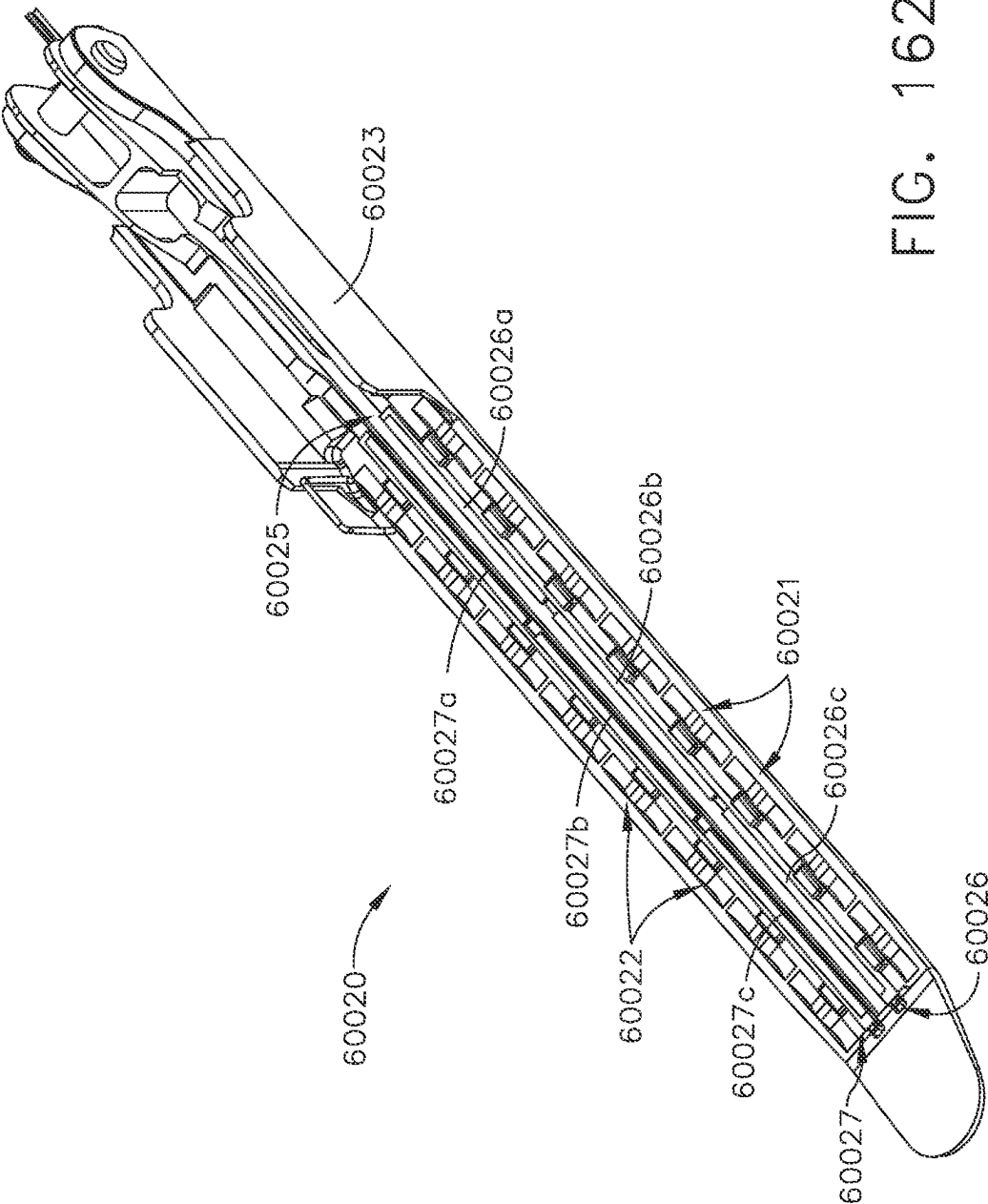
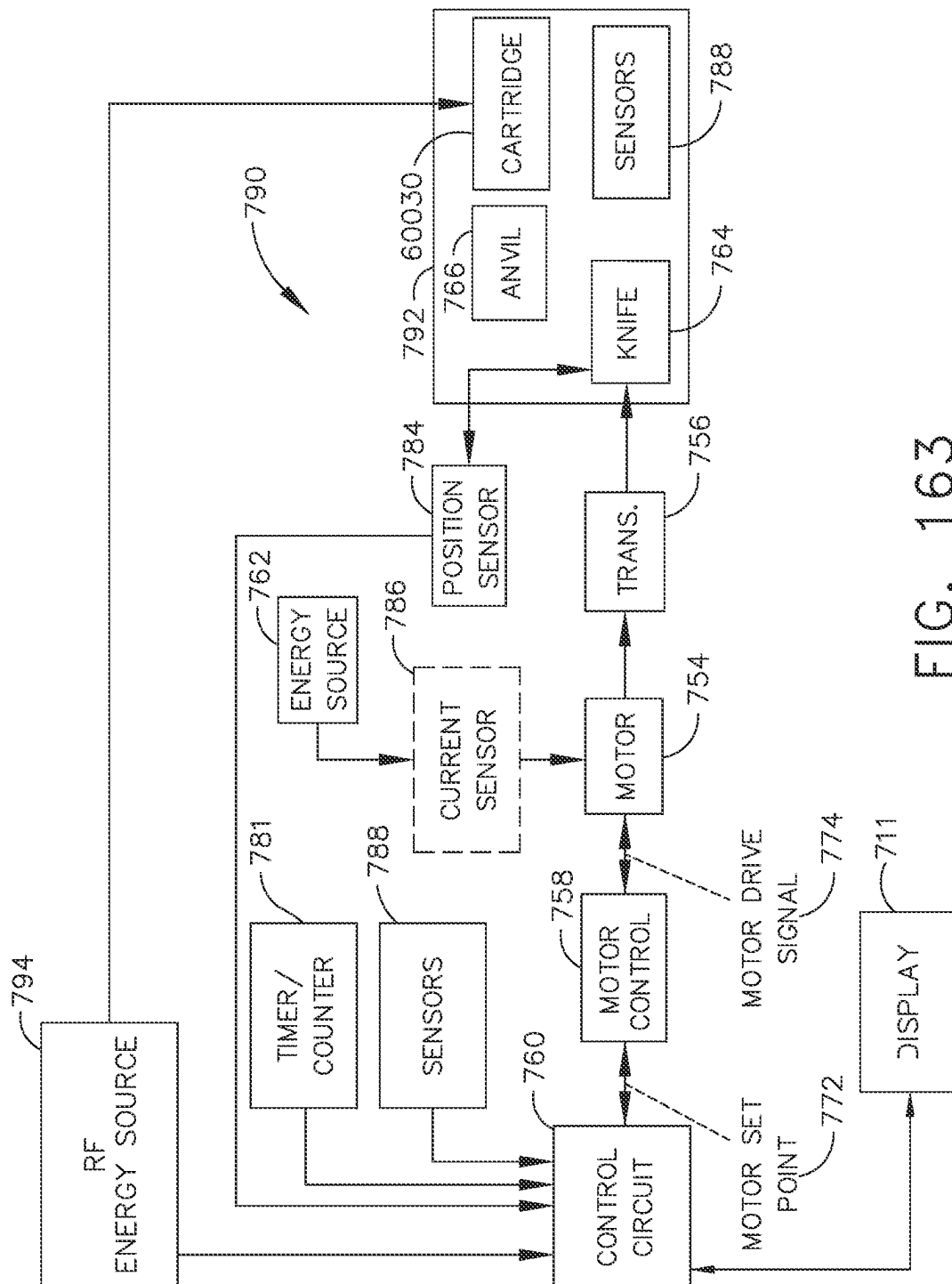


FIG. 162



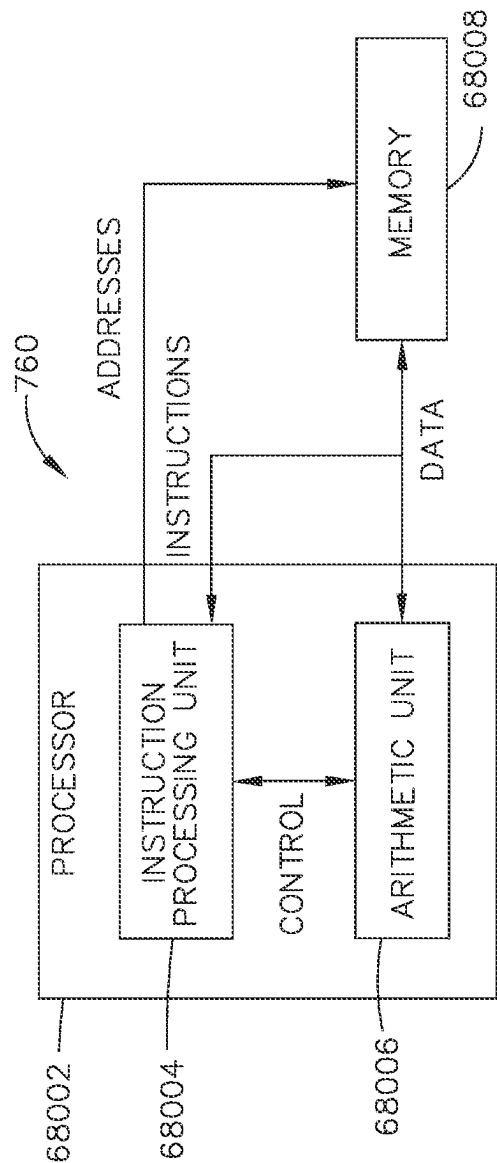


FIG. 164

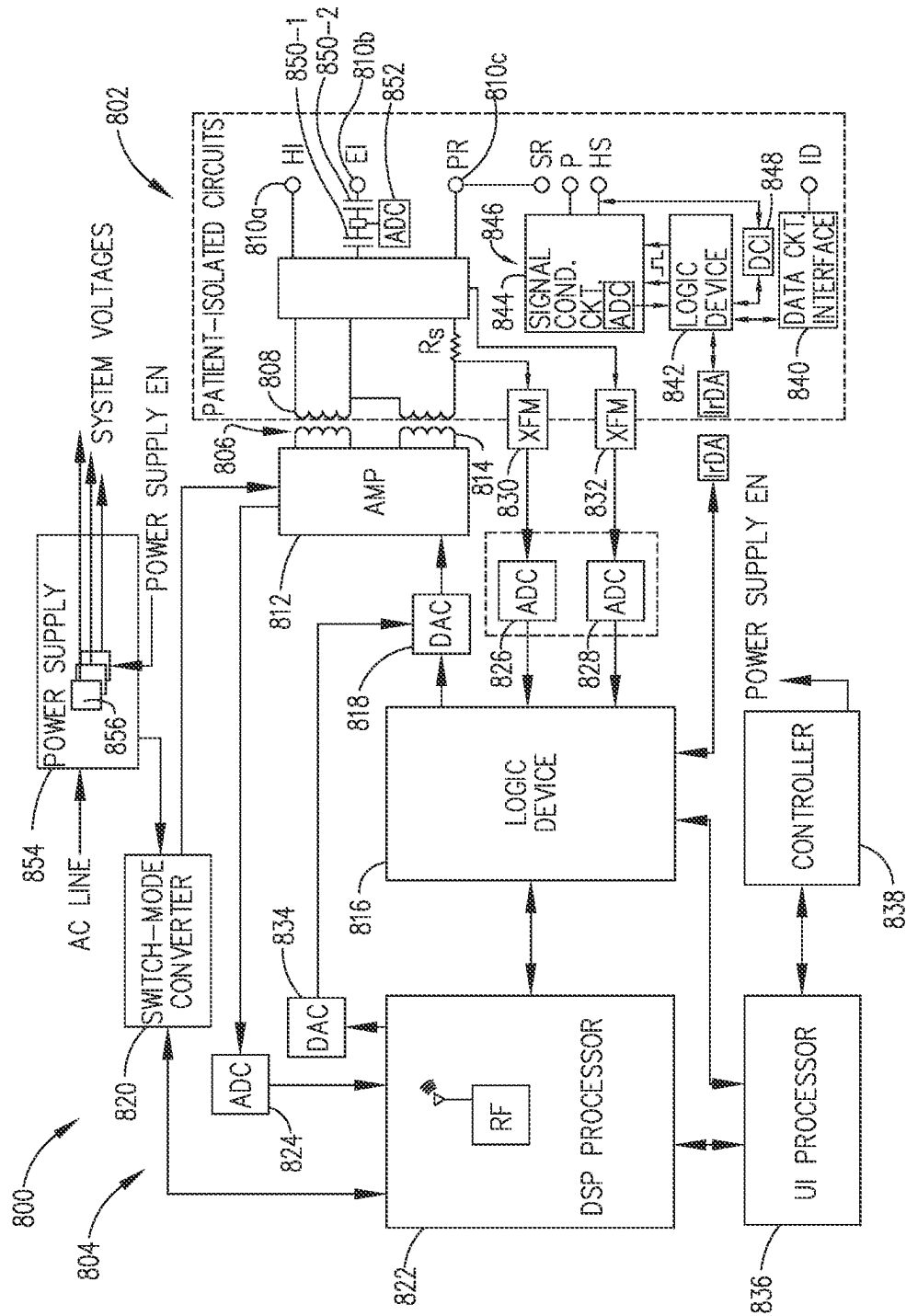
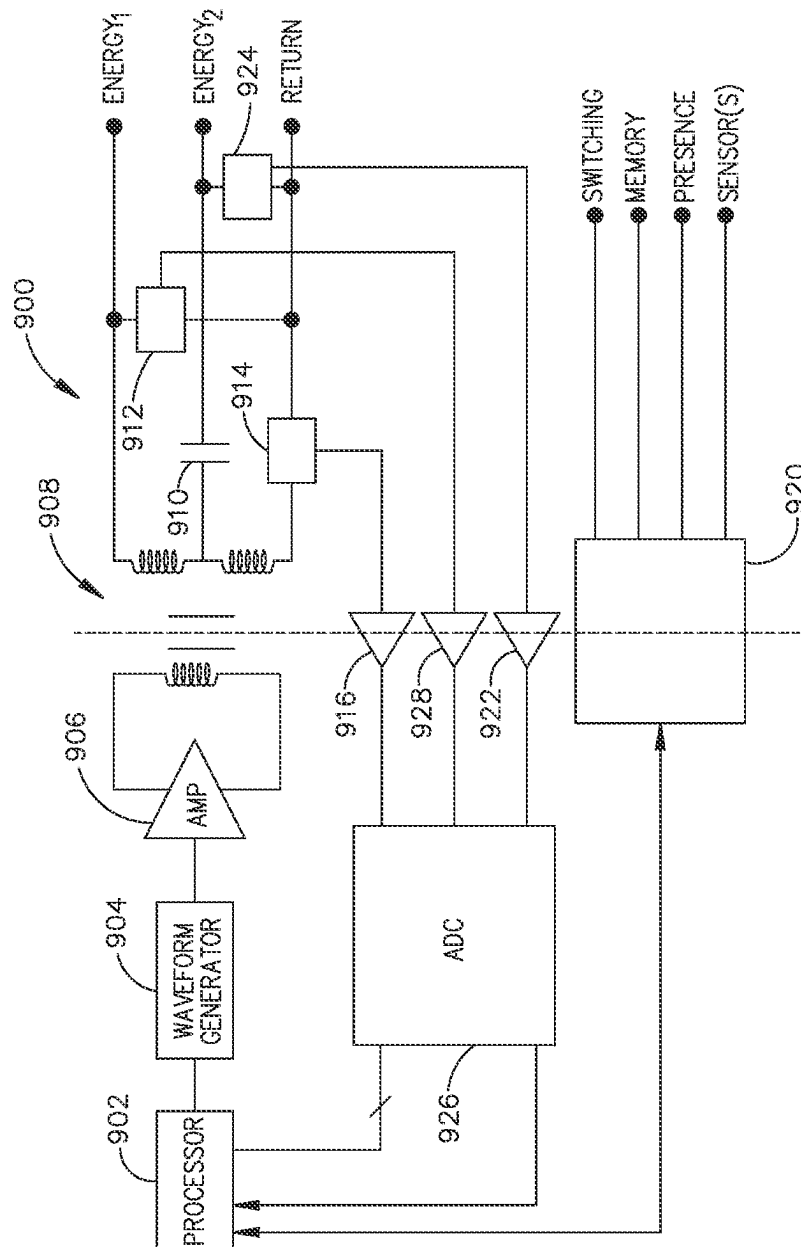


FIG. 165



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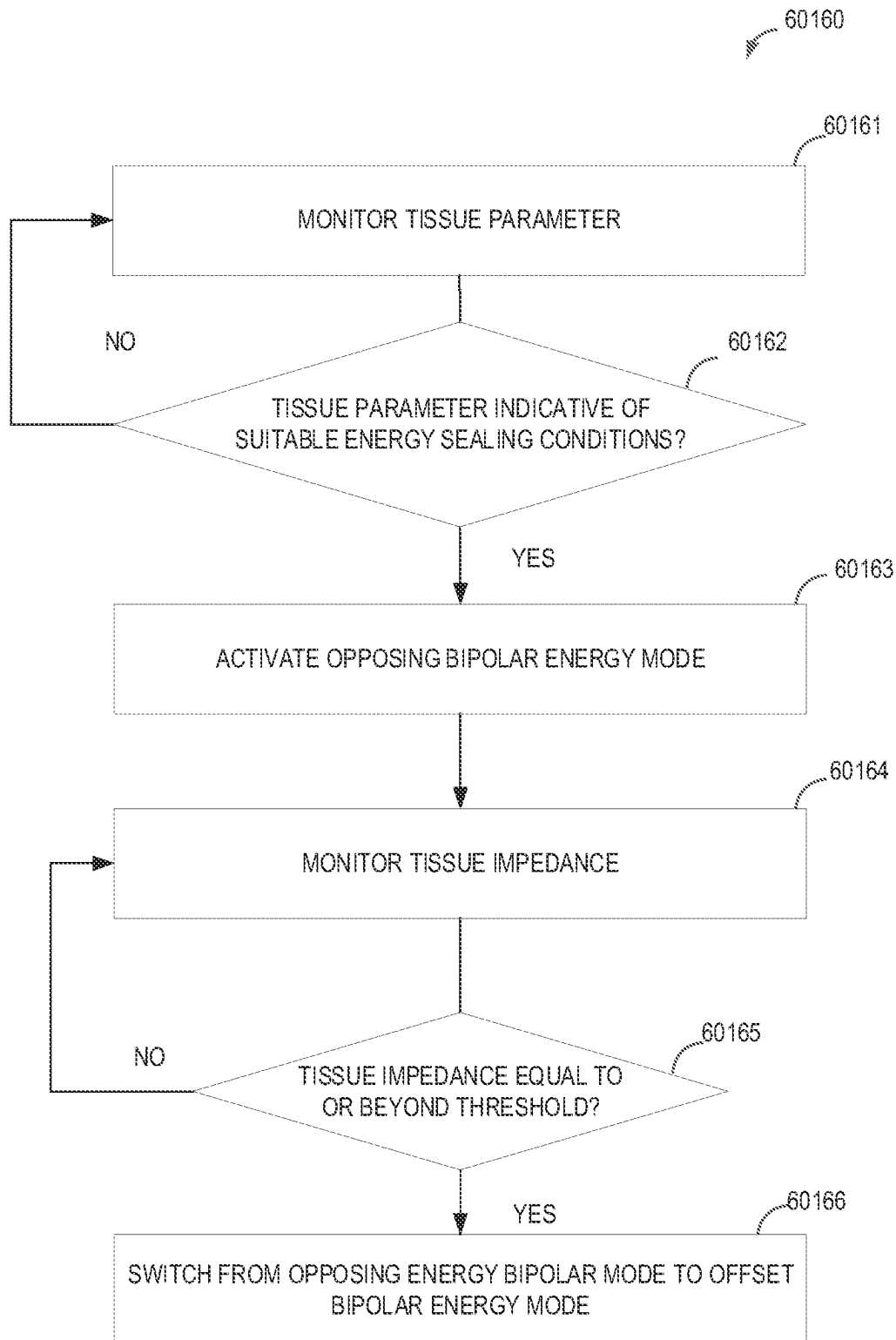


FIG. 167

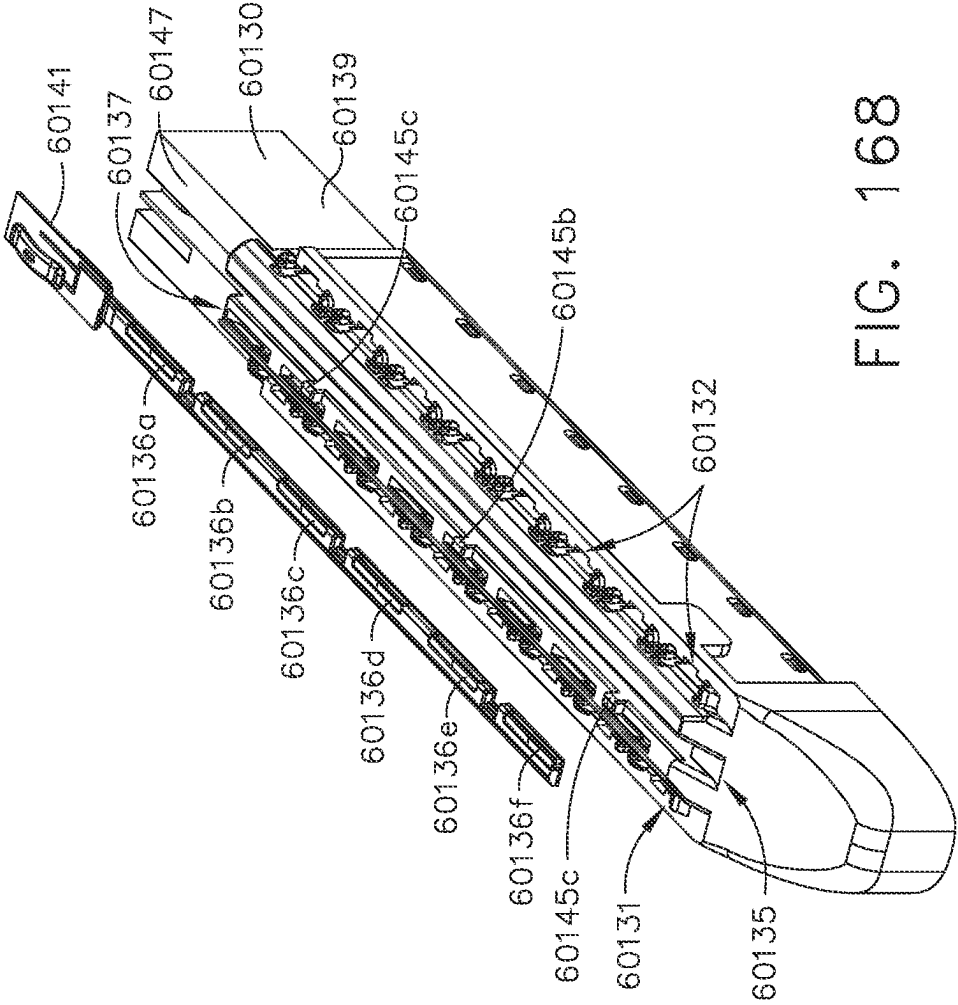
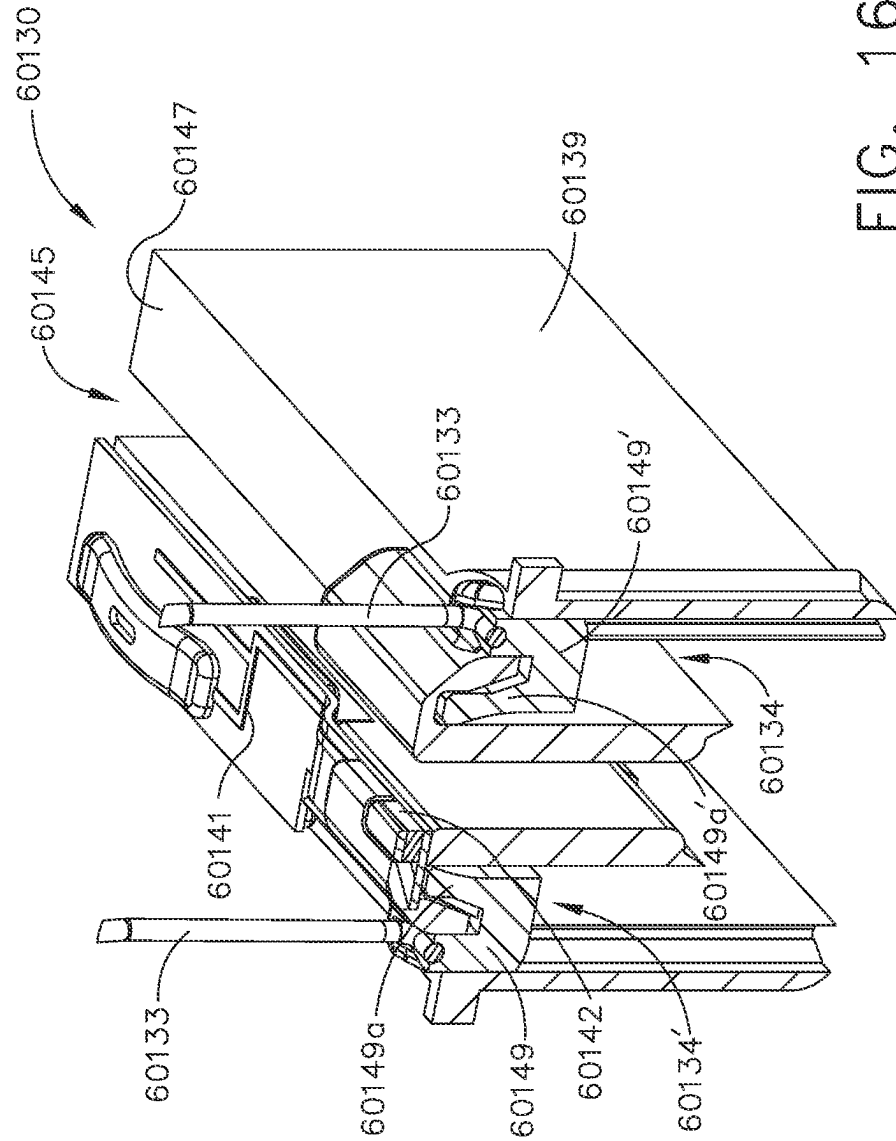


FIG. 168



907

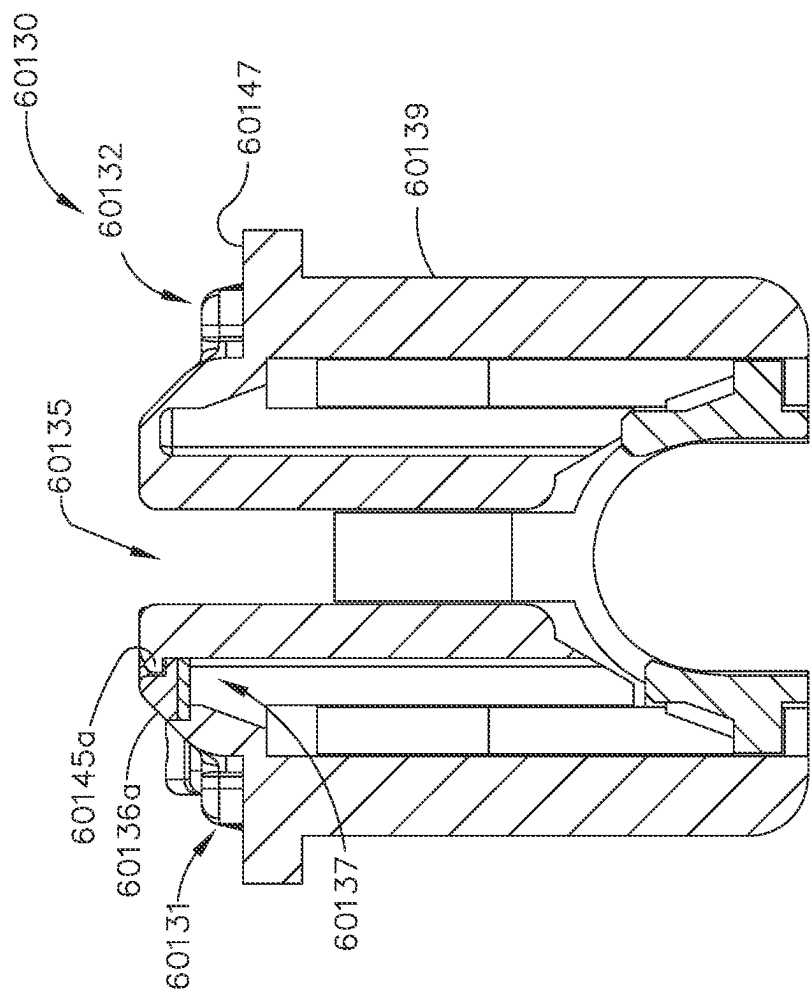


FIG. 170

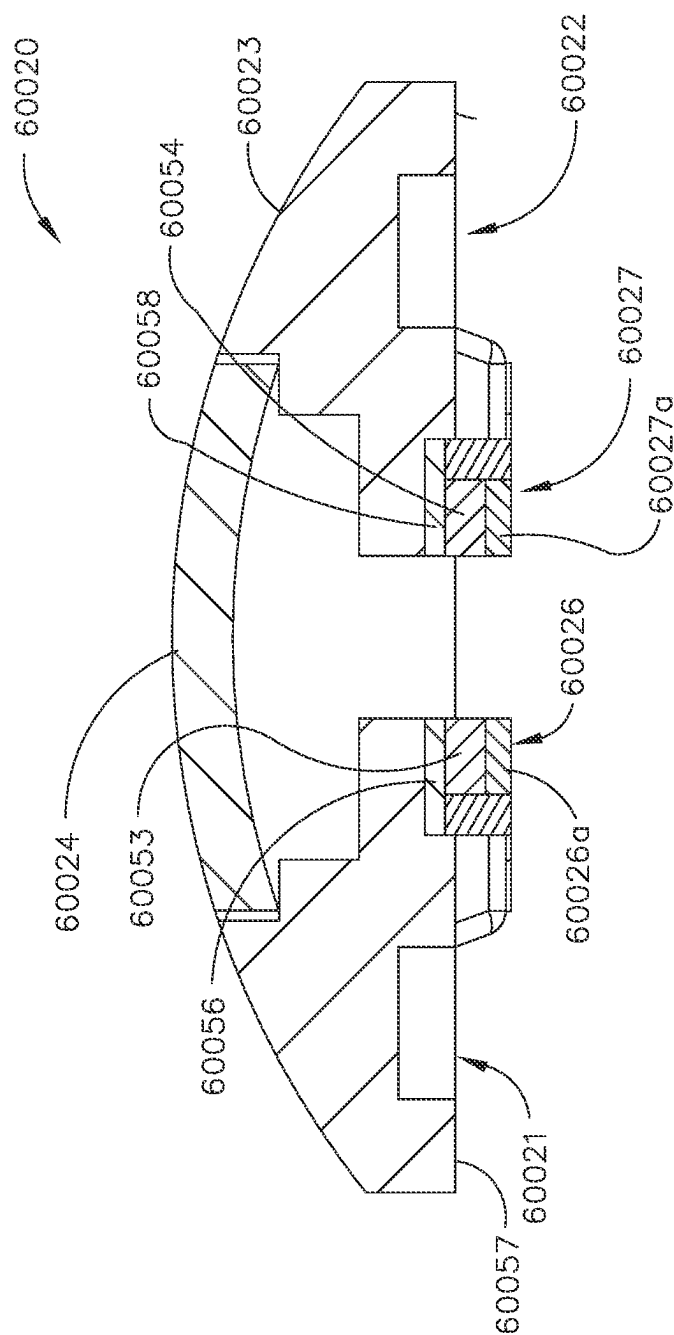


FIG. 171

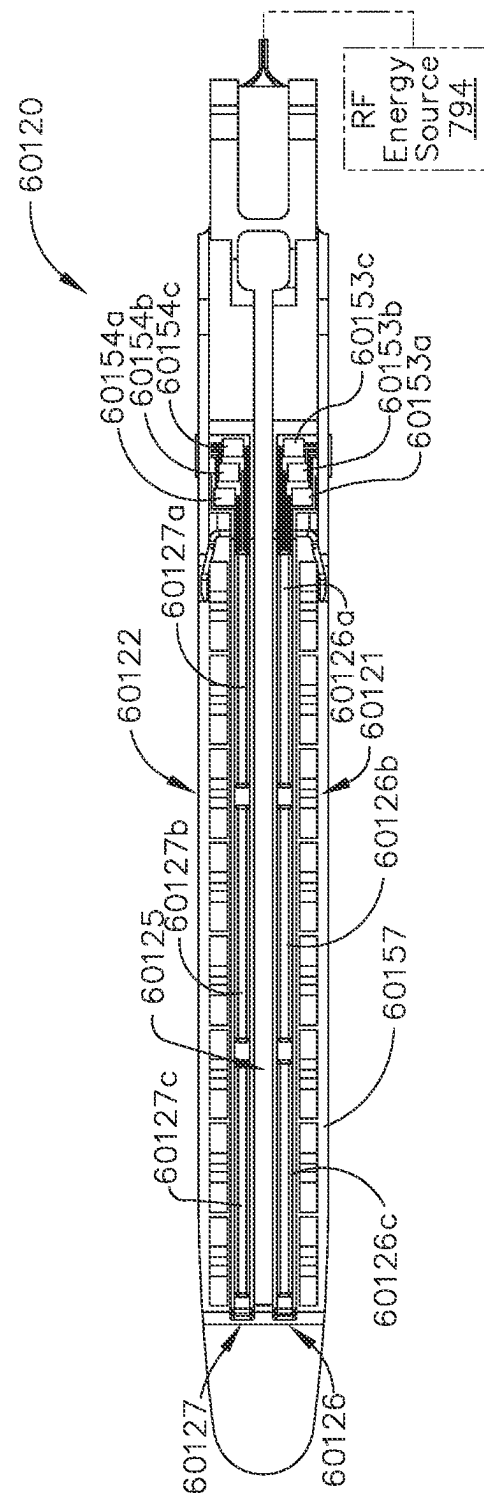


FIG. 172

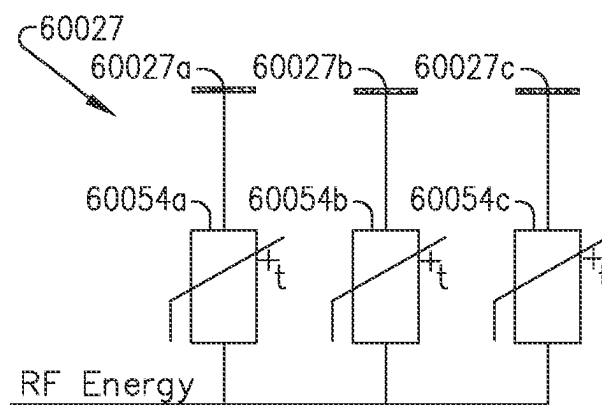
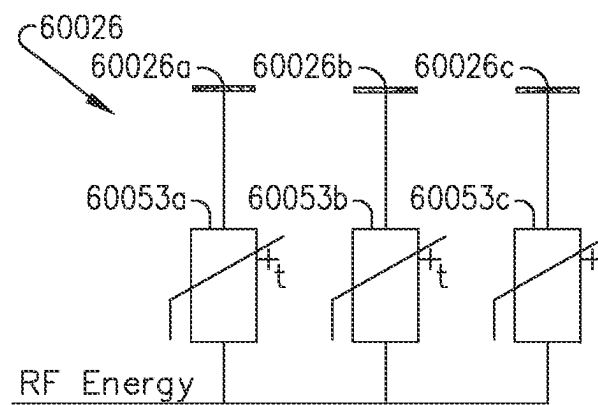
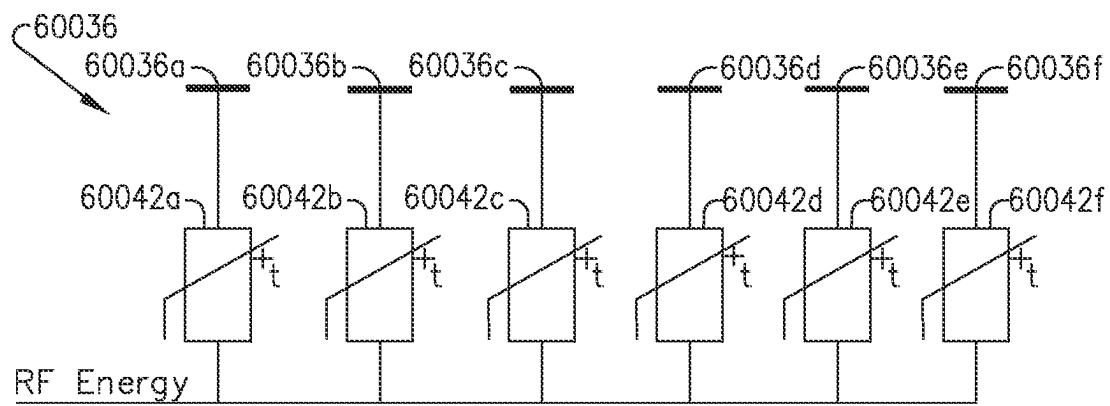


FIG. 173

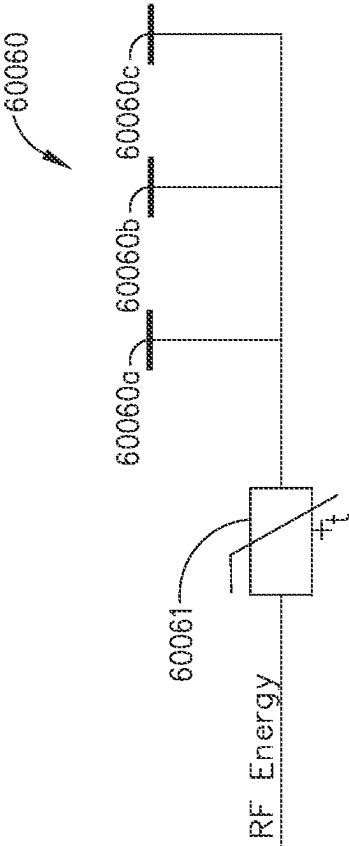


FIG. 174

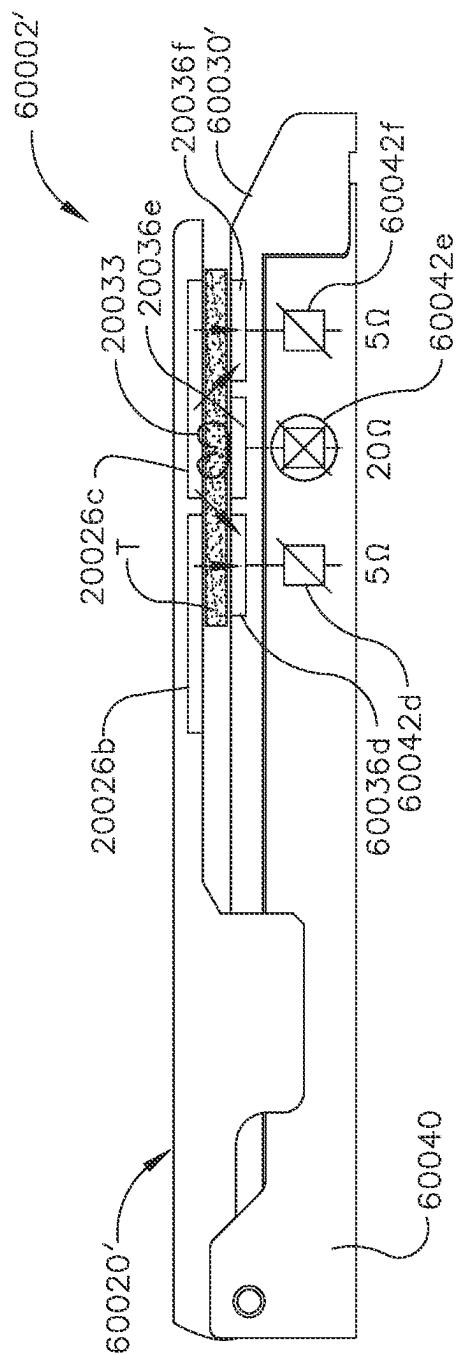


FIG. 175

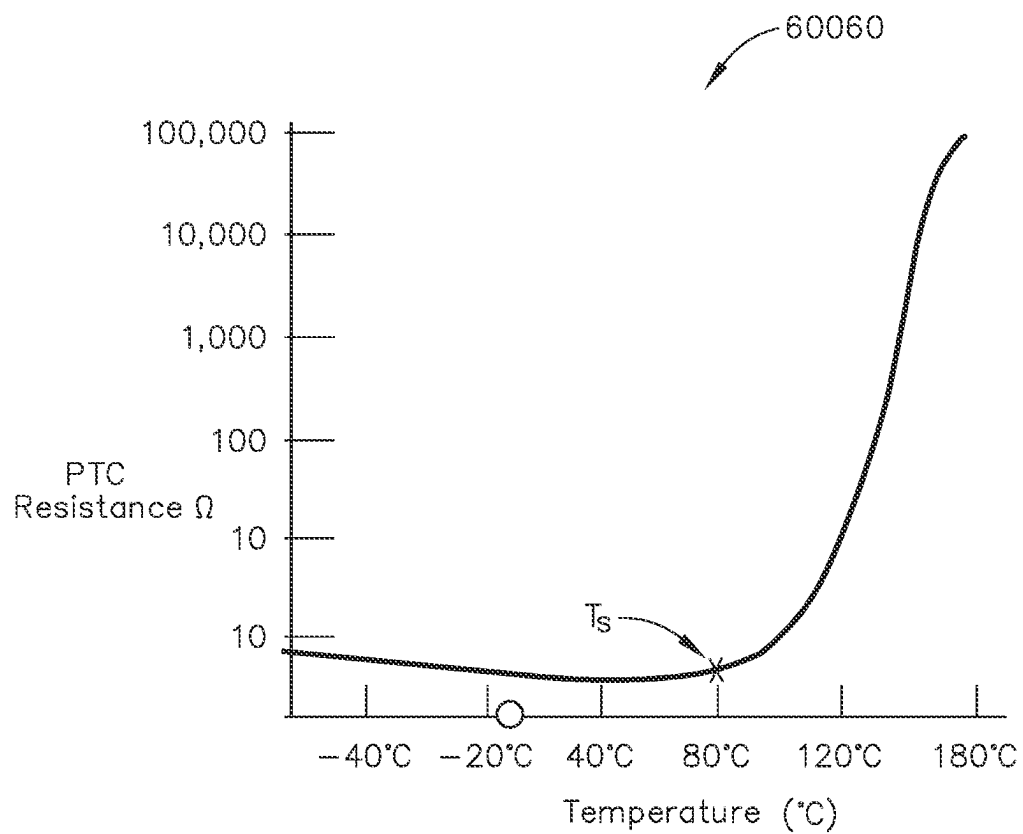


FIG. 176

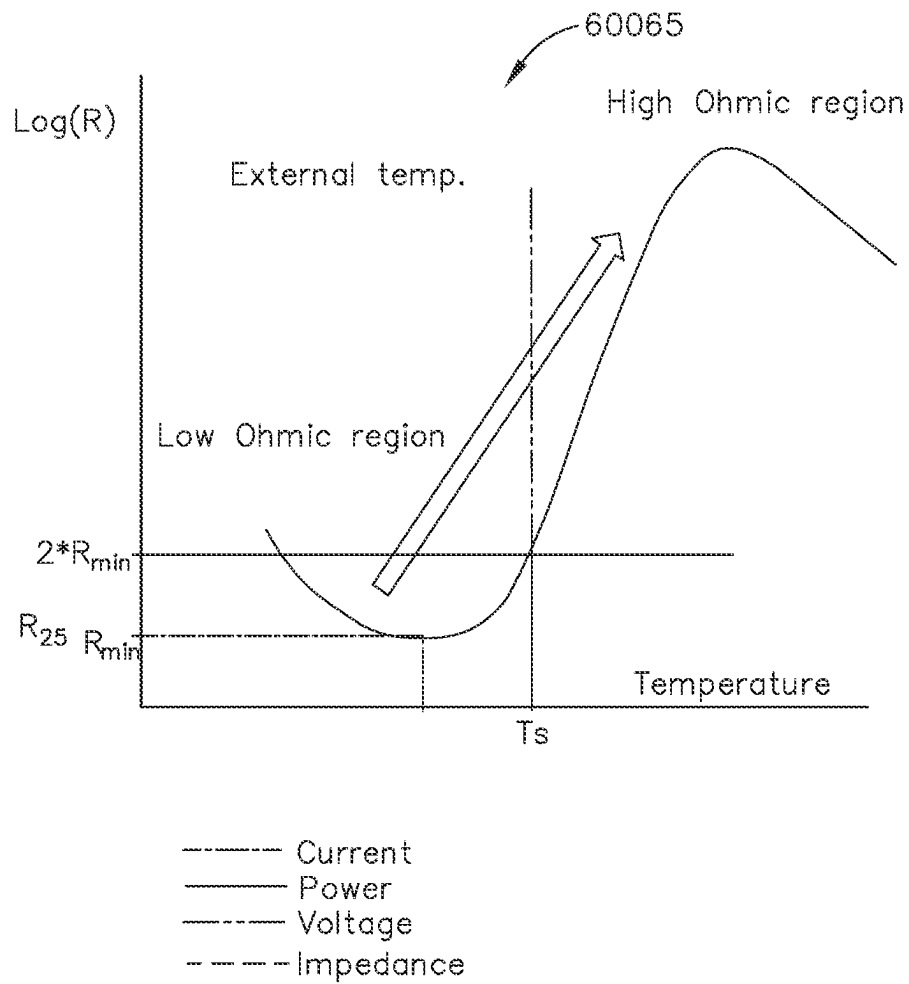


FIG. 177

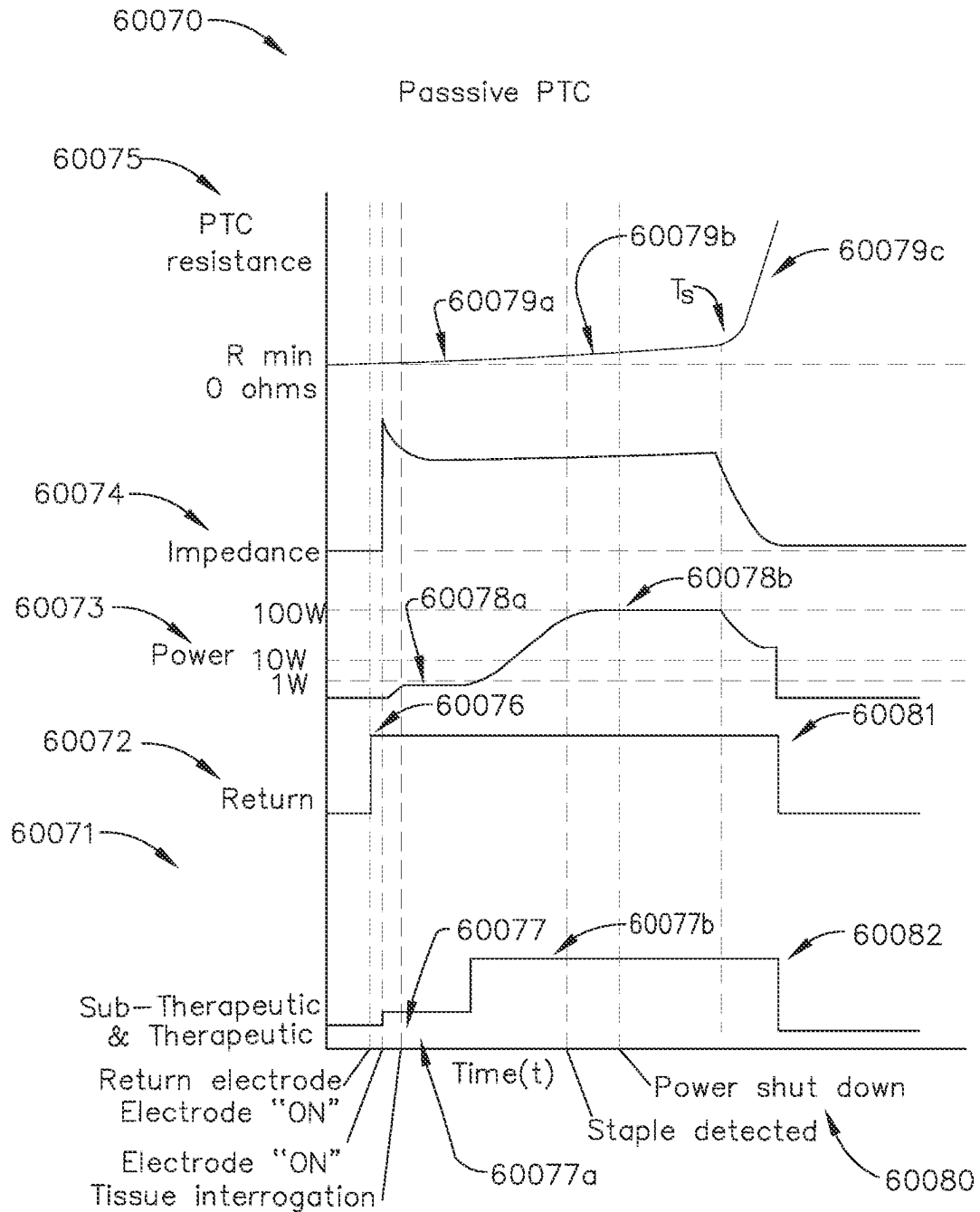


FIG. 178

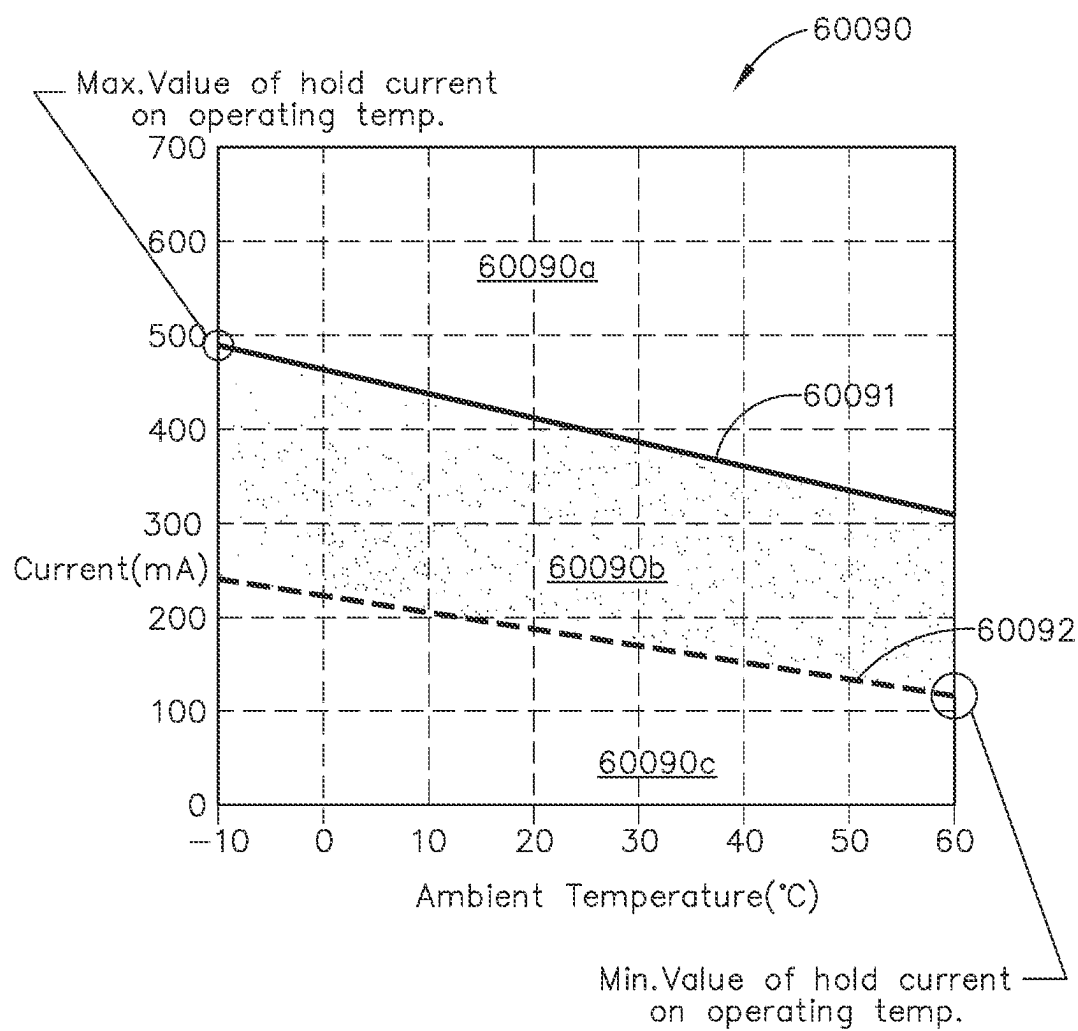


FIG. 179

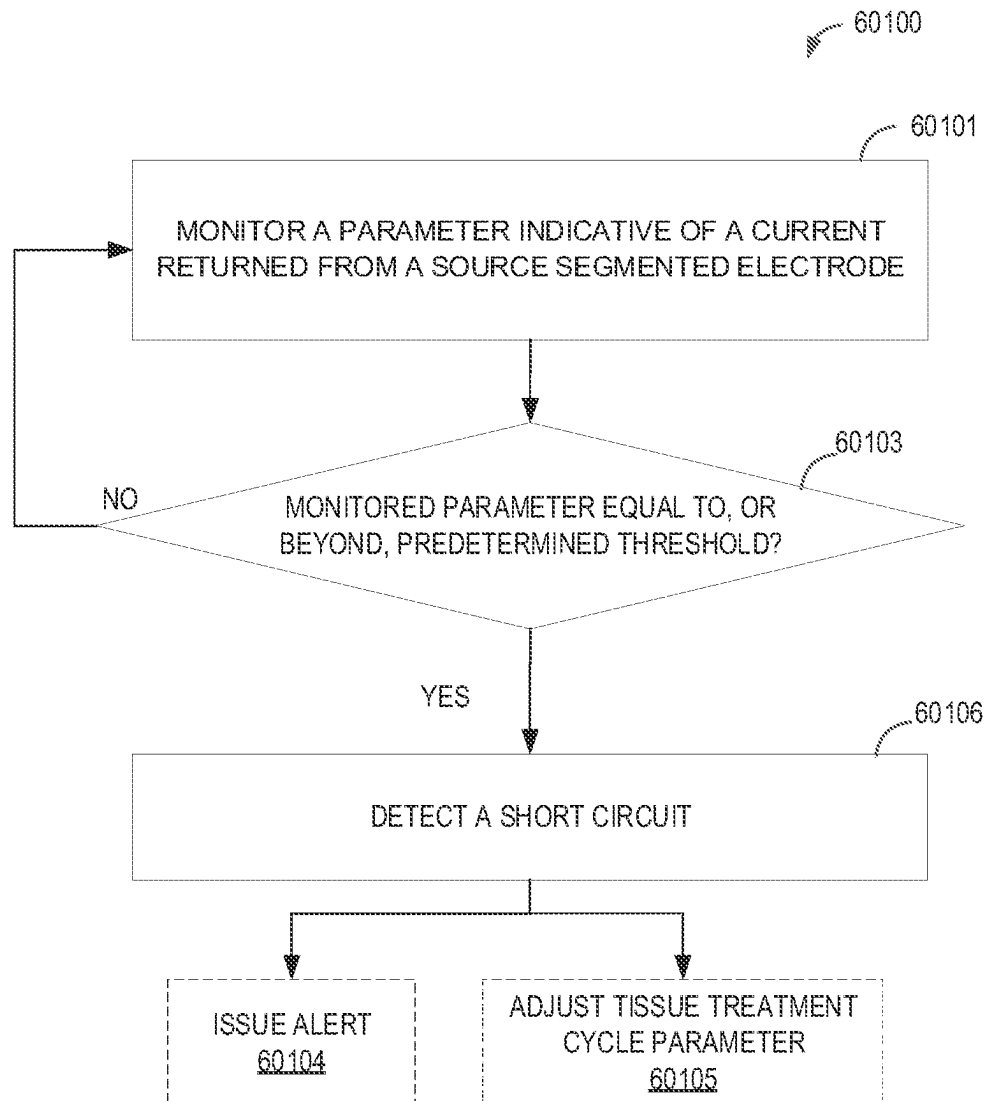


FIG. 180

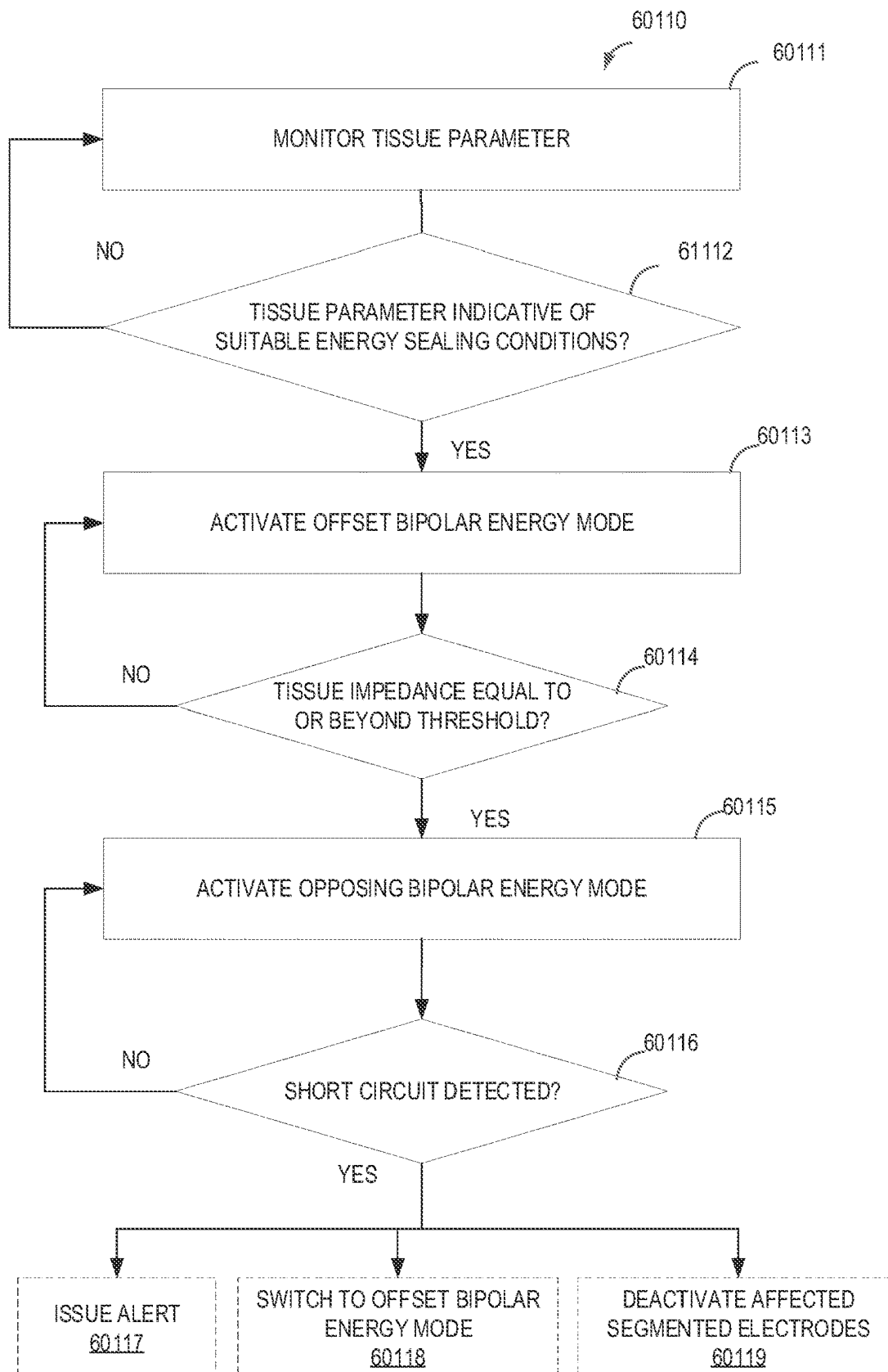


FIG. 181

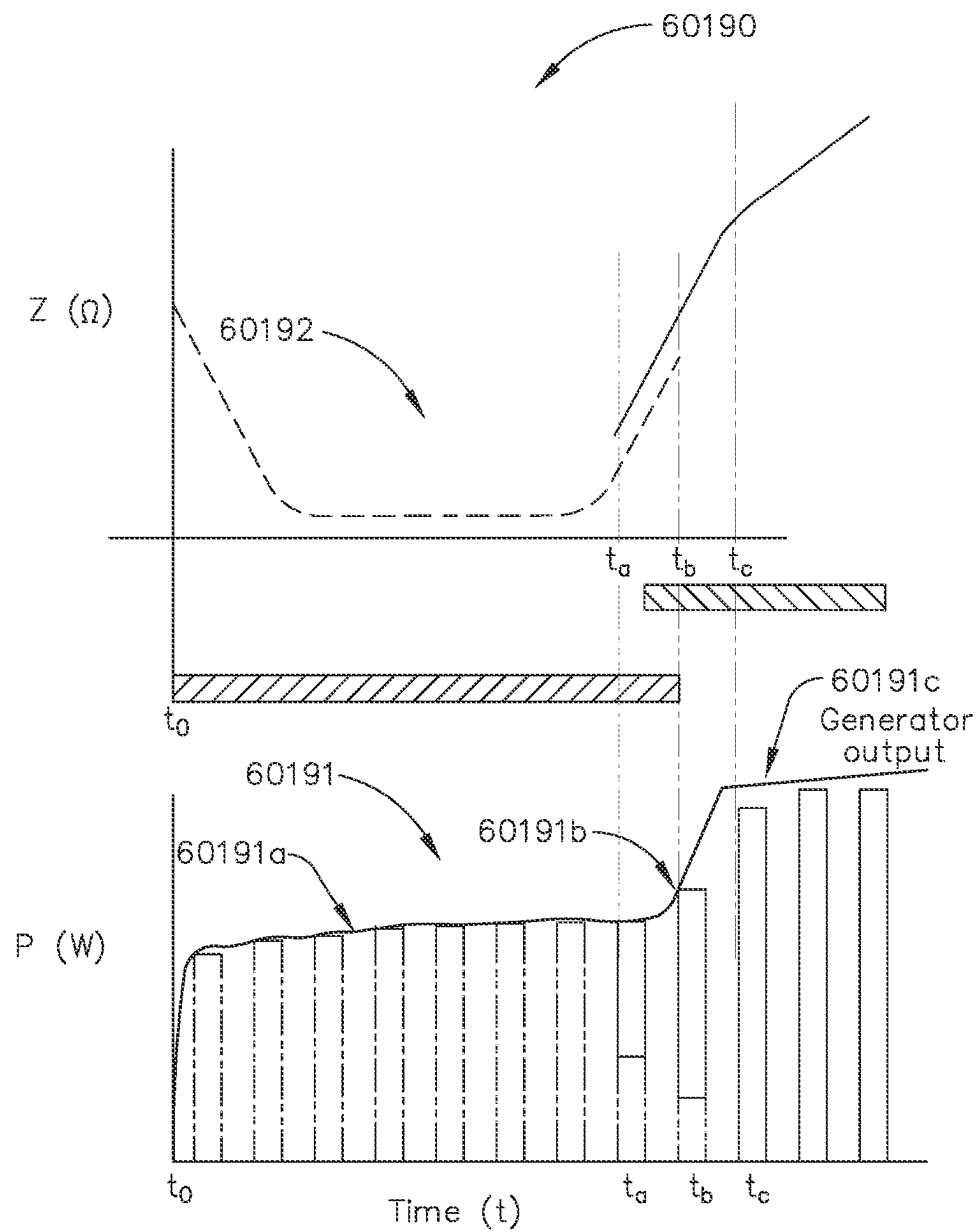


FIG. 182

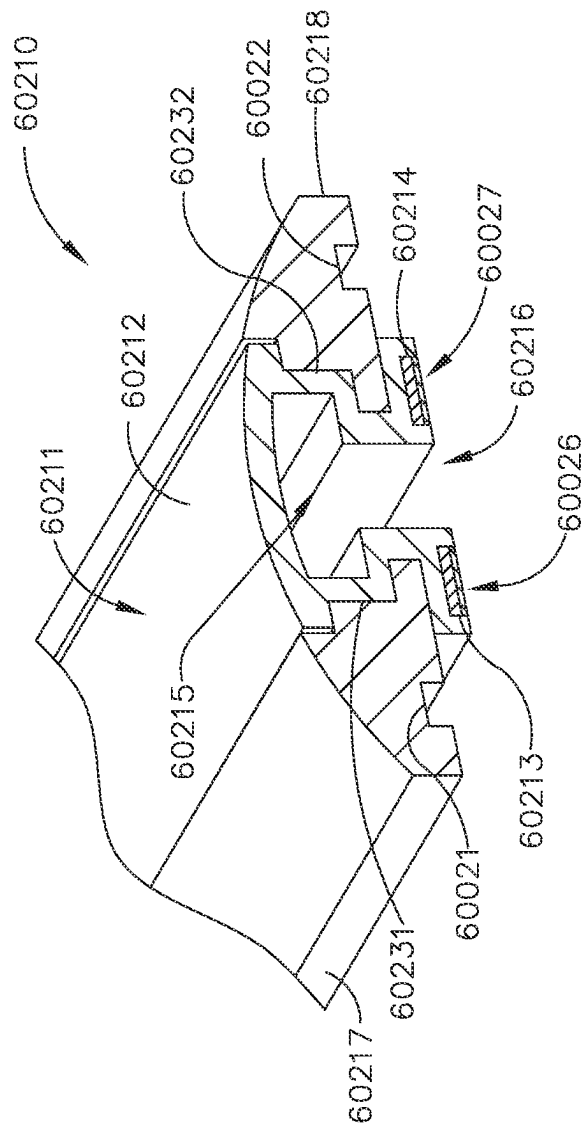


FIG. 183

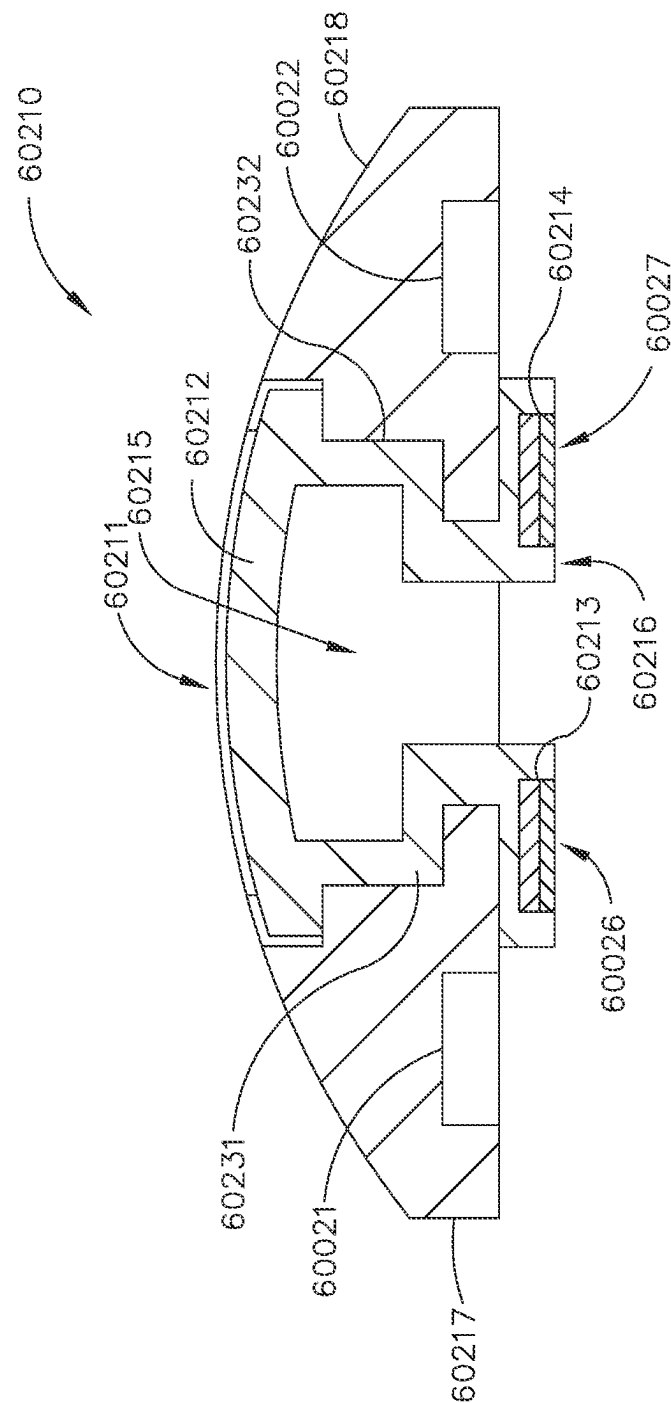


FIG. 184

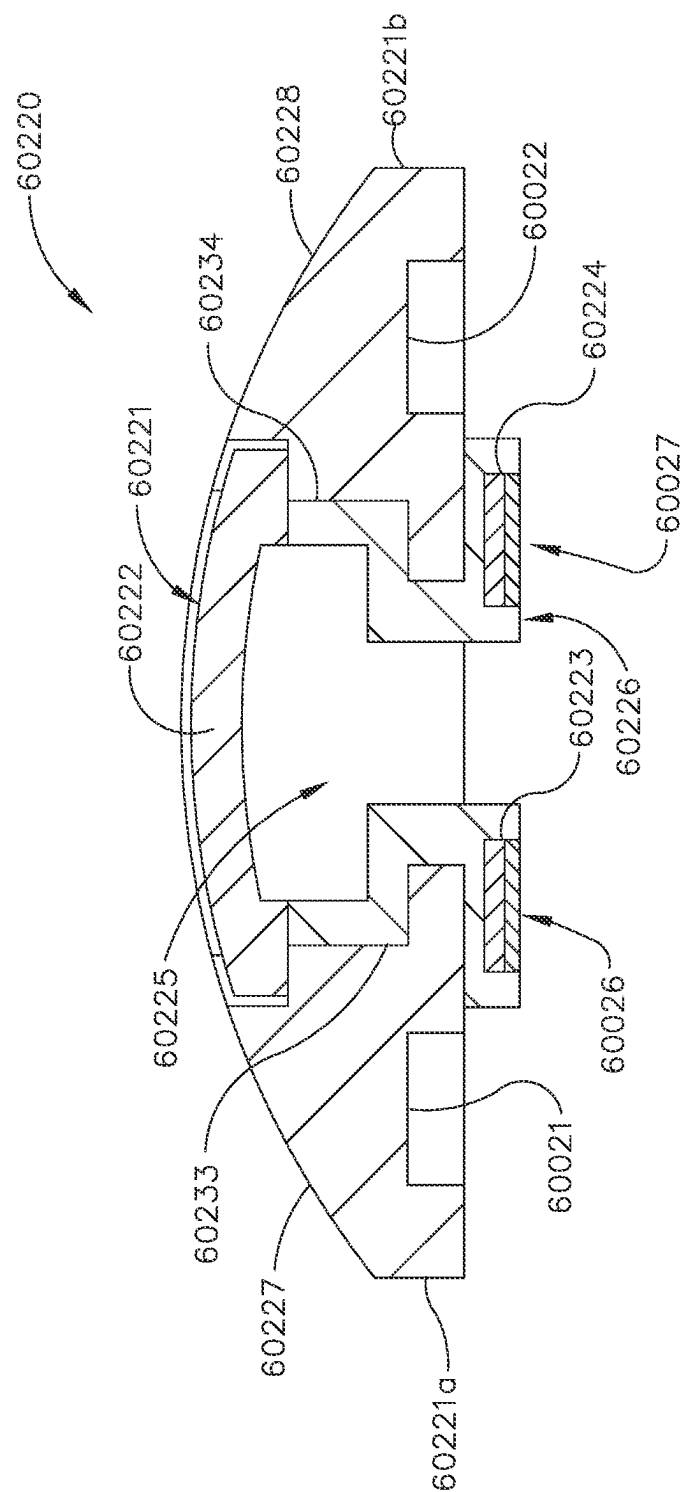


FIG. 186

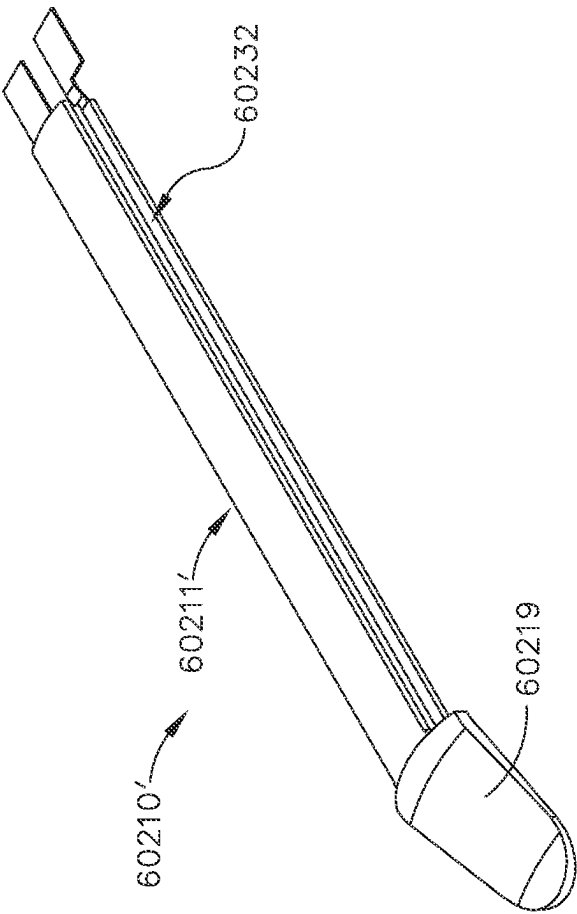


FIG. 187

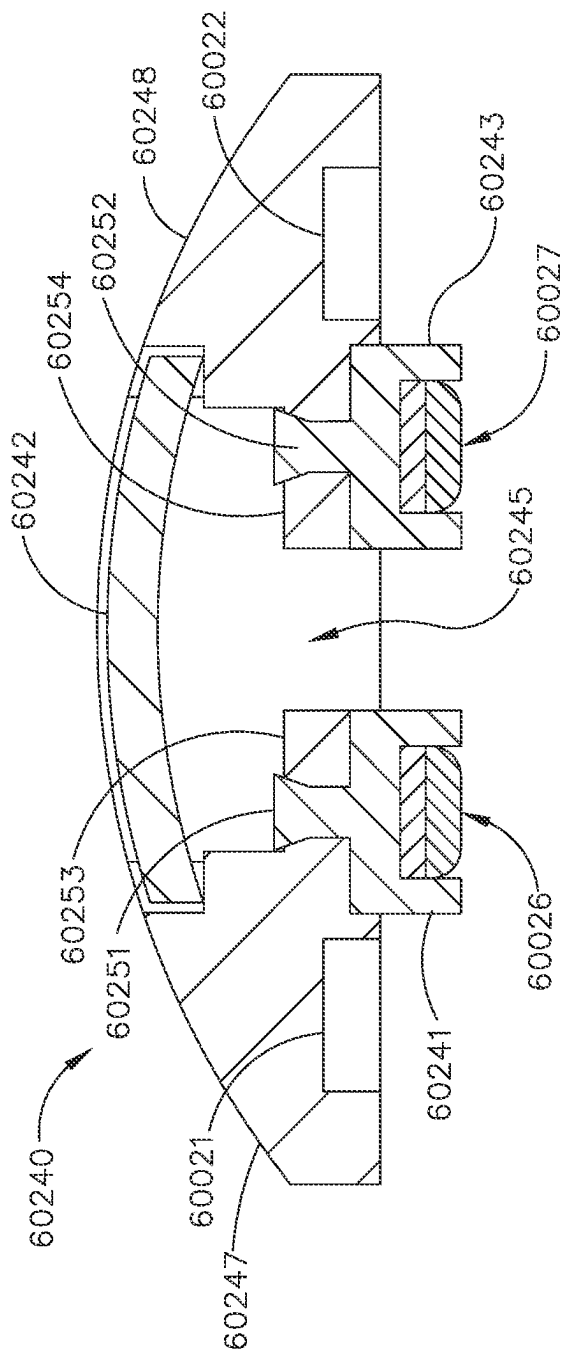


FIG. 188

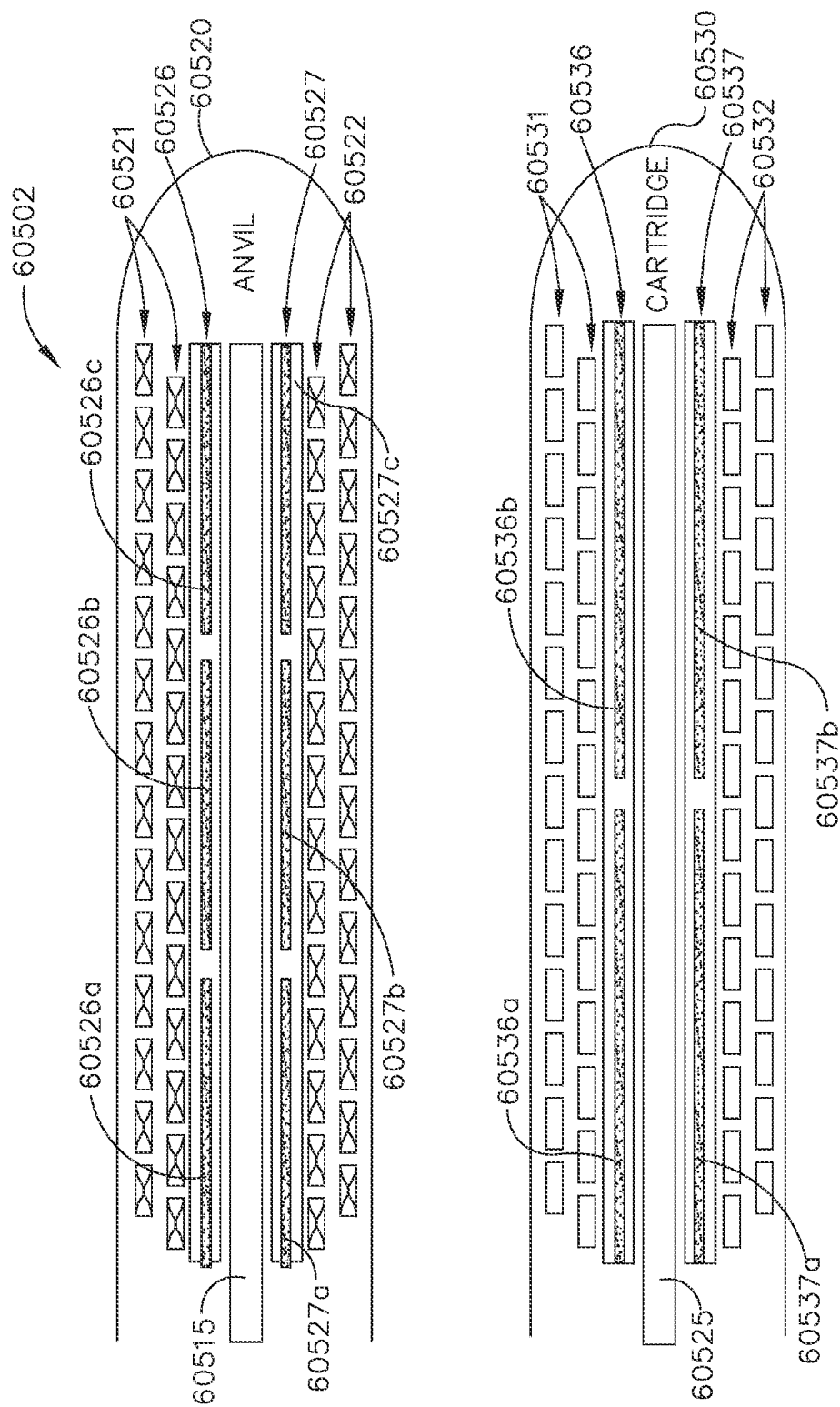


FIG. 189

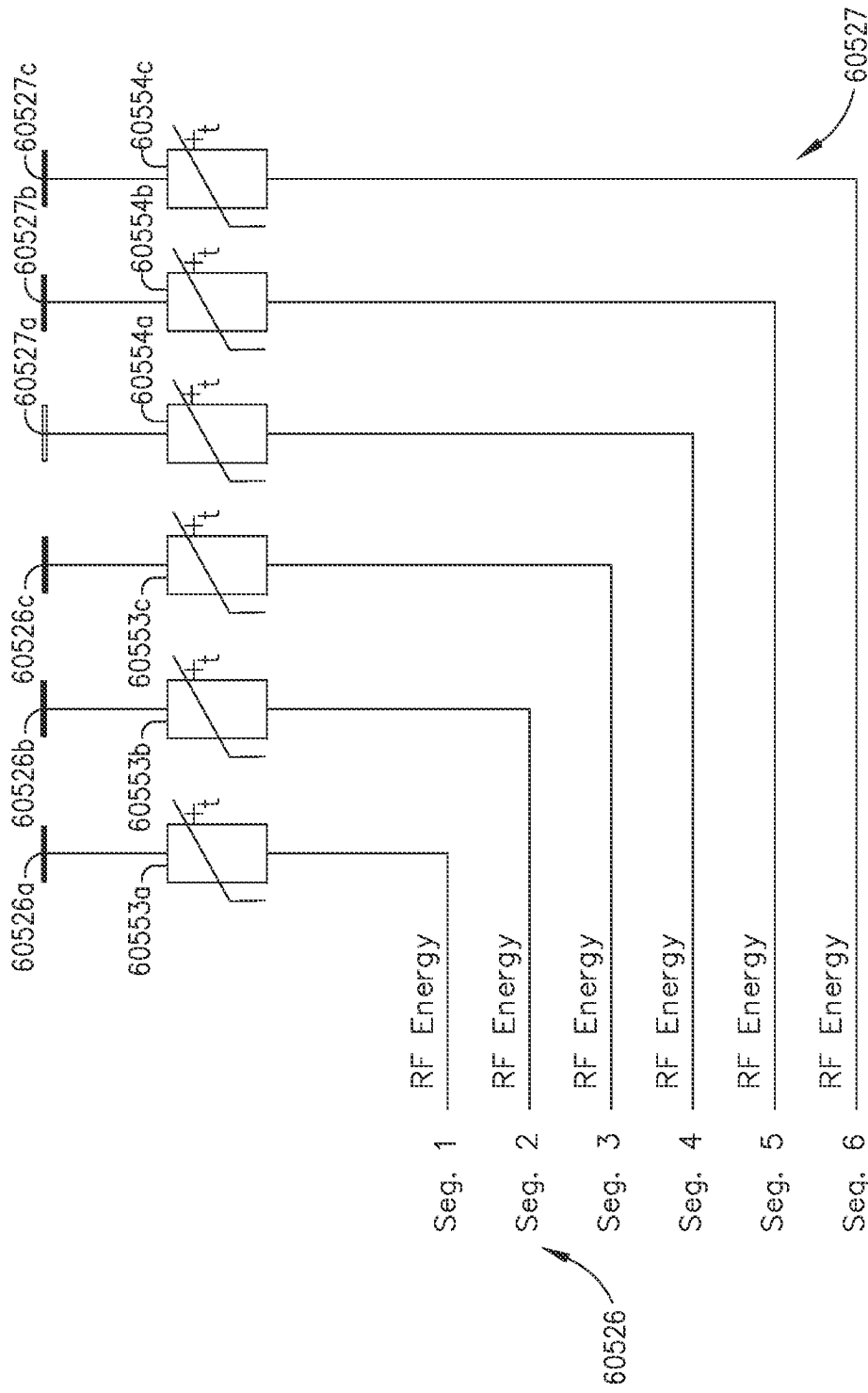


FIG. 190

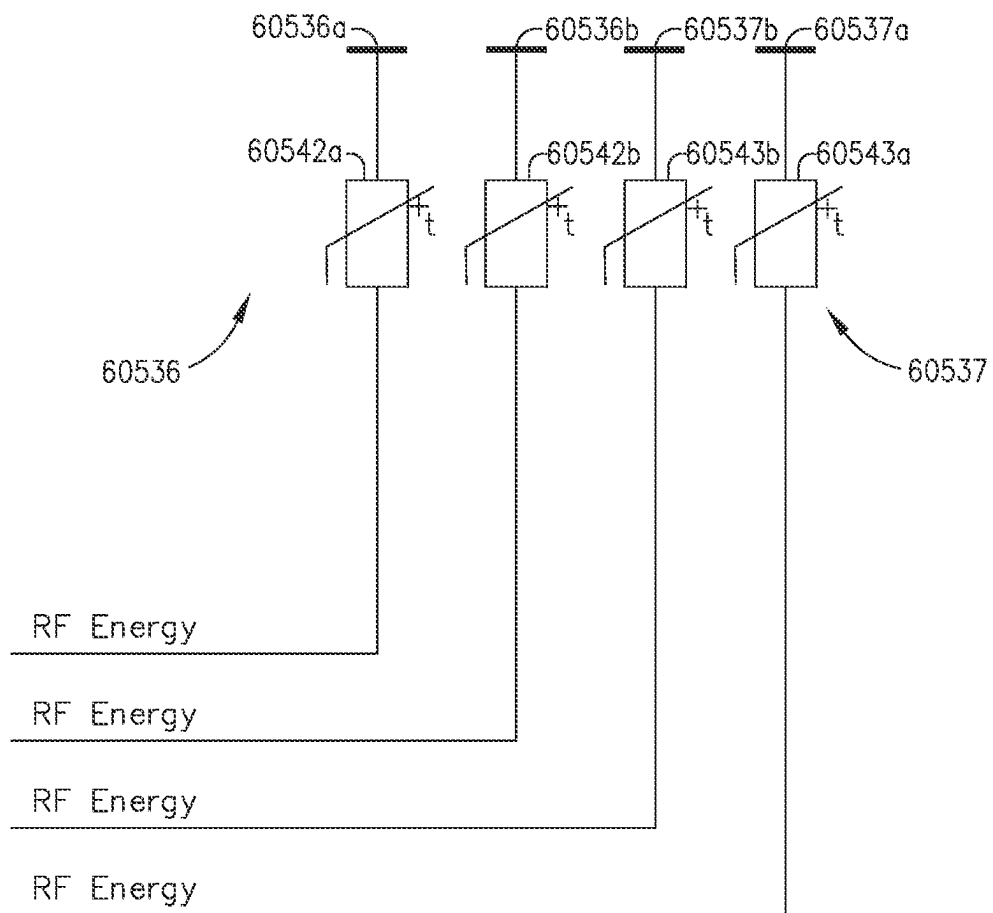


FIG. 191

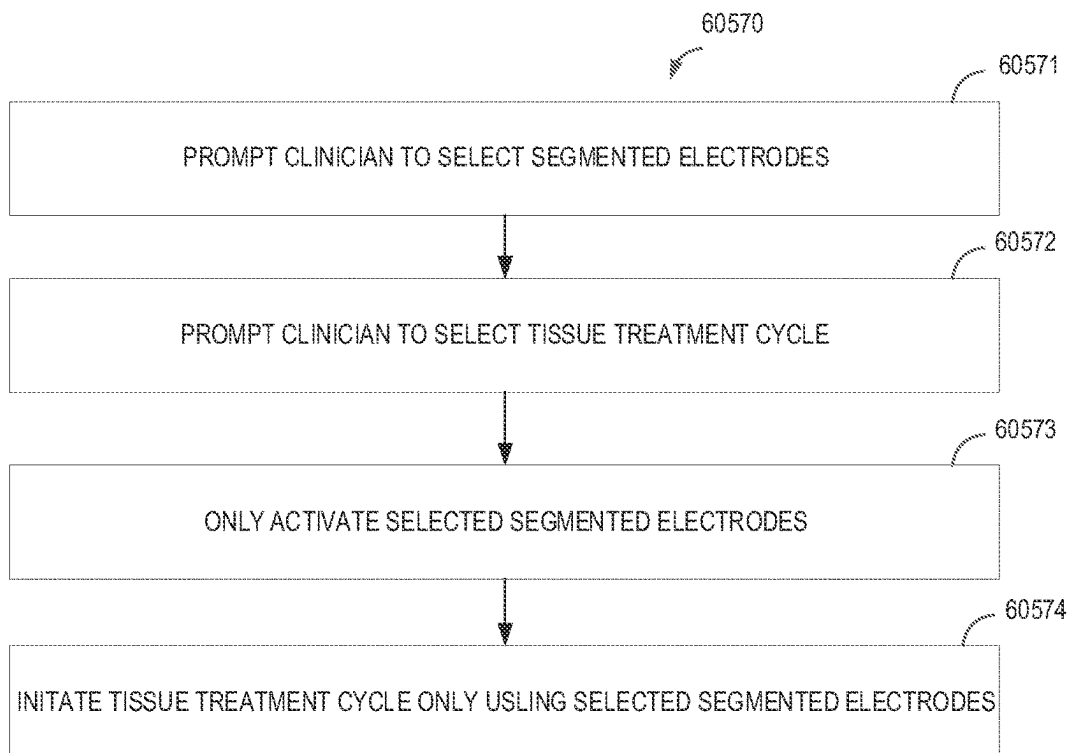


FIG. 192

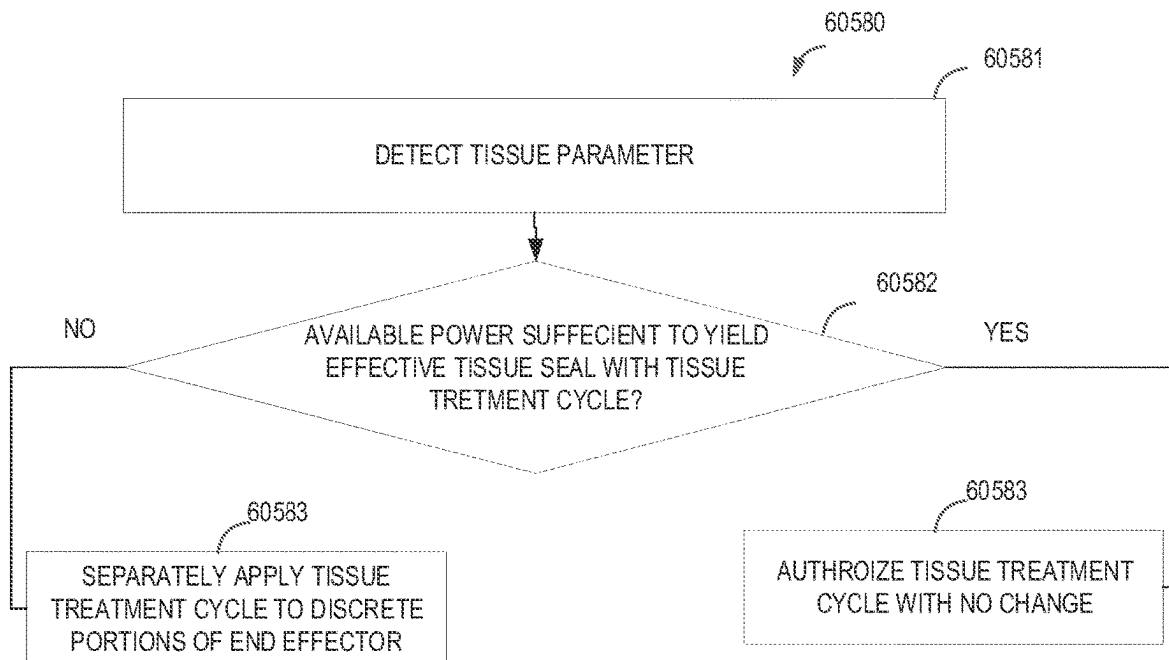


FIG. 193

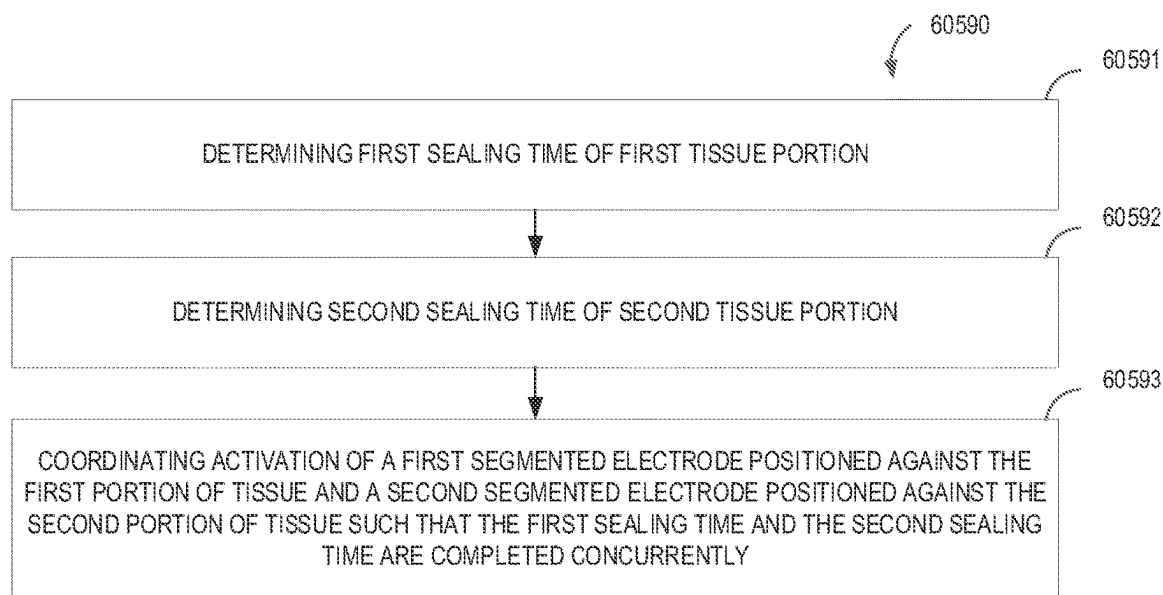


FIG. 194

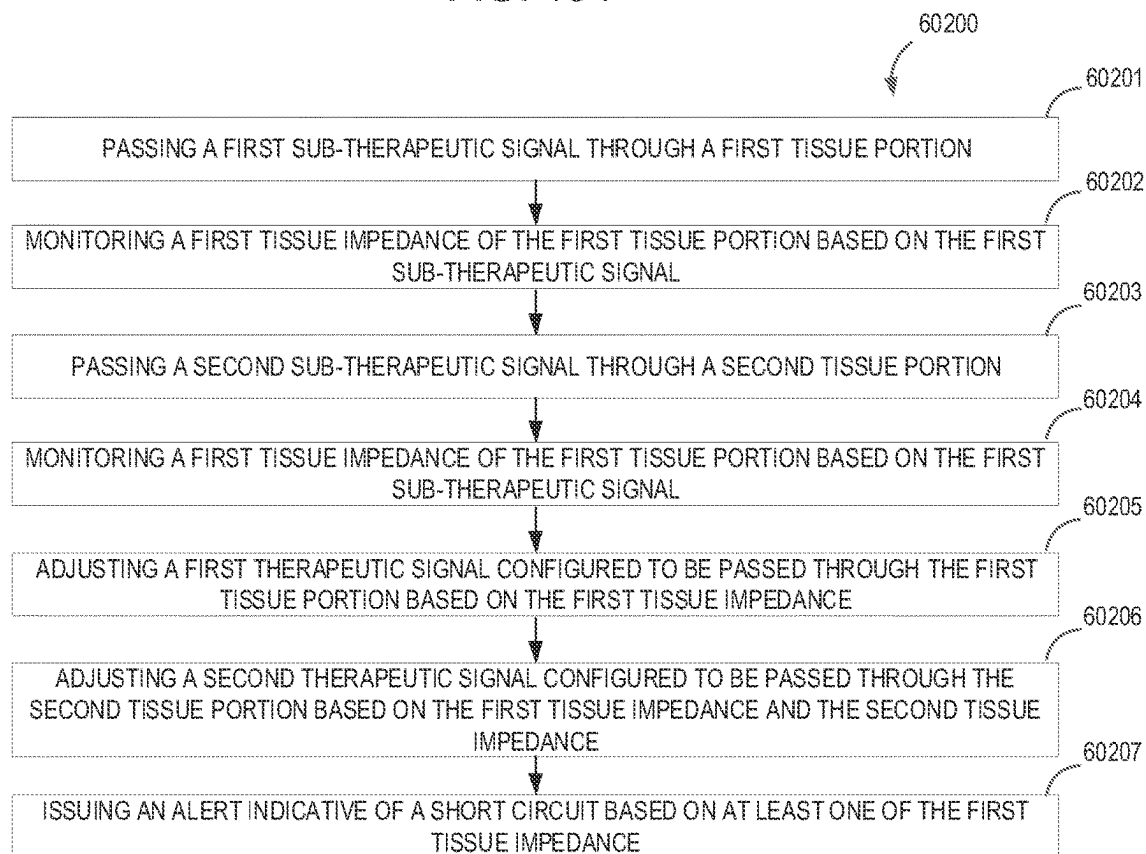


FIG. 195

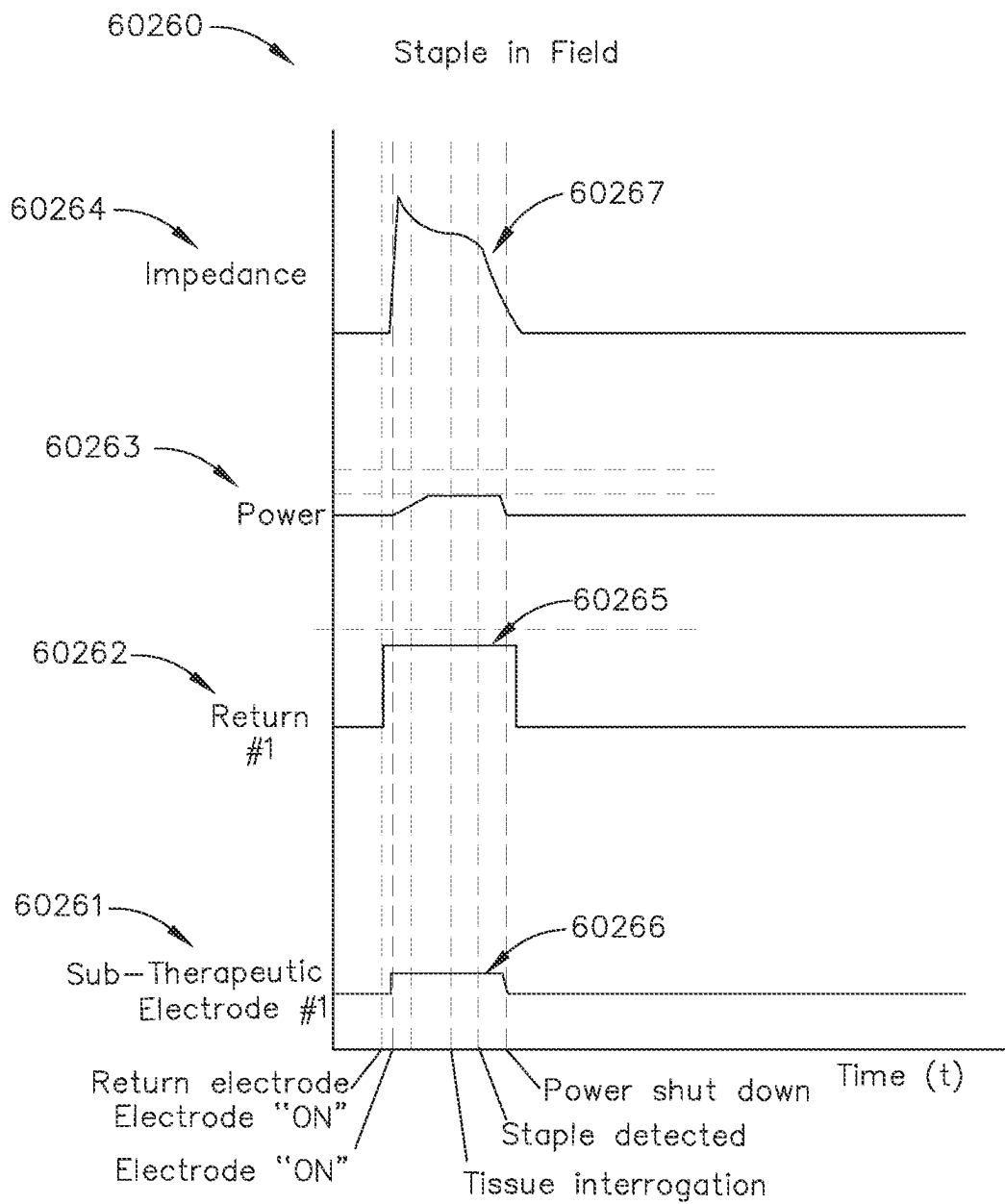


FIG. 196

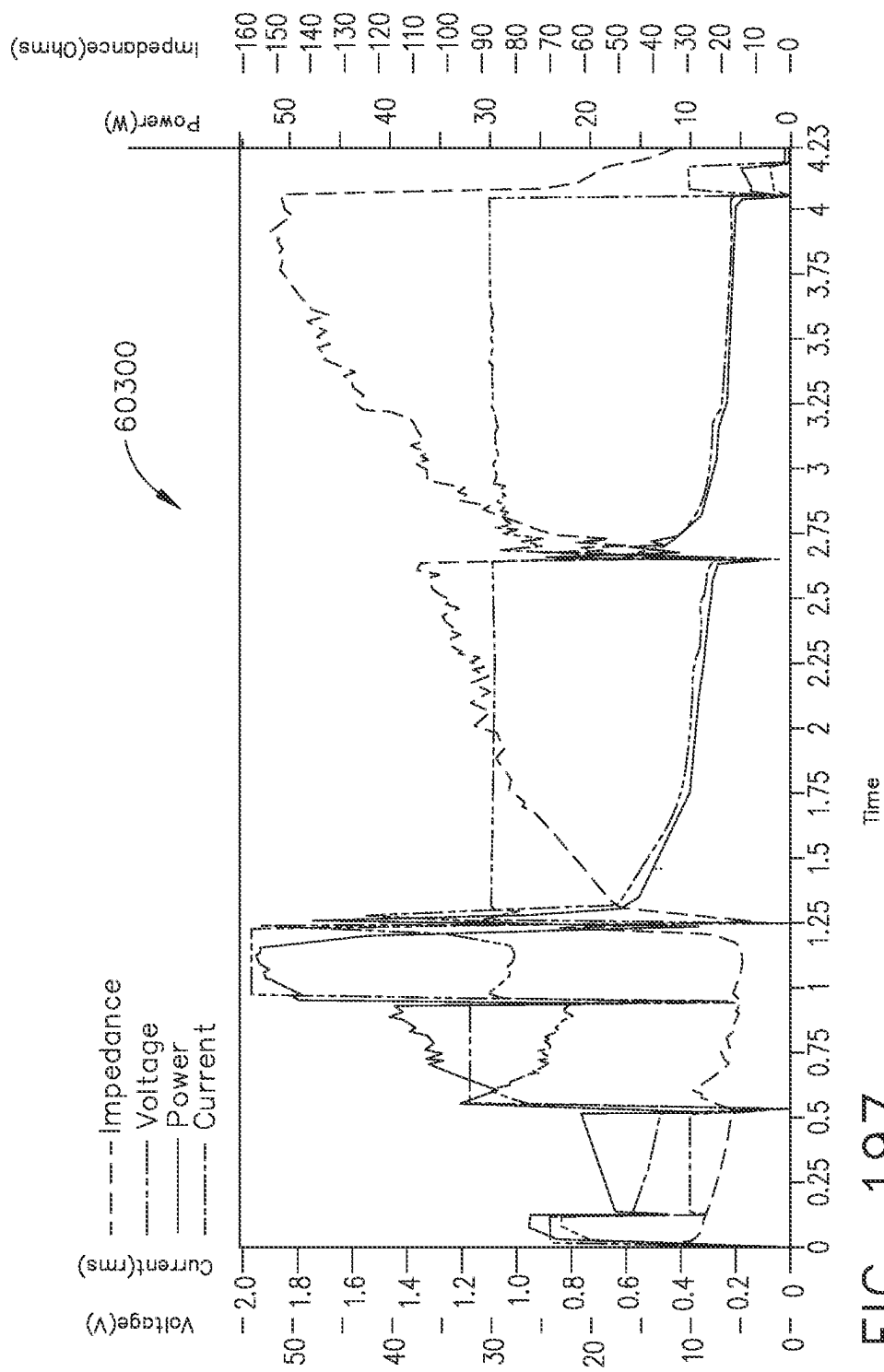


FIG. 197

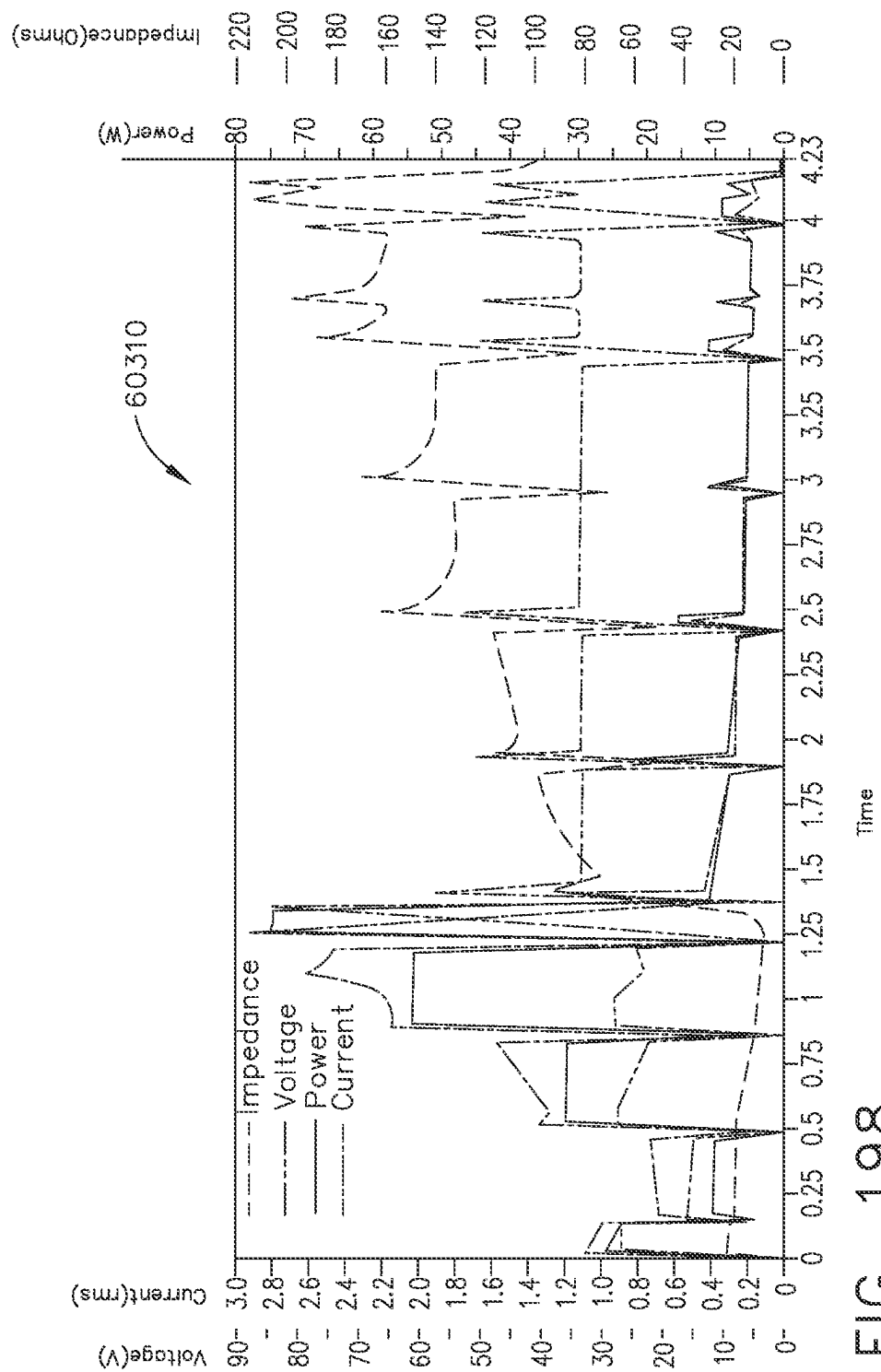


FIG. 198

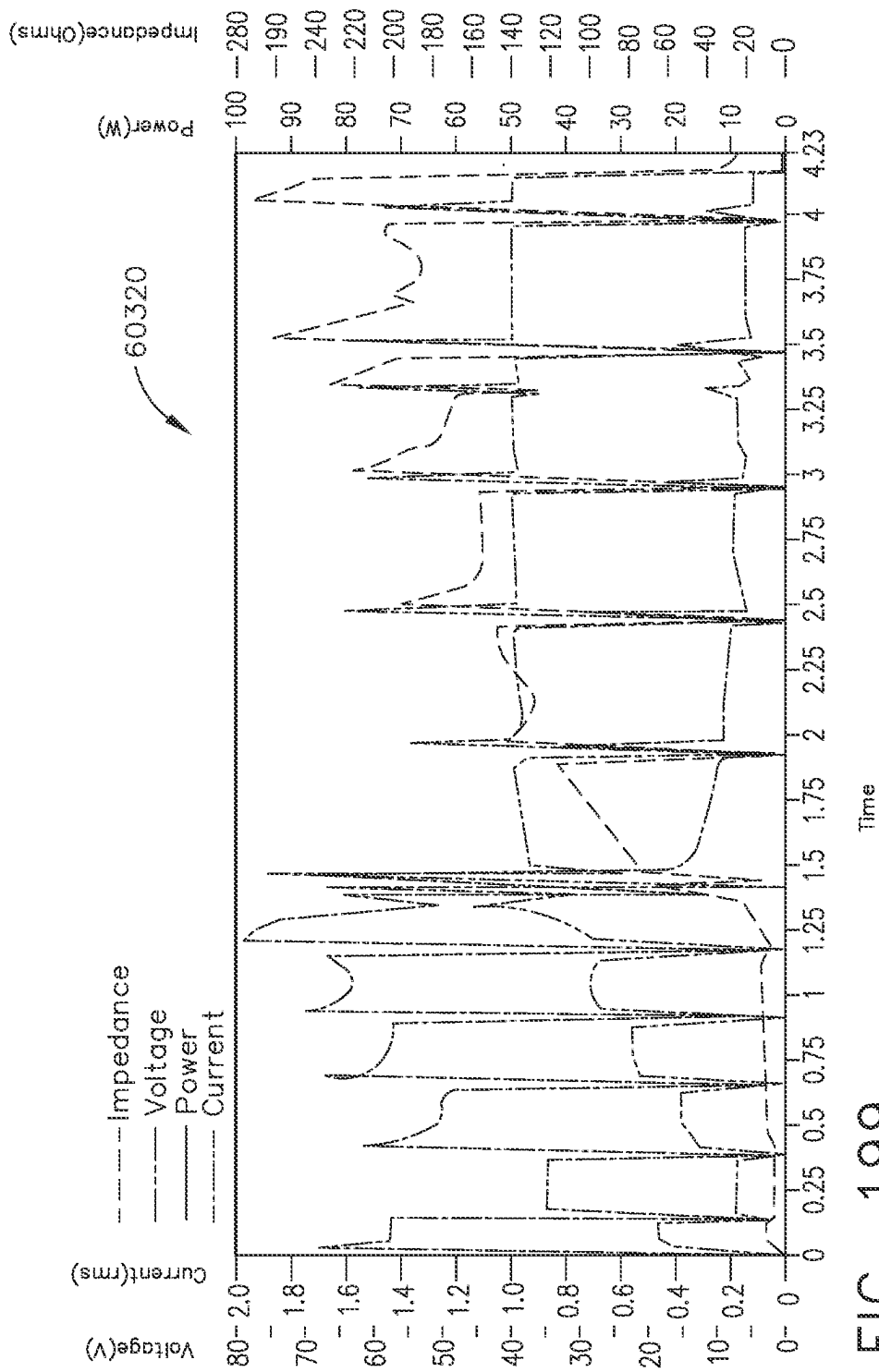


FIG. 199

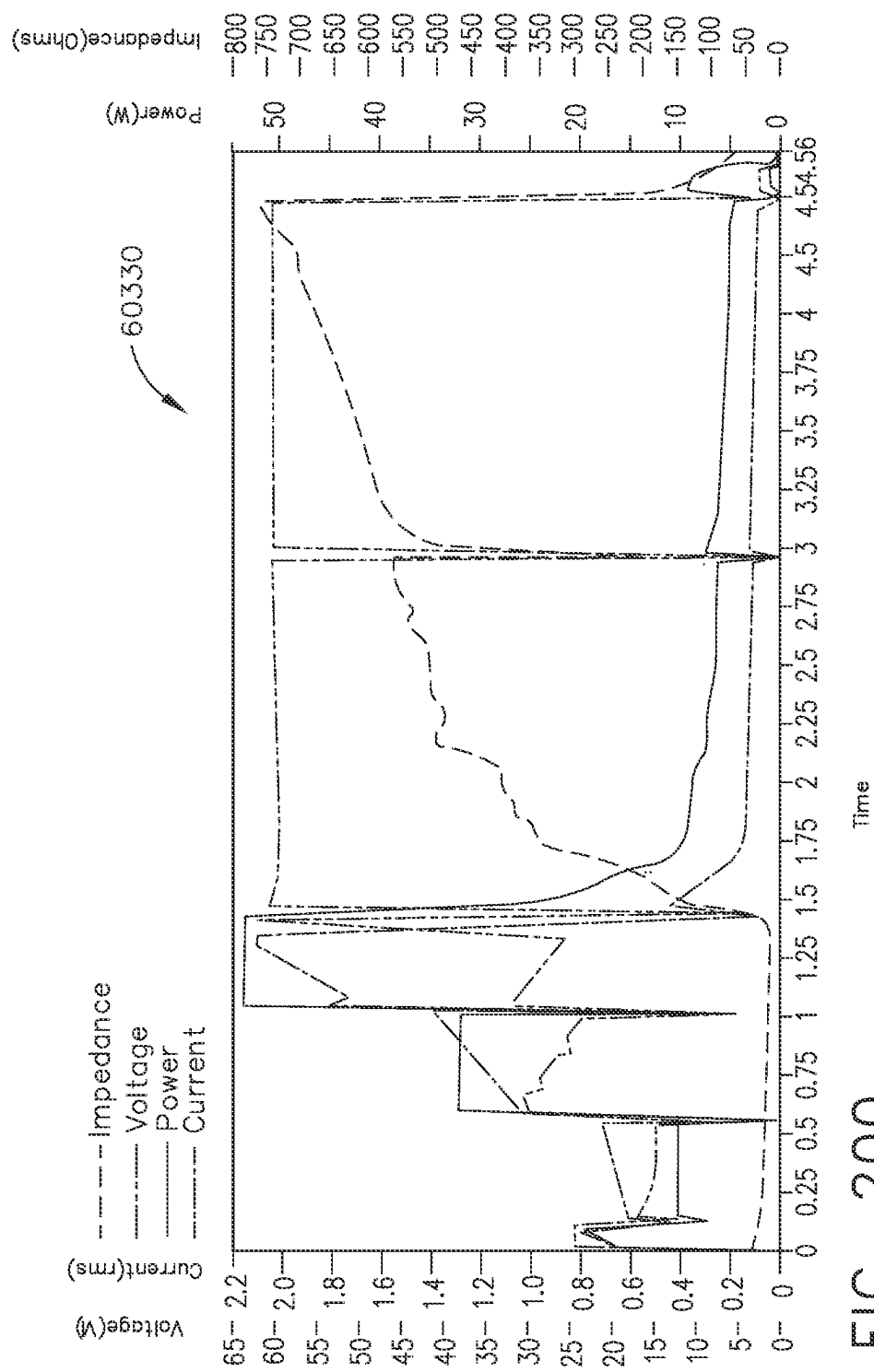


FIG. 200

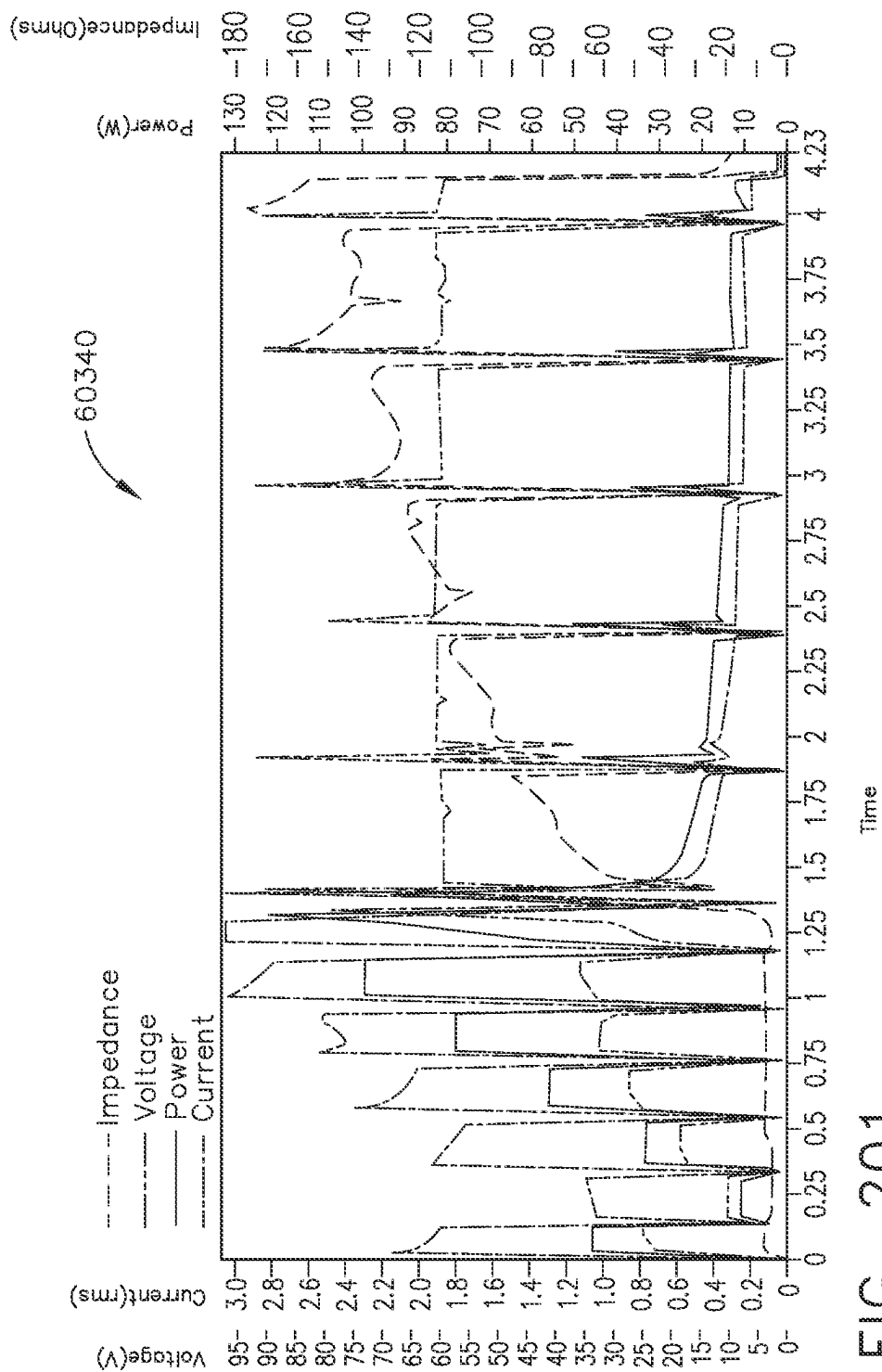


FIG. 201

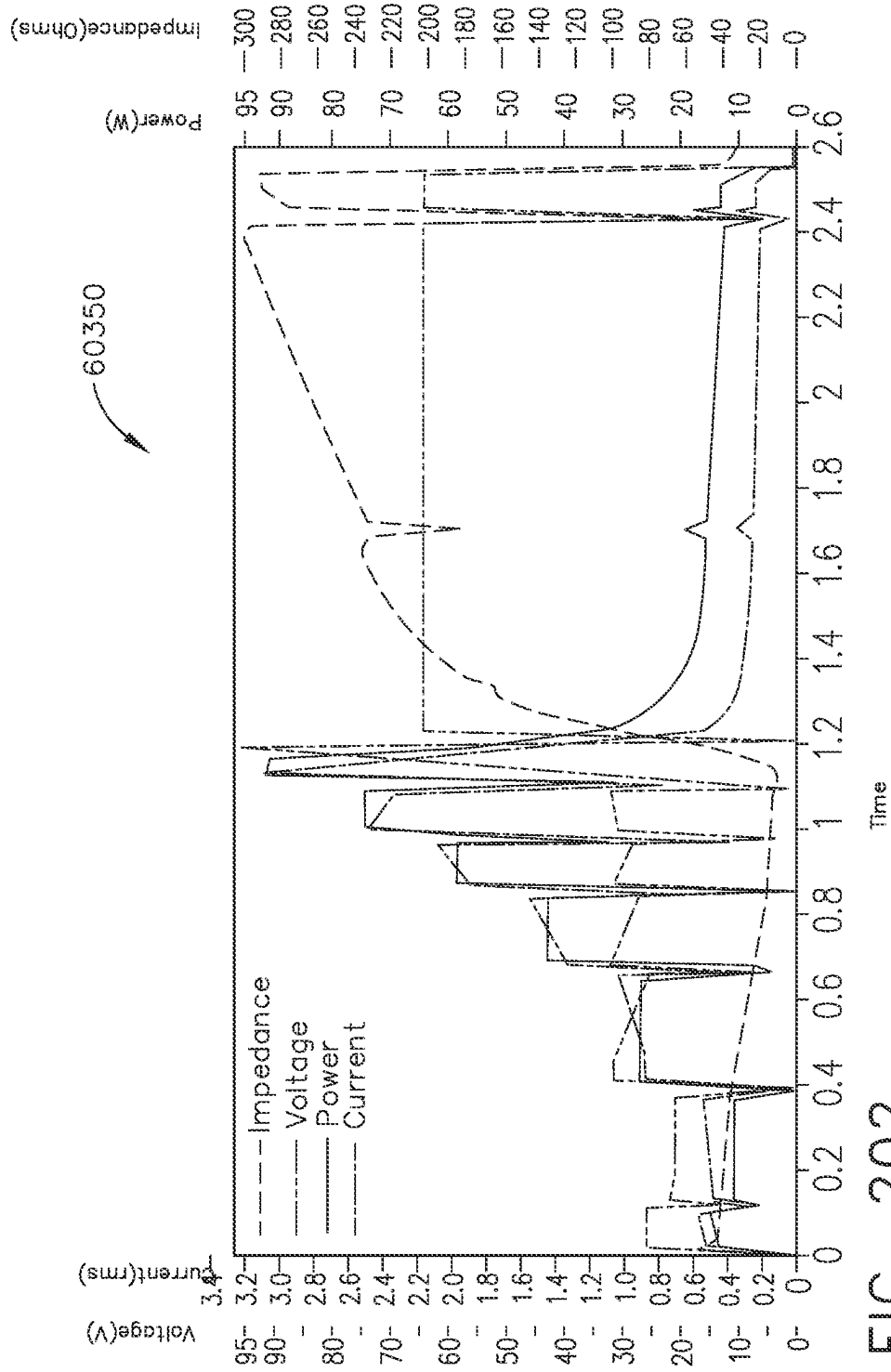


FIG. 202

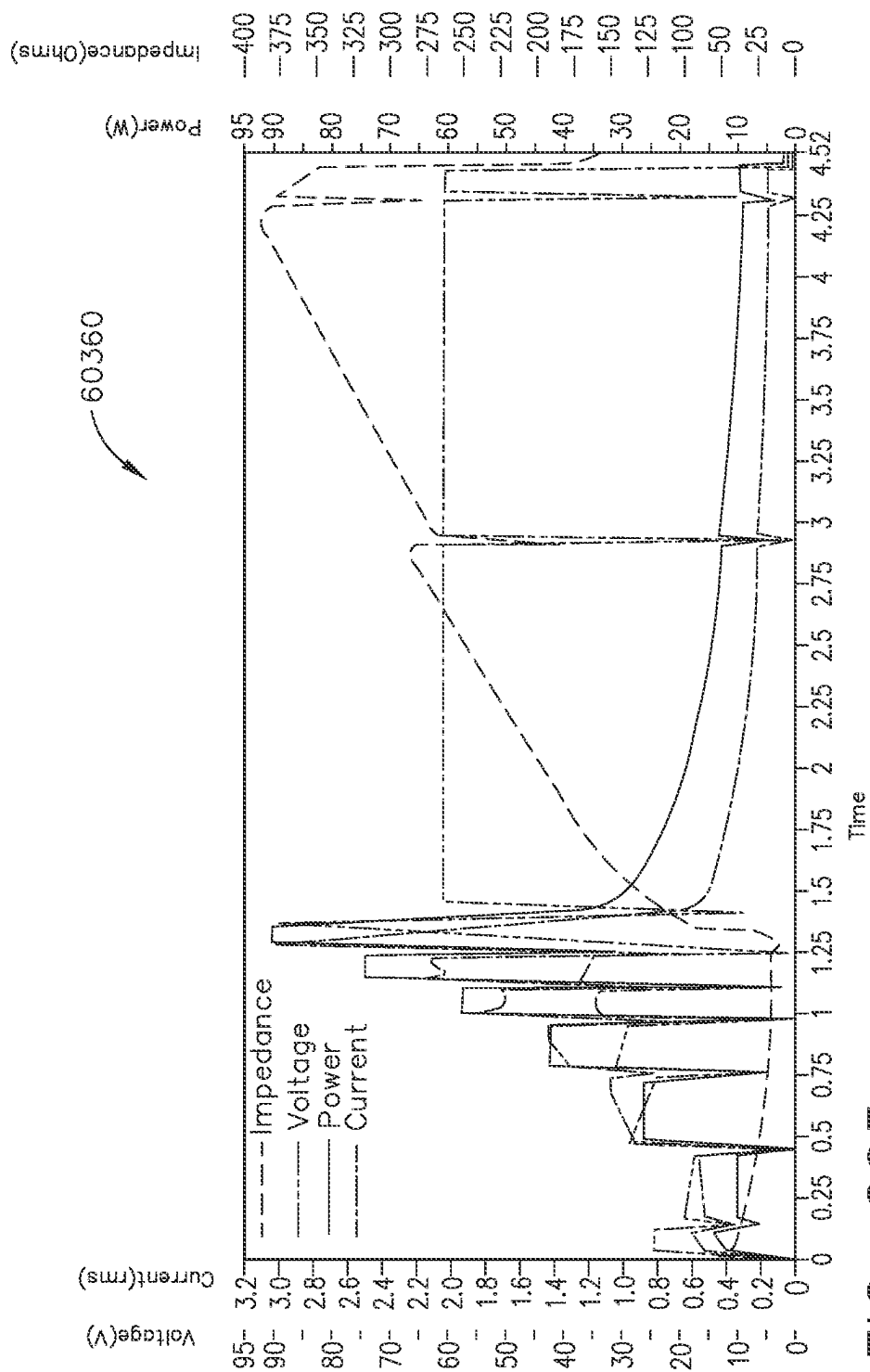


FIG. 203

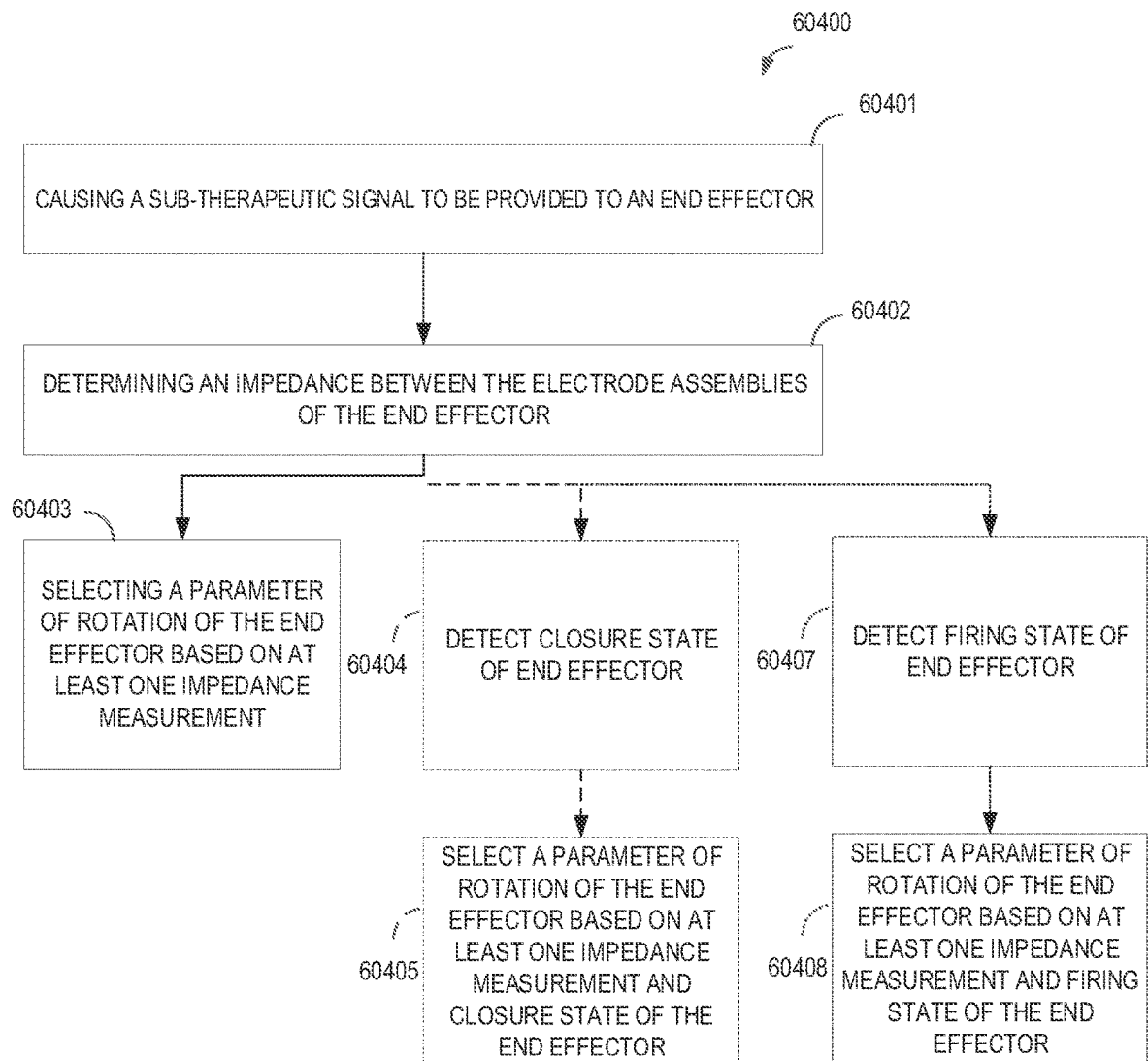


FIG. 204

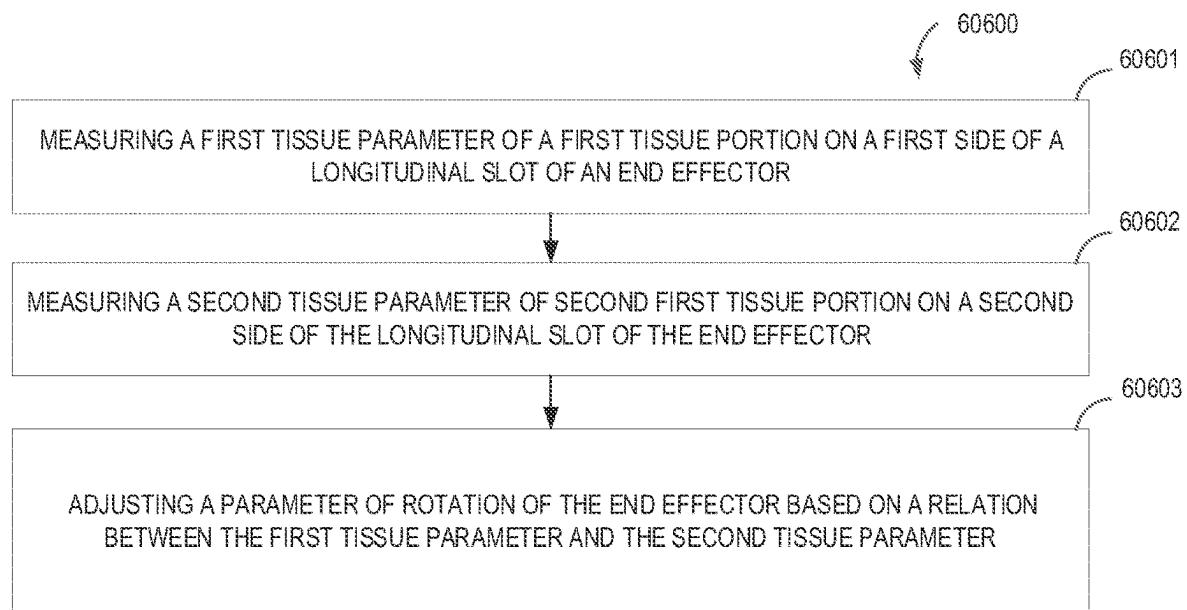


FIG. 205

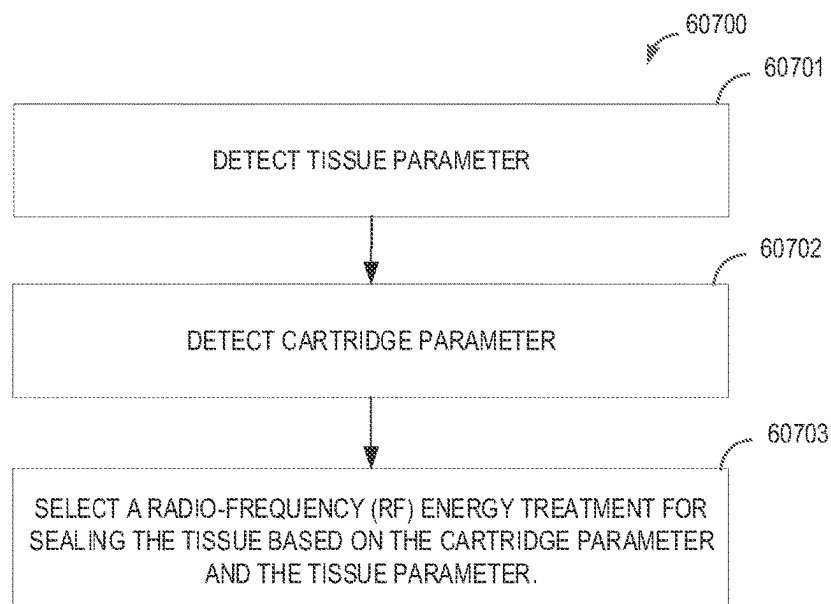


FIG. 206

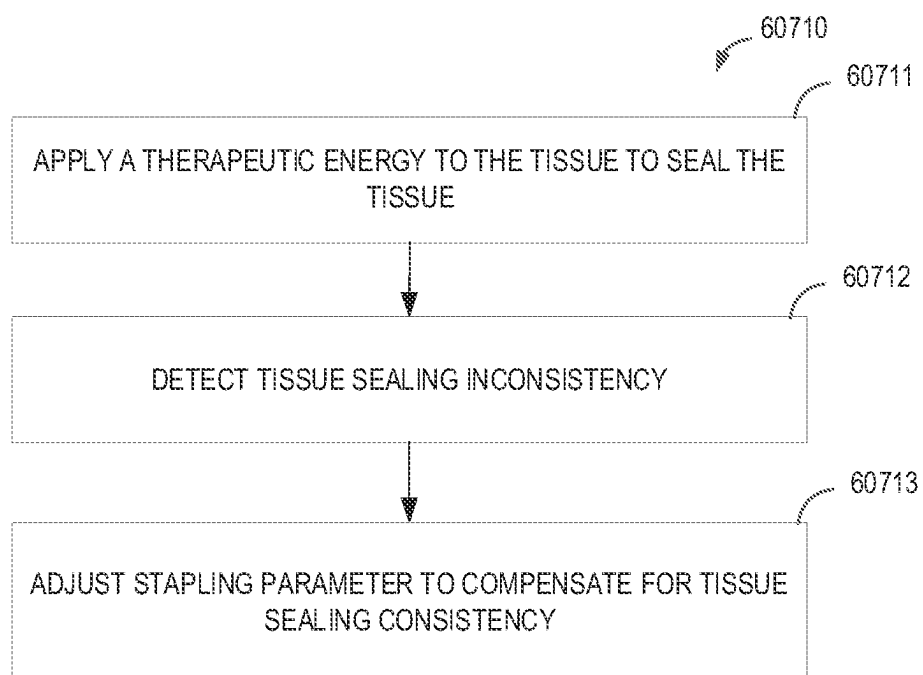


FIG. 207

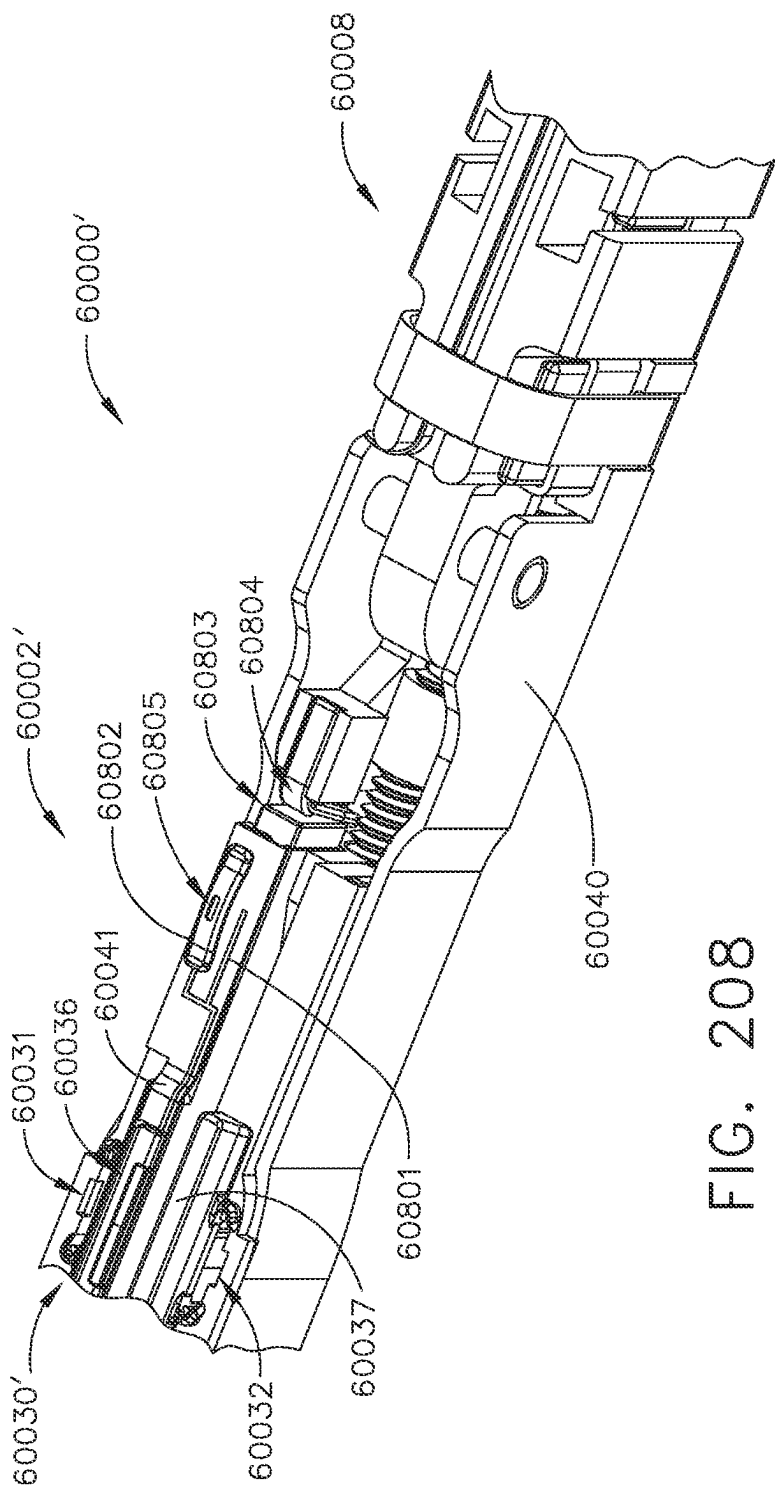


FIG. 208

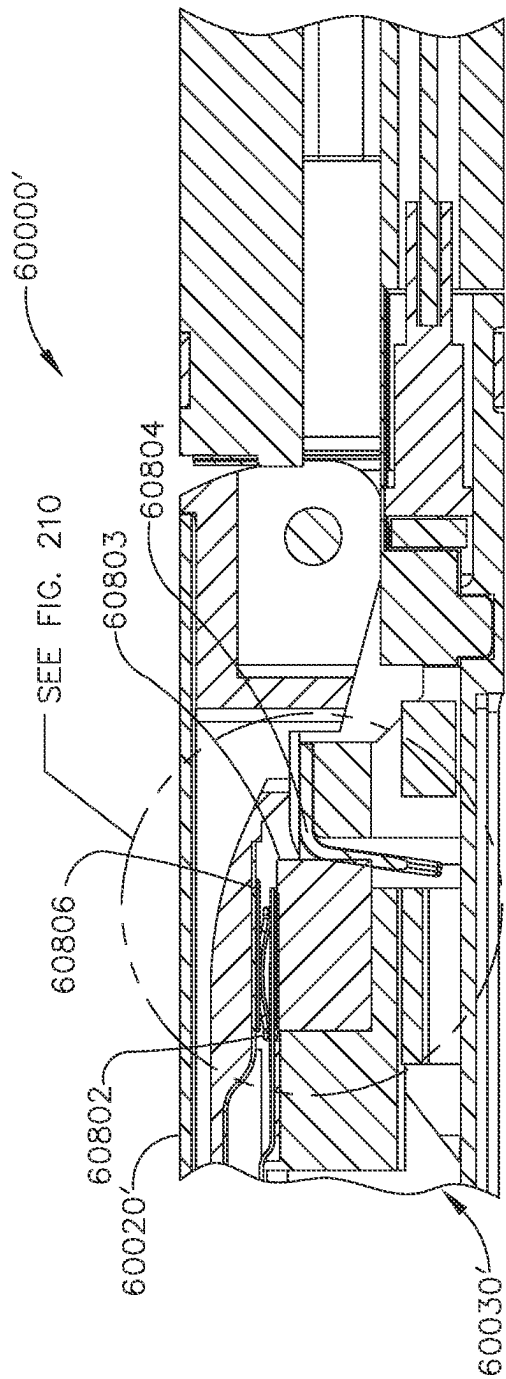


FIG. 209

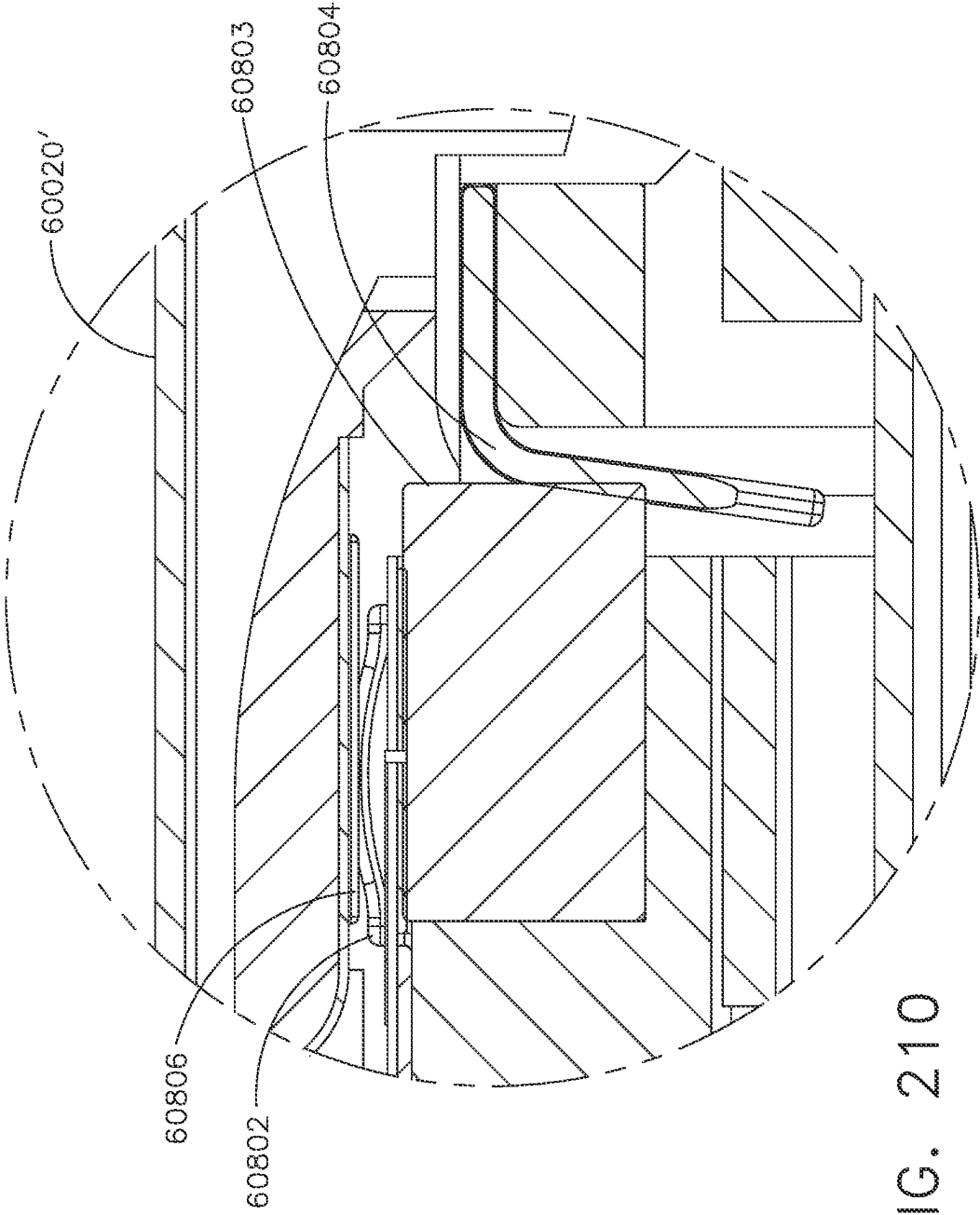


FIG. 210

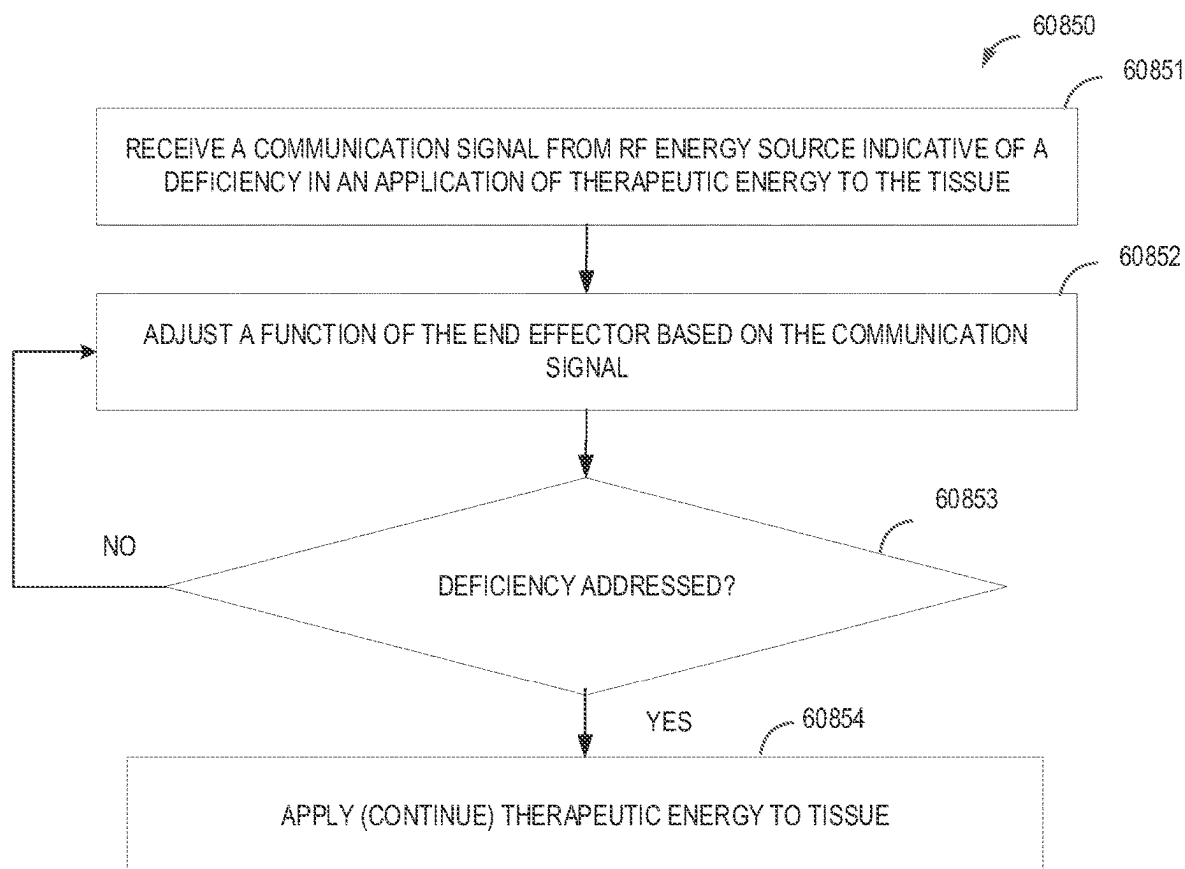


FIG. 211

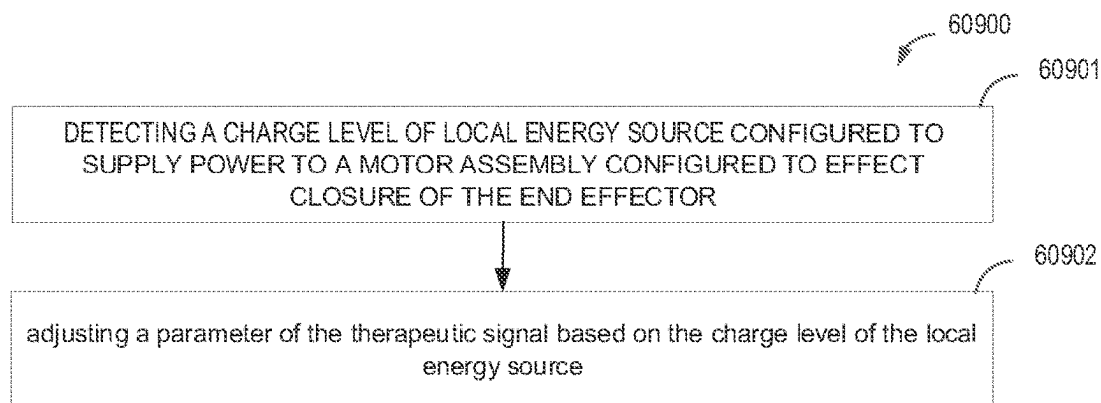


FIG. 212

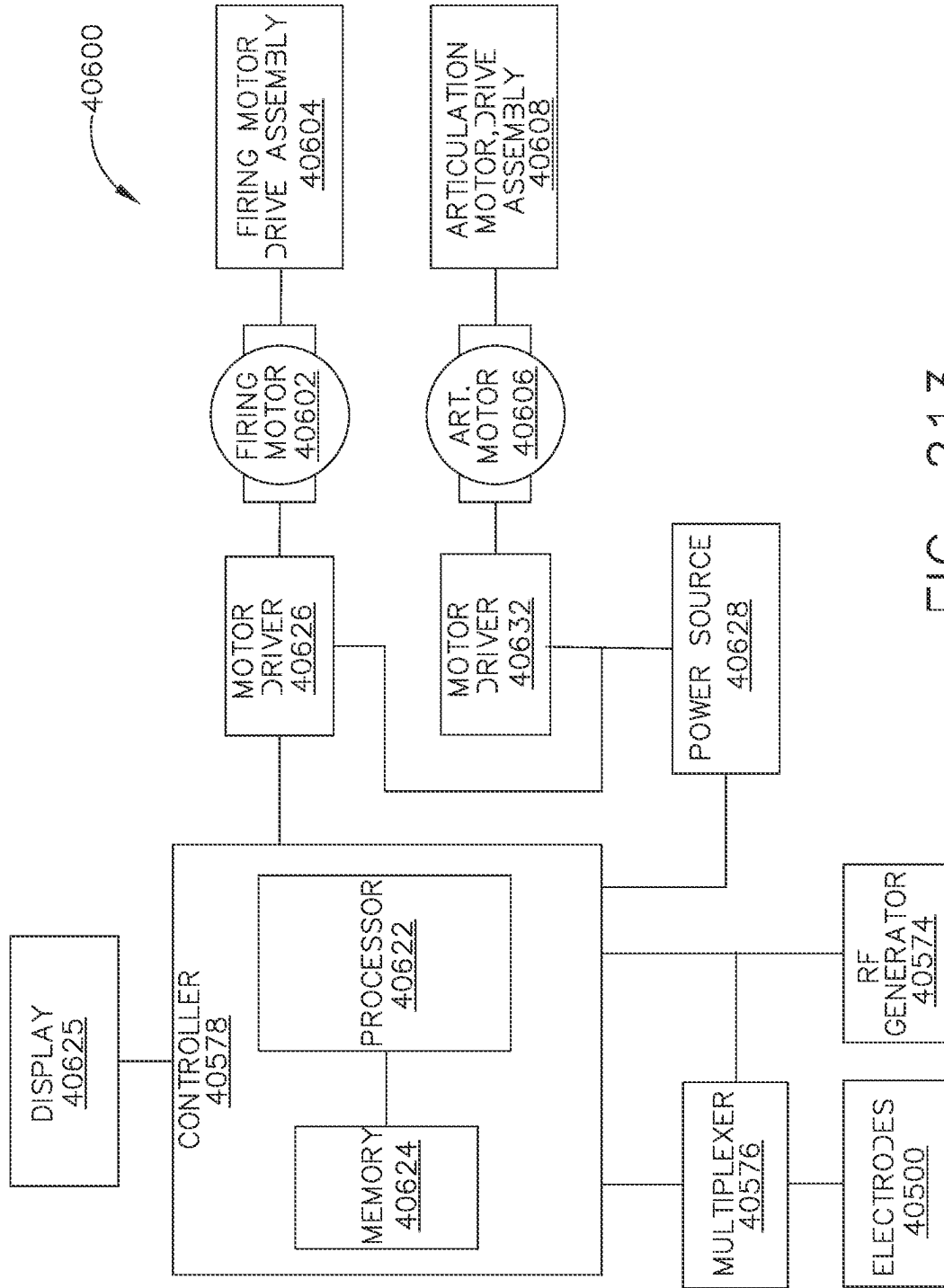


FIG. 213

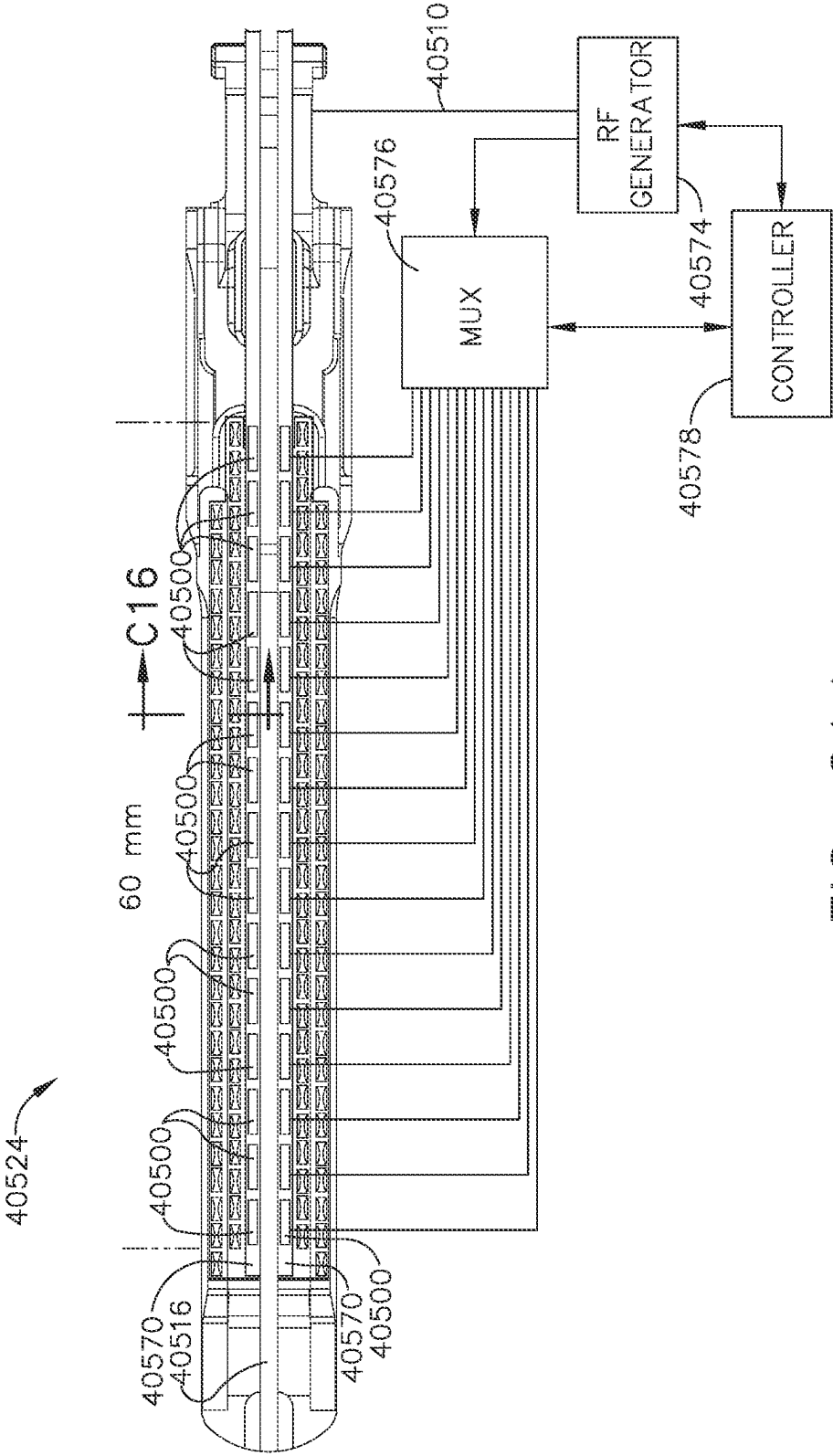


FIG. 214

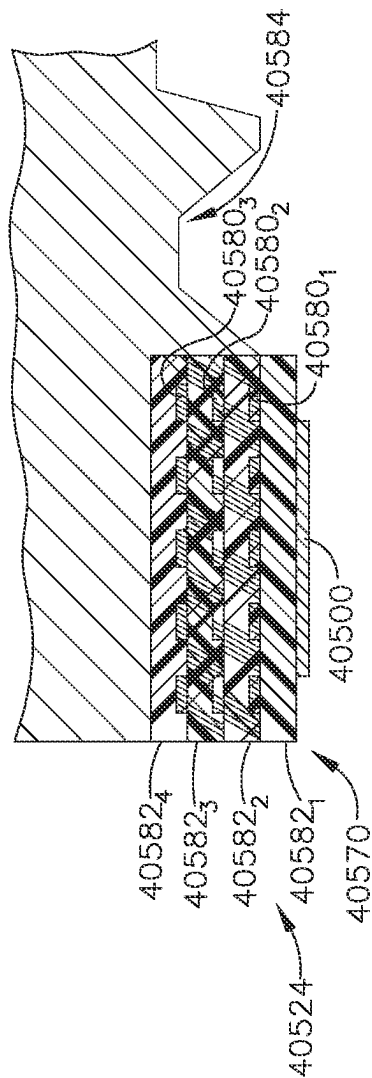


FIG. 215

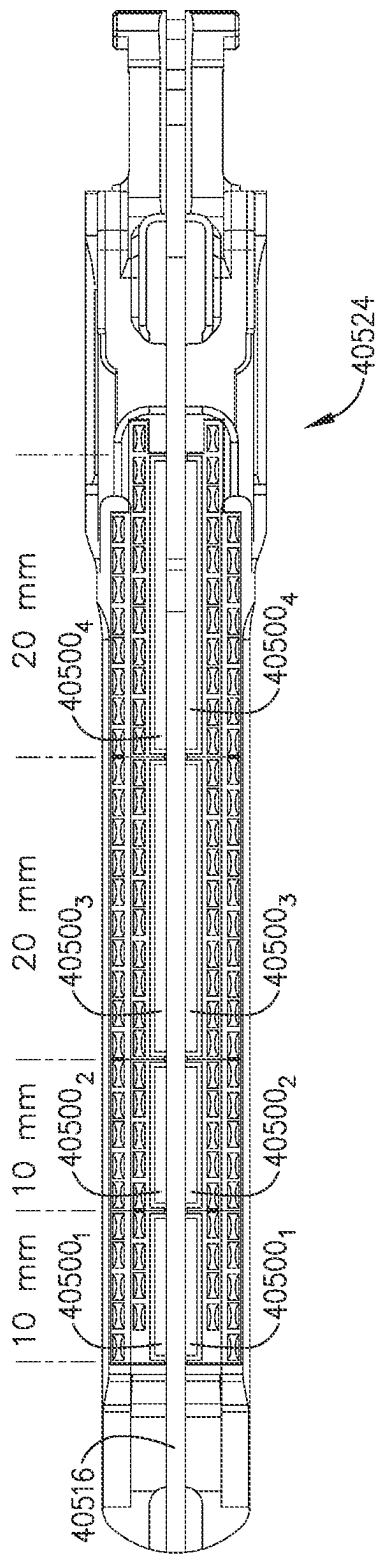


FIG. 216

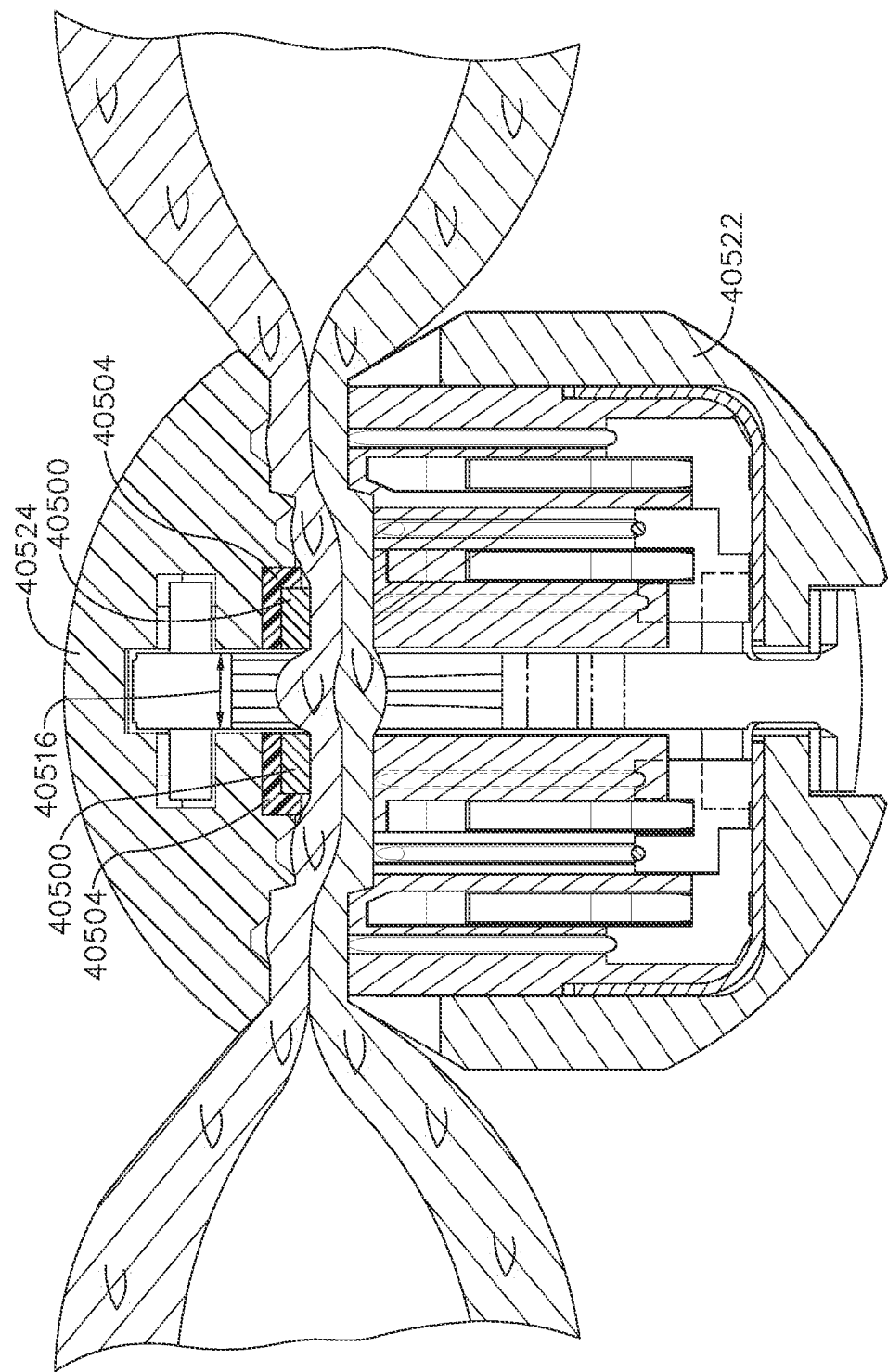
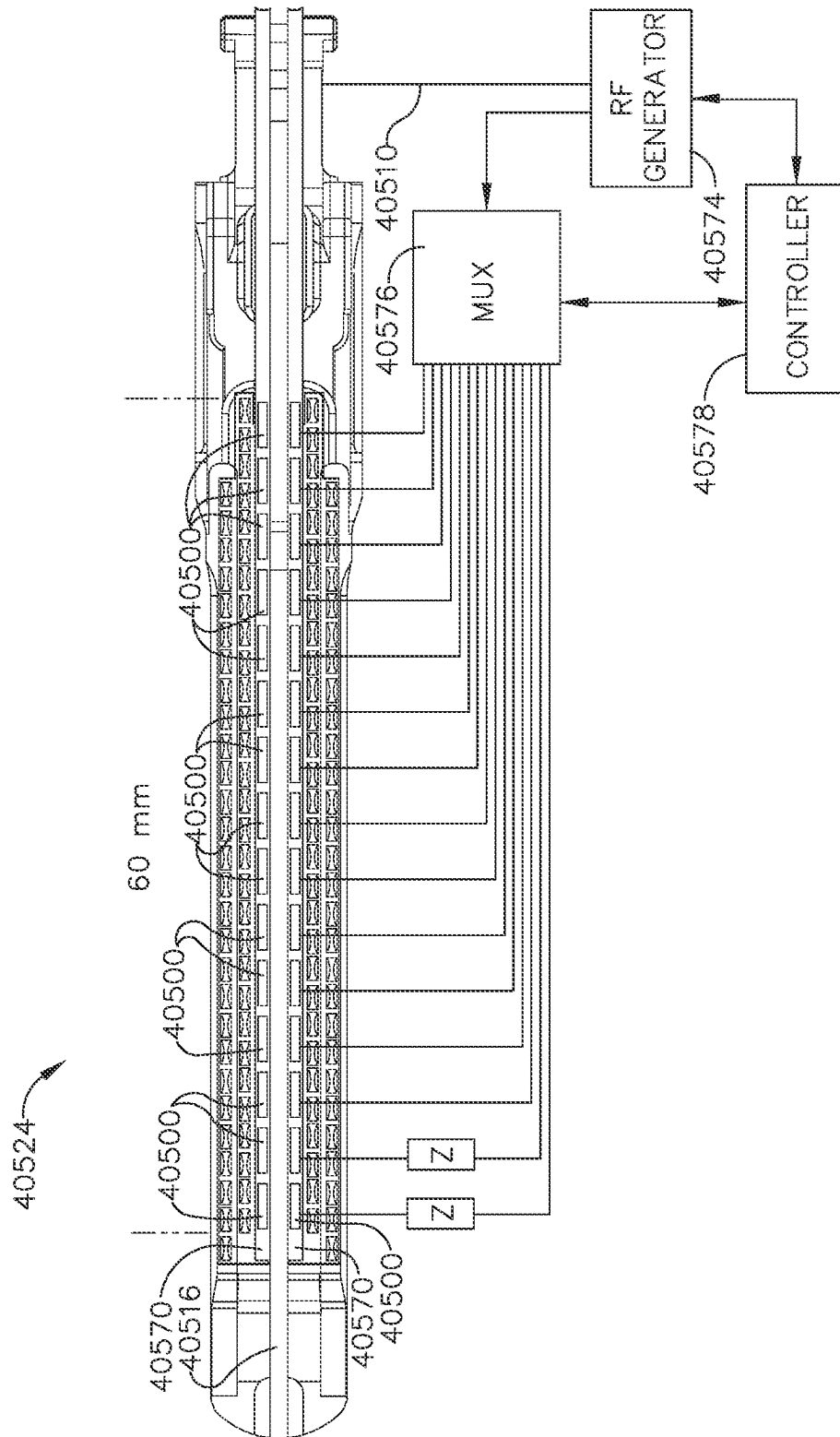


FIG. 217


$$\frac{\infty}{2} \cdot \frac{G}{F}$$

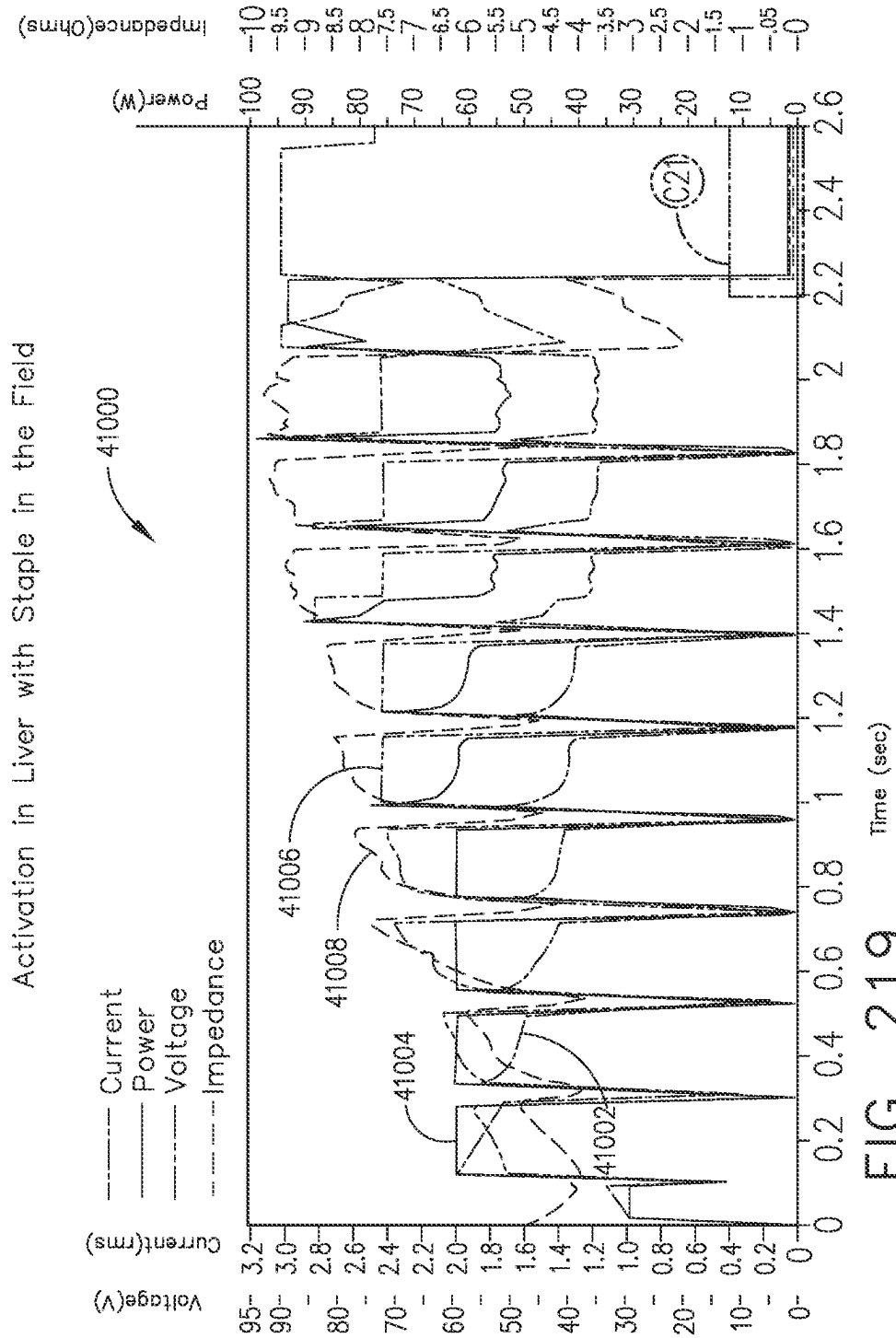
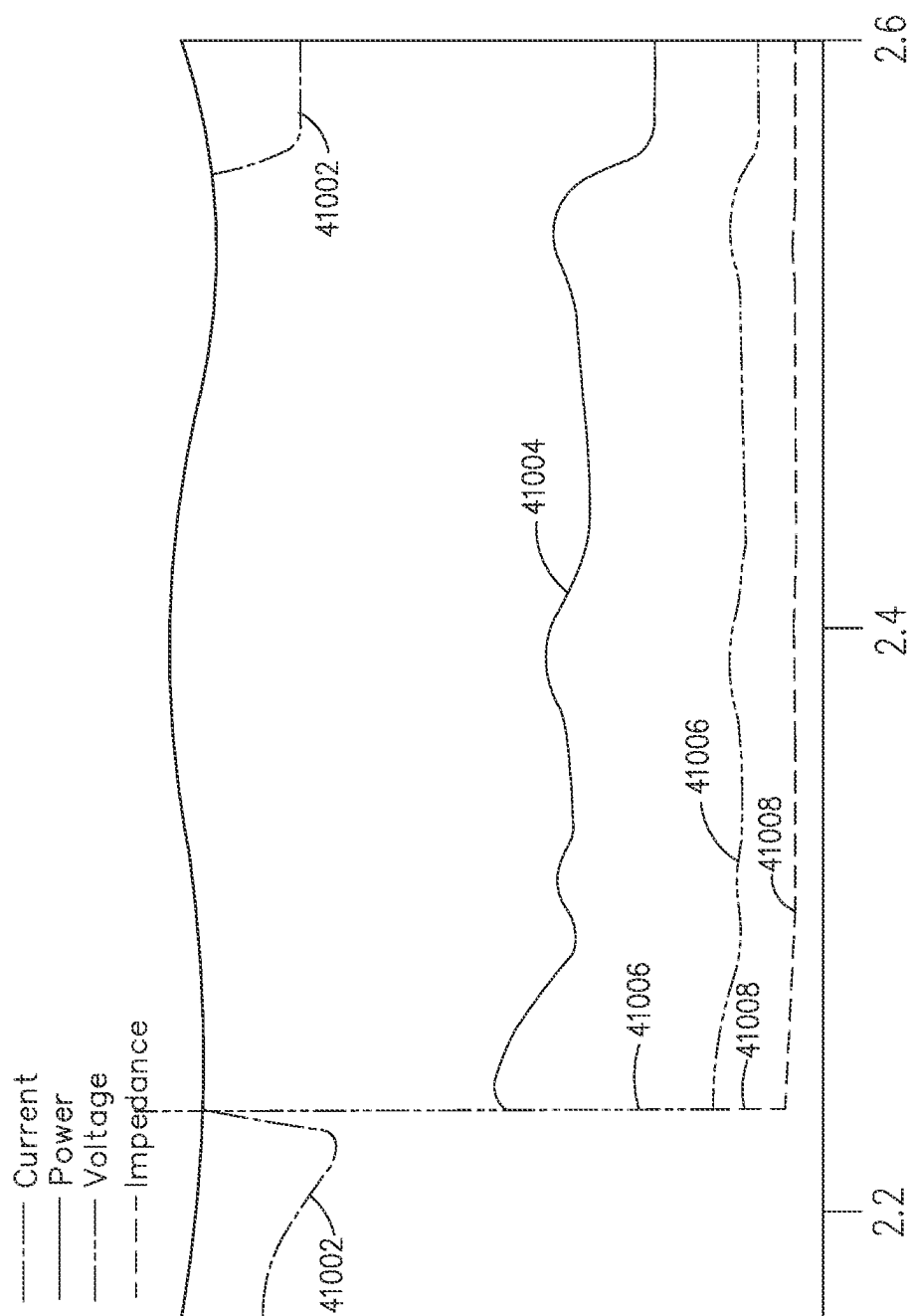


FIG. 219



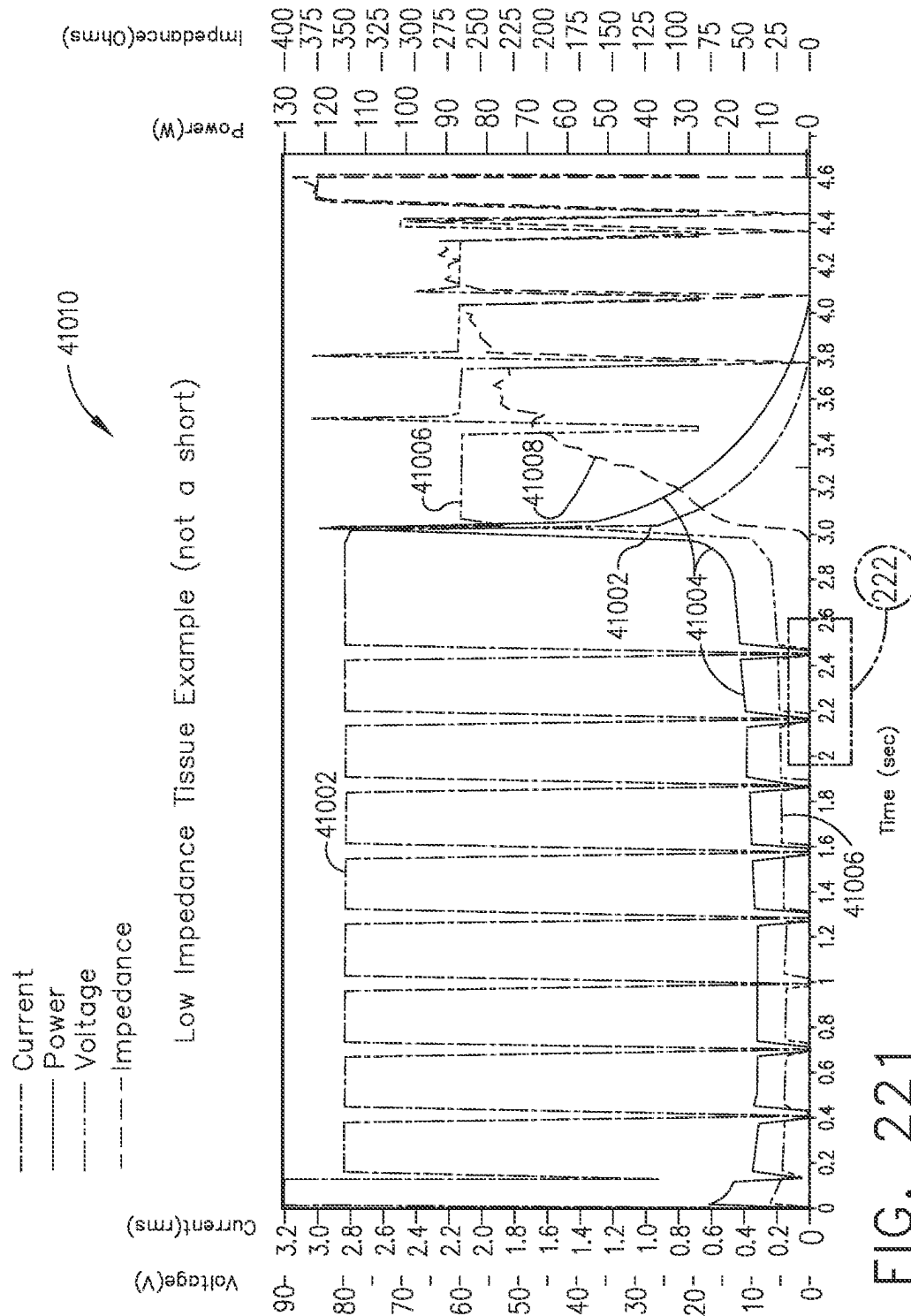


FIG. 221

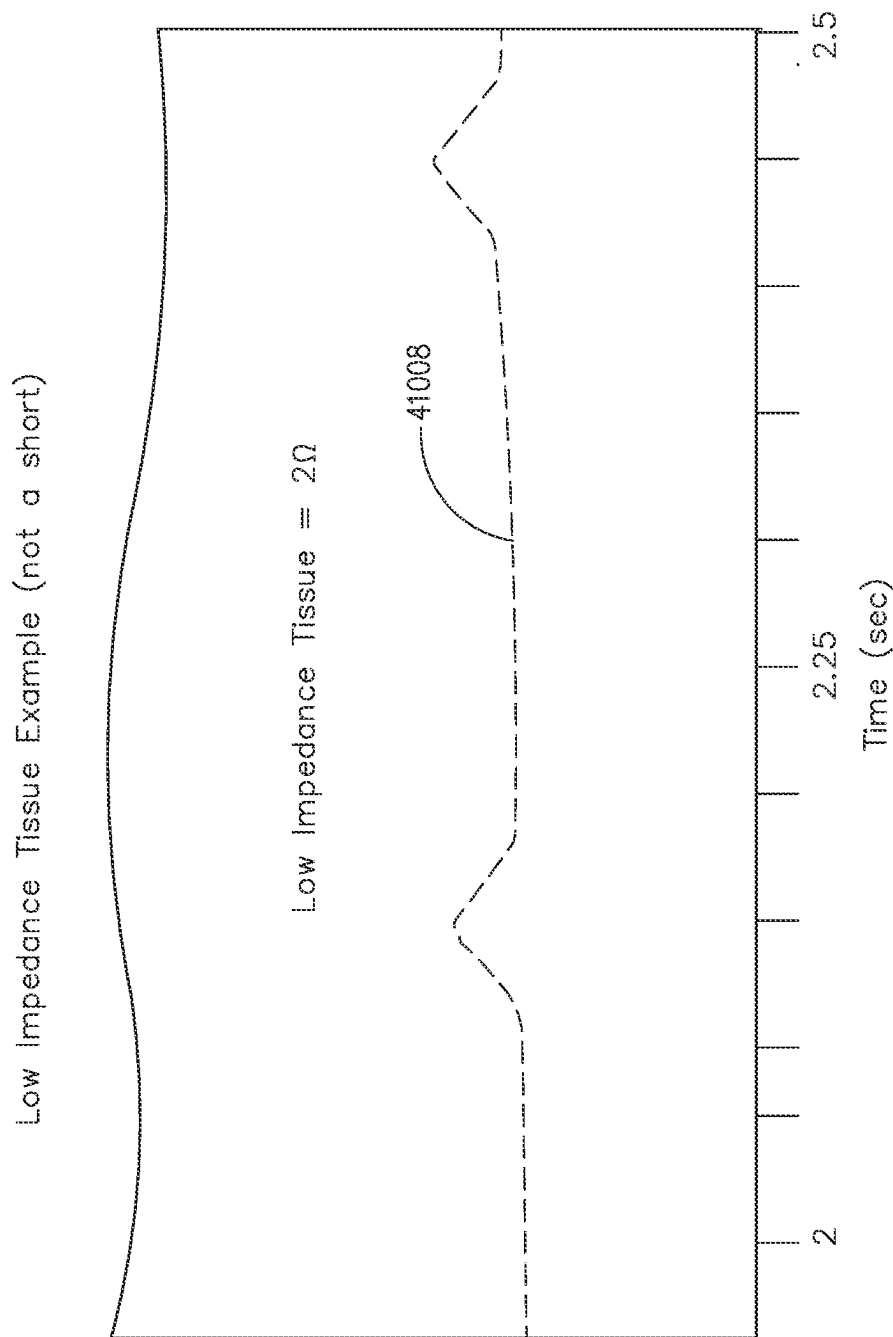
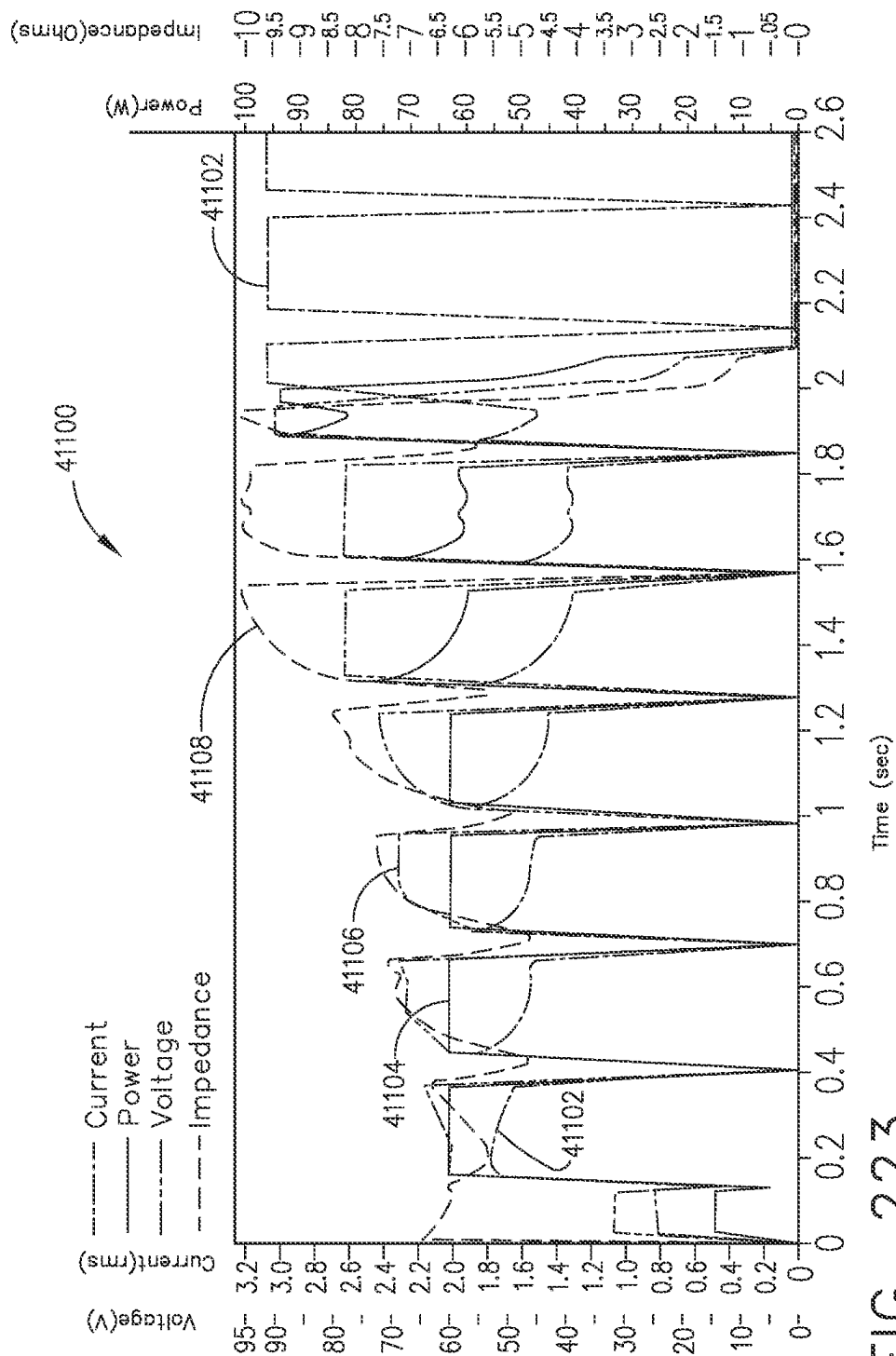


FIG. 222



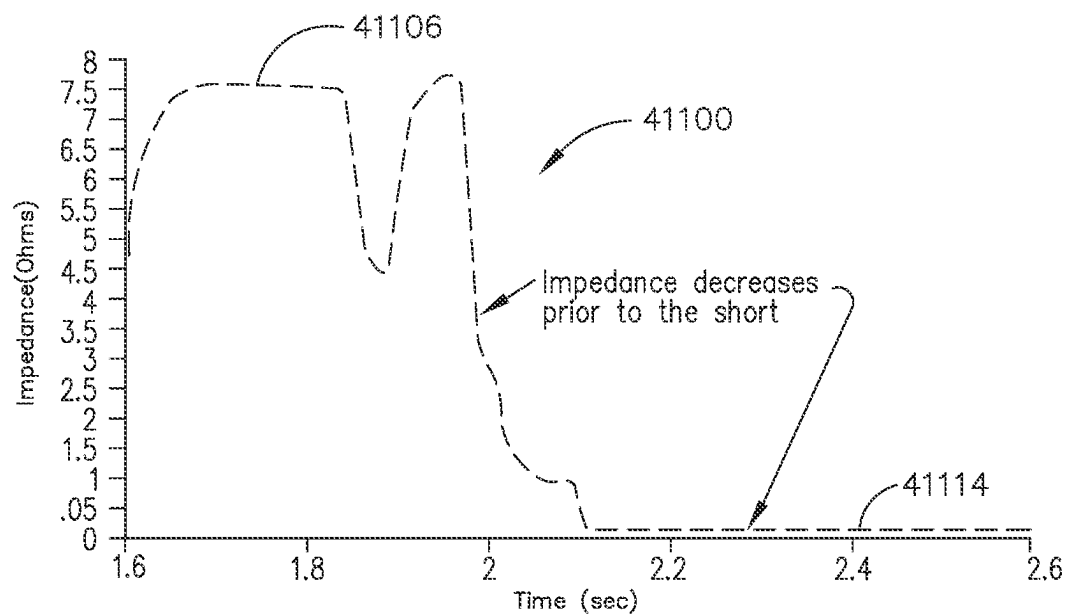


FIG. 224A

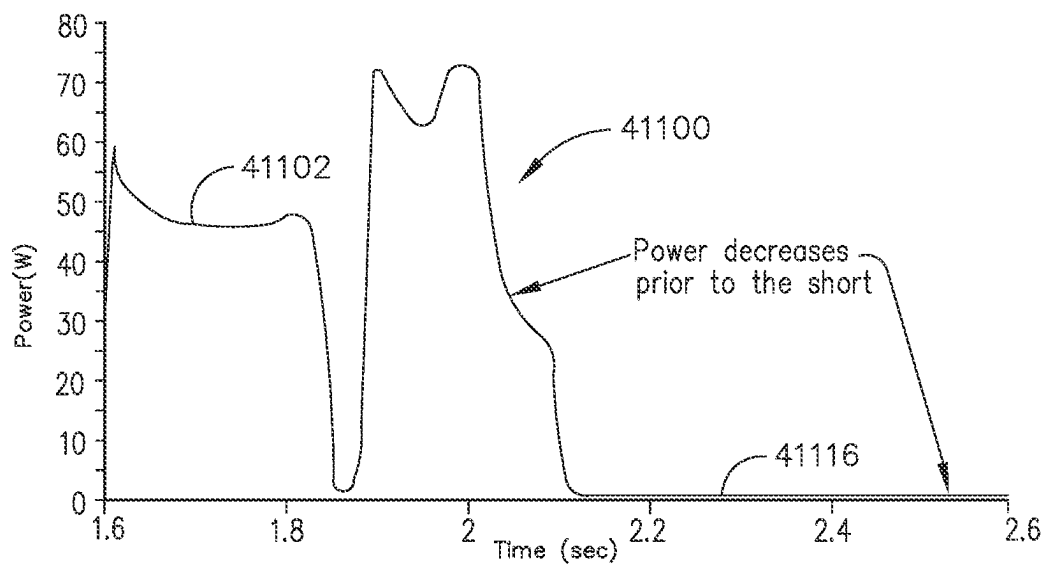


FIG. 224B

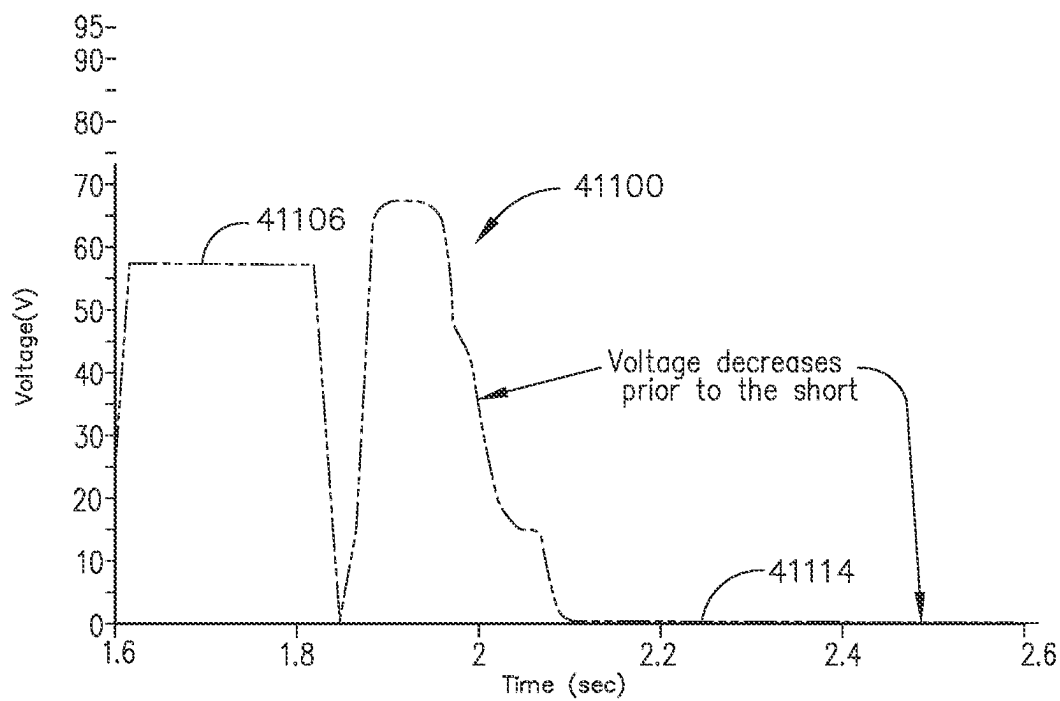


FIG. 224C

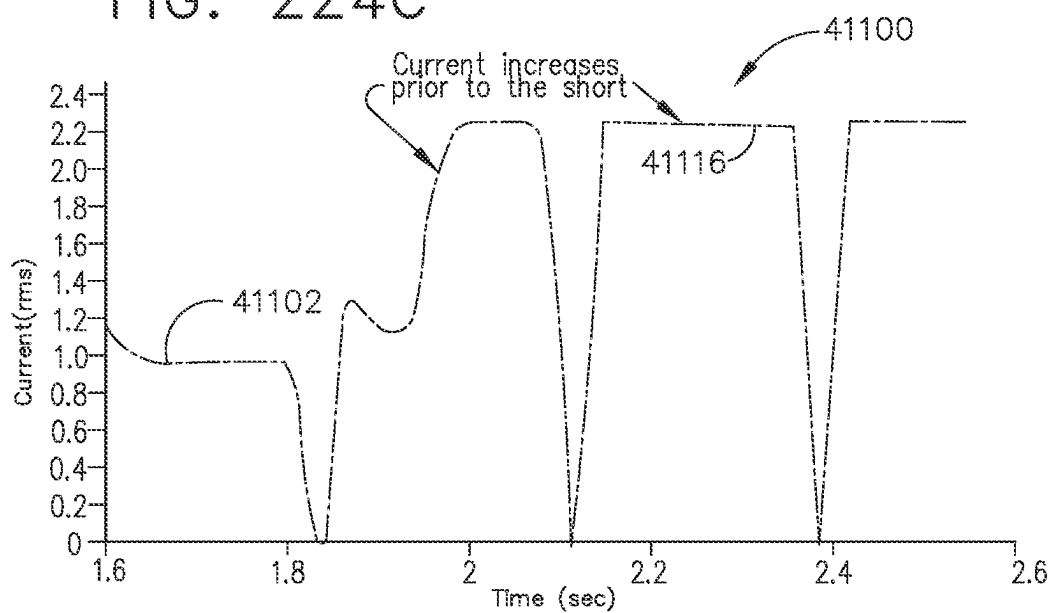


FIG. 224D

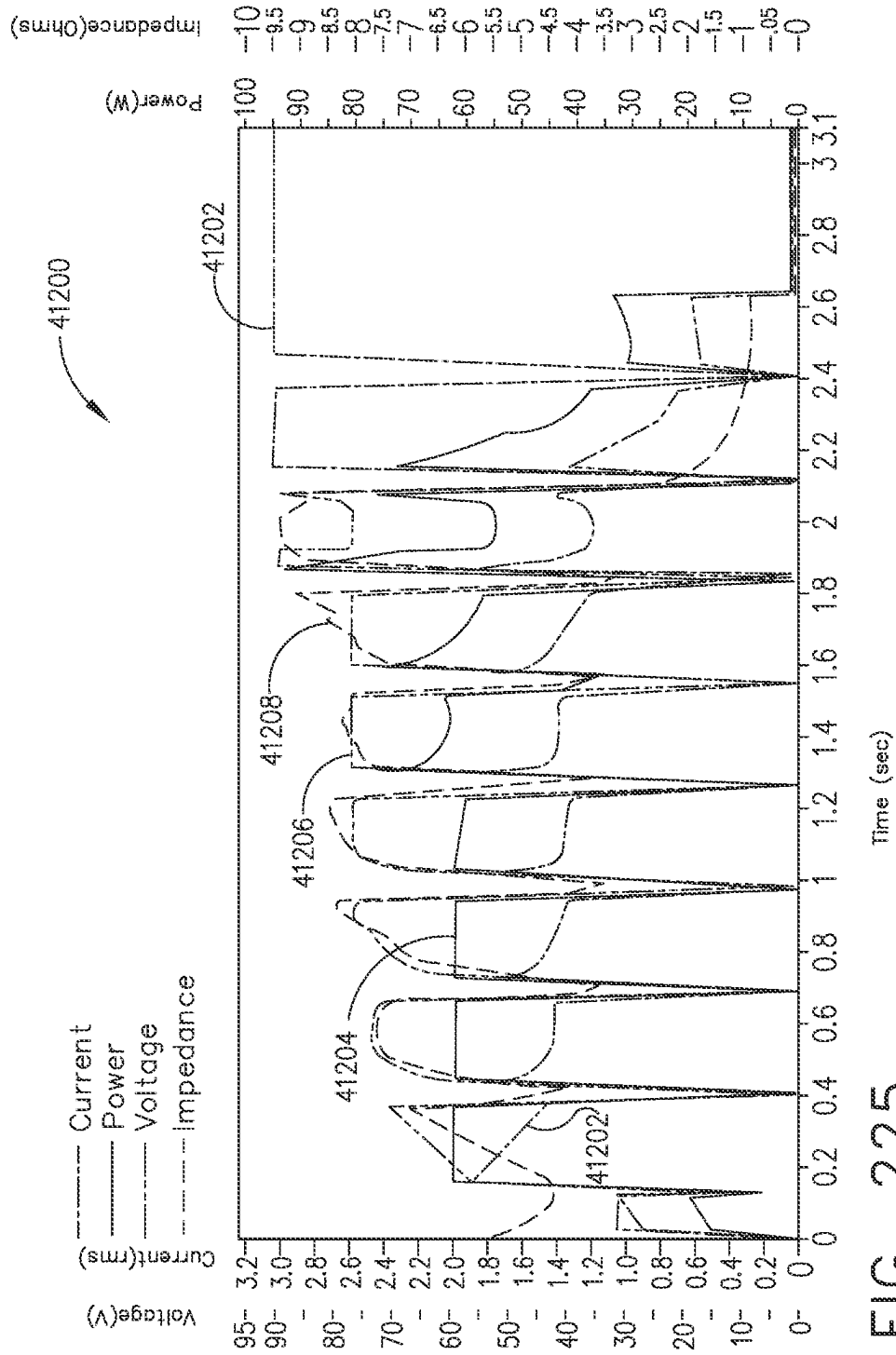


FIG. 225

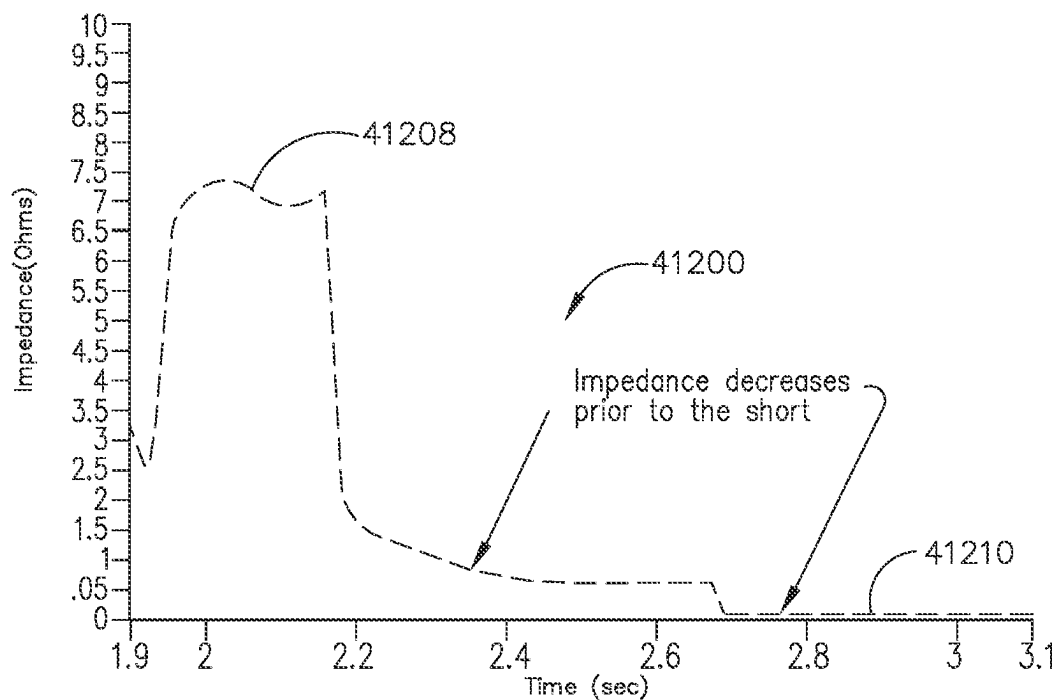


FIG. 226A

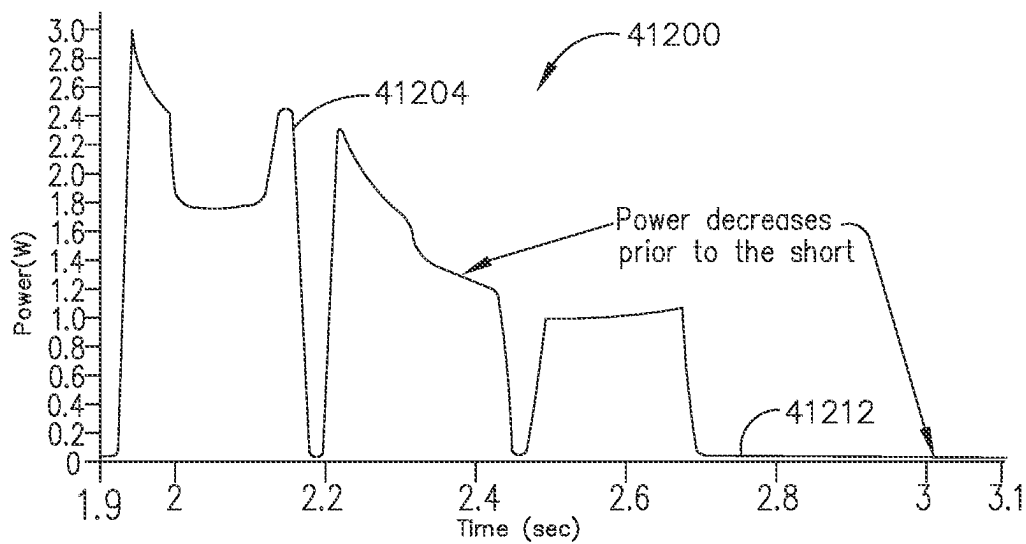


FIG. 226B

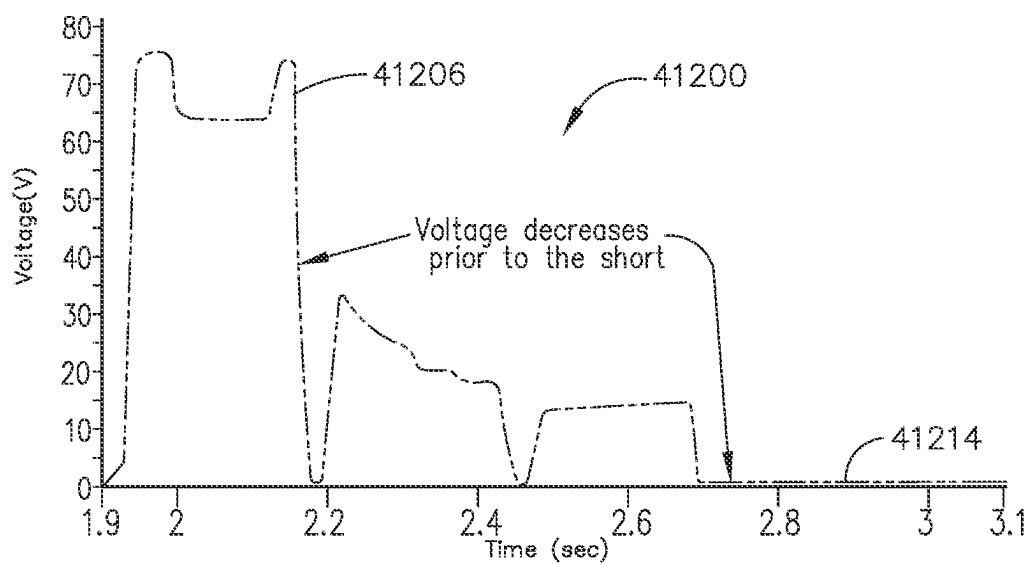


FIG. 226C

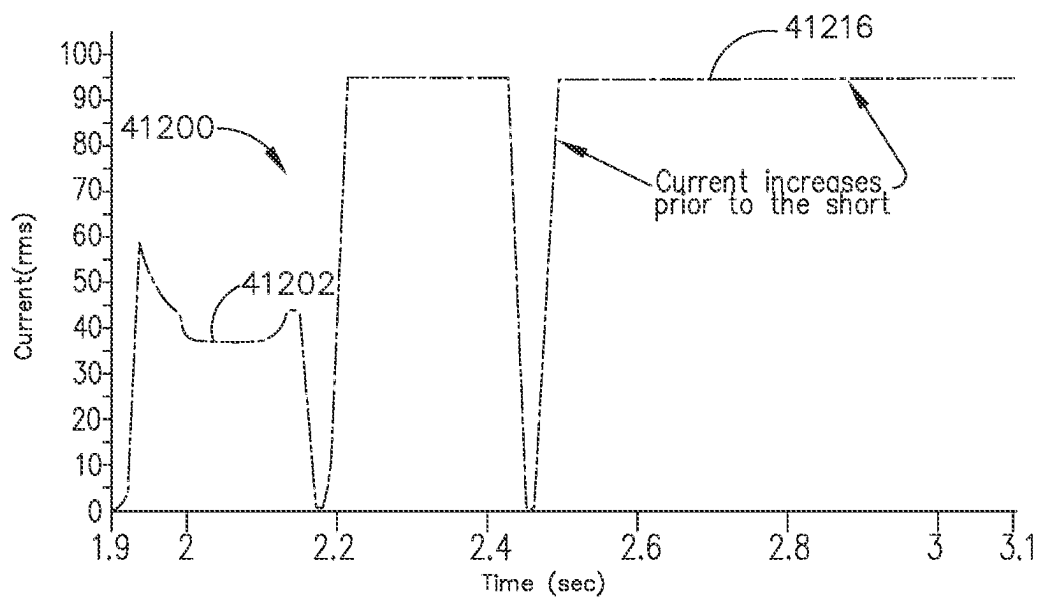


FIG. 226D

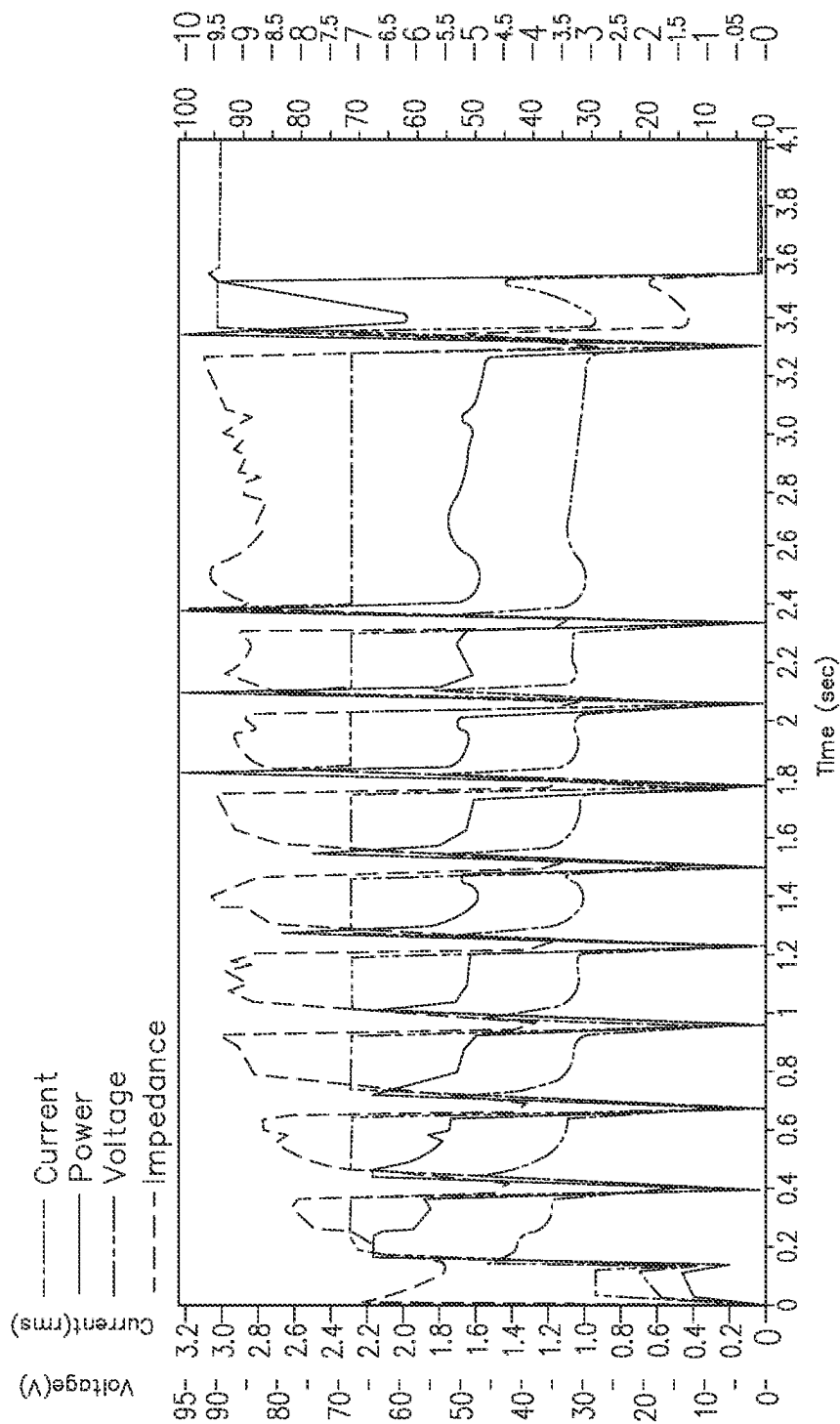


FIG. 227

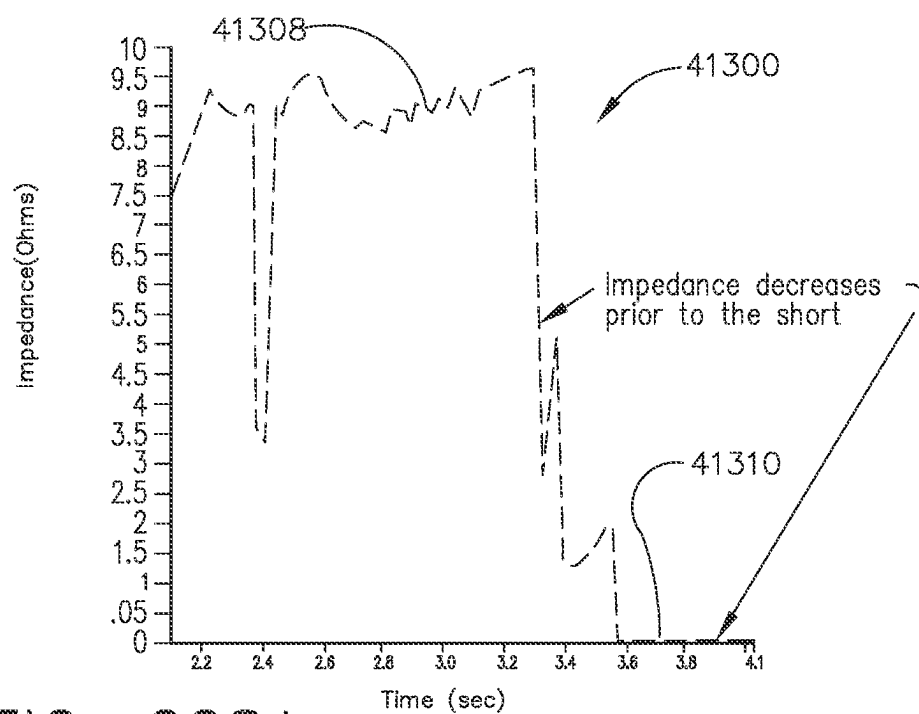


FIG. 228A

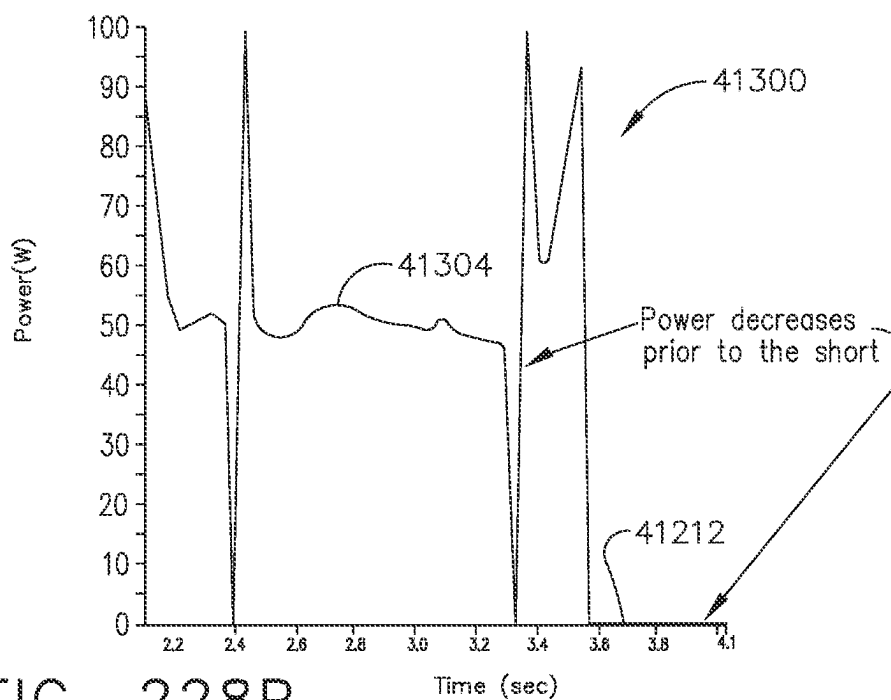


FIG. 228B

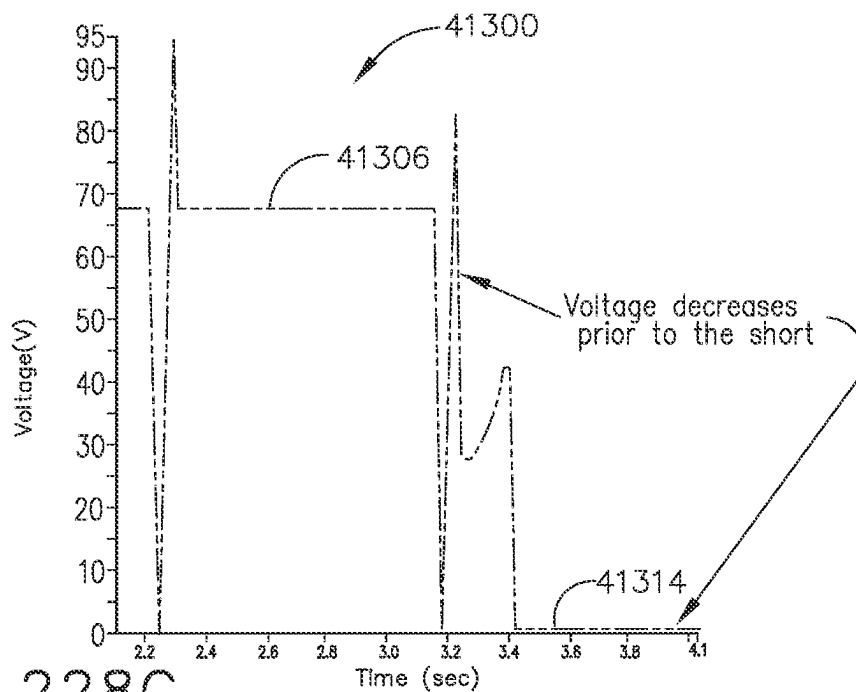


FIG. 228C

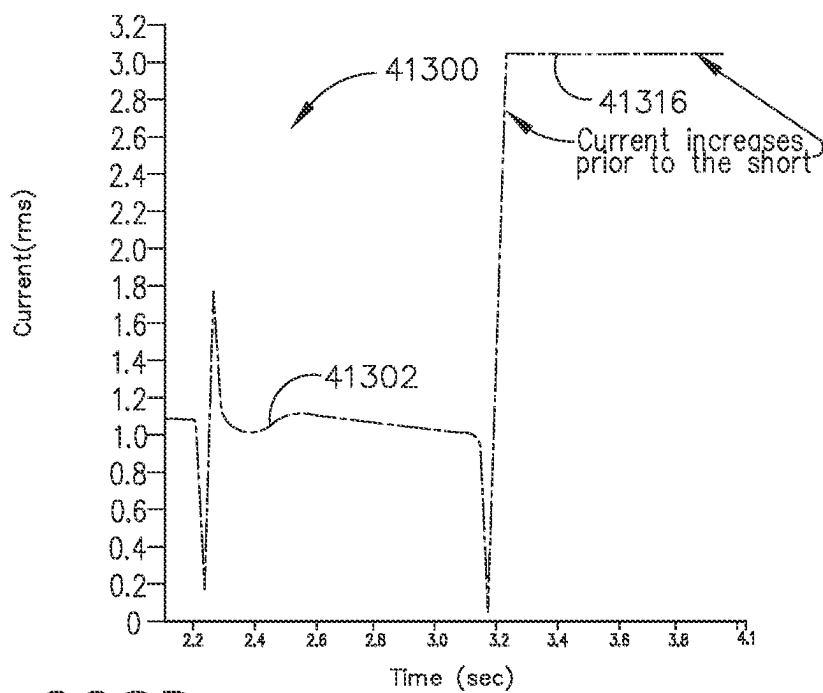


FIG. 228D

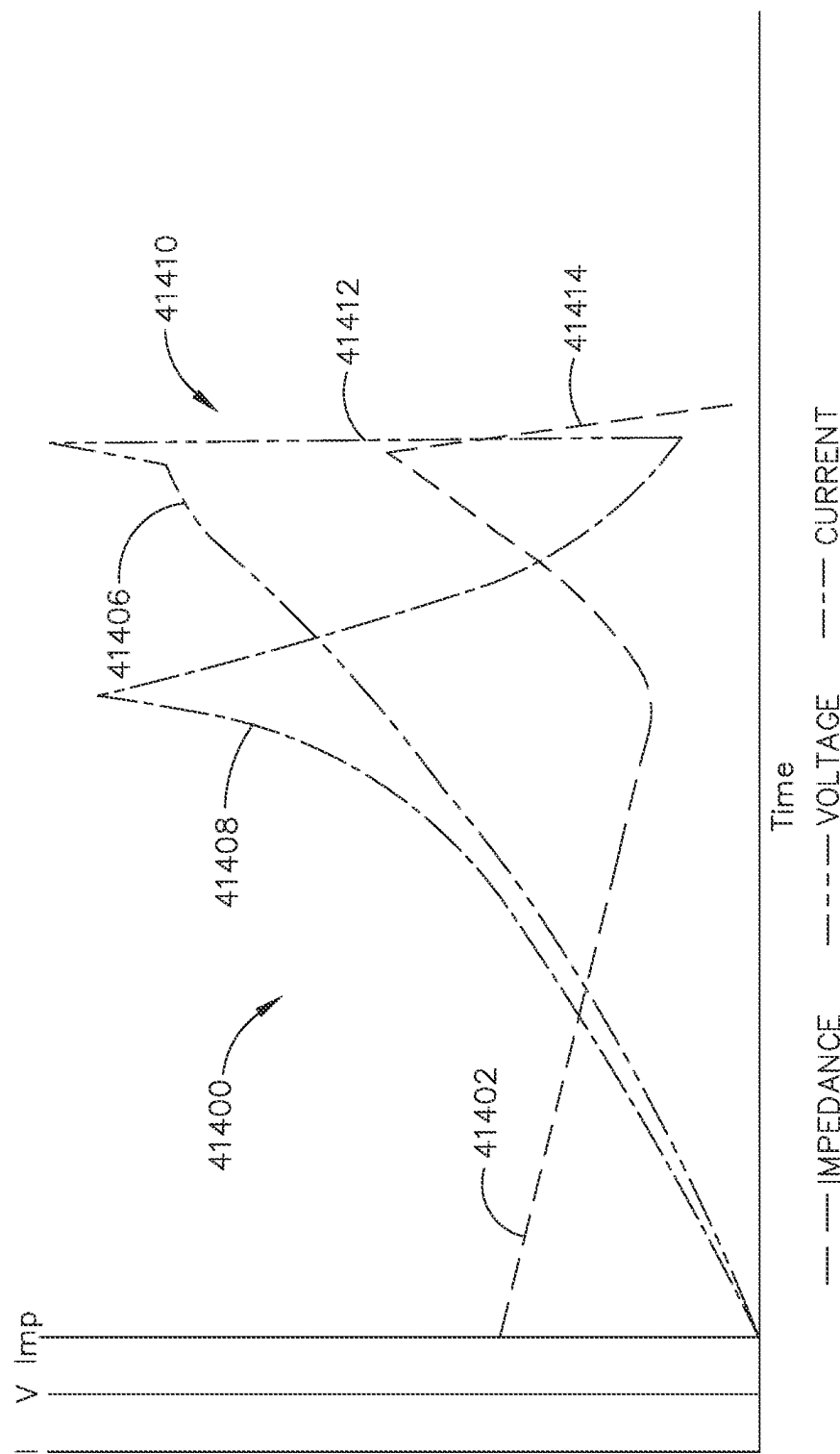


FIG. 229

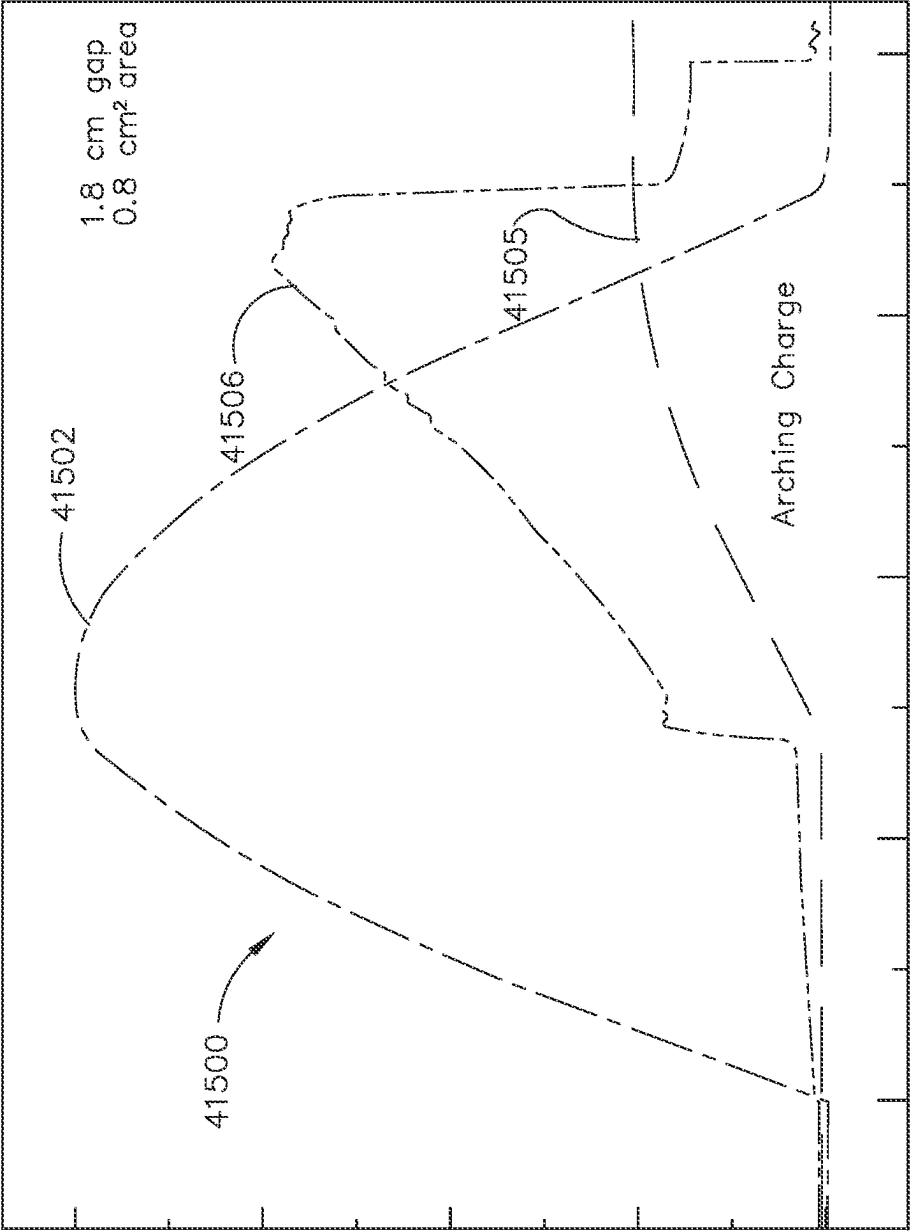


FIG. 230

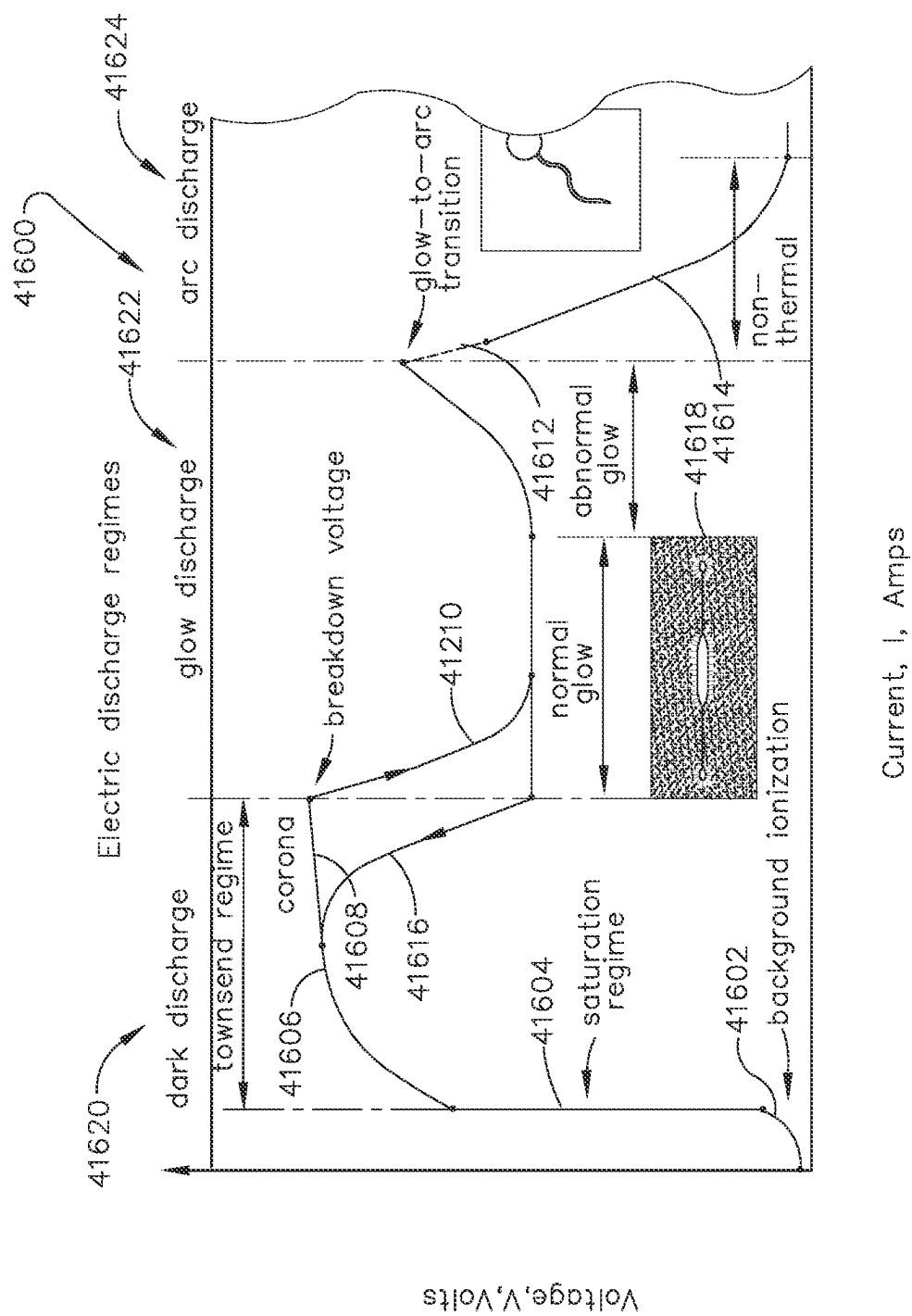


FIG. 231

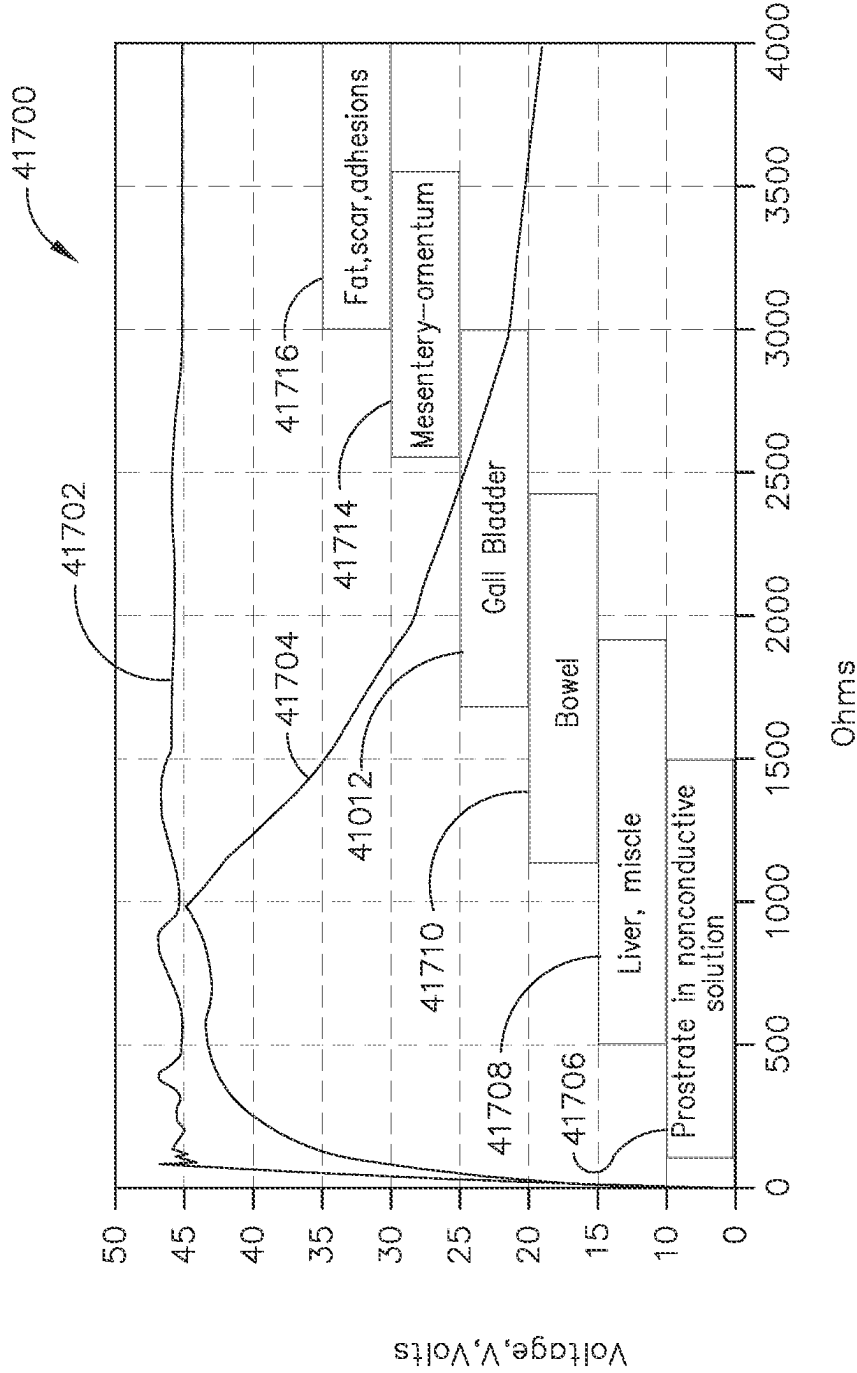


FIG. 232

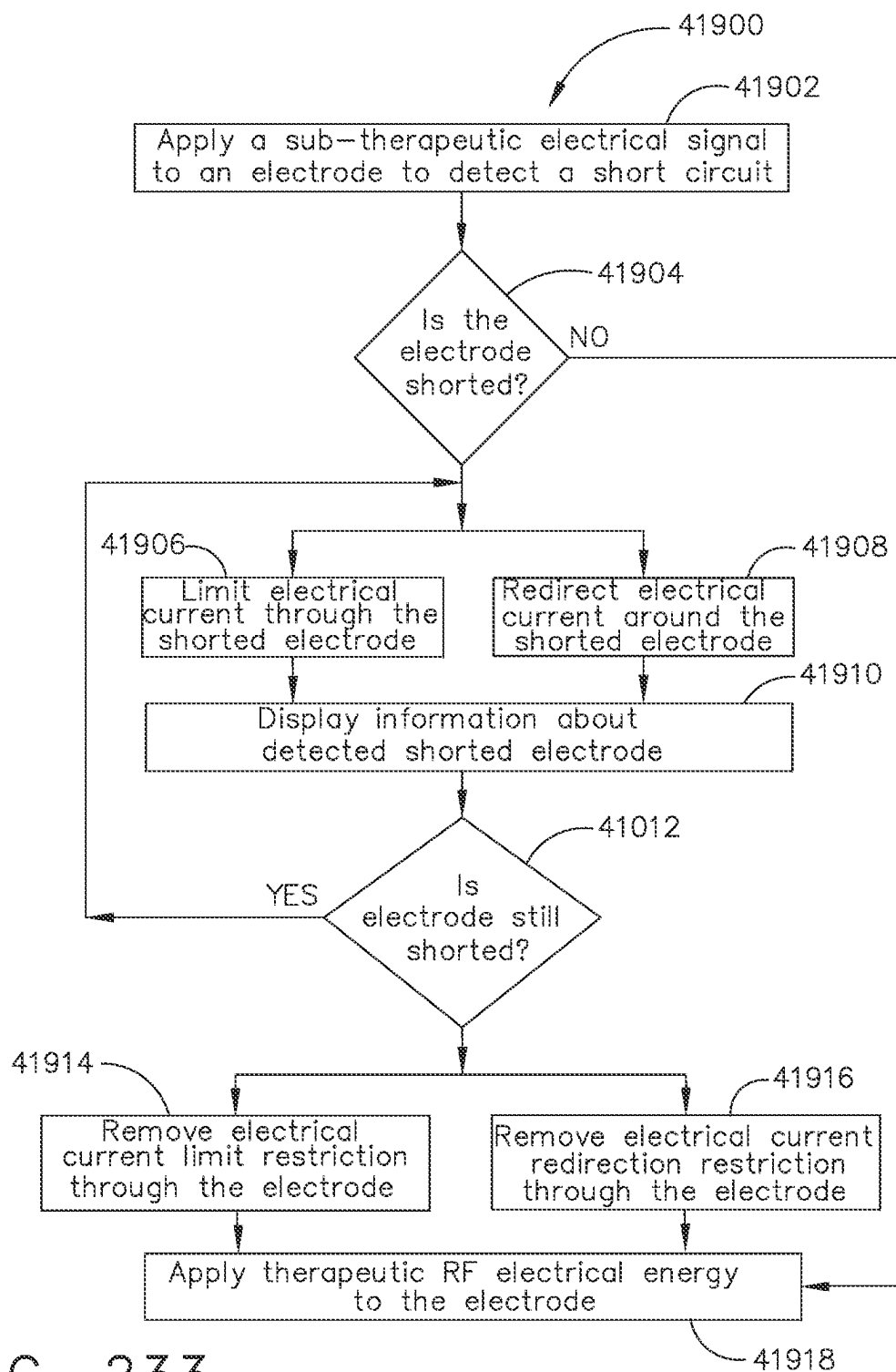


FIG. 233

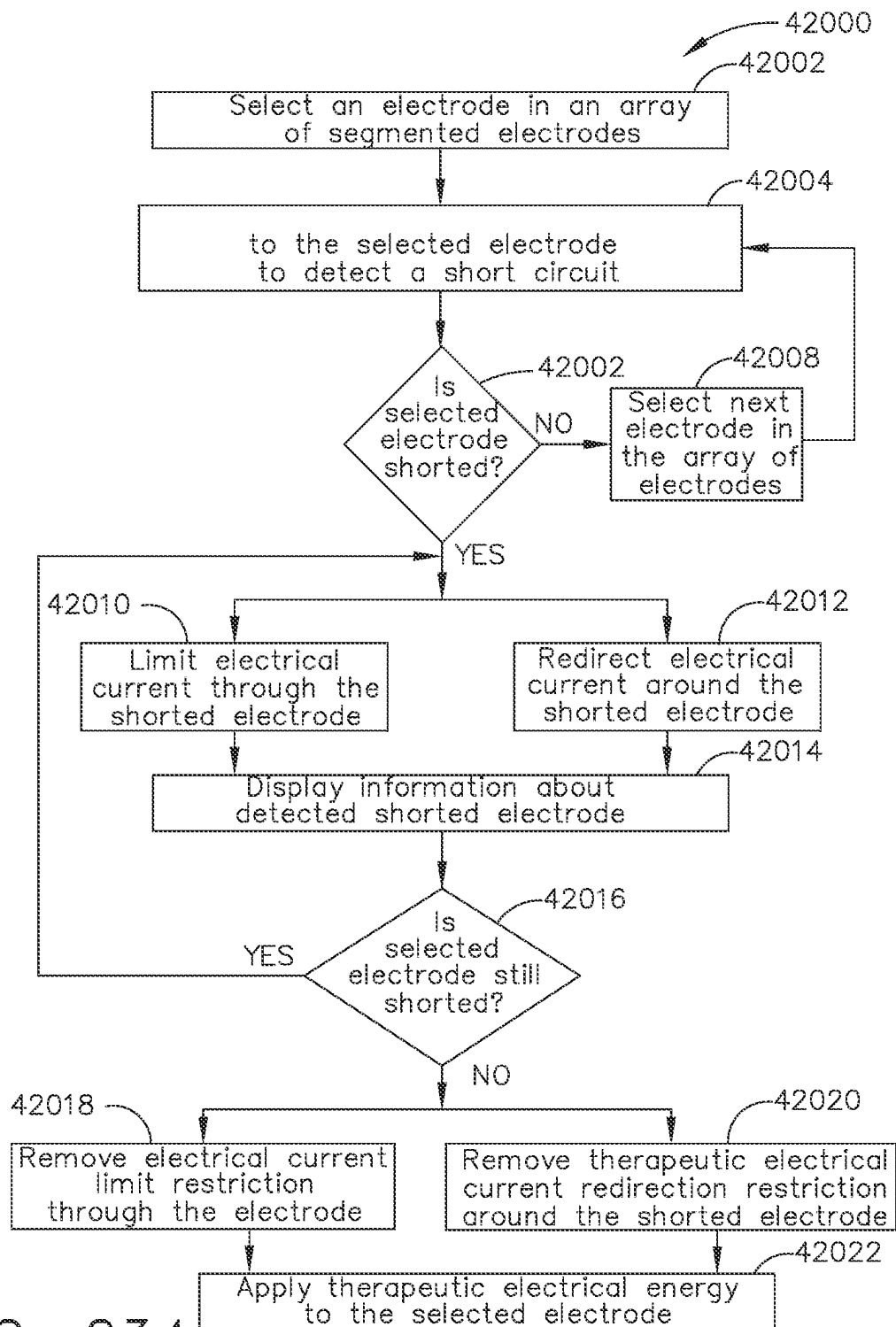


FIG. 234

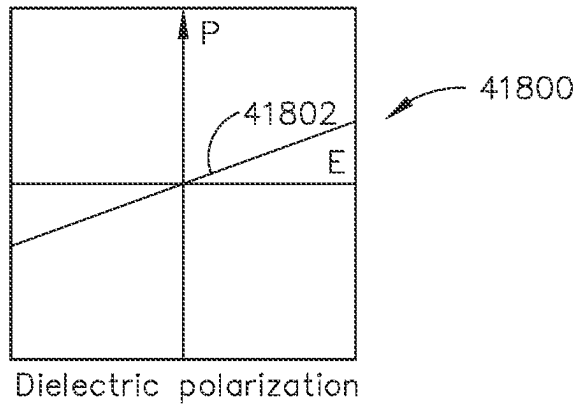


FIG. 235

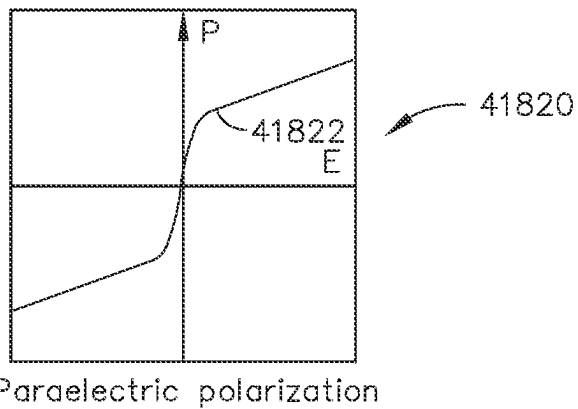


FIG. 236

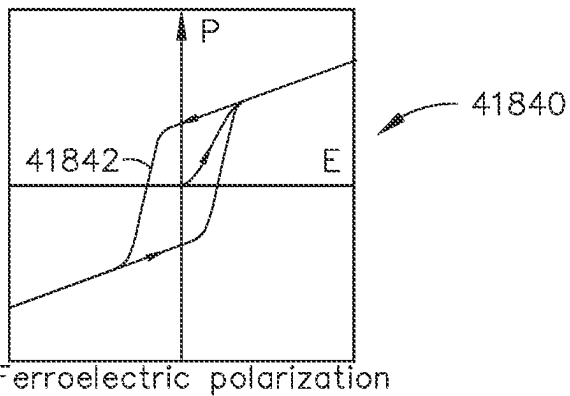


FIG. 237

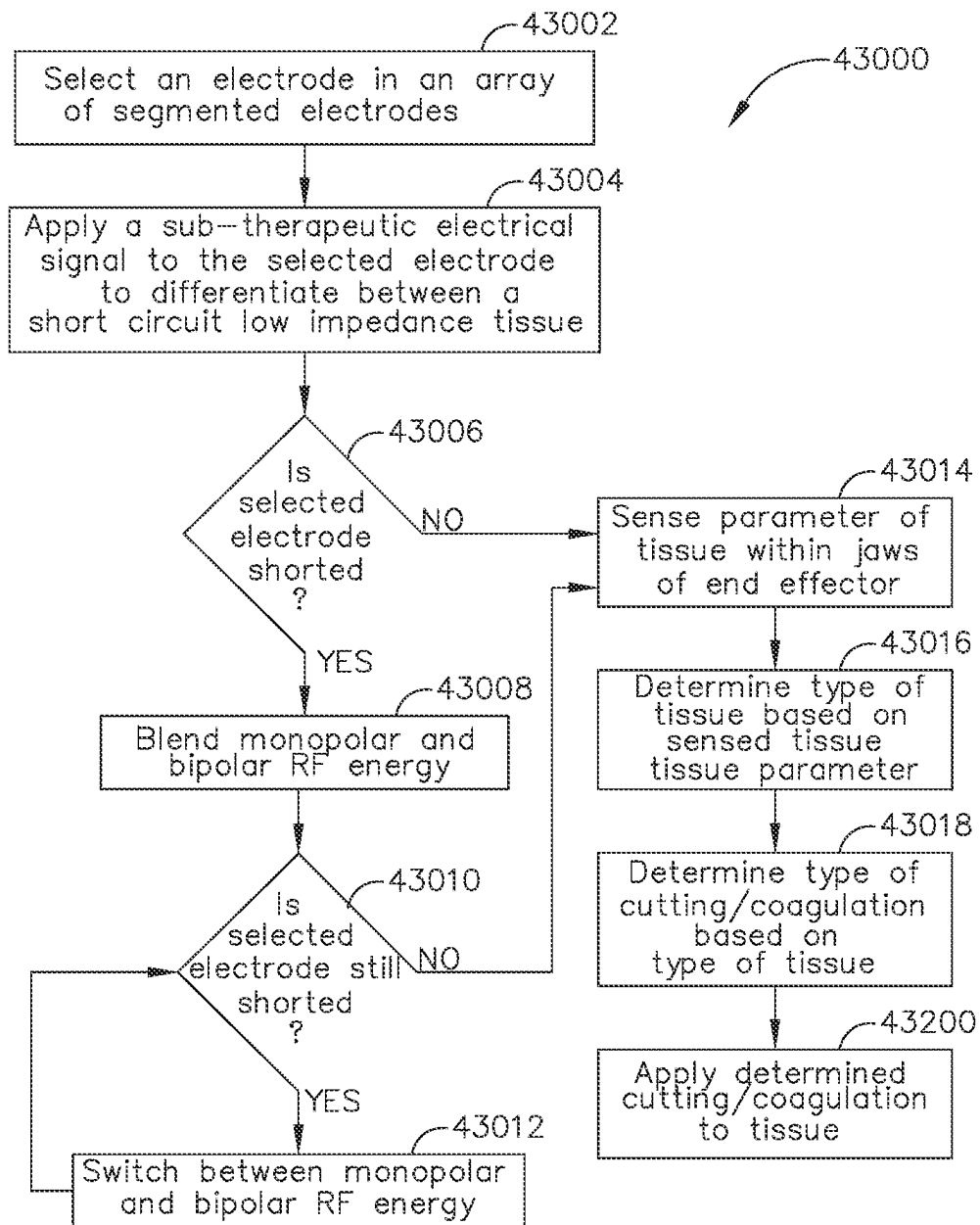


FIG. 238

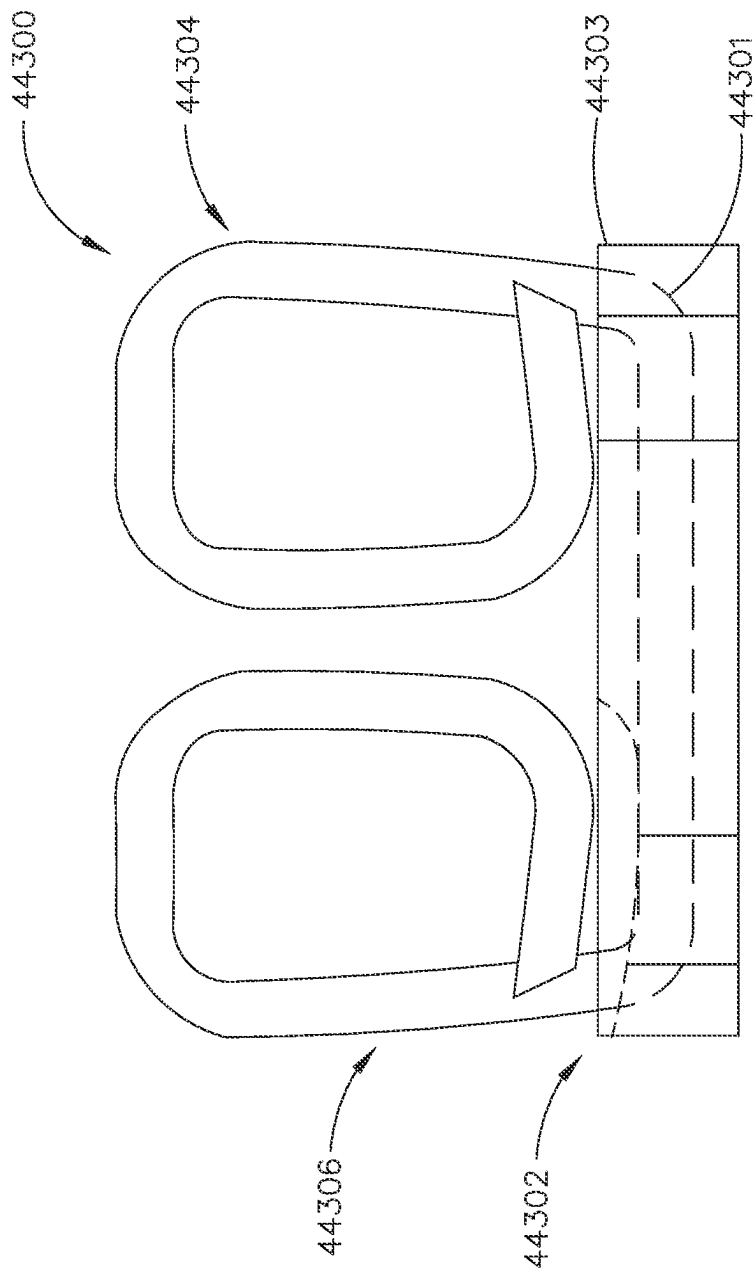


FIG. 239

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ELECTROSURGICAL ADAPTATION TECHNIQUES OF ENERGY MODALITY FOR COMBINATION ELECTROSURGICAL INSTRUMENTS BASED ON SHORTING OR TISSUE IMPEDANCE IRREGULARITY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of prior filed U.S. patent application Ser. No. 17/246,095 filed on Apr. 30, 2021 which is hereby incorporated by reference as set forth in full herein.

BACKGROUND

The present invention relates to surgical instruments and, in various arrangements, to surgical stapling and cutting instruments and staple cartridges for use therewith that are designed to staple and cut tissue.

BRIEF DESCRIPTION OF THE DRAWINGS

Various features of the embodiments described herein, together with advantages thereof, may be understood in accordance with the following description taken in conjunction with the accompanying drawings as follows:

FIG. 1 is a perspective view of surgical instrument in accordance with at least one embodiment;

FIG. 2 is a perspective view of a shaft of the surgical instrument of FIG. 1;

FIG. 3 is a perspective view of an end effector of the surgical instrument of FIG. 1;

FIG. 4 is a partial exploded view of the shaft of FIG. 2;

FIG. 5 is a partial exploded view of the end effector of FIG. 3;

FIG. 6 is an exploded view of the end effector of FIG. 3;

FIG. 7 is an elevational view of the end effector of FIG. 3 illustrating the end effector in an open configuration;

FIG. 8 is an elevational view of the end effector of FIG. 3 illustrating the end effector in a closed configuration;

FIG. 9 is a plan view of an articulation joint of the surgical instrument of FIG. 1 illustrating the end effector in an unarticulated position;

FIG. 10 is a plan view of an articulation joint of the surgical instrument of FIG. 1 illustrating the end effector in an articulated position;

FIG. 11 is a cross-sectional view of the end effector of FIG. 3 illustrated in a closed, unfired configuration;

FIG. 12 is a cross-sectional view of the end effector of FIG. 3 illustrated in a closed, fired configuration;

FIG. 12A is a cross-sectional view of the end effector of FIG. 3 illustrated in an open configuration;

FIG. 13 is a perspective view of a handle of the surgical instrument of FIG. 1 illustrated with some components removed;

FIG. 14 is a partial cross-sectional view of a surgical instrument in accordance with at least one embodiment;

FIG. 15 is a partial cross-sectional view of the surgical instrument of FIG. 14 illustrated in a partially-fired condition;

FIG. 16 is a cross-sectional view of an end effector of a surgical instrument in accordance with at least one embodiment;

FIG. 17 is a perspective view of an anvil of the end effector of FIG. 16;

FIG. 18 is an exploded view of the anvil of FIG. 17;

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FIG. 19 is a partial cross-sectional view of a closure drive and a firing drive in accordance with at least one embodiment;

FIG. 20 is another partial cross-sectional view of the closure drive and the firing drive of FIG. 19 taken at a different, or distal, location along the longitudinal length thereof;

FIG. 21 is a cross-sectional view of an end effector of a surgical instrument in accordance with at least one embodiment;

FIG. 22 is a detail view of a staple cavity of a staple cartridge including a staple positioned in the staple cavity in accordance with at least one embodiment;

FIG. 23 is a partial cross-sectional view of the staple cartridge of FIG. 22 illustrating the staple in an unfired position;

FIG. 24 is a partial cross-sectional view of the staple cartridge of FIG. 22 illustrating the staple in a partially-fired position;

FIG. 25 is a detail view of a staple cavity of a staple cartridge including a staple positioned in the staple cavity in accordance with at least one embodiment;

FIG. 26 is a partial cross-sectional view of the staple cartridge of FIG. 25 illustrating the staple in an unfired position;

FIG. 27 is a partial cross-sectional view of the staple cartridge of FIG. 25 illustrating the staple in a partially-fired position;

FIG. 28 is a partial cross-sectional view of a staple cartridge in accordance with at least one embodiment illustrating a staple positioned in a staple cavity in an unfired position;

FIG. 29 is a partial cross-sectional view of the staple cartridge of FIG. 28 illustrating the staple in a partially-fired position;

FIG. 30 is a detail view of a staple cavity of a staple cartridge including a staple positioned in the staple cavity in accordance with at least one embodiment;

FIG. 31 is a partial cross-sectional view of the staple cartridge of FIG. 30 illustrating the staple in an unfired position;

FIG. 32 is a partial cross-sectional view of the staple cartridge of FIG. 30 illustrating the staple in a partially-fired position;

FIG. 33 is a perspective view of a surgical staple in accordance with at least one embodiment;

FIG. 34 is a perspective view of a staple cartridge in accordance with at least one embodiment;

FIG. 35 is a bottom perspective view of the staple cartridge of FIG. 34;

FIG. 36 is an end view of the staple cartridge of FIG. 34;

FIG. 37 is a partial cross-sectional view of the staple cartridge of FIG. 34 illustrated in a partially-fired configuration;

FIG. 38 is a cross-sectional perspective view of the staple cartridge of FIG. 34;

FIG. 39 is a cross-sectional perspective view of a staple cartridge in accordance with at least one embodiment;

FIG. 40 is a cross-sectional elevational view of the staple cartridge of FIG. 39 illustrated in an unfired configuration;

FIG. 41 is a cross-sectional elevational view of the staple cartridge of FIG. 39 illustrated in a fired configuration;

FIG. 42 is a partial perspective view of a staple cartridge in accordance with at least one embodiment;

FIG. 43 is a perspective view of a staple driver of the staple cartridge of FIG. 42;

FIG. 44 is a plan view of the staple driver of FIG. 43;

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FIG. 45 is a partial bottom view of a staple cartridge in accordance with at least one embodiment;

FIG. 46 is a top view of a staple driver of the staple cartridge of FIG. 45;

FIG. 47 is a perspective view of the staple driver of FIG. 46;

FIG. 48 is an elevational view of the staple driver of FIG. 46;

FIG. 48A is a partial cross-sectional perspective view of an end effector including the staple cartridge of FIG. 45;

FIG. 48B is a partial cross-sectional perspective view of the end effector of FIG. 48A;

FIG. 49 is a partial cross-sectional perspective view of a staple cartridge in accordance with at least one embodiment;

FIG. 50 is a perspective view of a staple driver of the staple cartridge of FIG. 49;

FIG. 51 is a perspective view of a staple driver in accordance with at least one embodiment;

FIG. 52 is a top view of the staple driver of FIG. 51;

FIG. 53 is a partial perspective view of a staple cartridge in accordance with at least one embodiment;

FIG. 54 is an exploded view of the staple cartridge of FIG. 53;

FIG. 55 is a partial cross-sectional perspective view of a staple cartridge in accordance with at least one embodiment;

FIG. 56 illustrates the staple cartridge of FIG. 55 in a fired condition;

FIG. 57 is a partial elevational view of the staple cartridge of FIG. 55;

FIG. 58 is a perspective view of a staple driver of the staple cartridge of FIG. 55;

FIG. 59 is a partial cross-sectional view of the staple cartridge of FIG. 55 illustrated in an unfired condition;

FIG. 60 is a partial cross-sectional view of the staple cartridge of FIG. 55 illustrated in a fired condition;

FIG. 61 is a partial cross-sectional perspective view of a staple cartridge in accordance with at least one embodiment illustrated in an unfired condition;

FIG. 62 illustrates the staple cartridge of FIG. 61 in a fired condition;

FIG. 63 is a perspective view of a staple driver of the staple cartridge of FIG. 62;

FIG. 64 is a perspective view of a staple cartridge in accordance with at least one embodiment;

FIG. 65 is a cross-sectional perspective view of the staple cartridge of FIG. 64;

FIG. 66 is an end view of a staple cartridge in accordance with at least one embodiment;

FIG. 67 is a sled in accordance with at least one embodiment;

FIG. 68 is a perspective view of a sled of the staple cartridge of FIG. 64;

FIG. 69 is a cross-sectional perspective view of the staple cartridge of FIG. 64;

FIG. 70 is a partial cross-sectional perspective view of a staple cartridge in accordance with at least one embodiment;

FIG. 71 is a perspective view of supports of the staple cartridge of FIG. 70;

FIG. 72 is a perspective view of a staple driver and a staple of a staple cartridge in accordance with at least one embodiment;

FIG. 73 is a perspective view of the staple driver of FIG. 72;

FIG. 74 is a partial elevational view of the staple driver and staple of FIG. 72;

FIG. 75 is a top view of a sled in accordance with at least one embodiment;

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FIG. 76 is a perspective view of the sled of FIG. 75 and a staple driver;

FIG. 77 is a partial cross-sectional view of a staple cartridge including the sled of FIG. 75 and the driver of FIG. 76;

FIG. 78 is a partial perspective view of the staple cartridge of FIG. 77 illustrating the sled engaged with the staple driver;

FIG. 79 is a partial perspective view of the staple cartridge of FIG. 77 illustrating the staple driver in a fired position;

FIG. 80 is a perspective view of a drive system for use with a surgical instrument;

FIG. 81 is an exploded assembly view of the drive system of FIG. 80;

FIG. 82 is an end elevation view of the drive system of FIG. 80;

FIG. 83 is a side elevation view of the drive system of FIG. 80 in a first configuration;

FIG. 84 is a side elevation view of the drive system of FIG. 80 in a second configuration;

FIG. 85 is a side elevation view of the drive system of FIG. 80 in a third configuration;

FIG. 86 is a perspective view of another drive system for use with a surgical instrument;

FIG. 87 is a perspective view of the drive system of FIG. 86 with portions of cams removed for clarity;

FIG. 88 is an exploded assembly view of the drive system of FIG. 86;

FIG. 89 is an end elevation view of the drive system of FIG. 86;

FIG. 90 is a perspective view of the drive system of FIG. 86 in a first configuration;

FIG. 91 is a perspective view of the drive system of FIG. 86 in a second configuration;

FIG. 92 is a perspective view of the drive system of FIG. 86 in a third configuration;

FIG. 93 is an elevational view of an articulation joint for use with a surgical instrument, wherein the articulation joint comprises an articulation support pivot;

FIG. 94 is an elevational view of the articulation joint of FIG. 93 illustrated in a non-articulated configuration;

FIG. 95 is a cross-sectional view of the articulation joint of FIG. 93 taken along line 95-95 in FIG. 93;

FIG. 96 is an elevational view of the articulation joint of FIG. 93 illustrated in an articulated configuration;

FIG. 97 is a cross-sectional view of an articulation joint for use with a surgical instrument;

FIG. 98 is a perspective view of a surgical instrument assembly comprising an end effector cartridge, a firing member, and a plurality of flexible actuators;

FIG. 99 is an elevational view of the surgical instrument assembly of FIG. 98;

FIG. 100 is an elevational view of an articulation system for use with a surgical instrument assembly, wherein the articulation system comprises an articulation joint, an articulation actuation system, and a biasing system configured to bias the articulation joint into a non-articulated configuration, wherein the articulation joint is illustrated in the non-articulated configuration;

FIG. 101 is an elevational view of the articulation system of FIG. 100, wherein the articulation joint is illustrated in an articulated configuration;

FIG. 102 is a perspective view of a surgical instrument shaft assembly comprising a spine, an articulation joint, and a core insert positioned within the spine;

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FIG. 103 is a perspective view of the spine and core inset of FIG. 102, wherein the core insert comprises a proximal core member and a distal core member positioned within the spine;

FIG. 104 is a partial elevational view of the proximal core member and the distal core member of the surgical instrument assembly of FIG. 102;

FIG. 105 is an elevational view of a piezoelectric actuator for use with a surgical instrument, wherein the piezoelectric actuator comprises outer piezoelectric layers and an inner substrate layer;

FIG. 106 is an elevational view of a piezoelectric actuator for use with a surgical instrument, wherein the piezoelectric actuator is illustrated in an un-energized state;

FIG. 107 is an elevational view of the piezoelectric actuator of FIG. 106, wherein the piezoelectric actuator is illustrated in an energized state;

FIG. 108 is a perspective view of a piezoelectric actuator for use with a surgical instrument, wherein the piezoelectric actuator is illustrated in an un-energized state;

FIG. 109 is an elevational view of the piezoelectric actuator of FIG. 108, wherein the piezoelectric actuator is illustrated in an energized state;

FIG. 110 is a chart representing force generation vs. displacement of a piezoelectric actuator for use with a surgical instrument;

FIG. 111 is a perspective view of an electroactive polymer actuator for use with a surgical instrument;

FIG. 112 is a cross-sectional view of a shaft assembly comprising an outer shaft, a spine portion, and a rotational actuation system configured to rotate the spine portion relative to the outer shaft, wherein the rotational actuation system comprises a rotary drive shaft and a frictional interface between the rotary drive shaft and the spine portion;

FIG. 113 is a cross-sectional view of a shaft assembly comprising an outer shaft, a spine portion, and a rotational actuation system configured to rotate the spine portion relative to the outer shaft, wherein the rotational actuation system comprises a plurality of windings and magnets configured to cooperate to rotate the spine portion relative to the outer shaft;

FIG. 114 is a perspective cross-sectional view of a surgical instrument assembly comprising an outer shaft, a spine, and a piezoelectric rotary actuator configured to rotate the spine;

FIG. 115 is an elevational view of a limiter system for use with a surgical instrument assembly configured to limit rotational actuation of a rotary drive upon reaching a threshold position, wherein the limiter system is illustrated in a non-limited configuration;

FIG. 116 is an elevational view of the limiter system of FIG. 115, wherein the limiter system is engaged with the rotary drive in a partially limited state;

FIG. 117 is an elevational view of the limiter system of FIG. 115, wherein the limiter system is engaged with the rotary drive in a fully limited state;

FIG. 118 is an elevational view of a rotary actuation system for use with a surgical instrument, wherein the rotary actuation system comprises a limiting feature configured to prevent further over-rotation of a component, wherein the rotary actuation system is illustrated in a home position;

FIG. 119 is an elevational view of the rotary actuation system of FIG. 118, wherein the rotary actuation system is illustrated in a first threshold state;

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FIG. 120 is an elevational view of the rotary actuation system of FIG. 118, wherein the rotary actuation system is illustrated in a second threshold state;

FIG. 121 is a perspective view of a segmented ring contact system for use with a surgical instrument assembly;

FIG. 122 is an elevational view of a surgical instrument assembly comprising a first shaft, a second shaft, and an electrical transmission arrangement configured to transmit electrical signals between contacts positioned on the first shaft and contacts positioned on the second shaft;

FIG. 123 is an elevational view of a surgical instrument assembly comprising the first shaft, the second shaft, and the electrical transmission arrangement of FIG. 122, wherein the surgical instrument assembly further comprises grommets positioned within the electrical transmission arrangement;

FIG. 124 is an elevational view of a surgical instrument assembly comprising the first shaft, the second shaft, and the electrical transmission arrangement of FIG. 122, wherein the surgical instrument assembly further comprises a grommet positioned to prevent fluid ingress toward the electrical transmission arrangement;

FIG. 125 is an elevational view of a surgical instrument assembly comprising a first shaft, a second shaft, and an electrical transmission arrangement configured to transmit electrical signals between contacts positioned on the first shaft and contacts positioned on the second shaft;

FIG. 126 is an elevational view of a surgical instrument assembly comprising a first shaft, a second shaft, and an electrical transmission arrangement configured to transmit electrical signals between contacts positioned on the first shaft and contacts positioned on the second shaft;

FIG. 127 is an elevational view of a surgical instrument assembly comprising a first shaft, a second shaft, and an electrical transmission arrangement configured to transmit electrical signals between contacts positioned on the first shaft and contacts positioned on the second shaft;

FIG. 128 is a schematic representation of an inductive coil assembly for use with a surgical instrument assembly comprising a transmission coil and a receiver coil;

FIG. 129 is a schematic representation of an inductive coil assembly for use with a surgical instrument assembly comprising a transmission coil and a receiver coil;

FIG. 130 is a schematic representation of an electroactive polymer for use with a surgical instrument assembly, wherein the electroactive polymer is illustrated in a non-energized state;

FIG. 131 is a schematic representation of the electroactive polymer of FIG. 129, wherein the electroactive polymer is illustrated in an energized state;

FIG. 132 is a perspective view of an end effector having a channel and a replaceable assembly according to at least one aspect of the present disclosure;

FIG. 133 is a perspective view of the end effector of FIG. 132 prior to the replaceable assembly being seated in the channel;

FIG. 134 is a perspective view of the replaceable assembly of FIG. 132 having a staple cartridge and an anvil;

FIG. 135 is an elevational view of the end effector of FIG. 132 prior to the replaceable assembly being seated in the channel;

FIG. 136 is an elevational view of the end effector of FIG. 132 during a first stage of seating the replaceable assembly in the channel;

FIG. 137 is an elevational view of the end effector of FIG. 132 during a second stage of seating the replaceable assembly in the channel;

FIG. 138 is an elevational view of the end effector of FIG. 132 with the replaceable assembly fully seated in the channel;

FIG. 139 is a perspective view of a disposable end effector attached to an elongate shaft according to at least one aspect of the present disclosure;

FIG. 140 is a partial perspective view of the disposable end effector of FIG. 139 detached from the end effector;

FIG. 141 is a partial perspective view of flex circuits positioned within the disposable end effector of FIG. 139;

FIG. 142 is a partial perspective view of an attachment interface between the disposable end effector of FIG. 139 and the elongate shaft prior to the end effector being replaceably attached to the elongate shaft;

FIG. 143 is a partial perspective view of the disposable end effector and elongate shaft of FIG. 139 during a first stage of attaching the end effector to the elongate shaft;

FIG. 144 is a partial perspective view of the disposable end effector and elongate shaft of FIG. 139 during a second stage of attaching the end effector to the elongate shaft;

FIG. 145 is a partial perspective view of the disposable end effector fully attached to the elongate shaft;

FIG. 146 is a partial cross-sectional view of the disposable end effector fully attached to the elongate shaft;

FIG. 147 is a perspective view of a shaft and an end effector in a detached state with a firing member and a drive shaft shown in phantom according to at least one aspect of the present disclosure;

FIG. 148 is a perspective view of the shaft and the end effector of FIG. 147 in an attached state with the firing member and the drive shaft shown in phantom;

FIG. 149 is a perspective view of the firing member and the drive shaft of FIG. 147 in the detached state;

FIG. 150 is a partial cross-sectional view of the firing member and the drive shaft of FIG. 148 in the attached state;

FIG. 151 is a perspective view of a reinforced anvil according to at least one aspect of the present disclosure;

FIG. 152 is a perspective view of the reinforced anvil of FIG. 151 having an anvil and an anvil plate welded thereto;

FIG. 153 is an elevational view of an end effector having the reinforced anvil of FIG. 151;

FIG. 154 is a partial cross-sectional view of a channel having cartridge retention features and a cartridge seated therein according to at least one aspect of the present disclosure;

FIG. 155 is a perspective view of a surgical instrument including an end effector for use in a surgical procedure, in accordance with at least one aspect of the present disclosure;

FIG. 156 is a partial perspective view of a distal portion of the surgical instrument of FIG. 155;

FIG. 157 is an exploded view of an end effector of the surgical instrument of FIG. 155;

FIG. 158 is a cross-sectional view of the end effector of the surgical instrument of FIG. 155;

FIG. 159 is an exploded view of a cartridge, in accordance with at least one aspect of the present disclosure;

FIG. 160 is a close-up of the perspective view of the cartridge of FIG. 159;

FIG. 161 is a cross-sectional view of the cartridge of FIG. 159;

FIG. 162 is a perspective view of an anvil, in accordance with at least one aspect of the present disclosure;

FIG. 163 is a schematic diagram depicting components of a surgical instrument connected to a radio frequency (RF) energy source;

FIG. 164 is a schematic diagram depicting a control circuit, in accordance with at least one aspect of the present disclosure;

FIG. 165 is a schematic diagram of a surgical generator, in accordance with at least one aspect of the present disclosure;

FIG. 166 is a schematic diagram of a surgical generator, in accordance with at least one aspect of the present disclosure;

FIG. 167 is a logic flow diagram of a process depicting a control program or a logic configuration for sealing tissue grasped by an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 168 is an exploded view of a cartridge, in accordance with at least one aspect of the present disclosure;

FIG. 169 is a cross-sectional view of the cartridge of FIG. 168;

FIG. 170 is a cross-sectional view of the cartridge of FIG. 168;

FIG. 171 is a cross-sectional view of the anvil of FIG. 162;

FIG. 172 is a bottom view of an alternative anvil of the surgical instrument of FIG. 155;

FIG. 173 is an electrical diagram illustrating a simplified electrical layout of electrode assemblies of the surgical instrument of FIG. 155;

FIG. 174 is an electrical diagram illustrating an electrical layout of an alternative electrode assembly of the surgical instrument of FIG. 155;

FIG. 175 is a cross-sectional view of an alternative end effector of the surgical instrument of FIG. 155;

FIG. 176 is a graph illustrating the change in resistance (Ω) of a PTC segment in response to a change in temperature ($^{\circ}$ C.), in accordance with the at least one aspect of the present disclosure;

FIG. 177 is another graph illustrating the change in resistance (Ω) of a PTC segment in response to a change in temperature ($^{\circ}$ C.), in accordance with the at least one aspect of the present disclosure;

FIG. 178 is a graph depicting passive and independent control of a current through a tissue portion between electrode assemblies, in accordance with at least one aspect of the present disclosure;

FIG. 179 is a graph illustrating a PTC segment's trip response at different temperatures, in accordance with at least one aspect of the present disclosure;

FIG. 180 is a logic flow diagram of a process depicting a control program or a logic configuration for detecting and addressing a short circuit during a tissue treatment cycle applied to tissue grasped by an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 181 is a logic flow diagram of a process depicting a control program or a logic configuration for a tissue treatment cycle applied to tissue grasped by an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 182 is a graph representing a power scheme for a tissue treatment cycle and corresponding tissue impedance, in accordance with at least one aspect of the present disclosure;

FIG. 183 is a cross-sectional view of an alternative anvil, in accordance with at least one aspect of the present disclosure;

FIG. 184 is another cross-sectional view of the anvil of FIG. 183;

FIG. 185 is a cross-sectional view of an alternative anvil, in accordance with at least one aspect of the present disclosure;

FIG. 186 is another cross-sectional view of the anvil of FIG. 185;

FIG. 187 is a perspective of an electrode carrier of the anvil of FIG. 183;

FIG. 188 is a cross-sectional view of an alternative anvil, in accordance with at least one aspect of the present disclosure;

FIG. 189 is a schematic view of an alternative end effector of the surgical instrument of FIG. 155;

FIG. 190 is an electrical diagram illustrating an electrical layout of an electrode assembly of the end effector of FIG. 189;

FIG. 191 is an electrical diagram illustrating an electrical layout of an electrode assembly of the end effector of FIG. 189;

FIG. 192 is a logic flow diagram of a process depicting a control program or a logic configuration, in accordance with at least one aspect of the present disclosure;

FIG. 193 is a logic flow diagram of a process depicting a control program or a logic configuration, in accordance with at least one aspect of the present disclosure;

FIG. 194 is a logic flow diagram of a process depicting a control program or a logic configuration, in accordance with at least one aspect of the present disclosure;

FIG. 195 is a logic flow diagram of a process depicting a control program or a logic configuration, in accordance with at least one aspect of the present disclosure;

FIG. 196 is a graph representing an interrogation of a first tissue portion, in accordance with the process of FIG. 195;

FIG. 197 is a graph illustrating an energy profile, or therapeutic signal, the graph depicting tissue impedance, voltage, power, and current curves associated with application of the therapeutic signal to tissue grasped by an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 198 is a graph illustrating an energy profile, or therapeutic signal, the graph depicting tissue impedance, voltage, power, and current curves associated with application of the therapeutic signal to tissue grasped by an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 199 is a graph illustrating an energy profile, or therapeutic signal, the graph depicting tissue impedance, voltage, power, and current curves associated with application of the therapeutic signal to tissue grasped by an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 200 is a graph illustrating an energy profile, or therapeutic signal, the graph depicting tissue impedance, voltage, power, and current curves associated with application of the therapeutic signal to tissue grasped by an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 201 is a graph illustrating an energy profile, or therapeutic signal, the graph depicting tissue impedance, voltage, power, and current curves associated with application of the therapeutic signal to tissue grasped by an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 202 is a graph illustrating an energy profile, or therapeutic signal, the graph depicting tissue impedance, voltage, power, and current curves associated with application of the therapeutic signal to tissue grasped by an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 203 is a graph illustrating an energy profile, or therapeutic signal, the graph depicting tissue impedance,

voltage, power, and current curves associated with application of the therapeutic signal to tissue grasped by an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 204 is a logic flow diagram of a process depicting a control program or a logic configuration, in accordance with at least one aspect of the present disclosure;

FIG. 205 is a logic flow diagram of a process depicting a control program or a logic configuration, in accordance with at least one aspect of the present disclosure;

FIG. 206 is a logic flow diagram of a process depicting a control program or a logic configuration, in accordance with at least one aspect of the present disclosure;

FIG. 207 is a logic flow diagram of a process depicting a control program or a logic configuration, in accordance with at least one aspect of the present disclosure;

FIG. 208 is a partial perspective view of an end effector, in accordance with at least one aspect of the present disclosure;

FIG. 209 is a cross-sectional view of the end effector of FIG. 208;

FIG. 210 is a close-up of the cross-sectional view of FIG. 208;

FIG. 211 is a logic flow diagram of a process depicting a control program or a logic configuration, in accordance with at least one aspect of the present disclosure;

FIG. 212 is a logic flow diagram of a process depicting a control program or a logic configuration, in accordance with at least one aspect of the present disclosure;

FIG. 213 illustrates a control system for a surgical instrument comprising a plurality of motors which can be activated to perform various functions, in accordance with at least one aspect of the present disclosure;

FIG. 214 shows a jaw of an end effector for the surgical instrument described in FIGS. 1-13 where the electrode shown in FIG. 6 is configured with multiple pairs of segmented RF electrodes disposed on a circuit board, or other type of suitable substrate, on a lower surface of the jaw (i.e., the surface of the jaw facing tissue during operation), in accordance with at least one aspect of the present disclosure;

FIG. 215 illustrates a multi-layer circuit board, in accordance with at least one aspect of the present disclosure;

FIG. 216 shows segmented electrodes on either side of the knife slot in the jaw have different lengths, in accordance with at least one aspect of the present disclosure;

FIG. 217 is a cross-sectional view of an end effector comprising a plurality of segmented electrodes, in accordance with at least one aspect of the present disclosure;

FIG. 218 shows a jaw of an end effector for the surgical instrument described in FIGS. 1-13 and 214 where multiple pairs of segmented RF electrodes include a series current limiting element Z within the distal portion of the end effector for each electrode, in accordance with at least one aspect of the present disclosure;

FIG. 219 is a graphical representation of exploratory pulse waveforms applied by the RF generator under control of the controller to an electrode to detect a metallic object shorting the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 220 is a detailed view of the exploratory pulse waveforms applied to an electrode during a shorting event, in accordance with at least one aspect of the present disclosure;

FIG. 221 is a graphical representations of exploratory pulse waveforms applied to an electrode prior to firing or delivering therapeutic RF energy to seal tissue grasped

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between the jaws of the end effector, in accordance with at least one aspect of the present disclosure;

FIG. 222 is a detailed view depicting the pulsed impedance waveform applied to tissue having an impedance of approximately 2Ω , in accordance with at least one aspect of the present disclosure;

FIG. 223 depicts the application of a first example of low power exploratory pulse waveforms prior to firing or activating RF sealing energy in liver tissue that includes a metallic staple located in the field causing a short between an electrode and a return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 224A is a detailed view of the impedance waveform component of the exploratory pulse waveforms during a transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 224B is a detailed view of the power waveform component of the exploratory pulse waveforms during a transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 224C is a detailed view of the voltage waveform component of the exploratory pulse waveforms during a transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 224D is a detailed view of the current waveform component of the exploratory pulse waveforms during a transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 225 depicts the application of a second example of low power exploratory pulse waveforms prior to firing or activating RF sealing energy in liver tissue that includes a metallic staple located in the field causing a short between an electrode and a return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 226A is a detailed view of the impedance waveform component of the exploratory pulse waveforms during a transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 226B is a detailed view of the power waveform component of the exploratory pulse waveforms during a transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 226C is a detailed view of the voltage waveform component of the exploratory pulse waveforms during a transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 226D is a detailed view of the current waveform component of the exploratory pulse waveforms during a transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 227 depicts the application of a second example of low power exploratory pulse waveforms prior to firing or activating RF sealing energy in liver tissue that includes a metallic staple located in the field causing a short between an electrode and a return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 228A is a detailed view of the impedance waveform component of the exploratory pulse waveforms during a

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transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 228B is a detailed view of the power waveform component of the exploratory pulse waveforms during a transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 228C is a detailed view of the voltage waveform component of the exploratory pulse waveforms during a transition to a short circuit between the electrode and the return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 228D is a detailed view of a current waveform component of an exploratory pulse waveform during a transition to a short circuit between an electrode and a return path electrode, in accordance with at least one aspect of the present disclosure;

FIG. 229 is a graphical depiction of impedance, voltage, and current versus time (t), in accordance with at least one aspect of the present disclosure;

FIG. 230 is a graphical depiction of an electric arcing charge across a 1.8 cm gap in a 0.8 cm² area relative to current and voltage waveforms, in accordance with at least one aspect of the present disclosure;

FIG. 231 is a graphical depiction of electric discharge regimes as a function of voltage versus current, where current (Amps) is along the horizontal axis and voltage (Volts) is along the vertical axis, in accordance with at least one aspect of the present disclosure;

FIG. 232 is a graphical depiction of power (Watts) as a function of impedance (Ohms) of various tissue types, in accordance with at least one aspect of the present disclosure;

FIG. 233 is a logic flow diagram of a method of detecting a short circuit in the jaws of an end effector of a surgical instrument (see FIGS. 1-6 and 213-218), in accordance with at least one aspect of the present disclosure;

FIG. 234 is a logic flow diagram of a method of detecting a short circuit in the jaws of an end effector of a surgical instrument (see FIGS. 1-6 and 213-218), in accordance with at least one aspect of the present disclosure;

FIG. 235 shows a dielectric polarization plot where polarization (P) is a linear function of external electric field (E), in accordance with at least one aspect of the present disclosure;

FIG. 236 shows a paraelectric polarization plot where polarization (P) is a non-linear function of external electric field (E) exhibiting a sharp transition from negative to positive polarization at the origin, in accordance with at least one aspect of the present disclosure;

FIG. 237 shows ferroelectric polarization plot where polarization (P) is a non-linear function of external electric field (E) exhibiting hysteresis around the origin, in accordance with at least one aspect of the present disclosure;

FIG. 238 is logic flow diagram of a method of adapting energy modality due to a short circuit or tissue type grasped in the jaws of an end effector of a surgical instrument, in accordance with at least one aspect of the present disclosure; and

FIG. 239 illustrates a staple comprising a crown defining a base and deformable legs extending from each end of the base, in accordance with at least one aspect of the present disclosure.

Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate various embodiments of the inven-

tion, in one form, and such exemplifications are not to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION

Applicant of the present application also owns the following U.S. patent applications that were filed on even date herewith and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 17/246,010, entitled METHOD FOR OPERATING A SURGICAL INSTRUMENT INCLUDING SEGMENTED ELECTRODES;

U.S. patent application Ser. No. 17/246,017, entitled STAPLE CARTRIDGE COMPRISING STAPLE DRIVERS AND STABILITY SUPPORTS;

U.S. patent application Ser. No. 17/246,019, entitled STAPLE CARTRIDGE COMPRISING FORMATION SUPPORT FEATURES;

U.S. patent application Ser. No. 17/246,022, entitled INTERCHANGEABLE END EFFECTOR RELOADS;

U.S. patent application Ser. No. 17/246,030, entitled SURGICAL INSTRUMENT COMPRISING A ROTATION-DRIVEN AND TRANSLATION-DRIVEN TISSUE CUTTING KNIFE;

U.S. patent application Ser. No. 17/246,040, entitled SURGICAL INSTRUMENT COMPRISING A CLOSURE BAR AND A FIRING BAR;

U.S. patent application Ser. No. 17/246,055, entitled SURGICAL INSTRUMENT COMPRISING END EFFECTOR WITH LONGITUDINAL SEALING STEP;

U.S. patent application Ser. No. 17/246,067, entitled SURGICAL INSTRUMENT COMPRISING END EFFECTOR WITH ENERGY SENSITIVE RESISTANCE ELEMENTS;

U.S. patent application Ser. No. 17/246,073, entitled SURGICAL INSTRUMENT COMPRISING INDEPENDENTLY ACTIVATABLE SEGMENTED ELECTRODES;

U.S. patent application Ser. No. 17/246,080, entitled SURGICAL SYSTEMS CONFIGURED TO CONTROL THERAPEUTIC ENERGY APPLICATION TO TISSUE BASED ON CARTRIDGE AND TISSUE PARAMETERS;

U.S. patent application Ser. No. 17/246,089, entitled ELECTROSURGICAL TECHNIQUES FOR SEALING, SHORT CIRCUIT DETECTION, AND SYSTEM DETERMINATION OF POWER LEVEL;

U.S. patent application Ser. No. 17/246,101, entitled SURGICAL STAPLE FOR USE WITH COMBINATION ELECTROSURGICAL INSTRUMENTS;

U.S. patent application Ser. No. 17/246,118, entitled SURGICAL SYSTEMS CONFIGURED TO COOPERATIVELY CONTROL END EFFECTOR FUNCTION AND APPLICATION OF THERAPEUTIC ENERGY;

U.S. patent application Ser. No. 17/246,134, entitled ARTICULATION SYSTEM FOR SURGICAL INSTRUMENT;

U.S. patent application Ser. No. 17/246,141, entitled SHAFT SYSTEM FOR SURGICAL INSTRUMENT;

Applicant of the present application also owns the following U.S. patent applications that were filed on Feb. 26, 2021 and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 17/186,269, entitled METHOD OF POWERING AND COMMUNICATING WITH A STAPLE CARTRIDGE;

U.S. patent application Ser. No. 17/186,273, entitled METHOD OF POWERING AND COMMUNICATING WITH A STAPLE CARTRIDGE;

U.S. patent application Ser. No. 17/186,276, entitled ADJUSTABLE COMMUNICATION BASED ON AVAILABLE BANDWIDTH AND POWER CAPACITY;

U.S. patent application Ser. No. 17/186,283, entitled ADJUSTMENT TO TRANSFER PARAMETERS TO IMPROVE AVAILABLE POWER;

U.S. patent application Ser. No. 17/186,345, entitled MONITORING OF MANUFACTURING LIFE-CYCLE;

U.S. patent application Ser. No. 17/186,350, entitled MONITORING OF MULTIPLE SENSORS OVER TIME TO DETECT MOVING CHARACTERISTICS OF TISSUE;

U.S. patent application Ser. No. 17/186,353, entitled MONITORING OF INTERNAL SYSTEMS TO DETECT AND TRACK CARTRIDGE MOTION STATUS;

U.S. patent application Ser. No. 17/186,357, entitled DISTAL COMMUNICATION ARRAY TO TUNE FREQUENCY OF RF SYSTEMS;

U.S. patent application Ser. No. 17/186,364, entitled STAPLE CARTRIDGE COMPRISING A SENSOR ARRAY;

U.S. patent application Ser. No. 17/186,373, entitled STAPLE CARTRIDGE COMPRISING A SENSING ARRAY AND A TEMPERATURE CONTROL SYSTEM;

U.S. patent application Ser. No. 17/186,378, entitled STAPLE CARTRIDGE COMPRISING AN INFORMATION ACCESS CONTROL SYSTEM;

U.S. patent application Ser. No. 17/186,407, entitled STAPLE CARTRIDGE COMPRISING A POWER MANAGEMENT CIRCUIT;

U.S. patent application Ser. No. 17/186,421, entitled STAPLING INSTRUMENT COMPRISING A SEPARATE POWER ANTENNA AND A DATA TRANSFER ANTENNA;

U.S. patent application Ser. No. 17/186,438, entitled SURGICAL INSTRUMENT SYSTEM COMPRISING A POWER TRANSFER COIL; and

U.S. patent application Ser. No. 17/186,451, entitled STAPLING INSTRUMENT COMPRISING A SIGNAL ANTENNA.

Applicant of the present application also owns the following U.S. patent applications that were filed on Oct. 29, 2020 and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 17/084,179, entitled SURGICAL INSTRUMENT COMPRISING A RELEASABLE CLOSURE DRIVE LOCK;

U.S. patent application Ser. No. 17/084,190, entitled SURGICAL INSTRUMENT COMPRISING A STOWED CLOSURE ACTUATOR STOP;

U.S. patent application Ser. No. 17/084,198, entitled SURGICAL INSTRUMENT COMPRISING AN

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INDICATOR WHICH INDICATES THAT AN ARTICULATION DRIVE IS ACTUATABLE;
 U.S. patent application Ser. No. 17/084,205, entitled SURGICAL INSTRUMENT COMPRISING AN ARTICULATION INDICATOR;
 U.S. patent application Ser. No. 17/084,258, entitled METHOD FOR OPERATING A SURGICAL INSTRUMENT;
 U.S. patent application Ser. No. 17/084,206, entitled SURGICAL INSTRUMENT COMPRISING AN ARTICULATION LOCK;
 U.S. patent application Ser. No. 17/084,215, entitled SURGICAL INSTRUMENT COMPRISING A JAW ALIGNMENT SYSTEM;
 U.S. patent application Ser. No. 17/084,229, entitled SURGICAL INSTRUMENT COMPRISING SEALABLE INTERFACE;
 U.S. patent application Ser. No. 17/084,180, entitled SURGICAL INSTRUMENT COMPRISING A LIMITED TRAVEL SWITCH;
 U.S. Design patent application Ser. No. 29/756,615, Application entitled SURGICAL STAPLING ASSEMBLY;
 U.S. Design patent application Ser. No. 29/756,620, entitled SURGICAL STAPLING ASSEMBLY;
 U.S. patent application Ser. No. 17/084,188, entitled SURGICAL INSTRUMENT COMPRISING A STAGED VOLTAGE REGULATION START-UP SYSTEM; and
 U.S. patent application Ser. No. 17/084,193, entitled SURGICAL INSTRUMENT COMPRISING A SENSOR CONFIGURED TO SENSE WHETHER AN ARTICULATION DRIVE OF THE SURGICAL INSTRUMENT IS ACTUATABLE.

Applicant of the present application also owns the following U.S. patent applications that were filed on Apr. 11, 2020 and which are each herein incorporated by reference in their respective entireties:

U.S. patent application Ser. No. 16/846,303, entitled METHODS FOR STAPLING TISSUE USING A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/0345353; U.S. patent application Ser. No. 16/846,304, entitled ARTICULATION ACTUATORS FOR A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/0345354;
 U.S. patent application Ser. No. 16/846,305, entitled ARTICULATION DIRECTIONAL LIGHTS ON A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/0345446;
 U.S. patent application Ser. No. 16/846,307, entitled SHAFT ROTATION ACTUATOR ON A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/03453549;
 U.S. patent application Ser. No. 16/846,308, entitled ARTICULATION CONTROL MAPPING FOR A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/0345355;
 U.S. patent application Ser. No. 16/846,309, entitled INTELLIGENT FIRING ASSOCIATED WITH A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/0345356;
 U.S. patent application Ser. No. 16/846,310, entitled INTELLIGENT FIRING ASSOCIATED WITH A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/0345357;

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U.S. patent application Ser. No. 16/846,311, entitled ROTATABLE JAW TIP FOR A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/0345358;
 U.S. patent application Ser. No. 16/846,312, entitled TISSUE STOP FOR A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/0345359; and
 U.S. patent application Ser. No. 16/846,313, entitled ARTICULATION PIN FOR A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/0345360.

The entire disclosure of U.S. Provisional Patent Application Ser. No. 62/840,715, entitled SURGICAL INSTRUMENT COMPRISING AN ADAPTIVE CONTROL SYSTEM, filed Apr. 30, 2019, is hereby incorporated by reference herein.

Applicant of the present application owns the following U.S. patent applications that were filed on Feb. 21, 2019 and which are each herein incorporated by reference in their respective entireties:

U.S. patent application Ser. No. 16/281,658, entitled METHODS FOR CONTROLLING A POWERED SURGICAL STAPLER THAT HAS SEPARATE ROTARY CLOSURE AND FIRING SYSTEMS, now U.S. Patent Application Publication No. 2019/0298350;
 U.S. patent application Ser. No. 16/281,670, entitled STAPLE CARTRIDGE COMPRISING A LOCKOUT KEY CONFIGURED TO LIFT A FIRING MEMBER, now U.S. Patent Application Publication No. 2019/0298340;
 U.S. patent application Ser. No. 16/281,675, entitled SURGICAL STAPLERS WITH ARRANGEMENTS FOR MAINTAINING A FIRING MEMBER THEREOF IN A LOCKED CONFIGURATION UNLESS A COMPATIBLE CARTRIDGE HAS BEEN INSTALLED THEREIN, now U.S. Patent Application Publication No. 2019/0298354;
 U.S. patent application Ser. No. 16/281,685, entitled SURGICAL INSTRUMENT COMPRISING CO-OPERATING LOCKOUT FEATURES, now U.S. Patent Application Publication No. 2019/0298341;
 U.S. patent application Ser. No. 16/281,693, entitled SURGICAL STAPLING ASSEMBLY COMPRISING A LOCKOUT AND AN EXTERIOR ACCESS ORIFICE TO PERMIT ARTIFICIAL UNLOCKING OF THE LOCKOUT, now U.S. Patent Application Publication No. 2019/0298342;
 U.S. patent application Ser. No. 16/281,704, entitled SURGICAL STAPLING DEVICES WITH FEATURES FOR BLOCKING ADVANCEMENT OF A CAMMING ASSEMBLY OF AN INCOMPATIBLE CARTRIDGE INSTALLED THEREIN, now U.S. Patent Application Publication No. 2019/0298356;
 U.S. patent application Ser. No. 16/281,707, entitled STAPLING INSTRUMENT COMPRISING A DEACTIVATABLE LOCKOUT, now U.S. Patent Application Publication No. 2019/0298347;
 U.S. patent application Ser. No. 16/281,741, entitled SURGICAL INSTRUMENT COMPRISING A JAW CLOSURE LOCKOUT, now U.S. Patent Application Publication No. 2019/0298357;
 U.S. patent application Ser. No. 16/281,762, entitled SURGICAL STAPLING DEVICES WITH CARTRIDGE COMPATIBLE CLOSURE AND FIRING

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LOCKOUT ARRANGEMENTS, now U.S. Patent Application Publication No. 2019/0298343;
 U.S. patent application Ser. No. 16/281,666, entitled SURGICAL STAPLING DEVICES WITH IMPROVED ROTARY DRIVEN CLOSURE SYSTEMS, now U.S. Patent Application Publication No. 2019/0298352;
 U.S. patent application Ser. No. 16/281,672, entitled SURGICAL STAPLING DEVICES WITH ASYMMETRIC CLOSURE FEATURES, now U.S. Patent Application Publication No. 2019/0298353;
 U.S. patent application Ser. No. 16/281,678, entitled ROTARY DRIVEN FIRING MEMBERS WITH DIFFERENT ANVIL AND CHANNEL ENGAGEMENT FEATURES, now U.S. Patent Application Publication No. 2019/0298355; and
 U.S. patent application Ser. No. 16/281,682, entitled SURGICAL STAPLING DEVICE WITH SEPARATE ROTARY DRIVEN CLOSURE AND FIRING SYSTEMS AND FIRING MEMBER THAT ENGAGES BOTH JAWS WHILE FIRING, now U.S. Patent Application Publication No. 2019/0298346.

Applicant of the present application owns the following U.S. Provisional patent applications that were filed on Feb. 19, 2019 and which are each herein incorporated by reference in their respective entireties:

U.S. Provisional Patent Application Ser. No. 62/807,310, entitled METHODS FOR CONTROLLING A POWERED SURGICAL STAPLER THAT HAS SEPARATE ROTARY CLOSURE AND FIRING SYSTEMS;
 U.S. Provisional Patent Application Ser. No. 62/807,319, entitled SURGICAL STAPLING DEVICES WITH IMPROVED LOCKOUT SYSTEMS; and
 U.S. Provisional Patent Application Ser. No. 62/807,309, entitled SURGICAL STAPLING DEVICES WITH IMPROVED ROTARY DRIVEN CLOSURE SYSTEMS.

Applicant of the present application owns the following U.S. Provisional patent applications, filed on Mar. 28, 2018, each of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application Ser. No. 62/649,302, entitled INTERACTIVE SURGICAL SYSTEMS WITH ENCRYPTED COMMUNICATION CAPABILITIES;
 U.S. Provisional Patent Application Ser. No. 62/649,294, entitled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD;
 U.S. Provisional Patent Application Ser. No. 62/649,300, entitled SURGICAL HUB SITUATIONAL AWARENESS;
 U.S. Provisional Patent Application Ser. No. 62/649,309, entitled SURGICAL HUB SPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THEATER;
 U.S. Provisional Patent Application Ser. No. 62/649,310, entitled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS;
 U.S. Provisional Patent Application Ser. No. 62/649,291, entitled USE OF LASER LIGHT AND RED-GREEN-BLUE COLORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT;
 U.S. Provisional Patent Application Ser. No. 62/649,296, entitled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL DEVICES;

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U.S. Provisional Patent Application Ser. No. 62/649,333, entitled CLOUD-BASED MEDICAL ANALYTICS FOR CUSTOMIZATION AND RECOMMENDATIONS TO A USER;

U.S. Provisional Patent Application Ser. No. 62/649,327, entitled CLOUD-BASED MEDICAL ANALYTICS FOR SECURITY AND AUTHENTICATION TRENDS AND REACTIVE MEASURES;

U.S. Provisional Patent Application Ser. No. 62/649,315, entitled DATA HANDLING AND PRIORITIZATION IN A CLOUD ANALYTICS NETWORK;

U.S. Provisional Patent Application Ser. No. 62/649,313, entitled CLOUD INTERFACE FOR COUPLED SURGICAL DEVICES;

U.S. Provisional Patent Application Ser. No. 62/649,320, entitled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS;

U.S. Provisional Patent Application Ser. No. 62/649,307, entitled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; and

U.S. Provisional Patent Application Ser. No. 62/649,323, entitled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS.

Applicant of the present application owns the following U.S. Provisional patent application, filed on Mar. 30, 2018, which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application Ser. No. 62/650,887, entitled SURGICAL SYSTEMS WITH OPTIMIZED SENSING CAPABILITIES.

Applicant of the present application owns the following U.S. patent application, filed on Dec. 4, 2018, which is herein incorporated by reference in its entirety:

U.S. patent application Ser. No. 16/209,423, entitled METHOD OF COMPRESSING TISSUE WITHIN A STAPLING DEVICE AND SIMULTANEOUSLY DISPLAYING THE LOCATION OF THE TISSUE WITHIN THE JAWS, now U.S. Patent Application Publication No. 2019/0200981.

Applicant of the present application owns the following U.S. patent applications that were filed on Aug. 20, 2018 and which are each herein incorporated by reference in their respective entireties:

U.S. patent application Ser. No. 16/105,101, entitled METHOD FOR FABRICATING SURGICAL STAPLER ANVILS, now U.S. Patent Application Publication No. 2020/0054323;

U.S. patent application Ser. No. 16/105,183, entitled REINFORCED DEFORMABLE ANVIL TIP FOR SURGICAL STAPLER ANVIL, now U.S. Pat. No. 10,912,559;

U.S. patent application Ser. No. 16/105,150, entitled SURGICAL STAPLER ANVILS WITH STAPLE DIRECTING PROTRUSIONS AND TISSUE STABILITY FEATURES, now U.S. Patent Application Publication No. 2020/0054326;

U.S. patent application Ser. No. 16/105,098, entitled FABRICATING TECHNIQUES FOR SURGICAL STAPLER ANVILS, now U.S. Patent Application Publication No. 2020/0054322;

U.S. patent application Ser. No. 16/105,140, entitled SURGICAL STAPLER ANVILS WITH TISSUE STOP FEATURES CONFIGURED TO AVOID TISSUE PINCH, now U.S. Pat. No. 10,779,821;

U.S. patent application Ser. No. 16/105,081, entitled METHOD FOR OPERATING A POWERED

ARTICULATABLE SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2020/0054320;

U.S. patent application Ser. No. 16/105,094, entitled SURGICAL INSTRUMENTS WITH PROGRESSIVE JAW CLOSURE ARRANGEMENTS, now U.S. Patent Application Publication No. 2020/0054321;

U.S. patent application Ser. No. 16/105,097, entitled POWERED SURGICAL INSTRUMENTS WITH CLUTCHING ARRANGEMENTS TO CONVERT LINEAR DRIVE MOTIONS TO ROTARY DRIVE MOTIONS, now U.S. Patent Application Publication No. 2020/0054328;

U.S. patent application Ser. No. 16/105,104, entitled POWERED ARTICULATABLE SURGICAL INSTRUMENTS WITH CLUTCHING AND LOCKING ARRANGEMENTS FOR LINKING AN ARTICULATION DRIVE SYSTEM TO A FIRING DRIVE SYSTEM, now U.S. Pat. No. 10,842,492;

U.S. patent application Ser. No. 16/105,119, entitled ARTICULATABLE MOTOR POWERED SURGICAL INSTRUMENTS WITH DEDICATED ARTICULATION MOTOR ARRANGEMENTS, now U.S. Patent Application Publication No. 2020/0054330;

U.S. patent application Ser. No. 16/105,160, entitled SWITCHING ARRANGEMENTS FOR MOTOR POWERED ARTICULATABLE SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,856,870; and

U.S. Design patent application Ser. No. 29/660,252, entitled SURGICAL STAPLER ANVILS.

Applicant of the present application owns the following U.S. patent applications and U.S. Patents that are each herein incorporated by reference in their respective entireties:

U.S. patent application Ser. No. 15/386,185, entitled SURGICAL STAPLING INSTRUMENTS AND REPLACEABLE TOOL ASSEMBLIES THEREOF, now U.S. Pat. No. 10,639,035;

U.S. patent application Ser. No. 15/386,230, entitled ARTICULATABLE SURGICAL STAPLING INSTRUMENTS, now U.S. Patent Application Publication No. 2018/0168649;

U.S. patent application Ser. No. 15/386,221, entitled LOCKOUT ARRANGEMENTS FOR SURGICAL END EFFECTORS, now U.S. Pat. No. 10,835,247;

U.S. patent application Ser. No. 15/386,209, entitled SURGICAL END EFFECTORS AND FIRING MEMBERS THEREOF, now U.S. Pat. No. 10,588,632;

U.S. patent application Ser. No. 15/386,198, entitled LOCKOUT ARRANGEMENTS FOR SURGICAL END EFFECTORS AND REPLACEABLE TOOL ASSEMBLIES, now U.S. Pat. No. 10,610,224;

U.S. patent application Ser. No. 15/386,240, entitled SURGICAL END EFFECTORS AND ADAPTABLE FIRING MEMBERS THEREFOR, now U.S. Patent Application Publication No. 2018/0168651;

U.S. patent application Ser. No. 15/385,939, entitled STAPLE CARTRIDGES AND ARRANGEMENTS OF STAPLES AND STAPLE CAVITIES THEREIN, now U.S. Pat. No. 10,835,246;

U.S. patent application Ser. No. 15/385,941, entitled SURGICAL TOOL ASSEMBLIES WITH CLUTCHING ARRANGEMENTS FOR SHIFTING BETWEEN CLOSURE SYSTEMS WITH CLOSURE STROKE REDUCTION FEATURES AND ARTICULATION AND FIRING SYSTEMS, now U.S. Pat. No. 10,736,629;

U.S. patent application Ser. No. 15/385,943, entitled SURGICAL STAPLING INSTRUMENTS AND STAPLE-FORMING ANVILS, now U.S. Pat. No. 10,667,811;

U.S. patent application Ser. No. 15/385,950, entitled SURGICAL TOOL ASSEMBLIES WITH CLOSURE STROKE REDUCTION FEATURES, now U.S. Pat. No. 10,588,630;

U.S. patent application Ser. No. 15/385,945, entitled STAPLE CARTRIDGES AND ARRANGEMENTS OF STAPLES AND STAPLE CAVITIES THEREIN, now U.S. Pat. No. 10,893,864;

U.S. patent application Ser. No. 15/385,946, entitled SURGICAL STAPLING INSTRUMENTS AND STAPLE-FORMING ANVILS, now U.S. Patent Application Publication No. 2018/0168633;

U.S. patent application Ser. No. 15/385,951, entitled SURGICAL INSTRUMENTS WITH JAW OPENING FEATURES FOR INCREASING A JAW OPENING DISTANCE, now U.S. Pat. No. 10,568,626;

U.S. patent application Ser. No. 15/385,953, entitled METHODS OF STAPLING TISSUE, now U.S. Pat. No. 10,675,026;

U.S. patent application Ser. No. 15/385,954, entitled FIRING MEMBERS WITH NON-PARALLEL JAW ENGAGEMENT FEATURES FOR SURGICAL END EFFECTORS, now U.S. Pat. No. 10,624,635;

U.S. patent application Ser. No. 15/385,955, entitled SURGICAL END EFFECTORS WITH EXPANDABLE TISSUE STOP ARRANGEMENTS, now U.S. Pat. No. 10,813,638;

U.S. patent application Ser. No. 15/385,948, entitled SURGICAL STAPLING INSTRUMENTS AND STAPLE-FORMING ANVILS, now U.S. Patent Application Publication No. 2018/0168584;

U.S. patent application Ser. No. 15/385,956, entitled SURGICAL INSTRUMENTS WITH POSITIVE JAW OPENING FEATURES, now U.S. Pat. No. 10,588,631;

U.S. patent application Ser. No. 15/385,958, entitled SURGICAL INSTRUMENTS WITH LOCKOUT ARRANGEMENTS FOR PREVENTING FIRING SYSTEM ACTUATION UNLESS AN UNSPENT STAPLE CARTRIDGE IS PRESENT, now U.S. Pat. No. 10,639,034;

U.S. patent application Ser. No. 15/385,947, entitled STAPLE CARTRIDGES AND ARRANGEMENTS OF STAPLES AND STAPLE CAVITIES THEREIN, now U.S. Pat. No. 10,568,625;

U.S. patent application Ser. No. 15/385,896, entitled METHOD FOR RESETTING A FUSE OF A SURGICAL INSTRUMENT SHAFT, now U.S. Patent Application Publication No. 2018/0168597;

U.S. patent application Ser. No. 15/385,898, entitled STAPLE-FORMING POCKET ARRANGEMENT TO ACCOMMODATE DIFFERENT TYPES OF STAPLES, now U.S. Pat. No. 10,537,325;

U.S. patent application Ser. No. 15/385,899, entitled SURGICAL INSTRUMENT COMPRISING IMPROVED JAW CONTROL, now U.S. Pat. No. 10,758,229;

U.S. patent application Ser. No. 15/385,901, entitled STAPLE CARTRIDGE AND STAPLE CARTRIDGE CHANNEL COMPRISING WINDOWS DEFINED THEREIN, now U.S. Pat. No. 10,667,809;

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U.S. patent application Ser. No. 15/385,902, entitled SURGICAL INSTRUMENT COMPRISING A CUTTING MEMBER, now U.S. Pat. No. 10,888,322;

U.S. patent application Ser. No. 15/385,904, entitled STAPLE FIRING MEMBER COMPRISING A MISSING CARTRIDGE AND/OR SPENT CARTRIDGE LOCKOUT, now U.S. Pat. No. 10,881,401;

U.S. patent application Ser. No. 15/385,905, entitled FIRING ASSEMBLY COMPRISING A LOCKOUT, now U.S. Pat. No. 10,695,055;

U.S. patent application Ser. No. 15/385,907, entitled SURGICAL INSTRUMENT SYSTEM COMPRISING AN END EFFECTOR LOCKOUT AND A FIRING ASSEMBLY LOCKOUT, now U.S. Patent Application Publication No. 2018/0168608;

U.S. patent application Ser. No. 15/385,908, entitled FIRING ASSEMBLY COMPRISING A FUSE, now U.S. Patent Application Publication No. 2018/0168609;

U.S. patent application Ser. No. 15/385,909, entitled FIRING ASSEMBLY COMPRISING A MULTIPLE FAILED-STATE FUSE, now U.S. Patent Application Publication No. 2018/0168610;

U.S. patent application Ser. No. 15/385,920, entitled STAPLE-FORMING POCKET ARRANGEMENTS, now U.S. Pat. No. 10,499,914;

U.S. patent application Ser. No. 15/385,913, entitled ANVIL ARRANGEMENTS FOR SURGICAL STAPLERS, now U.S. Patent Application Publication No. 2018/0168614;

U.S. patent application Ser. No. 15/385,914, entitled METHOD OF DEFORMING STAPLES FROM TWO DIFFERENT TYPES OF STAPLE CARTRIDGES WITH THE SAME SURGICAL STAPLING INSTRUMENT, now U.S. Patent Application Publication No. 2018/0168615;

U.S. patent application Ser. No. 15/385,893, entitled BILATERALLY ASYMMETRIC STAPLE-FORMING POCKET PAIRS, now U.S. Pat. No. 10,682,138;

U.S. patent application Ser. No. 15/385,929, entitled CLOSURE MEMBERS WITH CAM SURFACE ARRANGEMENTS FOR SURGICAL INSTRUMENTS WITH SEPARATE AND DISTINCT CLOSURE AND FIRING SYSTEMS, now U.S. Pat. No. 10,667,810;

U.S. patent application Ser. No. 15/385,911, entitled SURGICAL STAPLERS WITH INDEPENDENTLY ACTUATABLE CLOSING AND FIRING SYSTEMS, now U.S. Pat. No. 10,448,950;

U.S. patent application Ser. No. 15/385,927, entitled SURGICAL STAPLING INSTRUMENTS WITH SMART STAPLE CARTRIDGES, now U.S. Patent Application Publication No. 2018/0168625;

U.S. patent application Ser. No. 15/385,917, entitled STAPLE CARTRIDGE COMPRISING STAPLES WITH DIFFERENT CLAMPING BREADTHS, now U.S. Patent Application Publication No. 2018/0168617;

U.S. patent application Ser. No. 15/385,900, entitled STAPLE-FORMING POCKET ARRANGEMENTS COMPRISING PRIMARY SIDEWALLS AND POCKET SIDEWALLS, now U.S. Pat. No. 10,898,186;

U.S. patent application Ser. No. 15/385,931, entitled NO-CARTRIDGE AND SPENT CARTRIDGE

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LOCKOUT ARRANGEMENTS FOR SURGICAL STAPLERS, now U.S. Patent Application Publication No. 2018/0168627;

U.S. patent application Ser. No. 15/385,915, entitled FIRING MEMBER PIN ANGLE, now U.S. Pat. No. 10,779,823;

U.S. patent application Ser. No. 15/385,897, entitled STAPLE-FORMING POCKET ARRANGEMENTS COMPRISING ZONED FORMING SURFACE GROOVES, now U.S. Patent Application Publication No. 2018/0168598;

U.S. patent application Ser. No. 15/385,922, entitled SURGICAL INSTRUMENT WITH MULTIPLE FAILURE RESPONSE MODES, now U.S. Pat. No. 10,426,471;

U.S. patent application Ser. No. 15/385,924, entitled SURGICAL INSTRUMENT WITH PRIMARY AND SAFETY PROCESSORS, now U.S. Pat. No. 10,758,230;

U.S. patent application Ser. No. 15/385,910, entitled ANVIL HAVING A KNIFE SLOT WIDTH, now U.S. Pat. No. 10,485,543;

U.S. patent application Ser. No. 15/385,903, entitled CLOSURE MEMBER ARRANGEMENTS FOR SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,617,414;

U.S. patent application Ser. No. 15/385,906, entitled FIRING MEMBER PIN CONFIGURATIONS, now U.S. Pat. No. 10,856,868;

U.S. patent application Ser. No. 15/386,188, entitled STEPPED STAPLE CARTRIDGE WITH ASYMMETRICAL STAPLES, now U.S. Pat. No. 10,537,324;

U.S. patent application Ser. No. 15/386,192, entitled STEPPED STAPLE CARTRIDGE WITH TISSUE RETENTION AND GAP SETTING FEATURES, now U.S. Pat. No. 10,687,810;

U.S. patent application Ser. No. 15/386,206, entitled STAPLE CARTRIDGE WITH DEFORMABLE DRIVER RETENTION FEATURES, now U.S. Patent Application Publication No. 2018/0168586;

U.S. patent application Ser. No. 15/386,226, entitled DURABILITY FEATURES FOR END EFFECTORS AND FIRING ASSEMBLIES OF SURGICAL STAPLING INSTRUMENTS, now U.S. Patent Application Publication No. 2018/0168648;

U.S. patent application Ser. No. 15/386,222, entitled SURGICAL STAPLING INSTRUMENTS HAVING END EFFECTORS WITH POSITIVE OPENING FEATURES, now U.S. Patent Application Publication No. 2018/0168647;

U.S. patent application Ser. No. 15/386,236, entitled CONNECTION PORTIONS FOR DEPOSABLE LOADING UNITS FOR SURGICAL STAPLING INSTRUMENTS, now U.S. Patent Application Publication No. 2018/0168650;

U.S. patent application Ser. No. 15/385,887, entitled METHOD FOR ATTACHING A SHAFT ASSEMBLY TO A SURGICAL INSTRUMENT AND, ALTERNATIVELY, TO A SURGICAL ROBOT, now U.S. Pat. No. 10,835,245;

U.S. patent application Ser. No. 15/385,889, entitled SHAFT ASSEMBLY COMPRISING A MANUALLY-OPERABLE RETRACTION SYSTEM FOR USE WITH A MOTORIZED SURGICAL INSTRUMENT SYSTEM, now U.S. Patent Application Publication No. 2018/0168590;

U.S. patent application Ser. No. 15/385,890, entitled SHAFT ASSEMBLY COMPRISING SEPARATELY ACTUATABLE AND RETRACTABLE SYSTEMS, now U.S. Pat. No. 10,675,025;

U.S. patent application Ser. No. 15/385,891, entitled 5 SHAFT ASSEMBLY COMPRISING A CLUTCH CONFIGURED TO ADAPT THE OUTPUT OF A ROTARY FIRING MEMBER TO TWO DIFFERENT SYSTEMS, now U.S. Patent Application Publication No. 2018/0168592;

U.S. patent application Ser. No. 15/385,892, entitled 10 SURGICAL SYSTEM COMPRISING A FIRING MEMBER ROTATABLE INTO AN ARTICULATION STATE TO ARTICULATE AN END EFFECTOR OF THE SURGICAL SYSTEM, now U.S. Pat. No. 10,918,385;

U.S. patent application Ser. No. 15/385,894, entitled SHAFT ASSEMBLY COMPRISING A LOCKOUT, now U.S. Pat. No. 10,492,785;

U.S. patent application Ser. No. 15/385,895, entitled 20 SHAFT ASSEMBLY COMPRISING FIRST AND SECOND ARTICULATION LOCKOUTS, now U.S. Pat. No. 10,542,982;

U.S. patent application Ser. No. 15/385,916, entitled 25 SURGICAL STAPLING SYSTEMS, now U.S. Patent Application Publication No. 2018/0168575;

U.S. patent application Ser. No. 15/385,918, entitled SURGICAL STAPLING SYSTEMS, now U.S. Patent Application Publication No. 2018/0168618;

U.S. patent application Ser. No. 15/385,919, entitled 30 SURGICAL STAPLING SYSTEMS, now U.S. Patent Application Publication No. 2018/0168619;

U.S. patent application Ser. No. 15/385,921, entitled 35 SURGICAL STAPLE CARTRIDGE WITH MOVABLE CAMMING MEMBER CONFIGURED TO DISENGAGE FIRING MEMBER LOCKOUT FEATURES, now U.S. Pat. No. 10,687,809;

U.S. patent application Ser. No. 15/385,923, entitled 40 SURGICAL STAPLING SYSTEMS, now U.S. Patent Application Publication No. 2018/0168623;

U.S. patent application Ser. No. 15/385,925, entitled JAW ACTUATED LOCK ARRANGEMENTS FOR PREVENTING ADVANCEMENT OF A FIRING MEMBER IN A SURGICAL END EFFECTOR UNLESS 45 AN UNFIRED CARTRIDGE IS INSTALLED IN THE END EFFECTOR, now U.S. Pat. No. 10,517,595;

U.S. patent application Ser. No. 15/385,926, entitled AXIALLY MOVABLE CLOSURE SYSTEM ARRANGEMENTS FOR APPLYING CLOSURE 50 MOTIONS TO JAWS OF SURGICAL INSTRUMENTS, now U.S. Patent Application Publication No. 2018/0168577;

U.S. patent application Ser. No. 15/385,928, entitled 55 PROTECTIVE COVER ARRANGEMENTS FOR A JOINT INTERFACE BETWEEN A MOVABLE JAW AND ACTUATOR SHAFT OF A SURGICAL INSTRUMENT, now U.S. Patent Application Publication No. 2018/0168578;

U.S. patent application Ser. No. 15/385,930, entitled 60 SURGICAL END EFFECTOR WITH TWO SEPARATE COOPERATING OPENING FEATURES FOR OPENING AND CLOSING END EFFECTOR JAWS, now U.S. Patent Application Publication No. 2018/0168579;

U.S. patent application Ser. No. 15/385,932, entitled 65 ARTICULATABLE SURGICAL END EFFECTOR

WITH ASYMMETRIC SHAFT ARRANGEMENT, now U.S. Patent Application Publication No. 2018/0168628;

U.S. patent application Ser. No. 15/385,933, entitled ARTICULATABLE SURGICAL INSTRUMENT WITH INDEPENDENT PIVOTABLE LINKAGE 5 DISTAL OF AN ARTICULATION LOCK, now U.S. Pat. No. 10,603,036;

U.S. patent application Ser. No. 15/385,934, entitled ARTICULATION LOCK ARRANGEMENTS FOR LOCKING AN END EFFECTOR IN AN ARTICULATED POSITION IN RESPONSE TO ACTUATION OF A JAW CLOSURE SYSTEM, now U.S. Pat. No. 10,582,928;

U.S. patent application Ser. No. 15/385,935, entitled 10 LATERALLY ACTUATABLE ARTICULATION LOCK ARRANGEMENTS FOR LOCKING AN END EFFECTOR OF A SURGICAL INSTRUMENT IN AN ARTICULATED CONFIGURATION, now U.S. Pat. No. 10,524,789;

U.S. patent application Ser. No. 15/385,936, entitled ARTICULATABLE SURGICAL INSTRUMENTS WITH ARTICULATION STROKE AMPLIFICATION FEATURES, now U.S. Pat. No. 10,517,596;

U.S. patent application Ser. No. 14/318,996, entitled 15 FASTENER CARTRIDGES INCLUDING EXTENSIONS HAVING DIFFERENT CONFIGURATIONS, now U.S. Patent Application Publication No. 2015/0297228;

U.S. patent application Ser. No. 14/319,006, entitled FASTENER CARTRIDGE COMPRISING FASTENER CAVITIES INCLUDING FASTENER CONTROL FEATURES, now U.S. Pat. No. 10,010,324;

U.S. patent application Ser. No. 14/318,991, entitled 20 SURGICAL FASTENER CARTRIDGES WITH DRIVER STABILIZING ARRANGEMENTS, now U.S. Pat. No. 9,833,241;

U.S. patent application Ser. No. 14/319,004, entitled SURGICAL END EFFECTORS WITH FIRING ELEMENT MONITORING ARRANGEMENTS, now U.S. Pat. No. 9,844,369;

U.S. patent application Ser. No. 14/319,008, entitled 25 FASTENER CARTRIDGE COMPRISING NON-UNIFORM FASTENERS, now U.S. Pat. No. 10,299,792;

U.S. patent application Ser. No. 14/318,997, entitled FASTENER CARTRIDGE COMPRISING DEPLOYABLE TISSUE ENGAGING MEMBERS, now U.S. Pat. No. 10,561,422;

U.S. patent application Ser. No. 14/319,002, entitled 30 FASTENER CARTRIDGE COMPRISING TISSUE CONTROL FEATURES, now U.S. Pat. No. 9,877,721;

U.S. patent application Ser. No. 14/319,013, entitled FASTENER CARTRIDGE ASSEMBLIES AND STAPLE RETAINER COVER ARRANGEMENTS, now U.S. Patent Application Publication No. 2015/0297233; and

U.S. patent application Ser. No. 14/319,016, entitled 35 FASTENER CARTRIDGE INCLUDING A LAYER ATTACHED THERETO, now U.S. Pat. No. 10,470,768.

Applicant of the present application owns the following U.S. patent applications that were filed on Jun. 24, 2016 and which are each herein incorporated by reference in their 40 respective entireties:

U.S. patent application Ser. No. 15/191,775, entitled 45 STAPLE CARTRIDGE COMPRISING WIRE

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STAPLES AND STAMPED STAPLES, now U.S. Patent Application Publication No. 2017/0367695;

U.S. patent application Ser. No. 15/191,807, entitled STAPLING SYSTEM FOR USE WITH WIRE STAPLES AND STAMPED STAPLES, now U.S. Pat. No. 10,702,270;

U.S. patent application Ser. No. 15/191,834, entitled STAMPED STAPLES AND STAPLE CARTRIDGES USING THE SAME, now U.S. Pat. No. 10,542,979;

U.S. patent application Ser. No. 15/191,788, entitled STAPLE CARTRIDGE COMPRISING OVER-DRIVEN STAPLES, now U.S. Pat. No. 10,675,024; and

U.S. patent application Ser. No. 15/191,818, entitled STAPLE CARTRIDGE COMPRISING OFFSET LONGITUDINAL STAPLE ROWS, now U.S. Pat. No. 10,893,863.

Applicant of the present application owns the following U.S. patent applications that were filed on Jun. 24, 2016 and which are each herein incorporated by reference in their respective entireties:

U.S. Design patent application Ser. No. 29/569,218, entitled SURGICAL FASTENER, now U.S. Design Pat. No. D826,405;

U.S. Design patent application Ser. No. 29/569,227, entitled SURGICAL FASTENER, now U.S. Design Pat. No. D822,206;

U.S. Design patent application Ser. No. 29/569,259, entitled SURGICAL FASTENER CARTRIDGE, now U.S. Design Pat. No. D847,989; and

U.S. Design patent application Ser. No. 29/569,264, entitled SURGICAL FASTENER CARTRIDGE, now U.S. Design Pat. No. D850,617.

Applicant of the present application owns the following patent applications that were filed on Apr. 1, 2016 and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 15/089,325, entitled METHOD FOR OPERATING A SURGICAL STAPLING SYSTEM, now U.S. Patent Application Publication No. 2017/0281171;

U.S. patent application Ser. No. 15/089,321, entitled MODULAR SURGICAL STAPLING SYSTEM COMPRISING A DISPLAY, now U.S. Pat. No. 10,271,851;

U.S. patent application Ser. No. 15/089,326, entitled SURGICAL STAPLING SYSTEM COMPRISING A DISPLAY INCLUDING A RE-ORIENTABLE DISPLAY FIELD, now U.S. Pat. No. 10,433,849;

U.S. patent application Ser. No. 15/089,263, entitled SURGICAL INSTRUMENT HANDLE ASSEMBLY WITH RECONFIGURABLE GRIP PORTION, now U.S. Pat. No. 10,307,159;

U.S. patent application Ser. No. 15/089,262, entitled ROTARY POWERED SURGICAL INSTRUMENT WITH MANUALLY ACTUATABLE BAILOUT SYSTEM, now U.S. Pat. No. 10,357,246;

U.S. patent application Ser. No. 15/089,277, entitled SURGICAL CUTTING AND STAPLING END EFFECTOR WITH ANVIL CONCENTRIC DRIVE MEMBER, now U.S. Pat. No. 10,531,874;

U.S. patent application Ser. No. 15/089,296, entitled INTERCHANGEABLE SURGICAL TOOL ASSEMBLY WITH A SURGICAL END EFFECTOR THAT IS SELECTIVELY ROTATABLE ABOUT A SHAFT AXIS, now U.S. Pat. No. 10,413,293;

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U.S. patent application Ser. No. 15/089,258, entitled SURGICAL STAPLING SYSTEM COMPRISING A SHIFTABLE TRANSMISSION, now U.S. Pat. No. 10,342,543;

U.S. patent application Ser. No. 15/089,278, entitled SURGICAL STAPLING SYSTEM CONFIGURED TO PROVIDE SELECTIVE CUTTING OF TISSUE, now U.S. Pat. No. 10,420,552;

U.S. patent application Ser. No. 15/089,284, entitled SURGICAL STAPLING SYSTEM COMPRISING A CONTOURABLE SHAFT, now U.S. Patent Application Publication No. 2017/0281186;

U.S. patent application Ser. No. 15/089,295, entitled SURGICAL STAPLING SYSTEM COMPRISING A TISSUE COMPRESSION LOCKOUT, now U.S. Pat. No. 10,856,867;

U.S. patent application Ser. No. 15/089,300, entitled SURGICAL STAPLING SYSTEM COMPRISING AN UNCLAMPING LOCKOUT, now U.S. Pat. No. 10,456,140;

U.S. patent application Ser. No. 15/089,196, entitled SURGICAL STAPLING SYSTEM COMPRISING A JAW CLOSURE LOCKOUT, now U.S. Pat. No. 10,568,632;

U.S. patent application Ser. No. 15/089,203, entitled SURGICAL STAPLING SYSTEM COMPRISING A JAW ATTACHMENT LOCKOUT, now U.S. Pat. No. 10,542,991;

U.S. patent application Ser. No. 15/089,210, entitled SURGICAL STAPLING SYSTEM COMPRISING A SPENT CARTRIDGE LOCKOUT, now U.S. Pat. No. 10,478,190;

U.S. patent application Ser. No. 15/089,324, entitled SURGICAL INSTRUMENT COMPRISING A SHIFTING MECHANISM, now U.S. Pat. No. 10,314,582;

U.S. patent application Ser. No. 15/089,335, entitled SURGICAL STAPLING INSTRUMENT COMPRISING MULTIPLE LOCKOUTS, now U.S. Pat. No. 10,485,542;

U.S. patent application Ser. No. 15/089,339, entitled SURGICAL STAPLING INSTRUMENT, now U.S. Patent Application Publication No. 2017/0281173;

U.S. patent application Ser. No. 15/089,253, entitled SURGICAL STAPLING SYSTEM CONFIGURED TO APPLY ANNULAR ROWS OF STAPLES HAVING DIFFERENT HEIGHTS, now U.S. Pat. No. 10,413,297;

U.S. patent application Ser. No. 15/089,304, entitled SURGICAL STAPLING SYSTEM COMPRISING A GROOVED FORMING POCKET, now U.S. Pat. No. 10,285,705;

U.S. patent application Ser. No. 15/089,331, entitled ANVIL MODIFICATION MEMBERS FOR SURGICAL STAPLERS, now U.S. Pat. No. 10,376,263;

U.S. patent application Ser. No. 15/089,336, entitled STAPLE CARTRIDGES WITH ATRAUMATIC FEATURES, now U.S. Pat. No. 10,709,446;

U.S. patent application Ser. No. 15/089,312, entitled CIRCULAR STAPLING SYSTEM COMPRISING AN INCISABLE TISSUE SUPPORT, now U.S. Patent Application Publication No. 2017/0281189;

U.S. patent application Ser. No. 15/089,309, entitled CIRCULAR STAPLING SYSTEM COMPRISING ROTARY FIRING SYSTEM, now U.S. Pat. No. 10,675,021; and

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U.S. patent application Ser. No. 15/089,349, entitled CIRCULAR STAPLING SYSTEM COMPRISING LOAD CONTROL, now U.S. Pat. No. 10,682,136.

Applicant of the present application also owns the U.S. patent applications identified below which were filed on Dec. 30, 2015 which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 14/984,488, entitled MECHANISMS FOR COMPENSATING FOR BATTERY PACK FAILURE IN POWERED SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,292,704;

U.S. patent application Ser. No. 14/984,525, entitled MECHANISMS FOR COMPENSATING FOR DRIVETRAIN FAILURE IN POWERED SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,368,865; and U.S. patent application Ser. No. 14/984,552, entitled SURGICAL INSTRUMENTS WITH SEPARABLE MOTORS AND MOTOR CONTROL CIRCUITS, now U.S. Pat. No. 10,265,068.

Applicant of the present application also owns the U.S. patent applications identified below which were filed on Feb. 9, 2016, which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 15/019,220, entitled SURGICAL INSTRUMENT WITH ARTICULATING AND AXIALLY TRANSLATABLE END EFFECTOR, now U.S. Pat. No. 10,245,029;

U.S. patent application Ser. No. 15/019,228, entitled SURGICAL INSTRUMENTS WITH MULTIPLE LINK ARTICULATION ARRANGEMENTS, now U.S. Pat. No. 10,433,837;

U.S. patent application Ser. No. 15/019,196, entitled SURGICAL INSTRUMENT ARTICULATION MECHANISM WITH SLOTTED SECONDARY CONSTRAINT, now U.S. Pat. No. 10,413,291;

U.S. patent application Ser. No. 15/019,206, entitled SURGICAL INSTRUMENTS WITH AN END EFFECTOR THAT IS HIGHLY ARTICULATABLE RELATIVE TO AN ELONGATE SHAFT ASSEMBLY, now U.S. Pat. No. 10,653,413;

U.S. patent application Ser. No. 15/019,215, entitled SURGICAL INSTRUMENTS WITH NON-SYMMETRICAL ARTICULATION ARRANGEMENTS, now U.S. Patent Application Publication No. 2017/0224332;

U.S. patent application Ser. No. 15/019,227, entitled ARTICULATABLE SURGICAL INSTRUMENTS WITH SINGLE ARTICULATION LINK ARRANGEMENTS, now U.S. Patent Application Publication No. 2017/0224334;

U.S. patent application Ser. No. 15/019,235, entitled SURGICAL INSTRUMENTS WITH TENSIONING ARRANGEMENTS FOR CABLE DRIVEN ARTICULATION SYSTEMS, now U.S. Pat. No. 10,245,030;

U.S. patent application Ser. No. 15/019,230, entitled ARTICULATABLE SURGICAL INSTRUMENTS WITH OFF-AXIS FIRING BEAM ARRANGEMENTS, now U.S. Pat. No. 10,588,625; and

U.S. patent application Ser. No. 15/019,245, entitled SURGICAL INSTRUMENTS WITH CLOSURE STROKE REDUCTION ARRANGEMENTS, now U.S. Pat. No. 10,470,764.

[Applicant of the present application also owns the U.S. patent applications identified below which were filed on Feb. 12, 2016, which are each herein incorporated by reference in their respective entirety:

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U.S. patent application Ser. No. 15/043,254, entitled MECHANISMS FOR COMPENSATING FOR DRIVETRAIN FAILURE IN POWERED SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,258,331;

U.S. patent application Ser. No. 15/043,259, entitled MECHANISMS FOR COMPENSATING FOR DRIVETRAIN FAILURE IN POWERED SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,448,948;

U.S. patent application Ser. No. 15/043,275, entitled MECHANISMS FOR COMPENSATING FOR DRIVETRAIN FAILURE IN POWERED SURGICAL INSTRUMENTS, now U.S. Patent Application Publication No. 2017/0231627; and

U.S. patent application Ser. No. 15/043,289, entitled MECHANISMS FOR COMPENSATING FOR DRIVETRAIN FAILURE IN POWERED SURGICAL INSTRUMENTS, now U.S. Patent Application Publication No. 2017/0231628.

Applicant of the present application owns the following patent applications that were filed on Jun. 18, 2015 and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 14/742,925, entitled SURGICAL END EFFECTORS WITH POSITIVE JAW OPENING ARRANGEMENTS, now U.S. Pat. No. 10,182,818;

U.S. patent application Ser. No. 14/742,941, entitled SURGICAL END EFFECTORS WITH DUAL CAM ACTUATED JAW CLOSING FEATURES, now U.S. Pat. No. 10,052,102;

U.S. patent application Ser. No. 14/742,933, entitled SURGICAL STAPLING INSTRUMENTS WITH LOCKOUT ARRANGEMENTS FOR PREVENTING FIRING SYSTEM ACTUATION WHEN A CARTRIDGE IS SPENT OR MISSING, now U.S. Pat. No. 10,154,841;

U.S. patent application Ser. No. 14/742,914, entitled MOVABLE FIRING BEAM SUPPORT ARRANGEMENTS FOR ARTICULATABLE SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,405,863;

U.S. patent application Ser. No. 14/742,900, entitled ARTICULATABLE SURGICAL INSTRUMENTS WITH COMPOSITE FIRING BEAM STRUCTURES WITH CENTER FIRING SUPPORT MEMBER FOR ARTICULATION SUPPORT, now U.S. Pat. No. 10,335,149;

U.S. patent application Ser. No. 14/742,885, entitled DUAL ARTICULATION DRIVE SYSTEM ARRANGEMENTS FOR ARTICULATABLE SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,368,861; and

U.S. patent application Ser. No. 14/742,876, entitled PUSH/PULL ARTICULATION DRIVE SYSTEMS FOR ARTICULATABLE SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,178,992.

Applicant of the present application owns the following patent applications that were filed on Mar. 6, 2015 and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 14/640,746, entitled POWERED SURGICAL INSTRUMENT, now U.S. Pat. No. 9,808,246;

U.S. patent application Ser. No. 14/640,795, entitled MULTIPLE LEVEL THRESHOLDS TO MODIFY OPERATION OF POWERED SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,441,279;

U.S. patent application Ser. No. 14/640,832, entitled ADAPTIVE TISSUE COMPRESSION TECHNIQUES TO ADJUST CLOSURE RATES FOR MULTIPLE TISSUE TYPES, now U.S. Pat. No. 10,687,806;

U.S. patent application Ser. No. 14/640,935, entitled OVERLAID MULTI SENSOR RADIO FREQUENCY (RF) ELECTRODE SYSTEM TO MEASURE TISSUE COMPRESSION, now U.S. Pat. No. 10,548,504;

U.S. patent application Ser. No. 14/640,831, entitled MONITORING SPEED CONTROL AND PRECISION INCREMENTING OF MOTOR FOR POWERED SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,895,148;

U.S. patent application Ser. No. 14/640,859, entitled TIME DEPENDENT EVALUATION OF SENSOR DATA TO DETERMINE STABILITY, CREEP, AND VISCOELASTIC ELEMENTS OF MEASURES, now U.S. Pat. No. 10,052,044;

U.S. patent application Ser. No. 14/640,817, entitled INTERACTIVE FEEDBACK SYSTEM FOR POWERED SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,924,961;

U.S. patent application Ser. No. 14/640,844, entitled CONTROL TECHNIQUES AND SUB-PROCESSOR CONTAINED WITHIN MODULAR SHAFT WITH SELECT CONTROL PROCESSING FROM HANDLE, now U.S. Pat. No. 10,045,776;

U.S. patent application Ser. No. 14/640,837, entitled SMART SENSORS WITH LOCAL SIGNAL PROCESSING, now U.S. Pat. No. 9,993,248;

U.S. patent application Ser. No. 14/640,765, entitled SYSTEM FOR DETECTING THE MIS-INSERTION OF A STAPLE CARTRIDGE INTO A SURGICAL STAPLER, now U.S. Pat. No. 10,617,412;

U.S. patent application Ser. No. 14/640,799, entitled SIGNAL AND POWER COMMUNICATION SYSTEM POSITIONED ON A ROTATABLE SHAFT, now U.S. Pat. No. 9,901,342; and

U.S. patent application Ser. No. 14/640,780, entitled SURGICAL INSTRUMENT COMPRISING A LOCKABLE BATTERY HOUSING, now U.S. Pat. No. 10,245,033.

Applicant of the present application owns the following patent applications that were filed on Feb. 27, 2015, and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 14/633,576, entitled SURGICAL INSTRUMENT SYSTEM COMPRISING AN INSPECTION STATION, now U.S. Pat. No. 10,045,779;

U.S. patent application Ser. No. 14/633,546, entitled SURGICAL APPARATUS CONFIGURED TO ASSESS WHETHER A PERFORMANCE PARAMETER OF THE SURGICAL APPARATUS IS WITHIN AN ACCEPTABLE PERFORMANCE BAND, now U.S. Pat. No. 10,180,463;

U.S. patent application Ser. No. 14/633,560, entitled SURGICAL CHARGING SYSTEM THAT CHARGES AND/OR CONDITIONS ONE OR MORE BATTERIES, now U.S. Patent Application Publication No. 2016/0249910;

U.S. patent application Ser. No. 14/633,566, entitled CHARGING SYSTEM THAT ENABLES EMERGENCY RESOLUTIONS FOR CHARGING A BATTERY, now U.S. Pat. No. 10,182,816;

U.S. patent application Ser. No. 14/633,555, entitled SYSTEM FOR MONITORING WHETHER A SURGICAL INSTRUMENT NEEDS TO BE SERVICED, now U.S. Pat. No. 10,321,907;

U.S. patent application Ser. No. 14/633,542, entitled REINFORCED BATTERY FOR A SURGICAL INSTRUMENT, now U.S. Pat. No. 9,931,118;

U.S. patent application Ser. No. 14/633,548, entitled POWER ADAPTER FOR A SURGICAL INSTRUMENT, now U.S. Pat. No. 10,245,028;

U.S. patent application Ser. No. 14/633,526, entitled ADAPTABLE SURGICAL INSTRUMENT HANDLE, now U.S. Pat. No. 9,993,258;

U.S. patent application Ser. No. 14/633,541, entitled MODULAR STAPLING ASSEMBLY, now U.S. Pat. No. 10,226,250; and

U.S. patent application Ser. No. 14/633,562, entitled SURGICAL APPARATUS CONFIGURED TO TRACK AN END-OF-LIFE PARAMETER, now U.S. Pat. No. 10,159,483.

Applicant of the present application owns the following patent applications that were filed on Dec. 18, 2014 and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 14/574,478, entitled SURGICAL INSTRUMENT SYSTEMS COMPRISING AN ARTICULATABLE END EFFECTOR AND MEANS FOR ADJUSTING THE FIRING STROKE OF A FIRING MEMBER, now U.S. Pat. No. 9,844,374;

U.S. patent application Ser. No. 14/574,483, entitled SURGICAL INSTRUMENT ASSEMBLY COMPRISING LOCKABLE SYSTEMS, now U.S. Pat. No. 10,188,385;

U.S. patent application Ser. No. 14/575,139, entitled DRIVE ARRANGEMENTS FOR ARTICULATABLE SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,844,375;

U.S. patent application Ser. No. 14/575,148, entitled LOCKING ARRANGEMENTS FOR DETACHABLE SHAFT ASSEMBLIES WITH ARTICULATABLE SURGICAL END EFFECTORS, now U.S. Pat. No. 10,085,748;

U.S. patent application Ser. No. 14/575,130, entitled SURGICAL INSTRUMENT WITH AN ANVIL THAT IS SELECTIVELY MOVABLE ABOUT A DISCRETE NON-MOVABLE AXIS RELATIVE TO A STAPLE CARTRIDGE, now U.S. Pat. No. 10,245,027;

U.S. patent application Ser. No. 14/575,143, entitled SURGICAL INSTRUMENTS WITH IMPROVED CLOSURE ARRANGEMENTS, now U.S. Pat. No. 10,004,501;

U.S. patent application Ser. No. 14/575,117, entitled SURGICAL INSTRUMENTS WITH ARTICULATABLE END EFFECTORS AND MOVABLE FIRING BEAM SUPPORT ARRANGEMENTS, now U.S. Pat. No. 9,943,309;

U.S. patent application Ser. No. 14/575,154, entitled SURGICAL INSTRUMENTS WITH ARTICULATABLE END EFFECTORS AND IMPROVED FIRING BEAM SUPPORT ARRANGEMENTS, now U.S. Pat. No. 9,968,355;

U.S. patent application Ser. No. 14/574,493, entitled SURGICAL INSTRUMENT ASSEMBLY COMPRISING A FLEXIBLE ARTICULATION SYSTEM, now U.S. Pat. No. 9,987,000; and

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U.S. patent application Ser. No. 14/574,500, entitled SURGICAL INSTRUMENT ASSEMBLY COMPRISING A LOCKABLE ARTICULATION SYSTEM, now U.S. Pat. No. 10,117,649.

Applicant of the present application owns the following patent applications that were filed on Mar. 1, 2013 and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 13/782,295, entitled ARTICULATABLE SURGICAL INSTRUMENTS WITH CONDUCTIVE PATHWAYS FOR SIGNAL COMMUNICATION, now U.S. Pat. No. 9,700,309;

U.S. patent application Ser. No. 13/782,323, entitled ROTARY POWERED ARTICULATION JOINTS FOR SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,782,169;

U.S. patent application Ser. No. 13/782,338, entitled THUMBWHEEL SWITCH ARRANGEMENTS FOR SURGICAL INSTRUMENTS, now U.S. Patent Application Publication No. 2014/0249557;

U.S. patent application Ser. No. 13/782,499, entitled ELECTROMECHANICAL SURGICAL DEVICE WITH SIGNAL RELAY ARRANGEMENT, now U.S. Pat. No. 9,358,003;

U.S. patent application Ser. No. 13/782,460, entitled MULTIPLE PROCESSOR MOTOR CONTROL FOR MODULAR SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,554,794;

U.S. patent application Ser. No. 13/782,358, entitled JOYSTICK SWITCH ASSEMBLIES FOR SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,326,767;

U.S. patent application Ser. No. 13/782,481, entitled SENSOR STRAIGHTENED END EFFECTOR DURING REMOVAL THROUGH TROCAR, now U.S. Pat. No. 9,468,438;

U.S. patent application Ser. No. 13/782,518, entitled CONTROL METHODS FOR SURGICAL INSTRUMENTS WITH REMOVABLE IMPLEMENT PORTIONS, now U.S. Patent Application Publication No. 2014/0246475;

U.S. patent application Ser. No. 13/782,375, entitled ROTARY POWERED SURGICAL INSTRUMENTS WITH MULTIPLE DEGREES OF FREEDOM, now U.S. Pat. No. 9,398,911; and

U.S. patent application Ser. No. 13/782,536, entitled SURGICAL INSTRUMENT SOFT STOP, now U.S. Pat. No. 9,307,986.

Applicant of the present application also owns the following patent applications that were filed on Mar. 14, 2013 and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 13/803,097, entitled ARTICULATABLE SURGICAL INSTRUMENT COMPRISING A FIRING DRIVE, now U.S. Pat. No. 9,687,230;

U.S. patent application Ser. No. 13/803,193, entitled CONTROL ARRANGEMENTS FOR A DRIVE MEMBER OF A SURGICAL INSTRUMENT, now U.S. Pat. No. 9,332,987;

U.S. patent application Ser. No. 13/803,053, entitled INTERCHANGEABLE SHAFT ASSEMBLIES FOR USE WITH A SURGICAL INSTRUMENT, now U.S. Pat. No. 9,883,860;

U.S. patent application Ser. No. 13/803,086, entitled ARTICULATABLE SURGICAL INSTRUMENT

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COMPRISING AN ARTICULATION LOCK, now U.S. Patent Application Publication No. 2014/0263541;

U.S. patent application Ser. No. 13/803,210, entitled SENSOR ARRANGEMENTS FOR ABSOLUTE POSITIONING SYSTEM FOR SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,808,244;

U.S. patent application Ser. No. 13/803,148, entitled MULTI-FUNCTION MOTOR FOR A SURGICAL INSTRUMENT, now U.S. Pat. No. 10,470,762;

U.S. patent application Ser. No. 13/803,066, entitled DRIVE SYSTEM LOCKOUT ARRANGEMENTS FOR MODULAR SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,629,623;

U.S. patent application Ser. No. 13/803,117, entitled ARTICULATION CONTROL SYSTEM FOR ARTICULATABLE SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,351,726;

U.S. patent application Ser. No. 13/803,130, entitled DRIVE TRAIN CONTROL ARRANGEMENTS FOR MODULAR SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,351,727; and

U.S. patent application Ser. No. 13/803,159, entitled METHOD AND SYSTEM FOR OPERATING A SURGICAL INSTRUMENT, now U.S. Pat. No. 9,888,919.

Applicant of the present application also owns the following patent application that was filed on Mar. 7, 2014 and is herein incorporated by reference in its entirety:

U.S. patent application Ser. No. 14/200,111, entitled CONTROL SYSTEMS FOR SURGICAL INSTRUMENTS, now U.S. Pat. No. 9,629,629.

Applicant of the present application also owns the following patent applications that were filed on Mar. 26, 2014 and are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 14/226,106, entitled POWER MANAGEMENT CONTROL SYSTEMS FOR SURGICAL INSTRUMENTS, now U.S. Patent Application Publication No. 2015/0272582;

U.S. patent application Ser. No. 14/226,099, entitled STERILIZATION VERIFICATION CIRCUIT, now U.S. Pat. No. 9,826,977;

U.S. patent application Ser. No. 14/226,094, entitled VERIFICATION OF NUMBER OF BATTERY EXCHANGES/PROCEDURE COUNT, now U.S. Patent Application Publication No. 2015/0272580;

U.S. patent application Ser. No. 14/226,117, entitled POWER MANAGEMENT THROUGH SLEEP OPTIONS OF SEGMENTED CIRCUIT AND WAKE UP CONTROL, now U.S. Pat. No. 10,013,049;

U.S. patent application Ser. No. 14/226,075, entitled MODULAR POWERED SURGICAL INSTRUMENT WITH DETACHABLE SHAFT ASSEMBLIES, now U.S. Pat. No. 9,743,929;

U.S. patent application Ser. No. 14/226,093, entitled FEEDBACK ALGORITHMS FOR MANUAL BAIL-OUT SYSTEMS FOR SURGICAL INSTRUMENTS, now U.S. Pat. No. 10,028,761;

U.S. patent application Ser. No. 14/226,116, entitled SURGICAL INSTRUMENT UTILIZING SENSOR ADAPTATION, now U.S. Patent Application Publication No. 2015/0272571;

U.S. patent application Ser. No. 14/226,071, entitled SURGICAL INSTRUMENT CONTROL CIRCUIT HAVING A SAFETY PROCESSOR, now U.S. Pat. No. 9,690,362;

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U.S. patent application Ser. No. 14/226,097, entitled SURGICAL INSTRUMENT COMPRISING INTER-ACTIVE SYSTEMS, now U.S. Pat. No. 9,820,738;

U.S. patent application Ser. No. 14/226,126, entitled INTERFACE SYSTEMS FOR USE WITH SURGI- 5 CAL INSTRUMENTS, now U.S. Pat. No. 10,004,497;

U.S. patent application Ser. No. 14/226,133, entitled MODULAR SURGICAL INSTRUMENT SYSTEM, now U.S. Patent Application Publication No. 2015/ 10 0272557;

U.S. patent application Ser. No. 14/226,081, entitled SYSTEMS AND METHODS FOR CONTROLLING A SEGMENTED CIRCUIT, now U.S. Pat. No. 9,804, 618;

U.S. patent application Ser. No. 14/226,076, entitled 15 POWER MANAGEMENT THROUGH SEGMENTED CIRCUIT AND VARIABLE VOLTAGE PROTECTION, now U.S. Pat. No. 9,733,663;

U.S. patent application Ser. No. 14/226,111, entitled SURGICAL STAPLING INSTRUMENT SYSTEM, 20 now U.S. Pat. No. 9,750,499; and

U.S. patent application Ser. No. 14/226,125, entitled SURGICAL INSTRUMENT COMPRISING A ROTATABLE SHAFT, now U.S. Pat. No. 10,201,364.

Applicant of the present application also owns the fol- 25 lowing patent applications that were filed on Sep. 5, 2014 and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 14/479,103, entitled 30 CIRCUITRY AND SENSORS FOR POWERED MEDICAL DEVICE, now U.S. Pat. No. 10,111,679;

U.S. patent application Ser. No. 14/479,119, entitled ADJUNCT WITH INTEGRATED SENSORS TO QUANTIFY TISSUE COMPRESSION, now U.S. Pat. No. 9,724,094;

U.S. patent application Ser. No. 14/478,908, entitled 35 MONITORING DEVICE DEGRADATION BASED ON COMPONENT EVALUATION, now U.S. Pat. No. 9,737,301;

U.S. patent application Ser. No. 14/478,895, entitled 40 MULTIPLE SENSORS WITH ONE SENSOR AFFECTING A SECOND SENSOR'S OUTPUT OR INTERPRETATION, now U.S. Pat. No. 9,757,128;

U.S. patent application Ser. No. 14/479,110, entitled 45 POLARITY OF HALL MAGNET TO IDENTIFY CARTRIDGE TYPE, now U.S. Pat. No. 10,016,199;

U.S. patent application Ser. No. 14/479,098, entitled SMART CARTRIDGE WAKE UP OPERATION AND DATA RETENTION, now U.S. Pat. No. 10,135,242;

U.S. patent application Ser. No. 14/479,115, entitled 50 MULTIPLE MOTOR CONTROL FOR POWERED MEDICAL DEVICE, now U.S. Pat. No. 9,788,836; and

U.S. patent application Ser. No. 14/479,108, entitled 55 LOCAL DISPLAY OF TISSUE PARAMETER STABILIZATION, now U.S. Patent Application Publication No. 2016/0066913.

Applicant of the present application also owns the fol- 60 lowing patent applications that were filed on Apr. 9, 2014 and which are each herein incorporated by reference in their respective entirety:

U.S. patent application Ser. No. 14/248,590, entitled MOTOR DRIVEN SURGICAL INSTRUMENTS WITH LOCKABLE DUAL DRIVE SHAFTS, now U.S. Pat. No. 9,826,976;

U.S. patent application Ser. No. 14/248,581, entitled SURGICAL INSTRUMENT COMPRISING A CLOS-

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ING DRIVE AND A FIRING DRIVE OPERATED FROM THE SAME ROTATABLE OUTPUT, now U.S. Pat. No. 9,649,110;

U.S. patent application Ser. No. 14/248,595, entitled 5 SURGICAL SYSTEM COMPRISING FIRST AND SECOND DRIVE SYSTEMS, now U.S. Pat. No. 9,844,368;

U.S. patent application Ser. No. 14/248,588, entitled POWERED LINEAR SURGICAL STAPLER, now U.S. Pat. No. 10,405,857;

U.S. patent application Ser. No. 14/248,591, entitled SURGICAL INSTRUMENT COMPRISING A GAP SETTING SYSTEM, now U.S. Pat. No. 10,149,680;

U.S. patent application Ser. No. 14/248,584, entitled 10 MODULAR MOTOR DRIVEN SURGICAL INSTRUMENTS WITH ALIGNMENT FEATURES FOR ALIGNING ROTARY DRIVE SHAFTS WITH SURGICAL END EFFECTOR SHAFTS, now U.S. Pat. No. 9,801,626;

U.S. patent application Ser. No. 14/248,587, entitled 15 POWERED SURGICAL STAPLER, now U.S. Pat. No. 9,867,612;

U.S. patent application Ser. No. 14/248,586, entitled DRIVE SYSTEM DECOUPLING ARRANGEMENT FOR A SURGICAL INSTRUMENT, now U.S. Pat. No. 10,136,887; and

U.S. patent application Ser. No. 14/248,607, entitled 20 MODULAR MOTOR DRIVEN SURGICAL INSTRUMENTS WITH STATUS INDICATION ARRANGEMENTS, now U.S. Pat. No. 9,814,460.

Applicant of the present application also owns the fol- 25 lowing patent applications that were filed on Apr. 16, 2013 and which are each herein incorporated by reference in their respective entirety:

U.S. Provisional Patent Application Ser. No. 61/812,365, 30 entitled SURGICAL INSTRUMENT WITH MULTIPLE FUNCTIONS PERFORMED BY A SINGLE MOTOR;

U.S. Provisional Patent Application Ser. No. 61/812,376, entitled LINEAR CUTTER WITH POWER;

U.S. Provisional Patent Application Ser. No. 61/812,382, 35 entitled LINEAR CUTTER WITH MOTOR AND PISTOL GRIP;

U.S. Provisional Patent Application Ser. No. 61/812,385, entitled SURGICAL INSTRUMENT HANDLE WITH MULTIPLE ACTUATION MOTORS AND MOTOR CONTROL; and

U.S. Provisional Patent Application Ser. No. 61/812,372, 40 entitled SURGICAL INSTRUMENT WITH MULTIPLE FUNCTIONS PERFORMED BY A SINGLE MOTOR.

Applicant of the present application owns the following 45 U.S. Provisional patent applications, filed on Dec. 28, 2017, the disclosure of each of which is herein incorporated by reference in its entirety:

U.S. Provisional Patent Application Ser. No. 62/611,341, entitled INTERACTIVE SURGICAL PLATFORM;

U.S. Provisional Patent Application Ser. No. 62/611,340, 50 entitled CLOUD-BASED MEDICAL ANALYTICS; and

U.S. Provisional Patent Application Ser. No. 62/611,339, entitled ROBOT ASSISTED SURGICAL PLATFORM.

Applicant of the present application owns the following 55 U.S. Provisional patent applications, filed on Mar. 28, 2018, each of which is herein incorporated by reference in its entirety:

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U.S. Provisional Patent Application Ser. No. 62/649,302, entitled INTERACTIVE SURGICAL SYSTEMS WITH ENCRYPTED COMMUNICATION CAPABILITIES;

U.S. Provisional Patent Application Ser. No. 62/649,294, 5 entitled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD;

U.S. Provisional Patent Application Ser. No. 62/649,300, 10 entitled SURGICAL HUB SITUATIONAL AWARENESS;

U.S. Provisional Patent Application Ser. No. 62/649,309, 15 entitled SURGICAL HUB SPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THEATER;

U.S. Provisional Patent Application Ser. No. 62/649,310, entitled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS;

U.S. Provisional Patent Application Ser. No. 62/649,291, 20 entitled USE OF LASER LIGHT AND RED-GREEN-BLUE COLORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT;

U.S. Provisional Patent Application Ser. No. 62/649,296, 25 entitled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL DEVICES;

U.S. Provisional Patent Application Ser. No. 62/649,333, 30 entitled CLOUD-BASED MEDICAL ANALYTICS FOR CUSTOMIZATION AND RECOMMENDATIONS TO A USER;

U.S. Provisional Patent Application Ser. No. 62/649,327, 35 entitled CLOUD-BASED MEDICAL ANALYTICS FOR SECURITY AND AUTHENTICATION TRENDS AND REACTIVE MEASURES;

U.S. Provisional Patent Application Ser. No. 62/649,315, 40 entitled DATA HANDLING AND PRIORITIZATION IN A CLOUD ANALYTICS NETWORK;

U.S. Provisional Patent Application Ser. No. 62/649,313, 45 entitled CLOUD INTERFACE FOR COUPLED SURGICAL DEVICES;

U.S. Provisional Patent Application Ser. No. 62/649,320, 50 entitled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS;

U.S. Provisional Patent Application Ser. No. 62/649,307, 55 entitled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; and

U.S. Provisional Patent Application Ser. No. 62/649,323, 60 entitled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS.

Applicant of the present application owns the following 65 U.S. patent applications, filed on Mar. 29, 2018, each of which is herein incorporated by reference in its entirety:

U.S. patent application Ser. No. 15/940,641, entitled INTERACTIVE SURGICAL SYSTEMS WITH ENCRYPTED COMMUNICATION CAPABILITIES, 55 now U.S. Patent Application Publication No. 2019/0207911;

U.S. patent application Ser. No. 15/940,648, entitled INTERACTIVE SURGICAL SYSTEMS WITH CON- 60 DITION HANDLING OF DEVICES AND DATA CAPABILITIES, now U.S. Patent Application Publication No. 2019/0206004;

U.S. patent application Ser. No. 15/940,656, entitled SURGICAL HUB COORDINATION OF CONTROL 65 AND COMMUNICATION OF OPERATING ROOM DEVICES, now U.S. Patent Application Publication No. 2019/0201141;

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U.S. patent application Ser. No. 15/940,666, entitled SPATIAL AWARENESS OF SURGICAL HUBS IN OPERATING ROOMS, now U.S. Patent Application Publication No. 2019/0206551;

U.S. patent application Ser. No. 15/940,670, entitled COOPERATIVE UTILIZATION OF DATA DERIVED FROM SECONDARY SOURCES BY INTELLIGENT SURGICAL HUBS, now U.S. Patent Application Publication No. 2019/0201116;

U.S. patent application Ser. No. 15/940,677, entitled SURGICAL HUB CONTROL ARRANGEMENTS, 5 now U.S. Patent Application Publication No. 2019/0201143;

U.S. patent application Ser. No. 15/940,632, entitled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD, now U.S. Patent Application Publication No. 2019/0205566;

U.S. patent application Ser. No. 15/940,640, entitled COMMUNICATION HUB AND STORAGE DEVICE FOR STORING PARAMETERS AND STATUS OF A SURGICAL DEVICE TO BE SHARED WITH CLOUD BASED ANALYTICS SYSTEMS, now U.S. Patent Application Publication No. 2019/0200863;

U.S. patent application Ser. No. 15/940,645, entitled SELF DESCRIBING DATA PACKETS GENERATED AT AN ISSUING INSTRUMENT, now U.S. Pat. No. 10,892,899;

U.S. patent application Ser. No. 15/940,649, entitled DATA PAIRING TO INTERCONNECT A DEVICE MEASURED PARAMETER WITH AN OUTCOME, now U.S. Patent Application Publication No. 2019/0205567;

U.S. patent application Ser. No. 15/940,654, entitled SURGICAL HUB SITUATIONAL AWARENESS, now U.S. Patent Application Publication No. 2019/0201140;

U.S. patent application Ser. No. 15/940,663, entitled SURGICAL SYSTEM DISTRIBUTED PROCESSING, now U.S. Patent Application Publication No. 2019/0201033;

U.S. patent application Ser. No. 15/940,668, entitled AGGREGATION AND REPORTING OF SURGICAL HUB DATA, now U.S. Patent Application Publication No. 2019/0201115;

U.S. patent application Ser. No. 15/940,671, entitled SURGICAL HUB SPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THE- 55 ATER, now U.S. Patent Application Publication No. 2019/0201104;

U.S. patent application Ser. No. 15/940,686, entitled DISPLAY OF ALIGNMENT OF STAPLE CARTRIDGE TO PRIOR LINEAR STAPLE LINE, now U.S. Patent Application Publication No. 2019/0201105;

U.S. patent application Ser. No. 15/940,700, entitled STERILE FIELD INTERACTIVE CONTROL DIS- 60 PLAYS, now U.S. Patent Application Publication No. 2019/0205001;

U.S. patent application Ser. No. 15/940,629, entitled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS, now U.S. Patent Application Publication No. 2019/0201112;

U.S. patent application Ser. No. 15/940,704, entitled USE OF LASER LIGHT AND RED-GREEN-BLUE COL- 65 ORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT, now U.S. Patent Application Publication No. 2019/0206050;

U.S. patent application Ser. No. 15/940,722, entitled CHARACTERIZATION OF TISSUE IRREGULARITIES THROUGH THE USE OF MONO-CHROMATIC LIGHT REFRACTIVITY, now U.S. Patent Application Publication No. 2019/0200905; and

U.S. patent application Ser. No. 15/940,742, entitled DUAL CMOS ARRAY IMAGING, now U.S. Patent Application Publication No. 2019/0200906.

Applicant of the present application owns the following U.S. patent applications, filed on Mar. 29, 2018, each of which is herein incorporated by reference in its entirety:

U.S. patent application Ser. No. 15/940,636, entitled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL DEVICES, now U.S. Patent Application Publication No. 2019/0206003;

U.S. patent application Ser. No. 15/940,653, entitled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL HUBS, now U.S. Patent Application Publication No. 2019/0201114;

U.S. patent application Ser. No. 15/940,660, entitled CLOUD-BASED MEDICAL ANALYTICS FOR CUSTOMIZATION AND RECOMMENDATIONS TO A USER, now U.S. Patent Application Publication No. 2019/0206555;

U.S. patent application Ser. No. 15/940,679, entitled CLOUD-BASED MEDICAL ANALYTICS FOR LINKING OF LOCAL USAGE TRENDS WITH THE RESOURCE ACQUISITION BEHAVIORS OF LARGER DATA SET, now U.S. Patent Application Publication No. 2019/0201144;

U.S. patent application Ser. No. 15/940,694, entitled CLOUD-BASED MEDICAL ANALYTICS FOR MEDICAL FACILITY SEGMENTED INDIVIDUALIZATION OF INSTRUMENT FUNCTION, now U.S. Patent Application Publication No. 2019/0201119;

U.S. patent application Ser. No. 15/940,634, entitled CLOUD-BASED MEDICAL ANALYTICS FOR SECURITY AND AUTHENTICATION TRENDS AND REACTIVE MEASURES, now U.S. Patent Application Publication No. 2019/0201138;

U.S. patent application Ser. No. 15/940,706, entitled DATA HANDLING AND PRIORITIZATION IN A CLOUD ANALYTICS NETWORK, now U.S. Patent Application Publication No. 2019/0206561; and

U.S. patent application Ser. No. 15/940,675, entitled CLOUD INTERFACE FOR COUPLED SURGICAL DEVICES, now U.S. Pat. No. 10,849,697.

Applicant of the present application owns the following U.S. patent applications, filed on Mar. 29, 2018, each of which is herein incorporated by reference in its entirety:

U.S. patent application Ser. No. 15/940,627, entitled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201111;

U.S. patent application Ser. No. 15/940,637, entitled COMMUNICATION ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201139;

U.S. patent application Ser. No. 15/940,642, entitled CONTROLS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201113;

U.S. patent application Ser. No. 15/940,676, entitled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201142;

U.S. patent application Ser. No. 15/940,680, entitled CONTROLLERS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201135;

U.S. patent application Ser. No. 15/940,683, entitled COOPERATIVE SURGICAL ACTIONS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201145;

U.S. patent application Ser. No. 15/940,690, entitled DISPLAY ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201118; and

U.S. patent application Ser. No. 15/940,711, entitled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201120.

Numerous specific details are set forth to provide a thorough understanding of the overall structure, function, manufacture, and use of the embodiments as described in the specification and illustrated in the accompanying drawings. Well-known operations, components, and elements have not been described in detail so as not to obscure the embodiments described in the specification. The reader will understand that the embodiments described and illustrated herein are non-limiting examples, and thus it can be appreciated that the specific structural and functional details disclosed herein may be representative and illustrative. Variations and changes thereto may be made without departing from the scope of the claims.

The terms “comprise” (and any form of comprise, such as “comprises” and “comprising”), “have” (and any form of have, such as “has” and “having”), “include” (and any form of include, such as “includes” and “including”) and “contain” (and any form of contain, such as “contains” and “containing”) are open-ended linking verbs. As a result, a surgical system, device, or apparatus that “comprises,” “has,” “includes” or “contains” one or more elements possesses those one or more elements, but is not limited to possessing only those one or more elements. Likewise, an element of a system, device, or apparatus that “comprises,” “has,” “includes” or “contains” one or more features possesses those one or more features, but is not limited to possessing only those one or more features.

The terms “proximal” and “distal” are used herein with reference to a clinician manipulating the handle portion of the surgical instrument. The term “proximal” refers to the portion closest to the clinician and the term “distal” refers to the portion located away from the clinician. It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical,” “horizontal,” “up,” and “down” may be used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

Various exemplary devices and methods are provided for performing laparoscopic and minimally invasive surgical procedures. However, the reader will readily appreciate that the various methods and devices disclosed herein can be used in numerous surgical procedures and applications including, for example, in connection with open surgical procedures. As the present Detailed Description proceeds, the reader will further appreciate that the various instruments disclosed herein can be inserted into a body in any way, such as through a natural orifice, through an incision or puncture hole formed in tissue, etc. The working portions or end effector portions of the instruments can be inserted directly into a patient's body or can be inserted through an

access device that has a working channel through which the end effector and elongate shaft of a surgical instrument can be advanced.

A surgical stapling system can comprise a shaft and an end effector extending from the shaft. The end effector comprises a first jaw and a second jaw. The first jaw comprises a staple cartridge. The staple cartridge is insertable into and removable from the first jaw; however, other embodiments are envisioned in which a staple cartridge is not removable from, or at least readily replaceable from, the first jaw. The second jaw comprises an anvil configured to deform staples ejected from the staple cartridge. The second jaw is pivotable relative to the first jaw about a closure axis; however, other embodiments are envisioned in which the first jaw is pivotable relative to the second jaw. The surgical stapling system further comprises an articulation joint configured to permit the end effector to be rotated, or articulated, relative to the shaft. The end effector is rotatable about an articulation axis extending through the articulation joint. Other embodiments are envisioned which do not include an articulation joint.

The staple cartridge comprises a cartridge body. The cartridge body includes a proximal end, a distal end, and a deck extending between the proximal end and the distal end. In use, the staple cartridge is positioned on a first side of the tissue to be stapled and the anvil is positioned on a second side of the tissue. The anvil is moved toward the staple cartridge to compress and clamp the tissue against the deck. Thereafter, staples removably stored in the cartridge body can be deployed into the tissue. The cartridge body includes staple cavities defined therein wherein staples are removably stored in the staple cavities. The staple cavities are arranged in six longitudinal rows. Three rows of staple cavities are positioned on a first side of a longitudinal slot and three rows of staple cavities are positioned on a second side of the longitudinal slot. Other arrangements of staple cavities and staples may be possible.

The staples are supported by staple drivers in the cartridge body. The drivers are movable between a first, or unfired position, and a second, or fired, position to eject the staples from the staple cavities. The drivers are retained in the cartridge body by a retainer which extends around the bottom of the cartridge body and includes resilient members configured to grip the cartridge body and hold the retainer to the cartridge body. The drivers are movable between their unfired positions and their fired positions by a sled. The sled is movable between a proximal position adjacent the proximal end and a distal position adjacent the distal end. The sled comprises a plurality of ramped surfaces configured to slide under the drivers and lift the drivers, and the staples supported thereon, toward the anvil.

Further to the above, the sled is moved distally by a firing member. The firing member is configured to contact the sled and push the sled toward the distal end. The longitudinal slot defined in the cartridge body is configured to receive the firing member. The anvil also includes a slot configured to receive the firing member. The firing member further comprises a first cam which engages the first jaw and a second cam which engages the second jaw. As the firing member is advanced distally, the first cam and the second cam can control the distance, or tissue gap, between the deck of the staple cartridge and the anvil. The firing member also comprises a knife configured to incise the tissue captured intermediate the staple cartridge and the anvil. It is desirable for the knife to be positioned at least partially proximal to the ramped surfaces such that the staples are ejected ahead of the knife.

A surgical instrument **1000** is illustrated in FIG. 1. As discussed in greater detail below, the surgical instrument **1000** is configured to clamp, incise, and seal patient tissue. The surgical instrument **1000** comprises an end effector **1300**, an articulation joint **1400**, and an articulation drive system **1700** (FIG. 13) configured to articulate the end effector **1300** about the articulation joint **1400**. The end effector **1300** comprises a first jaw **1310**, a second jaw **1320** movable between an open position and a closed position, and a drive system **1600** (FIG. 13) operable to close the second jaw **1320** during a closure stroke. After the end effector **1300** is closed, the drive system **1600** is operable once again to incise and staple the patient tissue captured between the first jaw **1310** and the second jaw **1320** during a firing stroke. Additionally, the surgical instrument **1000** comprises an energy delivery system **1900** which is also operable to seal the incised tissue.

The surgical instrument **1000** further comprises a handle **1100** and a shaft **1200** extending from the handle **1100**. The handle **1100** comprises a grip **1110** extending downwardly from a handle body **1120**. As discussed in greater detail below, the handle **1100** further comprises a closure actuator **1130** operable to close the end effector **1300** and an articulation actuator **1140** operable to articulate the end effector **1300** relative to the shaft **1200**. The shaft **1200** comprises an outer housing **1210** and an inner frame, or spine, **1230** (FIG. 4) which are rotatably mounted to the handle body **1120** about a rotation joint **1220**. Referring to FIG. 5, the articulation joint **1400** comprises a flexible outer housing **1410** affixed to the outer housing **1210** and a flexible articulation frame **1430** connected to the shaft frame **1230** (FIG. 4). The first jaw **1310** comprises a proximal end **1311** mounted to the flexible articulation frame **1430** via a pin forming a rotation joint **1330** between the first jaw **1310** and the second jaw **1320**. The distal end of the flexible articulation housing **1410** is also affixed to the first jaw **1310** via a clamp ring **1420** such that the end effector **1300** is affixed to the distal end of the articulation joint **1400**.

Referring primarily to FIGS. 1 and 13, the articulation drive system **1700** comprises an electric motor **1710** in communication with a control system of the surgical instrument **1000**. The control system is configured to supply power to the electric motor **1710** from a battery **1180** in response to an actuation of the articulation actuator **1140**. The articulation actuator **1140** comprises a switch that is actuatable in a first direction to operate the electric motor **1710** in a first direction and actuatable in a second direction to operate the electric motor **1710** in a second, or opposite, direction. The articulation drive system **1700** further comprises a transfer gear **1730** rotatably supported within the handle **1100** that is operably engaged with a gear output **1720** of the electric motor **1710**. Referring primarily to FIGS. 2 and 4, the articulation drive system **1700** also comprises a first articulation actuator **1740** and a second articulation actuator **1750** operably engaged with the transfer gear **1730**. More specifically, the transfer gear **1730** comprises a pinion gear portion **1735** intermeshed with a first drive rack **1745** defined on the proximal end of the first articulation actuator **1740** and a second drive rack **1755** defined on the proximal end of the second articulation actuator **1750**. Owing to the positioning of the first drive rack **1745** and the second drive rack **1755** on opposite sides of the transfer gear **1730**, the first articulation actuator **1740** and the second articulation actuator **1750** are driven proximally and distally in opposition to, or antagonistically with respect to, one another when the transfer gear **1730** is rotated. For instance, the first articulation actuator **1740** is

driven distally and the second articulation actuator **1750** is driven proximally to articulate the end effector **1300** in a first direction when the electric motor **1710** is operated in its first direction. Correspondingly, the first articulation actuator **1740** is driven proximally and the second articulation actuator **1750** is driven distally to articulate the end effector **1300** in a second, or opposite, direction when the electric motor **1710** is operated in its second direction.

Referring to FIGS. 4-6, the first jaw **1310** comprises a first articulation drive post **1317** extending upwardly on a first lateral side of the first jaw **1310** and a second articulation drive post **1319** extending upwardly on a second, or opposite, lateral side of the first jaw **1310**. The distal end of the first articulation actuator **1740** comprises a first drive mount **1747** engaged with the first articulation drive post **1317** and the distal end of the second articulation actuator **1750** comprises a second drive mount **1759** engaged with the second articulation drive post **1319** such that the proximal and distal movement of the articulation actuators **1740** and **1750** rotate the end effector **1300** about the articulation joint **1400**. When the end effector **1300** is articulated, the flexible outer housing **1410** and the flexible articulation frame **1430** of the articulation joint **1400** resiliently deflect to accommodate the rotation of the end effector **1300**. In various instances, the articulated position of the end effector **1300** is held in place due to friction within the articulation drive **1700**. In various embodiments, the articulation drive **1700** comprises an articulation lock configured to releasably hold the end effector **1300** in position.

As discussed above, the surgical instrument **1000** comprises a drive system **1600** operable to close the end effector **1300** and then operable once again to staple and incise the patient tissue captured between the first jaw **1310** and the second jaw **1320** of the end effector **1300**. Referring again to FIG. 13, the drive system **1600** comprises an electric motor **1610** in communication with the control system of the surgical instrument **1000**. The control system is configured to supply power to the electric motor **1610** from the battery **1180** in response to an actuation of the closure actuator **1130**. The closure actuator **1130** comprises a switch that is actuable in a first direction to operate the electric motor **1610** in a first direction to close the end effector **1300** and actuable in a second direction to operate the electric motor **1610** in a second, or opposite, direction to open the end effector **1300**. The closure drive system **1600** further comprises a transfer gear **1630** rotatably supported within the handle **1100** that is operably engaged with a gear output **1620** of the electric motor **1610**. The transfer gear **1630** is fixedly mounted to a rotatable drive shaft **1660** such that the drive shaft **1660** rotates with the transfer gear **1630**. The rotatable drive shaft **1660** extends through the shaft **1200** and comprises a flexible portion **1665** that extends through the articulation joint **1400** to accommodate the articulation of the end effector **1300**. The rotatable drive shaft **1660** further comprises a distal coupling **1661** that extends into a proximal end **1311** of the first jaw **1310**. In at least one embodiment, the distal coupling **1661** comprises a hex-shaped aperture, for example, but could comprise any suitable configuration.

Referring primarily to FIG. 6, the first jaw **1310** further comprises a channel **1312** extending between the proximal end **1311** and a distal end **1313**. The channel **1312** comprises two sidewalls extending upwardly from a bottom wall and is configured to receive a staple cartridge, such as a staple cartridge **1500**, for example, between the sidewalls. The staple cartridge **1500** comprises a cartridge body **1510** including a proximal end **1511**, a distal nose **1513**, and a

tissue-supporting deck **1512** extending between the proximal end **1511** and the distal nose **1513**. The cartridge body **1510** further comprises a longitudinal slot **1520** defined therein extending from the proximal end **1511** toward the distal nose **1513**. The cartridge body **1510** also comprises longitudinal rows of staple cavities **1530** extending between the proximal end **1511** and the distal nose **1513**. More specifically, the cartridge body **1510** comprises a single longitudinal row of staple cavities **1530** positioned on a first lateral side of the longitudinal slot **1520** and a single longitudinal row of staple cavities **1530** positioned on a second, or opposite, lateral side of the longitudinal slot **1520**. That said, a staple cartridge can comprise any suitable number of longitudinal rows of staple cavities **1530**. The staple cartridge **1500** further comprises a staple removably stored in each staple cavity **1530** which is ejected from the staple cartridge **1500** during a staple firing stroke, as discussed in greater detail further below.

Further to the above, referring again to FIG. 6, the staple cartridge **1500** further comprises a drive screw **1560** rotatably supported in the cartridge body. More specifically, the drive screw **1560** comprises a proximal end **1561** rotatably supported by a proximal bearing in the proximal end **1511** of the cartridge body **1510** and a distal end **1563** rotatably supported by a distal bearing in the distal end **1513** of the cartridge body **1510**. The proximal end **1561** of the drive screw **1560** comprises a hex coupling extending proximally with respect to the proximal end **1511** of the cartridge end **1510**. When the staple cartridge **1500** is seated in the first jaw **1310**, the proximal end **1511** of the cartridge body **1510** is slid into the proximal end **1311** of the first jaw **1310** such that the hex coupling of the proximal drive screw end **1561** is inserted into and is operably engaged with the distal drive end **1661** of the rotatable drive shaft **1660**. Once the drive screw **1560** is coupled to the rotatable drive shaft **1660** the distal nose **1513** of the staple cartridge **1500** is seated in the distal end **1313** of the first jaw **1310**. That said, the staple cartridge **1500** can be seated in the first jaw **1310** in any suitable manner. In various instances, the staple cartridge **1500** comprises one or more snap-fit and/or press-fit features which releasably engage the first jaw **1310** to releasably secure the staple cartridge **1500** within the first jaw **1310**. To remove the staple cartridge **1500** from the first jaw **1310**, an upward, or lifting, force can be applied to the distal nose **1513** of the staple cartridge **1500** to release the staple cartridge **1500** from the first jaw **1310**.

Referring again to FIG. 6, the drive screw **1560** further comprises a threaded portion **1565** extending between the proximal end **1561** and the distal end **1563** and the staple cartridge **1500** further comprises a firing member **1570** threadably engaged with the threaded portion **1565**. More specifically, the firing member **1570** comprises a threaded nut insert **1575** threadably engaged with the threaded portion **1565** which is constrained, or at least substantially constrained, from rotating such that the firing member **1570** translates within the staple cartridge **1500** when the drive screw **1560** is rotated. When the drive screw **1560** is rotated in a first direction, the firing member **1570** translates from a proximal unactuated position to a distal actuated position during a closure stroke to move the second jaw **1320** from its open, or unclamped, position (FIG. 7) to its distal, or clamped, position (FIGS. 8 and 11). More specifically, the firing member **1570** comprises a first cam **1572** that engages the first jaw **1310** and a second cam **1576** that engages the second jaw **1320** during the closure stroke which co-operatively position the second jaw **1320** relative to the first jaw **1310**. At the outset of the closure stroke, the second cam

1576 is not engaged with the second jaw 1320; however, the second cam 1576 comes into contact with a ramp 1326 (FIG. 6) during the closure stroke to rotate the second jaw 1320 toward the first jaw 1310.

Once the firing member 1570 has reached the end of its closure stroke (FIGS. 8 and 11), the controller of the surgical instrument 1000 stops the drive motor 1610 (FIG. 13). At such point, referring to FIG. 11, the firing member 1570 has not incised the patient tissue captured between the first jaw 1310 and the second jaw 1320 and/or stapled the patient tissue. If the clinician is unsatisfied with the positioning of the jaws 1310 and 1320 on the patient tissue, the clinician can release the closure actuator 1130 (FIG. 1) to operate the drive motor 1610 in its second, or opposite, direction and translate the firing member 1570 proximally out of engagement with the second jaw 1320. In at least one instance, the handle 1100 comprises a closure lock which releasably holds the closure actuator 1130 in its closed position and the clinician must deactivate the closure lock to reopen the closure actuator 1130. Referring to FIG. 1, the handle 1100 further comprises a closure lock release 1160 that, when actuated, unlocks the closure actuator 1130. Once the end effector 1300 is open, the clinician can re-position the end effector 1300 relative to the patient tissue and, once satisfied with the re-positioning of the end effector 1300 relative to the patient tissue, close the closure actuator 1130 once again to re-operate the drive motor 1610 in its first direction to re-close the second jaw 1320. At such point, the drive system 1600 can be operated to perform the staple firing stroke, as discussed further below.

Further to the above, as illustrated in FIG. 11, the firing member 1570 moves into contact with, or in close proximity to, a sled 1550 contained in the cartridge body 1510 at the end of the closure stroke. The surgical instrument 1000 further comprises a firing actuator in communication with the control system of the surgical instrument 1000 which, when actuated, causes the control system to operate the drive motor 1610 in its first direction to advance the firing member 1570 distally from its distal clamped position (FIGS. 8 and 11) and push the sled 1550 through the staple firing stroke, as illustrated in FIG. 12. In various embodiments, the staple cartridge 1500 comprises staple drivers which support and drive the staples out of the cartridge body 1510 when the staple drivers are contacted by the sled 1550 during the staple firing stroke. In other embodiments, as discussed in greater detail below, the staples comprise drivers integrally-formed thereon which are directly contacted by the sled 1550 during the staple firing stroke. In either event, the sled 1550 progressively ejects, or fires, the staples out of the cartridge body 1510 as the sled 1550 is moved from its distal clamped position (FIG. 11) to its distal fired position (FIG. 12) by the firing member 1570. Moreover, referring to FIG. 6, the firing member 1570 comprises a tissue cutting edge 1571 that moves through the longitudinal slot 1520 during the staple firing stroke to incise the tissue captured between the deck 1512 of the staple cartridge 1500 and the second jaw 1320.

Further to the above, the second jaw 1320 comprises a frame 1325 including a proximal end 1321 rotatably connected to the first jaw 1310, a longitudinal recess 1324, and a longitudinal slot 1329 configured to receive the firing member 1570 during the staple firing stroke. The frame 1325 further comprises a first longitudinal cam shoulder 1327 defined on a first side of the longitudinal slot 1329 and a second longitudinal cam shoulder 1328 defined on a second side of the longitudinal slot 1329. During the staple firing stroke, the second cam 1576 of the firing member 1570

slides along the first longitudinal cam shoulder 1327 and the second longitudinal cam shoulder 1328 which co-operates with the first cam 1572 to hold the second jaw 1320 in position relative to the first jaw 1310. The frame 1325 also comprises longitudinal rows of staple forming cavities defined therein which are registered with the staple cavities 1530 defined in the staple cartridge 1500 when the second jaw 1320 is in its closed position. The second jaw 1320 further comprises a cover, or cap, 1322 positioned in the longitudinal recess 1324 which is welded to the frame 1325 to enclose the longitudinal slot 1329 and extend over the second cam 1576. The cover 1322 comprises a distal end, or nose, 1323 which is angled toward the distal nose 1513 of the staple cartridge 1500.

In various instances, the clinician can depress and hold the firing actuator until the staple firing stroke is completed. When the firing member 1570 reaches the end of the staple firing stroke, in such instances, the control system can automatically switch the operation of the drive motor 1610 from its first direction to its second direction to retract the firing member 1570 proximally back into its distal clamped position (FIGS. 8 and 11). In such instances, the end effector 1300 remains in its closed, or clamped, configuration until the closure lock release 1160 (FIG. 1) is actuated by the clinician to re-open the closure actuator 1130 and the end effector 1300. In certain instances, the clinician can release the firing actuator prior to the end of the staple firing stroke to stop the drive motor 1610. In such instances, the clinician can re-actuate the firing actuator to complete the staple firing stroke or, alternatively, actuate a retraction actuator in communication with the control system to operate the drive motor 1610 in its second direction to retract the firing member 1570 back into its distal clamped position (FIGS. 8 and 11). In various alternative embodiments, the automatic retraction of the firing member 1570 and/or the actuation of the retraction actuator can retract the firing member 1570 back into its proximal unactuated position to automatically open the end effector 1300 without requiring the clinician to release the closure actuator 1130. To move the second jaw 1320 into an open position (FIG. 12A), the proximal end 1321 of the second jaw 1320 comprises a camming surface 1339 defined on the proximal end 1321 and the firing member 1570 comprises a cam portion 1579 defined on a proximal end of the firing member 1570. The cam portion 1579 is configured to pivot the second jaw 1320 into the open position upon contacting the camming surface 1339.

Once the staple cartridge 1500 has been at least partially expended, i.e., at least partially fired, and the end effector 1500 has been re-opened, the staple cartridge 1500 can be removed from the first jaw 1310 and replaced with another staple cartridge 1500, and/or any other suitable staple cartridge. If the expended staple cartridge 1500 is not replaced, the firing drive 1600 is locked out from performing another staple firing stroke. Such a lockout can comprise an electronic lockout that prevents the control system from operating the drive motor 1610 to perform another staple firing stroke until the spent staple cartridge 1500 is replaced with an unspent staple cartridge. In addition to or in lieu of an electronic lockout, the surgical instrument 1000 can include a mechanical lockout which blocks the distal advancement of the firing member 1570 through another staple firing stroke unless the spent staple cartridge 1500 is replaced. Notably, referring to FIG. 12, the sled 1550 is not retracted proximally with the firing member 1570 after the staple firing stroke. As such, the electronic and/or mechanical lockout can key off of the position of the sled 1550 at the outset of the staple firing stroke. Stated another way, the

staple firing stroke is prevented or blocked if the sled **1550** is not in its unfired position when the staple firing stroke is initiated.

The entire disclosures of U.S. Pat. No. 7,143,923, entitled SURGICAL STAPLING INSTRUMENT HAVING A FIRING LOCKOUT FOR AN UNCLOSED ANVIL, which issued on Dec. 5, 2006; U.S. Pat. No. 7,044,352, SURGICAL STAPLING INSTRUMENT HAVING A SINGLE LOCKOUT MECHANISM FOR PREVENTION OF FIRING, which issued on May 16, 2006; U.S. Pat. No. 7,000,818, SURGICAL STAPLING INSTRUMENT HAVING SEPARATE DISTINCT CLOSING AND FIRING SYSTEMS, which issued on Feb. 21, 2006; U.S. Pat. No. 6,988,649, SURGICAL STAPLING INSTRUMENT HAVING A SPENT CARTRIDGE LOCKOUT, which issued on Jan. 24, 2006; and U.S. Pat. No. 6,978,921, SURGICAL STAPLING INSTRUMENT INCORPORATING AN E-BEAM FIRING MECHANISM, which issued on Dec. 27, 2005, are incorporated by reference herein.

In various alternative embodiments, a surgical instrument can comprise separate and distinct closure and staple firing drive systems. In at least one such embodiment, the closure drive system comprises a closure actuator and a closure drive motor which, when actuated, moves the closure actuator distally through a closure stroke to close the end effector. In such embodiments, the staple firing drive system comprises a firing actuator and a separate firing drive motor which moves the firing actuator distally through a staple firing stroke. As discussed in greater detail below, the length of the closure stroke in such embodiments can be adjusted independently of the staple firing stroke to control the position of the second jaw **1320**. Moreover, the closure drive system can also be actuated during the staple firing stroke to further control the position of the second jaw **1320**.

As discussed above, the staples ejected from the staple cartridge **1500** can seal the incised patient tissue. That said, a single row of staples on each side of the incision may not be able to create a sufficient hemostatic seal in the incised patient tissue. To this end, as discussed in greater detail below, the surgical instrument **1000** is further configured to use electrical energy to seal the patient tissue. Referring to FIG. 1, the surgical instrument **1000** further comprises an energy delivery system **1900** including an off-board power supply and a cord **1990** in communication with the off-board power supply. In various alternative embodiments, the energy delivery system **1900** comprises an on-board power supply, such as the battery **1180**, for example. In either event, the control system of the surgical instrument **1000** is configured to control the delivery of energy from the energy delivery system **1900** to the patient tissue, as discussed in greater detail below. The energy delivery system **1900** comprises an electrical circuit extending through the shaft **1200** and the articulation joint **1400** and into the end effector **1300**. Referring to FIG. 6, the energy delivery system **1900** comprises an electrode supply circuit **1920** that extends into the second jaw **1320** and comprises a longitudinal electrode **1925** mounted to the frame **1325** of the second jaw **1320**. The longitudinal electrode **1925** is electrically insulated from the metal frame **1325** such that the current flowing through the longitudinal electrode **1925**, or at least a majority of the current flowing through the longitudinal electrode **1925**, flows into a longitudinal return electrode **1590** of the staple cartridge **1500**. The longitudinal return electrode **1590** is seated in the cartridge body **1510** and comprises a cartridge connector **1595** that engages, and makes an electrical connection with, a circuit connector **1915** of an electrode return circuit **1910** when the staple cartridge **1500** is

seated in the first jaw **1310**. In various alternative embodiments, the staple cartridge **1500** includes a supply electrode and the second jaw **1320** includes a return electrode.

Referring to FIG. 1, the surgical instrument **1000** further comprises a display **1190** in communication with the control system of the surgical instrument **1000**. In various instances, the control system is configured to display parameters and/or data regarding the staple firing system and/or the energy delivery system.

A surgical instrument **2000** is illustrated in FIG. 14. The surgical instrument **2000** comprises an end effector **2300** and a firing drive **2600**. The end effector **2300** comprises a first jaw **2310** and a second jaw **2320** rotatable relative to the first jaw **2310** about a pivot **2330**. The first jaw **2310** comprises a replaceable staple cartridge **2500** comprising staples removably stored therein and the second jaw **2320** comprises an anvil configured to deform the staples. The firing drive **2600** comprises a firing member **2570** that is advanced distally to push a sled contained in the staple cartridge **2500** during a staple firing stroke to drive the staples toward the second jaw **2320**. The firing member **2570** comprises a first cam **2572** configured to engage a longitudinal cam shoulder **2312** defined in the first jaw **2310** and a second cam **2576** configured to engage a longitudinal cam shoulder **2327** defined in the second jaw **2320** during the staple firing stroke which co-operatively hold the second jaw **2320** in position relative to the first jaw **2310**. The firing member **2570** further comprises a tissue cutting edge **2571** configured to incise the patient tissue captured between the staple cartridge **2500** and the second jaw **2320** during the staple firing stroke.

Further to the above, the firing drive **2600** further comprises a rotatable drive shaft **2660** engaged with a rotatable drive shaft **2560** extending within the staple cartridge **2500**. The rotatable drive shaft **2660** comprises a threaded portion **2665** that is rotatably supported in the first jaw **2310** by a threaded bearing **2315**. As a result of the interaction between the threaded portion **2665** of the drive shaft **2660** and the threaded bearing **2315**, the rotation of the drive shaft **2660** causes the drive shaft **2660** to translate proximally or distally relative to the end effector **2300** depending on the direction in which the drive shaft **2660** is rotated. When the drive shaft **2560** is rotated in a first direction, the drive shaft **2660** translates distally. When the drive shaft **2560** is rotated in a second, or opposite, direction, the drive shaft **2660** translates proximally. As discussed in greater detail below, the rotation and translation of the drive shaft **2660** is transmitted to the rotatable drive shaft **2560**.

Further to the above, the firing member **2570** comprises a threaded aperture **2575** defined therein that is threadably engaged with a threaded portion **2565** of the rotatable drive shaft **2560**. When the drive shaft **2560** is rotated in the first direction by the drive shaft **2660** as described above, referring to FIG. 15, the firing member **2570** translates distally relative to the drive shaft **2560**. Thus, the distal motion of the firing member **2570** relative to the end effector **2300** is a composition of two concurrent distal translations—the translation of the drive shaft **2560** relative to the end effector **2300** and the translation of the firing member **2570** relative to the drive shaft **2560**. When the drive shaft **2560** is rotated in the second, or opposite, direction by the drive shaft **2660** as also described above, the firing member **2570** translates proximally relative to the drive shaft **2560**. Thus, the proximal motion of the firing member **2570** relative to the end effector **2300** is a composition of two concurrent proximal translations—the translation of the drive shaft **2560** relative to the end effector **2300** and the translation of the firing member **2570** relative to the drive shaft **2560**. To achieve the

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above, the threaded portion **2565** and the threaded portion **2665** can comprise any suitable thread design including, for example, right-handed threads and/or left-handed threads.

As discussed above, the drive shaft **2560** translates distally relative to the end effector **2300** and, also, the firing member **2570** translates distally relative to the drive shaft **2560** during the staple firing stroke. In various instances, the drive shaft **2560** translates relative to the end effector **2300** at a first translational rate and the firing member **2570** translates distally relative to the drive shaft **2560** at a second translational rate. In at least one embodiment, the first translational rate and the second translational rate are the same, i.e., the firing member **2570** translates distally relative to the end effector **2300** at a speed which is twice that of the distal translation of the drive shaft **2560** relative to the end effector **2300**. In at least one such embodiment, the threaded portion **2665** of the drive shaft **2660** comprises a first thread pitch and the threaded portion **2565** of the drive shaft **2560** comprises a second thread pitch which is the same as the first thread pitch. In at least one such embodiment, the threaded portion **2665** of the drive shaft **2660** comprises a first threads-per-inch (TPI) and the threaded portion **2565** of the drive shaft **2560** comprises a second threads-per-inch which is the same as the first threads-per-inch.

In various embodiments, further to the above, the first translational rate of the drive shaft **2560** relative to the end effector **2300** is faster than the second translational rate of the firing member **2570** relative to the drive shaft **2560**. In other embodiments, the first translational rate of the drive shaft **2560** relative to the end effector **2300** is slower than the second translational rate of the firing member **2570** relative to the drive shaft **2560**. In either instance, however, the speed of the firing member **2570** relative to the end effector **2300** is faster than the speed of the drive shaft **2560** relative to the end effector **2300**. In various embodiments, the first thread pitch of the threaded portion **2665** and the second thread pitch of the threaded portion **2565** are different. Likewise, in various embodiments, the first threads-per-inch of the threaded portion **2665** is different than the second threads-per-inch of the threaded portion **2565**. When the second translational rate of the firing member **2570** relative to the drive shaft **2560** is faster than the first translational rate of the drive shaft **2560** relative to the end effector **2300**, the second threads-per-inch of the threaded portion **2565** is smaller than the first threads-per-inch of the threaded portion **2665**, for example. Likewise, the second threads-per-inch of the threaded portion **2565** is larger than the first threads-per-inch of the threaded portion **2665** when the second translational rate of the firing member **2570** relative to the drive shaft **2560** is slower than the first translational rate of the drive shaft **2560** relative to the end effector **2300**, for example.

In various embodiments, the first thread pitch of the threaded portion **2665** is constant along the length thereof. Thus, for a given speed of the electric motor driving the drive system **2600**, the drive shaft **2660** will translate at a constant speed relative to the end effector **2300**. In other embodiments, the first thread pitch of the threaded portion **2665** is not constant along the length thereof. In at least one such embodiment, the first thread pitch changes along the length of the threaded portion **2665** and, for a given speed of the electric motor driving the drive system **2600**, the translational speed of the drive shaft **2660** relative to the end effector **2300** changes during the staple firing stroke. Such an arrangement can be useful to create a soft start of the firing member **2570** at the beginning of the staple firing stroke and/or a soft stop of the firing member **2570** at the end

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of the staple firing stroke. In such instances, the first translational rate of the drive shaft **2660** is slower at the beginning and/or at the end of the staple firing stroke.

In various embodiments, further to the above, the second thread pitch of the threaded portion **2565** is constant along the length thereof. Thus, for a given speed of the electric motor driving the drive system **2600**, the firing member **2570** will translate at a constant speed relative to the drive shaft **2560**. In other embodiments, the second thread pitch of the threaded portion **2565** is not constant along the length thereof. In at least one such embodiment, the second thread pitch changes along the length of the threaded portion **2565** and, for a given speed of the electric motor driving the drive system **2600**, the translational speed of the firing member **2570** relative to the drive shaft **2560** changes during the staple firing stroke. Such an arrangement can be useful to create a soft start of the firing member **2570** at the beginning of the staple firing stroke and/or a soft stop of the firing member **2570** at the end of the staple firing stroke. In such instances, the second translational rate of the firing member **2570** is slower at the beginning and/or at the end of the staple firing stroke.

A surgical instrument **3000** is illustrated in FIG. 16. The surgical instrument **3000** comprises an end effector **3300**, a closure drive **3800**, and a firing drive **3600**. The end effector **3300** comprises a first jaw **3310** and a second jaw **3320** rotatable relative to the first jaw **3310** about a pivot **3330**. The second jaw **3320** is movable from an open position to a closed position by the closure drive **3800** during a closure stroke, as discussed in greater detail below. The first jaw **3310** comprises a staple cartridge **3500** including staples removably stored therein and the second jaw **3320** comprises forming pockets configured to deform the staples. Once the second jaw **3320** is in its closed position, as illustrated in FIG. 16, the firing drive **3600** is operable to fire the staples from the staple cartridge **3500** to staple the patient tissue captured between the staple cartridge **3500** and the second jaw **3320**, as also described in greater detail below.

Further to the above, the closure drive **3800** comprises a closure member **3870** which is movable distally to engage the second jaw **3320** and move the second jaw **3320** into its closed position during the closure stroke. The closure member **3870** comprises a first cam **3872** configured to engage a first longitudinal shoulder **3312** defined in the first jaw **3310** and a second cam **3876** configured to engage the second jaw **3320** during the staple firing stroke. Referring to FIGS. 17 and 18, the second jaw **3320** comprises an anvil plate **3325** and a cover, or cap, **3322** welded to the anvil plate **3325**. The anvil plate **3325** comprises a ramp **3326** and a longitudinal slot **3329** defined therein configured to receive the closure member **3870**. At the beginning of the closure stroke, the second cam **3876** of the closure member **3870** is not in contact with the ramp **3326**. Once the closure stroke is initiated, however, the second cam **3876** comes into contact with the ramp **3326** and begins to close the second jaw **3320**. As the closure stroke progresses, the second cam **3876** slides onto longitudinal shoulders **3327** and **3328** defined on the lateral sides of the longitudinal slot **3329**. At such point, the first cam **3872** and the second cam **3876** co-operatively hold the second jaw **3320** in its closed position.

Referring again to FIG. 16, the anvil **33200** comprises tissue stops **3340** extending downwardly therefrom which prevent, or at least inhibit, patient tissue from migrating into the proximal end of the end effector **3300**. Each tissue stop **3340** comprises a distal edge **3345** which co-operates with the lateral sides of the first jaw **3310** to prevent, or at least

inhibit, the patient tissue from moving proximally. At the end of the closure stroke, referring again to FIG. 16, the leading edge 3871 of the closure member 3870 is positioned proximally with respect to the distal edges 3345 of the tissue stops 3340.

After the closure stroke, as mentioned above, the firing drive 3600 can be actuated to fire the staples and incise the patient tissue during a staple firing stroke. The firing drive 3600 comprises a firing bar 3670 which is advanced distally to push a sled 3550 positioned in the staple cartridge 3500 through the staple firing stroke and drive the staples stored within the staple cartridge 3500 toward the second jaw 3320. The firing bar 3670 further comprises a tissue cutting edge 3675 which extends into the longitudinal slot 3329 defined in the second jaw 3320 and passes through the tissue gap defined between the second jaw 3320 and the staple cartridge 3500 during the staple firing stroke to incise the patient tissue as it is being stapled.

Notably, further to the above, the firing bar 3670 does not comprise cams which engage the first jaw 3310 and the second jaw 3320 to hold the second jaw 3320 in position during the staple firing stroke. In such embodiments, the position of the second jaw 3320 relative to the first jaw 3310 is controlled solely by the closure drive 3800 which is operated independently of the firing drive 3600. Thus, in various instances, the control system of the surgical instrument 3000 can modify the operation of the closure drive 3800 independently of modifying the operation of the firing drive 3600 to achieve a desired goal and/or therapeutic effect. For instance, the control system can operate the closure drive 3800 to further close the second jaw 3320 while the firing drive 3600 is being operated to perform the staple firing stroke. In such instances, the control system can increase the clamping force being applied to the patient tissue during the staple firing stroke to improve staple formation. In other instances, the control system can operate the closure drive 3800 to relax the clamping pressure being applied to the patient tissue during the staple firing stroke. In such instances, the control system can prevent the overcompression of the patient tissue and/or keep the forming pockets in the second jaw 3320 in registration with the staples being ejected from the staple cartridge 3500. Other embodiments are envisioned in which the firing bar 3670 comprises a first cam for engaging the first jaw 3310 during the staple firing stroke, but not a second cam engaged with the second jaw 3320. In various alternative embodiments, the firing bar 3670 can comprise both a first cam engaged with the first jaw 3310 and a second cam engaged with the second jaw 3320 during the staple firing stroke. In such embodiments, both the firing drive 3600 and the closure drive 3800 can be used to control the position of the second jaw 3320 but at different locations within the end effector 3300.

Further to the above, the surgical instrument 3000 comprises a handle including a jaw adjustment actuator and/or touch screen control in communication with the control system of the surgical instrument 3000. In various embodiments, the control system comprises one or more sensors configured to detect the firing load in the firing drive 3600 during the staple firing stroke. In at least one such embodiment, the control system comprises a current sensor configured to detect the magnitude of current through the electric motor of the firing drive 3600—which is a proxy for the firing load in the firing drive 3600—and adjust the position of the second jaw 3320 based on the magnitude of the current detected through the electric motor. In certain embodiments, the control system comprises a load cell

sensor and/or a strain gauge sensor, for example, configured to detect the clamping force being applied to the patient tissue and adjust the position of the second jaw 3320 based on the voltage potential output of the load cell sensor and/or strain gauge sensor. In various embodiments, the control system is configured to automatically adjust the position of the second jaw 3320. In other embodiments, the control system is configured to provide the clinician using the surgical instrument 3000 with the option of modifying the position of the second jaw 3320. In at least one such embodiment, the control system is configured to illuminate the jaw adjustment actuator, or present an actuatable input on the touch screen control, when the firing load in the firing drive 3600 and/or the clamping load in the closure drive 3800 has crossed a threshold and, when the actuator is actuated by the clinician, adjust the position of the second jaw 3320.

As discussed above, the closure drive 3800 is operable during the staple firing stroke to adjust the position of the closure member 3870. Thus, the closure member 3870 is movable distally during a first closure stroke to close the second jaw 3320 and then a second closure stroke to control the position of the second jaw 3320 during the staple firing stroke. Further to the above, referring again to FIG. 16, the closure member 3870 is movable distally into a first closed position as a result of the first closure stroke in which the closure member 3870 is positioned proximally with respect to the distal edges 3345 of the tissue stops 3340. As a result of the second closure stroke, at least a portion of the closure member 3870 is moved distally beyond the distal edges 3345 of the tissue stops 3340 to a second closed position. In such a second closed position, the closure member 3870 can better resist the upward movement and/or deflection of the second jaw 3320 during the staple firing stroke. The above being said, various embodiments are envisioned in which the closure member 3870 does not move distally beyond the distal edges 3345 of the tissue stops 3340 during the second closure stroke.

Referring again to FIG. 16, the firing bar 3670 extends distally past the closure member 3870. More specifically, the firing bar 3670 is positioned laterally with respect to the closure member 3870 along the length of the closure member 3870 and then extends distally in front of the closure member 3870 such that the firing bar 3670 is laterally centered, or at least substantially laterally centered, within the end effector 3300. As a result, the tissue cutting edge 3675 is aligned, or at least substantially aligned, with the longitudinal slot of the end effector 3300.

An alternative arrangement is illustrated in FIGS. 19 and 20 which comprises a firing drive including a first firing bar 3670a' that extends alongside a first lateral side of a closure member 3870' and a second firing bar 3670b' that extends alongside a second, or opposite, lateral side of the closure member 3870'. FIG. 19 is a cross-sectional view of this arrangement taken proximally with respect to the end effector of the surgical instrument and FIG. 20 is a cross-sectional view of this arrangement taken within the end effector. The closure member 3870' comprises a longitudinal bar extending between the firing bars 3670a' and 3670b' (FIG. 19) and, also, first and second cams 3872' and 3876' (FIG. 20) which are configured to engage the first and second jaws 3310 and 3320, respectively, during a firing stroke. Notably, the height of the firing bars 3670a' and 3670b' are shortened to fit between the first and second cams 3872' and 3876' such that the firing bars 3670a' and 3670b' extend further into the end effector. The distal ends of the firing bars 3670a' and 3670b' are connected at a location which is distal to the closure member 3870' such that the firing bars 3670a' and 3670b'

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co-operatively support a tissue cutting knife and/or firing member during the firing stroke.

A surgical instrument **4000** is illustrated in FIG. **21**. The surgical instrument **4000** comprises an end effector **4300** including a first jaw **4310** and a second jaw **4320**. The first jaw **4310** is rotatable relative to the second jaw **4320** between an open position and a closed position. The first jaw **4310** comprises a replaceable staple cartridge **4500** seated therein which comprises a cartridge body, a longitudinal slot **4520** defined in the cartridge body, a longitudinal row of staple cavities **4530** defined on each side of the longitudinal slot **4520**, and staples removably stored in the staple cavities **4530**. The staple cartridge **4500** further comprises an electrical circuit including one or more electrodes **4590** which are operable to seal the patient tissue as described in greater detail below.

Further to the above, the first jaw **4310** comprises a longitudinal cam slot **4312** defined therein and the second jaw **4320** comprises longitudinal cam shoulders **4327** and **4328** defined on opposite sides of a longitudinal slot **4329**. The closure drive of the surgical instrument **4000** comprises a closure member **4870** comprising a C-shaped channel including a base, or spine, **4877**, a first cam **4872** extending from the spine **4877**, and a second cam **4876** extending from the spine **4877**. The first cam **4872** is configured to extend into the cam slot **4312** of the first jaw **4310** and the second cam **4876** is configured to engage the longitudinal shoulder **4328** defined in the second jaw **4320** during a closure stroke to move the first jaw **4310** from an open, unclamped, position to a closed, clamped, position. The firing drive of the surgical instrument comprises a firing member **4670** comprising a C-shaped channel including a base, or spine, **4677**, a first cam **4672** extending from the spine **4677**, and a second cam **4676** extending from the spine **4677**. The first cam **4672** is configured to extend into the cam slot **4312** of the first jaw **4310** and the second cam **4676** is configured to engage the longitudinal shoulder **4327** defined in the second jaw **4320** during a staple firing stroke to eject the staples from the staple cartridge **4500**. Notably, the spines **4677** and **4877** both extend within the longitudinal slot **4520** defined in the staple cartridge **4500** and the longitudinal slot **4329** defined in the second jaw **4320** and are arranged in a back-to-back arrangement which permits the firing member **4670** and the closure member **4870** to slide relative to one another.

A portion of a staple cartridge **5500** is illustrated in FIGS. **22-24**. The staple cartridge **5500** comprises a cartridge body **5510** including staple cavities **5530** defined therein and staples **5540** positioned in the staple cavities **5530**. Each staple **5540** comprises a base **5541**, a first leg **5542** extending from the base **5541**, and a second leg **5544** extending from the base **5541**. Moreover, each staple **5540** comprises an integral driver portion that is directly engaged by a sled during a staple firing stroke which is discussed in greater detail below in connection with FIG. **33**. Each staple cavity **5530** comprises a central guide portion **5531** configured to guide the base **5541** of a staple **5540**, a first end **5532** configured to guide the first leg **5542** of the staple **5540**, and a second end **5534** configured to guide the second leg **5544** of the staple **5540** when the staple **5540** is lifted from an unfired position (FIG. **23**) to a fired position (FIG. **24**). As the staple **5540** is being fired, the first leg **5542** and the second leg **5544** contact forming pockets defined in an anvil positioned opposite the staple cartridge **5500** and are deformed generally inwardly, i.e., generally toward one another into a formed configuration, such as a B-shaped configuration, for example.

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In various instances, further to the above, the staple **5540** may become malformed during the staple forming process. For instance, one or both of the staple legs **5542** and **5544** may deform outwardly instead of inwardly during the staple forming process. While such outward deformation of the legs **5542** and **5544** may be acceptable in some circumstances, such malformation may not be desirable to some clinicians. To prevent, or at least inhibit, the malformation of the staple legs **5542** and **5544**, the staple **5540** and the staple cavity **5530** comprise co-operating features which bias the staple legs **5542** and **5544** inwardly during the staple forming process, as described in greater detail below.

Referring primarily to FIG. **24**, the staple cavity **5530** comprises a first cam **5533** and the staple **5540** comprises a first cam shoulder **5543** which engages the first cam **5533** as the staple **5540** is lifted upwardly toward the anvil. When the first cam shoulder **5543** contacts the first cam **5533**, the first leg **5542** is pushed inwardly, i.e., toward the second leg **5544**. In various instances, the first cam **5533** and the first cam shoulder **5543** are configured and arranged such that first cam shoulder **5543** contacts the first cam **5533** prior to the first leg **5542** contacting the anvil forming pocket registered with the first leg **5542**. In such instances, the first leg **5542** has inward momentum when the first leg **5542** contacts the anvil which, as a result, facilitates the proper deformation of the staple **5540**. In other instances, the first cam **5533** and the first cam shoulder **5543** are configured and arranged such that first cam shoulder **5543** contacts the first cam **5533** at the same time that the first leg **5542** contacts the anvil forming pocket registered with the first leg **5542**. In such instances, the anvil forming pocket and the first cam **5533** co-operatively provide two points of contact for the first staple leg **5542** as the first staple leg **5542** is being deformed. In various other instances, the first cam **5533** and the first cam shoulder **5543** are configured and arranged such that first cam shoulder **5543** contacts the first cam **5533** after the first leg **5542** contacts the anvil forming pocket registered with the first leg **5542**.

Referring primarily to FIG. **24**, the staple cavity **5530** further comprises a second cam **5535** and the staple **5540** further comprises a second cam shoulder **5545** which engages the second cam **5535** as the staple **5540** is lifted upwardly toward the anvil. When the second cam shoulder **5545** contacts the second cam **5535**, the second leg **5544** is pushed inwardly, i.e., toward the first leg **5542**. In various instances, the second cam **5535** and the second cam shoulder **5545** are configured and arranged such that second cam shoulder **5545** contacts the second cam **5535** prior to the second leg **5544** contacting the anvil forming pocket registered with the second leg **5544**. In such instances, the second leg **5544** has inward momentum when the second leg **5544** contacts the anvil which, as a result, facilitates the proper deformation of the staple **5540**. In other instances, the second cam **5535** and the second cam shoulder **5545** are configured and arranged such that second cam shoulder **5545** contacts the second cam **5535** at the same time that the second leg **5544** contacts the anvil forming pocket registered with the second leg **5544**. In such instances, the anvil forming pocket and the second cam **5535** co-operatively provide two points of contact for the second staple leg **5544** as the second staple leg **5544** is being deformed. In various other instances, the second cam **5535** and the second cam shoulder **5545** are configured and arranged such that second cam shoulder **5545** contacts the second cam **5535** after the second leg **5544** contacts the anvil forming pocket registered with the second leg **5544**.

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A portion of a staple cartridge **6500** is illustrated in FIGS. **25-27**. The staple cartridge **6500** comprises a cartridge body **5510** including staple cavities **5530** defined therein and staples **6540** positioned in the staple cavities **5530**. Each staple **6540** comprises a base **6541**, a first leg **6542** extending from the base **6541**, and a second leg **6544** extending from the base **6541**. Moreover, each staple **6540** comprises an integral driver portion that is directly engaged by a sled during a staple firing stroke. Further to the above, the first leg **6542** of the staple **6540** comprises an arcuate portion **6543** defined therein which is configured to contact the first cam **5533** as the staple **6540** is moved into its fired position. Similarly, the second leg **6544** of the staple **6540** comprises an arcuate portion **6545** defined therein which is configured to contact the second cam **5535** as the staple **6540** is moved into its fired position. The arcuate portion **6543** of the first leg **6542** and the arcuate portion **6545** of the second leg **6544** are cut-out during the stamping process. That said, various alternative embodiments are envisioned in which the arcuate portions **6543** and **6545** are bent into the staple legs **6542** and **6544**, respectively, during a secondary forming process.

A portion of a staple cartridge **7500** is illustrated in FIGS. **28** and **29**. The staple cartridge **7500** comprises a cartridge body **5510** including staple cavities **5530** defined therein and staples **7540** positioned in the staple cavities **5530**. Each staple **7540** comprises a base **7541**, a first leg **7542** extending from the base **7541**, and a second leg **7544** extending from the base **7541**. Moreover, each staple **7540** comprises an integral driver portion that is directly engaged by a sled during a staple firing stroke. Further to the above, the first leg **7542** of the staple **7540** comprises a bump **7543** defined therein which is configured to contact the first cam **5533** as the staple **7540** is moved into its fired position. Similarly, the second leg **7544** of the staple **7540** comprises a bump **7545** defined therein which is configured to contact the second cam **5535** as the staple **7540** is moved into its fired position. The bump **7543** of the first leg **7542** and the bump **7545** of the second leg **7544** are cut-out during the stamping process.

A portion of a staple cartridge **8500** is illustrated in FIGS. **30-32**. The staple cartridge **8500** comprises a cartridge body **5510** including staple cavities **5530** defined therein and staples **8540** positioned in the staple cavities **5530**. Each staple **8540** comprises a base **8541**, a first leg **8542** extending from the base **8541**, and a second leg **8544** extending from the base **8541**. Moreover, each staple **8540** comprises an integral driver portion that is directly engaged by a sled during a staple firing stroke. Further to the above, the first leg **8542** of the staple **8540** comprises an angled shoulder **8543** defined therein which is configured to contact the first cam **5533** as the staple **8540** is moved into its fired position. Similarly, the second leg **8544** of the staple **8540** comprises an angled shoulder **8545** defined therein which is configured to contact the second cam **5535** as the staple **8540** is moved into its fired position. The angled shoulder **8543** of the first leg **8542** and the angled shoulder **8545** of the second leg **8544** are cut-out during the stamping process. Moreover, the first leg **8542** and the second leg **8544** of the staple **8540** comprise notches or cut-outs **8549** defined therein which are configured to induce the legs **8542** and **8544** to bend inwardly, or at least substantially toward one another, during the staple forming process and assume a desired formed configuration.

A stamped staple **100** is depicted in FIG. **33**. The staple **100** comprises a proximal staple leg **110**, a distal staple leg **120**, and a staple base portion **130**. The staple **100** further comprises vertical transition portions, or bends, **118**, **128** and lateral transition portions, or bends, **116**, **126**. The

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vertical transition portions **118**, **128** bend, or extend, the legs **110**, **120** vertically, or upward, from the staple base portion **130**. The lateral transition portions **116**, **126** extend the staple legs **110**, **120** laterally outward, or at least substantially perpendicularly with respect to the staple base portion **130**. The staple legs **110**, **120** define a first plane and the staple base **130** defines a second plane. Together, the vertical transition portions **118**, **128** and the lateral transition portions **116**, **126** permit the staple legs **110**, **120** to be laterally offset and parallel with respect to the staple base portion **130**. Stated another way, the first plane is offset from and at least substantially parallel to the second plane. In FIG. **33**, the first plane is offset in a negative Y direction, which is orthogonal to a vertical Z direction. Other staples may be used in conjunction with a plurality of staples **100** where the other staples comprise a first plane which is offset in the positive Y direction. The use of both types of staples permits staple rows to be nested, or interwoven, where staple legs of neighboring rows may be at least substantially aligned and/or share a common longitudinal axis. In various instances, the staple rows can be nested to provide denser staple rows.

Further to the above, the proximal staple leg **110** comprises a generally rectangular cross-section including flat surfaces and corners. The corners of the cross-section comprise bevels, radiuses, and/or coined edges **114** which reduce the exposure of sharp edges to the patient tissue. That said, the proximal staple leg **110** comprises a sharp tip **112** configured to incise the patient tissue. Similarly, the distal staple leg **120** comprises a generally rectangular cross-section including flat surfaces **125** and corners **124** which are beveled, radiused, and/or coined to reduce the exposure of sharp edges to the patient tissue. Like the proximal leg **110**, the distal staple leg **120** comprises a sharp tip **122** configured to incise the patient tissue.

The staple base **130** comprises an upper portion **136** configured to contact and support patient tissue. The upper portion **136** of the staple base **130** comprises tissue contacting surfaces **137**, **138**, and **139** and edges **134** which are beveled, radiused, and/or coined to reduce the exposure of the sharp edges to the patient tissue. The staple base **130** further comprises a lower portion **135** which includes a drive cam **132** configured to be directly engaged by a sled. The lower portion **135** further comprises a bottom edge **131** which rides over the apex of a sled rail and a distal shoulder **133** which loses contact with the sled rail as the sled moves distally.

Further to the above, the legs **110** and **120** of the staple **100** extend in a first plane and the drive cam **132** of the staple **100** is defined in a second plane. The second plane is parallel to, or at least substantially parallel to, the first plane. When the legs **110** and **120** are deformed, the legs **110** and **120** capture patient tissue within the staple **100** outside of the second plane. Among other things, such an arrangement allows a larger volume of tissue to be captured within the staple **100** as compared to wire staples that are defined in a single plane. That said, such wire staples are desirable in many instances and, in some instances, can be used in conjunction with stamped staples. The entire disclosures of U.S. patent application Ser. No. 14/318,996, entitled FASTENER CARTRIDGES INCLUDING EXTENSIONS HAVING DIFFERENT CONFIGURATIONS, now U.S. Patent Application Publication No. 2015/0297228, U.S. patent application Ser. No. 15/385,907, entitled SURGICAL INSTRUMENT SYSTEM COMPRISING AN END EFFECTOR LOCKOUT AND A FIRING ASSEMBLY LOCKOUT, now U.S. Patent Application Publication No.

2018/0168608, and U.S. patent application Ser. No. 15/191,775, entitled STAPLE CARTRIDGE COMPRISING WIRE STAPLES AND STAMPED STAPLES, now U.S. Patent Application Publication No. 2017/0367695 are incorporated by reference herein.

A staple cartridge **9500** is illustrated in FIGS. **34-38**. The staple cartridge **9500** comprises a cartridge body **9510** including a proximal end **9511** and a distal nose **9513**. The cartridge body **9510** further comprises a deck **9512**, a longitudinal slot **9520** extending from said proximal end **9511** toward the distal nose **9513**, and longitudinal rows of staple cavities **9530** defined in the deck **9512** extending between the proximal end **9511** and the distal nose **9513**. The cartridge body **9510** also comprises longitudinal tissue compression rails **9515** and **9516** extending upwardly from the deck **9512**. The longitudinal compression rail **9515** extends along a first side of the longitudinal slot **9520** and the longitudinal compression rail **9516** extends along a second, or opposite, side of the longitudinal slot **9520**.

Further to the above, referring primarily to FIGS. **37** and **38**, the staple cartridge **9500** further comprises a staple **9540** stored in each staple cavity **9530** and staple drivers **9580** which support and drive the staples **9540** out of the staple cavities **9530** during a staple firing stroke. In this embodiment, each staple driver **9580** only supports and drives one staple **9540**, but other embodiments are envisioned in which a staple driver supports and drives more than one staple. The staple cartridge **9500** also comprises a sled **9550** which progressively contacts the staple drivers **9580** and lifts the staple drivers **9580** and staples **9540** within their respective staple cavities **9530** as the sled **9550** is moved distally during the staple firing stroke. Further to the above, the sled **9550** is pushed distally by a tissue cutting knife of a drive system during the staple firing stroke. After the staple firing stroke has been completed and/or otherwise stopped, the tissue cutting knife is retracted back into its unfired position. Notably, the sled **9550** is not retracted proximally and is instead left in its distal fired position. Such an arrangement can be used as part of a spent cartridge/missing cartridge firing lockout, as discussed above.

Further to the above, the staple cartridge **9500** comprises an electrode circuit **9590**. The electrode circuit **9590** comprises an electrical connector **9595** configured to engage a corresponding electrical connector in a surgical instrument when the staple cartridge **9500** is seated in the surgical instrument. The electrode circuit further comprises a longitudinal row of electrode contacts **9594** positioned in apertures defined in the longitudinal tissue compression rail **9516** and a flex circuit **9592** and conductor bar **9596** electrically connecting the electrode contacts **9594** to the electrical connector **9595**. As discussed herein, electrical power is supplied to the electrode circuit **9590** to seal the patient tissue in co-operation with the staples **9540**.

Further to the above, referring primarily to FIG. **38**, each staple **9540** of the staple cartridge **9500** comprises a base **9541** and legs **9542** extending from the base **9541**. Each staple driver **9580** comprises a seat **9581** slideably positioned in a staple cavity **9530** which is configured to receive and support the base **9541** of a staple **9540** positioned in the staple cavity **9530**. The seat **9581** of the staple driver **9580** is sized and configured such that it is closely received within its staple cavity **9530**. As a result, the movement of the staple driver **9580** is constrained, or at least substantially constrained, to upward movement toward the anvil positioned opposite the staple cartridge **9500** during the staple firing stroke. As such, the lateral movement, longitudinal movement, and/or rotation of the staple driver **9580** within the

staple cavity **9530** is prevented, or at least limited, owing to the close fit therebetween. In addition, each staple driver **9580** comprises a lateral support **9589** slideably positioned within a support cavity **9539** defined in the cartridge body **9510**. The lateral supports **9589** of the staple drivers **9580** extend inwardly and above the seats **9581** and are sized and configured such that the lateral supports **9589** are closely received within the support cavities **9539**. As a result, the lateral supports **9589** prevent, or at least limit, lateral movement, longitudinal movement, and/or rotation of the staple drivers **9580** within the staple cavities **9530**. In at least one embodiment, the lateral supports **9589** extend into cavities defined under the longitudinal compression rails **9515** and **9516** when the staple drivers **9580** are in their fired positions, as illustrated in FIG. **39**. Moreover, the lateral supports **9589** of one row of the staple drivers **9580** are positioned under the electrode contacts **9594** when the staple drivers **9580** are in their fired positions.

A staple cartridge **10500** is illustrated in FIGS. **39-41** and is similar to the staple cartridge **9500** in many respects which are not discussed herein for the sake of brevity. The staple cartridge **10500** comprises a cartridge body **10510** and longitudinal rows of staple cavities **10530** defined therein. The staple cartridge **10500** further comprises longitudinal rows of staple drivers **10580** configured to fire the staples positioned in the staple cavities **10530**. Each staple driver **10580** comprises a staple seat slideably positioned in a staple cavity **10530**, a lateral support **10539** slideably positioned in a support cavity **10589**, and a drive surface, or cam, **10585** positioned intermediate the staple seat and the lateral support **10539**. The drive cams **10585** of a longitudinal row of staple drivers **10580** are aligned, or at least substantially aligned, with one another longitudinally such that a ramp of a sled can sequentially engage all of the drive cams **10585** during the staple firing stroke. The staple drivers **10580** are driven from an unfired, or low, position (FIG. **40**) to a fired, or raised, position (FIGS. **39** and **41**) by the sled during the staple firing stroke. In various instances, the staple drivers **10580**, and the staples supported thereon, may accidentally be displaced upwardly within the staple cavities **10530** while the staple cartridge **10500** is being handled and/or inserted into the stapling instrument. To prevent, or at least inhibit, this from happening, each staple driver **10580** comprises a latch **10588** which is releasably engaged within a lock window **10517** defined in the cartridge body **10510** when the staple drivers **10580** are in their unfired positions. When the sled contacts the staple drivers **10580**, however, the latches **10588** release from the lock windows **10517** which permits the staple drivers **10580** to be lifted into their fired positions. Moreover, the latch **10588** can engage a lock shoulder **10518** defined in the cartridge body **10510** to hold the staple driver **10580** in its fired position so that the staple driver **10580** does not sink back down into its staple cavity **10530** after the sled passes thereby. Such an arrangement allows the staple drivers **10580** to hold the staples in their deformed shapes thereby reducing springback of the staples after the staple firing stroke, for example.

A staple cartridge **11500** is illustrated in FIGS. **42-44** and is similar to the staple cartridges **9500** and **10500** in many respects, most of which will not be discussed herein for the sake of brevity. The staple cartridge **11500** comprises a cartridge body **11510** including a deck **11512**, a longitudinal slot **11520** configured to receive a tissue cutting knife, and longitudinal rows of staple cavities **11530** defined in the deck **11512**. The cartridge body **11510** further comprises longitudinal tissue compression rails **11515** and **11516** extending upwardly from the deck **11512**. The staple car-

tridge **11500** further comprises staples removably stored in the staple cavities **11530**, staple drivers **11580** configured to support and drive the staples during a staple firing stroke, and a sled configured to sequentially drive the staple drivers **11580** and the staples from an unfired position to a fired position during the staple firing stroke. The staple cartridge **11500** also comprises an electrode circuit **11590** which, although not illustrated, includes electrode contacts on the longitudinal tissue compression rails **11515** and **11516**.

Further to the above, referring primarily to FIGS. **43** and **44**, each staple driver **11580** comprises a staple seat **11581** including a slot configured to support a staple, a lateral support **11589**, and a drive cam **11585** connecting the staple seat **11581** and lateral support **11589**. Notably, the lateral support **11589** of each staple driver **11580** is positioned laterally inwardly with respect to the staple seat **11581** and is closely received within a support cavity defined in the cartridge body **11510**. The support cavities on one side of the staple cartridge **11500** comprise openings **11519** defined in the longitudinal tissue compression rail **11516** which are sized and configured to permit the lateral supports **11589** of the drivers **11580** to protrude upwardly from the cartridge body **11510** when the staple drivers **11580** are lifted into their unfired positions. Such an arrangement allows the lateral supports **11589** to provide additional anti-roll stability to the staple drivers **11580**. In addition to or in lieu of the above, the longitudinal tissue compression rail **11515** can comprise openings **11519** which are configured to receive the lateral supports **11589** of the other row of staple drivers **11580**. Also, notably, the lateral support **11589** extends proximally relative to the staple seat **11581**. Such an arrangement also provides anti-roll stability to the staple drivers **11580**. In various alternative embodiments, the lateral supports **11589** extend distally relative to the staple seats **11581**. Similar to the above, each staple driver **11580** comprises a latch arm **11588** which releasably secures the staple driver **11580** in its unfired and fired positions and provides additional stability support in those positions.

A staple cartridge **12500** is illustrated in FIGS. **45-48B** and is similar to the staple cartridges **9500**, **10500**, and **11500** in many respects, most of which will not be discussed herein for the sake of brevity. The staple cartridge **12500** comprises a cartridge body **12510** including a longitudinal slot **12520** defined therein which is configured to receive a tissue cutting knife. The cartridge body **12510** also includes a longitudinal row of staple cavities **12530** defined on each side of the longitudinal slot **12520**. The staple cartridge **12500** further comprises staples removably stored in the staple cavities **12530**, longitudinal rows of staple drivers **12580** configured to support and drive the staples, a sled **12550** moveable from a proximal unfired position (FIG. **45**) to a distal fired position to engage and drive the staple drivers **12580** during a staple firing stroke, and a pan **12505** that is attached to and extends at least partially under the cartridge body **12510**. The pan **12505** prevents, or at least inhibits, the staple drivers **12580** from being accidentally dislodged from their unfired positions and/or falling out of the bottom of the cartridge body **12510** until the staple cartridge **12500** is seated in a surgical instrument **12000** (FIG. **48A**), for example.

Further to the above, referring primarily to FIGS. **46-48**, each staple driver **12580** comprises a staple seat **12581**, two lateral supports **12589**, and a drive cam **12585**. One of the lateral supports **12589** is laterally-aligned with the staple seat **12581** and the other lateral support **12589** is positioned proximally with respect to the staple seat **12581**. Each staple driver **12580** further comprises staple supports **12582** which

limit the movement of the staple supported thereon. The staple supports **12582** have a sufficient height to control the movement of the staple and prevent the staple from sliding laterally off of the staple seat **12581**. In at least one embodiment, the staple supports **12582** extend above the base of the staple positioned in the staple seat **12581**. Notably, the staple supports **12582** have open longitudinal ends. That said, the longitudinal movement of the staples within the staple cavities **12530** can be constrained by the longitudinal ends of the staple cavities **12530**. In any event, referring to FIG. **48**, the overall height of the staple seat **12581** is defined between the top of the staple supports **12582** and a bottom surface **12583**. As illustrated in FIG. **48**, the overall height of the lateral supports **12589** is taller than the overall height of the staple seat **12581**. Moreover, the lateral supports **12589** extend vertically above the staple seat **12581**. Also, the lateral supports **12589** extend vertically below the staple seat **12581**. Such an arrangement stabilizes the staple seat **12581** during the staple forming process. Notably, referring to FIG. **48B**, the pan **12505** comprises clearance openings **12509** defined therein for the lateral supports **12589** when the staple drivers **12580** are in their unfired position.

A staple cartridge **13500** is illustrated in FIGS. **49** and **50** and is similar to the staple cartridges **9500**, **10500**, **11500**, and **12500** in many respects, most of which will not be discussed herein out of the sake of brevity. The staple cartridge **13500** comprises a cartridge body **13510** including a longitudinal slot **13520** configured to receive a tissue cutting knife. The cartridge body **13510** further comprises longitudinal rows of staple cavities **13530** defined therein. The staple cartridge **13500** further comprises staples removably stored in the staple cavities **13530** and longitudinal rows of staple drivers **13580** configured to support and drive the staples from an unfired position to a fired position during a staple firing stroke. Each staple driver **13580** comprises a staple seat **13581**, two lateral supports **13589** positioned laterally with respect to the staple seat **13581**, and a drive cam **13585** positioned between the staple seat **13581** and the lateral supports **13589**. The staple seat **13581** further comprises staple supports **13582** which define a groove configured to receive the base of a staple and enclosed longitudinal ends **13586** which co-operatively limit the lateral and longitudinal movement of the staple relative to the staple driver **13580**.

Further to the above, each staple driver **13580** comprises a guide slot **13584** defined in the staple seat **13581** which is slideably engaged with a guide rail **13514** defined in the cartridge body **13510**. The guide rails **13514** and the guide slots **13584** comprise co-operating features which permit the staple drivers **13580** to move upwardly within the staple cavities **13530** but prevent, or at least limit, lateral translation, longitudinal translation, and/or rotation of the staple drivers **13580** within the staple cavities **13530**. In various instances, the guide rails **13514** are closely received within guide slots **13584** to prevent, or limit, such relative movement. In at least one such embodiment, the guide rails **13514** and the guide slots **13584** comprise a dovetail arrangement, for example.

Further to the above, the staple cartridge **13500** further comprises electrode contacts positioned on longitudinal rails **13515** extending upwardly from the upper surface, or deck, of the cartridge body **13510**. During use, the current flows from and/or through the electrode contacts and into the patient tissue to heat, cauterize, and/or seal the patient tissue. In some instances, the patient tissue may stick to the electrode contacts. The cartridge body **13510** further comprises longitudinal rows of openings **13519** defined therein

which are configured to permit the lateral supports **13589** to extend above the cartridge body **13510** when the staple drivers **13580** are in their fired positions. In such instances, the lateral supports **13589** can lift the cauterized tissue away from the electrode contacts and free the patient tissue from the staple cartridge **13500**. In such instances, the patient tissue is at least partially cauterized before the tissue is incised and lifted away from the cartridge body **13510** during the staple firing stroke.

A staple driver **14580** is illustrated in FIGS. **51** and **52**. The staple driver **14580** comprises two staple seats **14581**, a lateral support **14589**, and a driver cam **14585** which connects the staple seats **14581** and the lateral support **14589** together. One of the staple seats **14581** is positioned in a first staple cavity defined in a staple cartridge and the other staple seat **14581** is positioned in a second staple cavity defined in a staple cartridge. The staple seats **14581** are aligned longitudinally with one another and aligned longitudinally with other staple seats **14581** of other staple drivers **14580** in the staple cartridge. Each staple seat **14581** comprises a groove configured to support the base of a staple and staple supports **14582** configured to limit the relative movement of the staple base relative to the staple seat **14581**. Moreover, each staple seat **14581** comprises guide end rails **14586** which extend into corresponding guide slots defined in the staple cavities which co-operatively prevent, or at least limit, lateral translation, longitudinal translation, and rotation of the staple seats **14581** within their staple cavities. Further to the above, each staple seat **14581** comprises a latch **14588** configured to releasably hold the staple driver **14580** in its unfired position and/or fired position.

A staple cartridge **15500** is illustrated in FIGS. **53** and **54** and is similar to the other staple cartridges disclosed herein in many respects, most of which will not be discussed herein for the sake of brevity. The staple cartridge **15500** comprises a cartridge body **15510** including a deck, a longitudinal slot **15520** defined therein which is configured to receive a tissue cutting knife, and, also, a longitudinal row of staple cavities **15530** defined on each side of the longitudinal slot **15520**. The cartridge body **15510** further comprises a deck and longitudinal tissue compression rails **15515** extending upwardly from the deck. Further to the above, one or both of the tissue compression rails **15515** is configured to support and/or house one or more electrodes. As discussed in greater detail further below, the cartridge body **15510** further comprises pocket extenders **15537** extending upwardly from the deck. When patient tissue is clamped against the staple cartridge **15500**, the pocket extenders **15537** atraumatically grip the patient tissue and prevent, or at least inhibit, the patient tissue from sliding relative to the staple cartridge **15500**.

Further to the above, the staple cartridge **15500** further comprises staples **15540** stored in the staple cavities **15530**, staple drivers **15580** configured to support and drive the staples **15540**, and a sled **15550** configured to sequentially engage the staple drivers **15580** during a staple firing stroke. Similar to the above, each staple **15540** comprises a base and legs **15542** extending from the base. Each staple driver **15580** comprises a seat configured to receive and support the base of a staple **15540** positioned in a staple cavity **15530**. Each staple driver **15580** further comprises lateral supports **15589** which provide stability to the seat and a guide slot **15584** defined in the seat which co-operates with a vertical guide rail **15534** defined in the staple cavity **15530** to control the movement of the staple driver **15580**. The sled **15550** comprises a central portion **15554** positioned in the longitudinal slot **15520** and projections **15552** extending from the

opposite sides of the central portion **15554** which are configured to engage the sidewalls of the longitudinal slot **15520**. The interaction between the projections **15552** and the sidewalls of the longitudinal slot **15520** inhibits the sled **15550** from being accidentally moved distally prior to the staple firing stroke but permits the sled **15550** to be moved distally by the firing drive of a surgical instrument during the staple firing stroke. When the sled **15550** is not being pushed distally by the firing drive, the sled **15550** is held in position. The sled **15550** further comprises two ramps **15555**—one on each side of the central portion **15554**—which are each configured to engage and drive a longitudinal row of staple drivers **15580**.

A staple driver **21580** and a staple **21540** of a staple cartridge are illustrated in FIGS. **72-74**. The staple **21540** is comprised of wire and includes a base **21541** and legs **21542** extending upwardly from the base **21541**. The staple **21540** is depicted in its unfired configuration in FIG. **72** and is substantially V-shaped, for example. In at least one embodiment, the legs **21542** of the staple **21540** are engaged with the longitudinal ends of a staple cavity which resiliently bias the legs **21542** inwardly when the staple **21540** is positioned in the staple cavity. When the staple **21540** is moved from its unfired position to its fired position by the staple driver **21540**, the legs **21542** emerge from the staple cavity and contact the anvil forming pockets positioned opposite the staple cavity. In some instances, the legs **21542** begin to splay outwardly as the staple **21540** is lifted upwardly into its fired position. The pocket extenders **15537** (FIG. **53**) mentioned above in connection with the staple cartridge **15500** can limit the outward splay of the staple legs **21542** and assist in maintaining the alignment between the staple legs **21542** and the anvil forming pockets.

Further to the above, the staple driver **21580** comprises a staple seat **21581** including a groove defined therein which supports the base **21541** of the staple **21540** and enclosed ends **21582** which co-operatively prevent, or at least limit, the lateral translation and/or longitudinal translation of the staple base **21541** relative to the staple seat **21581**. Notably, the enclosed ends **21582** of the staple seat **21581** extend above the base **21541** of the staple **21540** when the staple **21540** is positioned in the staple seat **21581**. The staple driver **21580** further comprises a drive cam **21585** positioned laterally inwardly with respect to the staple seat **21581** and a stability support **21589** extending from the drive cam **21585**. As the staple **21540** is pushed upwardly into its fired position by the staple driver **21580**, the enclosed ends **21582** of the staple driver **21580** and the pocket extenders **15537** of the cartridge body **15510** co-operatively support the staple legs **21582** as the staple **21580** is being deformed into its fired configuration.

A staple cartridge **16500** is illustrated in FIGS. **55-60** and is similar to the other staple cartridges disclosed herein in many respects, most of which will not be discussed herein out of the sake of brevity. The staple cartridge **16500** comprises a cartridge body **16510** including a deck **16512**, a longitudinal slot **16520** defined therein which is configured to receive a tissue cutting knife, and a longitudinal row of staple cavities **16530** defined on each side of the longitudinal slot **16520**. The cartridge body **16510** further comprises longitudinal tissue compression rails **16515** extending upwardly from the deck **16512** where one or both of the tissue compression rails **16515** is configured to support and/or house one or more electrodes. The cartridge body **16510** further comprises pocket extenders **16537** extending upwardly from the deck **16512**. When patient tissue is clamped against the staple cartridge **16500**, the pocket

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extenders **16537** atraumatically grip the patient tissue and prevent, or at least inhibit, the patient tissue from sliding relative to the staple cartridge **16500**.

Further to the above, the staple cartridge **16500** further comprises staples stored in the staple cavities **16530**, staple drivers **16580** configured to support and drive the staples **16540**, and a sled configured to sequentially engage the staple drivers **16580** during a staple firing stroke. Referring primarily to FIG. **58**, each staple driver **16580** comprises a staple seat **16581**, lateral supports **16589**, and a drive cam connecting the lateral supports **16589**. Each staple cavity **16530** comprises a lateral support cavity **16539** within which the lateral supports **16589** are closely received to resist unwanted lateral and longitudinal translation and/or unwanted rotation of the staple driver **16580**. Notably, referring primarily to FIGS. **59** and **60**, the top of the lateral support cavities **16539** are enclosed and provide an upward stop for the staple drivers **16580** during the staple firing stroke. In addition, referring to FIGS. **57** and **58**, each staple driver **16580** further comprises a latch, or lock arm, **16588** which releasably engages a sidewall of a lock window **16517** defined in the cartridge body **16510** to releasably hold the staple driver **16580** in its unfired position (FIG. **59**) until the staple driver **16580** is driven upwardly by the sled. The lock arm **16588** comprises a cantilever which flexes inwardly when the staple driver **16580** is lifted upwardly by the sled and then resiliently flexes outwardly when the staple driver **16580** reaches its fired position (FIG. **60**). In such instances, the lock arm **16588** engages the deck **16512** and holds the staple driver **16580** in its fired position.

A staple cartridge **17500** is illustrated in FIGS. **61-63**, and is similar to the other staple cartridges disclosed herein in many respects, most of which will not be discussed herein out of the sake of brevity. The staple cartridge **17500** comprises a cartridge body **17510** including a deck **17512**, a longitudinal slot **17520** defined therein which is configured to receive a tissue cutting knife, and a longitudinal row of staple cavities **17530** defined on each side of the longitudinal slot **17520**. The cartridge body **17510** further comprises longitudinal tissue compression rails **17515** extending upwardly from the deck **17512** where one or both of the tissue compression rails **17515** is configured to support and/or house one or more electrodes. The cartridge body **17510** further comprises pocket extenders **17537** extending upwardly from the deck **17512**. When patient tissue is clamped against the staple cartridge **17500**, the pocket extenders **17537** atraumatically grip the patient tissue and prevent, or at least inhibit, the patient tissue from sliding relative to the staple cartridge **17500**.

Further to the above, the staple cartridge **17500** further comprises staples stored in the staple cavities **17530**, staple drivers **17580** configured to support and drive the staples, and a sled configured to sequentially engage the staple drivers **17580** during a staple firing stroke. Referring primarily to FIG. **63**, each staple driver **17580** comprises a staple seat **17581** which defines a groove configured to receive the base of a staple, staple supports **17582** extending to the lateral sides of the groove, lateral supports **17589**, and a drive cam **17585** connecting the lateral supports **17589** to the staple seat **17581**. Each staple cavity **17530** comprises a lateral support cavity within which the lateral supports **17589** are closely received to resist unwanted lateral and longitudinal translation and/or unwanted rotation of the staple driver **17580**. Notably, the lateral supports **17589** of each staple driver **17580** define a guide slot **17584** therebetween which closely receives a guide rail **17534** defined in a staple cavity **17530**. The guide slot **17584** and guide rail

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17534 co-operatively constrain the movement of the staple driver **17580** to vertical movement within the staple cavity **17530**. Also, notably, the lateral supports **17589** are positioned laterally outwardly with respect to the staple seat **17581** and do not extend under the longitudinal tissue compression rails **17515**. In addition, each staple driver **17580** further comprises a latch, or lock arm, **17588** which is releasably engaged with a sidewall of an internal lock window defined in the cartridge body **17510** to releasably hold the staple driver **17580** in its unfired position until the staple driver **17580** is driven upwardly by the sled. The lock arm **17588** comprises a cantilever which flexes inwardly when the staple driver **17580** is lifted upwardly by the sled and then resiliently flexes outwardly when the staple driver **17580** reaches its fired position. In such instances, the lock arm **17588** engages the deck **17512** and holds the staple driver **17580** in its fired position. A lock shoulder of the lock arm **17588** faces outwardly toward the lateral supports **17589** but could extend in any suitable direction.

A staple cartridge **18500** is illustrated in FIGS. **64** and **65** and is similar to the other staple cartridges disclosed herein in many respects, most of which are not discussed herein for the sake of brevity. The staple cartridge **18500** comprises a cartridge body **18510** including a longitudinal slot **18520** configured to receive a tissue cutting knife and a longitudinal row of staple cavities **18530** defined on each side of the longitudinal slot **18520**. The cartridge body **18510** further comprises an upper portion, or deck, **18512** and longitudinal tissue compression rails **18515** and **18516** extending upwardly from the deck **18512**. The staple cartridge **18500** further comprises a staple **18540** positioned in each staple cavity **18530**, staple drivers **18580** configured to support and drive the staples **18540** during a staple firing stroke, and a sled configured to contact and drive the staple drivers **18580**. The staple cartridge **18500** further comprises an electrode circuit **18590** including electrode contacts **18594** housed within the longitudinal tissue compression rail **18516** and a conductor **18596** electrically connecting the electrode contacts **18594**. As illustrated in FIG. **64**, each electrode contact **18594** extends longitudinally and the electrode contacts **18594** collectively extend along a substantial majority of the longitudinal tissue compression rail **18516**. In at least one embodiment, the electrode contacts **18594** extend along at least 90% of the longitudinal length of the tissue compression rail **18516**, for example. In at least one embodiment, the electrode contacts **18594** cover at least 95% of the longitudinal length of the tissue compression rail **18516**, for example.

A staple cartridge **19500** is illustrated in FIGS. **66-69** and is similar to the other staple cartridges disclosed herein in many respects, most of which will not be discussed herein out of the sake of brevity. The staple cartridge **19500** comprises a cartridge body **19510** including a deck, a longitudinal slot **19520** configured to receive the firing member **1570** (FIG. **69**) of the firing drive **1600**, and longitudinal rows of staple cavities **19530**. The staple cartridge **19500** further comprises a staple **19540** positioned in each staple cavity **19530**, staple drivers **19580** configured to support and drive the staples **19540** during a staple firing stroke, and a sled **19550** configured to sequentially contact and push the staple drivers **19580** and **19540** upwardly within the staple cavities **19530** during the staple firing stroke. Referring primarily to FIG. **67**, the sled **19550** comprises a central portion **19554** which slides within the longitudinal slot **19520**, and lateral ramps **19555** which slide within longitudinal ramp slots defined in the cartridge body **19510** and engage the staple drivers **19580**. When the staple

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cartridge **19500** is seated in the cartridge jaw **1310**, referring primarily to FIG. **69**, the sled **19550** is positioned over, but not operably engaged with, the drive screw **1560**. Notably, the drive screw **1560** is closely received within a clearance slot **19553** defined in the bottom of the sled **19550** such that there is little gap between the drive screw **1560** and the sled **19550**. During the staple firing stroke, the drive screw **1560** is rotated to drive the firing member **1570** distally which pushes the sled **19550** distally.

Further to the above, the firing member **1570** is configured to pull the anvil jaw toward the staple cartridge **19500** during the staple firing stroke. In many instances, as a result, the staple cartridge **19500** can experience a significant compressive load—especially around the staples **19540** being deformed against the anvil jaw. Notably, the sled **19550** is positioned directly under the staple drivers **19580** being lifted by the sled **19550** and can support the cartridge body **19510** if it deflects downwardly as a result of the compressive load. Referring again to FIGS. **66** and **67**, the sled **19550** comprises angled support shoulders **19551** defined on opposite sides thereof. The angled support shoulders **19551** of the sled **19550** are directly adjacent to and/or are in abutting contact with angled shoulders **19511** defined in the cartridge body **19510** which extend along the longitudinal length thereof. As a result, the cartridge body **19510** can be directly supported by the sled **19550** and limit the deflection of the cartridge body **19510** during the staple firing stroke. In some instances, the sled **19550** can be pushed downwardly against the drive screw **1560** by the cartridge body **19510**. As such, the surface of the clearance aperture **19553** in the sled **19550** is smooth such that the sled **19550** can slide over and relative to the drive screw **1560** even though the drive screw **1560** is rotating.

Further to the above, each staple driver **19580** comprises a lateral stability support **19589** configured to slide within a support slot **19539** defined in the cartridge body **19510**. Each staple driver **19580** further comprises a clearance recess **19583** defined therein which is configured to closely receive the drive screw **1560** when the staple drivers **19580** are in their unfired positions. Such an arrangement allows for a staple cartridge **19500** that is vertically compact.

A staple cartridge **20500** is illustrated in FIGS. **70** and **71**. The staple cartridge **20500** comprises a cartridge body **20510** comprising staple cavities, a pan **20505** attached to the cartridge body **20510**, staples removably stored in the staple cavities, and staple drivers. The pan **20505** comprises a plurality of latches and/or lock windows engaged with features defined on the cartridge body **20510** which secure the pan **20505** to the cartridge body **20510**. Further to the above, the pan **20505** at least partially extends under the cartridge body **20510** and prevents, or at least inhibits, the staple drivers and staples stored within the cartridge body **20510** from being accidentally dislodged from their unfired positions when the staple cartridge **20500** is loaded into a cartridge jaw.

Further to the above, the cartridge body **20510** further comprises supports **20501** embedded therein. In at least one embodiment, the cartridge body **20510** is comprised of a plastic material which is injection molded around the supports **20501** such that the supports **20501** are integrally formed with the cartridge body **20510**. Referring to FIG. **71**, each support **20501** comprises an upper portion **20502** embedded in the deck of the cartridge body **20510** and a lower portion **20503** which extends out of the bottom of the cartridge body **20510**. When the pan **20505** is assembled to the cartridge body **20510**, the lower portions **20503** of the supports **20501** are engaged with and/or directly adjacent to

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the pan **20505**. When a compression load is applied to the staple cartridge **20500** as a result of the end effector being closed, further to the above, the supports **20501** resist the downward deflection of the cartridge body **20510** by transmitting at least a portion of the compression load into the pan **20505**. During the staple firing stroke, in at least one embodiment, the supports **20501** yield, or give way, under the compressive load and/or as the result of the sled contacting the supports **20501** and bending them out of contact with the pan **20505**. As a result of the above, the staple cartridge **20500** is able to resist the compressive loading during use but is not re-usable.

A staple cartridge **22500** is illustrated in FIGS. **75-79** and is similar to other staple cartridge disclosed herein in many respects, most of which will not be discussed herein for the sake of brevity. The staple cartridge **22500** comprises a cartridge body **22510** including staple cavities **22530** defined therein, a staple positioned in each staple cavity **22530**, staple drivers **22580** configured to drive the staples upwardly within the staple cavities **22530**, and a sled **22550** movable from a proximal unfired position (FIG. **77**) to a distal fired position (FIG. **79**) to engage the staple drivers **22580** during a staple firing stroke. Referring primarily to FIGS. **75** and **76**, the sled **22550** comprises lateral angled drive plane surfaces **22555** configured to engage and lift the staple drivers **22580** during the staple firing stroke. Each angled drive plane surface **22555** extends from the distal, or wedge tip, end of the sled **22550** to the proximal, apex, end of the sled **22550**. Each staple driver **22580** comprises a corresponding angled cam plane surface which slides upwardly on one of the angled drive plane surfaces **22555** as the sled **22550** slides under the staple drivers **22580**. Each staple driver **22580** comprises a guide key **22859** extending therefrom which is slideably received in a key slot defined in the cartridge body **22510** which constrains the motion of the staple drivers **22580** to vertical movement within the cartridge body **22510**.

FIGS. **80-85** illustrate a drive system **23000** for use with a surgical instrument, such as those described herein. The drive system **23000** comprises a shift motor **23100**, a drive motor **23300**, and a lock bar, or brake, **23400**. Referring primarily to FIG. **81**, the shift motor **23100** comprises a rotary output shaft **23110** including an external thread portion **23120**. The shift motor **23100** may be a stepper motor or any suitable motor configured to actuate the rotary output shaft **23110** between a plurality of set rotated positions. The threaded portion **23120** is threadably engaged with a motor carrier **23200**. Specifically, internal threads of the motor carrier **23200** are threadably engaged with the external thread portion **23120** of the rotary output shaft **23110**. As such, when the rotary output shaft **23110** is rotated in a first direction, the motor carrier **23200** is translated distally. Notably, the motor carrier **23200** does not rotate with the rotary output shaft **23110**. Correspondingly, when the rotary output shaft **23110** is rotated in a second direction opposite the first direction, the motor carrier **23200** is translated proximally.

Further to the above, the motor carrier **23200** comprises an opening **23220** configured to receive the drive motor **23300**. The drive motor **23300** is fixed and/or attached to the motor carrier **23200** such that the drive motor **23300** translates with the motor carrier **23200**. Any suitable method may be utilized to affix the drive motor **23300** within the opening **23220** of the motor carrier **23200** such as welding, and/or adhesives, and/or fasteners, for example. Other embodiments are envisioned where the drive motor **23300** is press fit into the opening **23220** of the motor carrier **23200**.

Further, other embodiments are envisioned where the motor carrier **23200** and the drive motor **23300** are one unitary component. In any event, the motor carrier **23200** and the drive motor **23300** translate together between a plurality of positions in response to the actuation of the rotary output shaft **23110** of the shift motor **23100** between a plurality of radial positions.

Further to the above, the drive motor **23300** comprises a rotary output shaft, or drive motor shaft **23310**. The drive motor shaft **23310** extends distally from a body portion **23305** of the drive motor **23300**. The drive motor shaft **23310** comprises a proximal radial groove **23320** and a distal radial groove **23330** spaced apart from one another along the drive motor shaft **23310**. The radial grooves **23320**, **23330** define narrower shaft portions compared to the remainder of the drive motor shaft **23310**. Further, the drive system **23000** comprises a main drive gear **23340** fixed to the drive motor shaft **23310** intermediate the proximal radial groove **23320** and the distal radial groove **23330**. The main drive gear **23340** may be fixed to the drive motor shaft **23310** using any suitable means such as welding, and/or fasteners, and/or adhesives, for example. Other embodiments are envisioned where the main drive gear **23340** is press fit onto the drive motor shaft **23310**, for example. In any event, rotation of the drive motor shaft **23310** via the drive motor **23300** will result in the rotation of the main drive gear **23340**. Further, the main drive gear **23340** is configured to rotate one of a plurality of output drive gears and their respective output shafts depending upon the longitudinal position of the drive motor **23300**, as discussed in greater detail below.

Further to the above, the drive system **23000** further comprises a lock bar, or brake **23400**, a first output gear **23500**, a second output gear **23600**, and a third output gear **23700**. Referring primarily to FIG. **81**, the brake **23400** comprises a body portion **23405** including a clevis portion **23407** extending laterally from the body portion **23405**. The clevis portions **23407** comprises a proximal collar **23410** and a distal collar **23420** spaced apart from one another. The proximal collar **23410** is configured to be received around the proximal radial groove **23320**, and the distal collar **23420** is configured to be received around the distal radial groove **23330**. Specifically, the proximal collar **23410** comprises a proximal opening **23412** which receives the drive motor shaft **23310** in the region of the proximal radial groove **23320**. Further, the distal collar **23420** comprises a distal opening **23422** which receives the drive motor shaft **23310** in the region of the distal recess **23330**. Further, the brake **23400** is free to rotate about the drive motor shaft **23310**. As such, the brake **23400**, the drive motor **23300**, and the drive motor shaft **23310** translate together when the shift motor **23100** is actuated; however, the brake **23400** does not rotate with the drive shaft **23310**. Other embodiments are envisioned where the brake **23400** is operably attached to the handle or housing of the instrument such that the brake **23400** translates with the drive motor **23300** without the brake **23400** being attached to the drive motor shaft **23310**. In any event, the brake **23400** translates with the drive motor shaft **23310** to selectively engage two of the three output gears **23500**, **23600**, and **23700** to prevent their rotation while permitting one of the three output gears **23500**, **23600**, and **23700** to rotate, as discussed in greater detail below.

Referring primarily to FIG. **81**, the first output gear **23500** comprises a first output shaft **23510** extending distally therefrom, the second output gear **23600** comprises a second output shaft **23610** extending distally therefrom, and the third output gear **23700** comprises a third output shaft **23710**

extending distally therefrom. The output drive shafts **23510**, **23610**, **23710** are rotatably supported within the handle or housing of the instrument and are configured to effectuate different motions within an end effector or stapling attachment of a surgical instrument. Further, the output drive shafts **23510**, **23610**, **23710** are nested within one another. Specifically, the first output drive shaft **23510** is received within an opening **23620** in the second output drive shaft **23610**, and the first and second output drive shafts **23510**, **23610** are received within an opening **23720** in the third output drive shaft **23710**. As such, the output drive shafts **23510**, **23610**, **23710** are rotatable relative to one another about the same longitudinal axis.

Referring primarily to FIG. **82**, the brake **23400** comprises a pair of longitudinal teeth **23430** extending laterally from the body portion **23405**. The pair of longitudinal teeth **23430** extend longitudinally along the entire body portion **23405** except for a gap **23440** defined in the pair of longitudinal teeth **23430**. Specifically, FIGS. **83-85** illustrate the gap **23440** in the pair of longitudinal teeth **23430**. The longitudinal teeth **23430** are configured to meshingly engage with teeth of the output gears **23500**, **23600**, **23700** to selectively prevent their rotation depending upon the longitudinal position of the brake **23400**. Specifically, the longitudinal position of the brake **23400**, which is translatable by the shift motor **23100**, determines which of the output gears **23500**, **23600**, **23700** can be freely rotated, as discussed in greater detail below.

In use, when the shift motor **23100** positions the drive motor **23300** and brake **23400** in a first position, as illustrated in FIG. **83**, teeth on the main drive gear **23340** are meshingly engaged with teeth on the first output gear **23500**. As such, rotation of the main drive gear **23340** will rotate the first output gear **23500** and the first output drive shaft **23510** to perform a first end effector function. Further, the gap **23440** of the brake **23400** is positioned such that the pair of longitudinal teeth **23430** of the brake **23400** are only engaged with the second output gear **23600** and the third output gear **23700** and, thus, the second output gear **23600** and the third output gear **23700** are prevented from rotating—thereby locking out a second end effector function and a third end effector function.

In various embodiments, the first end effector function comprises the articulation of the end effector, for example. In at least one such embodiment, the end effector of the surgical instrument is rotatable about an articulation joint. In at least one embodiment, the second end effector function comprises rotating the end effector about a longitudinal axis, for example. In at least one such embodiment, the surgical instrument comprises a rotation joint proximal to the articulation joint which permits at least a portion of the shaft and the end effector of the surgical instrument to rotate about the longitudinal axis. In at least one embodiment, the surgical instrument comprises a rotation joint distal to the articulation joint which permits the end effector to rotate relative to the shaft about a longitudinal axis. In at least one embodiment, the third end effector function comprises advancing a tissue cutting knife distally through the end effector, for example.

Further to the above, when the shift motor **23100** positions the drive motor **23300** and the brake **23400** in a second position, as illustrated in FIG. **84**, the teeth of the main drive gear **23340** are meshingly engaged with the teeth of the second output gear **23600**. As such, rotation of the main drive gear **23340** will rotate the second output gear **23600** and the second output drive shaft **23610**. Further, the gap **23440** of the brake **23400** is positioned such that the pair of

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longitudinal teeth **23430** of the brake **23400** are only engaged with the first output gear **23500** and the third output gear **23700**—and not the second output gear **23600**—and, thus, the first output gear **23500** and the third output gear **23700** are prevented from rotating.

Further to the above, when the shift motor **23100** positions the drive motor **23300** and the brake **23400** in a third position, as illustrated in FIG. **85**, the teeth of the main drive gear **23340** are meshingly engaged with teeth of the third output gear **23700**. As such, rotation of the main drive gear **23340** will rotate the third output gear **23700** and the third output drive shaft **23710**. Further, the gap **23440** of the brake **23400** is positioned such that the pair of longitudinal teeth **23430** of the brake **23400** are only engaged with the first output gear **23500** and the second output gear **23600**—and not the third output gear **23700**—and, thus, the first output gear **23500** and the second output gear **23600** are prevented from rotating.

FIGS. **86-92** illustrate a drive system **24000** for use with a surgical instrument, such as those described herein. The drive system **24000** comprises a drive motor **24100** and a shift motor **24200**. Referring primarily to FIG. **89**, the drive motor **24100** comprises a rotary input shaft **24110** and a drive motor gear **24120** mounted onto the rotary input shaft **24110**. The drive motor gear **24120** is operably engaged with a first idler gear **24130**, a second idler gear **24140**, and a third idler gear **24150**. Specifically, the teeth of the drive motor gear **24120** are meshingly engaged with only the teeth of the first idler gear **24130** while the teeth of the first idler gear **24130** are meshingly engaged with the teeth of the second idler gear **24140** and the teeth of the third idler gear **24150**. The second idler gear **24140** and the third idler gear **24150** are positioned on opposite sides of the first idler gear **24130**. As such, rotation of the drive motor gear **24120** via the drive motor **24100** results in simultaneous rotation of the first idler gear **24130**, the second idler gear **24140**, and the third idler gear **24150**. The above being said, other embodiments are envisioned where the drive motor gear **24120** is positioned in between all three idler gears **24130**, **24140**, **24150** and meshingly engaged with all three idler gears **24130**, **24140**, **24150**.

Referring primarily to FIG. **88**, the first idler gear **24130** is mounted to a first rotary input shaft **24132**, the second idler gear **24140** is mounted to a second rotary input shaft **24142**, and the third idler gear **24150** is mounted to a third rotary input shaft **24152**. In the illustrated embodiment, the drive motor gear **24120** and the idler gears **24130**, **24140**, **24150** are attached to their respective shafts **24132**, **24142**, **24152** via a pin, or screw. However, other embodiments are envisioned where the drive motor gear **24120** and the idler gears **24130**, **24140**, **24150** are fixed and/or attached to their respective shafts **24132**, **24142**, **24152** using any suitable means such as welding, adhesives, press fitting, etc., for example. In any event, the first rotary input shaft **24132** comprises a first input clutch **24134** extending from its distal end, the second rotary input shaft **24142** comprises a second input clutch **24144** extending from its distal end, and the third rotary input shaft **24152** comprises a third input clutch **24154** extending from its distal end. The input clutches **24134**, **24144**, **24154** are configured to be selectively engageable with three different output clutches, as discussed in greater detail below.

Referring primarily to FIG. **88**, the shift motor **24200** comprises a shift motor shaft **24210** comprising a rotary index shaft **24220**. The rotary index shaft **24220** defines a longitudinal axis LA and is configured to rotate about its longitudinal axis LA when the shift motor **24200** is actuated.

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The shift motor **24200** may be a stepper motor or any suitable motor configured to actuate the rotary index shaft **24220** between a plurality of set rotated positions, for example. Further, the rotary index shaft **24220** comprises three separate cam profiles **24222**, **24224**, **24226** extending all the way around the rotary index shaft **24220**, as discussed in greater detail below.

Further to the above, the rotary index shaft **24220** comprises a first cam profile **24222**, a second cam profile **24224**, and a third cam profile **24226**. Each of the first, second, and third cam profiles **24222**, **24224**, **24226** define a radial groove in the rotary index shaft **24220**. Further, each cam profile **24222**, **24224**, **24226** is different when viewed in reference to the longitudinal axis LA. Specifically, the first cam profile **24222** is identical to the second cam profile **24224**; however, the second cam profile **24224** is rotated approximately 60 degrees relative to the first cam profile **24222** about the longitudinal axis LA. Further, the second cam profile **24224** is identical to the third cam profile **24226**; however, the third cam profile is rotated approximately 60 degrees relative to the second cam profile **24225**. It shall be understood that any suitable orientation of the cam profiles **24222**, **24224**, **24226** relative to one another are contemplated. As discussed in greater detail below, each of the cam profiles **24222**, **24224**, **24226** are distinctly defined in the rotary index shaft **24220** relative to the longitudinal axis LA to effectuate different movements of three separate cams.

Referring primarily to FIG. **88**, a first cam **24300** comprises an opening **24310** configured to receive the rotary index shaft **24220**. The first cam **24300** comprises a first cam pin **24320** (see FIG. **87**) extending through the opening **24310** and into the first cam profile **24222** such that the first cam pin **24320** rides within and along the first cam profile **24222** when the rotary index shaft **24220** is rotated. Further, a second cam **24400** comprises an opening **24410** configured to receive the rotary index shaft **24220**. The second cam **24400** comprises a second cam pin **24420** (see FIG. **87**) extending through the opening **24410** and into the second cam profile **24224** such that the second cam pin **24420** rides within and along the second cam profile **24224** when the rotary index shaft **24220** is rotated. Further, a third cam **24500** comprises an opening **24510** configured to receive the rotary index shaft **24220**. The third cam **24500** comprises a third cam pin **24520** extending through the opening **24510** and into the third cam profile **24226** such that the third cam pin **24520** rides within and along the third cam profile **24226** when the rotary index shaft **24220** is rotated. As discussed in greater detail below, each of the cams **24300**, **24400**, **24500** can translate longitudinally relative to the longitudinal axis LA when the rotary index shaft **24220** is rotated about the longitudinal axis LA.

Referring primarily to FIG. **88**, the first cam **24300** comprises a first lateral flange **24330** and a first opening **24340** defined in the first lateral flange **24330**. The second cam **24400** comprises a second lateral flange **24430** and a second opening **24440** defined in the second lateral flange **24430**. The third cam **24500** comprises a third lateral flange **24530** and a third opening **24540** defined in the third lateral flange **24530**. A first rotary output shaft **24600** extends through the first opening **24340**, a second rotary output shaft **24700** extends through the second opening **24440**, and a third rotary output shaft **24800** extends through the third opening **24540**, as discussed in greater detail below.

Further to the above, the first rotary output shaft **24600**, the second rotary output shaft **24700**, and the third rotary output shaft **24800** are rotatably mounted to the surgical instrument. The output shafts **24600**, **24700**, **24800** are

rotatably supported within the instrument by thrust bearings, for example, and/or any other suitable means. A first output clutch **24610** is slideably mounted on the proximal end of the first rotary output shaft **24600**. The first output clutch **24610** comprises a protrusion, or key, **24630** positioned in a groove **24640** defined in the first output shaft **24600**. The protrusion and groove arrangement **24630**, **24640** permits the first output clutch **24610** to slide, or translate, relative to the first output shaft **24600** and also rotate with the first output shaft **24600**. Further, a second output clutch **24710** is slideably mounted on the proximal end of the second rotary output shaft **24700**. The second output clutch **24710** comprises a protrusion, or key, **24730** positioned in a groove **24740** defined in the second output shaft **24700**. The protrusion and groove arrangement **24730**, **24740** permits the second output clutch **24710** to slide, or translate, relative to the second output shaft **24700** and also rotate with the second output shaft **24700**. Further, a third output clutch **24810** is slideably mounted on the proximal end of the third rotary output shaft **24800**. The third output clutch **24810** comprises a protrusion, or key, **24830** positioned in a groove **24840** in the third output shaft **24800**. The protrusion and groove arrangement **24830**, **24840** permits the third output clutch **24810** to slide, or translate, relative to the third output shaft **24800** and also rotate with the third output shaft **24800**.

Referring primarily to FIG. **88**, the first output clutch **24610** comprises a first radial groove **24620** that is received in—and rotatable within—the first opening **24340** of the first cam **24300**. The second output clutch **24710** comprises a second radial groove **24720** that is received in—and rotatable within—the second opening **24440** of the second cam **24400**. The third output clutch **24810** comprises a third radial groove **24820** that is received in—and rotatable within—the third opening **24540** of the third cam **24500**. As such, the first output clutch **24610** is rotatable relative to the first cam **24300**, the second output clutch **24710** is rotatable relative to the second cam **24400**, and the third output clutch **24810** is rotatable relative to the third cam **24500**. Further, the sidewalls of the radial grooves **24620**, **24720**, **24820** of the output clutches **24610**, **24710**, **24810**, respectively, provide bearing surfaces for the cam members **24300**, **24400**, **24500** to translate the output clutches **24610**, **24710**, **24810** relative to their respective output shafts **24600**, **24700**, **24800**. As discussed in greater detail below, such translation of the output clutches **24610**, **24710**, **24810** relative to their respective output shafts **24600**, **24700**, **24800** allows the output clutches **24610**, **24710**, **24810** to be selectively engaged with and disengaged from their respective input clutches **24134**, **24144**, **24154**.

Referring to FIG. **90**, the rotary index shaft **24220** of the shift motor **24200** is in a first radial position relative to the longitudinal axis LA. The cams **24300**, **24400**, **24500** are in a first configuration when the rotary index shaft **24200** is in its first radial position. In the first configuration, the first cam **24300** and the first output clutch **24610** are in a distal position where the first output clutch **24610** is not engaged with the first input clutch **24134**. Further, in the first configuration, the second cam **24400** and the second output clutch **24710** are in a proximal position where the second output clutch **24710** is engaged with the second input clutch **24144**. Further, in the first configuration, the third cam **24500** and the third output clutch **24810** are in a distal position where the third output clutch is not engaged with the third input clutch **24154**. As such, in the first configuration, only the second output clutch **24710** is engaged with its respective input clutch **24400**. Therefore, when the cams **24300**, **24400**, **24500** are in their first configuration (FIG.

90), rotation of drive motor gear **24120** will result in rotation of the second output shaft **24700**.

Referring to FIG. **91**, the rotary index shaft **24220** has been rotated into a second radial position about the longitudinal axis LA from the first radial position in FIG. **90**. The cams **24300**, **24400**, **24500** are in a second configuration when the rotary index shaft **24220** is in its second radial position. Specifically, the first cam **24300** and the second cam **24400** have moved toward one another while the third cam **24500** remains in the same longitudinal position as in the first configuration of FIG. **90**. The first cam **24300** and the second cam **24400** are translated toward one another due to the first and second cam profiles **24222**, **24224** of the rotary index shaft **24220** cammingly engaging the first and second cam pins **24320**, **24420** of the first and second cams **24300**, **24400** when the rotary index shaft **24220** is rotated from its first radial position to its second radial position. A dwell of the third cam profile **24226** is radially oriented relative to the first and second cam profiles **24222**, **24224** such that the third cam pin **24520** is not translated when the rotary index shaft **24200** rotates from its first radial position to its second radial position. As such, the third cam **24500** and the third output clutch **24810** do not translate when the rotary index shaft **24220** rotates from its first radial position to its second radial position.

Further to the above, when the cams **24300**, **24400**, **24500** are in the second configuration as illustrated in FIG. **91**, the first cam **24300** and the first output clutch **24610** are in a proximal position where the first output clutch **24610** is engaged with the first input clutch **24134**. Further, in the second configuration, the second cam **24400** and the second output clutch **24710** are in a distal position where the second output clutch **24710** is not engaged with the second input clutch **24144**. Further, in the second configuration, the third cam **24500** and the third output clutch **24810** remain in their distal position where the third output clutch **24810** is not engaged with the third input clutch **24154**. As such, in the second configuration, only the first output clutch **24610** is engaged with its respective input clutch **24134**. Therefore, when the cams **24300**, **24400**, **24500** are in their second configuration (FIG. **91**), rotation of drive motor gear **24120** will result in rotation of first output shaft **24600**.

Referring to FIG. **92**, the rotary index shaft **24220** has been rotated into a third radial position about the longitudinal axis LA from its second radial position in FIG. **91**. The cams **24300**, **24400**, **24500** are in a third configuration when the rotary index shaft **24220** is in its third radial position. Specifically, the first cam **24300** and the third cam **24500** move away from one another while the second cam **24400** remains in the same longitudinal position as the second configuration (FIG. **91**). The first cam **24300** and the third cam **24500** are translated away from one another due to the first and third cam profiles **24222**, **24226** of the rotary index shaft **24220** cammingly engaging the first and third cam pins **24320**, **24520** of the first and third cams **24300**, **24500** when the rotary index shaft **24220** is rotated from its second radial position to its third radial position. A dwell of the second cam profile **24224** is radially oriented relative to the first and third cam profiles **24222**, **24226** such that the second cam pin **24420** is not translated when the rotary index shaft **24200** rotates from its second radial position to its third radial position. As such, the second cam **24400** and the second output clutch **24710** do not translate when the rotary index shaft **24220** rotates from its second radial position to its third radial position.

Further to the above, when the cams **24300**, **24400**, **24500** are in the third configuration as illustrated in FIG. **92**, the

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first cam **24300** and the first output clutch **24610** are in the distal position where the first output clutch **24610** is not engaged with the first input clutch **24134**. Further, in the third configuration, the second cam **24400** and the second output clutch **24710** remain in the distal position where the second output clutch **24710** is not engaged with the second input clutch **24144**. Further, in the third configuration, the third cam **24500** and the third output clutch **24810** are in a proximal position where the third output clutch **24810** is engaged with the third input clutch **24154**. As such, in the third configuration, only the third output clutch **24810** is engaged with its respective input clutch **24154**. Therefore, when the cams **24300**, **24400**, **24500** are in their third configuration (FIG. 92), rotation of drive motor gear **24120** will result in rotation of third output shaft **24800**.

Referring to FIG. 86, the rotary index shaft **24220** is in a fourth radial position that is different than the first radial position (FIG. 90), the second radial position (FIG. 91), and the third radial position (FIG. 92). When the rotary index shaft **24220** is in the fourth radial position, the cam members **24300**, **24400**, **24500** are in a fourth configuration. In the fourth configuration, the cams **24300**, **24400**, **24500** and their respective output clutches **24610**, **24710**, **24810** are in their distal positions where the output clutches **24610**, **24710**, **24810** are not engaged with their respective input clutches **24134**, **24144**, **24154**. As such, when the cams **24300**, **24400**, **24500** are in the fourth configuration (FIG. 86), rotation of drive motor gear **24120** will not result in the rotation of any of the output shafts **24600**, **24700**, **24800**.

FIGS. 93-96 depict a surgical instrument assembly **25000** comprising a shaft **25010**, an end effector **25020**, and an articulation joint, or region, **25030**. The surgical instrument assembly **25000** further comprises a primary drive shaft **25060** configured to actuate a function of the end effector **25020** and articulation actuators **25050** configured to articulate the end effector **25020** relative to the shaft **25010** about pivot axis PA. The shaft **25010** comprises a distal end **25011** comprising tabs **25012** extending from the distal end **25011** of the shaft **25010**. The shaft **25010** further comprises a central cavity **25014** configured to receive the primary drive shaft **25060** and articulation actuators **25050** therethrough. The central cavity **25014** may also receive other drive shafts, frame components, and/or electrical components therethrough, for example. The end effector **25020** comprises a proximal end **25021** comprising tabs **25023** extending from the proximal end **25021** of the end effector **25020**. The tabs **25012** are pivotally coupled to the tabs **25023** to pivotally couple the shaft **25010** and the end effector **25020** together to enable articulation of the end effector **25020** relative to the shaft **25010**. The tabs **25012** and the tabs **25023** are pivotally coupled to each other by way of pins **25031**. The pivot axis PA is defined by the pins **25031**.

The articulation joint **25030** comprises an articulation support pivot **25040**. The articulation support pivot **25040** comprises a cylindrical member positioned within a cavity **25022** defined between the tabs **25012** and the tabs **25023** and is configured to pivot when actuated by articulation actuators **25050**. While the term 'cylindrical' is used, the articulation support pivot need not resemble a perfect cylinder. Each articulation actuator **25050** comprises a distal end **25051**. The distal ends **25051** are pinned to the articulation support pivot **25040** by way of actuation pin **25035**. The articulation actuators **25050** may comprise any suitable type of actuator such as, for example, flexible actuators, cables, flexible plastic plates, electroactive polymer actuators, and/or piezoelectric bimorph actuators. The articulation support pivot **25040** comprises a central cavity **25041**

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defined therethrough along a longitudinal axis LA. The primary drive shaft **25060** is configured to be received through the central cavity **25041**. In at least one instance, the primary drive shaft **25060** is flexible and is configured to bend, or flex, as the end effector **25030** is articulated relative to the shaft **25010**. In at least one instance, the primary drive shaft **25060** comprises a flexible actuator. In at least one instance, the primary drive shaft **25060** comprises a linearly translatable actuator. In at least one instance, the primary drive shaft **25060** comprises rotary drive shaft. In at least one instance, the primary drive shaft **25060** is flexible, is configured to be rotated to actuate a function of the end effector, and is configured to be translated to actuate a function of the end effector **25020**.

In at least one instance, the articulation support pivot comprises a prism structure, a spherical structure, and/or a rectangular structure.

To articulate the end effector **25020**, the articulation actuators **25050** are configured to be pushed and pulled in an antagonistic manner to articulate the end effector **25030** relative to the shaft **25010**. For example, a first actuator **25050** is configured to push a first side of the pin **25035** distally and a second actuator **25050** is configured to pull a second side of the pin **25035** proximally resulting in the rotation, or pivoting, of the articulation support pivot **25040** to articulate the end effector **25020** in a first direction. Similarly, the first actuator **25050** is configured to pull a first side of the pin **25035** proximally and the second actuator **25050** is configured to push a second side of the pin **25035** distally resulting in the rotation, or pivoting, of the articulation support pivot **25040** to articulate the end effector **25020** in a second direction opposite the first direction. In at least one instance, the primary drive shaft **25060** is bent, or pivoted, by the central cavity **25041** of the articulation support pivot **25040**. As a result, the primary drive shaft **25060** is configured to apply a pivot force to the end effector **25020** to articulate the end effector **25020** in the desired direction.

In at least one instance, a first articulation actuator **25050** is actively actuated and passive movement of a second articulation actuator **25050** is dependent on the actuation of the first actuator **25050**. In at least one instance, only one articulation actuator **25050** is provided.

In at least one instance, the end effector **25020** is fixedly attached to the articulation support pivot **25040** such that, as the articulation support pivot **25040** is rotated by the actuators **25050** and actuation pin **25035**, the articulation support pivot **25040** directly articulates the end effector **25020** relative to the shaft **25010** by virtue of the fixed relationship between the end effector **25020** and the articulation support pivot **25040**. In such an instance, the end effector **25020** may aid in flexing the primary drive shaft **25060** when the end effector **25020** is articulated relative to the shaft **25010**.

In at least one instance, the articulation support pivot **25040** defines a central axis which is transverse to the longitudinal axis LA. In at least one instance, the central axis is aligned with the pivot axis PA. In such an instance, the articulation support pivot **25040** rotates about the pivot axis PA. In at least one instance, the articulation support pivot **25040** is configured float laterally within the articulation joint **25030**. In such an instance, the axis about which the articulation support pivot **25040** rotates is not fixed relative to the end effector **25020** and/or the shaft **25010** and, rather, moves laterally and/or longitudinally relative to the end effector **25020** and/or the shaft **25010**. Such a configuration may provide a degree of flexibility within the articulation

joint **25030** by removing a fixed pivot axis and providing a semi-movable, or floatable, pivot axis.

As can be seen in FIG. **95**, the articulation support pivot **25040** is configured to prevent the primary drive shaft **25060** from blowing out of the articulation joint **25030**. The central cavity **25041** is configured to restrain the primary drive shaft **25060** within the articulation joint **25030** as the end effector **25020** is articulated relative to the shaft **25010**. In at least one instance, the central cavity **25041** laterally and vertically supports the primary drive shaft **25060** through the articulation joint **25030**. In at least one instance, the articulation pin **25035** provides a vertical support limit within the central cavity **25041**.

In at least one instance, the articulation support pivot **25040** is assembled with the shaft **25010** and end effector **25020** and then the primary drive shaft **25060** is inserted through the shaft **25010** and central cavity **25041** and into the end effector **25020**. As a result, the primary drive shaft **25060** itself is configured to prevent disassembly of the articulation joint **25030**. In such an instance, the primary drive shaft **25060** itself holds one or more components of the articulation joint **25030** together.

FIG. **97** depicts a surgical instrument assembly **25100** comprising many of the same components of the surgical instrument assembly **25000**. The surgical instrument assembly **25100** comprises an articulation joint **25130** comprising pivot pins **25131** which, unlike the surgical instrument assembly **25100**, pin the tabs **25012**, **25023** to each other in addition to an articulation support pivot **25140**. The articulation support pivot **25140** may comprise the same and/or similar functions of the articulation support pivot **25040**. The articulation support pivot **25140** comprises a central cavity **25141** defined therethrough configured to receive a portion of the pin **25035** and the primary drive shaft **25060**. The articulation joint **25130** may allow for a more distinct pivot by pivotally coupling the shaft **25010** to the articulation support pivot **25140**. In at least one instance, the end effector **25020** is fixedly attached to the articulation support pivot **25140**. In at least one instance, the end effector **25020** is pivotally attached to the articulation support pivot **25140**.

FIGS. **98** and **99** depict a surgical instrument assembly **25200** comprising an end effector cartridge **25210**, a firing member **25270**, and a plurality of flexible actuators **25220**. The actuators **25220** comprise a plurality of first actuators **25260** and a tube **25230**. The tube **25230** may comprise a linearly translatable member configured to push and/or pull the firing member **25270**. In at least one instance, the tube **25230** acts only as a jacket to the actuator **25260** to allow a flex circuit **25240** to be wrapped therearound. The surgical instrument assembly **25200** may comprise an articulation joint through which the actuators **25220** are configured to extend. To this end, each actuator **25260** comprises a plurality of slits **25261** configured to allow the actuators **25260** to flex, or bend, in a first predetermined direction. The direction may correspond to the plane of articulation through which the end effector cartridge **25210** is articulated. In at least one instance, each actuator **25260** comprises additional slits to allow the actuators **25260** to flex, or bend, in a second predetermined direction in addition to the first predetermined direction. Such a configuration would permit the use of the actuators **25260** in a multi-axis articulation joint where the end effector cartridge **25210** may be articulated in two distinct planes.

In at least one instance, the actuators **25260** are provided to articulate the end effector cartridge **25210** by applying an articulation force to the end effector cartridge **25210** through the firing member **25270**. The actuators **25260** may com-

prise an electroactive polymer and/or a piezoelectric bimorph configured to be energized to bend the actuators **25260** into a desired bent configuration thereby causing the firing member **25270** to which the actuators **25260** are attached to be moved in a predetermined direction. The actuators **25260** may also be advanced and/or rotated to effect one or more functions of the end effector cartridge **25210** and/or end effector assembly comprising the end effector cartridge **25210**. For example, the actuators **25260** may be translated linearly to push the firing member **25270** distally and/or pull the firing member **25270** proximally. In at least one instance, the actuators **25260** are configured to apply a rotational force to the firing member **25270** to rotate the end effector cartridge **25210** relative to a shaft, for example. In such an instance, the actuators **25260** may be actuated by a planetary gear train, for example.

The slits **25260** may be formed in the actuators **25260** by way of any suitable method. For example, the slits **25260** may be laser cut into the actuators **25260**. The actuators **25260** may comprise of any suitable material and/or materials. For example, the actuators **25260** may comprise of a metal material and may be actuated by way of additional articulation bands, cables, and/or plates, for example. In at least one instance, the actuators **25260** comprise of an electroactive polymer and are configured to be energized and de-energized to bend and/or advance/retract the actuators **25260**.

Referring to FIG. **99**, the flex circuit **25240** is attached to the firing member **25270**. The flex circuit **25240** is spiral wrapped, or coiled, around the tube **25230**. In at least one instance, the coiling of the flex circuit **25240** is configured to reduce capacitive coupling between various electrical components within a shaft, for example, by fluctuating the position of the flex circuit **25240** radially within the shaft.

In at least one instance, a control circuit is provided configured to actively mitigate capacitive coupling. An active inductor tunable impedance system can be employed to monitor and mitigate capacitive coupling within a surgical instrument assembly.

In at least one instance, a control circuit is configured to provide active power management to electrical systems within a surgical instrument assembly. In such an instance, the control circuit is configured to detect capacitive coupling and actively adjust power delivery to reduce capacitive coupling between various electrical components within the surgical instrument assembly.

In at least one instance, the flex circuit **25240** is wrapped around one or more components of a shaft assembly such that in a neutral, un-rotated state, the flex circuit **25240** is in a minimum tension state. In such an instance, rotation of components which would cause the flex circuit **25240** to rotate as well would cause the flex circuit **25240** to increase in tension as the flex circuit **25240** twists. The flex circuit **25240** can be configured to experience a maximum amount of twist-induced tension before a control circuit stops rotation. In various instances, rotation in a first direction causes the flex circuit **25240** to tighten around the shaft and rotation in the opposite direct causes the flex circuit **25240** to loosen around the shaft. Such a configuration provides a magnitude of slack in the system prior to rotation of components of the shaft assembly. In at least one instance, the flex circuit **25240** is manufactured in a coiled state. In at least one instance, the flex circuit **25240** is manufactured in a non-coiled state and is assembled into a neutral coiled state. Manufacturing the flex circuit **25240** in a coiled state can permit a thicker and/or wider flex circuit allowing for more signal transmission, for example. In at least one instance, the coiled

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configuration of the flex circuit **25240** reduces capacitive coupling between various signal transmission lines. In at least one instance, multiple ground layers or planes can be employed to surround radio frequency signals and/or isolate any stray fields generated within the surgical instrument assembly.

FIGS. **100** and **101** depict an articulation system **25300** configured to be used with a surgical instrument assembly. The articulation system **25300** comprises a shaft **25301**, a biasing system **25310**, and an articulation joint **25330** comprising a plurality of electromagnets **25351** and a plurality of shaft segments **25360** configured to flex the articulation joint **25330** and, thus, the shaft **25301**, in an articulation plane. The biasing system **25310** is configured to bias the articulation system into a non-articulated configuration.

The biasing system **25310** comprises a ratchet fork **25311**, translatable rack members **25320** and slave cables **25340** attached to the translatable rack members **25320** and a proximal shaft segment **25360**. The ratchet fork **25311** comprises toothed prongs **25312** configured to flex inwardly relative to each other when a spring force of the ratchet fork **25311** is overcome owing to translation of one or more of the translatable rack members **25320**. The toothed prongs **25312** are engaged with the translatable rack members **25320** such that, as the translatable rack members **25320** are pushed and/or pulled by the articulation joint **25350**, the toothed prongs **25312** ride against teeth **25321** of the rack members **25320** to provide a predetermined holding force to the rack members **25320**. The slave cables **25340** are attached to a distal end **25322** of each rack member **25320** to translate the pushing and/or pulling force of the articulation joint **25350** to the rack members **25320**. The rack members **25320** are attached to coil springs **25330** within a shaft assembly, for example, such that as the articulation joint **25350** is articulated, the coil springs **25330** are configured to push the rack members **25320** away from the articulation joint **25350** as slack is introduced to a corresponding slave cable **25340** and pulled toward the articulation joint **25350** as tension is applied to a corresponding slave cable **25340** by the articulation joint **25350**.

To articulate the shaft **25301** with the articulation joint **25350**, the shaft segments **25360** are actuated in an accordion-like manner such that the electromagnets **25351** on one side of the articulation joint **25350** are energized to attract the electromagnets **25351** to each other to contract this side of the articulation joint **25350** and bend the shaft **25301** in a first direction. In at least one instance, the electromagnets **25351** on the other side of the articulation joint **25350** are de-energized, or not energized, so as to allow the electromagnets to move away from each other with the expansion of this other side of the articulation joint **25350** owing to the direction of articulation caused by the electromagnets **25351** which are energized. In at least one instance, the electromagnets **25351** on the expansion side of the articulation joint **25350** are energized in such a manner so as to repel the electromagnets on the expansion side of the articulation joint **25350** so as to aid expansion of this side of the articulation joint **25350**. Similarly, the articulation joint **25350** may be bent in the other direction by energizing the electromagnets in a manner opposite to the manner described above.

In at least one instance, each electromagnet **25351** is energized simultaneously to attract and repel the desired electromagnets **25351**. In at least one instance, the proximal electromagnet **25351** attached to the slave cable is energized to activate contraction and/or expansion of the entire chain of electromagnets distal to the proximal electromagnet **25351** on one side of the articulation joint **25350**. In at least

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one instance, both sides of the articulation joint **25350** are energized corresponding to the desired configuration (expanded or contracted). Cables **25352** may contract and expand according to the desired configuration of the articulation joint **25350**. In at least one instance, the cables **25352** are configured to bias the articulation joint **25350** into a non-articulated configuration and are only compressed, or relaxed, and/or stretched, or pulled into tension, upon energizing the corresponding electromagnets **25351**.

As discussed above, the biasing system **25310** is configured to bias the articulation joint **25350** into a non-articulated configuration. In at least one instance, the electromagnets **25351** are de-energized to allow the biasing system **25310** to push and pull the articulation joint **25350** into the non-articulated configuration. Referring to FIG. **101**, once the electromagnets **25351** are de-energized, the expanded coil spring **25330** will pull its corresponding rack member **25320** toward the articulation joint **25350** and the compressed coil spring **25330** will push its corresponding rack member **25320** away from the articulation joint **25350**. This pushing and pulling motion is applied to the slave cables **25340** and is configured to aid in moving the articulation joint **25350** into the non-articulated configuration. In at least one instance, the teeth **25321** and toothed prongs **25312** provide an audible sound to a user to indicate when the articulation joint **25350** has attained a fully non-articulation configuration.

In at least one instance, the power supplied to the electromagnets **25351** can be varied to vary the articulation angle. For example, the more the user wants an end effector to articulate, the power supplied to the electromagnets **25351** can be progressively increased. In various instances, the cables **25352**, **25340** comprise a conductive thread, for example. The conductive thread can be monitored to detect the articulation angle of the articulation joint by monitoring the resistance and/or conductivity of the thread in real time and correlating the monitored resistance and/or conductivity to the articulation angle. In at least one instance, another set of electromagnets can be employed to allow for multi-axis articulation rather than single plane articulation.

FIGS. **102-104** depict a surgical instrument shaft assembly **25400** configured for use with a surgical instrument such as those disclosed herein, for example. The shaft assembly **25400** comprises many of the same components as the surgical instrument **1000**. The shaft assembly **25400** may comprise various drive members configured to articulate an end effector, rotate an end effector about a longitudinal axis, and/or fire an end effector, for example. One or more of these drive members and/or components within a shaft assembly may be subject to tension and/or compression owing to the interaction of such drive members and/or components with other drive members and/or components of a surgical instrument employing the shaft assembly **25400**. In at least one instance, articulation of an end effector may cause a spine member to which an articulation joint may be attached to stretch and/or compress upon articulation of the end effector. This can be attributed to the attachment of the articulation joint to the spine member and the bending, or articulation, of the articulation joint. A core insert may aid in strengthening the shaft assembly **25400** and/or help define a maximum system stretch of the shaft assembly **25400**. The maximum system stretch may be defined by a maximum load and/or a maximum stretch length, for example. A core insert may prevent a member of a shaft assembly from prematurely failing. A core insert may also predefine the maximum system stretch of a shaft assembly so as to provide a

predictable amount of stretch of one or more components of the shaft assembly and/or surgical instrument with which the shaft assembly is used.

The shaft assembly **25400** comprises a spine member **25410** and the articulation joint **1400**. The shaft assembly **25400** further comprises a proximally extending articulation joint portion **25430** comprising pin apertures **25431**. The spine member **25410** comprises lateral slots **25411** defined therein each configured to receive an articulation actuator. The lateral slots **25411** can provide space between an outer shaft tube and the spine member **25410** for the articulation actuators. The spine member **25410** further comprises a primary slot **25412** configured to receive a drive member therethrough such as, for example, a primary drive shaft.

The shaft assembly **25400** further comprises a core insert **25420** positioned with the spine member **25410**. The core insert **25420** may be insert molded and/or overmolded into the spine member **25410**. Other suitable manufacturing techniques are contemplated. The core insert **25420** comprises a proximal core member **25421** comprising a distal hook end **25422**. The distal hook end **25422** comprises a hook tab **25423** extending from the proximal core member **25421**. The core insert **25420** further comprises a distal core member **25425** comprising a proximal hook end **25426**. The proximal hook end **25426** comprises a hook tab **25427** extending from the distal core member **25425**. The hook tabs **25423**, **25427** face each other and cooperate to transmit stretching forces to each other through the spine member **25410**. The distal core member **25425** further comprises a distal mounting portion **25428** extending distally out of the spine member **25410**. The distal mounting portion **25428** comprises pin apertures **25429** defined therethrough. The shaft assembly **25400** further comprises pins **2543** configured to pin the articulation joint portion **25430** to the distal mounting portion **25428** by way of apertures **25429**. The pinned engagement between the distal mounting portion **25428** and the articulation joint portion **25430** may result in stretching, or tensile, forces being applied to the spine member **25410**. The core insert **25420** may help prevent the spine member **25410** from overstretching, for example.

In at least one instance, the spine member **25410** comprises a first material and the core insert **25420** comprises a second material which is different than the first material. The first material may comprise a polymer material and the second material may comprise a metallic material. In at least one instance, the tensile strength of the second material is greater than the tensile strength of the first material. Such an arrangement can reduce weight, for example, of a surgical instrument which employs the shaft assembly **25400** while maintaining a desired system and/or shaft strength of the surgical instrument where significant actuation forces are present. For example, as an articulation actuator articulates the end effector and bends the articulation joint **1400**, stretching forces are applied to the spine member **25410** and the core insert **25420** can serve to counter these stretching forces. The material of the spine member **25410** positioned between the proximal core member **25421** and the distal core member **25425** may reduce capacitive coupling and/or electrically isolate the proximal core member **25421** from the distal core member **25425**.

FIGS. **105-111** depict a plurality of articulation actuators configured for use with a surgical instrument. In at least one instance, the actuators discussed herein can be used for any suitable system requiring an actuator. FIG. **105** depicts a piezoelectric actuator **25600** comprising an energizing circuit **25601** and a piezoelectric bimorph polymer **25610**. The piezoelectric bimorph **25610** comprises an inner substrate

layer **25611** and outer piezoelectric layers **25612**. The outer piezoelectric layers **25612** are configured to be energized in such a manner so as to bend the bimorph **25610** in the desired direction. The actuator **25600** may be used to articulate an end effector in an articulation plane. The substrate may comprise any suitable material. In at least one instance, the substrate comprises a material selected specifically for its rigidity and/or one or more other material properties, for example. In response to an electrical field, the layers **25612** are configured to bend in a desired direction. In at least one instance, the layers **25611**, **25612** are configured to splay relative to each other to compensate for radial differences in length upon bending within an articulation joint, for example.

FIGS. **106** and **107** depict a piezoelectric bimorph actuator **25800** configured to be used with a surgical instrument. In at least one instance, one or more of the actuator **25800** is configured to be used to articulate an end effector. The actuator **25800** comprises an inner substrate layer **25810** and piezoelectric outer layers **25820** configured to be energized to bend the actuator **25800** in a desired direction. The polarization direction of the actuator **25800** can be pre-determined in order to predictable bend the actuator **25800** in the desired direction. The actuator **25800** further comprises an input circuit **25801** configured to actuate, or energize, the actuator **25800**. In at least one instance, both piezoelectric layers comprise the same polarization direction. In at least one instance, the same voltage signal is connected to the exposed outer surfaces of the piezoelectric layers. In at least one instance, the substrate layer is grounded.

FIGS. **108** and **109** depicts a piezoelectric bimorph actuator **25900** configured to be used with a surgical instrument. In at least one instance, one or more of the actuator **25900** is configured to be used to articulate an end effector. The actuator **25900** comprises an input circuit **25910** and an actuation member **25920**. The actuator **25900** is configured to be energized to bend a bendable length **25923** of the actuator **25900** a pre-determined displacement amount **25924** and direction. In at least one instance, a portion **25921** of the actuator **25900** is inactive. In at least one instance, the actuator **25900** is energized in such a manner so as to bend the actuator **25900** in multiple directions to be able to articulate an end effector in multiple directions.

Any suitable combination of the actuators described herein may be combined for use with a surgical instrument. For example, a piezoelectric bimorph actuator may be used in addition to an electroactive polymer actuator. In at least one instance, the circuit employed to energize various actuators disclosed herein can be specifically tuned depending on the desired amount of flexion of the actuator and/or depending on the force required to actuate the function of the end effector such as, for example, articulating an end effector. A chart **25650** is provided in FIG. **112** illustrating force generation vs. displacement of a piezoelectric actuator for use with a surgical instrument.

FIG. **111** depicts an electroactive polymer (EAP) actuator **25700** configured to be used with a surgical instrument. In at least one instance, one or more of the actuator **25700** is configured to be used to articulate an end effector. In at least one instance, the actuator **25700** comprises a PVDF material (polyvinylidene fluoride). The actuator **25700** comprises an input mounting circuit **25701** and a bendable member **25710**. The bendable member **25710** comprises conductive layer **25722** (such as gold, for example), substrate layer **25721** (such as a PVDF layer, for example), and polypyrrole layers **25723**.

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FIG. 112 depicts a shaft assembly **26000** configured to permit distal end effector rotation within a surgical instrument. The shaft assembly **26000** comprises an outer shaft **26010**, a spine shaft **26020**, a primary drive shaft **26030**, and a distal head rotation drive shaft **26040**. In at least one instance, an end effector extends distally from the spine shaft **26020** so that the end effector can be rotated by the spine shaft **26020**. In at least one instance, the spine shaft **26020** rotates independently of the outer shaft **26010**. To rotate the spine shaft **26020**, a driving engagement surface **26050** is employed on the drive shaft **26040** and the inner diameter of the spine shaft **26020** such that, as the drive shaft **26040** is rotated, the spine shaft **26020** is rotated. In at least one instance, an elastomeric, friction-inducing material is positioned around the drive shaft **26040** and positioned around the inner diameter of the spine shaft **26020**. In at least one instance, the spine shaft **26020** comprises spline grooves and the drive shaft **26040** comprises teeth configured to engage the spine grooves.

FIG. 113 depicts a shaft assembly **26100** configured to permit distal end effector rotation within a surgical instrument. The shaft assembly **26100** comprises an outer shaft **26110**, a spine shaft **26120**, a primary drive shaft **26140**, and a drive system configured to rotate the spine shaft **26120**. In at least one instance, an end effector extends distally from the spine shaft **26120** so that the end effector can be rotated by the spine shaft **26120**. In at least one instance, the spine shaft **26120** rotates independently of the outer shaft **26110**. To rotate the spine shaft **26120**, the drive system comprises windings **26160** positioned around shaft **26150** and magnets **26130** positioned on an inner diameter of the spine shaft **26120**. To rotate the spine shaft **26120** the windings **26160** are energized to cause the magnets **26130** to move around the windings **26160**.

Various methods of locking rotational drive mechanisms are contemplated. For example, a system can rely on the resonant position holding torque of the magnets **26130** to hold an end effector in position relative to a shaft. In at least one instance, a mechanical ratchet is employed to hold an end effector in position relative to a shaft. In at least one instance, a sprung clutch system is employed to require a motor to overcome the sprung clutch system to unlock end effector rotation.

In at least one instance, a ring gear is locked and unlocked to effect rotation of an end effector and to effector closure of a jaw relative to a fixed jaw. A planetary gear system can be employed to rotate different elements of a shaft assembly to effect different functions of a surgical instrument assembly, for example.

FIG. 114 depicts a surgical instrument assembly **26200** comprising an outer shaft **26210**, a proximal spine member **26220** positioned within the outer shaft **26210**, and a distal spine member **26230** positioned within the outer shaft **26210** and configured to be rotated relative to the proximal spine member **26220** and, in at least one instance, the outer shaft **26210**. Rotation of the distal spine member **26230** can be employed to rotate an end effector of a surgical instrument, for example. The surgical instrument assembly **26200** further comprises a drive shaft **26250** configured to actuate a function of an end effector such as, for example, firing staples and/or cutting tissue. The proximal spine member **26220** comprises an annular flange portion **26221** and the distal spine member **26230** comprises an annular flange portion **26231**. The surgical instrument assembly **26200** further comprises one or more bearings **26245** positioned between the annular flange portions **26221**, **26231** such that

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the distal spine member **26230** can be rotated relative to the proximal spine member **26220**.

The surgical instrument assembly **26200** further comprises a piezoelectric rotary motor. The piezoelectric rotary motor comprises a rotary piezoelectric member **26240** fixed within the assembly **26200** and one or more drive members **26241** configured to be actuated by the piezoelectric member **26240**. The surgical instrument assembly **26200** further comprises an electrical trace **26260** configured to energize the piezoelectric member **26240** to actuate the drive members **26241** in such a manner so as to apply a rotational torque to an inner drive surface **26233**. As a rotational torque is applied to the surface **26233**, the distal spine member **26230** is rotated to rotate an end effector, for example. In at least one instance, the piezoelectric rotary motor is configured to rotate the distal spine member **26230** in a clockwise direction and in a counter clockwise direction.

In various instances, shaft assemblies for use with surgical instruments can contain electrical traces and/or wires, for example, extending through the shaft assembly from a proximal end to a distal end. The electrical traces may extend into an end effector attached to the distal end of the shaft assembly. In various instances, end effectors can be configured to rotate relative to the shaft. In various instances, end effectors and the shaft assembly to which the end effector is attached are configured to rotate relative to a proximal attachment interface and/or surgical instrument handle, for example. In such instances, the rotation of the end effector and/or shaft assembly may cause electrical traces to bind if the end effector and/or shaft assembly is over-rotated. Various ways of handling binding issues and/or contact issues with electrical traces positioned within shaft assemblies, which may be caused by rotation of an end effector and/or shaft assembly, are discussed herein.

FIGS. 115-117 depict a limiter system **26300** configured to be used with a surgical instrument. The limiter system **26300** is configured to cease over-rotation of a drive train. In at least one instance, the limiter system **26300** is automatic and does not require input from a user to cease over-rotation of a drive train. In at least one instance, the limiter system **26300** requires input from a user. The limiter system **26300** comprises an actuator **26310** comprising a solenoid, for example. The actuator **26310** comprises a shaft **26311** comprising a spring **26312** and a distal end **26313**. The limiter system **26300** further comprises a gear **26320**. In at least one instance, the gear **26320** is part of a rotational drive train configured to actuate a function of an end effector. For example, the gear **26320** may be a part of a rotation drive train configured to articulate an end effector in multiple directions, rotate an end effector about a longitudinal axis relative to a shaft, clamp jaws of an end effector, and/or actuate a firing member of an end effector.

As can be seen in FIG. 115, the gear **26320** is free to rotate because the actuator **26310** is not actuated. In at least one instance, the actuator **26310** comprises a brake applied only in certain instances. The actuator **26310** may only be activated, or triggered, when a user desires and/or a surgical robot is programmed to limit movement of the gear **26320**. For example, as discussed above, the gear **26320** may comprise a component of a rotational drive train configured to articulate an end effector. In at least one instance, a user may activate an articulation drive train thereby rotating the gear **26320**. At any point, the user and/or a surgical robot may activate the actuator **26310** to stop articulation of an end effector. FIG. 116 illustrates the actuator **26310** in an actuated position. The distal end **26313** comprises teeth **26314** configured to engage teeth **26321** of the gear **26320**. How-

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ever, at this point of rotation of the gear **26320**, a braking force applied to the gear **26320** may not be sufficient to cease rotation of the rotational drive train. The rotational drive train may be motorized and/or manual. Both can be ceased using the limiter system **26300**. In the position illustrated in FIG. **116**, an audible ratcheting noise may be heard during rotation of the gear **26320**. The spring **26312** is not fully compressed and will not apply a full braking force until the gear **26320** rotates to the position illustrated in FIG. **117**.

As can be seen in FIG. **117**, the spring **26312** is fully compressed. In this position, the limiter system **26300** is configured to apply a maximum braking force to a rotational drive train by engaging teeth **26323** of the gear **26320**. Engagement between the teeth **26314** and the teeth **26323** results in maximum braking force because the teeth **26323** comprise the greatest radius of all of the teeth **26321** of the gear **26320** resulting in maximum compression of the spring **26312**. As the gear **26320** rotates from the position illustrated in FIG. **116** to the threshold position illustrated in FIG. **117**, an audible ratchet sound may increase in volume and/or slow in frequency. This may indicate to a user and/or a control circuit that maximum braking force is being approached. In at least one instance, a control circuit is configured to detect the braking force as it is applied and is configured to automatically shut off a motor actuating the rotational drive train connected to the gear **26320**.

In at least one instance, a limiter system is applied with a substantially circular gear. In such an instance, an actuator may be progressively actuated to advance a shaft progressively toward the circular gear. In such an instance, a gradually increasing braking force may be applied to the gear. A control circuit may be configured to monitor and actively adjust the braking force during use of the limiter system. In at least one instance, a control circuit is configured to actuate the limiter system upon receiving input from one or more other control systems and/or circuits indicating that one or more systems of a surgical instrument are to be shut down during operation.

In at least one instance, the limiter system **26300** is configured to be overridden such that the gear **26320** may be rotated past the threshold position where the maximum braking force is applied. In at least one instance, the limiter system **26300** is configured to be automatically activated upon an end of stroke for the function configured to be actuated by the rotational drive train. For example, as an end effector nears a maximum articulation angle, the limiter system **26300** may be activated to apply a braking force thereto. The maximum articulation angle may be detected by an encoder on an articulation motor and/or a sensor configured to detect directly the angle of articulation, for example. In various instances, the limiter system **26300** may be deactivated at any point a user and/or control circuit seeks to continue uninterrupted actuation of the rotational drive train. In at least one instance, audible ratcheting noises may be heard during rotational of the gear **26320** in both the counterclockwise direction and the clockwise direction. If the actuator **26310** is actuated, an audible ratchet noise is heard during rotation of the gear **26320** in either direction.

In at least one instance, the limiter system **26300** is configured to provide only feedback of the threshold position being reached and is not configured to affect actuation of a rotational drive train for which it provides feedback. In other words, the limiter system **26300** is only an indicator system and does not apply braking force to the function of the end effector being monitored.

In at least one instance, the limiter system **26300** provides a hard stop for the function of the end effector. Once the

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threshold position is reached, a motor actuating the rotational drive system cannot overcome the braking force applied thereto by the limiter system **26300**.

In various instances, a control circuit configured to actuate the limiter system **26300** comprises a counter rotation feature. Once the gear **23620** reaches the threshold position, the control circuit may deactivate the actuator **26310** and counter rotate the gear **26320** to a non-threshold position. Once the gear **26320** is counter rotated, a user may regain control of actuation of the rotational drive train. In at least one instance, a user may indicate the need for rotation of the rotational drive train beyond the threshold position. In such an instance, a user may indicate that further rotational is desired. If the user indicates that further rotation is desired, the actuator **26310** may be automatically deactivated and the rotational drive train is free to rotate. In at least one instance, an absolute maximum rotation is predetermined and cannot be surpassed. In such instance, a soft maximum threshold may be predetermined allowing for some rotation passed the soft maximum threshold but not beyond the absolute maximum rotation. The absolute maximum rotation may be defined by mechanical limits, for example. The soft maximum threshold may be defined by an operational limit which does not overstress any components, for example. In at least one instance, the counter rotation feature is inhibited if jaws of an end effector sense a fully clamped state onto tissue. This can reduce the likelihood of accidentally opening the jaws and losing grip on targeted tissue.

In at least one instance, braking force may be applied during several rotations of the gear **26320**. In such an instance, shaft rotation of the rotational drive train may be tracked and the braking force applied by the actuator **26310** is gradually increased as the gear **26320** rotates.

FIGS. **118-120** depict a rotary actuation system **26400** for use with a surgical instrument. The rotary actuation system comprises a mechanical limiting system configured to prevent over-rotation, or actuation, of a drive system. The drive system may comprise an articulation drive system, an end effector rotation drive system, a jaw clamping and/or unclamping drive system, and/or a firing member drive system, for example. The rotary actuation system **26400** comprises a motor **26410**, a variable screw **26420** configured to be rotated by the motor **26410**, and a drive nut **26430** configured to be linearly actuated by the screw **26420**. The motor **26410** is configured to rotate the screw **26420** to actuate the drive nut **26430** to actuate a function of a surgical instrument. The drive nut **26430** may be connected to a drive member configured to actuate a function of the surgical instrument. While any suitable function may be actuated by the rotary actuation system, **26400**, the rotary actuation system **26400** will be described in connection with an articulation system.

The screw **26420** comprises variable threads **26425**, an inner section **26421**, and outer sections **26422** extending from the inner section **26421**. The outer sections **26422** extend from the inner section **26421** gradually increasing a thread diameter of the threads **26425**. In at least one instance, the thread diameter is varied along a screw axis defined by the screw **26420**. In at least one instance, the thread pitch is varied along the screw axis defined by the screw **26420**. In at least one instance, the thread diameter and the thread pitch are varied along the screw axis. In at least one instance, a thread profile varies along the length of the screw **26420**. The varied thread profile is engaged with the drive nut **26430** such that engagement of threads **26431** of the drive nut **26430** and threads **26425** of the screw **26420** varies along the length of the screw **26420**.

As the screw **26420** is rotated in a first direction, the drive nut **26430** is configured to move in a corresponding first direction toward an outer section **26422** of the screw **26420**. In at least one instance, movement of the drive nut **26430** toward an outer section **26422** corresponds to articulation of an end effector. As the drive nut **26430** moves toward an outer section **26422**, the threaded engagement between the nut **26430** and the screw **26420** tightens owing to the varied thread profile. This tightened engagement may cause increased load on the motor **26410**. This increased load can be monitored and detected. The detected load can be conveyed to a user and/or a control circuit to indicate to a user and/or a control circuit that the drive nut **26430** is nearing an end of stroke position. In at least one instance, the motor **26410** is automatically slowed so as to slow the velocity of the drive nut **26430** near the end of stroke position. In at least one instance, the motor **26410** is automatically stopped upon detecting a threshold load. In at least one instance, the drive nut **26430** is automatically counter-rotated at least partially to decrease load on the motor **26410**. In at least one instance, the outer ends **26422** provide a hard stop for an actuation stroke, such as an articulation stroke, for example. In at least one instance, the distance capable of being traveled by the drive nut **26430** corresponds to mechanical limitations by the corresponding actuation stroke such as, for example, maximum articulation angle.

In at least one instance, the threads **26431** comprise a non-variable thread profile while the threads **26425** comprise a variable thread profile. In at least one instance, the threads **26431** also comprise a variable thread profile in addition to the threads **26425** of the screw **26420**. In at least one instance, the motor is configured to stall upon reaching a maximum rotational limit. In at least one instance, the threaded engagement locks the nut **26430** into place upon reaching the maximum rotational limit. In at least one instance, a control circuit is configured to unlock the drive nut **26430** after reaching the maximum rotational limit by re-activating the motor **26410** to rotate the screw **26420** in an opposite direction. In at least one instance, a larger torque may be required to unlock the drive nut **26430** from its maximum rotational limit position.

In at least one instance, feedback is provided as the maximum rotational limit position is approached. For example, a control circuit may provide audio and/or tactile feedback to a user, based on detected increase motor load, as the drive nut **26430** approaches the maximum rotational limit position. In at least one instance, a control circuit is configured to automatically adjust a control motion of actuation of the motor **26410** before, during, and/or after the drive nut **26430** reaches the maximum rotational limit position. The drive nut **26430** comprises a maximum rotational limit position on both outer sections **26422** of the screw **26420**. In at least one instance, a hard stop is provided to prevent irreversible binding of the nut **26430** and the screw **26420**.

FIG. **121** depicts a segmented ring contact system **26500** for use with a surgical instrument assembly. The segmented ring contact system **26500** may be employed between two or more components where electrical transmission is desired between two or more of the components and one or more of the components are configured to be rotated relative to one or more other components. The segmented ring contact system **26500** is configured to provide redundant slip ring contacts within a shaft assembly for a surgical instrument, for example. The segmented ring contact system **26500** comprises an outer segmented contact system **26510** comprising a plurality of slip ring contact segments **26511** and an

inner segmented contact system **26520** comprising a plurality of slip ring contact segments **26521**. As can be seen in FIG. **121**, the slip ring contact segments **26511** span gaps defined between the slip ring contact segments **26521** and the slip ring contact segments **26521** span gaps defined between the slip ring contact segments **26511**. In at least one instance, the contact system **26500** may mitigate fluid shorting between contacts by providing multiple segments as opposed to a single slip ring contact spanning a 360 degree length of a shaft, for example. If one segment shorts out, another segment may provide a redundant means for transmitting electrical signals.

In at least one instance, the segments **26511** and the segments **26521** comprise different resistance values which can be detected and monitored by a control circuit. Such an arrangement may indicate to a user and/or control circuit, for example, which contacts are transmitting electrical signals and which contacts are not transmitting electrical signals. Such an arrangement may also allow a control circuit to determine rotational shaft position.

FIGS. **122-127** depict various electrical transmission arrangements for use with surgical instrument assemblies. In various instances, the electrical transmission arrangements are configured to transmit electrical signals between a first shaft and a second shaft. The first shaft may be attached to a surgical robot and/or handle, for example, and the second shaft may comprise an end effector attached to a distal end thereof. In at least one instance, the electrical transmission arrangements are configured to transmit electrical signals between sensors, processors, and/or power sources, etc., of the first shaft assembly and the second shaft assembly. For example, the second shaft may comprise a motor requiring power from the first shaft and/or a component upstream of the first shaft. Another example may include receiving electrical signals from sensors positioned on the second shaft and/or end effector attached to the second shaft. Other systems requiring electrical transmission between the first shaft assembly and second shaft assembly are contemplated. The electrical transmission arrangements disclosed herein can be configured to help prevent fluid shorting of the transmission arrangement, for example.

FIG. **122** depicts a surgical instrument assembly **26600** comprising a first shaft **26610**, a second shaft **26620**, and an electrical transmission arrangement **26640**. The second shaft **26620** is rotatable relative to the first shaft **26610**. In at least one instance, the first shaft **26610** is rotatable relative to the second shaft **26620**. In at least one instance, the first shaft **26610** and the second shaft **26620** are rotatable relative to each other. In at least one instance, the second shaft **26620** comprises an end effector attached to a distal end thereof. The electrical transmission arrangement **26640** comprises electrical traces **26611** and first contacts **26612** connected to the electrical traces **26611** and positioned in an inner channel **26613** of the first shaft **26610**. The first contacts **26612** may comprise slip ring contacts, for example, extending around the entire inner diameter of the channel **26613**. In at least one instance, the first contacts **26612** comprise isolated contact segments.

The electrical transmission arrangement **26640** further comprises electrical traces **26621** and second contacts **26622** connected to the electrical traces **26621** and positioned on an outer surface **26623** of the second shaft **26620**. The second contacts **26622** may comprise slip ring contacts, for example, extending around the entire outer diameter of the outer surface **26623** of the second shaft **26620**. The second contacts **26622** are configured to contact the first contacts **26612** to transmit electrical signals therebetween. The sec-

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ond contacts **26622** are configured to maintain electrical contact with the first contacts **26612** during rotation of the second shaft **26620** relative to the first shaft **26610**.

The surgical instrument assembly **26600** further comprises a channel **26630** between the first shaft **26610** and the second shaft **26620**. Fluid and/or debris from a patient may flow into the channel **26630** during an operation. The electrical transmission arrangement **26640** may help prevent fluid and/or debris from flowing into the channel **26630**. In at least one instance, each contact **26612** is configured to supply and/or receive different electrical signals for different electrical systems. In at least one instance, the contacts **26612**, **26622** act as redundant contacts.

FIG. **123** depicts a surgical instrument assembly **26700**. The surgical instrument assembly **26700** comprises many of the same components of the surgical instrument assembly **26600**. The surgical instrument assembly **26700** further comprises grommets **26710** positioned between each set of contacts **26612**, **26622**. The grommets **26710** may comprise of a rubber material, for example. The grommets **26710** may help prevent fluid and/or debris from flowing into the channel **26630**.

FIG. **124** depicts a surgical instrument assembly **26800**. The surgical instrument assembly **26800** comprises many of the same components of the surgical instrument assembly **26600**. The surgical instrument assembly **26800** further comprises a grommet **26810** positioned away from the contacts **26612**, **26622**. The grommet **26810** may help prevent fluid and/or debris from flowing into the channel **26630** and toward the contacts **26612**, **26622** well away from the contacts **26612**, **26622**.

FIG. **125** depicts a surgical instrument assembly **26900** comprising a first shaft **26910**, a second shaft **26920**, and an electrical transmission arrangement **26940**. The second shaft **26920** is rotatable relative to the first shaft **26910**. In at least one instance, the first shaft **26910** is rotatable relative to the second shaft **26920**. In at least one instance, the first shaft **26910** and the second shaft **26920** are rotatable relative to each other. In at least one instance, the second shaft **26920** comprises an end effector attached to a distal end thereof. The electrical transmission arrangement **26940** comprises electrical traces **26911** and first contacts **26912A**, **26912B** connected to the electrical traces **26911** and positioned in an inner channel **26913** of the first shaft **26910**. The first contacts **26912A**, **26912B** comprise isolated contact segments. The contacts **26912A** and the contacts **26912B** are positioned opposite each other. This positioning may help prevent contacts **26912A**, **26912B** from shorting out where fluid flows into an upper portion of the channel **26930** and not a lower portion of the channel **26930**.

The electrical transmission arrangement **26940** further comprises electrical traces **26921** and second contacts **26922** connected to the electrical traces **26921** and positioned on an outer surface **26923** of the second shaft **26920**. The second contacts **26922** may comprise slip ring contacts, for example, extending around the entire outer diameter of the outer surface **26923** of the second shaft **26920**. The second contacts **26922** are configured to contact the first contacts **26912A**, **26912B** to transmit electrical signals therebetween. The second contacts **26922** are configured to maintain electrical contact with the first contacts **26912A**, **26912B** during rotation of the second shaft **26920** relative to the first shaft **26910**.

The surgical instrument assembly **26900** further comprises a channel **26930** between the first shaft **26910** and the second shaft **26920**. Fluid and/or debris from a patient may flow into the channel **26930** during an operation. The

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surgical instrument assembly **26900** further comprises a grommet **26931** configured to prevent fluid and/or debris from flowing into the channel **26930**.

FIG. **126** depicts a surgical instrument assembly **27000** comprising a first shaft **27010**, a second shaft **27020**, and an electrical transmission arrangement **27040**. The second shaft **27020** is rotatable relative to the first shaft **27010**. In at least one instance, the first shaft **27010** is rotatable relative to the second shaft **27020**. In at least one instance, the first shaft **27010** and the second shaft **27020** are rotatable relative to each other. In at least one instance, the second shaft **27020** comprises an end effector attached to a distal end thereof. The electrical transmission arrangement **27040** comprises first electrical contacts **27012** positioned within an annular slot **27011** defined in an inner diameter **27013** of the first shaft **27010**. The electrical transmission arrangement **27040** further comprises a second electrical contact **27021**, such as a slip ring contact, for example, positioned on an outer diameter **27022** of the shaft **27020**. The first electrical contacts **27012** are configured to maintain electrical contact as one of the shafts **27010**, **27020** rotates relative to the other shaft **27010**, **27020**. This contact arrangement may be referred to as a blade-style electrical contact arrangement. The second electrical contact **27021** is configured to be positioned at least partially within the annular slot **27011** and may be referred to as a blade contact.

FIG. **127** depicts a surgical instrument assembly **27100** comprising a first shaft **27110**, a second shaft **27120**, and an electrical transmission arrangement **27140**. The second shaft **27120** is rotatable relative to the first shaft **27110**. In at least one instance, the first shaft **27110** is rotatable relative to the second shaft **27120**. In at least one instance, the first shaft **27110** and the second shaft **27120** are rotatable relative to each other. In at least one instance, the second shaft **27120** comprises an end effector attached to a distal end thereof. The electrical transmission arrangement **27140** comprises first electrical contacts **27113** positioned within annular slots **27112** defined in an inner diameter **27111** of the first shaft **27110**. The electrical transmission arrangement **27140** further comprises second electrical contacts **27123** positioned on blade wheels **27122** positioned on an outer diameter **27121** of the shaft **27120**. The first electrical contacts **27113** and second electrical contacts **27123** are configured to maintain electrical contact with each other as one of the shafts **27110**, **27120** rotates relative to the other shaft **27110**, **27120**. The second electrical contacts **27123** are configured to be positioned at least partially within the annular slots **27112**. The blade wheels **27122** may help alleviate shorting of the contacts **27123**, **27113** by reducing the amount of exposed electrical contact area exists within the electrical transmission arrangement **27140**.

FIGS. **128** and **129** depict inductive coil systems **28000**, **28100** configured to be used with a surgical instrument shaft assembly. Employing wired electrical traces between components configured to rotate relative to each other such as, for example, a shaft assembly and an end effector. The inductive coil system **28000** comprises a first inductive coil **28010** and a second inductive coil **28020**. In at least one instance, the coil **28010** comprises a transmitter coil and the coil **28020** comprises a receiver coil. The coils **28010**, **28020** can be configured to transmit electrical signals therebetween. In at least one instance, one of the coils **28010**, **28020** is positioned on a first component and the other of the coils **28010**, **28020** is positioned on a second component configured to rotate relative to the first component. In at least one instance, the distance between the coils **28010** is less than the diameter of each coil **28010**, **28020**. The coil system

28100 comprises a first inductive coil **28110** and a second inductive coil **28120**. In at least one instance, the coil **28110** comprises a transmitter coil and the coil **28120** comprises a receiver coil. The coil **28120** comprises a diameter which is less than the diameter of the coil **28110**. In at least one instance, multiple coil systems are employed with a surgical instrument assembly. For example, one or more coil systems can be utilized to transmit power and one or more coil systems can be utilized to transmit data.

FIGS. **130** and **131** depict an electroactive polymer system **29000** for use with a surgical instrument assembly. The system **29000** comprises an electroactive polymer **29010** and an input circuit **29020**. The system **29000** can be used as an actuator for a surgical instrument assembly such as, for example, an articulation actuator. FIG. **130** illustrates the polymer **29010** in an energized state. FIG. **131** illustrates the polymer **29010** in an un-energized state. In at least one instance, the polymer **29010** is employed to rotate an end effector relative to a shaft. One end of the polymer **29010** can be fixed to the shaft and the bendable end of the polymer **29010** can be attached to the end effector. The polymer **29010** can be configured to be twisted to cause rotation of an end effector relative to a shaft. The material selected for the system **29000** can be selected based on material limitations to predefine the amount of deflection required for the actuation.

End effectors of surgical instruments, including the components thereof, experience significant forces upon them during a single firing stroke. Such forces lead to equipment wear, which can ultimately lead to ineffective tissue treatment, for example. In various instances, a clinician may want to use a new cutting element for each tissue cutting stroke during a particular surgical procedure. The disposable end effector assemblies described herein allow for a clinician to interchangeably replace one or more components of the end effector from a particular surgical instrument.

FIGS. **132-138** depict an end effector **30000** for use with a surgical instrument. The end effector **30000** comprises a channel **30100**, an anvil **30200**, and a cartridge **30300**. In various instances, the channel **30100** is configured to fixedly, or non-replaceably, extend from an elongate shaft **30500** of the surgical instrument. In other instances, the channel **30100** is configured to be replaceably attached to the elongate shaft **30500**. In any event, the channel **30100** is configured to extend from the elongate shaft **30500** at a point distal to an articulation joint.

The anvil **30200** comprises an elongate slot **30280** defined therein. The elongate slot **30280** extends from a proximal end **30202** toward a distal end **30204** of the anvil **30200** and is configured to receive a first camming member **30406** of a firing member **30400**. An anvil projection **30210** extends from a sidewall, or tissue stop, **30208** near the proximal end **30202** of the anvil **30200**. The anvil projection **30210** defines a pivot joint about which the anvil **30200** is movable relative to the cartridge **30300**. The anvil projection **30210** comprises an aperture **30212** defined therein. The aperture **30212** is sized to fittingly receive a cartridge projection **30310** therein. The cartridge projection **30310** extending through at least a portion of the anvil projection **30210** establishes a coupling and/or attachment between the cartridge **30300** and the anvil **30200** while also maintaining component alignment. In various instances, the cartridge **30300** and the anvil **30200** are coupled together during the manufacturing and/or packaging process. In other instances, a clinician is able to selectively choose between various combinations of compatible anvils and cartridges prior to use of the assembly with a surgical instrument.

As shown in FIG. **134**, the cartridge **30300** comprises a cartridge pivot member **30350** from which the cartridge projection **30310** extends. The cartridge pivot member **30350** serves as an electronics interface to the channel **30100** when the cartridge **30300** is seated therein. In various instances, the cartridge pivot member **30350** is comprised of metal while the remaining cartridge body is comprised of a plastic material, for example. In various instances, the cartridge **30300** comprises a plurality of staple cavities **30360** defined therein, at least one electrode **30370**, and a longitudinal slot **30380** extending from a proximal end towards a distal end. The at least one electrode **30370** is similar to longitudinal electrode **1925**, whose functionality is described in greater detail with respect to FIGS. **1** and **6**. In various instances, the at least one electrode **30370** is an RF electrode. The longitudinal slot **30380** is configured to receive a portion of the firing member **30400** as the firing member **30400** translates through the end effector **30000** during a firing stroke. Staples are removably positioned in the staple cavities **30360**. In other instances, the cartridge may only comprise staple cavities or may only comprise an RF electrode. In any event, the cartridge **30300** is configured to be removably seated in the channel **30100**. The cartridge **30300** further comprises a lateral projection **30320** extending from a cartridge sidewall.

Sidewalls of the channel **30100** comprise a notch **30120** defined in a distal portion thereof. The notch **30120** is sized to receive the lateral projection **30320** of the cartridge **30300** therein as the cartridge **30300** is seated in the channel **30100**. In addition to securing the cartridge **30300** to the channel **30100**, the notch **30120** ensures that the assembly comprised of the cartridge **30300** and the anvil **30200** is appropriately aligned with the channel **30100**, and thus the elongate shaft **30500**. The act of installing the assembly comprised of the cartridge **30300** and the anvil **30200** into the channel **30100** also serves to connect various electrical components **30700** throughout the end effector **30000**.

The sidewalls of the channel **30100** further comprise a pivot notch **30110** defined therein. The pivot notch **30110** comprises a size and/or geometry configured to receive the anvil projection **30210** therein. As shown in FIGS. **132** and **133**, the pivot notch **30110** is angled in an effort to prevent the assembly of the anvil **30200** and cartridge **30300** from unwantedly detaching from the channel **30100**, for example. When the disposable assembly comprised of the anvil **30200** and the cartridge **30300** are fully seated in the channel **30100**, the anvil **30200** is not physically, or directly, attached to the elongate shaft **30500**. Stated another way, the anvil **30200** is only physically, or directly, coupled to the cartridge **30300** and the channel **30100**.

As shown in FIG. **133**, the channel **30100** further comprises a drive screw **30150** positioned therein prior to the attachment of the staple cartridge **30300** thereto. A distal end **30104** of the channel base **30108** comprises a mounting interface **30130** for securing a distal end **30154** of the drive screw **30150**. A firing member **30400** is mounted on the drive screw **30150** prior to the staple cartridge **30300** being seated in the channel **30100**.

FIGS. **135-138** show the progression of seating the disposable assembly comprising the anvil **30200** and the cartridge **30300** into the channel **30100**. As shown in FIG. **135**, the anvil **30200** is coupled to the cartridge **30300** as an assembly, and the channel **30100** is attached to the elongate shaft **30500** at a point distal to any articulation joint; however, the anvil **30200** and the cartridge **30300** are completely detached from the channel **30100**.

A first stage of seating the disposable assembly in the channel 30100 is shown in FIG. 136. As the disposable assembly is brought toward the channel 30100, the proximal end 30202 of the anvil 30200 is tilted toward the base 30106 of the channel 30100, while the distal end 30204 of the anvil 30200, and thus the disposable assembly, is tilted slightly away from the base 30106 of the channel 30100. Initial contact is made between the anvil projection 30210 and the pivot notch 30110 of the channel 30100. Notably, the lateral projection 30320 is not yet aligned with the notch 30120 of the channel 30100. In the first stage, the first camming member 30406 of the firing member 30400 is slid into a proximal portion of the elongate slot 30280 of the anvil 30200. Additionally, the drive screw 30150 is not yet aligned with the longitudinal slot 30380 of the cartridge 30300. Such misalignment prevents the cartridge 30300 from being fully seated in the channel 30100.

A second stage of seating the disposable assembly in the channel 30100 is shown in FIG. 137. As the anvil projection 30210 is slid completely into the pivot notch 30110 of the channel 30100, the disposable assembly moves distally within the channel 30100, disengaging the first camming member 30406 of the firing member 30400 from the elongate slot 30280 of the anvil 30200. Such distal movement brings the lateral projection 30320 in line with the notch 30120; however, the distal end of the cartridge 30300 remains elevated.

FIG. 138 depicts the disposable assembly fully seated in the channel 30100. At such a point, the anvil projection 30210 is completely housed within the pivot notch 30110, the lateral projection 30320 is completely housed within the notch 30120, and the drive screw 30150 is completely housed within the longitudinal slot 30380 of the cartridge 30300. Such alignment between the cartridge 30300 and the drive screw 30150 allows for the cartridge 30300 and the anvil 30200 disposable assembly to be fully seated in the channel 30100. When the disposable assembly is fully seated in the channel 30100, all electrical components, such as flex circuits and/or sensor arrays, 30700 are coupled and in communication with the channel 30100 and/or the elongate shaft 30500 of the surgical instrument.

In various instances, the disposable assembly comprised of the cartridge 30300, the anvil 30200, and various flex circuits 30700 is only intended for a single use. Stated another way, upon completion of a single firing stroke, the cartridge 30300, the anvil 30200, and the associated flex circuits 30700 are removed, or unseated, from the channel 30100 leaving behind the drive screw 30150 and the firing member 30400. In such instances, the drive screw 30150 and the firing member 30400 are intended to be used for more than one firing stroke. The channel 30100, including the drive screw 30150 and the firing member 30400, can be detached from the elongate shaft 30500 and disposed of after being used for a pre-determined number of firing strokes, or upon becoming defective, for example.

FIGS. 139-146 depict an end effector 31000 for use with a surgical instrument. Similar to the end effector 30000, the end effector 31000 comprises a channel 31100, an anvil 31200, and a cartridge 31300. The end effector 31000 is detachably, or replaceably, coupled to an elongate shaft 31500 of the surgical instrument at a point distal to any articulation joint. Stated another way, the end effector 31000 is configured to be disposed of after a pre-determined number of firing strokes, such as one, for example. As shown in FIG. 146, the end effector 31000 comprises a firing member 31400 as part of a disposable portion. In such

instances, a new cutting element, for example, is present every time the end effector 31000 is replaced.

Similar to the cartridge 30300, the cartridge 31300 can comprise staple cavities, an RF electrode, and/or any suitable combination of features. The cartridge 31300 further comprises a lateral projection 31320 extending from a cartridge sidewall. Sidewalls of the channel 31100 comprise a notch 31120 defined in a distal portion thereof. The notch 31120 is sized to receive the lateral projection 31320 of the cartridge 31300 therein as the cartridge 31300 is seated in the channel 31100. In addition to securing the cartridge 31300 to the channel 31100, the notch 31120 ensures that the cartridge 31300 is appropriately aligned with the channel 31100. While the end effector 31000 is shown as being detachably coupled to the elongate shaft 31500, the cartridge 31300 is also replaceably seated in the channel 31100. As shown in FIG. 141, the anvil 31200 comprises a flex circuit 31700 having traces arranged on a sidewall, or tissue stop, of the anvil 31200 near a proximal end. When pivotally coupled to the channel 31100, the anvil traces are in electrical contact with a flex circuit comprising traces 31151 positioned on the channel.

A coupling member 31800 serves as an attachment interface between elongate shaft 31500 and the assembly formed of the channel 31100, the anvil 31200, and the cartridge 31300. A distal end of the elongate shaft 31500 is shown in FIG. 140 prior to attachment of the coupling member 31800 and end effector 31000 thereto. The distal end of the elongate shaft 31500 comprises various attachment members configured to secure and/or align the elongate shaft 31500 with the end effector 31000 in addition to coupling the drive systems and/or electrical connections. A proximal end of the coupling member 31800 is configured to interface with the distal end of the elongate shaft 31500. The proximal end of the coupling member 31800 is shown in FIG. 142 prior to attachment to the elongate shaft 31500. The coupling member 31800 comprises complementary features to those of the elongate shaft 31500.

More specifically, the distal end of the elongate shaft 31500 comprises a drive shaft 31600 extending therefrom. A channel 31860 is defined in the coupling member 31800 that is sized to closely receive the drive shaft 31600 therein. The drive shaft 31600 extends through the channel 31860 for ultimate attachment to a drive screw within the channel 31100 and/or cartridge 31300 of the end effector 31000. As described in greater detail throughout, a flex circuit, or electrical traces, 31550 extend through the elongate shaft 31500 to a control circuit and/or processor within a proximal housing, for example. The flex circuit 31550 of the elongate shaft 31500 is electrically coupled to a flex circuit 31850 on the coupling member 31800. The flex circuit 31850 on the coupling member 31800 is in electrical communication with the flex circuit 31700 on the anvil 31200. Sensed parameters and/or component statuses can be communicated through the chain of flex circuits when the end effector 31000 is coupled to the elongate shaft 31500 via the coupling member 31800.

The distal end of the elongate shaft 31500 further comprises an attachment member 31570 and an alignment pin 31580. The proximal end of the coupling member 31800 comprises an attachment groove 31870 sized to receive the attachment member 31570 and an alignment groove 31880 sized to receive the alignment pin 31580 when the end effector 31000 is attached to the elongate shaft 31500.

FIGS. 142-146 show the progression of attaching the disposable end effector 31000 comprising the channel 31100, the anvil 31200, and the cartridge 31300 to the

elongate shaft **31500**. As shown in FIG. **142**, the cartridge **31300** is fully seated in the channel **31100** and the anvil **30200** is coupled thereto as an assembly. The end effector **31000** further comprises a coupling member **31800** for replaceably attaching the end effector **31000** to the elongate shaft **31500** at a point distal to any articulation joint.

A first stage of attaching the disposable end effector assembly to the elongate shaft **31500** is shown in FIG. **143**. As the coupling member **31800** of the disposable assembly is brought toward the distal end of the elongate shaft **31500**, initial contact is made between the attachment member **31570** extending from the elongate shaft **31500** and the attachment groove **31870** defined in the coupling member **31800**. The drive shaft **31600** is initially received within the channel **31860**; however, the drive shaft **31600** has not yet been coupled to a drive screw **31150**. Notably, the flex circuits **31850**, **31750** are misaligned and out of physical contact in the first stage of attachment. Furthermore, the alignment pin **31580** is out of alignment with the alignment groove **31880** defined in the coupling member **31800**. Such misalignment prevents the disposable end effector **31000** from being fully attached to the elongate shaft **31500**.

A second stage of attaching the disposable end effector assembly to the elongate shaft **31500** is shown in FIG. **144**. Contact between the proximal end of the coupling member **31800** and the alignment pin **31580** causes the alignment pin **31580** to be spring biased away from the coupling member **31800** thereby allowing the disposable end effector **31000** and/or the elongate shaft **31500** to be freely rotated with respect to one another. Such rotation of the disposable end effector **31000** and/or the elongate shaft **31500** with respect to one another begins to rotatably attach the drive shaft **31600** to the drive screw **31150**; however, the flex circuits **31850**, **31550** are still out of physical contact and the alignment pin **31580** has not yet been received by the alignment groove **31880** defined in the coupling member **31800**.

FIG. **145** depicts the disposable end effector assembly fully attached to the elongate shaft **31500**. At such a point, the alignment pin **31580** is biased back toward the coupling member **31800** and is completely housed within the alignment groove **31880** defined within the coupling member **31800**. As shown in FIG. **146**, complete operational coupling between the drive shaft **31600** and the drive screw **31150** is achieved when the disposable end effector assembly is fully attached to the elongate shaft **31500**. Furthermore, such alignment between the end effector **31000** and the elongate shaft **31500** also ensures alignment and/or physical contact between the flex circuits **31850**, **31550**. When the disposable assembly is fully attached to the elongate shaft **31500**, all electrical components, including flex circuits **30700** and/or sensor arrays positioned in the anvil **31200**, cartridge **31300**, and/or channel **31100** are coupled and in communication with the elongate shaft **31500** of the surgical instrument.

In various instances, the disposable assembly comprised of the channel **31100**, the anvil **31200**, the cartridge **31300**, and various flex circuits **31700** is only intended for a single use. Stated another way, upon completion of a single firing stroke, the end effector **31000**, including the firing member **31400**, and the associated flex circuits **31700** are removed, or detached, from the elongate shaft **31500**. Detachment can occur after being used for a pre-determined number of firing strokes, or upon becoming defective, for example.

FIGS. **147** and **148** depict an end effector **32050** configured to be replaceably attached to an elongate shaft **32500** of a surgical instrument. The end effector **32050** has a channel

32100, an anvil **32200**, and a cartridge **32300**. The cartridge **32300** is sized and/or configured to be seated in the channel **32100**. As described in greater detail with respect to FIGS. **132-138**, in various instances, the anvil **32200** and the cartridge **32300** are pivotally attached to one another about a pivot joint **32210** prior to the cartridge **32300** being seated in the channel **32100**. In various instances, the anvil **32200** is configured to be pivotally attached to the channel **32100**.

The channel **32100** comprises a proximal end **32052** and a distal end **32054**. The proximal end **32052** of the channel **32100** comprises an attachment member **32056** extending proximally therefrom. The attachment member **32056** is configured to releasably secure the end effector **32050** to an elongate shaft **32500** of the surgical instrument. While FIGS. **147** and **148** show the attachment member **32056** extending from the proximal end of the channel **32100**, the attachment member **32056** can extend from any suitable component of the end effector **32050** such as the anvil **32200** or the cartridge **32300**. In various instances, the attachment member **32056** is integrally formed with the particular end effector component. In other instances, the end effector **32050** comprises an adapter attached to a proximal end of the end effector **32050**. The adapter comprises the attachment member **32056** for securement of the end effector **32050** to the elongate shaft **32500**.

A distal end of the elongate shaft **32500** comprises a securement door **32510** movable between an open position and a closed position about a pivot joint **32520**. In various instances, the securement door **32510** remains in the closed position until motivated into the open position. In such instances, the securement door **32510** is in the closed position prior to attachment of an end effector **32050** thereto. The attachment member **32056** of the end effector **32050** can be used to bias the securement door **32510** into an open position. Alternatively, a clinician can motivate the securement door **32510** into the open position prior to attaching the end effector **32050** to the elongate shaft **32500**. The securement door **32510** can remain biased open in its open position until an attachment member **32056** is appropriately positioned in the groove and/or until the securement door **32510** is motivated into the closed position.

In its open position, as shown in FIG. **147**, the securement door **32510** exposes a groove sized to receive the attachment member **32056** of the end effector **32050** therein. Stated another way, when the securement door **32510** is in the open position, a path is cleared for the attachment member **32056** to be positioned in the groove of the elongate shaft **32500**. In various instances, the securement door **32510** can return to its closed position when the attachment member **32056** is appropriately positioned in the groove. In other instances, a clinician can motivate the securement door **32510** into the closed position. A sensor assembly can communicate a status and/or position of the securement door **32510** to a processor. In such instances, the processor is configured to prevent use of the surgical instrument while the securement door **32510** is in the open position and/or defective.

The securement door **32510** has a distal end **32512** with a latch geometry. The attachment member **32056** comprises a proximal portion having a first thickness and a distal portion having a second thickness. As shown in FIGS. **147** and **148**, the first thickness is greater than the second thickness. Such a geometry allows for the distal end **32512** of the securement door **32510** and/or the corresponding geometry of the groove to retain the attachment member **32056** therein. The geometry of the groove prevents unwanted movement of the attachment member **32056** and/or maintains alignment of the end effector **32050** and the

elongate shaft **32500**, for example. In various instances, the attachment member **32056** has a press-fit relationship with the groove; however, any suitable mechanism that maintains attachment and/or alignment between the components is envisioned.

In various instances, a geometry and/or size of the attachment member **32056** does not correspond to a geometry and/or size of the channel. Such a mismatch in geometry and/or size prevents the end effector **32050** from being fully attached to and/or aligned with the elongate shaft **32500**. In such instances, the firing drive(s) and/or electronic components are not connected and the surgical instrument is non-operable. Should the attachment member **32056** be too large to fit within the groove, the securement door **32510** will be unable to reach its fully closed position, and an alert can be sent to a processor as described in greater detail herein. Similarly, a sensor assembly can detect an absence of contact between the attachment member **32056** and the barriers of the groove, suggestive of an attachment member **32056** comprising an inappropriately small geometry for use with the surgical instrument. In such instances, the processor prevents the use of the surgical instrument.

The end effector **32050** further comprises a firing member **32400** mounted on a drive screw. A drive shaft **32600**, similar to drive shaft **1660**, extends through the elongate shaft **32500** and is coupled with the drive screw of the end effector **32050** upon attachment of the end effector **32050** to the elongate shaft **32500**. Subsequent rotation of the drive screw causes the firing member **32400** to translate through the end effector **32050**. The firing member **32400** comprises a first camming member **32406** configured to engage the anvil **32200** as the firing member **32400** translates through the end effector **32050**, a second camming member **32408** configured to engage the channel **32100** as the firing member **32400** translates through the end effector **32050**, and a cutting element **32410**. As discussed in greater detail throughout, the firing member **32400** can be mounted on a drive screw in the channel **32100** prior to attachment of the cartridge **32300** thereto or the firing member **32400** can be an integral component with the cartridge **32300** prior to seating the cartridge **32300** in the channel **32100**.

In any event, as shown in FIGS. **149** and **150**, the firing member **32400** comprises a projection **32420** having a keyed profile **32425** on a proximal end **32402** of the firing member **32400**. The keyed profile **32425** is configured to be received within a corresponding groove **32610** formed in a distal end **32604** of the drive shaft **32600**. As the end effector **32050** is brought into alignment with the elongate shaft **32500**, the keyed profile **32425** of the projection **32420** is configured to be positioned in the groove **32610**. In various instances, the groove **32610** comprises a larger geometry than the keyed profile **32425** of the firing member **32400**. However, the groove **32610** comprises a notch configured to catch the keyed profile **32425** of the firing member **32400** and prevent the firing member **32400** from translating distally out of connection with the drive shaft **32600**. In various instances, the width of the groove **32610** is similar to the width of the keyed profile **32425**. Such a similarity in width allows for the keyed profile **32425** to comfortably fit into the groove **32610** yet prevents unwanted proximal translation and/or rotation of the keyed profile **32425** within the groove **32610**.

FIGS. **151** and **152** depict a reinforce anvil **33200** having an anvil **33250** and an anvil plate **33260** circumferentially welded thereto. The anvil **33250** comprises a projection **33210** for pivotal attachment to a cartridge and/or a channel as described in greater detail herein. The anvil plate **33260** bridges, or crosses, at least partially over top of the pivot

joint formed about the projection **33210**. While the anvil plate **33260** is described as being welded to the anvil **33200**, any attachment method that provides suitable reinforcement to the anvil **33200** is envisioned. The reinforced anvil **33200** provides increased strength to allow the reinforced anvil **33200** to withstand greater loads experienced during closure and/or firing strokes, especially over the pivotal attachment joint, for example.

As shown in FIG. **153**, the reinforced anvil **33200** is pivotally attached to a channel **33100** of an end effector **33000**. The end effector **33000** further comprises a cartridge **33300** seated in the channel **33100**. A firing member **33400** is positioned in the end effector **33000**. The firing member **33400** has a first camming member **33406** configured to engage an elongate slot **33220** of the anvil **33200** as the firing member **33400** translates through the end effector **33000**, a second camming member **33408** configured to engage the channel **33100** as the firing member **33400** translates through the end effector **33000**, and a cutting element **33410**.

The anvil plate **33260** comprises a first thickness **A** at a proximal end **33262** and a second thickness **a** at a distal end **33264** thereof. In various instances, the first thickness **A** can range from 0.03 inches to 0.035 inches, while the second thickness **a** can range from 0.01 inches to 0.015 inches, for example. The first thickness **A** is larger than the second thickness to provide an increased strength to the reinforced anvil **33200** at the pivot joint formed about projection **33210**, for example. The reinforced anvil **33200** comprises a tissue-compressing surface. The tissue-compressing surface has a curved topography, wherein the distance between the tissue-compressing surface and a tissue-supporting surface of the cartridge **33300** is smaller at a point closer to the pivot joint about projection **33210**. The curved topography prevents patient tissue from becoming trapped and/or pinched between the reinforced anvil **33200** and the cartridge **33300** and/or the channel **33100**, for example. Welding the anvil plate **33260** to the anvil **33250** allows for the reinforced anvil **33200** to have an increased stiffness along the elongate slot **33220** of the anvil **33250** where substantial loads are applied by the firing member **33400** in addition to the portion of the anvil **33250** surrounding the projection **33210**. Such an increase in stiffness improves tissue manipulation and/or tissue clamping loads, for example.

FIG. **154** depicts an assembly comprised of a cartridge **34300** and a channel **34100**. Such an assembly is configured to be replaceably coupled to an elongate shaft of a surgical instrument distal to an articulation joint. In an effort to, for example, form a more rigid, disposable assembly, the assembly comprises an interlock system molded into the walls of the channel **34100** and the cartridge **34300**. The channel **34100** comprises a base **34120** with an elongate slot **34110** defined therein for receiving a portion of a firing member. The channel **34100** further comprises a pair of sidewalls **34130** extending from the base **34120**. A notch **34150** is defined in the sidewall **34130**.

As defined in greater detail throughout, the cartridge **34300** is configured to be seated in the channel **34100**. The cartridge **34300** comprises a plurality of staples removably positioned within staple cavities, a longitudinal slot **34310** extending from a proximal end toward a distal end of the cartridge **34300**, and a wedge sled **34600** configured to motivate the staples out of the respective staple cavities as the wedge sled **34600** translates through the longitudinal slot **34310** during a firing stroke. The cartridge **34300** further comprises a projection **34350** configured to be received within the notch **34150** when the cartridge **34300** is fully

seated in the channel **34100**. A portion of a cartridge deck **34320** is configured to rest upon a top portion **34140** of the channel sidewall **34130**. While the cartridge is described as having a projection and the channel is described as having a notch, any suitable attachment mechanism or combination of attachment mechanisms are envisioned to releasably secure the cartridge in the channel.

When the wedge sled **34600** is inserted into a proximal end of the cartridge **34300**, the cartridge **34300** is pushed laterally, causing the projection **34350** to nest within the notch **34150** of the channel **34100**. Such an interlocking engagement enables the channel **34100** to provide additional support to the cartridge deck **34320** and cartridge body than from the base **34120** alone. Lateral motivation of the cartridge **34300** diverts a tissue compression load from the cartridge deck **34320** into the sidewalls **34130** of the channel **34100** rather than allowing the tissue compression load to be transmitted through the body of the cartridge alone.

Similar to the reinforced anvil **33200** shown in FIGS. **151-154**, the channel **34100** can be reinforced with a channel cap that bridges, or crosses, at least partially over top of the elongate slot **34110**. The base **34120** of the channel **34100** can range in thickness from 0.025 inches to 0.035 inches, for example. A channel cap with a thickness of between 0.01 inches and 0.015 inches, for example, can be welded to the base **34120** of the channel **34100**. The addition of the channel cap allows for a more robust cartridge and channel assembly.

Various aspects of the present disclosure are directed to methods, devices, and systems for sealing tissue using a combination of energy and stapling modalities. The hybrid approach improves upon, and compensates for, the shortcomings of using the energy and stapling modalities separately.

Referring now to FIG. **155**, a surgical instrument **60000** is configured to seal tissue using a combination of energy and stapling modalities or phases. In certain instances, is also configured to cut the tissue. The surgical instrument **60000** is similar in many respects to other surgical instruments (e.g. surgical instrument **1000**) described elsewhere herein, which are not repeated herein in the same level of detail for brevity. The surgical instrument **60000** includes an end effector **60002**, an articulation assembly **60008**, a shaft assembly **60004**, and a housing assembly **60006**. In the illustrated example, the articulation assembly **60008** permits the end effector **60002** to be articulated about a central longitudinal axis **60005** relative to the shaft assembly **60006**.

In the illustrated example, the housing assembly **60006** is in the form of a handle that includes a trigger **60010** movable relative to a handle portion **60012** to effect a motion at the end effector **60002**. In other examples, however, the housing assembly **60006** can be incorporated into a robotic system. It will be understood that the various unique and novel arrangements of the various forms of the surgical instruments disclosed herein may also be effectively employed in connection with robotically-controlled surgical systems. Thus, the term "housing" may also encompass a housing or similar portion of a robotic system that houses or otherwise operably supports at least one drive system that is configured to generate and apply at least one control motion which could be used to actuate shaft assemblies disclosed herein and their respective equivalents. For example, the surgical instruments disclosed herein may be employed with various robotic systems, instruments, components and methods disclosed in U.S. patent application Ser. No. 13/118,241, entitled SURGICAL STAPLING INSTRUMENTS WITH

MENTS, now U.S. Patent Application Publication No. US 2012/0298719. U.S. patent application Ser. No. 13/118,241, entitled SURGICAL STAPLING INSTRUMENTS WITH ROTATABLE STAPLE DEPLOYMENT ARRANGEMENTS, now U.S. Patent Application Publication No. US 2012/0298719, is incorporated by reference herein in its entirety. In certain aspects, the housing assembly **60006** is detachably couplable to an interchangeable assembly that includes the shaft assembly **60004**, the articulation assembly **60008**, and the end effector **60002**, for example.

Referring to FIG. **156-162**, the end effector **60002** extends distally from the articulation assembly **60008**, and includes an anvil **60020** and a cartridge support channel **60040** configured to accommodate a cartridge **60030**. In the illustrated example, the anvil **60020** defines a first jaw, while the support channel **60040** and the cartridge **60030** define a second jaw. At least one of the first jaw and the second jaw is movable relative to the other jaw to grasp tissue therebetween. In the illustrated example, rotation of a drive member, which can be in the form of a drive screw, causes a firing member, which can be in the form of an I-beam **764**, to move distally to pivot the anvil **60020** toward the cartridge **60030** in a closure motion to grasp tissue therebetween.

Further rotation of the drive member causes of the I-beam **764** to engage and motivate a sled, in a firing motion, to deploy staples **60033** (FIG. **159**) from the anvil **60020** into the grasped tissue. The staples are generally stored in rows of staple cavities **60031**, **60032** extending longitudinally on opposite sides of a longitudinal slot **60035** defined in a cartridge body **60039** of the cartridge **60030**. The sled is configured to deploy the staples **60033** by pushing upward staple drivers in the rows of staple cavities **60031**, **60032**. Upward motion of the staple drivers deploys the staples **60033** from the rows of staple cavities **60031**, **60032** into the tissue. Staple legs of the staples **60033** are then deformed by corresponding rows of anvil pockets **60021**, **60022** (FIG. **162**) on opposite sides of a longitudinal slot **60025** defined in an anvil plate **60024** of the anvil **60020**.

Referring primarily to FIG. **163**, a control circuit **760** may be programmed to control one or more functions of the surgical instrument **750** such as, for example, closure of the end effector **60002**, activation of the at least one electrode, and/or firing the cartridge **60030**. The control circuit **760**, in some examples, may comprise one or more microcontrollers, microprocessors, or other suitable processors for executing instructions that cause the processor or processors to control the one or more functions of the surgical instrument **60000**. In one aspect, a timer/counter **781** provides an output signal, such as the elapsed time or a digital count, to the control circuit **760**. The timer/counter **781** may be configured to measure elapsed time, count external events, or time external events.

The control circuit **760** may generate a motor set point signal **772**. The motor set point signal **772** may be provided to a motor controller **758**. The motor controller **758** may comprise one or more circuits configured to provide a motor drive signal **774** to the motor **754** to drive a motor **754** as described herein. In some examples, the motor **754** may be a brushed DC electric motor. For example, the velocity of the motor **754** may be proportional to the motor drive signal **774**. In some examples, the motor **754** may be a brushless DC electric motor and the motor drive signal **774** may comprise a PWM signal provided to one or more stator windings of the motor **754**. Also, in some examples, the motor controller **758** may be omitted, and the control circuit **760** may generate the motor drive signal **774** directly.

The motor **754** may receive power from an energy source **762**. The energy source **762** may be or include a battery, a super capacitor, or any other suitable energy source. The motor **754** may be mechanically coupled to the drive member **751** via a transmission **756**. The transmission **756** may include one or more gears or other linkage components to couple the motor **754** to the drive member **751**.

In certain instances, a current sensor **786** can be employed to measure the current drawn by the motor **754**. The force required to advance the drive member **751** corresponds to the current drawn by the motor **754**. The force is converted to a digital signal and provided to the control circuit **760**. The current drawn by the motor **754** can represent tissue compression.

Referring to FIG. **164**, a schematic of a control circuit **760** is depicted. According to the non-limiting aspect, the control circuit **760** can include a microcontroller comprising one or more processors **68002** (e.g., microprocessor, microcontroller) coupled to at least one memory circuit **68008**. The memory circuit **68008** can be configured to store machine-executable instructions that, when executed by the processor **68002**, can cause the processor **68002** to execute machine instructions to implement the various processes described herein. The processor **68002** can be any one of a number of single-core or multicore processors known in the art. Alternatively and/or additionally, the microcontroller can include a logic board, such as a Field Programmable Gate Array, for example. The memory circuit **8008** can comprise volatile and non-volatile storage media. The processor **68002** may include an instruction processing unit **68004** and an arithmetic unit **68006**. The instruction processing unit **68004** can be configured to receive instructions from the memory circuit **68008** of this disclosure.

The control circuit **760** may employ a position sensor **784** to determine the position of the I-beam **764**. Position information is provided to a processor **68002** of the control circuit **760**, which can be programmed or configured to determine the position of the I-beam **764** based on the position information. In one aspect, the position information is indicative of the rotational position of the drive member **751**, and the processor **68002** is configured to calculate the position of the I-beam **764** based on the rotational position of the drive member **751**.

A display **711** displays a variety of operating conditions of the surgical instrument **60000** and may include touch screen functionality for data input. Information displayed on the display **711** may be overlaid with images acquired via imaging modules.

The control circuit **760** may be in communication with one or more sensors **788**. The sensors **788** may be positioned on the end effector **752** and adapted to operate with the surgical instrument **750** to measure the various derived parameters such as gap distance versus time, tissue compression versus time, and anvil strain versus time. The sensors **788** may comprise a magnetic sensor, a magnetic field sensor, a strain gauge, a pressure sensor, a force sensor, an inductive sensor such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor for measuring one or more parameters of the end effector **752**.

In one aspect, sensors **788** may be implemented as a limit switch, electromechanical device, solid-state switches, Hall-effect devices, MR devices, GMR devices, magnetometers, among others. In other implementations, the sensors **788** may be solid-state switches that operate under the influence of light, such as optical sensors, IR sensors, ultraviolet sensors, among others. Still, the switches may be solid-state

devices such as transistors (e.g., FET, junction FET, MOS-FET, bipolar, and the like). In other implementations, the sensors **788** may include electrical conductorless switches, ultrasonic switches, accelerometers, and inertial sensors, among others. The sensors **788** may include one or more sensors.

The control circuit **760** can be configured to simulate the response of the actual system of the instrument in the software of a controller. The drive member **751** can move one or more elements in the end effector **752** at or near a target velocity. The surgical instrument **750** can include a feedback controller, which can be one of any feedback controllers, including, but not limited to a PID, a state feedback, LQR, and/or an adaptive controller, for example. The surgical instrument **750** can include a power source to convert the signal from the feedback controller into a physical input such as case voltage, PWM voltage, frequency modulated voltage, current, torque, and/or force, for example.

In addition to stapling tissue grasped between the anvil **60020** and the cartridge **60030**, the surgical instrument **60000** is further configured to apply an RF energy treatment to the tissue. An RF energy source **794** (FIG. **163**) is coupled to the end effector **60002**. As illustrated in FIG. **162**, the anvil **60020** includes a first electrode assembly **60026** on a first side of the longitudinal slot **60025**, and a second electrode assembly **60027** on a second side of the longitudinal slot **60025** opposite the first side. The electrode assemblies **60026**, **60027** can be separately, or commonly, connected to the RF energy source, and are configured to deliver RF energy to the tissue separately, or simultaneously.

In the illustrated example, the first electrode assembly **60026** includes three segmented electrodes **60026a**, **60026b**, **60026c** arranged in a first row on the first side of the longitudinal slot **60025**. Likewise, the second electrode assembly **60027** includes three segmented electrodes **60027a**, **60027b**, **60027c** arranged in a second row on the second side of the longitudinal slot **60025**. It is, however, understood that the number of segmented electrodes in the anvil **60020** can be varied to accommodate various applications, for example. The segmented electrodes **60026a-c** and the segmented electrodes **60027a-c** can be separately, or simultaneously, activated to deliver an RF energy treatment to the tissue in accordance with one or more RF energy algorithms, as discussed in greater detail below.

In the illustrated example, the first electrode assembly **60026** is stepped up from the row of staple cavity **60021**. Likewise, the second electrode assembly **60027** is stepped up from the row of staple cavity **60022**. In various aspects, the segmented electrodes **60026a-c** and/or the segmented electrodes **60027a-c** comprise the same, or substantially the same, height. In other examples, the segmented electrodes **60026a-c** and/or the segmented electrodes **60027a-c** comprise different heights. In one arrangement, the segmented electrodes **60026a-c** and/or the segmented electrodes **60027a-c** are arranged such that their heights gradually decrease from the most distal to the most proximal. In another arrangement, the segmented electrodes **60026a-c** and/or the segmented electrodes **60027a-c** are arranged such that their heights gradually increase from the most distal to the most proximal.

Further to the above, the cartridge **60030** includes an asymmetric cartridge body **60034** which accommodates a third electrode assembly **60036** including a row of segmented electrodes **60036a**, **60036b**, **60036c**, **60036d**, **60036e**, **60036f** on a first side of the longitudinal slot **60035**. The third electrode assembly **60036** is configured to oppose

the first electrode assembly **60026** of the anvil **60020** in a closed configuration, as illustrated in FIG. **158**. The cartridge **60030** lacks an electrode assembly on the second side of the longitudinal slot **60035**. Instead, the second electrode assembly **60027** of the anvil **60020** is opposed by a longitudinal step **60037** extending alongside the third electrode assembly **60036**. In certain instances, the longitudinal step **60037** extends in parallel, or at least substantially in parallel, with the third electrode assembly **60036**.

Although the third electrode assembly **60036** is depicted with six segmented electrodes **60036a-f**, more or less segmented electrodes could be utilized. The segmented electrodes **60036a-f** can be separately, or commonly, connected to the RF energy source **794**, and can be activated separately, or simultaneously. In the illustrated example, the electrode assemblies **60026**, **60027** define source electrodes, while the third electrode assembly **60036** defines a return electrode such that bipolar RF energy is configured to flow from the electrode assemblies **60026**, **60027** to the third electrode assembly **60036**. In other examples, however, the third electrode assembly **60036** can be configured as a source electrode, and one or both of the electrode assemblies **60026**, **60027** can be configured as return electrodes.

Further to the above, the segmented electrodes **60036a-f** of the third electrode assembly **60036** are arranged in a longitudinal row, and are spaced apart from one another. The third electrode assembly **60036** further includes insulators **60039a-e** disposed in spaces between the segmented electrodes **60036a-f** along the longitudinal row, as best illustrated in FIG. **159**. In one example, the insulators **60039a-e** comprise a uniform length and/or shape. In other examples, the insulators **60039a-e** comprise different lengths and/or shapes.

Referring primarily to FIGS. **160** and **161**, a support wall **60048** extends between and separates, or at least partially separates, the third electrode assembly **60036** and the row of staple cavities **60031**. The third electrode assembly **60036** and the support wall **60048** are stepped up from the row of staple cavities **60031** on the first side of the longitudinal slot **60035**. Likewise, the longitudinal step **60037** is stepped up from the row of staple cavities **60032** on the second side of the longitudinal slot **60035**. The longitudinal step **60037** and third electrode assembly **60036** cooperatively define an interior tissue sealing zone stepped up from exterior tissue stapling zones defined by the rows of staple cavities **60031**, **60032** defined on opposite sides of the interior tissue sealing zone. In the illustrated example, the longitudinal step **60037** and third electrode assembly **60036** define, or at least partially define, opposite side walls of the longitudinal slot **60035**. The I-beam **764** is configured to pass between the longitudinal step **60037** and third electrode assembly **60036** in a firing motion of the surgical instrument **60000**.

FIGS. **160** and **161** are a close-up view of the cartridge **60030** illustrating an example composition of the third electrode assembly **60036**. A flex circuit **60041** extends longitudinally behind the segmented electrodes **60036a-f** and insulators **60039a-c**. The flex circuit **60041** is positioned against the cartridge deck **60047**. In the illustrated example, the segmented electrodes **60036a-f** are electrically connected to the flex circuit **60041** via passive switches, current limiting elements, energy-sensitive resistance elements, or locally-adjustable resistance elements, which can be in the form of positive temperature coefficient (PTC) segments **60042a-f**. In other examples, the segmented electrodes **60036a-f** can be directly connected to the flex circuit **60041**. In any event, the flex circuit **60041** is configured to connect the segmented electrodes **60036a-f** to the energy source **794**.

In the illustrated example, the segmented electrodes **60036a-f** are separately connected in series with corresponding PTC segments **60042a-f**, respectively, as illustrated in FIG. **173**. In other words, there are an equal number of segmented electrodes and PTC segments. In other examples, however, two or more segmented electrodes can be connected to one PTC segment.

In various aspects, the insulators **60039a-e** extend in gaps between the segmented electrodes **60036a-f**, and comprise the same, or at least substantially the same, height as the segmented electrodes **60036a-f** permitting a uniform, or at least substantially uniform, tissue contacting surface along the tissue sealing zone defined by the third electrode assembly **60036**. Alternatively, as best illustrated in FIG. **160**, the segmented electrodes **60036a-f** and the insulators **60039a-e** may comprise different heights. The height disparity may allow the segmented electrodes **60036a-f** to act as conductive tissue gripping features.

In the illustrated example, the insulator **60039e** extends longitudinally between the segmented electrode **60036f** and the segmented electrode **60036e**, and extends vertically to a first height slightly lesser than second and third heights of the segmented electrode **60036f** and the segmented electrode **60036e**. In other examples, the second and third heights are greater than the first height. In other examples, the first, second, and third heights are the same, or at least substantially the same.

In various aspects, the segmented electrodes **60036a-f** comprise a uniform, or at least substantially uniform, height. In other examples, the segmented electrodes **60036a-f** comprise different heights. In one arrangement, the segmented electrodes **60036a-f** are arranged such that their heights gradually decrease from the most distal (segmented electrode **60036f**) to the most proximal (segmented electrode **60036a**). In another arrangement, the segmented electrodes **60036a-f** are arranged such that their heights gradually increase from the most distal (segmented electrode **60036f**) to the most proximal (segmented electrode **60036a**).

In various aspects, the third electrode assembly **60036** can be secured to the cartridge body **60039** via posts **60043** extending through holes in third electrode assembly **60036**. As illustrated in FIG. **159**, in certain examples, the holes are defined in the insulators **60039a-e**. The posts **60043** can also function as heat stakes, for example. Additionally, or alternatively, the third electrode assembly **60036** can be secured to the cartridge body **60039** using any suitable locking, or mating, features, for example.

In various aspects, the longitudinal step **60037** and the third electrode assembly **60036** comprise the same, or at least substantially the same, height. In other examples, the longitudinal step **60037** and the third electrode assembly **60036** comprise different heights. As illustrated in FIG. **158**, the anvil **60020** and the cartridge **60030** cooperatively define a tissue sealing gap **60044** including a first gap portion **60044a** defined between the first electrode assembly **60026** and the third electrode assembly **60036**, and a second gap portion **60044b** defined between the second electrode assembly **60027** and the longitudinal step **60037**. In various aspects, the gap portions **60044a**, **60044b** comprise the same, or at least substantially the same, size and/or height. In other aspects, the gap portions **60044a**, **60044b** comprise different sizes and/or heights.

Further to the above, the anvil **60020** and the cartridge **60030** cooperatively define a tissue-stapling gap **60045** therebetween. The tissue stapling gap **60045** includes a first gap portion **60045a** defined between the row of staple pockets **60021** and the row of staple cavities **60031**, and a

second gap portion **60045b** defined between the row of staple pockets **60022** and the row of staple cavities **60032**. The tissue sealing gap **60044** extends between the first gap portion **60045a** and the second gap portion **60045b**.

In the illustrated example, the tissue sealing gap **60044** comprises a different height than the tissue stapling gap **60045**. For effective tissue sealing, the tissue sealing gap **60044** comprises a height selected from a range of about 0.005" to about 0.02", a range of about 0.008" to about 0.018", or a range of about 0.009" to about 0.011", for example. For effective tissue stapling, the tissue stapling gap **60045** comprises a height selected from a range of about 0.04" to about 0.08", a range of about 0.05" to about 0.07", or a range of about 0.055" to about 0.065. In at least one example, the anvil **60020** and the cartridge **60030** cooperate to define a tissue sealing gap **60044** and a tissue stapling gap **60045** therebetween, wherein the tissue sealing gap comprises a height of about 0.01", and the tissue stapling gap comprises a height of about 0.06".

In various aspects, as best illustrated in FIG. 160, the rows of staple cavities **60031**, **60032** include pocket extenders **60046** that protrude from a cartridge deck **60047** of the cartridge **60030**. The pocket extenders **60046** ensure a proper deployment of the staples **60033** into tissue positioned against the cartridge deck **60047**. In certain examples, the tissue sealing gap **60044** is raised above the pocket extenders **60046**. In such examples, the longitudinal step **60037** and/or the third electrode assembly **60036** comprise a height, or heights, greater than that of the pocket extenders **60046**, for example.

In various aspects, as best illustrated in FIG. 160, the longitudinal step **60037** and the support wall **60048** include distal ramps **60037a**, **60048a** that facilitate insertion of the cartridge **60030** beneath a target tissue. The distal ramps **60037a**, **60048a** gradually protrude from the cartridge deck **60047** toward top edges that are coplanar, or at least substantially coplanar, with tissue contacting surfaces of the longitudinal step **60037** and the third electrode assembly **60036**, for example.

Referring primarily to FIG. 158, the end effector **60002** is shown in a closed configuration suitable for application of a therapeutic energy treatment to a tissue portion between the electrode assemblies **60026**, **60027** and the third electrode assembly **60036** and the longitudinal step **60037**, and application of a tissue stapling treatment to tissue portions between rows of staple pockets **60021**, **60022** and rows of staple cavities **60031**, **60032**. In the closed configuration, a tissue sealing centerline is defined through the tissue sealing gap **60044**, and a tissue stapling center line is defined through the tissue stapling gap **60045**, wherein the tissue sealing centerline is higher than the tissue stapling centerline. In other words, the tissue sealing centerline is further away from the cartridge deck **60047** than the tissue stapling centerline. In the illustrated example, the tissue sealing gap **60044** is higher, or further away from the cartridge deck **60047**, than the tissue stapling centerline.

In other configurations of the end effector **60002**, the tissue sealing centerline and the tissue stapling centerline are collinear. In various aspects, the tissue sealing centerline is equidistant from the first electrode assembly **60026** and the third electrode assembly **60036**, and/or equidistant from the second electrode assembly **60027** and the longitudinal step **60037**. In various aspects, the tissue stapling centerline is equidistant from the first row of staple pockets **60021** and the first row of staple cavities **60031**, and/or equidistant from the second row of staple cavities **60022** and the second row of staple cavities **60032**.

In various aspects, the RF energy device **794**, which is configured to supply RF energy to the end effector **60002**, can be in the form of a generator such as, for example, generators **800**, **900**, which are described in greater detail below in connection with FIGS. 165, 166. In various aspects, the RF energy device **794** is electrically coupled to the electrode assemblies **60026**, **60027**, **60036**, and the control circuit **760** is configured to cause the RF energy source **794** to selectively switch one or more of the segmented electrodes of the electrode assemblies **60026**, **60027**, **60036** between an active mode and an inactive mode. In certain instances, one or more switching mechanisms can be employed to transition one or more of the segmented electrodes of the electrode assemblies **60026**, **60027**, **60036** between the active mode and inactive mode. In the active mode, the segmented electrodes of electrode assemblies **60026**, **60027**, **60036** can be utilized as source electrodes or return electrodes, depending on polarity, to implement various tissue sealing algorithms defined by the control circuit **760**, for example.

In various aspects, the control circuit **760** may cause the RF energy source **794** to adaptively alternate, or switch, between an opposing bipolar energy mode and an offset bipolar energy mode. In the opposing bipolar energy mode the control circuit **760** is configured to cause the RF energy source **794** to pass a first therapeutic signal between the first electrode assembly **60026** and the third electrode assembly **60036**. In the offset bipolar energy mode, the control circuit **760** is configured to cause the RF energy source **794** to pass a second therapeutic signal between the second electrode assembly **60027** and the third electrode assembly **60036**.

The cartridge **60030**, on one side of the longitudinal slot **60035**, includes the longitudinal step **60037** which is configured to cooperate with the second electrode assembly **60027** of the anvil **60020** to achieve a tissue compression suitable for energy sealing, but does not act as a return/source electrode. Alternating between the opposing energy mode and the offset energy mode permits a proper sealing of tissue in the second gap portion **60044b** of the tissue sealing gap **60044** where an electrode assembly is lacking due to the presence of the longitudinal step **60037**. In various aspects, as described in greater detail elsewhere herein, the longitudinal step **60037** includes a cavity **60049** therein configured to accommodate a driver support that resists driver roll. The longitudinal step **60037** permits the driver support to be extended above the cartridge deck **60047** to resist driver roll.

In various aspects, the control circuit **760** may cause the RF energy source **794** to adaptively alternate, or switch, between the opposing bipolar energy mode and the offset bipolar energy mode based on a tissue parameter, or condition, such as, for example, tissue impedance. FIG. 167 is a logic flow diagram of a process **60160** depicting a control program or a logic configuration for sealing tissue grasped by an end effector by alternating, or switching, between the opposing energy mode and the offset energy mode. In certain instances, the process **60160** can be implemented by the surgical instrument **60000**, for example. In certain instances, the memory **68008** stores program instructions that, when executed by the processor **68002**, cause the processor **68002** to perform one or more aspects of the process **60160**.

The process **60160** includes monitoring **60161** a tissue parameter of a tissue grasped by the end effector **60002**. In certain examples, the tissue parameter is a tissue compression. The control circuit **760** may monitor **60161** the tissue compression based on sensor signals from one or more sensors **788**.

If **60162** the tissue parameter indicates suitable energy sealing conditions, the process **60160** activates **60163** one of the opposing energy mode and the offset energy mode. To determine whether the tissue parameter is indicative of suitable energy sealing conditions, the control circuit **760** may, for example, compare detected values of the tissue parameter to a predetermined threshold indicative of suitable energy sealing conditions, which can be stored in a storage medium accessible by the processor **68002** such as, for example, the memory **68008**.

Following detection of a tissue parameter indicative of suitable energy sealing conditions, only the opposing energy mode is activated **60163**, while the offset energy mode remains inactive. In the opposing energy mode, the control circuit **760** may activate the electrode assemblies **60026**, **60036**, while the electrode assembly **60027** remains inactive. The process **60160** further includes monitoring tissue impedance **60164** to determine when to alternate, or switch, between the opposing energy mode and the offset energy mode. As described elsewhere herein in greater detail, tissue impedance of a tissue portion can be detected, for example by a control circuit **760**, by causing a sub-therapeutic signal to be passed through the tissue portion, receiving measurements from a voltage sensing circuit **924** and the current sensing circuit **914**, and dividing the measurements from the voltage sensing circuit **924**, by the corresponding measurements from the current sensing circuit **914**, for example.

In the illustrated example, if **60165** a tissue impedance equal to, or beyond, a predetermined threshold is detected, the process **60160** switches **60166** from the opposing energy mode to the offset energy mode. To switch to the offset energy mode, the control circuit **760** may deactivate the electrode assembly **60026**, and activate the electrode assembly **60027**. In other instances, the offset energy mode is activated before activation of the opposing energy mode, and deactivated with, or after, activation of the opposing energy mode.

Generator Hardware

FIG. **165** is a simplified block diagram of a generator **800** configured to provide inductorless tuning, among other benefits. Additional details of the generator **800** are described in U.S. Pat. No. 9,060,775, titled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, which issued on Jun. 23, 2015, which is herein incorporated by reference in its entirety. The generator **800** may comprise a patient isolated stage **802** in communication with a non-isolated stage **804** via a power transformer **806**. A secondary winding **808** of the power transformer **806** is contained in the isolated stage **802** and may comprise a tapped configuration (e.g., a center-tapped or a non-center-tapped configuration) to define drive signal outputs **810a**, **810b**, **810c** for delivering drive signals to different surgical instruments, such as, for example, an ultrasonic surgical instrument, an RF electrosurgical instrument, and a multifunction surgical instrument which includes ultrasonic and RF energy modes that can be delivered alone or simultaneously. In particular, drive signal outputs **810a**, **810c** may output an ultrasonic drive signal (e.g., a 420V root-mean-square (RMS) drive signal) to an ultrasonic surgical instrument, and drive signal outputs **810b**, **810c** may output an RF electrosurgical drive signal (e.g., a 100V RMS drive signal) to an RF electrosurgical instrument (e.g. surgical instrument **60000**), with the drive signal output **810b** corresponding to the center tap of the power transformer **806**.

It will be appreciated that the electrosurgical signal, provided to the surgical instrument **60000**, may be either a therapeutic or sub-therapeutic level signal where the sub-therapeutic signal can be used, for example, to monitor tissue or instrument conditions and provide feedback to the generator **800**. In certain instances, a sub-therapeutic signal can be employed, for example, to detect an impedance of a tissue grasped by the end effector **60002**.

The non-isolated stage **804** may comprise a power amplifier **812** having an output connected to a primary winding **814** of the power transformer **806**. In certain forms, the power amplifier **812** may comprise a push-pull amplifier. For example, the non-isolated stage **804** may further comprise a logic device **816** for supplying a digital output to a digital-to-analog converter (DAC) circuit **818**, which in turn supplies a corresponding analog signal to an input of the power amplifier **812**. In certain forms, the logic device **816** may comprise a programmable gate array (PGA), a FPGA, programmable logic device (PLD), among other logic circuits, for example. The logic device **816**, by virtue of controlling the input of the power amplifier **812** via the DAC circuit **818**, may therefore control any of a number of parameters (e.g., frequency, waveform shape, waveform amplitude) of drive signals appearing at the drive signal outputs **810a**, **810b**, **810c**. In certain forms and as discussed below, the logic device **816**, in conjunction with a processor (e.g., a DSP discussed below), may implement a number of DSP-based and/or other control algorithms to control parameters of the drive signals output by the generator **800**.

Power may be supplied to a power rail of the power amplifier **812** by a switch-mode regulator **820**, e.g., a power converter. In certain forms, the switch-mode regulator **820** may comprise an adjustable buck regulator, for example. The non-isolated stage **804** may further comprise a first processor **822**, which in one form may comprise a DSP processor such as an Analog Devices ADSP-21469 SHARC DSP, available from Analog Devices, Norwood, MA, for example, although in various forms any suitable processor may be employed. In certain forms the DSP processor **822** may control the operation of the switch-mode regulator **820** responsive to voltage feedback data received from the power amplifier **812** by the DSP processor **822** via an ADC circuit **824**. In one form, for example, the DSP processor **822** may receive as input, via the ADC circuit **824**, the waveform envelope of a signal (e.g., an RF signal) being amplified by the power amplifier **812**. The DSP processor **822** may then control the switch-mode regulator **820** (e.g., via a PWM output) such that the rail voltage supplied to the power amplifier **812** tracks the waveform envelope of the amplified signal. By dynamically modulating the rail voltage of the power amplifier **812** based on the waveform envelope, the efficiency of the power amplifier **812** may be significantly improved relative to a fixed rail voltage amplifier schemes.

In certain forms, the logic device **816**, in conjunction with the DSP processor **822**, may implement a digital synthesis circuit such as a direct digital synthesizer control scheme to control the waveform shape, frequency, and/or amplitude of drive signals output by the generator **800**. In one form, for example, the logic device **816** may implement a DDS control algorithm by recalling waveform samples stored in a dynamically updated lookup table (LUT), such as a RAM LUT, which may be embedded in an FPGA. This control algorithm is particularly useful for ultrasonic applications in which an ultrasonic transducer, such as an ultrasonic transducer, may be driven by a clean sinusoidal current at its resonant frequency. Because other frequencies may excite parasitic resonances, minimizing or reducing the total dis-

tortion of the motional branch current may correspondingly minimize or reduce undesirable resonance effects. Because the waveform shape of a drive signal output by the generator **800** is impacted by various sources of distortion present in the output drive circuit (e.g., the power transformer **806**, the power amplifier **812**), voltage and current feedback data based on the drive signal may be input into an algorithm, such as an error control algorithm implemented by the DSP processor **822**, which compensates for distortion by suitably pre-distorting or modifying the waveform samples stored in the LUT on a dynamic, ongoing basis (e.g., in real time). In one form, the amount or degree of pre-distortion applied to the LUT samples may be based on the error between a computed motional branch current and a desired current waveform shape, with the error being determined on a sample-by-sample basis. In this way, the pre-distorted LUT samples, when processed through the drive circuit, may result in a motional branch drive signal having the desired waveform shape (e.g., sinusoidal) for optimally driving the ultrasonic transducer. In such forms, the LUT waveform samples will therefore not represent the desired waveform shape of the drive signal, but rather the waveform shape that is required to ultimately produce the desired waveform shape of the motional branch drive signal when distortion effects are taken into account.

The non-isolated stage **804** may further comprise a first ADC circuit **826** and a second ADC circuit **828** coupled to the output of the power transformer **806** via respective isolation transformers **830**, **832** for respectively sampling the voltage and current of drive signals output by the generator **800**. In certain forms, the ADC circuits **826**, **828** may be configured to sample at high speeds (e.g., 80 mega samples per second (MSPS)) to enable oversampling of the drive signals. In one form, for example, the sampling speed of the ADC circuits **826**, **828** may enable approximately 200× (depending on frequency) oversampling of the drive signals. In certain forms, the sampling operations of the ADC circuit **826**, **828** may be performed by a single ADC circuit receiving input voltage and current signals via a two-way multiplexer. The use of high-speed sampling in forms of the generator **800** may enable, among other things, calculation of the complex current flowing through the motional branch (which may be used in certain forms to implement DDS-based waveform shape control described above), accurate digital filtering of the sampled signals, and calculation of real power consumption with a high degree of precision. Voltage and current feedback data output by the ADC circuits **826**, **828** may be received and processed (e.g., first-in-first-out (FIFO) buffer, multiplexer) by the logic device **816** and stored in data memory for subsequent retrieval by, for example, the DSP processor **822**. As noted above, voltage and current feedback data may be used as input to an algorithm for pre-distorting or modifying LUT waveform samples on a dynamic and ongoing basis. In certain forms, this may require each stored voltage and current feedback data pair to be indexed based on, or otherwise associated with, a corresponding LUT sample that was output by the logic device **816** when the voltage and current feedback data pair was acquired. Synchronization of the LUT samples and the voltage and current feedback data in this manner contributes to the correct timing and stability of the pre-distortion algorithm.

In certain forms, the voltage and current feedback data may be used to control the frequency and/or amplitude (e.g., current amplitude) of the drive signals. In one form, for example, voltage and current feedback data may be used to determine impedance phase. The frequency of the drive

signal may then be controlled to minimize or reduce the difference between the determined impedance phase and an impedance phase setpoint (e.g., 0°), thereby minimizing or reducing the effects of harmonic distortion and correspondingly enhancing impedance phase measurement accuracy. The determination of phase impedance and a frequency control signal may be implemented in the DSP processor **822**, for example, with the frequency control signal being supplied as input to a DDS control algorithm implemented by the logic device **816**.

In another form, for example, the current feedback data may be monitored in order to maintain the current amplitude of the drive signal at a current amplitude setpoint. The current amplitude setpoint may be specified directly or determined indirectly based on specified voltage amplitude and power setpoints. In certain forms, control of the current amplitude may be implemented by control algorithm, such as, for example, a proportional-integral-derivative (PID) control algorithm, in the DSP processor **822**. Variables controlled by the control algorithm to suitably control the current amplitude of the drive signal may include, for example, the scaling of the LUT waveform samples stored in the logic device **816** and/or the full-scale output voltage of the DAC circuit **818** (which supplies the input to the power amplifier **812**) via a DAC circuit **834**.

The non-isolated stage **804** may further comprise a second processor **836** for providing, among other things user interface (UI) functionality. In one form, the UI processor **836** may comprise an Atmel AT91SAM9263 processor having an ARM 926EJ-S core, available from Atmel Corporation, San Jose, California, for example. Examples of UI functionality supported by the UI processor **836** may include audible and visual user feedback, communication with peripheral devices (e.g., via a USB interface), communication with a foot switch, communication with an input device (e.g., a touch screen display) and communication with an output device (e.g., a speaker). The UI processor **836** may communicate with the DSP processor **822** and the logic device **816** (e.g., via SPI buses). Although the UI processor **836** may primarily support UI functionality, it may also coordinate with the DSP processor **822** to implement hazard mitigation in certain forms. For example, the UI processor **836** may be programmed to monitor various aspects of user input and/or other inputs (e.g., touch screen inputs, foot switch inputs, temperature sensor inputs) and may disable the drive output of the generator **800** when an erroneous condition is detected.

In certain forms, both the DSP processor **822** and the UI processor **836**, for example, may determine and monitor the operating state of the generator **800**. For the DSP processor **822**, the operating state of the generator **800** may dictate, for example, which control and/or diagnostic processes are implemented by the DSP processor **822**. For the UI processor **836**, the operating state of the generator **800** may dictate, for example, which elements of a UI (e.g., display screens, sounds) are presented to a user. The respective DSP and UI processors **822**, **836** may independently maintain the current operating state of the generator **800** and recognize and evaluate possible transitions out of the current operating state. The DSP processor **822** may function as the master in this relationship and determine when transitions between operating states are to occur. The UI processor **836** may be aware of valid transitions between operating states and may confirm if a particular transition is appropriate. For example, when the DSP processor **822** instructs the UI processor **836** to transition to a specific state, the UI processor **836** may verify that requested transition is valid. In the event that a

requested transition between states is determined to be invalid by the UI processor **836**, the UI processor **836** may cause the generator **800** to enter a failure mode.

The non-isolated stage **804** may further comprise a controller **838** for monitoring input devices (e.g., a capacitive touch sensor used for turning the generator **800** on and off, a capacitive touch screen). In certain forms, the controller **838** may comprise at least one processor and/or other controller device in communication with the UI processor **836**. In one form, for example, the controller **838** may comprise a processor (e.g., a Meg168 8-bit controller available from Atmel) configured to monitor user input provided via one or more capacitive touch sensors. In one form, the controller **838** may comprise a touch screen controller (e.g., a QT5480 touch screen controller available from Atmel) to control and manage the acquisition of touch data from a capacitive touch screen.

In certain forms, when the generator **800** is in a “power off” state, the controller **838** may continue to receive operating power (e.g., via a line from a power supply of the generator **800**, such as the power supply **854** discussed below). In this way, the controller **838** may continue to monitor an input device (e.g., a capacitive touch sensor located on a front panel of the generator **800**) for turning the generator **800** on and off. When the generator **800** is in the power off state, the controller **838** may wake the power supply (e.g., enable operation of one or more DC/DC voltage converters **856** of the power supply **854**) if activation of the “on/off” input device by a user is detected. The controller **838** may therefore initiate a sequence for transitioning the generator **800** to a “power on” state. Conversely, the controller **838** may initiate a sequence for transitioning the generator **800** to the power off state if activation of the “on/off” input device is detected when the generator **800** is in the power on state. In certain forms, for example, the controller **838** may report activation of the “on/off” input device to the UI processor **836**, which in turn implements the necessary process sequence for transitioning the generator **800** to the power off state. In such forms, the controller **838** may have no independent ability for causing the removal of power from the generator **800** after its power on state has been established.

In certain forms, the controller **838** may cause the generator **800** to provide audible or other sensory feedback for alerting the user that a power on or power off sequence has been initiated. Such an alert may be provided at the beginning of a power on or power off sequence and prior to the commencement of other processes associated with the sequence.

In certain forms, the isolated stage **802** may comprise an instrument interface circuit **840** to, for example, provide a communication interface between a control circuit of a surgical instrument (e.g., a control circuit comprising handpiece switches) and components of the non-isolated stage **804**, such as, for example, the logic device **816**, the DSP processor **822**, and/or the UI processor **836**. The instrument interface circuit **840** may exchange information with components of the non-isolated stage **804** via a communication link that maintains a suitable degree of electrical isolation between the isolated and non-isolated stages **802**, **804**, such as, for example, an IR-based communication link. Power may be supplied to the instrument interface circuit **840** using, for example, a low-dropout voltage regulator powered by an isolation transformer driven from the non-isolated stage **804**.

In one form, the instrument interface circuit **840** may comprise a logic circuit **842** (e.g., logic circuit, program-

mable logic circuit, PGA, FPGA, PLD) in communication with a signal conditioning circuit **844**. The signal conditioning circuit **844** may be configured to receive a periodic signal from the logic circuit **842** (e.g., a 2 kHz square wave) to generate a bipolar interrogation signal having an identical frequency. The interrogation signal may be generated, for example, using a bipolar current source fed by a differential amplifier. The interrogation signal may be communicated to a surgical instrument control circuit (e.g., by using a conductive pair in a cable that connects the generator **800** to the surgical instrument) and monitored to determine a state or configuration of the control circuit. The control circuit may comprise a number of switches, resistors, and/or diodes to modify one or more characteristics (e.g., amplitude, rectification) of the interrogation signal such that a state or configuration of the control circuit is uniquely discernable based on the one or more characteristics. In one form, for example, the signal conditioning circuit **844** may comprise an ADC circuit for generating samples of a voltage signal appearing across inputs of the control circuit resulting from passage of interrogation signal therethrough. The logic circuit **842** (or a component of the non-isolated stage **804**) may then determine the state or configuration of the control circuit based on the ADC circuit samples.

In certain forms, the first data circuit may store information pertaining to the particular surgical instrument with which it is associated. Such information may include, for example, a model number, a serial number, a number of operations in which the surgical instrument has been used, and/or any other type of information. This information may be read by the instrument interface circuit **840** (e.g., by the logic circuit **842**), transferred to a component of the non-isolated stage **804** (e.g., to logic device **816**, DSP processor **822**, and/or UI processor **836**) for presentation to a user via an output device and/or for controlling a function or operation of the generator **800**. Additionally, any type of information may be communicated to the first data circuit for storage therein via the first data circuit interface **846** (e.g., using the logic circuit **842**). Such information may comprise, for example, an updated number of operations in which the surgical instrument has been used and/or dates and/or times of its usage.

As discussed previously, a surgical instrument may be detachable from a handpiece (e.g., the multifunction surgical instrument may be detachable from the handpiece) to promote instrument interchangeability and/or disposability. In such cases, conventional generators may be limited in their ability to recognize particular instrument configurations being used and to optimize control and diagnostic processes accordingly. The addition of readable data circuits to surgical instruments to address this issue is problematic from a compatibility standpoint, however. For example, designing a surgical instrument to remain backwardly compatible with generators that lack the requisite data reading functionality may be impractical due to, for example, differing signal schemes, design complexity, and cost. Forms of instruments discussed herein address these concerns by using data circuits that may be implemented in existing surgical instruments economically and with minimal design changes to preserve compatibility of the surgical instruments with current generator platforms.

Additionally, forms of the generator **800** may enable communication with instrument-based data circuits. For example, the generator **800** may be configured to communicate with a second data circuit contained in an instrument (e.g., the multifunction surgical instrument). In some forms, the second data circuit may be implemented in a many

similar to that of the first data circuit described herein. The instrument interface circuit **840** may comprise a second data circuit interface **848** to enable this communication. In one form, the second data circuit interface **848** may comprise a tri-state digital interface, although other interfaces may also be used. In certain forms, the second data circuit may generally be any circuit for transmitting and/or receiving data. In one form, for example, the second data circuit may store information pertaining to the particular surgical instrument with which it is associated. Such information may include, for example, a model number, a serial number, a number of operations in which the surgical instrument has been used, and/or any other type of information.

In some forms, the second data circuit may store information about the electrical and/or ultrasonic properties of an associated ultrasonic transducer, end effector, or ultrasonic drive system. For example, the first data circuit may indicate a burn-in frequency slope, as described herein. Additionally or alternatively, any type of information may be communicated to second data circuit for storage therein via the second data circuit interface **848** (e.g., using the logic circuit **842**). Such information may comprise, for example, an updated number of operations in which the instrument has been used and/or dates and/or times of its usage. In certain forms, the second data circuit may transmit data acquired by one or more sensors (e.g., an instrument-based temperature sensor). In certain forms, the second data circuit may receive data from the generator **800** and provide an indication to a user (e.g., a light emitting diode indication or other visible indication) based on the received data.

In certain forms, the second data circuit and the second data circuit interface **848** may be configured such that communication between the logic circuit **842** and the second data circuit can be effected without the need to provide additional conductors for this purpose (e.g., dedicated conductors of a cable connecting a handpiece to the generator **800**). In one form, for example, information may be communicated to and from the second data circuit using a one-wire bus communication scheme implemented on existing cabling, such as one of the conductors used transmit interrogation signals from the signal conditioning circuit **844** to a control circuit in a handpiece. In this way, design changes or modifications to the surgical instrument that might otherwise be necessary are minimized or reduced. Moreover, because different types of communications implemented over a common physical channel can be frequency-band separated, the presence of a second data circuit may be “invisible” to generators that do not have the requisite data reading functionality, thus enabling backward compatibility of the surgical instrument.

In certain forms, the isolated stage **802** may comprise at least one blocking capacitor **850-1** connected to the drive signal output **810b** to prevent passage of DC current to a patient. A single blocking capacitor may be required to comply with medical regulations or standards, for example. While failure in single-capacitor designs is relatively uncommon, such failure may nonetheless have negative consequences. In one form, a second blocking capacitor **850-2** may be provided in series with the blocking capacitor **850-1**, with current leakage from a point between the blocking capacitors **850-1**, **850-2** being monitored by, for example, an ADC circuit **852** for sampling a voltage induced by leakage current. The samples may be received by the logic circuit **842**, for example. Based changes in the leakage current (as indicated by the voltage samples), the generator **800** may determine when at least one of the blocking

capacitors **850-1**, **850-2** has failed, thus providing a benefit over single-capacitor designs having a single point of failure.

In certain forms, the non-isolated stage **804** may comprise a power supply **854** for delivering DC power at a suitable voltage and current. The power supply may comprise, for example, a 400 W power supply for delivering a 48 VDC system voltage. The power supply **854** may further comprise one or more DC/DC voltage converters **856** for receiving the output of the power supply to generate DC outputs at the voltages and currents required by the various components of the generator **800**. As discussed above in connection with the controller **838**, one or more of the DC/DC voltage converters **856** may receive an input from the controller **838** when activation of the “on/off” input device by a user is detected by the controller **838** to enable operation of, or wake, the DC/DC voltage converters **856**.

FIG. **166** illustrates an example of a generator **900**, which is one form of the generator **800** (FIG. **165**). The generator **900** is configured to deliver multiple energy modalities to a surgical instrument. The generator **900** provides RF and ultrasonic signals for delivering energy to a surgical instrument either independently or simultaneously. The RF and ultrasonic signals may be provided alone or in combination and may be provided simultaneously. As noted above, at least one generator output can deliver multiple energy modalities (e.g., ultrasonic, bipolar or monopolar RF, irreversible and/or reversible electroporation, and/or microwave energy, among others) through a single port, and these signals can be delivered separately or simultaneously to the end effector to treat tissue.

The generator **900** comprises a processor **902** coupled to a waveform generator **904**. The processor **902** and waveform generator **904** are configured to generate a variety of signal waveforms based on information stored in a memory coupled to the processor **902**, not shown for clarity of disclosure. The digital information associated with a waveform is provided to the waveform generator **904** which includes one or more DAC circuits to convert the digital input into an analog output. The analog output is fed to an amplifier **1106** for signal conditioning and amplification. The conditioned and amplified output of the amplifier **906** is coupled to a power transformer **908**. The signals are coupled across the power transformer **908** to the secondary side, which is in the patient isolation side. A first signal of a first energy modality is provided to the surgical instrument between the terminals labeled ENERGY1 and RETURN. A second signal of a second energy modality is coupled across a capacitor **910** and is provided to the surgical instrument between the terminals labeled ENERGY2 and RETURN. It will be appreciated that more than two energy modalities may be output and thus the subscript “n” may be used to designate that up to n ENERGYn terminals may be provided, where n is a positive integer greater than 1. It also will be appreciated that up to “n” return paths RETURNn may be provided without departing from the scope of the present disclosure.

A first voltage sensing circuit **912** is coupled across the terminals labeled ENERGY1 and the RETURN path to measure the output voltage therebetween. A second voltage sensing circuit **924** is coupled across the terminals labeled ENERGY2 and the RETURN path to measure the output voltage therebetween. A current sensing circuit **914** is disposed in series with the RETURN leg of the secondary side of the power transformer **908** as shown to measure the output current for either energy modality. If different return paths are provided for each energy modality, then a separate

current sensing circuit should be provided in each return leg. The outputs of the first and second voltage sensing circuits **912**, **924** are provided to respective isolation transformers **916**, **922** and the output of the current sensing circuit **914** is provided to another isolation transformer **918**. The outputs of the isolation transformers **916**, **928**, **922** in the on the primary side of the power transformer **908** (non-patient isolated side) are provided to a one or more ADC circuit **926**. The digitized output of the ADC circuit **926** is provided to the processor **902** for further processing and computation. The output voltages and output current feedback information can be employed to adjust the output voltage and current provided to the surgical instrument and to compute output impedance, among other parameters. Input/output communications between the processor **902** and patient isolated circuits is provided through an interface circuit **920**. Sensors also may be in electrical communication with the processor **902** by way of the interface circuit **920**.

In one aspect, the impedance may be determined by the processor **902** by dividing the output of either the first voltage sensing circuit **912** coupled across the terminals labeled ENERGY1/RETURN or the second voltage sensing circuit **924** coupled across the terminals labeled ENERGY2/RETURN by the output of the current sensing circuit **914** disposed in series with the RETURN leg of the secondary side of the power transformer **908**. The outputs of the first and second voltage sensing circuits **912**, **924** are provided to separate isolations transformers **916**, **922** and the output of the current sensing circuit **914** is provided to another isolation transformer **916**. The digitized voltage and current sensing measurements from the ADC circuit **926** are provided the processor **902** for computing impedance. As an example, the first energy modality ENERGY1 may be ultrasonic energy and the second energy modality ENERGY2 may be RF energy. Nevertheless, in addition to ultrasonic and bipolar or monopolar RF energy modalities, other energy modalities include irreversible and/or reversible electroporation and/or microwave energy, among others. Also, although the example illustrated in FIG. **166** shows a single return path RETURN may be provided for two or more energy modalities, in other aspects, multiple return paths RETURNn may be provided for each energy modality ENERGYn. Thus, as described herein, the ultrasonic transducer impedance may be measured by dividing the output of the first voltage sensing circuit **912** by the current sensing circuit **914** and the tissue impedance may be measured by dividing the output of the second voltage sensing circuit **924** by the current sensing circuit **914**.

As shown in FIG. **166**, the generator **900** comprising at least one output port can include a power transformer **908** with a single output and with multiple taps to provide power in the form of one or more energy modalities, such as ultrasonic, bipolar or monopolar RF, irreversible and/or reversible electroporation, and/or microwave energy, among others, for example, to the end effector depending on the type of treatment of tissue being performed. For example, the generator **900** can deliver energy with higher voltage and lower current to drive an ultrasonic transducer, with lower voltage and higher current to drive RF electrodes for sealing tissue, or with a coagulation waveform for spot coagulation using either monopolar or bipolar RF electrosurgical electrodes. The output waveform from the generator **900** can be steered, switched, or filtered to provide the frequency to the end effector of the surgical instrument. The connection of an ultrasonic transducer to the generator **900** output would be preferably located between the output labeled ENERGY1 and RETURN as shown in FIG. **166**. In one example, a

connection of RF bipolar electrodes to the generator **900** output would be preferably located between the output labeled ENERGY2 and RETURN. In the case of monopolar output, the preferred connections would be active electrode (e.g., pencil or other probe) to the ENERGY2 output and a suitable return pad connected to the RETURN output.

Additional details are disclosed in U.S. Patent Application Publication No. 2017/0086914, titled TECHNIQUES FOR OPERATING GENERATOR FOR DIGITALLY GENERATING ELECTRICAL SIGNAL WAVEFORMS AND SURGICAL INSTRUMENTS, which published on Mar. 30, 2017, which is herein incorporated by reference in its entirety.

Referring to FIGS. **168-170**, a cartridge **60130** is similar in many respects to the cartridge **60030**. For example, the cartridge **60130** can also be utilized with the surgical instrument **60000** to seal and staple tissue. Also, the cartridge **60130** includes rows of staple cavities **60131**, **60132** extending on opposite sides of a longitudinal slot **60135** defined in a cartridge body **60139**, and housing staples **60133**. The cartridge **60130** also includes a third electrode assembly **60136** coupled to the cartridge body **60139**. In the illustrated example, the third electrode assembly **60136** includes segmented electrodes **60136a-f** and a flex circuit **60141** extending longitudinally behind the segmented electrodes, and configured to connect the segmented electrodes to the RF energy source **794**. The flex circuit **60141** is positioned against the cartridge deck **60147**. Sandwiched between the segmented electrodes and the flex circuit **60141** are passive switches, current limiting elements, energy-sensitive resistance elements, or locally-adjustable resistance elements, which can be in the form of positive temperature coefficient (PTC) segments **60142**.

Further to the above, the third electrode assembly **60136** is configured to cover exposed cavities **60134** in the cartridge body **60139**, wherein the exposed cavities **60134** are configured to accommodate driver supports **60149a** of staple drivers **60149**. The driver supports **60149a** are configured to resist driver roll. The exposed cavities **60134** permit the driver support **60149a** to be extended above the cartridge deck **60147** to resist driver roll. Like the cartridge **60030**, the cartridge **60130** includes a longitudinal step **60137**—similar to the longitudinal stop **60037**, which are not repeated herein for brevity. The longitudinal step **60137** is configured to cover exposed cavities **60134'** in the cartridge body **60139**, wherein the exposed cavities **60134'** are configured to accommodate driver supports **60149a'** of staple drivers **60149'**. The driver supports **60149a'** are configured to resist driver roll. The exposed cavities **60134'** permit the driver supports **60149a'** to be extended above the cartridge deck **60147** to resist driver roll. Additional details about driver supports are disclosed elsewhere in the present disclosure, and are not repeated herein for brevity.

Securing members **60145a-c** protrude from a cartridge deck **60147** of the cartridge **60130**. The securing members **60145a-c** are configured to lockingly engage the third electrode assembly **60136**. In the illustrated example, the securing members **60145a-c** define right-angled brackets that are configured to matingly engage portions of the third electrode assembly **60136**. During assembly, insulative segments of the third electrode assembly **60136** are configured to snap into a locking engagement with the securing members **60145a-c**.

FIG. **171** illustrates a cross-sectional view of the anvil **60020** of FIG. **162**. In the illustrated example, the electrode assemblies **60026**, **60027** include segmented electrodes **60026a-c** and **60027a-c**, respectively. Further, Flex circuits

60056, **60058** extend longitudinally between the segmented electrodes **60026a-c** and **60027a-c**, respectively, and an anvil deck **60057**. The flex circuits **60056**, **60058** are configured to connect the segmented electrodes to the RF energy source **794**, as illustrated in FIG. 173.

Sandwiched between the segmented electrodes and the flex circuits **60056**, **60058** are passive switches, current limiting elements, energy-sensitive resistance elements, or locally-adjustable resistance elements, which can be in the form of positive temperature coefficient (PTC) segments **60053a-c**, **60054a-c**. In the illustrated example, the segmented electrodes **60026a-c**, **60027a-c** of the electrode assemblies **60026**, **60027** are separately connected in series with corresponding PTC segments **60053a-c**, **60054a-c**, respectively, as illustrated in FIG. 173. In other words, there are an equal number of segmented electrodes and PTC segments. In other examples, however, two or more segmented electrodes can be connected to one PTC segment.

FIG. 172 illustrates an anvil **60120** similar in many respects to the anvil **60020**. For example, the anvil **60120** can also be utilized with the surgical instrument **60000** to seal and staple tissue. Also, the anvil **60120** includes rows of staple pockets **60121**, **60122** defined in an anvil deck **60157**, and electrode assemblies **60126**, **60127** extending on opposite sides of a longitudinal slot **60125**. In the illustrated example, the electrode assemblies **60126**, **60127** include segmented electrodes **60126a-c** and **60127a-c**, respectively. Furthermore, flex circuits **60156**, **60158** extend longitudinally behind the segmented electrodes **60126a-c** and **60127a-c**, respectively, and are configured to connect the segmented electrodes **60126a-c** and **60127a-c** to the RF energy source **794**.

Further, the electrode assemblies **60126**, **60127** include passive switches, current limiting elements, energy-sensitive resistance elements, or locally-adjustable resistance elements, which can be in the form of positive temperature coefficient (PTC) segments **60153a-c**, **60154a-c**. In the illustrated example, the segmented electrodes **60126a-c** and **60127a-c** are connected in series to the PTC segments **60153a-c**, **60154a-c**, respectively, but the PTC segments **60153a-c**, **60154a-c** are not located directly behind the segmented electrodes **60126a-c** and **60127a-c**. Instead, the PTC segments **60153a-c**, **60154a-c** are disposed at a proximal portion of the anvil **60120**. The flex Circuits **60156**, **60158** extend between the PTC segments **60153a-c**, **60154a-c** and the segmented electrodes **60126a-c** and **60127a-c**. In the illustrated example, each of segmented electrodes **60126a-c** and **60127a-c** is connected to a dedicated PTC segment. In other, examples, however, two or more segmented electrodes can share a PTC segment.

Referring still to FIG. 172, PTC segments **60153a-c** and the PTC segments **60154a-c** are arranged on opposite sides of the longitudinal slot **60125** at a proximal portion of the anvil **60120**. In the illustrated example, the PTC segments **60153a-c** and **60154a-c** are coupled to the anvil **60120** proximal to the rows of staple pockets **60121**, **60122**. In addition, the PTC segments **60153a-c** and **60154a-c** are arranged in two rows. Other arrangements are contemplated by the present disclosure.

FIG. 173 is an electrical diagram illustrating a simplified electrical layout of the electrode assemblies **60036**, **60026**, **60027**, in accordance with at least one aspect of the present disclosure. In the illustrated example, the segmented electrodes **60036a-f**, **60026a-c**, **60027a-c** are separately connected to passive switches, current limiting elements, energy-sensitive resistance elements, or locally-adjustable

resistance elements, which can be in the form of positive temperature coefficient (PTC) segments **60042a-f**, **60053a-c**, **60054a-c**, respectively.

The passive switches, current limiting elements, energy-sensitive resistance elements, or locally-adjustable resistance elements, which can be in the form of positive temperature coefficient (PTC) segments **60042a-f**, **60053a-c**, **60054a-c**, are configured to adaptively and independently control current through their respective segmented electrodes **60036a-f**, **60026a-c**, **60027a-c**. In certain instances, the passive switches, current limiting elements, energy-sensitive resistance elements, or locally-adjustable resistance elements, which can be in the form of positive temperature coefficient (PTC) segments **60042a-f**, **60053a-c**, **60054a-c**, are configured to passively and independently deactivate, or reduce, energy flow through their respective segmented electrodes **60036a-f**, **60026a-c**, **60027a-c**, in response to a short circuit, for example.

In the example illustrated in FIG. 173, each of the electrode assemblies **60026**, **60027**, **60036** includes PTC segments. In other examples, however, PTC segments can be limited to any one or two of the electrode assemblies **60026**, **60027**, **60036**. As described in greater detail below, the PTC segments utilized with the electrode assemblies **60026**, **60027**, **60036** can be different in one or more aspects. In certain examples, the PTC segments can be different in transition temperatures, material composition, and/or response to short circuits.

FIG. 174 is an electrical diagram illustrating an electrical layout of an alternative electrode assembly **60060** that can be implemented with one or more of the electrode assemblies **60036**, **60026**, **60027**. In the illustrated example, the electrode assembly **60060** includes segmented electrodes **60060a-c**. In other examples, however, the electrode assembly **60060** may include more, or less, than three segmented electrodes. In any event, the segmented electrodes **60060a-c** are commonly connected to a passive switch, current limiting element, energy-sensitive resistance element, or locally-adjustable resistance element, which can be in the form of a positive temperature coefficient (PTC) segment **60061**. In such example, the PTC segment **60061** is configured to adaptively and independently control current through segmented electrodes **60060a-c**. In certain instances, the PTC segment **60061** is configured to passively and independently deactivate energy flow through the segmented electrodes **60060a-c** in response to a short circuit, for example.

Referring primarily to FIGS. 163 and 173, the control circuit **760** may cause the RF energy source **794** to adaptively alternate, or switch, between an opposing bipolar energy mode and an offset bipolar energy mode. In the opposing bipolar energy mode the control circuit **760** is configured to cause the RF energy source **794** to pass a first therapeutic signal between the first electrode assembly **60026** and the third electrode assembly **60036**. In the offset bipolar energy mode, the control circuit **760** is configured to cause the RF energy source **794** to pass a second therapeutic signal between the second electrode assembly **60027** and the third electrode assembly **60036**.

Further to the above, in the opposing energy mode, a current pathway can be defined through the PTC segment **60053a**, the segmented electrode **60026a**, the tissue (T) (between the anvil **60020** and the staple cartridge **60030**), the segmented electrode **60036a**, and the PTC segment **60042a**, for example. When the opposing energy mode is switched to the offset energy mode, the current pathway is also switched. For example, in the offset energy mode, the current pathway can be defined through the PTC segment

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60054a, the segmented electrode 60024a, the tissue (T), the segmented electrode 60036a, and the PTC segment 60042a.

Furthermore, depending on the size of the segmented electrodes and circuit polarity, various other current pathways can be established. For example, current pathways can be established between the segmented electrode 60026a and the segmented electrodes 60036a, 60036b.

FIG. 175 is a cross-sectional view of an end effector 60002' similar in many respects to the end effector 60002, which are not repeated herein for brevity. For example, the end effector 60002' can also be used with the surgical instrument 6000 in a similar manner to the end effector 60002. Various components of the end effector 60002', which are similar to the end effector 60002 are removed for clarity. Unlike the end effector 60002, the end effector 60002' includes an anvil 60020' that lacks the PTC segments 60054a-f, 60053a-f.

In the illustrated example, tissue (T) is captured between an anvil 60020' and the cartridge 60030'. The control circuit 760 then activates the electrode assemblies 60026, 60036 to apply a tissue treatment cycle to the tissue (T) utilizing an opposing bipolar energy mode. In the illustrated example, the RF energy source 794 causes current to flow from the electrode assembly 60026 to the electrode assembly 60036. Accordingly, the segmented electrodes 60026b, 60026c define source electrodes and the segmented electrodes 60036d, 60036e, 60036f define return electrodes. In other examples, the RF energy source 794 may reverse the circuit polarity such that the segmented electrodes 60026b, 60026c define return electrodes and the segmented electrodes 60036d, 60036e, 60036f define source electrodes.

In any event, the tissue (T) includes a staple 60033 previously-deployed into the tissue. The presence of the staple 60033 causes a short circuit to form between the segmented electrodes 60026c, 60036c. As described in greater detail below, the short circuit causes the resistance of the PTC segment 60042e to increase (e.g. from 5 Ω to 20 Ω) thereby effectively deactivating the energy flow between the segmented electrodes 60026c and 60036e, or at least reducing the energy flow to a sub-therapeutic level.

Accordingly, the PTC segment 60042e is configured to passively and independently control the current flow through the tissue (T). In certain instances, the PTC segment 60042e is configured to passively deactivate the segmented electrode 60036e without processor-based communication or control, in response to a short circuit between the segmented electrodes 60026c, 60036e. In the illustrated example, current is automatically diverted to adjacent segmented electrode 60036d, 60036f, as the resistances of the PTC segments 60042d, 60042f have not been affected by the short circuit.

PTC segments, in accordance with at least one aspect of the present disclosure, generally comprise a low-resistance condition at a temperature during a normal operation. However, on exposure to high temperature due to, for example, unusually large current resulting from the formation of a short circuit or excessive discharge, the PTC segments switch into an extremely high-resistance mode. Simply put, when the PTC segments are included in a circuit and an abnormal current passes through the circuit, such as in the instance of a short circuit caused by a staple 60033, the resulting higher temperature condition switches the PTC segments to a higher resistance condition to decrease the current passing through the circuit to a minimal level and, thus, protect electric elements of the circuit.

FIG. 176 is a graph 60060 illustrating the change in resistance (Ω) of a PTC segment in response to a change in

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temperature ($^{\circ}$ C.), in accordance with the at least one aspect of the present disclosure. In the illustrated example, the PTC segment comprises a transition temperature T_s , also referred to elsewhere herein as a switching temperature or a threshold temperature. In operation, a short circuit, possibly caused by the presence of a previously-fired staple 60033 in the tissue (T), may cause an increase in the temperature of the PTC segment. At the transition temperature T_s , the resistance (Ω) of the PTC segment increases significantly, effectively deactivating the segmented electrode affected by the short circuit, as described supra. Accordingly, the PTC segment acts as a resettable fuse for overcurrent protection.

FIG. 177 is another graph 60065 illustrating the change in resistance (Ω) of a PTC segment in response to a change in temperature ($^{\circ}$ C.), in accordance with the at least one aspect of the present disclosure. The resistance (Ω) is shown on a logarithmic scale. The resistance (Ω) of the PTC segment is generally maintained at a steady state during normal operation, but the resistance (Ω) begins to increase exponentially at the transition temperature T_s . As the temperature rises toward the transition temperature T_s , the resistance (Ω) increases slightly from a minimum resistance R_{min} to a higher resistance (e.g. double R_{min}) at the transition temperature T_s . However, beyond the transition temperature T_s , the increase in the resistance (Ω) is exponential, effectively resulting in a deactivation of the current through the PTC segment.

Referring again to FIG. 175, the PTC segments 60042a-f can be configured to locally sense shorted electrical pathways, where overlap with previously-fired staples occurs, and deactivate affected segmented electrodes 60036a-f, 60026a-c, 60027a-c, based on the same principles discussed in connection with FIGS. 176 and 177. In the exemplification illustrated in FIG. 175, the segmented electrodes 60036a-f are commonly connected to the RF energy source 794 by individual connections originating from individual bifurcations of a common connection. As such, deactivation of the segmented electrode 60036e by the PTC segment 60042e does not stop current flow to the other segmented electrodes in the electrode assembly 60036. Instead, the current from the segmented electrode 60026c is automatically diverted from the segmented electrode 60036e, and away from the staple 60033, toward the segmented electrodes 60036f, 60036d. In result, the segmented electrode 60036e is passively and independently deactivated in response to the short circuit without processor based communication or control, while the remaining segmented electrodes of the third electrode assembly 60036, which are not affected by the short circuit, uninterruptedly continue to deliver energy to the tissue to seal the tissue.

FIG. 178 is a graph 60070 depicting passive and independent control of a current through a tissue portion including a staple 60033 between the electrode assemblies 60026, 60036 (e.g. between the segmented electrodes 60026c, 60036f) during a tissue treatment cycle employing an opposing energy mode, in accordance with at least one aspect of the present disclosure. In the illustrated example, PTC segments 60042c, 60053f are configured to perform a passive and independent control of the current during the tissue treatment cycle. The Graph 60070 includes multiple graphs depicting time (t) on the X-axis vs source electrode (e.g. 60026c) active status 60071, return electrode (e.g. segmented electrode 60036f) active status 60072, power level 60073, tissue impedance 60074, and resistance 60075 of at least one of the PTC segments 60042c, 40053f, on the Y-axis.

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In the illustrated example, the control circuit **760** is configured to execute a tissue treatment cycle including a sub-therapeutic signal **60077a** and a therapeutic signal **60077b** applied to a tissue grasped by the end effector **60002**. The control circuit **760** is configured to cause the RF energy source **794** to activate **70076** the return electrode, and then activate **70077** the source electrode. In certain instances, the RF energy source **794** may include one or more switching mechanisms for transitioning one or more of the segmented electrodes of the electrode assemblies **60026**, **60027**, **60036** between active and inactive modes, for example.

Further to the above, during application of the sub-therapeutic signal **60077a**, the power level is maintained at a first level **60078a** too low to effect a significant change (first portion **70079a** of the resistance line) in the resistance of the PTC segment(s) in the presence of the staple **60033**. However, during application of the therapeutic signal **60077b**, the power level is increased to a second level **60078b**, which begins to increase (second portion **70079b** of the resistance line) the resistance of the PTC segment(s) due to an increase in temperature caused by the short circuit created by the staple **60033**. In certain instances, the change in the resistance of the PTC segment(s) can be detected by the control circuit **760** by monitoring current and voltage parameters. In certain instances, if the change in resistance is greater than or equal to a predetermined threshold, the control circuit **760** concludes the presence of a short circuit, which can be reported to a user through the display **711**, for example. Furthermore, in certain instances, the control circuit **760** may be further configured to shut down **60080** power delivery to the end effector **60002** at this point.

If power delivery continues, the temperature of the PTC segment(s) will eventually reach the transition temperature T_s of the PTC segment(s). At such point, the resistance of the PTC segment(s) increases (third portion **60079c** of the resistance line) exponentially, effectively deactivating **60081**, **60081** the source and return electrodes.

In various instances, PTC segments in accordance with at least one aspect of the present disclosure are ceramic PTC segments, with resistance-temperature characteristics that are attributed to the electronic properties of ceramic grain boundaries. In certain aspects, one or more of the PTC segments **60042a-f**, **60053a-c**, **60054a-c** can be selected to operate as a quick-trip fuse or a slow-trip fuse in response to a short circuit, such as one caused by a previously fired staple **60033**, based on their temperature-resistance characteristics.

FIG. **179** is a graph **60090** illustrating a PTC segment's trip response at different temperatures. The Y-axis represents current through the PTC segment, and the X-axis represents temperature of the PTC segment. In the illustrated example, a line **60091** is a hold-current line, which represents a maximum value of hold current at operating temperature. Hold current is a maximum current value which can be flowed in normal operation. Further, a line **60092** is a trip-current line, which represents a minimum current value which is necessary for PTC segment to move to high-resistance state. Hold current and trip current have temperature dependence which features decreasing current value with increasing temperature. The lines **60091**, **60092** define three distinct regions. A first region **60090a** identifies where the PTC segment operates as a quick-trip fuse, a second region **60090b** identifies where the PTC segment operates at a low/normal resistance, and a third region **60090c** identifies where the PTC segment operates as a slow-trip fuse.

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Accordingly, one or more of the PTC segments **60042a-f**, **60053a-c**, **60054a-c** can be selected to operate as a quick-trip fuse or a slow-trip fuse based on resistance-temperature characteristics. In certain instances, the slow-trip PTC segments comprise a higher transition temperature T_s than the fast-trip PTC segments. In one example, one or more of the PTC segments **60042a-f** can be selected to operate as a quick-trip fuse, and one or more of the PTC segments **60053a-c**, **60054a-c** can be selected to operate as a slow-trip fuse, in response to the short circuit. In at least one example, one or more of the PTC segments **60042a-f** may include a first transition temperature T_s , and one or more of the PTC segments **60053a-c**, **60054a-c** may include a second transition temperature T_s higher than the first transition temperature.

In certain instances, the quick-trip fuse characteristic of the PTC segments **60042a-f** ensure that current flow to the tissue is stopped during a short circuit. Meanwhile, the slow-trip fuse characteristic of the PTC segments **60053a-c**, **60054a-c** may still permit current to flow through the corresponding segmented electrodes **60026a-c**, **60027a-c**. In certain instances, the electrode assemblies **60026**, **60027** are configured such that, while the slow-trip fuse of the PTC segments **60053a-c**, **60054a-c** is triggered, each of the segmented electrodes **60026a-c**, **60027a-c** allows the current to flow back to the RF energy source **794**, or control electronics thereof (e.g. control circuit **760**), in an isolated manner, to allow the control electronics to detect which of the segmented electrodes **60026a-c**, **60027a-c** is associated with a PTC segment that has changed its resistance. The control circuit **760** may then alert a user, for example through the display **711**, of the portion of the end effector **60002** affected by the short circuit. Furthermore, the control circuit **760** may also adjust an ongoing tissue treatment cycle to address the detected short circuit.

FIG. **180** is a logic flow diagram of a process **60100** depicting a control program or a logic configuration for detecting and addressing a short circuit during a tissue treatment cycle applied to tissue grasped by the end effector **60002**. In certain instances, the process **60100** can be implemented by the surgical instrument **60000**, for example. In at least one example, the process **60100** can be executed by the control circuit **760**. In certain instances, the memory **68008** stores program instructions that, when executed by the processor **68002**, cause the processor **68002** to perform one or more aspects of the process **60100**.

The process **60100** includes monitoring **60101** a parameter indicative of a current returned from a source segmented electrode (e.g. the segmented electrodes **60026a-c**, **60027a-c**). The control circuit **760** may receive signals from a current sensor indicative of one or more current values of the returned current. The process **60100** detects **60106** a short circuit at the segmented electrode if **60103** a change in the monitored parameter is equal to, or beyond, a predetermined value. In certain instances, the predetermined value can be stored in a storage medium such as, for example, the memory **68008**. The process **60100** may further issue **60104** an alert and/or adjust **60105** at least one parameter of the tissue treatment cycle in response to the short circuit.

FIG. **181** is a logic flow diagram of a process **60110** depicting a control program or a logic configuration for a tissue treatment cycle applied to tissue grasped by the end effector **60002**, in accordance with at least one aspect of the present disclosure. In certain instances, the process **60110** can be implemented by the surgical instrument **60000**, for example. The process **60110** is similar in many respects to the process **60150**. For example, the process **60110** can also

be executed by the control circuit 760. In certain instances, the memory 68008 stores program instructions that, when executed by the processor 68002, cause the processor 68002 to perform one or more aspects of the process 60110.

The process 60110 includes monitoring 60111 a tissue parameter of a tissue grasped by the end effector 60002. In certain examples, the tissue parameter is a tissue compression. The control circuit 760 may monitor 60111 the tissue compression based on sensor signals from one or more sensors 788. If 60112 the tissue parameter indicates suitable energy sealing conditions, the process 60110 activates 60113 the offset energy mode, while the opposing energy mode remains inactive. To determine whether the tissue parameter is indicative of suitable energy sealing conditions, the control circuit 760 may, for example, compare detected values of the tissue parameter to a predetermined threshold indicative of suitable energy sealing conditions, which can be stored in a storage medium accessible by the processor 68002 such as, for example, the memory 68008.

In the offset energy mode, the control circuit 760 may activate the electrode assemblies 60026, 60036, while the electrode assembly 60027 remains inactive. The process 60110 further includes monitoring tissue impedance 60114 to determine when to switch from the offset energy mode to the opposing energy mode. As described elsewhere herein in greater detail, tissue impedance of a tissue portion can be detected, for example by a control circuit 760, by causing a sub-therapeutic signal to be passed through the tissue portion, receiving measurements from a voltage sensing circuit 924 and the current sensing circuit 914, and dividing the measurements from the voltage sensing circuit 924, by the corresponding measurements from the current sensing circuit 914, for example.

In the illustrated example, if 60114 a tissue impedance equal to, or beyond, a predetermined threshold is detected, the process 60110 switches 60115 from the opposing energy mode to the offset energy mode. To switch to the offset energy mode, the control circuit 760 may deactivate the electrode assembly 60026, and activate the electrode assembly 60027. In other instances, the offset energy mode is activated before activation of the opposing energy mode, and deactivated with, or after, activation of the opposing energy mode.

Further to the above, if 60116 a short circuit is detected, the process 60110 may issue an alert 60117, switch to the offset energy mode 60118, and/or deactivate 60119 the affected segmented electrodes of one or more of the electrode assemblies 60026, 60027, 60036, for example. A short circuit can be detected, as described in greater detail in connection with the process 60100, by monitoring a parameter indicative of a current returned from a source segmented electrode, for example.

In certain instances, the tissue is a thick tissue such as a liver tissue, and the end effector 60002 is configured to grasp the liver tissue and apply a tissue treatment cycle thereto in accordance with the process 60110. For example, the control circuit 760 may utilize the opposing energy modes to apply a feathering load technique, which causes the end effector 60002 to maintain a predetermined compression onto the grasped tissue at a constant, or substantially constant, value while applying the one of the offset and opposing energy modes to the grasped tissue. As the grasped tissue thins during welding, a short circuit may be detected 60116 by the control circuit 760. For example, the thinning tissue may expose a pre-existing metallic object (e.g. staple or clip) in the tissue, which may cause the short circuit. In response, the

control circuit 760 may switch to the offset energy mode 60118 to mitigate the short circuit.

In certain instances, the process 60150 and/or the process 60100 can be modified to begin with the offset energy mode instead of the opposing energy mode. In such instances, the process 60150 and/or the process 60100 may switch from the offset energy mode to the opposing energy mode to mitigate the short circuit. In certain instances, one of the offset and opposing energy modes can be utilized in an initial tissue warming portion of a tissue treatment cycle, while the other one of the offset and opposing energy modes can be utilized in a tissue welding portion of the tissue treatment cycle, which follows the tissue warming portion.

In various aspects, The control circuit 760 can be configured to cause the RF energy source 794 to adjust power levels in segmented electrodes that are kept in an active state following the detection of a short circuit. The adjustments may include increasing the power levels to compensate for the segmented electrodes deactivated in response to the short circuit.

In various aspects, the short circuit between one or more segmented electrodes of the electrode assemblies 60036, 60026, 60027 can be detected by incorporating temperature sensors such as, for example, thermocouples at, or near, the segmented electrodes to detect PTC transition temperatures T_s . A short circuit due to a pre-existing metallic object such as a staple 60033 between a segmented electrode 60026c and a segmented electrode 60036e, for example, causes a temperature increase in the PTC segment 60042e to, or beyond the PTC transition temperature T_s . The increase is detectable by the control circuit 760 based on signals generated by the temperature sensors. In response, the control circuit 760 may adjust one or more parameters of the RF energy source 794 to mitigate the short circuit.

In various aspects, a short circuit between segmented electrodes of the electrode assemblies 60036, 60026, 60027, due for example to a pre-existing metallic object in the tissue, abnormally changes an electric output of such segmented electrodes beyond the electric output expected during normal operating conditions. Further, the magnetic field induced by the electric output in the event of a short circuit is different than magnetic field induced during normal operations.

In various instances, the short circuit between segmented electrodes of the electrode assemblies 60036, 60026, 60027 such as, for example, segmented electrode 60026c, 60036e can be detected by incorporating magnetic sensors at, or near, the segmented electrodes 60026c, 60036e to measure a parameter of the magnetic field induced by the electric output of the segmented electrodes 60026c, 60036e. If a measured value of the parameter is equal to, or is beyond, a predetermined threshold, the control circuit 760 detects a short circuit between the segmented electrodes 60026c, 60036e. Accordingly, the control circuit 760 can be configured to detect the short circuit based on the signals from magnetic sensors configured to monitor the magnetic field induced by the electric output of segmented electrodes of the electrode assemblies 60036, 60026, 60027.

Referring to FIG. 182, a graph 60190 represents a power scheme 60191 for a tissue treatment cycle and corresponding tissue impedance 60192, in accordance with at least one aspect of the present disclosure. The power scheme 60191 can be implemented by the RF energy source 794, for example. In certain instances, the control circuit 760 is configured to cause the RF energy source 794 to apply an energy treatment cycle, in accordance with the power scheme 60191, to tissue grasped by the end effector 60002.

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In the illustrated example, the power scheme 60191 includes a first segment 60121a (between times t_0 and t_a), a second segment 60121b (between times t_a and t_b), and a third segment 60121c (between times t_b and t_c). In the first segment 60121a, the control circuit 760 is configured to cause the RF energy source 794 to apply a therapeutic energy to the tissue in an offset bipolar energy mode. The RF energy source 794 may activate the electrode assemblies 60027, 60036 to effect the offset bipolar energy mode. The electrode assembly 60026 remains inactive during the first segment 60121a. In at least one example, the electrode assembly 60027 is configured as a source electrode, while the electrode assembly 60036 is configured as a return electrode. In other examples, the RF energy source 794 may reverse the circuit polarity such that the electrode assembly 60036 becomes the source electrode, and the electrode assembly 60027 becomes the return electrode.

In the second segment 60121b, the control circuit 760 is configured to cause the RF energy source 794 to apply a therapeutic energy to the tissue in a hybrid mode using a combination of the opposing and offset bipolar energy modes. The RF energy source 794 may activate the electrode assembly 60026 to effect the opposing bipolar energy mode during the second segment 60121b. In the hybrid mode, the offset bipolar energy mode is gradually decreased while the opposing bipolar energy mode is gradually increased, as time transitions toward t_b . In other words, energy flow through the electrode assembly 60026 is gradually increased, while energy flow through the electrode assembly 60027 is gradually decreased.

In the third segment 60121c, the control circuit 760 is configured to cause the RF energy source 794 to apply a therapeutic energy to the tissue in an opposing bipolar energy mode. The RF energy source 794 may cause the electrode assemblies 60026, 60036 to effect the opposing bipolar energy mode. The electrode assembly 60027 is deactivated during the third segment 60121c. In at least one example, the electrode assembly 60026 is configured as a source electrode, while the electrode assembly 60036 is configured as a return electrode. In other examples, the RF energy source 794 may reverse the circuit polarity such that the electrode assembly 60036 becomes the source electrode, and the electrode assembly 60026 becomes the return electrode.

Referring to FIGS. 183, 184, 185, and 186, anvils 60210, 60220 are similar in many respects to the anvil 60020. For example, anvils 60210, 60220 can be readily utilized with the end effector 60002, and similarly include rows of staple pockets 60021, 60022 and electrode assemblies 60026, 60027, with or without PTC segments 60053a-c, 60054a-c. In the examples illustrated in FIGS. 183-186, the electrode assemblies 60026, 60027 lack PTC segments. In other examples, however, the electrode assemblies 60026, 60027 may include PTC segments 60053a-c, 60054a-c, as described in connection with the anvil 60020.

Furthermore, the anvils 60210, 60220 are assembled in different ways, and from different components. For example, the anvils 60210, 60220 include different electrode carriers 60211, 60221 configured to accommodate the electrode assemblies 60026, 60027. The electrode assemblies 60026, 60027 are manufactured separately, then assembled with the electrode carriers 60211, 60221. In at least one example, the electrode assemblies 60026, 60027 are assembled with the electrode carriers 60211, 60221 by press-fitting, snap-fitting, or interference-fitting, for example, into corresponding longitudinal slots 60213, 60214 and 60223, 60224, defined in tissue-contacting surfaces 60216, 60226 of the anvil carriers

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60211, 60221, respectively. In other instances, an adhering agent such as any suitable glue can be utilized to fix the electrode assemblies 60026, 60027 to the anvil carriers 60211, 60221, respectively.

The anvil carrier 60211 of the anvil 60210 defines an anvil cap 60212 integrated therewith. On the other hand, the anvil carrier 60221 of the anvil 60220 includes separate carrier portions 60221a, 60221b that are manufactured separately, and configured to be assembled with a separate anvil cap 60222. The anvil caps 60212, 60222 bridge longitudinal anvil slots 60215, 60225 configured to slidably accommodate an I-beam 764.

In various aspects, the electrode carriers 60211, 60221 are configured to provide structural support to the anvils 60210, 60220, reduce I-beam friction in the longitudinal slots 60215, 60225, and/or insulate the metallic anvil components from the electrode assemblies 60026, 60027, and ease assembly thereof into the anvils 60210, 60220, respectively.

Further to the above, the anvil carriers 60211, 60221 include sidewalls 60231, 60232, 60233, 602234 keyed for mating engagement with sidewalls of staple pocket carriers 60217, 60218, 60227, 60228, respectively. In certain examples, the anvil carriers 60211, 60221 are configured to slidably enter into a locking engagement with staple pocket carriers 60217, 60218, 60227, 60228, respectively. In certain instances, the locking engagements can be in the form of press-fit engagements, snap-fit engagements, interference-fit engagements, or any other suitable engagement. Further, in certain instances, the anvil carriers 60211, 60221 and corresponding staple pocket carriers 60217, 60218, 60227, 60228, can be welded using any suitable welding technique.

In certain instances, as illustrated in FIGS. 183-186, the sidewalls 60231, 60232, 60233, 602234 of the electrode carriers 60211, 60221 define longitudinally-extending lateral slots configured to slidably receive longitudinally-extending lateral portions of the staple pocket carriers 60217, 60218, 60227, 60228, respectively, for assembly therewith. In certain examples, mating portions of the staple pocket carriers 60217, 60218, 60227, 60228 are insertable into the slots of the sidewalls 60231, 60232, 60233, 602234 in a distal to proximal direction. In such examples, nose, or distal, portions of the anvils 60210, 60220 are attached to distal portions of the electrode carriers 60211, 60221, respectively, after assembly of the staple pocket carriers 60217, 60218, 60227, 60228 with the electrode carriers 60211, 60221.

In other examples, as illustrated in FIG. 187, an electrode carrier 60211' of an anvil 60210', which is similar in many respects to the anvil 60210, may include an integral nose, or distal, portion 60219. In such instances, staple pocket carriers 60217, 60218 can be assembled with the electrode carrier 60211' by laterally inserting mating portions of the staple pocket carriers 60217, 60218 into the slots defined by the side walls 60231, 60232 of the electrode carrier 60211'.

In various aspects, one or more surfaces of the electrode carriers 60211, 60221 are covered, or coated, with an insulative material to isolate metallic components of the anvils 60210, 60220 from the electrode assemblies 60026, 60027. The insulative coatings on internal surfaces of the electrode carriers 60211, 60221, which interact with the I-beam 764 during staple firing, also act as friction-reducing coatings. In such instances, the anvil longitudinal slots 60215, 60225 can be manufactured cheaply, using looser tolerances, while manufacturing the insulating/friction-reducing coatings to tighter specifications to compensate for the discrepancies/defects in the anvil longitudinal slots 60215, 60225.

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FIG. 188 illustrates an anvil **60240** similar in many respects to the anvils **60210**, **60220**. In the illustrated example, the anvil **60240** includes separately-manufactured anvil cap **60242**, staple pocket carriers **60247**, **60248**, and electrode carriers **60241**, **60243**. The staple pocket carriers **60247**, **60248** include longitudinal openings defined in ledges **60253**, **60254** configured to receive, and matingly engage with, securing features **60251**, **60252** of the electrode carriers **60241**, **60243**.

In various instances, electrode sticking/charring is associated with energy application to tissue by an end effector such as, for example, the end effector **60002**. Energy travel through tissue between the electrode assemblies **60026**, **60027** of the anvil **60020** and the electrode assembly **60036** of the cartridge **60030** may damage the electrode assemblies by sticking and/or charring. To improve the life cycle of a surgical instrument **60000**, the end effector **60002** is configured to concentrate the energy near disposable components of the end effector **60002** to protect against, or reduce the damage cause by, sticking and/or charring in non-disposable components.

In the illustrated example, the cartridge **60030** is disposable, and the anvil **60020** is non-disposable. Consequently, the cartridge **60030** is replaced with a new cartridge **60030** after every firing, while the same anvil **60020** is utilized throughout the life cycle of the surgical instrument **60000**. Accordingly, the end effector **60002** is configured to protect against, or reduce the damage cause by, sticking and/or charring in the electrode assemblies **60026**, **60027** by concentrating the energy near the electrode assembly **60036**. In the illustrated example, this is achieved by designing the surface area of the segmented electrodes **60026a-c**, **60027a-c** to be greater than the surface area of corresponding segmented electrodes **60036a-f**. In other instances, energy concentration can also be achieved by designing the disposable segmented electrodes **60036a-f** to include raised portions, which can be spine-like portions, for example.

FIG. 189 is a schematic view of an end effector **60502** including an anvil **60520** and a staple cartridge **60530**. The end effector **60502** is similar in many respects to the anvil **60020**, which are not repeated herein for brevity. For example, the end effector **60502** can be utilized with the surgical instrument **60000**, and is configured to apply a tissue treatment cycle to tissue grasped between the anvil **60520** and the cartridge **60530**. The tissue treatment cycle may include a tissue sealing phase and a tissue stapling phase, for example, which can be administered simultaneously, sequentially, or in a staggered manner.

The anvil **60520** includes a longitudinal slot **60525** configured to slidably accommodate the I-beam **764**. Rows of staple pockets **60521**, **60522** are disposed on opposite sides of the longitudinal slot **60525**. The anvil **60520** further includes electrode assemblies **60526**, **60527** also disposed on opposite sides of the longitudinal slot **60525**. The electrode assemblies **60526**, **60527** include rows of segmented electrodes **60526a-c**, **60527a-c**. In the illustrated example, two rows of staple pockets and one row of segmented electrodes are depicted on each side of the longitudinal slot **60525**. Further, each row of segmented electrodes includes three segmented electrodes. These numbers, however, are for illustrative purposes, and should not be construed as limiting. In other examples, the anvil **60520** may include two, three, five, or six rows of staple pockets, and/or one, three, or four rows of segmented electrodes each includes two, four, five, or six segmented electrodes, for example.

Further to the above, the cartridge **60530** includes a longitudinal slot **60535** configured to slidably accommodate

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the I-beam **764**. Rows of staple cavities **60531**, **60532** on opposite sides of the longitudinal slot **60535**. The cartridge **60530** further includes electrode assemblies **60536**, **60537** also disposed on opposite sides of the longitudinal slot **60535**. The electrode assemblies **60536**, **60537** include rows of segmented electrodes **60536a-b**, **60537a-b**. In the illustrated example, two rows of staple cavities and one row of segmented electrodes are depicted on each side of the longitudinal slot **60535**. Further, each row of segmented electrodes includes two segmented electrodes. These numbers, however, are for illustrative purposes, and should not be construed as limiting. In other examples, the cartridge **60530** may include two, three, five, or six rows of staple pockets, and/or one, three, or four rows of segmented electrodes each includes three, four, five, or six segmented electrodes, for example.

In the illustrated example, the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b** are separately connected to the RF energy source **794**. This configuration permits the control circuit **760** to cause the RF energy source **794** to selectively activate and deactivate the individual electrode segments in accordance with predetermined tissue treatment cycles and/or in response to certain events such as, for example, the detection of a short circuit in connection with one or more of the segmented electrodes, as described in greater detail elsewhere herein. In various aspects, a multiplexer may distribute the RF energy from the RF energy source **794** to the various segmented electrodes as desired under the control of the control circuit **760**, for example.

In certain instances, one or more of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b** is separately connected to a separate conductor configured to separately connect to the RF energy source **794**, which may include individual power sources for one or more of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b**, for example. The conductors can be responsible for transmitting control, sensing, communication, and/or other signals. The individual conductors may originate at a processor. In certain instances, the processor can reside locally in the end effector **60502**. In other instances, the processor can be located proximally such as, for example, in a proximal housing of the surgical instrument **60000**, or at the RF energy source **794**. In various instances, a multiplexer can be employed, for example, at the end effector **60502** to control the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b**.

In certain instances, where a proximal processor (e.g. at a proximal housing of the surgical instrument **60000** or at the RF energy source **794**) is involved, a single conductor may extend from the processor to the end effector **60502**, which may split into separate connections for each of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b**, for example. Alternatively, individual conductors, which can be incorporated into a flex circuit for example, may extend from the processor to each of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b**.

In other instances, as illustrated in FIGS. 190 and 191, an end effector **60502'**, which is similar in many respects to the end effector **60502**, which are not repeated herein for brevity, may further include one or more passive switches, current limiting elements, energy-sensitive resistance elements, or locally-adjustable resistance elements, which can be in the form of positive temperature coefficient (PTC) segments, for example. In the illustrated example, the end effector **60502'** includes anvil electrode assemblies **60526'**, **60527'** and cartridge electrode assemblies **60536'**, **60537'**,

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wherein each of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b** is connected in series, or alternatively in parallel, to a PTC segment. In the illustrated example, each of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b** is connected in series to one of the PTC segments **60553a-c**, **60554a-c**, **60542a-b**, **60543a-b**.

Further to the above, since the RF energy source **794** is independently connected to each of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b**, the control circuit **760** can be configured to detect a location of a short circuit by detecting an increase in the resistance of a PTC segment through a measured change current and/or voltage. In response, the control circuit **760** may issue an alert, for example through the display **711**, indicating the location of the effected segmented electrodes. Additionally, or alternatively, the control circuit **760** can be configured to deactivate the affected segmented electrodes at the determined location of the short circuit.

Also, since the RF energy source **794** is independently connected to each of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b**, the control circuit **760** can be configured to prompt a clinician for instructions on which of the segmented electrodes to activate for a tissue treatment cycle. FIG. **192** is a logic flow diagram of a process **60570** depicting a control program or a logic configuration for applying a tissue treatment cycle to a tissue grasped by an end effector such as the end effector **60502** exclusively using segmented electrodes selected by a clinician. In certain instances, the process **60570** can be implemented by the surgical instrument **60000**, for example. The process **60570** can be executed by a control circuit **760**. In certain instances, the memory circuit **68008** stores machine-executable instructions that, when executed by the processor **68002**, cause the processor **68002** to execute machine instructions to implement the process **60570**, for example.

In the illustrated example, the process **60570** includes prompting **60571** a clinician to select segmented electrode. In at least one example, the control circuit **760** causes the display **711** to present the schematic diagram of FIG. **189**. The clinician may then select segmented electrodes to be activated, for example by pressing onto the display **711**. The process **60570** further includes prompting **60572** the clinician to select a tissue treatment cycle, only activating **60573** the selected segmented electrodes, and initiating **60574** the tissue treatment cycle only using the selected electrodes.

In certain instances, the control circuit **760** is configured to automatically initiate the tissue treatment cycle, once clinician selections are made. In certain examples, the automatic tissue treatment cycle initiation can be further based on a tissue parameter. In such examples, initiation of the tissue treatment cycle is triggered by (i) receipt of the clinician selection(s), and (ii) detecting that a measurement of a tissue parameter is within a predefined range, or is equal to, or beyond, a predetermined threshold. In certain instances, the tissue parameter is an impedance of the tissue grasped by the end effector **60502**, for example.

As described elsewhere herein in greater detail, tissue impedance of a tissue portion can be detected, for example by a control circuit **760**, by causing a sub-therapeutic signal to be passed through the tissue portion, receiving measurements from a voltage sensing circuit **924** and the current sensing circuit **914**, and dividing the measurements from the voltage sensing circuit **924**, by the corresponding measurements from the current sensing circuit **914**, for example.

The control circuit **760** is then configured to only activate the selected segmented electrodes. Accordingly, only the

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selected segmented electrodes are utilized in treating the tissue. This approach causes energy to flow in specific, preselected, portions of the jaws, while maintaining the remainder of the jaws in a cooler state.

In various aspects, the power requirements for effectively sealing a tissue grasped by the end effector **60502** may vary depending, for example, on the thickness and/or type of the grasped tissue. In certain instances, impedance of the grasped tissue can be indicative of the power required to effectively seal the tissue. Attempting to seal a grasped tissue while available power is less than the required power can yield an ineffective, incomplete, tissue seal. This can yield undesirable consequences particularly if the grasped tissue includes a blood vessel.

FIG. **193** is a logic flow diagram of a process **60580** depicting a control program or a logic configuration for addressing situations where power available to apply in a tissue treatment cycle is less than the power requirements for an effective tissue seal. In certain instances, the process **60580** can be implemented by the surgical instrument **60000**, for example. The process **60580** can be executed by a control circuit **760**. In certain instances, the memory circuit **68008** stores machine-executable instructions that, when executed by the processor **68002**, cause the processor **68002** to execute machine instructions to implement the process **60580**, for example.

In the illustrated example, the process **60580** includes detecting **60581** a tissue parameter of a tissue grasped by an end effector such as, for example, the end effector **60502**, and determining based on the measured tissue parameter if **60582** available power is sufficient to yield an effective tissue seal via a tissue treatment cycle. In certain instances, the control circuit **760** is configured to detect **60581** based on signals from one or more sensors, e.g. current sensors, indicative of the tissue parameter. The tissue parameter can be a tissue impedance or a tissue thickness, for example.

In such instances, the control circuit **760** can further be configured to ascertain the power required to achieve an effective tissue seal via a tissue treatment cycle based on the detected tissue parameter from information stored in a storage medium such as, for example, the memory circuit **68008**. The information can be in the form of a database, equation, formula, and/or table relating various values of the tissue parameter to corresponding values of the power requirement. Furthermore, the control circuit **760** can be configured to compare the ascertained power requirement to available power to determine whether the available power is sufficient to yield an effective tissue seal. In other instances, the information stored in the memory circuit **68008** can be in the form of a range, or a listing, of values of the tissue parameter suitable for achieving an effective tissue seal via the tissue treatment cycle.

In any event, If **60581** it is determined that the available power is sufficient to yield an effective tissue seal, the process **60580** authorizes **60583** the tissue treatment cycle with no change. For example, the process **60580** may apply the tissue treatment cycle simultaneously to all portions of the end effector **60502**.

If **60582**, however, the process **60580** determines that the available power is insufficient to yield an effective tissue seal via the tissue treatment cycle, the process **60581** may separately apply **60584** the tissue treatment cycle in discrete portions of the end effector **60502**. The control circuit **760** may be configured to implement the separate application **60584** of the tissue treatment cycle to discrete portions of the end effector **60502** by separately activating groups of the segmented electrodes of the electrode assemblies **60526**,

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60527, 60536, 60537 along the length of the end effector to separately effect a tissue seal in discrete portions of the end effector **60502**. Accordingly, all of the available power will be fully directed to achieving an effective tissue seal at a first tissue portion in a first discrete portion of the end effector **60502**, then at a second tissue portion in a second discrete portion the end effector **60502**, and so forth, until all tissue portions in all discrete portions of the end effector **60502** are treated in accordance with the tissue treatment cycle.

As discussed supra, the RF energy source **794** is independently connected to each of the segmented electrodes **60526a-c, 60527a-c, 60536a-b, 60537a-b**. Accordingly, the control circuit **760** may cause RF energy source **794** to exclusively activate the segmented electrodes **60526a, 60536a** to apply a tissue treatment cycle to a first tissue portion between the segmented electrodes **60526a, 60536a** yielding an effective tissue seal of the first tissue portion with an available power lesser than the power required to achieve an effective tissue seal for all of the tissue grasped by the end effector **60502** simultaneously. Then, the control circuit **760** may cause RF energy source **794** to exclusively activate the segmented electrodes **60526b, 60536b** to apply the tissue treatment cycle to a second tissue portion between the segmented electrodes **60526b, 60536b**, and so forth until the tissue treatment cycle is applied to all the tissue grasped by the end effector **60502**.

In various aspects, different portions of a tissue grasped by the end effector **60502** may require different amounts of time for achieving an effective tissue seal via a tissue treatment cycle. In certain instances, the amount of time required for achieving an effective tissue seal can be a function of a tissue parameter of the tissue portion such as, for example, an impedance of the tissue portion. As described elsewhere herein in greater detail, tissue impedance of a tissue portion can be detected, for example by a control circuit **760**, by causing a sub-therapeutic signal to be passed through the tissue portion, receiving measurements from a voltage sensing circuit **924** and the current sensing circuit **914**, and dividing the measurements from the voltage sensing circuit **924**, by the corresponding measurements from the current sensing circuit **914**, for example.

FIG. **194** is a logic flow diagram of a process **60590** depicting a control program or a logic configuration for balancing different sealing times for different tissue portions exposed to a tissue treatment cycle by an end effector. In certain instances, the process **60590** can be implemented by the surgical instrument **60000**, for example. The process **60590** can be executed by a control circuit **760**. In certain instances, the memory circuit **68008** stores machine-executable instructions that, when executed by the processor **68002**, cause the processor **68002** to execute machine instructions to implement the process **60590**, for example.

The process **60590** includes determining **60591** a first sealing time associated with applying a tissue treatment cycle to a first portion of a tissue grasped by the end effector **60502**, determining **60592** a second sealing time associated with applying a tissue treatment cycle to a second portion of the tissue grasped by the end effector **60502**, staggering/ coordinating **60593** activation of a first segmented electrode positioned against the first portion of tissue and a second segmented electrode positioned against the second portion of tissue such that the first sealing time and the second sealing time are completed concurrently. In other words, begin the longer sealing time prior to the shorter sealing to ensure a concurrent completion of both sealing times.

Further to the above, determining **60591** the first sealing time and determining **60592** the second sealing times can be

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achieved by measuring a first tissue impedance of the first portion of the tissue and measuring a second tissue impedance of the second portion of the tissue. The control circuit **760** may cause segmented electrodes of the end effector **60502**, which are positioned against the first and second tissue portions, to pass sub-therapeutic signals through the first and second tissue portions for the purposes of determining their tissue impedances. Further, the control circuit **760** can further be configured to ascertain a sealing time of a tissue portion based on its tissue impedance from information stored in a storage medium such as, for example, the memory circuit **68008**. The information may include a correlation between tissue impedance values and corresponding sealing time values, which can be in the form a database, equation, formula, and/or table relating various values of the tissue impedance to corresponding values of the sealing time.

In other instances, a process similar in many respects to the process **60590**, which are not repeated herein for brevity, may stagger/coordinate activation of the first segmented electrode positioned against the first portion of tissue and the second segmented electrode positioned against the second portion of tissue for other purposes. For example, the process may stagger/coordinate the activations to avoid an occurrence of a particular event in the tissue treatment cycle simultaneously at the first tissue portion and the second tissue portion. In certain instances, the particular event can be a point during the first and second sealing times where a maximum power is applied to the first and second tissue portions, for example.

Accordingly, the control circuit **760** can be configured to cause the RF energy source **794** to activate the first segmented electrode prior to activation of the second segmented electrode. In certain instances, where the first sealing time is greater than the second sealing time, the control circuit **760** can be configured to cause the RF energy source **794** to activate the second segmented electrode after completion of the maximum power event by the first segmented electrode, for example. In certain examples, the maximum power event is defined by a power level greater than or equal to a predetermined threshold. In certain examples, the maximum power event is defined by a minimum tissue impedance threshold.

Further to the above, in various aspects, the control circuit **760** can be configured to rapidly alternate activation of segmented electrodes to concurrently seal different portions of a tissue grasped by the end effector **60502**. For example, the control circuit **760** may cause the RF energy source **794** to rapidly alternate activation of groups of segmented electrodes positioned against different tissue portions, wherein only one of the groups is active at any point of time, until a complete application of a tissue treatment cycle is achieved in all the tissue portions.

Further to the above, in various aspects, the control circuit **760** can be configured to sequentially activate segmented electrodes to seal different portions of a tissue grasped by the end effector **60502**. For example, the control circuit **760** may cause the RF energy source **794** to activate a proximal subset of segmented electrodes to apply a tissue treatment cycle to a proximal portion of the tissue grasped by the end effector **60502**, prior to activation of a distal subset of segmented electrodes to apply a tissue treatment cycle to a distal portion of the tissue grasped by the end effector **60502**.

FIG. **195** is a logic flow diagram of a process **60200** depicting a control program or a logic configuration for detecting and addressing a short circuit during a tissue treatment cycle applied to tissue grasped by the end effector

60502. The process **60200** includes passing **60201** a first sub-therapeutic signal through a first tissue portion of the tissue grasped by the end effector **60502**, and monitoring **60202** a first tissue impedance of the first tissue portion based on the first sub-therapeutic signal. The process **60200** further includes passing **60203** a second sub-therapeutic signal through a second tissue portion of the tissue grasped by the end effector **60502**, wherein the second tissue portion is different than the first tissue portion. The process **60200** further includes monitoring **60204** a second tissue impedance of the second tissue portion based on the second sub-therapeutic signal. In addition, the process **60200** includes adjusting **60205** a first therapeutic signal configured to be passed through the first tissue portion based on the first tissue impedance, and adjusting **60206** a second therapeutic signal configured to be passed through the second tissue portion based on the first tissue impedance and the second tissue impedance. Furthermore, the process **60200** includes issuing **60207** an alert indicative of a short circuit based on the first tissue impedance.

In certain instances, the first tissue portion is proximal to the second tissue portion. For example, the first tissue portion may be positioned between the segmented electrodes **60526a**, **605236a**, while second tissue portion can be positioned between the segmented electrodes **60526b**, **60536b**.

FIG. **196** is a graph **60260** representing an interrogation of the first tissue portion, in accordance with the process **60200**. The graph **60260** includes multiple graphs depicting time (t) on the X-axis vs source electrode (e.g. **60526a**) active status **60261**, return electrode (e.g. segmented electrode **60536a**) active status **60262**, power level **60263**, and tissue impedance **60264**, on the Y-axis. In the illustrated example, the control circuit **760** is configured to cause the RF energy source **794** to selectively activate **60265**, **60266** segmented electrodes **60526a**, **60536a** abutting the first tissue portion to pass **60201** the first sub-therapeutic signal between activated segmented electrodes **60526a**, **60536a**, for example. The control circuit **760** is further configured to cause the RF energy source **794** to monitor **60202** the first tissue impedance curve **60267** of the first tissue portion.

In the illustrated example, the monitored first tissue impedance curve **60267** is indicative of a short circuit between segmented electrodes **60526a**, **60536a** due to the presence of a metallic object such as, for example, a previously fired staple in the first tissue portion. The first tissue impedance curve **60267** shows an abnormal, or premature, decrease prior to stopping the first sub-therapeutic signal, which is indicative of the short circuit. In certain instances, a storage medium such as, for example, the memory circuit **68008** stores information representing an expected tissue impedance in response to a sub-therapeutic signal. The information can be in the form of one or more curves, tables, databases, equations, or any suitable medium. A deviation from the expected tissue impedance, as shown in the curve **60267** indicates a short circuit.

Further to the above, the control circuit **760** may be configured to similarly interrogate the second tissue portion abutting segmented electrodes **60526b**, **60536b**. In addition, the control circuit **760** may cause the RF energy source **794** to adjust a first therapeutic signal configured to be passed between the segmented electrodes **60526a**, **60536a**, and a second therapeutic signal configured to be passed between the segmented electrodes **60526b**, **60536b**, to address the detected short circuit.

In certain instances, adjusting the first therapeutic signal includes a reduction in a power parameter of the first therapeutic signal. In certain instances, adjusting the first

therapeutic signal includes reducing the first therapeutic signal to a sub-therapeutic level. In other instances, adjusting the first therapeutic signal includes reducing the first therapeutic signal to a tissue warm-up only level. In other instances, adjusting the first therapeutic signal comprises deactivating at least one of the segmented electrodes **60526a**, **60536a**.

In certain instances, adjusting the second therapeutic signal includes any modification suitable for extending a thermal effect of the second therapeutic signal to the first tissue portion to compensate for the decrease in the power parameter of the first therapeutic signal. In certain instances, adjusting the second therapeutic signal includes an increase in a power parameter of the second therapeutic signal. In other instances, adjusting the second therapeutic signal includes an increase in the time the second therapeutic signal is applied to the tissue, which can be at the same voltage, or at a lower voltage. In other instances, the second therapeutic signal is adjusted to cause an over-sealing of the second tissue portion, in response to the short circuit associated with the adjacent first tissue portion.

In various instances, the adjustments to the first and second therapeutic signals are performed in accordance with predetermined tissue treatment cycles stored in a storage medium such as, for example, the memory circuit **68008**. The control circuit **760** may, in response to detecting the short circuit between the segmented electrodes **60526a**, **60526a**, select a tissue treatment cycle with first and second therapeutic signals adjusted, as described supra, for addressing the short circuit situation.

In various instances, the control circuit **760** may respond to the detection of the short circuit by causing the RF energy source **794** to actively cycle both source and return segmented electrodes to seal around the tissue portion with a detected short circuit. Furthermore, various neighboring segmented electrodes can also be utilized in offset and/or opposing energy delivery modes to seal around the tissue portion with a detected short circuit including activation/cycling surrounding electrode segments in crisscross configurations, for example. In certain instances, depending on a location of short circuit, the control circuit **760** may cause the RF energy source **794** to selectively activate specific segmented electrodes as source electrodes, and simultaneously activating others as return electrodes. Such activations can be cycled or alternated to achieve an effective seal of the entire tissue grasped by the end effector **60502**, while avoiding the location of the detected short circuit.

Electrical arcing is a phenomenon that may occur during application of a sealing energy to a tissue grasped by an end effector **60502**, for example. In certain instances, the presence of a metallic object such as previously-fired staples adjacent active segmented electrodes may yield electrical arcing. The efficacy of a tissue treatment cycle can be negatively influenced by electrical arcing due to a diversion/leap of the sealing energy away from the intended tissue target. The energy diversion may also cause unintended injury to neighboring tissue. In various instances, the ability of the control circuit **760** to separately control activation, deactivation, and polarity of each of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b** further allows the control circuit **760** to manage an arcing event (predicted and/or active) in a localized manner by selectively adjusting various parameters of segmented electrodes near the arcing event, for example. In certain instances, the adjusted parameters are power parameters. In certain instances, the control circuit **760** may selectively cause the RF energy source **794** to reduce the voltage across selected

pairs of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b** to address an arcing event.

In situations where an active arc has occurred, for example due to the presence of an adjacent metallic object such as previously-fired staples, the control circuit **760** can be configured to respond by exclusively deactivating the segmented electrodes responsible for the arcing event. Then, the deactivated segmented electrodes can be reactivated to complete a tissue treatment cycle after adjustments are made to the gap between the affected segmented electrodes and/or the voltage level. In certain instances, voltage levels can be reduced, while increasing the tissue treatment time, to still achieve an effective tissue seal with the reduced voltage. In various aspects, the control circuit **760** is configured to employ a sub-therapeutic signal to test if the power and/or gap adjustments are effective at avoiding recurrence of arcing, prior to restarting the tissue treatment cycle.

In various instances, a control circuit **760** is configured to address an arcing event by increasing an overall tissue gap between jaws of the end effector **60502**, for example. However, to ensure an effective sealing the tissue with an increased tissue gap, the control circuit **760** may further increase at least one of a power parameter, for example voltage, or a sealing time of the tissue. The increased power parameter and/or the increased sealing time can be limited to selected pairs of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b**, for example.

In various instances, the control circuit **760** can be configured to detect an arcing event by analyzing imaging data of the end effector **60502** during a tissue treatment cycle. Additionally, or alternatively, the arcing event can be detected through a clinician input via the display **711**, for example. Additionally, or alternatively, the arcing event can be detected by monitoring one or more parameter of the RF energy source **794**, for example. Additionally, or alternatively, the arcing event can be detected by monitoring tissue temperature during the tissue treatment cycle via temperature sensors in the end effector **60502**, for example. A deviation from an expected correlation between the energy supplied to the tissue portion and the temperature of the tissue portion can indicate an arcing event associated with segmented electrodes configured to supply energy to the tissue portion.

In various instances, the ability of the control circuit **760** to separately control activation, deactivation, and polarity of each of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b** further allows the control circuit **760** to manage capacitive coupling issues which may occur within the shaft of the surgical instrument **60000**. In certain instances, the capacitive coupling may result in reducing power supply to the end effector **60502**, for example.

The reduction may render available power ineffective in simultaneously applying a tissue treatment cycle to an entire tissue grasped by the end effector **60502**. In response, the control circuit **760** can be configured to separately apply the tissue treatment cycle to portions of the grasped tissue. This can be achieved, for example, by selectively activating subsets of the segmented electrodes of the end effector **60502**, one subset at a time, to separately apply the tissue treatment cycle to the tissue portions.

For example, a first subset (e.g. proximal subset) of the segmented electrodes of the end effector **60502** can be activated to apply the tissue treatment cycle to a first portion (e.g. proximal portion) of the grasped tissue. The first subset is then deactivated, and a second subset (e.g. distal subset positioned distally with respect to the proximal subset) can be activated to apply the tissue treatment cycle to a second

portion (e.g. distal portion positioned distally with respect to the proximal portion) of the tissue.

In other instances, the control circuit **760** can be configured to address a reduction in power supply by alternating activation of the subsets of the segmented electrodes. In such instances, only one subset of the segmented electrodes is activated at each point of time. In other instances, the control circuit **760** can be configured to address a reduction in power supply by selecting a different tissue treatment cycle, for example one with a reduced power requirement and an increased sealing time.

In various instances, the ability of the control circuit **760** to separately control activation, deactivation, and polarity of each of the segmented electrodes **60526a-c**, **60527a-c**, **60536a-b**, **60537a-b** further allows the control circuit **760** to dynamically adjust energy modalities in a tissue treatment cycle applied to a tissue grasped by the end effector **60502**. The different energy modalities can be applied to different tissue portions, or can be applied to the same tissue portion, or the entire grasped tissue, in a predetermined sequence. In certain instances, the control circuit **760** is configured to selectively activate one or more segmented electrodes to apply a monopolar energy modality, a bipolar energy modality, and/or a combination, or blended, bipolar/monopolar energy modality to a tissue portion abutting the activated segmented electrodes.

Further to the above, a number of factors can be considered in the energy modality selection by the control circuit **760** including, but not limited to, closure load response, the percentage of jaw closure, tissue impedance, tissue location and/or type, and/or the presence of a short circuit. In certain instances, detecting a blood vessel may cause the control circuit **760** to select the bipolar modality. In certain instances, detecting a tissue thickness beyond a predetermined threshold, for example, may cause the control circuit **760** to select a tissue treatment cycle with an initial bipolar energy modality, to reduce the thickness of the tissue, an intermediate monopolar energy modality to increase the sealing speed, and then a final bipolar energy modality to complete the tissue seal.

Further to the above, if a short circuit is detected, due for example to the presence of a previously-fired staple, the control circuit **760** can be configured to select a tissue treatment cycle with a bipolar energy modality, and a monopolar energy modality, especially modified to address the short circuit. Further, the control circuit **760** can be configured to selectively apply the bipolar energy modality only to a subset of the segmented electrodes that are not affected by the detected short circuit, and then apply the monopolar energy modality to all the segmented electrodes. For example, the control circuit **760** may cause the RF energy source **794** to deactivate the segmented electrodes where the short circuit is detected, and then apply the bipolar energy modality to the remaining segmented electrodes. Next, the control circuit **760** may cause the RF energy source **794** to reactivate the previously-deactivated segmented electrodes for application of the monopolar energy modality to the tissue.

FIGS. 197-203 illustrate a number of energy profiles, or therapeutic signals, **60300**, **60310**, **60320**, **60330**, **60340**, **60350**, **60360** depicted in graphs representing Tissue impedance, Voltage, Power, and Current curves associated with application of the therapeutic signals **60300**, **60310**, **60320**, **60330**, **60340**, **60350**, **60360** to tissue grasped by the end effector **60502**, for example. It is understood that the therapeutic signals **60300**, **60310**, **60320**, **60330**, **60340**, **60350**, **60360** are for illustrative purposes only and, as such, are not

limiting. Other high, medium, and low energy profiles can be utilized in tissue treatment cycles effected by the control circuit 760. In certain instances, two or more of the therapeutic signals 60300, 60310, 60320, 60330, 60340, 60350, 60360 can be delivered to different tissue portions in different zones along a length of the end effector 60502 in a tissue treatment cycle effected by the control circuit 760. The different zones can be defined by different subsets of the segmented electrodes 60526a-c, 60527a-c, 60536a-b, 60537a-b.

In certain instances, the two or more of the therapeutic signals 60300, 60310, 60320, 60330, 60340, 60350, 60360 can be delivered simultaneously in the different zones in a tissue treatment cycle. In certain instances, the different zones include a proximal zone and a distal zone. In other instances, the different zones include a proximal zone, one or more intermediate zones, and a distal zone.

In various instances, various parameters of the therapeutic signals 60300, 60310, 60320, 60330, 60340, 60350, 60360 can be stored in a storage medium such as, for example, the memory circuit 68008, which can be accessed to implement a tissue treatment cycle, for example. The control circuit 760 can be configured to select one or more of the therapeutic signals 60300, 60310, 60320, 60330, 60340, 60350, 60360 for execution in a tissue treatment cycle applied to one or more zones of the end effector 60502 based on one or more conditions of the grasped tissue in the one or more zone including tissue thickness, tissue type, tissue location, and/or tissue impedance, for example.

Referring primarily to FIGS. 1 and 155, a surgical instrument (e.g. surgical instruments 1000, 60000) can include an end effector (e.g. end effectors 1300, 60002, 60502). One or more motor assemblies can be motivated by a control circuit (e.g. control circuit 760) to effect one or more functions/motions of the end effector including closure of the jaws, firing of the staples, and/or rotation and/or articulation of the end effector about a central longitudinal axis (e.g. axis 60005) of the surgical instrument. Various mechanisms for articulation, rotation, closure, and firing of an end effector are described in greater details elsewhere in the present disclosure, and are not repeated herein for brevity.

In various aspects, the control circuit 760 can be configured to cause one or more motor assemblies to effect various rotation and/or articulation motions of an end effector (e.g. end effectors 1300, 60002, 60502) in response to inputs from a clinician to align the jaws of the end effector with respect to a tissue. The clinician may then position one of the jaws behind the tissue. Further, the control circuit 760 can also be configured to cause one or more motor assemblies to motivate the jaws to grasp the tissue in a closure motion, in response to another clinician input. In certain instances, closure of the jaws can be reversed multiple times until a satisfactory tissue bite is achieved. At such point, the control circuit 760 can be configured to cause a firing driver such as, for example, the I-beam 764 to be advanced distally to fire staples stored in staple cavities of a staple cartridge into the grasped tissue.

In certain instances, the clinician may elect to perform additional rotational adjustments of the end effector in the vicinity of the tissue such as, for example, prior to end effector closure, during end effector closure, and following end effector closure. In certain instances, the clinician may elect to perform additional rotational adjustments of the end effector after a successful end effector closure, or tissue bite, has been achieved by prior to applying a therapeutic energy to the tissue, or prior to firing staples into the tissue. The additional rotational adjustments can be fine rotational

adjustments with different rotational parameters than standard rotational adjustments to protect the tissue and/or aid less-experienced clinicians.

FIG. 204 is a logic flow diagram of a process 60400 depicting a control program or a logic configuration for adjusting a parameter of rotation of an end effector of a surgical instrument based on whether a tissue is being grasped by the end effector as determined based on at least one impedance measurement, in accordance with at least one aspect of the present disclosure. In various instances, the process 60400 can be implemented by any suitable surgical instrument such as, for example, surgical instruments 1000, 60000 including any suitable end effector such as, for example, end effectors 1300, 60002, 60502. However, for brevity, the following description of the process 60400 will focus on its implementation in the surgical instrument 60000 and the end effector 60502, for example. In certain instances, the memory 68008 stores program instructions that, when executed by the processor 68002, cause the processor 68002 to perform one or more aspects of the process 60400.

The process 60400 includes causing 60401 a sub-therapeutic signal to be provided to the end effector 60502. For example, the control circuit 760 may cause the RF energy source 794 to attempt to pass a sub-therapeutic signal between the electrode assemblies 60526, 60536. The process 60400 further includes determining 60402 an impedance between the electrode assemblies 60526, 60536 in response to the sub-therapeutic signal to assess whether tissue is being grasped by the end effector 60502. The process 60400 further includes selecting 60403 a parameter of rotation of the end effector based on at least one impedance measurement. The parameter of rotation of the end effector includes rotation speed, rotation distance, rotation direction, and/or rotation time, for example.

As described elsewhere herein in greater detail, the control circuit 760 is configured to determine 60402 the impedance between the electrode assemblies 60526, 60536, in response to the sub-therapeutic signal, based on measurements from a voltage sensing circuit 924 and the current sensing circuit 914, for example. The control circuit 760 can be configured to divide the measurements from the voltage sensing circuit 924, by the corresponding measurements from the current sensing circuit 914, for example, to determine the impedance.

Further to the above, the control circuit 760 can be configured to select 60403 a parameter of rotation of the end effector 60502 based on a comparison of the impedance measurement to a predetermined threshold. The impedance measurements can be indicative of the presence or absence of tissue in contact with the end effector 60502. The control circuit 760 can be configured to detect an absence of tissue if the impedance measurement is greater than, or equal, to a predetermined threshold, for example due to an open circuit. On the contrary, the control circuit 760 can be configured to detect a presence of tissue if the impedance measurement is below the predetermined threshold. In certain instances, the predetermined threshold can be stored in a storage medium such as, for example, the memory circuit 68008, and can be utilized by the processor 68002 to determine whether tissue is in contact with the end effector 60502.

Further to the above, selecting 60403 a parameter of rotation of the end effector 60502 can include selecting a speed of rotation, or a distance of rotation of the end effector 60502. In certain instances, selecting 60403 a parameter of rotation of the end effector 60502 comprises selecting between a first rotational profile and a second rotational profile. The first and second rotational profiles can be stored

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in a storage medium such as, for example, the memory circuit **68008**. The control circuit **760** can be configured to select the first rotational profile in the absence of tissue, and the first rotational profile in the absence of tissue, as determined based on the comparison of the impedance measurements and the predetermined threshold.

Further to the above, the first rotational profile may include a first speed of rotation greater than a second speed of rotation of the second rotational profile. In certain examples, the first speed of rotation may be a maximum speed of rotation. In certain example, the second speed of rotation can be a percentage of the first speed of rotation. The percentage can, for example, be selected from a range of about 1% to about 50%. In certain instances, the first rotational profile comprises a greater initial acceleration to a predetermined speed of rotation than the second rotational profile.

In certain instances, the first rotational profile may include a first distance of rotation greater than a second distance of rotation of the second rotational profile. In certain examples, the first distance of rotation may be a maximum distance of rotation. In certain example, the second distance of rotation can be a percentage of the first distance of rotation. The percentage can, for example, be selected from a range of about 1% to about 50%.

In certain instances, selecting **60403** a parameter of rotation of the end effector **60502** includes selecting a parameter of the power supplied to a motor to effect the rotation of the end effector **60502**. As described elsewhere herein in greater detail, a motor assembly may include a motor and a motor control circuit configured to supply power to the motor in accordance with power parameters selected by the control circuit **760**, for example. The motor can be configured to cause a rotation of the shaft **60004** and the end effector **60502** relative to the housing assembly **60006**, for example.

In certain instances, current supplied to the motor by the motor control circuit can be selected based on the impedance measurement. The control circuit **760** can be configured to select a first current in the absence of tissue and a second current in the presence of tissue, wherein the first current is greater than the second current.

In certain instances, the second current comprises a value of zero. Accordingly, the control circuit **760** can be configured to deactivate the motor to seize all rotational motions if tissue is detected between the jaws of the end effector **60502**.

Further to the above, if tissue is no longer detected, based on impedance measurements, the control circuit **760** can be configured to readjust the power parameter of the motor. For example, the control circuit **760** can be configured to reselect the first current, or reselect the first rotational profile.

In other embodiments, as illustrated in FIG. **204**, the parameter of rotation of the end effector **60502** can be selected **60405** based on a closure state of the end effector **60502** in addition to impedance measurements. Alternatively, the parameter of rotation of the end effector **60502** can be selected solely based on a closure state of the end effector **60502**.

In certain examples, the parameter of rotation of the end effector **60502** is adjusted to different values associated with different closure states. For example, the control circuit **760** can be configured to select a first value for the parameter of rotation of the end effector **60502** for a fully-open state, select a second value for the parameter of rotation of the end effector **60502** for a partially-open state, and/or select a third value for the parameter of rotation of the end effector **60502** for a fully-closed state. In certain instances,

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the first value is greater than the second value, and the second value is greater than third value.

The closure state of the end effector **60502** can be detected **60404** by the control circuit **760** based on sensor signals of one or more sensors. For example, sensor signals from the position sensor **784** (FIG. **163**) can be indicative of the position of a drive member (e.g. I-beam **764** or closure drive **3800**) movable by the motor **754** to effect a closure of the end effector **60502**. The position of the drive member can be correlated to the different closure states of the end effector **60502**. Other sensors **788** (FIG. **163**) can also be utilized by the control circuit **760** to determine the closure states of the end effector **60502** such as, for example, sensors configured to detect the gap between the jaws of the end effector **60502**.

In other embodiments, the parameter of rotation of the end effector **60502** can be selected based on a closure load of the end effector **60502** instead of tissue impedance, or in addition to tissue impedance. In certain examples, the parameter of rotation of the end effector **60502** is adjusted to different closure loads. For example, the control circuit **760** can be configured to select a first value for the parameter of rotation of the end effector **60502** for a first closure load, select a second value for the parameter of rotation of the end effector **60502** for a second closure load, and/or select a third value for the parameter of rotation of the end effector **60502** for a third closure load. In certain instances, the third closure load is greater than the second closure load which is greater than the first closure load. In such instances, the third value is less than the second value, and the second value is less than first value. In various instances, the control circuit **760** is configured to detect a closure load of the end effector **60502** based on current draw by the motor effecting the closure load. A current sensor **786** can be configured to measure the current draw of the motor.

In other embodiments, as illustrated in FIG. **204**, the parameter of rotation of the end effector **60502** can be selected **60408** based on a firing state of the end effector **60502** in addition to impedance measurements. Alternatively, the parameter of rotation of the end effector **60502** can be selected solely based on a firing state of the end effector **60502**. In certain examples, the parameter of rotation of the end effector **60502** is adjusted to different values associated with different firing states. For example, the control circuit **760** can be configured to select a first value for the parameter of rotation of the end effector **60502** for an unfired state, select a second value for the parameter of rotation of the end effector **60502** for a partially-fired state, and/or select a third value for the parameter of rotation of the end effector **60502** for a fully-fired state. In certain instances, the first value is greater than the second value. In certain instances, the third value is greater than the second value.

The firing state of the end effector **60502** can be detected **60404** by the control circuit **760** based on sensor signals of one or more sensors. For example, sensor signals from the position sensor **784** (FIG. **163**) can be indicative of the position of a drive member (e.g. I-beam **764**) movable by the motor **754** to effect a firing of the staples from the end effector **60502**. The position of the drive member can be correlated to the different firing states of the end effector **60502**.

In various instances, tissue impedance measurements of a tissue grasped by the end effector **60502**, as described supra in connection with the process **60400** of FIG. **204**, can be useful in assessing tissue tension cause by over-rotation, or unintended rotation, of the end effector **60502**. A rotation of the end effector **60502** about the longitudinal axis **60005**

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increases tension on a first tissue portion on a first side of the longitudinal slot **60535**, while reducing tension on a second tissue portion on a second side, opposite the first side, of the longitudinal slot **60535**. Consequently, a first tissue thickness of the first tissue portion may be reduced, while a second tissue thickness of the second tissue portion may be increased. Furthermore, the changes in tissue thickness may be accompanied by changes in tissue impedances of the first and second tissue portions due to a change in the fluid content of the tissue portions.

FIG. **205** is a logic flow diagram of a process **60600** depicting a control program or a logic configuration for adjusting a parameter of rotation of an end effector of a surgical instrument based a detected over-rotation of the end effector, in accordance with at least one aspect of the present disclosure. In various instances, the process **60600** can be implemented by any suitable surgical instrument such as, for example, surgical instruments **1000**, **60000** including any suitable end effector such as, for example, end effectors **1300**, **60002**, **60502**. However, for brevity, the following description of the process **60600** will focus on its implementation in the surgical instrument **60000** and the end effector **60502**, for example. In certain instances, the memory **68008** stores program instructions that, when executed by the processor **68002**, cause the processor **68002** to perform one or more aspects of the process **60600**.

The process **60600** includes measuring **60601** a first tissue parameter of a first tissue portion on a first side of a longitudinal slot of an end effector, measuring **60602** a second tissue parameter of a second tissue portion on a second side of the longitudinal slot of the end effector, adjusting **60603** a parameter of rotation of the end effector based on a relation between the first tissue parameter and the second tissue parameter. The first and second tissue parameters can, for example, be tissue impedance, or tissue thickness.

The control circuit **760** can be configured to monitor tissue impedance of the first tissue portion and the second tissue portion grasped by the end effector **60502**. For example, the control circuit **760** may cause the RF energy source **794** to pass sub-therapeutic signals between the electrode assemblies **60526**, **60536** and between the electrode assemblies **60527**, **60537**. The control circuit **760** may then calculate a first tissue impedance of the first tissue portion and a second tissue impedance of the second tissue portion, based on the sub-therapeutic signals. Furthermore, the control circuit **760** can be configured to adjust a parameter of rotation of the end effector **60502** based on the difference between the first and second tissue impedances. In certain examples, the control circuit **760** can be configured to slow, deactivate, or reverse, end effector rotation if the difference between the first and second tissue impedances is greater than or equal to a predetermined threshold.

Further to the above, various adjustments can be made to one or more parameters of rotation of the end effector **60502** include a rotational position, rotational distance, a rotational speed, a rotational time, and/or a rotational direction to avoid, or mitigate, a detected obstacle. In various aspects, a control circuit **760** can be configured to adjust a parameter of rotation of the end effector **60502** in response to detecting a rotation obstacle. The control circuit **760** can be configured to detect a rotation obstacle if a current draw of the motor effecting a rotation of the end effector **60502** is greater than, or equal to a predetermined threshold, for example.

In various aspects, the control circuit **760** can be configured to perform a predictive analysis to assess whether a previously-detected obstacle will be reached based on a

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requested movement by the clinician. Furthermore, the control circuit **760** may be configured to issue an alert, for example through the display **711**, and/or seize further rotation of the end effector **60502**, if it is determined that the requested movement will cause the end effector to reach the obstacle. In certain instances, the a previously-detected obstacle can be in the form of a system constraint such as, for example, a maximum rotation angle, which can be a predetermined maximum rotation angle that will be reached, or exceeded, if the requested movement is complied with.

Referring primarily to FIG. **189**, the end effector **60502** is can be configured to apply a hybrid tissue treatment cycle to a tissue grasped between the cartridge **60530** and the anvil **60520**. The hybrid tissue treatment cycle includes an RF energy phase and a stapling phase, which can be applied separately, or sequentially, to tissue portions along a length of the end effector **60502**. In the hybrid tissue treatment cycle, RF energy can be applied to the grasped tissue by the electrode assemblies **60526**, **60527**, **60536**, **60537**. An RF energy zone may be cooperatively defined by segmented electrodes of the electrode assemblies **60526**, **60527**, **60536**, **60537**, for example. Further, the hybrid tissue treatment cycle also includes deploying staples into the grasped tissue from rows of staple cavities **60531**, **60532**, which are deformed by rows of staple pockets **60521**, **60522**. A stapling zone may be cooperatively defined by staple cavities **60531**, **60532** and corresponding staple pockets **60521**, **60522**. In the instance of the end effector **60502**, the RF zone is laterally surrounded by portions of the stapling zone due to the arrangement of the electrode assemblies **60526**, **60527**, **60536**, **60537**, the rows of staple cavities **60531**, **60532** and rows of staple pockets **60521**, **60522**.

FIG. **206** is a logic flow diagram of a process **60700** depicting a control program or a logic configuration for cooperatively applying the RF energy phase and the stapling phase to tissue portions of a tissue grasped by an end effector **60502**, for example, in a hybrid tissue treatment cycle. In certain instances, the RF energy phase may be utilized to mitigate, counterbalance, compensate for, and/or offset defects in the stapling phase. In other instances, the stapling phase may be utilized mitigate, counterbalance, compensate for, and/or offset defects in the RF energy phase.

In various instances, the process **60700** can be implemented by any suitable surgical instrument such as, for example, surgical instruments **1000**, **60000** including any suitable end effector such as, for example, end effectors **1300**, **60002**, **60502**. However, for brevity, the following description of the process **60700** will focus on its implementation in the surgical instrument **60000** and the end effector **60502**, for example. In certain instances, the memory **68008** stores program instructions that, when executed by the processor **68002**, cause the processor **68002** to perform one or more aspects of the process **60700**.

In the illustrated example, the process **60700** includes detecting **60701** a tissue parameter. The tissue parameter can, for example, be a tissue thickness of the tissue grasped by the end effector **60502**. The process **60700** further includes detecting **60702** a cartridge parameter. The cartridge parameter can be a staple height of staples stored in the rows of staple cavities **60531**, **60532** of the end effector **60502**, for example. In addition, the process **60700** includes selecting **60703** a radio-frequency (RF) energy treatment for sealing the tissue based on the cartridge parameter and the tissue parameter.

The process **60700** may utilize the RF energy phase to compensate for a discrepancy between a tissue thickness of a tissue grasped by the end effector **60502**, for example, and

a staple height of the cartridge **60530**, for example. The discrepancy may arise when the grasped tissue is thicker than can be successfully accommodated by the staple height of the cartridge **60530**. In such instances, the RF energy phase can be utilized to thin the grasped tissue—through warming or drying out beyond an RF zone of the end effector **60502** and into a tissue stapling zone of the end effector **60502**—to yield a tissue thickness that can be successfully accommodated by the staple height of the cartridge **60530**.

In other embodiments, the discrepancy between the tissue thickness and the staple height may arise when the grasped tissue is thinner than can be successfully stapled the cartridge **60530** due to the staple height being too tall. Consequently, the formed staples may not be able to apply sufficient compression to effectively seal the tissue. In such instances, the RF energy phase can be adjusted to expand a thermal spread through the tissue beyond the RF zone, and into the stapling to support energy sealing of tissue portions where staples will be too tall to effectively seal the tissue. Alternatively, in instances where thermal spread beyond the RF zone may reduce the thickness of the grasped tissue below what can be successfully stapled, the RF energy phase can be adjusted to minimize, or prevent, a thermal spread beyond the RF zone.

In various instances, adjusting the thermal spread can be achieved by adjusting one or more parameters of the RF energy phase such as, for example, power level and/or activation time of the RF energy. In certain instances, adjusting parameters of the RF energy phase can be applied to individual segmented electrodes, or subsets of segmented electrodes, of the electrode assemblies **60526**, **60527**, **60536**, **60537**.

In certain instances, the tissue thickness can be determined based on tissue impedance, for example. As described supra, the control circuit **760** can be configured to determine tissue impedance by causing the RF energy source **794** to pass one or more sub-therapeutic signals through the grasped tissue, utilizing for example the electrode assemblies **60526**, **60527**, **60536**, **60536**. The tissue thickness can then be determined based on a correlation between tissue impedance and tissue thickness, which can be stored in a storage medium such as, for example, the memory circuit **86006**. The correlation can be stored in any suitable form including a table, equation, or database, for example. In other embodiments, the tissue thickness can be determined by measuring a gap between the cartridge **60530** and the anvil **60520** abutting the grasped tissue. The gap can be measured by one or more of the sensors **788**, for example, and is representative of the tissue thickness.

In certain instances, staple height, and other parameters of the cartridge **60530** can be stored in a storage medium such as, for example, a memory circuit, which can locally reside on, or within, the cartridge **60530**. The control circuit **760** can be configured to interrogate the storage medium of the cartridge **60530** to detect **60702** the cartridge parameter.

In certain embodiments, the stapling phase of a hybrid tissue treatment cycle may be utilized to mitigate, counter-balance, compensate for, and/or offset defects in the RF energy phase, for example. FIG. **207** is a logic flow diagram of a process **60710** depicting a control program or a logic configuration for cooperatively applying the RF energy phase and the stapling phase to tissue portions of a tissue grasped by an end effector **60502**, for example, in a hybrid tissue treatment cycle.

In various instances, the process **60710** can be implemented by any suitable surgical instrument such as, for

example, surgical instruments **1000**, **60000** including any suitable end effector such as, for example, end effectors **1300**, **60002**, **60502**. However, for brevity, the following description of the process **60710** will focus on its implementation in the surgical instrument **60000** and the end effector **60502**, for example. In certain instances, the memory **68008** stores program instructions that, when executed by the processor **68002**, cause the processor **68002** to perform one or more aspects of the process **60710**.

In the illustrated example, the process **60710** includes applying **60711** a therapeutic energy to a tissue grasped by an end effector **60502**, for example, to seal the tissue in an RF phase of a hybrid tissue treatment cycle. The control circuit **760** can be configured to cause the RF energy source **794** to activate segmented electrodes of one or more of the electrode assemblies **60526**, **60527**, **60536**, **60536** to apply the therapeutic energy to one or more tissue portions of the grasped tissue, in accordance with predetermined parameters of the hybrid tissue treatment cycle.

Further to the above, the process **60710** includes detecting **60712** a tissue sealing inconsistency in the grasped tissue. The tissue sealing inconsistency can be an inadequate tissue seal due, for example, to a short circuit, which can result from the presence of a previously-fired staple. In certain examples, the control circuit **760** can be configured to cause the RF energy source **794** to pass one or more interrogation signals, which can be in the form of sub-therapeutic signals, through different tissue portions of the grasped tissue to detect inconsistencies in the tissue seal. The sub-therapeutic signals can be passed between pairs of segmented electrodes of the electrode assemblies **60526**, **60527**, **60536**, **60536**, for example. Tissue impedance of the different tissue portions can be determined following the RF energy phase. Since an inadequate tissue seal comprises different tissue impedance characteristics than those associated with an adequate seal, detecting tissue seal inconsistencies in the tissue portions can be achieved by comparing determined tissue impedance of such portions to a predetermined threshold, for example.

Further to the above, the process **60700** may include adjusting **60713** a stapling parameter to compensate for the tissue sealing inconsistencies. In certain instances, adjusting the stapling parameter includes adjusting a tissue gap between the cartridge **60530** and an anvil **60520**. In certain instances, adjusting the stapling parameter includes adjusting a tissue compression of the grasped tissue, or a closure load of the end effector, for example. The control circuit **760** may be configured to cause a motor assembly to increase or decrease the closure load applied to the end effector by a closure driver such as, for example, the I-beam **764**, or closure drive **3800** in instances where closure and firing are driven separately.

In certain instances, adjusting the stapling parameter includes adjusting a staple height of formed staples of the cartridge **60530**. In certain instances, adjusting the stapling parameter includes adjusting a firing speed for fine tuning the formed-staple height. The control circuit **760** may be configured to cause a motor assembly to increase or decrease the speed of a firing driver (e.g. I-beam **764**) to adjust the formed-staple height to compensate for the tissue sealing inconsistencies.

In one example, the control circuit **760** can be configured to detect an inadequate seal in a first tissue portion between segmented electrodes **60536a**, **60526a**, for example, based on a comparison of the first tissue impedance to a predetermined threshold, or threshold range. The first tissue impedance can be measured by passing a first sub-therapeutic signal between the segmented electrodes **60536b**, **60526c**.

Further, the control circuit **760** can be also configured to detect an adequate seal in a second tissue portion between segmented electrodes **60536b**, **60526c**, for example, based on a comparison of the second tissue impedance to the predetermined threshold, or threshold range. The second tissue impedance can be measured by passing a second sub-therapeutic signal between the segmented electrodes **60536b**, **60526c**.

Furthermore, the control circuit **760** can be configured to select a firing speed of the firing driver (e.g. I-beam) in a tissue portion based on adequacy of the tissue seal in the tissue portion. Accordingly, the control circuit **760** can be configured to select a first firing speed of the firing driver (e.g. I-beam) in the first tissue portion with the inadequate tissue seal, and a second firing speed of the firing driver (e.g. I-beam) in the second tissue portion with the adequate tissue seal, wherein the first firing speed is less than the second firing speed, for example. In certain instances, the control circuit **760** can be configured to pause firing of the staple at a tissue portion with an inadequate tissue seal.

In various aspects, a hybrid tissue treatment cycle can be applied to discrete tissue portions of a tissue grasped by an end effector **60502** by alternating between the RF energy phase and the stapling phase. The RF energy phase may lead the stapling phase to avoid circuit shorting conditions, which may occur if there are staples in the tissue during application of the RF energy phase. In other words, the stapling phase may follow the RF energy phase.

In certain instances, the RF energy is applied to a proximal tissue portion, for example a tissue portion between the electrode assemblies **60536a**, **60526a**. Then, staples are fired from rows of staple cavities **60221**, **60222** into the proximal tissue portion by advancing the firing driver through the first tissue portion. The firing driver is then paused until the RF energy is applied to a proximal tissue portion, for example a tissue portion between the electrode assemblies **60536b**, **60526b**. Following application of the RF energy to the second tissue portion, the movement of the firing driver is reactivated to advance the firing driver through the second tissue portion thereby firing staples from rows of staple cavities **60231**, **60232** into the second tissue portion. Alternating between the RF phase and the stapling phase can be repeated for additional tissue portions until all the tissue portions of the grasped tissue are treated.

Referring now to FIGS. **208-210**, a surgical instrument **60000'** is configured to seal tissue using a combination of energy and stapling modalities or phases. The surgical instrument **60000'** is similar in many respects to other surgical instruments such as, for example, the surgical instruments **1000**, **60000**, which are not repeated herein for brevity. For example, the surgical instrument **60000'** includes an end effector **60002'**, the articulation assembly **60008**, the shaft assembly **60004**, and the housing assembly **60006**.

Further to the above, the surgical instrument **60000'** mainly differs from the surgical instrument **60000** in the electrical wiring associated with the electrode assembly **60036**. The surgical instrument **60000'** comprises electrical wiring that defines two separate RF return paths **60801**, **60802** for the electrode assembly **60036**, while in the surgical instrument **60000** comprises electrical wiring that defines a single RF return path **60801** for the electrode assembly **60036**. For brevity, the following description focuses on the dual RF return paths **60801**, **60802** of the surgical instrument **60000'**.

In the illustrated example, the staple cartridge **60030'** comprises a proximal electrical contact **60803** define in a

proximal wall of the staple cartridge **60030'**. A leaf-spring contact **60804** is connected to the proximal electrical contact **60803**, when the staple cartridge **60030'** is properly inserted into the cartridge channel **60040** of the end effector **60002'**, as best illustrated in FIG. **208**. Additional wiring extends proximally from the leaf-spring contact **60804** to connect the electrical assembly **60036** to proximal electronics such as, for example, the control circuit **760** and/or the RF energy source **794**.

Further to the above, the RF return path **60801** extends proximally from the electrode assembly **60036**, from the flex circuit **60041**, and penetrates the cartridge deck **60047** terminating at the proximal electrical contact **60803**. Similarly, the RF return path **60802** extends proximally from the electrode assembly **60036**, from the flex circuit **60041**, and penetrates the cartridge deck **60047** terminating at the proximal electrical contact **60803**. However, the RF return path **60802** comprises a gap **60805** configured to be bridged by an isolated return pad of anvil **60020'** of the end effector **60002'**, when the end effector **60002'** is in a closed, or partially-closed, configuration, as illustrated in FIG. **209**.

Accordingly, the RF return path **60802** remains open until the gap **60508** is bridged by the isolated return pad of the anvil **60020'**. In certain instances, the RF return paths **60801**, **60802** are utilized simultaneously, which ensures adequate connections through redundancy. In other instances, the RF return paths **60801**, **60802** define separate electrical pathways for separately connecting first and second electrical elements of the end effector **600**, respectively, to proximal electronics such as, for example, the RF energy source **794** and/or the control circuit **760**. In such instances, the first electrical elements, connected via the first RF return path **60801**, can be activated while the anvil **60020'** remains in an open, or partially open, configuration, while the second electrical elements, connected via the second RF return path **60802**, can only be activated while the anvil **60020'** remains in the closed configuration, as illustrated in FIG. **210**.

FIG. **211** is a logic flow diagram of a process **60850** depicting a control program or logic configuration for cooperatively controlling application of a therapeutic signal to a tissue grasped by an end effector (e.g. end effector **60502**) and controlling a function of the end effector. The function includes at least one of an articulation of the end effector, a rotation of the end effector, a closure of the end effector about the tissue, and a firing of the staples into the tissue. In various instances, the process **60850** can be implemented by any suitable RF energy source (e.g. RF energy source **794**) and any suitable surgical instrument such as, for example, surgical instruments **1000**, **60000** that include any suitable end effector such as, for example, end effectors **1300**, **60002**, **60502**. However, for brevity, the following description of the process **60850** will focus on its implementation in a surgical system that includes the RF energy source **794**, the surgical instrument **60000**, and the end effector **60502**, for example.

As described supra, the end effector **60502** is configured to grasp tissue in a closure motion of one, or both, of the jaws of the end effector **60502**. Further, the end effector **60502** is also configured to apply a tissue treatment cycle to the grasped tissue. The tissue treatment cycle includes an RF energy phase where the RF energy source **794** is configured to cause a therapeutic signal to be passed through the tissue to seal the tissue, and a stapling phase where staples are deployed into the tissue from a cartridge **60530** in a firing stroke.

In the illustrated example, the process **60850** includes receiving **60851** a communication signal from the RF energy

source **794** indicative of a deficiency in an application of the therapeutic signal to the grasped tissue, and adjusting **60852** a function of the end effector **60502** based on the communication signal to address the deficiency. The function includes at least one of an articulation of the end effector, a rotation of the end effector, a closure of the end effector about the tissue, and a firing of the staples into the tissue.

The deficiency may, for example, be a power insufficiency to complete an effective tissue seal of grasped tissue via the therapeutic signal. The power insufficiency may result from an inadequacy of pressure applied to the tissue by the jaws of the end effector **60502**. Inadequate pressure may change the amount of fluid in the grasped tissue, which can change tissue impedance to a level that hinders a proper transfer of the therapeutic signal through the grasped tissue, by changing the power required to complete an effective seal beyond the safe capabilities of the RF energy source **794**.

The RF energy source **794** may detect the power insufficiency based on impedance of the grasped tissue, for example. As described elsewhere herein in greater detail, the RF energy source **794** can measure tissue impedance of tissue portions between opposite segmented electrodes of the electrode assemblies **60526**, **60527**, **50536**, **50536**. Tissue impedance can then be compared to a threshold to determine whether sufficient power is available for an effective tissue seal. The threshold can be stored in a storage medium such as, for example, a memory circuit. In certain instances, the comparison can be performed by a processing unit at the RF energy source **794**. A communication signal can then be sent to the control circuit **760** to communicate the result of the comparison. In other instances, the communication signal may represent the value of the measured tissue impedance. In such instances, the comparison is performed by the control circuit **760**, and the threshold can be stored in the memory circuit **68008**, for example.

In any event, if power insufficiency is detected, the control circuit **760** can be configured to adjust **60852** on or more function of the end effector **60502** to change the pressure applied onto the tissue by the jaws, which changes fluid levels in the grasped tissue, which changes the tissue impedance. If **60853** the change in tissue impedance addresses the deficiency, the control circuit **760** authorities application **60804** of the therapeutic signal to the tissue.

In at least one example, various aspects of the process **60850** can be executed via the control circuit **760**. In certain instances, the memory **68008** stores program instructions that, when executed by the processor **68002**, cause the processor **68002** to perform one or more aspects of the process **60800** such as, for example, adjusting **60852** a function of the end effector **60502**. The control circuit **760** may cause one or more motor assemblies to change a degree of articulation and/or rotation of the end effector **60502** to adjust the pressure applied by the end effector **60502** onto the grasped tissue to address **60853** the deficiency. Additionally, or alternatively, the control circuit **760** may cause a motor assembly to move one or both of the jaws of the end effector **60502** to adjust a drive force of a closure drive (e.g. I-beam **764**, closure drive **3800**), which adjusts the clamp pressure applied by the end effector **60502** onto the grasped tissue, to address **60853** the deficiency. Additionally, or alternatively, the control circuit **760** may a motor assembly to adjust a parameter of motion of the I-beam **764** to address **60853** the deficiency.

Referring primarily to FIG. **212**, in certain instances, the deficiency to be addressed can be in the end effector function rather than the application of the therapeutic signal. In one example, the deficiency can be a power insufficiency to

perform the end effector function. As described supra, end effector functions are driven by one or more motor assemblies that can be powered by a local energy source such as, for example, the energy source **762** (FIG. **163**) which can be in the form of a battery, for example. A power insufficiency may result where a charge level of the local energy source **762** is less than the power requirement to complete one or more of the end effector functions, for example

FIG. **212** is another logic flow diagram of a process **60900** depicting a control program or logic configuration for cooperatively controlling application of a therapeutic signal to a tissue grasped by an end effector (e.g. end effector **60502**) and controlling a function of the end effector in an application of a tissue treatment cycle. More specifically, the process **60900** is focused on addressing a deficiency in an end effector function such as, for example, a local power insufficiency to complete an end effector closure.

In various instances, the process **60850** can be implemented by any suitable RF energy source (e.g. RF energy source **794**) and any suitable surgical instrument such as, for example, surgical instruments **1000**, **60000** that include any suitable end effector such as, for example, end effectors **1300**, **60002**, **60502**. However, for brevity, the following description of the process **60850** will focus on its implementation in a surgical system that includes the RF energy source **794**, the surgical instrument **60000**, and the end effector **60502**, for example. In certain instances, the memory **68008** stores program instructions that, when executed by the processor **68002**, cause the processor **68002** to perform one or more aspects of the process **60900**.

In the illustrated example, the process **60900** is relevant to an application of an RF energy to a tissue grasped by the end effector **60502** below an optimal closure threshold due to a power insufficiency to complete the closure of the end effector **60502**. The RF energy source **794** can be configured to cause one or more of the electrode assemblies **60526**, **60527**, **60536**, **60536** to apply the RF energy to the tissue by passing a therapeutic signal through the tissue. The process **60900** includes detecting **60901** a charge level of a local energy source (e.g. energy source **794**) configured to supply power to a motor assembly configured to effect closure of the end effector **60502**. The process **60900** further includes adjusting **60902** a parameter of the therapeutic signal based on the charge level of the local energy source.

In certain instances, the control circuit **760** is configured to monitor a charge level of the local energy source **762**. In at least one example, the control circuit **760** employs a charge meter to monitor the charge level. If the charge level is below a predetermined threshold associated with an end effector function such as, for example, closure of the end effector, the control circuit may cause the RF energy source **794** to adjust a parameter of the therapeutic signal to compensate for the inability of the motor assembly responsible for the closure of the end effector to fully complete the closure function. In certain instances, the adjusted parameter of the therapeutic signal is power. The control circuit **760** can be configured to cause the RF energy source **794** to increase a power level of the therapeutic signal, for example, in response to determining that a charge level of the local energy source is below the predetermined threshold.

Energy Sealing, Sensing, and Algorithms Therefor

The surgical instrument **1000**, as described above in connection with FIGS. **1-13**, may be adapted and configured for energy sealing and sensing under the control of various algorithm as described hereinbelow in connection with

FIGS. 213-233. The surgical instrument 1000 comprises an energy delivery system 1900 and control circuit configured to seal tissue with electrical energy and to execute algorithms for sensing short circuits in the end effector 1300 jaws 1310, 1320. In particular, the following description is directed generally to algorithms for detecting RF short circuits in the end effector 1300 jaws 1310, 1320, determining system RF power levels (including deactivation) from the energy delivery system 1900, determining which portions of an electrode 1925 in the end effector 1300 jaws 1310, 1320 are energized, and indicating to a user, by way of the display 1190 in communication with the control system of the surgical instrument 1000, the status of the surgical instrument 1000 and an explanation of what is occurring within the surgical instrument 1000. Prior to describing the various algorithms that may be executed by the control circuit of the surgical instrument 1000, the description first turns to an explanation of the electrical/electronic operating environment in which the algorithms are executed for energy sealing and sensing operations.

FIG. 213 illustrates a control system 40600 for the surgical instrument 1000 described in connection with FIGS. 1-13 comprising a plurality of motors 40602, 40606 which can be activated to perform various functions, in accordance with at least one aspect of the present disclosure. It will be appreciated that the surgical instrument 1000 may comprise electronic control circuits having different configurations without limiting the scope of the present disclosure in this context. In certain instances, a first motor 40602 can be activated to perform a first function and a second motor 40606 can be activated to perform a second function, and so on. In certain instances, the plurality of motors 40602, 40606 of the control system 40600 can be individually activated to cause firing, closure, and/or articulation motions in the end effector. The firing, closure, and/or articulation motions can be transmitted to the end effector through a shaft assembly, for example.

In certain aspects, the control system 40600 may include a firing motor 40602. The firing motor 40602 may be operably coupled to a firing motor drive assembly 40604 which can be configured to transmit firing motions, generated by the motor 40602 to the end effector, in particular to displace the knife element. In certain instances, the firing motions generated by the motor 40602 may cause the staples to be deployed from the staple cartridge into tissue grasped by the end effector and/or the cutting edge of the knife element to be advanced to cut the grasped tissue, for example. The knife element may be retracted by reversing the direction of the motor 40602.

In certain aspects, the control system 40600 may include an articulation motor 40606, for example. The articulation motor 40606 may be operably coupled to an articulation motor drive assembly 40608, which can be configured to transmit articulation motions generated by the articulation motor 40606 to the end effector. In certain instances, the articulation motions may cause the end effector to articulate relative to the shaft, for example.

As described above, the control system 40600 may include a plurality of motors which may be configured to perform various independent functions. In certain aspects, the plurality of motors 40602, 40606 of the control system 40600 can be individually or separately activated to perform one or more functions while the other motors remain inactive. For example, the articulation motor 40606 can be activated to cause the end effector to be articulated while the firing motor 40602 remains inactive. Alternatively, the firing motor 40602 can be activated to fire the plurality of staples,

and/or to advance the cutting edge, while the articulation motor 40606 remains inactive.

Each of the motors 40602, 40606 may comprise a torque sensor to measure the output torque on the shaft of the motor. The force on an end effector may be sensed in any conventional manner, such as by force sensors on the outer sides of the jaws or by a torque sensor for the motor actuating the jaws.

In various aspects, as illustrated in FIG. 213, the control system 40600 may comprise a first motor driver 40626 to drive the firing motor 40602 and a second motor driver 40632 to drive the articulation motor 40606. In other aspects, a single motor driver may be employed to drive the firing and articulation motors 40602, 40606. In one aspect, the motor drivers 40626, 40632 each may comprise one or more H-Bridge field effect transistors (FETs). The firing motor driver 40626 may modulate the power transmitted from a power source 40628 to the firing motor 40602 based on input from a microcontroller 40578 (the "controller" or "control circuit"), for example. In certain instances, the microcontroller 40578 can be employed to determine the current drawn by the firing motor 40602, for example, while the firing motor 40602 is coupled to microcontroller 40578, as described above.

In certain aspects, the microcontroller 40578 may include a microprocessor 40622 (the "processor") and one or more non-transitory computer-readable mediums or memory units 40624 (the "memory") coupled to the processor 40622. In certain aspects, the memory 40624 may store various program instructions, which when executed may cause the processor 40622 to perform a plurality of functions and/or calculations described herein. In certain aspects, one or more of the memory units 40624 may be coupled to the processor 40622, for example.

In certain instances, the power source 40628 can be employed to supply power to the microcontroller 40578, for example. In certain instances, the power source 40628 may comprise a battery (or "battery pack" or "power pack"), such as a lithium-ion battery, for example. In certain instances, the battery pack may be configured to be releasably mounted to a handle for supplying power to the control system 40600. A number of battery cells connected in series may be used as the power source 40628. In certain instances, the power source 628 may be replaceable and/or rechargeable, for example.

In various instances, the processor 40622 may control the firing motor driver 40626 to control the position, direction of rotation, and/or velocity of the firing motor 40602. Similarly, the processor 40622 may control the articulation motor driver 40632 to control the position, direction of rotation, and/or velocity of the articulation motor 40606. In certain aspects, the processor 40622 can signal the motor drivers 40626, 40632 to stop and/or disable the firing or articulation motor 40602, 40606 coupled to the processor 40622. It should be understood that the term "processor" as used herein includes any suitable microprocessor, microcontroller, or other basic computing device that incorporates the functions of a computer's central processing unit (CPU) on an integrated circuit or, at most, a few integrated circuits. The processor 40622 is a multipurpose, programmable device that accepts digital data as input, processes it according to instructions stored in its memory 40624, and provides results as output. It is an example of sequential digital logic, as it has internal memory. Processors 40622 operate on numbers and symbols represented in the binary numeral system. In other aspects, the controller 40578 or control circuit may comprise analog or digital circuits such pro-

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grammable logic devices (PLD), field programmable gate arrays (FPGA), discrete logic, or other hardware circuits, software, and/or firmware, or other machine executable instructions to perform the functions explained in the following description.

In one aspect, the processor **40622** may be any single-core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In certain aspects, the microcontroller **40578** may be an LM 4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising an on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHZ, a prefetch buffer to improve performance above 40 MHZ, a 32 KB single-cycle SRAM, an internal ROM loaded with Stellaris Ware® software, a 2 KB EEPROM, one or more PWM modules, one or more QEI analogs, one or more 12-bit ADCs with 12 analog input channels, among other features that are readily available for the product datasheet. Other microcontrollers may be readily substituted for use with the control system **40600**. Accordingly, the present disclosure should not be limited in this context.

In certain aspects, the memory **40624** may include program instructions for controlling each of the firing and articulation motors **40602**, **40606** of the control system **40600** that are couplable to the processor **40622**. For example, the memory **40624** may include program instructions for controlling the firing motor **40602** and the articulation motor **40606**. Such program instructions may cause the processor **40622** to control the firing, closure, and articulation functions in accordance with inputs from algorithms or control programs of the surgical instrument or tool.

In certain aspects, the controller **40578** may be coupled to an RF generator **40574** and a plurality of electrodes **40500** disposed in the end effector via a multiplexer **40576**. The RF generator **40574** is configured to supply bipolar or monopolar RF energy either individually or in combination. In one aspect, the RF generator **40574** is configured to drive the segmented RF electrodes **40500** with an in-series current limiting element Z within the distal portion of the instrument for each electrode **40500**. The RF generator **40574** may be configured to sense a short circuit between the electrode **40550** and the return path **40510** by monitoring the output current, voltage, power, and impedance characteristics of the segmented electrode **40500**. In one aspect, the RF generator **40574** may be configured to actively limit the current through or redirect the current around a shorted electrode **40500** when a short is detected. This function also may be accomplished by the controller **40578** in combination with a switching element such as the multiplexer **40576**. The redirection or current limiting function may be controlled by the RF generator **40574** in response to a detected short circuit or electrode **40500** irregularity. If the RF generator **40574** is equipped with a display, the RF generator **40574** can display information to the user when a restricted electrode **40500** has been detected and the restriction on current may be removed when sensing of the short circuit is removed. The RF generator **40574** can engage the sensing and limiting functions as the tissue welding operation continues or at the start of a tissue welding operation. In one aspect, the RF generator **40574** may be a stand alone generator. In another aspect, the RF generator **40574** may be contained within the surgical instrument housing.

In one aspect, the RF generator may be configured to adapt the energy modality (monopolar/bipolar) RF applied to the end effector **1300** of the surgical instrument **1000**

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based on shorting or other tissue resistance, impedance, or irregularity. The RF monopolar/bipolar energy modality may be adapted by the RF generator **40574** or by the controller **40578** in combination with a switching element such as the multiplexer **40576**. In one aspect, the present disclosure provides a dual energy mode RF endocutter surgical instrument **1000** configured to apply monopolar or bipolar RF energy. Further, the RF generator **40574** can be configured to adjust the power level and percentage of each monopolar or bipolar RF energy modality based on tissue impedance conditions detected either by the RF generator **40574** or the controller **40578**. The energy modality adjustment function may comprise switching between bipolar and monopolar RF energy modalities, blending the bipolar and monopolar RF energy modalities, or blending of certain electrode segments **40500**₁₋₄. In one aspect, the independently controlled electrode segments **40500**₁₋₄ could be switched together as a group or as individual electrode segment-by-segment **40500**₁₋₄.

With reference now also to FIG. **239**, in various aspects, the dual energy mode RF endocutter surgical instrument **1000** may be employed with staples **44300** that have variable electrical conductivity along their body. In one aspect, the variable electrical conductivity staple **44300** may comprise a portion of the staple **44300** such as the deformable legs **44304**, **44306** that are electrically conductive and a portion of the staple **44300** such as the crown **44320** that has a different electrical conductivity from the deformable legs **44304**, **44306**. The electrical conductivity of the staple **44300** may vary based on its geometry or material composition such that when the staple **44300** is grasped in a shorting condition between the RF electrode **40500** and the return path **40510** of a dual mode RF energy/stapling combination surgical instrument **1000**, the variable conductivity of the staple **44300** may be advantageously exploited to prevent the staple **44300** from shorting one electrode **40500** to the other. In one aspect, the conductivity of the staple **44300** may be based on the temperature of the staple **44300**, current through the staple **44300**, or a portion of the staple **44300** having a high dielectric breakdown coefficient.

In one aspect, the surgical staple **44300** for a combination energy stapler surgical instrument **1000** comprises a crown **44302** defining a base **44301** and first and second deformable legs **44304**, **44306** extending from the each end of the base **44301**. A first electrically conductive material disposed on at least a first portion of the base **44301** and a second electrically conductive material disposed on at least a second portion of the base **44301**. The electrical conductivity of the first electrically conductive material is different from the electrical conductivity of the second electrically conductive material. In one aspect, the first and second electrically conductive materials are the same and the electrical conductivity varies based on different geometries of the first and second electrically conductive materials deposited on the first and second portions of the base **44301**. In one aspect, the first and second electrically conductive materials have different compositions and the electrical conductivity varies based on the different compositions of the first and second electrically conductive materials deposited on the first and second portions of the base **44301**. In one aspect, the first and second electrically conductive materials have similar geometries and different material compositions to provide different electrical conductivities.

With reference back to FIG. **213**, in other aspects, the controller **40578** may be coupled to one or more mechanisms and/or sensors to alert the processor **40622** to the program instructions that should be used in a particular

setting. For example, the sensors may alert the processor **40622** to use the program instructions associated with firing, closing, and articulating the end effector. In one aspect, the memory **40624** may store executable instructions to cause the processor **40622** to detect RF shorting in the end effector by monitoring one or more than one electrode **406234**. In another aspect, the memory **40624** may store executable instructions to cause the processor **40622** to determine RF power level (including deactivation). In other aspects, memory **40624** may store executable instructions to cause the processor **40622** to determine which portions of the electrodes **40500** are energized and indicate to the user of why and what is happening via a display **40625** coupled to the controller **40578**.

In one aspect, the memory **40624** may comprise executable instructions that when executed cause the processor **40622** to detect short circuits in the end effector and predict the electrode **40500** by the controller **40578** and in response adapt the RF energy path of the RF energy generated by the RF generator **40574**. In one aspect, the electrodes **40500** may be segmented RF electrodes with an in-series current limiting element within the distal portion of the control system **40600** for each electrode. Aspects of segmented electrodes are described hereinbelow in connection with FIGS. **214-217**. In other aspects, the memory **40624** may store executable instructions to cause the processor **40622** to sense a short circuit between an electrode **40500** and the return path **40510**. In other aspects, the memory **40624** may store executable instructions that when executed cause the processor **40622** to actively limit the current through or redirect the current around a shorted electrode **40500** when a short circuit is detected. In various aspects, the redirection or current limiting is performed by the controller **40578** or the RF generator **40574** in response to a detected short circuit or electrode **40500** irregularity. In various aspects, the controller **40578** can detect when an electrode **40574** has been restricted and can display that information to the user via the display **40625**. In various aspects, the current restriction function may be removed when the sensing of the short circuit is removed. In various aspects, the sensing and limiting functions can be engaged as the tissue welding process continues or at the start of a tissue welding process.

In various aspects, prior to applying therapeutic energy, the controller **40578** may apply a pre-sealing energy cycle to the electrode **40500** array at a lower than therapeutic level to provide an initial screen to scan for short circuits between the electrodes **40500** or between the electrodes **40500** and the return electrode **40510**. The multiplexer **40576** may cycle through the electrode **40500** array by sending low level signals out to determine if faults are present. This could be reported back to the RF generator **405774**. A contained system could then exclude channels where faults are or shorts are present and cycle through the remaining channels without the RF generator **40574** needing to adapt its output to the surgical instrument **1000**. In one aspect, coils may be employed as miniature metal detectors to determine presence of existing staples in a proposed energy path. In this example, the system is passive and does not pass an electric current through the tissue.

In certain aspects, the controller **40578** may be coupled to various sensors. The sensors may comprise position sensors which can be employed to sense the position of switches, for example. Accordingly, the processor **40622** may use the program instructions associated with firing the knife of the end effector upon detecting, through the sensors, for example, that the switch is in the first position; the processor **40622** may use the program instructions associated with

closing the anvil upon detecting, through the sensors for example, that the switch is in the second position; and the processor **40622** may use the program instructions associated with articulating the end effector upon detecting, through the sensors for example, that the switch is in the third or fourth position.

Additional sensors include, without limitation, arc detection sensors to measure AC ripple on the base RF waveform and to measure current $\Delta i/dt$. Other sensors include optical detectors and/or laparoscopic cameras to monitor specific frequencies or wavelengths in the visible, infrared (IR), or other portions of the electromagnetic spectrum. In one aspects, sensors to detect negative incremental resistance and RF arc temperature may be coupled to the controller **40578**. Other sensors, include environmental sensors to measure humidity, atmospheric pressure, temperature, or combinations thereof.

FIG. **214** shows a jaw **40524** of an end effector for the surgical instrument **1000** described in FIGS. **1-13** where the electrode **1925** shown in FIG. **6** is configured with multiple pairs of segmented RF electrodes **40500** disposed on a circuit board **40570**, or other type of suitable substrate, on a lower surface of the jaw **40524** (i.e., the surface of the jaw **40524** facing tissue during operation), in accordance with at least one aspect of the present disclosure. The various pairs of segmented RF electrodes **40500** are energized by an RF source (or generator) **40574**. A multiplexer **40576** may distribute the RF energy to the various pairs of segmented RF electrodes **40500** as desired under the control of a controller **40578**. According to various aspects, the RF source **40574**, the multiplexer **40576**, and the controller **40578** may be located in the energy delivery system **1900** extending through the shaft **1200** and the articulation joint **1400** and into the end effector **1300** of the surgical instrument **1000** as described in connection with FIGS. **1** and **6**. The RF energy is coupled between the electrodes **40500** and a return path **40510** back to the RF generator **40574**.

In the example of the pairs of segmented electrodes **40500** shown in FIG. **214**, the circuit board **40570** may comprise multiple layers that provide electrical connections between the multiplexer **40576** and the various pairs of segmented electrodes **40500**. For example, the circuit board **40570** may comprise multiple layers providing connections to the pairs of segmented electrodes **40500**. In one example, an upper most layer may provide connections to the most proximate pairs of segmented electrodes **40500**; a middle layer may provide connections to middle pairs of segmented electrodes **40500**; and a lowest layer may provide connections to most distal pairs of segmented electrodes **40500**. The pairs of segmented electrodes **40500** configuration, however, is not limited in this context.

FIG. **215** illustrate a multi-layer circuit board **40570**, in accordance with at least one aspect of the present disclosure. FIG. **215** shows a cross-sectional end view of the jaw **40524**. The circuit board **40570**, adjacent to staple pockets **50584**, comprises three conducting layers **40580**₁₋₃, having insulating layers **40582**₁₋₄ therebetween, showing how the various layers **40580**₁₋₃ may be stacked to connect back to the multiplexer **40576**.

An advantage of having multiple RF electrodes **40500** in the end effector **1300**, as shown in FIG. **6**, is that, in the case of a metal staple line or other electrically conductive object left in the tissue from a previous instrument firing or surgical procedure that may cause a short circuit of the electrodes **40500**, such a short situation could be detected by the RF generator **40574**, the multiplexer **40576**, and/or the control-

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ler 40578, and the energy may be modulated in a manner appropriate for the short circuit or adaptation of the energy path in response.

FIG. 216 shows segmented electrodes 40500 on either side of the knife slot 40516 in the jaw 40524 have different lengths, in accordance with at least one aspect of the present disclosure. In the illustrated example, there are four co-linear segmented electrodes, but the most distal electrodes 40500₁, 40500₂ are 10 mm in length, and the two proximate electrodes 40500₃, 40500₄ are 20 mm in length. Having shorter distal electrodes 40500₁, 40500₂ may provide the advantage of concentrating the therapeutic energy applied to the tissue.

FIG. 217 is a cross-sectional view of an end effector comprising a plurality of segmented electrodes 40500, in accordance with at least one aspect of the present disclosure. As shown in the example of FIG. 217, the segmented electrodes 40500 are disposed on the upper jaw 40524 (or anvil) of the end effector. In the illustrated example, the active segmented electrodes 40500 are positioned adjacent the knife slot 40516. A metal anvil portion of the jaw 40524 may serve as the return electrode. Insulators 40504, which may be made of ceramic, insulate the segmented electrodes 40500 from the metallic jaw 40524.

FIG. 218 shows a jaw 40524 of an end effector for the surgical instrument 1000 described in FIGS. 1-13 and 214 where multiple pairs of segmented RF electrodes 40500 include a series current limiting element Z within the distal portion of the end effector for each electrode, in accordance with at least one aspect of the present disclosure. The current limiting element Z is shown schematically in series with the multiplexer 40576, but may be disposed on the circuit board 40570 where the electrode elements are disposed. Accordingly, the controller 40578 or the RF generator 40574 may be configured to sense a short between an electrode 40500 and the return path 40510 and actively limit the current through or redirect the current around the shorted electrode 40500 when a short circuit is detected. In one aspect, the redirection or current limiting is done by the controller 40578 electronics in the surgical instrument 1000 (FIGS. 1-13) or the RF generator 40574 in response to a detected short circuit or electrode irregularity. In one aspect, the controller 40578 or RF generator 40574 can detect when an electrode 40500 has been restricted and can display that information to the user on the display 40625 (FIG. 213). In one aspect, the restriction on current is removed when the sensing of the short circuit is removed. In one aspect, the sensing and limiting can be engaged as the tissue welding process continues or at the start of the tissue welding process.

RF Shorting Detection Methods and Systems Therefor

With reference to FIGS. 1-13, the present disclosure now turns to a description of systems and methods for detecting RF shorting in the jaws 1310, 1320 of the end effector 1300 and determining the RF power level (including deactivation), and which portions of the electrode 1925 are energized and indicating to the user of why and what is happening via the display 1190. The systems and methods comprise detecting RF short circuiting in the jaws 1310, 1320 of the end effector 1300 employing algorithmic differentiation and detecting RF arching by monitoring the surgical instrument 1000. In one general aspect, a system and method comprises detecting an RF short circuit in the end effector 1300 by algorithmic differentiation between low impedance tissue

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grasped in the jaws 1310, 1320 of the end effector 1300 and a metallic short circuit between the electrode 1925 and the return path defined by the return electrode 1590. A detection/warning to surgeons of shorting risk is provided to the user via the display 1190. Algorithms utilized low power exploratory pulses prior to firing and can differentiate clips/staples and acceptable compared to unacceptable amounts of metal in the jaws 1310, 1320.

With reference also to FIGS. 213-218 above and 219-234 hereinbelow, the systems and methods for detecting RF short circuiting in the jaws 1310, 1320 of the end effector 1300 employing algorithmic differentiation and detecting RF arching by monitoring the surgical instrument 1000, will be described in connection with the control system 40600 for the surgical instrument 1000 shown in FIG. 213 and the segmented electrodes 40500 described in connection with FIGS. 213-218 and the graphical representations shown in FIGS. 219-232. Finally, the method will be further described in connection with the method 41900, 42000 described in connection with FIGS. 233 and 234.

In one general aspect, the present disclosure provides a system 40600 and method 41900 for detecting and predicting shorting of the of the electrode 1925 and/or the segmented electrode 40500 by the controller 40578 electronics and adaptation of the energy path in response thereto. In one aspect, the segmented RF electrodes 40500 may comprises an in-series current limiting element Z (FIG. 218) within the distal portion of the instrument 1000 jaw 40524 for each electrode 40500₁₋₄, among others, for example. In another aspect, the controller 40578 is configured to sense a short between the electrode 1925 and the return path defined by the return electrode 1590 or the electrode 40500 and the return path 40510. In yet another aspect, the controller 40578 may be configured to actively limit the current through or redirect the current around a shorted electrode 1925, 40500, when a short is detected. In yet another aspect, the controller 40578 or RF generator 40574 may be configured to redirect or limit the current through a shorted electrode element in response to a detected short or electrode irregularity. In yet another aspect, the controller 40578 may be configured to detect when an electrode 1925, 40500 has been restricted and can display that information to the user via the display 1190, 40625. Further, in yet another aspect, the controller 40578 may be configured to remove the restriction on current when the sensing of the short is removed. Further, in yet another aspect, the controller 40578 may be configured to engage short circuit sensing and current limiting as the tissue welding process continues or at the start of the tissue welding process.

Algorithmic Differentiation

With reference to FIGS. 1-13 and 213-228D, the present disclosure now turns to a description of one aspect of algorithmic differentiation between low impedance tissue conditions and a metallic short between the electrode 1925 and the return path electrode 1590 or the electrode 40500 and the return path 40510. Upon detecting a short circuit, the controller 40578 provides a warning to the surgeon of shorting risk. The algorithms utilize low power exploratory pulses prior to firing, differentiate clips/staples, acceptable compared to unacceptable amounts of metal, and detection of metal in the jaws 1320, 40524 causing energy control adjustments.

FIGS. 219-222 illustrate various graphical representations of low power exploratory pulse waveforms 41000, e.g., current, power, voltage, and impedance, applied to an elec-

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trodes 1925 or a segmented electrode 40500 to illustrate the algorithmic differentiation between low impedance tissue conditions and a metallic short between electrodes 1925, 40500 and the return path electrodes 1590, 40510. FIG. 219 is a graphical representation of exploratory pulse waveforms 41000 applied by the RF generator 40574 under control of the controller 40578 to an electrode 1925, 40500 to detect a metallic object shorting the electrode 1925, 40500 and the return path electrode 1590, 40510, in accordance with at least one aspect of the present disclosure. In particular, FIG. 219 depicts the application of low power exploratory pulse waveforms 41000 prior to firing or activating RF sealing energy in liver tissue that includes a metallic staple located in the field causing a short between an electrode 1925, 40500 and a return path electrode 1590, 40510. The exploratory pulse waveforms 41000 comprise a pulsed current waveform 41002, a pulsed power waveform 41004, a pulsed voltage waveform 41006, and a pulsed impedance waveform 41008 measured between the electrode 1925, 40500 and the return path electrode 1590, 40510 before and during a shorting event, which is shown in the detailed view in FIG. 220.

FIG. 220 is a detailed view of the exploratory pulse waveforms 41000 applied to an electrode 1925, 40500 during a shorting event, in accordance with at least one aspect of the present disclosure. The exploratory pulse waveforms 4100 are applied prior to firing or delivering therapeutic RF energy to seal tissue grasped between the jaws 1320 (40524), 1310 of the end effector 1300. As shown, during the shorting event period, the pulsed current waveform 41002 increases to a maximum value (e.g., $i_{max} \geq 3$ A) and at the same time the pulsed power waveform 41004 decreases to a minimum value (e.g., $p_{min} \leq 2$ W), the pulsed voltage waveform 41006 decreases to a minimum value (e.g., $v_{min} \leq 0.6$ V), and the pulsed impedance waveform 41008 decreases to a minimum value (e.g., $Z_{min} \leq 0.2$ Ohms). In one aspect, the shorting detection algorithm applies exploratory energy pulses monitors the values of the pulsed waveforms 41002, 41004, 41006, 41008 and compares them to predetermined values to determine if a short circuit is present between the jaws 1320 (40524), 1310 of the end effector 1300. The algorithm, then determines whether the exploratory pulse waveforms 41000 are due to a short circuit or low impedance tissue grasped between the jaws 1320 (40524), 1310 of the end effector 1300.

FIG. 221 is a graphical representations of exploratory pulse waveforms 41010 applied to an electrode 1925, 40500 prior to firing or delivering therapeutic RF energy to seal tissue grasped between the jaws 1320 (40524), 1310 of the end effector 1300, in accordance with at least one aspect of the present disclosure. The exploratory pulse waveforms 41010 are applied to low impedance tissue without the presence of a short circuit between the electrode 1925, 40500 and the return electrode 1590, 40510. The low impedance tissue exploratory pulse waveforms 41010 comprise a current waveform 41012, a power waveform 41014, a voltage waveform 41016, and an impedance waveform 41018.

FIG. 222 is a detailed view depicting the pulsed impedance waveform 41018 applied to tissue having an impedance of approximately 2Ω , in accordance with at least one aspect of the present disclosure. It has been determined that low tissue impedance is approximately in the range of 1Ω to 3Ω . As shown in FIG. 221, the value of the exploratory pulsed current waveform 41012 applied the low impedance tissue increases to about 2.8 A while the exploratory pulsed voltage waveform 41016 drops to about 5V and the exploratory

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pulsed power waveform 41014 drops to about 20 W. Testing of the RF generator 40574 identified tissue impedance $Z < 1\Omega$ as a short circuit compared to low impedance tissue impedance, which has been identified as $\sim 2\Omega$ and in the range of 1Ω to 3Ω .

With reference to FIGS. 213-222, in one aspect, the present disclosure provides a method of detecting shorting in a jaw 50524 of an end effector prior to initiating a tissue sealing (welding) cycle. Accordingly, the memory 40624 stores executable instructions that when executed by the processor 40622 cause the processor 40622 control the RF generator 40574 to generate a series of pre-cycle exploratory pulses as shown in FIGS. 219-222 to determine whether there is a short in the jaw 40524 of the end effector or whether tissue in contact with the jaw 40524 has a low impedance. Under the control of the processor 40622, the RF generator 40574 delivers pulses of non-therapeutic RF energy levels to the electrodes 40500₁ located at the distal end (nose) of the jaw 40524 at the initiation of an energy activation cycle. The nose pulse(s) is not detectable to a surgeon and it is part of the activation sequence. In one aspect, the pulse/detection period may be selected in the range of 0.1 to 1.0 seconds in duration. In other aspects, the pulse/detection period is less than 0.5 seconds in duration.

In other aspects, under control of the processor 40622 the RF generator 40574 generates nose pulse(s) with non-therapeutic energy level at the initiation of energy activation to provide shorting detection specificity. In one aspect, the RF generator 40574 generates a single pulse that is applied to all active/return electrodes 40500 simultaneously. In another aspect, the RF generator 40574 generates multiple pulses, each with different combinations of segments of active/return electrodes 40500 to enable extremely specific targeting of active/return electrodes 40500 when in therapeutic mode in order to seal around a detected short.

Still with reference to FIGS. 213-222, in one aspect, the present disclosure provides a method of detecting shorting in a jaw 50524 of an end effector during the tissue sealing cycle. Detection of shorts in the jaw 50524 within the tissue sealing cycle may be necessary when staples/clips are protected by the tissue and shorting may not occur until the tissue sealing cycle has begun. Such tissue protected staples/clips are not detectable using the pre sealing cycle nose pulse as described above. In this aspect, the controller 40578 of the control system 40600 is configured to react in real time to manage activation of one or more electrode segments 40500₁-40500₄. Accordingly, the memory 40624 may store executable instructions that when executed by the processor 40622 cause the processor 40622 to control the RF generator 40574 to generate and apply a continuous non-pulsed energy to the electrodes 40500 and determines real time rate changes or real time level thresholds of the current, power, voltage, and/or impedance. In one aspect, the processor 40622 is configured to detect decreased voltage, impedance, and/or power. In another aspect, the processor 40622 is configured to detect increased current.

In various other aspect, the memory 40624 may store executable instructions that when executed by the processor 40622 cause the processor 40622 to execute alternative detection techniques which are not based on energy flow. In one aspect, segmented thermocouples may be located at each active and/or return electrode 40500 location and the processor 40622 is configured to read the temperature of each thermocouple and to employ a heat signature at a location of a segmented electrode 40500 to determine the presence of a short.

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In another alternative detection aspect of the present disclosure, a coil pickup may be located at each active and/or return electrode **40500** location and the processor **40622** is configured to detect a magnetic field induced from electric output by the electrode **40500** segment. The coils may be employed as miniature metal detectors to determine the presence of existing staples in a proposed energy path. The coil detection system is passive and does include passing a current through the tissue to enable detection of a short.

In another alternative detection aspect of the present disclosure, a single frequency detector is employed to sense if a short has occurred in the jaw **40524**. In one aspect, the single frequency detector comprises two coils to detect a very low frequency (VLF) inductance or resistance. In another aspect, the single frequency detector employs pulse induction (PI) utilizing one coil for both transmit and receive functions and is good in saline environments. In yet another aspect, the single frequency detector comprises two coils and is configured to detect beat frequency oscillations (BFO).

In another alternative detection aspect of the present disclosure, a multiple frequency detector is employed to determine to sense if a short has occurred in the jaw **40524**. In one aspect, the multiple frequency detector may be configured short depth frequency (shallow target) or long depth frequency (deep target).

In another alternative detection aspect of the present disclosure, a balance device may be employed to remove unwanted signal of background environment (tissue, fluids). In one aspect, the balance device may employ manual or automatic adjustments. In automatic adjustments, the balance device determines the best balance settings. In one aspect, the balance device provides tracking adjustments where the balance device continuously adjusts based on current conditions of surrounding environment.

In another alternative detection aspect of the present disclosure, staple material may be selected for specific identification of shorts. In one aspect, the staple material composition may be made unique to the manufacturer. This technique may be employed to identify specific competitor staples.

In another alternative detection aspect of the present disclosure, coils may be positioned horizontal and/or vertical where gains/losses in signal depend upon the position of a foreign object relative to the coil. In one aspect, each coil is positioned surrounding each electrode **40500** on the deck or circuit board **40570**. In another aspect, the coils may be positioned/molded into a plastic cartridge wall surrounding the electrode **40500**.

In various other aspect, the memory **40624** may store executable instructions that when executed by the processor **40622** cause the processor **40622** to predict a short, prior to full shorting, by interrogating data in a pulsed energy application. In one aspect, the pulsing may enable prediction of shorting verse reaction to shorting. In one aspect, the energy profile may be a pulsed application rather than a continuous application of energy. Pulsing may provide an extra layer of information than non-pulsed energy techniques based on needing to ramp up energy repeatedly throughout a cycle.

FIGS. 223-228D illustrate several examples of energy activation in liver tissue that includes a metallic staple in the field as described in connection with FIGS. 219-222. FIG. 223 is a graphical representation of a first example of exploratory pulse waveforms **41100** applied by the RF generator **40574** under control of the controller **40578** to an

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electrode **1925**, **40500** to detect a metallic object shorting the electrode **1925**, **40500** and the return path electrode **1590**, **40510**.

FIG. 223 depicts the application of a first example of low power exploratory pulse waveforms **41100** prior to firing or activating RF sealing energy in liver tissue that includes a metallic staple located in the field causing a short between an electrode **1925**, **40500** and a return path electrode **1590**, **40510**, in accordance with at least one aspect of the present disclosure. The exploratory pulse waveforms **41100** comprise a pulsed current waveform **41102**, a pulsed power waveform **41104**, a pulsed voltage waveform **41106**, and a pulsed impedance waveform **41108** measured between the electrode **1925**, **40500** and the return path electrode **1590**, **40510** before and during a shorting event.

FIG. 224A is a detailed view of the impedance waveform **41108** component of the exploratory pulse waveforms **41100** during a transition to a short circuit between the electrode **1925**, **40500** and the return path electrode **1590**, **40510**, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the impedance **41108** decreases prior to reaching the short circuit impedance **41110** during the shorting event.

FIG. 224B is a detailed view of the power waveform **41104** component of the exploratory pulse waveforms **41100** during a transition to a short circuit between the electrode **1925**, **40500** and the return path electrode **1590**, **40510**, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the power **41104** decreases prior to reaching the short circuit power **41112** during the shorting event.

FIG. 224C is a detailed view of the voltage waveform **41106** component of the exploratory pulse waveforms **41100** during a transition to a short circuit between the electrode **1925**, **40500** and the return path electrode **1590**, **40510**, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the voltage **41106** decreases prior to reaching the short circuit voltage **41114** during the shorting event.

FIG. 224D is a detailed view of the current waveform **41102** component of the exploratory pulse waveforms **41100** during a transition to a short circuit between the electrode **1925**, **40500** and the return path electrode **1590**, **40510**, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the current **41102** increases prior to reaching the short circuit current **41116** during the shorting event.

FIG. 225 depicts the application of a second example of low power exploratory pulse waveforms **41200** prior to firing or activating RF sealing energy in liver tissue that includes a metallic staple located in the field causing a short between an electrode **1925**, **40500** and a return path electrode **1590**, **40510**, in accordance with at least one aspect of the present disclosure. The exploratory pulse waveforms **41200** comprise a pulsed current waveform **41202**, a pulsed power waveform **41204**, a pulsed voltage waveform **41206**, and a pulsed impedance waveform **41208** measured between the electrode **1925**, **40500** and the return path electrode **1590**, **40510** before and during a shorting event.

FIG. 226A is a detailed view of the impedance waveform **41208** component of the exploratory pulse waveforms **41200** during a transition to a short circuit between the electrode **1925**, **40500** and the return path electrode **1590**, **40510**, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the impedance **41208** decreases prior to reaching the short circuit impedance **41210** during the shorting event.

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FIG. 226B is a detailed view of the power waveform 41204 component of the exploratory pulse waveforms 41200 during a transition to a short circuit between the electrode 1925, 40500 and the return path electrode 1590, 40510, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the power 41204 decreases prior to reaching the short circuit power 41212 during the shorting event.

FIG. 226C is a detailed view of the voltage waveform 41206 component of the exploratory pulse waveforms 41200 during a transition to a short circuit between the electrode 1925, 40500 and the return path electrode 1590, 40510, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the voltage 41206 decreases prior to reaching the short circuit voltage 41214 during the shorting event.

FIG. 226D is a detailed view of the current waveform 41202 component of the exploratory pulse waveforms 41200 during a transition to a short circuit between the electrode 1925, 40500 and the return path electrode 1590, 40510, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the current 41202 increases prior to reaching the short circuit current 41216 during the shorting event.

FIG. 227 depicts the application of a second example of low power exploratory pulse waveforms 41300 prior to firing or activating RF sealing energy in liver tissue that includes a metallic staple located in the field causing a short between an electrode 1925, 40500 and a return path electrode 1590, 40510, in accordance with at least one aspect of the present disclosure. The exploratory pulse waveforms 41200 comprise a pulsed current waveform 41302, a pulsed power waveform 41304, a pulsed voltage waveform 41306, and a pulsed impedance waveform 41308 measured between the electrode 1925, 40500 and the return path electrode 1590, 40510 before and during a shorting event.

FIG. 228A is a detailed view of the impedance waveform 41308 component of the exploratory pulse waveforms 41300 during a transition to a short circuit between the electrode 1925, 40500 and the return path electrode 1590, 40510, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the impedance 41308 decreases prior to reaching the short circuit impedance 41310 during the shorting event.

FIG. 228B is a detailed view of the power waveform 41304 component of the exploratory pulse waveforms 41300 during a transition to a short circuit between the electrode 1925, 40500 and the return path electrode 1590, 40510, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the power 41304 increases prior to reaching the short circuit power 41312 during the shorting event.

FIG. 228C is a detailed view of the voltage waveform 41306 component of the exploratory pulse waveforms 41300 during a transition to a short circuit between the electrode 1925, 40500 and the return path electrode 1590, 40510, in accordance with at least one aspect of the present disclosure. As shown, prior to the shorting event, the voltage 41306 decreases prior to reaching the short circuit voltage 41314 during the shorting event.

FIG. 228D is a detailed view of the current waveform 41302 component of the exploratory pulse waveforms 41300 during a transition to a short circuit between the electrode 1925, 40500 and the return path electrode 1590, 40510. As shown, prior to the shorting event, the current 41302 increases prior to reaching the short circuit current 41316 during the shorting event.

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In one aspect, the exploratory waveforms define a ramp. The controller 40578 may be configured to compare an actual pulse ramp to a specified pulse ramp. Each pulse of energy application has a specified pulse ramp. The controller 40578 may be configured to identify a short circuit risk when the actual pulse ramp is different from the specified pulse ramp for a predefined voltage, current, or impedance exploratory waveform. In one aspect, the controller 40578 may be configured to compare present pulse data to pulse data of previous pulses including for example, moving average, etc. As previously described, the controller 40578 may identify a short circuit risk by detecting a decrease in voltage, impedance, or power or detecting an increase in current.

In one aspect, the controller 40578 may be configured to monitor the electrode 40500 or segments of the electrode 40500₁₋₄ to determine a level of shorting risk based on predicted or actual shorting conditions in the jaws 1310, 1320 (40524) of the end effector 1300. The controller 40578 may be configured to differentiate clips/staples from a short circuit condition, acceptable in contrast to unacceptable amounts of metal, and the location of metallic objects in the jaws 1310, 1320 (40524) of the end effector 1300.

A higher risk of shorting may be determined by the controller 40578 by differentiating between clips, staples, and unknown metallic objects in the sealing zone of the jaws 1310, 1320 (40524) of the end effector 1300. The controller 40578, however, may be configured to differentiate a clip by measuring the resistance where a clip has a lower resistance (micro ohms) based on the amount of metal in the clip compared to the staple. The controller 40578 may be configured to measure electrode 40500₁₋₄ segment temperature where the clip most likely has a lower temperature, based on lower resistance, compared to a staple. In one aspect, segmented thermocouples may be incorporated in the jaws of the end effector to measure the temperature at different locations in the jaws. In one aspect, a clip impedance ramp is different than a staple impedance ramp and the controller 40578 may be configured to determine the difference. For example, clips and staples have different inductive or capacitive reactance that can be monitored by the controller 40578 to determine risk of shorting. In one aspect, the controller 40578 may be configured to detect a short circuit between two or more adjacent segments of the segmented electrode 40500₁₋₄ to be indicative of a clip or multiple staples. In other aspects, the controller 40578 may be configured to detect a short circuit distributed over more than 25% of the segments in a segmented electrode 40500₁₋₄.

A lower risk may be determined by the controller 40578 when the RF sealing pathway is defined through a non-optimal path, i.e., opposed sealing path compared to offset sealing path. In other aspects, switching between bipolar to monopolar sealing presents a lower risk of shorting. Accordingly, the controller 40578 may be configured to assess a lower risk of shorting when the surgical instrument switches from bipolar to monopolar sealing. Also, a staple located in the sealing zone presents a lower risk of shorting as well as the presence of known metallic objects in the sealing zone. Also, lower risk may be determined when the staples are made of a unique metal for a particular RF/endocutter device such as the surgical instrument 1000. In such instance, the controller 40578 may be configured to differentiate how the surgical instrument 1000 responds to the presence of a known unique metal.

Upon detection of a short circuit condition in the jaws 1310, 1320 (40524) of the end effector 1300, the controller 40578 may be configured to output a warning to surgeons of a shorting risk through one or more than one user interfaces

that may be audible, visual, tactile, or combinations thereof. In one aspect, the controller **40578** may output a warning on the display **40625**. In addition to communicating the presence of a shorting risk to the surgeon, the controller **40578** may be configured to identify and inform of the location of the risk, the risk level, device changes in response to risk, and/or provide recommendations for surgeon action. In other aspects, the controller **40578** may be configured to communicate aggregated information to a simple yes/no, good/bad, ready to fire, etc. type concise communication. In other aspects, the controller **40578** may be configured to not communicate to the surgeon, but rather manage the shorting events or potential risks of sorting appropriately.

RF Arcing Detection

In one aspect, the controller **40578** may be configured to monitor the electrode **40500** or segments of the electrode **40500₁₋₄** to detect RF arcing. Arc detection may be implemented by the controller **40578** monitoring the operation and functionality of the surgical instrument **1000**. Potential arc detection/risk factors include excessive AC ripple on the base RF waveform. Accordingly, in one aspect, the controller **40578** or the RF generator **40574** may be configured to monitor excessive AC ripple on the base RF waveform. In another aspect, the controller **40578** may be configured to measure the RF current and determine the potential arc detection/risk by measuring an increasing current $\Delta di/dt$. In addition to or alternatively, the controller **40578** may be configured to monitor corona glow by employing an optical detector to make optical measurements. In one aspect, optical measurements may be made using a laparoscopic camera and monitoring specific frequencies in the visible, infrared (IR), or other portions of the electromagnetic spectrum. In one aspect, such optical measurement techniques may be integration into a surgical hub architecture. Other RF arcing detection techniques include, without limitation, configuring the controller **40578** to detect negative incremental resistance, which causes the electrical resistance to decrease as the arc temperature increases. Other environmental factors that may cause or exacerbate RF arcing that the controller **40578** may be taken into account in the configuration of the controller **40578** include monitoring a variety of sensors coupled to the controller to measure humidity, atmospheric pressure, temperature, or combinations thereof. Potential measurement tools include probes on the surgical instrument **1000** or the end effector **1300**, other devices or measurements taken in the operating room or coupled to a surgical hub, laparoscopes, etc.

The description now turns to several plots that depict electrical parameters associated with RF arcing. FIG. **229** is a graphical depiction **41400** of impedance **41402**, voltage **41406**, and current **41408** versus time (t), in accordance with at least one aspect of the present disclosure. At the time of the arc point **41410** an excess di/dt (current versus time) results in a steep rising current **41408** versus time (t) slope **41412** and a rapid decrease in impedance $-dZ/dt$ (negative impedance versus time) results in a steep falling impedance **41402** versus time (t) slope **41414**. This may be referred to as a negative incremental resistance that produces an electric arc. In one aspect, as previously described, the controller **40578** may be configured to monitor either the current **41408** versus time (t) slope **41412** (di/dt), the impedance **41402** versus time slope **41414** ($-dZ/dt$), or a combination thereof to predict the occurrence of the risk of an electric arc.

FIG. **230** is a graphical depiction **41500** of an electric arcing charge **41505** across a 1.8 cm gap in a 0.8 cm² area

relative to current **41502** and voltage **41506** waveforms, in accordance with at least one aspect of the present disclosure. As shown, the current **41502** rapidly increases until the arcing charge **41505** starts to rise. The voltage **41506** rises rapidly and the current **41502** drops. After the electric arc discharge, the voltage **41506** drops rapidly and the current **41502** decreases to zero.

FIG. **231** is a graphical depiction **41600** of electric discharge regimes as a function of voltage versus current, where current (Amps) is along the horizontal axis and voltage (Volts) is along the vertical axis, in accordance with at least one aspect of the present disclosure. As shown, the electric discharge starts in the dark regime **41620**, transitions to the glow discharge regime **41622**, and then transitions to the arc discharge regime **41624**. In the dark discharge regime **41620**, the voltage curve transitions from background ionization **41602** through a saturation regime **41604** to a corona region **41608**. In the glow discharge regime **41622**, the voltage **41610** drops after it reaches a breakdown voltage point and transitions from a normal glow **41618** region to an abnormal glow region. The voltage **41612** rises until it transitions into the arc discharge regime **41624**, at which point there is glow-to-arc transition where the voltage **41614** rapidly drops and creates first a non-thermal arc and then a thermal arc.

FIG. **232** is a graphical depiction **41700** of power (Watts) as a function of impedance (Ohms) of various tissue types, in accordance with at least one aspect of the present disclosure. As current **41702** is applied into low impedance tissue, the power **41704** is relatively low. As the tissue impedance starts to increase, the power **41704** increases until the impedance reaches ~1000 Ohms. At which point the power **41704** starts to decrease exponentially with increasing tissue impedance. As shown, the tissue impedance of prostate tissue **41706** in nonconductive solution is in the range of ~10 Ohms to ~1500 Ohms as energy is applied. The impedance of liver and muscle tissue **41708** is in the range of ~500 Ohms to ~1900 Ohms as energy is applied. The impedance of bowel tissue **41710** is in the range of ~1200 Ohms to ~2400 Ohms as energy is applied. The impedance of gall bladder tissue **41712** is in the range of ~1700 to ~3000 Ohms as energy is applied. The impedance of mesentery momentum tissue **41714** is in the range of ~2600 Ohms to ~3600 Ohms as energy is applied. The impedance of fat, scar, or adhesion tissue **41716** is in the range of ~3000 Ohms to ~4000 Ohms as energy is applied.

In various aspects, the controller **40578** may be configured to detect an electric arc discharge in real time. Multiple frequencies may be employed to detect the tissue state as indicated in FIG. **232**. Real time detection of electric arc discharges in real time can speed up diagnostics and can be configured to provide real time diagnostics in a tissue environment (pressure, moisture content) for different tissues as shown in FIG. **232**, for example.

FIG. **233** is a logic flow diagram of a method **41900** of detecting a short circuit in the jaws **1310**, **1320** (**40524**) of an end effector **1300** of a surgical instrument **1000** (see FIGS. **1-6** and **213-218**), in accordance with at least one aspect of the present disclosure. With reference also to FIGS. **6** and **213-218**, in accordance with the method **41900**, the memory **40624** may store a set of executable instructions that when executed cause to processor **40622** to execute the method **41900**. In accordance with the method **41900**, the processor **40622** causes the RF generator **40574** to apply **41902** a sub-therapeutic electrical signal to an electrode **40500** located in the jaw **1320** (**40524**) of the end effector **1300** to detect a short circuit. If the jaw **1320** comprises a

single longitudinal electrode 1925, the RF generator 40574 can apply the sub-therapeutic electrical signal directly to the single longitudinal electrode 1925. If the jaw 40524 comprises a segmented electrode 40500, the processor 40622 selects one of the segmented electrodes 40500 through the multiplexer 40576 and then causes the RF generator 40574 to apply the sub-therapeutic electrical signal to the selected electrode 40500. It should be appreciated that the sub-therapeutic electrical signal is a signal used to detect a short circuit between the electrode 1925 (40500) and the return electrode 1590 (40510) without causing any therapeutic effects on the tissue grasped in the end effector 1300.

In accordance with the method 41900, based on the signals received by the processor 40622 after applying the sub-therapeutic electrical signals, the processor 40622 determines 41904 if the electrode 1925 (40500) is shorted to the return electrode 1590 (40510). If the electrode 1925 (40500) is not shorted, the method 41900 continues along the NO path and the processor 40622 causes the RF generator 40574 to apply 41918 therapeutic RF electrical energy to the electrode 1925 (40500). In contrast, if the electrode 1925 (40500) is shorted, the method 41900 continues along the YES path and the processor 40622 modifies the electrical current through the shorted electrode 1925 (40500). In one aspect, the processor 40622 limits 41906 the electrical current through the shorted electrode 1925 (40500). In one aspect, if the jaw 1320 comprises a single electrode 1925, the processor 40622 causes the RF generator 40574 to limit 41906 the output current. In another aspect, if the jaw 40524 comprises a segmented electrode 40500, the processor 40622 through the multiplexer 40576 selectively redirects 41908 the current path around the shorted electrode 40500 through the current limiter Z coupled to a distal electrode segment. In either case, the processor 40622 causes the display 40625 to display 41910 information about the detected shorted electrode 1925 (40500) to the surgeon or other members of the surgical team.

In accordance with the method 41900, the processor 40622 determines 41912 if the electrode 1925 (40500) is still shorted. If the electrode 1925 (40500) is still shorted, the method 41900 continues along the YES path and the processor 40622 continues to limit 41906 or redirect 41908 the electrical current applied to the shorted electrode 1925 (40500). If the electrode 1925 (40500) is no longer shorted, the method 41900 continues along the NO branch and the processor 40622 removes 41914 the electrical current limit restriction through the electrode 1925 (40500) or removes 41918 the electrical current redirection around the electrode 1925 (40500). The processor 40622 then causes the RF generator 40574 to apply 41918 therapeutic RF electrical energy to the electrode 1925 (40500).

FIG. 234 is a logic flow diagram of a method 42000 of detecting a short circuit in the jaws 1310, 1320 (40524) of an end effector 1300 of a surgical instrument 1000 (see FIGS. 1-6 and 213-218), in accordance with at least one aspect of the present disclosure. With reference also to FIGS. 6 and 213-218, in accordance with the method 42000, the memory 40624 may store a set of executable instructions that when executed cause to processor 40622 to execute the method 42000. In accordance with the method 42000, the processor 40622 causes the multiplexer 40576 to select 42002 an electrode 40500₁₋₄ in an array of segmented electrodes 40500. The processor 40622 causes the RF generator 40574 to apply 42004 a sub-therapeutic electrical signal to the selected electrode 40500₁₋₄ located in the jaw 40524 of the end effector 1300 to detect a short circuit.

In accordance with the method 42000, based on the signals received by the processor 40622 after applying the sub-therapeutic electrical signals, the processor 40622 determines 42006 if the selected electrode 40500₁₋₄ is shorted to the return electrode 40510. If the selected electrode 40500₁₋₄ is not shorted, the method 42000 continues along the NO path and the processor 40622 selects 42008 the next electrode 40500₁₋₄ in the array of electrodes 40500 through the multiplexer 40576 and then tests the newly selected electrode 40500₁₋₄ until all segmented electrodes 40500₁₋₄ have been tested for shorts. If any one of the selected electrodes 40500₁₋₄ is shorted, the method 42000 continues along the YES path and the processor 40622 modifies the electrical current through the shorted electrode 40500₁₋₄. In one aspect, the processor 40622 selectively modifies the current through the shorted electrode 40500₁₋₄ through the multiplexer 40576 either to limit 42010 the electrical current through the shorted selected electrode 40500₁₋₄ or redirect 42012 electrical current around the shorted electrode 40500₁₋₄, the causes the RF generator 40574 to apply 41918 therapeutic RF electrical energy to the selected electrode 40500₁₋₄. In one aspect, the processor 40622 through the multiplexer 40576 redirects 41908 the current path around the shorted electrode 40500 through the current limiter Z coupled to a distal electrode segment. In either case, the processor 40622 causes the display 40625 to display 42014 information about the detected shorted electrode 40500₁₋₄ to the surgeon or other members of the surgical team.

In accordance with the method 42000, the processor 40622 determines 42016 if the selected electrode 40500₁₋₄ is still shorted. If the selected electrode 40500₁₋₄ is still shorted, the method 42000 continues along the YES path and the processor 40622 continues to limit 42010 or redirect 42012 the electrical current to the shorted selected electrode 40500₁₋₄. If the selected electrode 40500₁₋₄ is no longer shorted, the method 42000 continues along the NO branch and the processor 40622 removes 42018 the electrical current limit restriction through the selected electrode 40500₁₋₄ or removes 42020 the electrical current redirection around the selected electrode 40500₁₋₄. The processor 40622 then causes the RF generator 40574 to apply 42022 therapeutic RF electrical energy to the selected electrode 40500₁₋₄.

In various aspects, the processor 40622 determines 42006 if the electrode 1925 (40500) is shorted to the return electrode 1590 (40510) as described in FIG. 233 or determines 42006 if the selected electrode 40500₁₋₄ is shorted to the return electrode 40510 as described in FIG. 234 using the techniques described in FIGS. 219-228D. For example, with reference to FIGS. 219-222, the processor 40622 may be configured to distinguish a shorted electrode from low impedance tissue. In one aspect, with reference to 219-220, the processor 40622 controls the RF generator 40574 to apply a sequence of exploratory pulse waveforms 41000 to the electrode 1925 (40500). In one aspect, the exploratory pulses are applied prior to firing or delivering therapeutic RF energy to seal tissue grasped between the jaws 1320 (40524), 1310 of the end effector 1300. The processor 40622 monitors the exploratory waveforms 41000 and determines that an electrode 1925 (40500) is shorted when the pulsed current waveform 41002 increases to a maximum value (e.g., $i_{max} \geq 3$ A) and at the same time the pulsed power waveform 41004 decreases to a minimum value (e.g., $p_{min} \leq 2$ W), the pulsed voltage waveform 41006 decreases to a minimum value (e.g., $v_{min} \leq 0.6$ V), and the pulsed impedance waveform 41008 decreases to a minimum value less than 1 Ohm (e.g., $Z_{min} \leq 0.2$ Ohms). The processor 40622 distinguishes a shorted electrode 1925 (40500) from low

impedance tissue as described in connection with FIGS. 221-222 when the tissue impedance is approximately in the range of 1Ω to 3Ω . As shown in FIG. 221, the value of the exploratory pulsed current waveform 41012 applied the low impedance tissue increases to about 2.8 A while the exploratory pulsed voltage waveform 41016 drops to about 5V and the exploratory pulsed power waveform 41014 drops to about 20 W. Testing of the RF generator 40574 identified tissue impedance $Z < 1\Omega$ as a short circuit compared to low impedance tissue impedance, which has been identified as $\sim 2\Omega$ and in the range of 1Ω to 3Ω .

Dual Energy Modality Combination Surgical Instrument

With reference to FIGS. 1-6 and 213-218, in one aspect, the surgical instrument 1000 may be configured as a dual energy modality combination energy device with switchable/blendable energy modalities. The controller 40578 is configured to adapt energy modality (monopolar/bipolar) RF endocutter based on shorting or other tissue resistance, impedance, or irregularity. In one aspect, the surgical instrument 1000 RF endocutter may be configured to apply monopolar or bipolar RF energy to the segmented electrodes 40500. In one aspect, the power level and percentage of each energy modality may be adjusted based on the low resistance tissue conditions detected by the controller 40578. As previously described, the controller 40578 comprises a memory 40624 storing executable instructions and a processor 40622 configured to execute the instructions and adjust energy modalities. In one aspect, the processor 40622 may be configured to interchange the energy modalities from bipolar to monopolar RF, blend the two energy modalities, or blend certain electrode segments 40500₁₋₄ only. In one aspect, the processor 40622 may be configured to independently control the electrode segments 40500₁₋₄ to switch together as a group or as individual electrode segments 40500₁₋₄ on a segment-by-segment basis.

As described in connection with FIGS. 213-228D, the controller 40578 may be configured to determine the difference between a shorting event and a low impedance tissue event for use in controlling or switching the energy modalities. In one aspect, the controller 40578 may be configured to blend or switch the energy modality to prevent shorting of the segmented electrodes 40500₁₋₄ by metallic contact. In one aspect, the controller 40578 may be configured to determine that a segment of an electrode 40500₁₋₄ is shorted by the presence of a metallic staple within the jaws 1310, 1320 (40524) of the end effector 1300 or by the jaws 1310, 1320 (40524) physically touching one another. Upon determining that there is a shorting event, the controller 40578 first blends the energy modality and then switches the energy modality from bipolar to monopolar if blending the energy modalities does not resolve the short circuit event below an arcing discharge (e.g., sparking) threshold as described above in connection with FIGS. 229-232.

In one aspect, for example, if the controller 40578 determines that monopolar and bipolar return paths 40510 are open simultaneously, blending will not occur as desired because energy will take the path of least resistance, which may bypass the desired energy modality path. Therefore, the processor 40622 may be configured to selectively control the multiplexer 40576 to switch between monopolar/bipolar energy paths as necessary, such that the energy modality return paths are not open simultaneously. The processor 40622 can consider blending the energy modalities at the time in which active switching is occurring.

In one aspect, the processor 40622 may alternate active switching (energy modality blending) between bipolar and monopolar in accordance with the following technique. During bipolar energization, the processor 40622 through the multiplexer 40576 opens the bipolar energy return path 40510 with all the electrodes 40500₁₋₄ turned on and any shorted electrodes 40500₁₋₄ off. During monopolar energization, the processor 40622 through the multiplexer 40576 opens the monopolar return path with only the shorted active electrode 40500₁₋₄ turned on. In another aspect, during monopolar energization, the processor 40622 through the multiplexer 40576 opens the monopolar return path with all electrodes 40550₁₋₄ turned on.

In another aspect, the processor 40622 may be configured to adjust energy modality or balance based on sensed tissue impedance limit. The processor 40622 may be configured to sense a parameter of the tissue grasped within the jaws 1310, 1320 (40524) of the end effector 1300 to interrogate if a conductive element or other metallic object is located within the tissue in the jaws 1310, 1320 (40524). As discussed above in connection with FIGS. 213-228D, the controller 40578 may be configured to apply several exploratory pulse waveforms of non-therapeutic energy to the electrodes 40500₁₋₄ during a pre-energy activation cycle. The exploratory pulses may be applied prior to firing or delivering therapeutic RF energy to seal the tissue. The RF exploratory pulse waveforms may comprise multiple high frequency waves transmitted through the electrodes 40500₁₋₄. The return signal may be employed to determine various tissue parameters including the type of cutting/coagulation desired such as electrosurgical cutting, fulguration, desiccation, or time based. In one aspect, the processor 40622 may employ the impedance readings to determine the tissue type as described above in connection with FIG. 232. In one aspect, the initial power settings may be based on known tissue parameters and subsequent power setting may be adapted based on measurements or readings of tissue impedance, for example.

In one aspect, tissue parameters may be sensed utilizing ferroelectric ceramic materials. Ferroelectricity is a characteristic of certain materials that have a spontaneous electric polarization (P) that can be reversed by the application of an external electric field (E). Three examples of spontaneous electric polarization by ferroelectric ceramic materials are shown in FIGS. 235-237. FIG. 235 shows a dielectric polarization plot 41800 where polarization (P) is a linear 41802 function of external electric field (E), in accordance with at least aspect of the present disclosure. FIG. 236 shows a paraelectric polarization plot 41820 where polarization (P) is a non-linear 41822 function of external electric field (E) exhibiting a sharp transition from negative to positive polarization at the origin, in accordance with at least aspect of the present disclosure. FIG. 237 shows ferroelectric polarization plot 41840 where polarization (P) is a non-linear 41842 function of external electric field (E) exhibiting hysteresis around the origin, in accordance with at least aspect of the present disclosure. Examples of ferroelectric ceramic materials include, barium titanate, ceramics incorporated with a metallic wire, or ceramic coatings applied on staples. Barium titanate is a ceramic with dielectric constant values as high as 7,000. Over a narrow temperature range, values as high as 15,000 are possible.

FIG. 238 is logic flow diagram of a method 43000 of adapting energy modality due to a short circuit or tissue type grasped in the jaws 1310, 1320 (40524) of an end effector 1300 of a surgical instrument 1000, in accordance with at least one aspect of the present disclosure. With reference

also to FIGS. 6 and 213-218, in one aspect, the processor 40622 selects 43002 an electrode 40500₁₋₄ in an array of segmented electrodes 40500 through the multiplexer 40576. During a pre-energy activation cycle, the processor 40622 causes the RF generator 40574 to apply 43004 a sub-therapeutic electrical signal to the selected electrode 40500₁₋₄ to differentiate between a shorted electrode and low impedance tissue grasped in the jaws 1310, 1320 (40524) of the end effector 1300. Based on a measured parameter received by the processor 40622 after applying the sub-therapeutic electrical signal, the processor 40622 determines 43006 if the selected electrode 40500₁₋₄ is shorted.

In accordance with the method 43000, if the selected electrode 40500₁₋₄ is shorted, the method 43000 proceeds along the YES path and the processor 40622 causes the RF generator 40574 to blend 43008 monopolar and bipolar RF energy. After a period of time of applying blended monopolar and bipolar RF energy, the processor 40622 determines 43010 if the selected electrode 40500₁₋₄ is still shorted. If the selected electrode 40500₁₋₄ is still shorted the method 43000 proceeds along the YES path and the processor 40622 switches 43012 the output energy of the RF generator 40574 between monopolar and bipolar RF energy through the multiplexer 40576 and continues determining 43010 if the selected electrode is still shorted.

In accordance with the method 43000, when the processor 40622 determines 43006, 43010 that the selected electrode 40500₁₋₄ is no longer shorted, the method 43000 proceeds along the NO path and the processor 40622 senses 43014 parameters of tissue grasped within the jaws 1310, 1320 (40524) of the end effector 1300. As described above in connection with FIG. 232, the processor 40622 determines 43016 the type of tissue based on the sensed tissue parameter such as impedance or other measured parameters. As described in connection with FIG. 33, the tissue impedance of prostate tissue 41706 in nonconductive solution is in the range of ~10 Ohms to ~1500 Ohms as energy is applied. The impedance of liver and muscle tissue 41708 is in the range of ~500 Ohms to ~1900 Ohms as energy is applied. The impedance of bowel tissue 41710 is in the range of ~1200 Ohms to ~2400 Ohms as energy is applied. The impedance of gall bladder tissue 41712 is in the range of ~1700 to ~3000 Ohms as energy is applied. The impedance of mesentery momentum tissue 41714 is in the range of ~2600 Ohms to ~3600 Ohms as energy is applied. The impedance of fat, scar, or adhesion tissue 41716 is in the range of ~3000 Ohms to ~4000 Ohms as energy is applied. Once the type of tissue is determined 43016, the processor 40622 determines 443018 a suitable procedure for cutting/coagulation based on the tissue type and applies 43200 the determined cutting/coagulation procedure to the tissue.

Accordingly, during the execution of the method 4300 and the application of the RF monopolar or bipolar RF energy, the processor 40622 controls the power level and/or percentage of each energy modality and adjusts the power level and percentage of each energy modality based on the low resistance tissue conditions detected. The processor 40622 may adjust energy modality by switching between bipolar to monopolar, blending of the two energy modalities, or blending a subset of the electrode segments 40500₁₋₄. In other aspects, the processor 40622 is configured to independently control the electrode segments 40500₁₋₄ to switch together as a group or as an individual segment-by-segment process.

Controlled Reaction to RF Shorting from Previous Staple Line

FIG. 239 illustrates a staple 44300 comprising a crown 44302 defining a base 44301 and deformable legs 44304,

44306 extending from each end of the base 44301, in accordance with at least one aspect of the present disclosure. Similar to the above, the staple cartridge recesses can be configured to guide and/or deform the legs 44304, 44306 when they contact the stapler cartridge. In one aspect, the crown 44302 includes a material 44303 disposed on the base 44301, where the material may be overmolded or coated onto the base 44301. As discussed in greater detail below, the material 44303 can be comprised of a material such as, for example, an electrically insulative material, a material having variable electrical resistance that increases resistance as the staple 44300 become heated, or a variable resistance thermally sensitive material, each of which is described in detail hereinbelow. In at least one of these aspects, the material 44303 may be formed around a single continuous wire comprising base 44301 and deformable legs 44304, 44306. In other aspects, the deformable legs 44304, 44306 can include separate deformable members embedded in a material 44303. Further, in various aspects, the wire comprising the base 44301 can be deformed to provide the recesses and anvils described above.

In one aspect, the controller 40578 may be configured for monitor a controlled reaction of the staple 44300 due to RF shorting from a previous staple line. In one aspect, the present disclosure provides a RF endocutter surgical instrument 1000 for use with staples 44300 that have variable electrical conductivity along their body and in one aspect along the crown 44302 or base 44301 portion of the crown 44302. In one aspect the staple 44300 may comprise a portion having a first electrical conductivity and another portion having a second electrical conductivity, where the first and second electrical conductivities are different. In one aspect, the electrical conductivity of the staple 44300 may vary based on geometry or material aspects. For example, when the staple 44300 is grasped in a shorting condition between the RF electrode 40524 and the return path 40510 of a energy/stapling combination device such as the RF endocutter surgical instrument 1000, the variable conductivity prevents the staple 44300 from shorting electrodes 40500 against each other. In various other aspects, the electrical conductivity of the staple 44300 may be based on the temperature of the staple 44300, electrical current through the staple 44300, or a portion of the staple 44300, such as the base 44301 or other portion of the crown 44302, having a high dielectric breakdown coefficient.

In various aspects, the present disclosure provides various staple configurations to minimize the chance of shorting by modifying or adjusting the electrical conductivity of the staple 44300. In one aspect, the staple 44300 may be configured such that a non-bendable crown 44302 portion of the staple 44300 is electrically insulated to minimize the likelihood that the next firing results in shorting while the end effector 1300 is clamped across a previously deployed staple 44300 embedded in the tissue. In one aspect, the non-bendable crown 44302 portion of the staple 44300 may be formed of an electrically insulative material or may comprise an absorbable polymer to minimize shorting. In other aspects, an absorbable insulating material may double as a driver to eliminate the driver in the cartridge stack of the end effector 1300.

In various aspects of the present disclosure, the crown 4132 portion of or the entire staple 44300 may comprise electrically insulative portions formed of electrically insulative materials or may comprise an electrically insulative material 44303 overmolded onto the base 44301 of the staple 44300. In one aspect, the electrically insulative material 44303 may be overmolded over the crown 44302, or

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base **44301**, of the staple **44300**. In one aspect, the electrically insulative material **44303** may be overmolded or applied to the staple **44300** in the form of a coating having a thickness in the range of 0.0005 inches to 0.0015 inches and typically about 0.001 inches. In one aspect, the staple **44300** may be overmolded onto the crown **44302** portion or base portion **44301** of the staple **44300** with a lactide and glycolide copolymer plus calcium stearate coating similar to the material known under the common name Vikryl. The thickness of the coating material **44303** may be in the range of 0.0005 inches to 0.0015 inches and typically about 0.001 inches.

In various aspects of the present disclosure, the crown **44302** portion of the staple **44300** or the entire staple **44300** may be dipped or coated in a polyimide material **44303** such as, for example, a polyimide film known under the common name Kapton developed by DuPont. Polyimide provides high dielectric strength to resist shorts. Various polyimide materials **44303** that are suitable candidates for coating or dipping the staple **44300** are described in U.S. Pat. No. 6,686,437 titled MEDICAL IMPLANTS MADE OF WEAR-RESISTANT, HIGH-PERFORMANCE POLYIMIDE, PROCESS OF MAKING SAME AND MEDICAL USE OF SAME, which is herein incorporated by reference. Other polyimide materials for dipping or coating the staple **44300** include, without limitation, a polymer known under the common name Parylene C, which has a high dielectric strength of approximately 6800 V and may be applied by vapor deposition.

In various aspects of the present disclosure, the crown **44302** portion, or base **44301** portion, of the staple **44300** or the entire staple **44300** may comprise ferroelectric ceramic materials **44303**. Ferroelectric materials **44303** may be characterized as having a spontaneous electric polarization that can be reversed by the application of an external electric field, as described in FIGS. 235-237, for example. In one aspect, metal detector coils may be embedded in the jaw **1310** of the end effector **1300** that is opposed to the jaw **1320** (40524) comprising the electrodes **1925**, **40500**. In this configuration, the embedded metal detector coils may be energized to induce an electric field in the staple to cause a polarization change of the ferroelectric material. The polarization change of the ferroelectric material lowers the electrical conductivity of the staple **44300** and thereby prevent the staple **44300** from short circuiting. Other ferroelectric materials **44303** include, without limitation, barium titanate and lead zirconate titanate. Barium titanate is a ceramic material **44303** having dielectric constant values as high as approximately 7,000. Over a narrow temperature range, dielectric constant values as high as 15,000 may be achievable.

In various aspects of the present disclosure, the crown **44302** portion, or the base **44301** portion, of or the entire staple **44300** may comprise a Polyisobutene material **44303**, a class of organic polymers prepared by polymerization of isobutene. Examples of Polyisobutene materials **44303** that may be employed are described in U.S. Pat. No. 8,927,660 titled CROSSLINKABLE POLYISOBUTYLENE-BASED POLYMERS AND MEDICAL DEVICES CONTAINING THE SAME, which is incorporated herein by reference.

In various aspects, the present disclosure provides a staple **44300** made of a wire material having an electrical resistance that is temperature dependent such that electrical resistivity increases as its temperature increases to minimize shorting of staples from previous firings. Accordingly, when the staple wire is placed in a short circuit condition, its temperature increases. The increase in temperature increases

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the electrical resistivity of the staple wire. Accordingly, in one aspect, the staple **44300** may be characterized as a variable electrical resistance where the resistance increase as the temperature of the staple increases when the staple is under a short circuit condition. This characteristic may be realized by making the staple wire from a metal/material hybrid such as, for example, the materials used to make resistance temperature devices (RTDs) or any metal/material that employs the resistance/temperature relationship of metals. Accordingly, the variable electrical resistance staple may be made of a length of wire wrapped around a ceramic core, for example. The temperature resistive wire may be made of a material, such as platinum, nickel, or copper, for example. The temperature/resistance relationship of the material can be used to increase the electrical resistance of the staple **44300** as its temperature increases under a short circuit condition. The temperature resistive wire may be housed in a protective layer of material.

In various aspects, the present disclosure provides a staple wire material that increases its electrical resistance based on the temperature of the staple wire to minimize shorting of staples from previous firings. Variable resistance thermally sensitive staples **44300** may be employed in the RF endocutter surgical instrument **1000** described herein. In one aspect, the temperature resistive wire material may be made of a metallic alloy which has a positive temperature coefficient where the electrical resistance increases as a function of temperature. Therefore, as the staple **44300** heats up under a short circuit condition, the electrical resistance of the staple wire increases to minimize the effects of a short circuit. In addition, the staple wire material has the material properties of staples. The staple wire material may be similar to a light bulb filament where the resistance to electrical current of the metal wire increases as the metal wire gets hotter so it does not short and melt. In one aspect, the staple wire may be a cobalt-nickel-chromium-molybdenum-tungsten-iron alloy, for example.

The entire disclosures of U.S. Pat. No. 8,070,034, entitled SURGICAL STAPLER WITH ANGLED STAPLE BAYS, which issued on Dec. 6, 2011, U.S. Pat. No. 10,143,474, entitled SURGICAL STAPLER, which issued on Dec. 4, 2018, and U.S. Pat. No. 7,611,038, entitled DIRECTIONALLY BIASED STAPLE AND ANVIL ASSEMBLY FOR FORMING THE STAPLE, which issued on Nov. 3, 2009, are incorporated by reference herein.

The entire disclosures of U.S. Pat. No. 8,424,735, entitled VARIABLE COMPRESSION SURGICAL FASTENER CARTRIDGE, which issued on Apr. 23, 2013, U.S. Pat. No. 7,722,610, entitled MULTIPLE COIL STAPLE AND STAPLE APPLIER, which issued on May 25, 2010, and U.S. Pat. No. 8,056,789, entitled STAPLE AND FEEDER BELT CONFIGURATIONS FOR SURGICAL STAPLER, which issued on Nov. 15, 2011, are incorporated by reference herein.

The entire disclosure of U.S. Pat. No. 6,843,403, entitled SURGICAL CLAMPING, CUTTING AND STAPLING DEVICE, which issued on Jan. 18, 2005, is incorporated by reference herein.

The entire disclosures of U.S. Pat. No. 8,070,034, entitled SURGICAL STAPLER WITH ANGLED STAPLE BAYS, which issued on Dec. 6, 2011, U.S. Pat. No. 10,143,474, entitled SURGICAL STAPLER, which issued on Dec. 4, 2018, and U.S. Pat. No. 7,611,038, entitled DIRECTIONALLY BIASED STAPLE AND ANVIL ASSEMBLY FOR FORMING THE STAPLE, which issued on Nov. 3, 2009, are incorporated by reference herein. The entire disclosures of U.S. Pat. No. 8,424,735, entitled VARIABLE COM-

PRESSION SURGICAL FASTENER CARTRIDGE, which issued on Apr. 23, 2013, U.S. Pat. No. 7,722,610, entitled MULTIPLE COIL STAPLE AND STAPLE APPLIER, which issued on May 25, 2010, and U.S. Pat. No. 8,056,789, entitled STAPLE AND FEEDER BELT CONFIGURATIONS FOR SURGICAL STAPLER, which issued on Nov. 15, 2011, are incorporated by reference herein. The entire disclosure of U.S. Pat. No. 6,843,403, entitled SURGICAL CLAMPING, CUTTING AND STAPLING DEVICE, which issued on Jan. 18, 2005, is incorporated by reference herein.

The surgical instrument systems described herein have been described in connection with the deployment and deformation of staples; however, the embodiments described herein are not so limited. Various embodiments are envisioned which deploy fasteners other than staples, such as clamps or tacks, for example. Moreover, various embodiments are envisioned which utilize any suitable means for sealing tissue. For instance, an end effector in accordance with various embodiments can comprise electrodes configured to heat and seal the tissue. Also, for instance, an end effector in accordance with certain embodiments can apply vibrational energy to seal the tissue.

The entire disclosures of:

U.S. Pat. No. 5,403,312, entitled ELECTROSURGICAL HEMOSTATIC DEVICE, which issued on Apr. 4, 1995;

U.S. Pat. No. 7,000,818, entitled SURGICAL STAPLING INSTRUMENT HAVING SEPARATE DISTINCT CLOSING AND FIRING SYSTEMS, which issued on Feb. 21, 2006;

U.S. Pat. No. 7,422,139, entitled MOTOR-DRIVEN SURGICAL CUTTING AND FASTENING INSTRUMENT WITH TACTILE POSITION FEEDBACK, which issued on Sep. 9, 2008;

U.S. Pat. No. 7,464,849, entitled ELECTRO-MECHANICAL SURGICAL INSTRUMENT WITH CLOSURE SYSTEM AND ANVIL ALIGNMENT COMPONENTS, which issued on Dec. 16, 2008;

U.S. Pat. No. 7,670,334, entitled SURGICAL INSTRUMENT HAVING AN ARTICULATING END EFFECTOR, which issued on Mar. 2, 2010;

U.S. Pat. No. 7,753,245, entitled SURGICAL STAPLING INSTRUMENTS, which issued on Jul. 13, 2010;

U.S. Pat. No. 8,393,514, entitled SELECTIVELY ORIENTABLE IMPLANTABLE FASTENER CARTRIDGE, which issued on Mar. 12, 2013;

U.S. patent application Ser. No. 11/343,803, entitled SURGICAL INSTRUMENT HAVING RECORDING CAPABILITIES, now U.S. Pat. No. 7,845,537;

U.S. patent application Ser. No. 12/031,573, entitled SURGICAL CUTTING AND FASTENING INSTRUMENT HAVING RF ELECTRODES, filed Feb. 14, 2008;

U.S. patent application Ser. No. 12/031,873, entitled END EFFECTORS FOR A SURGICAL CUTTING AND STAPLING INSTRUMENT, filed Feb. 15, 2008, now U.S. Pat. No. 7,980,443;

U.S. patent application Ser. No. 12/235,782, entitled MOTOR-DRIVEN SURGICAL CUTTING INSTRUMENT, now U.S. Pat. No. 8,210,411;

U.S. patent application Ser. No. 12/235,972, entitled MOTORIZED SURGICAL INSTRUMENT, now U.S. Pat. No. 9,050,083.

U.S. patent application Ser. No. 12/249,117, entitled POWERED SURGICAL CUTTING AND STAPLING

APPARATUS WITH MANUALLY RETRACTABLE FIRING SYSTEM, now U.S. Pat. No. 8,608,045;

U.S. patent application Ser. No. 12/647,100, entitled MOTOR-DRIVEN SURGICAL CUTTING INSTRUMENT WITH ELECTRIC ACTUATOR DIRECTIONAL CONTROL ASSEMBLY, filed Dec. 24, 2009, now U.S. Pat. No. 8,220,688;

U.S. patent application Ser. No. 12/893,461, entitled STAPLE CARTRIDGE, filed Sep. 29, 2012, now U.S. Pat. No. 8,733,613;

U.S. patent application Ser. No. 13/036,647, entitled SURGICAL STAPLING INSTRUMENT, filed Feb. 28, 2011, now U.S. Pat. No. 8,561,870;

U.S. patent application Ser. No. 13/118,241, entitled SURGICAL STAPLING INSTRUMENTS WITH ROTATABLE STAPLE DEPLOYMENT ARRANGEMENTS, now U.S. Pat. No. 9,072,535;

U.S. patent application Ser. No. 13/524,049, entitled ARTICULATABLE SURGICAL INSTRUMENT COMPRISING A FIRING DRIVE, filed on Jun. 15, 2012, now U.S. Pat. No. 9,101,358;

U.S. patent application Ser. No. 13/800,025, entitled STAPLE CARTRIDGE TISSUE THICKNESS SENSOR SYSTEM, filed on Mar. 13, 2013, now U.S. Pat. No. 9,345,481;

U.S. patent application Ser. No. 13/800,067, entitled STAPLE CARTRIDGE TISSUE THICKNESS SENSOR SYSTEM, filed on Mar. 13, 2013, now U.S. Patent Application Publication No. 2014/0263552;

U.S. Patent Application Publication No. 2007/0175955, entitled SURGICAL CUTTING AND FASTENING INSTRUMENT WITH CLOSURE TRIGGER LOCKING MECHANISM, filed Jan. 31, 2006; and

U.S. Patent Application Publication No. 2010/0264194, entitled SURGICAL STAPLING INSTRUMENT WITH AN ARTICULATABLE END EFFECTOR, filed Apr. 22, 2010, now U.S. Pat. No. 8,308,040, are hereby incorporated by reference herein.

Examples

Example 1—A method of adapting energy modality due to a short circuit or tissue type grasped in the jaws of an end effector of a surgical instrument, the method comprising selecting, by a control circuit through a multiplexer, an electrode in an array of segmented electrodes during a pre-energy activation cycle, applying, by the control circuit through a radio frequency (RF) generator, a sub-therapeutic electrical signal to the selected electrode to differentiate between a shorted electrode and low impedance tissue grasped in the jaws of the end effector, determining, by the control circuit, the selected electrode is shorted based on a measured electrical parameter received by the control circuit after applying the sub-therapeutic electrical signal, blending, by control circuit through the RF generator, monopolar and bipolar RF energy, determining, by the control circuit, that the selected electrode is shorted, and switching, by the control circuit through the multiplexer, output energy of the RF generator between monopolar and bipolar RF energy.

Example 2—The method of Example 1 comprising determining, by the control circuit, that the selected electrode is not shorted, and sensing, by the control circuit, an electrical parameter of the tissue grasped within the jaws of the end effector.

Example 3—The method of Examples 1 or 2, comprising determining, by the control circuit, tissue type based on the sensed electrical parameter of the tissue.

Example 4—The method of Examples 1, 2, or 3, comprising determining, by the control circuit, an energy sealing and cutting procedure based on the tissue type, and applying, by the control circuit, the determined energy sealing and cutting procedure to the tissue.

Example 5—The method of Examples 1, 2, 3, or 4, wherein blending comprises adjusting, by the control circuit, a power level of each energy modality applied to the selected electrode based on the measured electrical parameter.

Example 6—The method of Examples 1, 2, 3, 4, or 5, wherein blending comprises adjusting, by the control circuit, a percentage of each energy modality applied to the selected electrode based on the measured electrical parameter.

Example 7—The method of Examples 1, 2, 3, 4, 5, or 6, wherein the measured electrical parameter is any one of impedance, tissue impedance, current, power, or voltage, or any combinations thereof.

Example 8—The method of Examples 1, 2, 3, 4, 5, 6, or 7, wherein blending comprises blending, by the control circuit, a subset of electrode segments in the array of segmented electrodes as a group.

Example 9—The method of Examples 1, 2, 3, 4, 5, 6, 7, or 8, wherein blending comprises blending, by the control circuit, a subset of electrode segments in the array of segmented electrodes individually.

Example 10—A surgical instrument comprising an end effector comprising a first and second jaw configured to grasp tissue therebetween, wherein one jaw comprises an electrode and the other jaw comprises a return electrode, a control circuit coupled to the end effector electrode and return electrode, wherein the control circuit is configured to selecting, by a control circuit through a multiplexer, an electrode in an array of segmented electrodes during a pre-energy activation cycle, applying, by the control circuit through a radio frequency (RF) generator, a sub-therapeutic electrical signal to the selected electrode to differentiate between a shorted electrode and low impedance tissue grasped in the jaws of the end effector, determining, by the control circuit, the selected electrode is shorted based on a measured electrical parameter received by the control circuit after applying the sub-therapeutic electrical signal, blending, by control circuit through the RF generator, monopolar and bipolar RF energy, determining, by the control circuit, that the selected electrode is shorted, and switching, by the control circuit through the multiplexer, output energy of the RF generator between monopolar and bipolar RF energy.

Example 11—The surgical instrument of Example 10, wherein the control circuit is configured to determine that the selected electrode is not shorted and sense an electrical parameter of the tissue grasped within the jaws of the end effector.

Example 12—The surgical instrument of Examples 10 or 11, wherein the control circuit is configured to determine tissue type based on the sensed electrical parameter of the tissue.

Example 13—The surgical instrument of Examples 10, 11, or 12, wherein the control circuit is configured to determine an energy sealing and cutting procedure based on the tissue type and apply the determined energy sealing and cutting procedure to the tissue.

Example 14—The surgical instrument of Examples 10, 11, 12, or 13, wherein to blend the monopolar and bipolar RF energy, the control circuit is configured to adjust a power level of each energy modality applied to the selected electrode based on the measured electrical parameter.

Example 15—The surgical instrument of Examples 10, 11, 12, 13, or 14, wherein to blend the monopolar and

bipolar RF energy, the control circuit is configured to adjust a percentage of each energy modality applied to the selected electrode based on the measured electrical parameter.

Example 16—The surgical instrument of Examples 10, 11, 12, 13, 14, or 15, wherein the measured electrical parameter is any one of impedance, tissue impedance, current, power, or voltage, or any combinations thereof.

Example 17—The surgical instrument of Examples 10, 11, 12, 13, 14, 15, or 16, wherein to blend the monopolar and bipolar RF energy, the control circuit is configured to blend a subset of electrode segments in the array of segmented electrodes as a group.

Example 18—The surgical instrument of Examples 10, 11, 12, 13, 14, 15, 16, or 17, wherein to blend the monopolar and bipolar RF energy, the control circuit is configured to blend a subset of electrode segments in the array of segmented electrodes individually.

Although various devices have been described herein in connection with certain embodiments, modifications and variations to those embodiments may be implemented. Particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment may be combined in whole or in part, with the features, structures or characteristics of one or more other embodiments without limitation. Also, where materials are disclosed for certain components, other materials may be used. Furthermore, according to various embodiments, a single component may be replaced by multiple components, and multiple components may be replaced by a single component, to perform a given function or functions. The foregoing description and following claims are intended to cover all such modification and variations.

The devices disclosed herein can be designed to be disposed of after a single use, or they can be designed to be used multiple times. In either case, however, a device can be reconditioned for reuse after at least one use. Reconditioning can include any combination of the steps including, but not limited to, the disassembly of the device, followed by cleaning or replacement of particular pieces of the device, and subsequent reassembly of the device. In particular, a reconditioning facility and/or surgical team can disassemble a device and, after cleaning and/or replacing particular parts of the device, the device can be reassembled for subsequent use. Those skilled in the art will appreciate that reconditioning of a device can utilize a variety of techniques for disassembly, cleaning/replacement, and reassembly. Use of such techniques, and the resulting reconditioned device, are all within the scope of the present application.

The devices disclosed herein may be processed before surgery. First, a new or used instrument may be obtained and, when necessary, cleaned. The instrument may then be sterilized. In one sterilization technique, the instrument is placed in a closed and sealed container, such as a plastic or TYVEK bag. The container and instrument may then be placed in a field of radiation that can penetrate the container, such as gamma radiation, x-rays, and/or high-energy electrons. The radiation may kill bacteria on the instrument and in the container. The sterilized instrument may then be stored in the sterile container. The sealed container may keep the instrument sterile until it is opened in a medical facility. A device may also be sterilized using any other technique known in the art, including but not limited to beta radiation, gamma radiation, ethylene oxide, plasma peroxide, and/or steam.

While this invention has been described as having exemplary designs, the present invention may be further modified within the spirit and scope of the disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles.

The foregoing detailed description has set forth various forms of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, and/or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. Those skilled in the art will recognize that some aspects of the forms disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as one or more program products in a variety of forms, and that an illustrative form of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution.

Instructions used to program logic to perform various disclosed aspects can be stored within a memory in the system, such as dynamic random access memory (DRAM), cache, flash memory, or other storage. Furthermore, the instructions can be distributed via a network or by way of other computer readable media. Thus a machine-readable medium may include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer), but is not limited to, floppy diskettes, optical disks, compact disc, read-only memory (CD-ROMs), and magneto-optical disks, read-only memory (ROMs), random access memory (RAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic or optical cards, flash memory, or a tangible, machine-readable storage used in the transmission of information over the Internet via electrical, optical, acoustical or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.). Accordingly, the non-transitory computer-readable medium includes any type of tangible machine-readable medium suitable for storing or transmitting electronic instructions or information in a form readable by a machine (e.g., a computer).

As used in any aspect herein, the term “control circuit” may refer to, for example, hardwired circuitry, programmable circuitry (e.g., a computer processor including one or more individual instruction processing cores, processing unit, processor, microcontroller, microcontroller unit, controller, digital signal processor (DSP), programmable logic device (PLD), programmable logic array (PLA), or field programmable gate array (FPGA)), state machine circuitry, firmware that stores instructions executed by programmable circuitry, and any combination thereof. The control circuit may, collectively or individually, be embodied as circuitry

that forms part of a larger system, for example, an integrated circuit (IC), an application-specific integrated circuit (ASIC), a system on-chip (SoC), desktop computers, laptop computers, tablet computers, servers, smart phones, etc. Accordingly, as used herein “control circuit” includes, but is not limited to, electrical circuitry having at least one discrete electrical circuit, electrical circuitry having at least one integrated circuit, electrical circuitry having at least one application specific integrated circuit, electrical circuitry forming a general purpose computing device configured by a computer program (e.g., a general purpose computer configured by a computer program which at least partially carries out processes and/or devices described herein, or a microprocessor configured by a computer program which at least partially carries out processes and/or devices described herein), electrical circuitry forming a memory device (e.g., forms of random access memory), and/or electrical circuitry forming a communications device (e.g., a modem, communications switch, or optical-electrical equipment). Those having skill in the art will recognize that the subject matter described herein may be implemented in an analog or digital fashion or some combination thereof.

As used in one or more aspects of the present disclosure, a microcontroller may generally comprise a memory and a microprocessor (“processor”) operationally coupled to the memory. The processor may control a motor driver circuit generally utilized to control the position and velocity of a motor, for example. In certain instances, the processor can signal the motor driver to stop and/or disable the motor, for example. In certain instances, the microcontroller may be an LM 4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), internal read-only memory (ROM) loaded with Stellaris Ware® software, 2 KB electrically erasable programmable read-only memory (EEPROM), one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QEI) analog, one or more 12-bit Analog-to-Digital Converters (ADC) with 12 analog input channels, among other features that are readily available for the product datasheet.

It should be understood that the term processor as used herein includes any suitable microprocessor, or other basic computing device that incorporates the functions of a computer’s central processing unit (CPU) on an integrated circuit or at most a few integrated circuits. The processor is a multipurpose, programmable device that accepts digital data as input, processes it according to instructions stored in its memory, and provides results as output. It is an example of sequential digital logic, as it has internal memory. Processors operate on numbers and symbols represented in the binary numeral system. In at least one instance, the processor may be any single core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. Nevertheless, other suitable substitutes for microcontrollers and safety processor may be employed, without limitation.

As used in any aspect herein, the term “logic” may refer to an app, software, firmware and/or circuitry configured to perform any of the aforementioned operations. Software may be embodied as a software package, code, instructions, instruction sets and/or data recorded on non-transitory computer readable storage medium. Firmware may be embodied

as code, instructions or instruction sets and/or data that are hard-coded (e.g., nonvolatile) in memory devices.

As used in any aspect herein, the terms “component,” “system,” “module” and the like can refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution.

As used in any aspect herein, an “algorithm” refers to a self-consistent sequence of steps leading to a desired result, where a “step” refers to a manipulation of physical quantities and/or logic states which may, though need not necessarily, take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It is common usage to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like. These and similar terms may be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities and/or states.

A network may include a packet switched network. The communication devices may be capable of communicating with each other using a selected packet switched network communications protocol. One example communications protocol may include an Ethernet communications protocol which may be capable permitting communication using a Transmission Control Protocol/Internet Protocol (TCP/IP). The Ethernet protocol may comply or be compatible with the Ethernet standard published by the Institute of Electrical and Electronics Engineers (IEEE) titled “IEEE 802.3 Standard”, published in December, 2008 and/or later versions of this standard. Alternatively or additionally, the communication devices may be capable of communicating with each other using an X.25 communications protocol. The X.25 communications protocol may comply or be compatible with a standard promulgated by the International Telecommunication Union-Telecommunication Standardization Sector (ITU-T). Alternatively or additionally, the communication devices may be capable of communicating with each other using a frame relay communications protocol. The frame relay communications protocol may comply or be compatible with a standard promulgated by Consultative Committee for International Telegraph and Telephone (CCITT) and/or the American National Standards Institute (ANSI). Alternatively or additionally, the transceivers may be capable of communicating with each other using an Asynchronous Transfer Mode (ATM) communications protocol. The ATM communications protocol may comply or be compatible with an ATM standard published by the ATM Forum titled “ATM-MPLS Network Interworking 2.0” published August 2001, and/or later versions of this standard. Of course, different and/or after-developed connection-oriented network communication protocols are equally contemplated herein.

As used in any aspect herein, a wireless transmission such as, for example, a wireless communication or a wireless transfer of a data signal can be achieved, by a device including one or more transceivers. The transceivers may include, but are not limited to cellular modems, wireless mesh network transceivers, Wi-Fi® transceivers, low power wide area (LPWA) transceivers, and/or near field communications transceivers (NFC). The device may include or may be configured to communicate with a mobile telephone, a sensor system (e.g., environmental, position, motion, etc.) and/or a sensor network (wired and/or wireless), a computing system (e.g., a server, a workstation computer, a desktop computer, a laptop computer, a tablet computer (e.g., iPad®, GalaxyTab® and the like), an ultraportable computer, an ultramobile computer, a netbook computer and/or a subnote-

book computer; etc. In at least one aspect of the present disclosure, one of the devices may be a coordinator node.

The transceivers may be configured to receive serial transmit data via respective universal asynchronous receiver-transmitters (UARTs) from a processor to modulate the serial transmit data onto an RF carrier to produce a transmit RF signal and to transmit the transmit RF signal via respective antennas. The transceiver(s) can be further configured to receive a receive RF signal via respective antennas that includes an RF carrier modulated with serial receive data, to demodulate the receive RF signal to extract the serial receive data and to provide the serial receive data to respective UARTs for provision to the processor. Each RF signal has an associated carrier frequency and an associated channel bandwidth. The channel bandwidth is associated with the carrier frequency, the transmit data and/or the receive data. Each RF carrier frequency and channel bandwidth is related to the operating frequency range(s) of the transceiver(s). Each channel bandwidth is further related to the wireless communication standard and/or protocol with which the transceiver(s) may comply. In other words, each transceiver may correspond to an implementation of a selected wireless communication standard and/or protocol, e.g., IEEE 802.11 a/b/g/n for Wi-Fi® and/or IEEE 802.15.4 for wireless mesh networks using Zigbee routing.

One or more drive systems or drive assemblies, as described herein, employ one or more electric motors. In various forms, the electric motors may be a DC brushed driving motor, for example. In other arrangements, the motor may include a brushless motor, a cordless motor, a synchronous motor, a stepper motor, or any other suitable electric motor. The electric motors may be powered by a power source that in one form may comprise a removable power pack. Batteries may each comprise, for example, a Lithium Ion (“LI”) or other suitable battery. The electric motors can include rotatable shafts that operably interface with gear reducer assemblies, for example. In certain instances, a voltage polarity provided by the power source can operate an electric motor in a clockwise direction wherein the voltage polarity applied to the electric motor by the battery can be reversed in order to operate the electric motor in a counter-clockwise direction. In various aspects, a microcontroller controls the electric motor through a motor driver via a pulse width modulated control signal. The motor driver can be configured to adjust the speed of the electric motor either in clockwise or counter-clockwise direction. The motor driver is also configured to switch between a plurality of operational modes which include an electronic motor braking mode, a constant speed mode, an electronic clutching mode, and a controlled current activation mode. In electronic braking mode, two terminal of the drive motor 200 are shorted and the generated back EMF counteracts the rotation of the electric motor allowing for faster stopping and greater positional precision.

Unless specifically stated otherwise as apparent from the foregoing disclosure, it is appreciated that, throughout the foregoing disclosure, discussions using terms such as “processing,” “computing,” “calculating,” “determining,” “displaying,” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system’s registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

One or more components may be referred to herein as “configured to,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that “configured to” can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

Those skilled in the art will recognize that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flow diagrams are presented in a sequence(s), it should be understood that the various operations may be

performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

It is worthy to note that any reference to “one aspect,” “an aspect,” “an exemplification,” “one exemplification,” and the like means that a particular feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. Thus, appearances of the phrases “in one aspect,” “in an aspect,” “in an exemplification,” and “in one exemplification” in various places throughout the specification are not necessarily all referring to the same aspect. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner in one or more aspects.

In this specification, unless otherwise indicated, terms “about” or “approximately” as used in the present disclosure, unless otherwise specified, means an acceptable error for a particular value as determined by one of ordinary skill in the art, which depends in part on how the value is measured or determined. In certain embodiments, the term “about” or “approximately” means within 1, 2, 3, or 4 standard deviations. In certain embodiments, the term “about” or “approximately” means within 50%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, or 0.05% of a given value or range.

In this specification, unless otherwise indicated, all numerical parameters are to be understood as being prefaced and modified in all instances by the term “about,” in which the numerical parameters possess the inherent variability characteristic of the underlying measurement techniques used to determine the numerical value of the parameter. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter described herein should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques.

Any numerical range recited herein includes all sub-ranges subsumed within the recited range. For example, a range of “1 to 10” includes all sub-ranges between (and including) the recited minimum value of 1 and the recited maximum value of 10, that is, having a minimum value equal to or greater than 1 and a maximum value equal to or less than 10. Also, all ranges recited herein are inclusive of the end points of the recited ranges. For example, a range of “1 to 10” includes the end points 1 and 10. Any maximum numerical limitation recited in this specification is intended to include all lower numerical limitations subsumed therein, and any minimum numerical limitation recited in this specification is intended to include all higher numerical limitations subsumed therein. Accordingly, Applicant reserves the right to amend this specification, including the claims, to expressly recite any sub-range subsumed within the ranges expressly recited. All such ranges are inherently described in this specification.

Any patent application, patent, non-patent publication, or other disclosure material referred to in this specification and/or listed in any Application Data Sheet is incorporated by reference herein, to the extent that the incorporated materials is not inconsistent herewith. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by

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reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more forms has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more forms were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various forms and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

What is claimed is:

1. A method comprising:
 - determining, by a control circuit, that a first electrode of an array of segmented electrodes is at least temporarily shorted to a second electrode in the array of segmented electrodes;
 - blending, by control circuit through an RF generator, monopolar and bipolar RF energy to the array of segmented electrodes after determining that the first electrode is at least temporarily shorted to the second electrode; and
 - determining, by the control circuit, that the first electrode remains at least temporarily shorted to the second electrode in the array of segmented electrodes after blending the monopolar and bipolar RF energy to the array of segmented electrodes.
2. The method of claim 1, wherein determining, by the control circuit, that the first electrode is at least temporarily shorted to the second electrode is based on a measured electrical parameter received by the control circuit after applying a sub-therapeutic electrical signal to the first electrode.
3. The method of claim 2, wherein the measured electrical parameter comprises any one of: impedance, tissue impedance, current, power, or voltage, or any combinations thereof.
4. The method of claim 2, wherein blending comprises adjusting, by the control circuit, a power level of the monopolar RF energy applied to the first electrode based on the measured electrical parameter and a power level of the bipolar RF energy applied to the first electrode based on the measured electrical parameter.
5. The method of claim 2, wherein blending comprises adjusting, by the control circuit, a percentage of the monopolar RF energy applied to the first electrode based on the measured electrical parameter and a percentage of the bipolar RF energy applied to the first electrode based on the measured electrical parameter.
6. The method of claim 2, wherein the sub-therapeutic electrical signal is configured to differentiate between a shorted electrode and low impedance tissue grasped in jaws of an end effector.
7. The method of claim 1, comprising:
 - switching, by the control circuit, output energy between monopolar and bipolar RF energy to the array of segmented electrode while the first electrode remains at least temporarily shorted to the second electrode in the array of segmented electrodes.

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8. The method of claim 1, comprising:
 - determining, by the control circuit, that the first electrode is not shorted to any other electrode in array of segmented electrodes; and
 - sensing, by the control circuit, an electrical parameter of tissue grasped within jaws of an end effector.
9. The method of claim 8, comprising determining, by the control circuit, tissue type based on the sensed electrical parameter of the tissue.
10. The method of claim 9, comprising:
 - determining, by the control circuit, an energy sealing and cutting procedure based on the tissue type; and
 - applying, by the control circuit, the determined energy sealing and cutting procedure to the tissue.
11. The method of claim 1, wherein blending comprises applying, by the control circuit, the monopolar RF energy to a subset of shorted electrode segments in the array of segmented electrodes as a group and applying, by the control circuit, the bipolar RF energy to a subset of non-shorted electrode segments in the array of segmented electrodes as a group.
12. The method of claim 1, wherein blending comprises applying, by the control circuit, the monopolar RF energy and/or the bipolar RF energy to a subset of electrode segments in the array of segmented electrodes individually.
13. A surgical instrument, comprising:
 - an end effector comprising a first and second jaw configured to grasp tissue therebetween, and an array of segmented electrodes coupled to the first and/or second jaw;
 - a control circuit coupled to the array of segmented electrodes, wherein the control circuit is configured to:
 - determine that a first electrode of the array of segmented electrodes is at least temporarily shorted to a second electrode of the array of segmented electrodes;
 - blend monopolar and bipolar RF energy to the array of segmented electrodes after determining that the first electrode is at least temporarily shorted to the second electrode; and
 - determine that the first electrode remains at least temporarily shorted to the second electrode after blending the monopolar and bipolar RF energy to the array of segmented electrodes.
14. The surgical instrument of claim 13,
 - wherein the control circuit is configured to determine that the first electrode is at least temporarily shorted to the second electrode based on a measured electrical parameter received by the control circuit after applying a sub-therapeutic electrical signal to the first electrode, and
 - wherein the measured electrical parameter comprises any one of: impedance, tissue impedance, current, power, or voltage, or any combinations thereof.
15. The surgical instrument of claim 14, wherein to blend the monopolar and bipolar RF energy, the control circuit is configured to adjust a power level of the monopolar RF energy applied to the first electrode based on the measured electrical parameter and adjust a power level of the bipolar RF energy applied to the first electrode based on the measured electrical parameter.
16. The surgical instrument of claim 14, wherein to blend the monopolar and bipolar RF energy, the control circuit is configured to adjust a percentage of the monopolar RF energy applied to the first electrode based on the measured

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electrical parameter and adjust a percentage of the bipolar RF energy applied to the first electrode based on the measured electrical parameter.

17. The surgical instrument of claim 13, wherein the control circuit is configured to switch output energy between monopolar and bipolar RF energy to the array of segmented electrode while the first electrode remains at least temporarily shorted to the second electrode in the array of segmented electrodes.

18. The surgical instrument of claim 13, wherein to blend the monopolar and bipolar RF energy, the control circuit is configured to apply the monopolar RF energy to a subset of shorted electrode segments in the array of segmented electrodes as a group and apply the bipolar RF energy to a subset of non-shortened electrode segments in the array of segmented electrodes as a group.

19. The surgical instrument of claim 13, wherein to blend the monopolar and bipolar RF energy, the control circuit is configured to apply the monopolar RF energy and/or the bipolar RF energy to a subset of electrode segments in the array of segmented electrodes individually.

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20. A surgical instrument, comprising:
an end effector comprising a first and second jaw configured to grasp tissue therebetween, and an array of segmented electrodes coupled to the first and/or second jaw;

a control circuit coupled to the array of segmented electrodes, wherein the control circuit is configured to:
determine that a first electrode of the array of segmented electrodes is at least temporarily shorted to a second electrode of the array of segmented electrodes;

blend monopolar and bipolar RF energy to the array of segmented electrodes after determining that the first electrode is at least temporarily shorted to the second electrode;

determine that the first electrode is not shorted to the second electrode after blending monopolar and bipolar RF energy to the array of segmented electrodes;
sense an electrical parameter of tissue grasped within the jaws of the end effector; and

determine an energy sealing and cutting procedure based on the sensed electrical parameter.

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